# IGCSE 0607 Practice Questions

**Total Questions:** 1636 — **Total Marks:** 10052 Marks

## Table of Contents & Quick Navigation


- **E1.1** Types of number
  → **Paper 2 (Non-Calculator)** *(14 Qs)* · **Paper 4 (Calculator)** *(7 Qs)*
- **E1.2** Sets
  → **Paper 2 (Non-Calculator)** *(9 Qs)* · **Paper 4 (Calculator)** *(9 Qs)*
- **E1.3** Powers and roots
  → **Paper 2 (Non-Calculator)** *(12 Qs)* · **Paper 4 (Calculator)** *(6 Qs)*
- **E1.4** Fractions, decimals and percentages
  → **Paper 2 (Non-Calculator)** *(12 Qs)* · **Paper 4 (Calculator)** *(6 Qs)*
- **E1.5** Ordering
  → **Paper 2 (Non-Calculator)** *(11 Qs)* · **Paper 4 (Calculator)** *(7 Qs)*
- **E1.6** The four operations
  → **Paper 2 (Non-Calculator)** *(12 Qs)* · **Paper 4 (Calculator)** *(6 Qs)*
- **E1.7** Indices I
  → **Paper 2 (Non-Calculator)** *(14 Qs)* · **Paper 4 (Calculator)** *(1 Q)*
- **E1.8** Standard form
  → **Paper 2 (Non-Calculator)** *(11 Qs)* · **Paper 4 (Calculator)** *(7 Qs)*
- **E1.9** Estimation
  → **Paper 2 (Non-Calculator)** *(15 Qs)* · **Paper 4 (Calculator)** *(6 Qs)*
- **E1.10** Ratio and proportion
  → **Paper 2 (Non-Calculator)** *(8 Qs)* · **Paper 4 (Calculator)** *(10 Qs)*
- **E1.11** Rates
  → **Paper 2 (Non-Calculator)** *(7 Qs)* · **Paper 4 (Calculator)** *(14 Qs)*
- **E1.12** Percentages
  → **Paper 2 (Non-Calculator)** *(7 Qs)* · **Paper 4 (Calculator)** *(14 Qs)*
- **E1.13** Using a calculator
  → **Paper 4 (Calculator)** *(21 Qs)*
- **E1.14** Time
  → **Paper 2 (Non-Calculator)** *(7 Qs)* · **Paper 4 (Calculator)** *(14 Qs)*
- **E1.15** Money
  → **Paper 2 (Non-Calculator)** *(6 Qs)* · **Paper 4 (Calculator)** *(15 Qs)*

- **E1.16** Exponential growth and decay
  → **Paper 2 (Non-Calculator)** *(2 Qs)* · **Paper 4 (Calculator)** *(19 Qs)*

- **E1.17** Surds
  → **Paper 2 (Non-Calculator)** *(39 Qs)*

- **E2.1** Introduction to algebra
  → **Paper 2 (Non-Calculator)** *(18 Qs)* · **Paper 4 (Calculator)** *(12 Qs)*

- **E2.2** Algebraic manipulation
  → **Paper 2 (Non-Calculator)** *(24 Qs)* · **Paper 4 (Calculator)** *(6 Qs)*

- **E2.3** Algebraic fractions
  → **Paper 2 (Non-Calculator)** *(19 Qs)* · **Paper 4 (Calculator)** *(11 Qs)*

- **E2.4** Indices II
  → **Paper 2 (Non-Calculator)** *(25 Qs)* · **Paper 4 (Calculator)** *(8 Qs)*

- **E2.5** Equations
  → **Paper 2 (Non-Calculator)** *(28 Qs)* · **Paper 4 (Calculator)** *(20 Qs)*

- **E2.6** Inequalities
  → **Paper 2 (Non-Calculator)** *(29 Qs)* · **Paper 4 (Calculator)** *(22 Qs)*

- **E2.7** Sequences
  → **Paper 2 (Non-Calculator)** *(12 Qs)* · **Paper 4 (Calculator)** *(15 Qs)*

- **E2.8** Proportion
  → **Paper 2 (Non-Calculator)** *(12 Qs)* · **Paper 4 (Calculator)** *(14 Qs)*

- **E3.1** Graphs of functions
  → **Paper 2 (Non-Calculator)** *(22 Qs)* · **Paper 4 (Calculator)** *(20 Qs)*

- **E3.2** Sketching graphs on a calculator
  → **Paper 4 (Calculator)** *(18 Qs)*

- **E3.3** Functions
  → **Paper 2 (Non-Calculator)** *(22 Qs)* · **Paper 4 (Calculator)** *(17 Qs)*

- **E3.4** Finding a quadratic function using given information
  → **Paper 2 (Non-Calculator)** *(21 Qs)* · **Paper 4 (Calculator)** *(18 Qs)*

- **E3.5** Asymptotes
  → **Paper 2 (Non-Calculator)** *(9 Qs)* · **Paper 4 (Calculator)** *(9 Qs)*

- **E3.6** Transforming graphs of functions
  → **Paper 2 (Non-Calculator)** *(10 Qs)* · **Paper 4 (Calculator)** *(8 Qs)*

- **E3.7** The logarithmic function
  → **Paper 2 (Non-Calculator)** *(13 Qs)* · **Paper 4 (Calculator)** *(26 Qs)*

- **E4.1** Coordinates
  → **Paper 2 (Non-Calculator)** *(10 Qs)* · **Paper 4 (Calculator)** *(5 Qs)*

- **E4.2** Gradient of linear graphs
  → **Paper 2 (Non-Calculator)** *(12 Qs)* · **Paper 4 (Calculator)** *(6 Qs)*

- **E4.3** Length and midpoint
  → **Paper 2 (Non-Calculator)** *(11 Qs)* · **Paper 4 (Calculator)** *(7 Qs)*

- **E4.4** Equations of linear graphs
  → **Paper 2 (Non-Calculator)** *(12 Qs)* · **Paper 4 (Calculator)** *(6 Qs)*

- **E4.5** Parallel lines
  → **Paper 2 (Non-Calculator)** *(10 Qs)* · **Paper 4 (Calculator)** *(8 Qs)*

- **E4.6** Perpendicular lines
  → **Paper 2 (Non-Calculator)** *(12 Qs)* · **Paper 4 (Calculator)** *(6 Qs)*

- **E5.1** Geometrical terms
  → **Paper 2 (Non-Calculator)** *(15 Qs)* · **Paper 4 (Calculator)** *(9 Qs)*

- **E5.2** Angle measurement in degrees
  → **Paper 2 (Non-Calculator)** *(12 Qs)* · **Paper 4 (Calculator)** *(12 Qs)*

- **E5.3** Similarity
  → **Paper 2 (Non-Calculator)** *(16 Qs)* · **Paper 4 (Calculator)** *(23 Qs)*

- **E5.4** Symmetry
  → **Paper 2 (Non-Calculator)** *(12 Qs)* · **Paper 4 (Calculator)** *(9 Qs)*

- **E5.5** Angles
  → **Paper 2 (Non-Calculator)** *(11 Qs)* · **Paper 4 (Calculator)** *(10 Qs)*

- **E5.6** Circle theorems I
  → **Paper 2 (Non-Calculator)** *(24 Qs)* · **Paper 4 (Calculator)** *(21 Qs)*

- **E5.7** Circle theorems II
  → **Paper 2 (Non-Calculator)** *(10 Qs)* · **Paper 4 (Calculator)** *(14 Qs)*

- **E6.1** Units of measure
  → **Paper 2 (Non-Calculator)** *(7 Qs)* · **Paper 4 (Calculator)** *(11 Qs)*

- **E6.2** Area and perimeter
  → **Paper 2 (Non-Calculator)** *(7 Qs)* · **Paper 4 (Calculator)** *(11 Qs)*

- **E6.3** Circles, arcs and sectors
  → **Paper 2 (Non-Calculator)** *(8 Qs)* · **Paper 4 (Calculator)** *(10 Qs)*

- **E6.4** Surface area and volume
  → **Paper 2 (Non-Calculator)** *(7 Qs)* · **Paper 4 (Calculator)** *(11 Qs)*

- **E6.5** Compound shapes and parts of shapes
  → **Paper 2 (Non-Calculator)** *(13 Qs)* · **Paper 4 (Calculator)** *(23 Qs)*

- **E7.1** Pythagoras' theorem
  → **Paper 2 (Non-Calculator)** *(10 Qs)* · **Paper 4 (Calculator)** *(8 Qs)*

- **E7.2** Right-angled triangles
  → **Paper 2 (Non-Calculator)** *(1 Q)* · **Paper 4 (Calculator)** *(17 Qs)*

- **E7.3** Exact trigonometric values
  → **Paper 2 (Non-Calculator)** *(15 Qs)*

- **E7.4** Trigonometric functions
  → **Paper 2 (Non-Calculator)** *(10 Qs)* · **Paper 4 (Calculator)** *(11 Qs)*

- **E7.5** Non-right-angled triangles
  → **Paper 2 (Non-Calculator)** *(3 Qs)* · **Paper 4 (Calculator)** *(36 Qs)*

- **E7.6** Pythagoras' theorem and trigonometry in 3D
  → **Paper 2 (Non-Calculator)** *(2 Qs)* · **Paper 4 (Calculator)** *(7 Qs)*

- **E8.1** Transformations
  → **Paper 2 (Non-Calculator)** *(13 Qs)* · **Paper 4 (Calculator)** *(11 Qs)*

- **E8.2** Vectors in two dimensions
  → **Paper 2 (Non-Calculator)** *(9 Qs)* · **Paper 4 (Calculator)** *(12 Qs)*

- **E8.3** Magnitude of a vector
  → **Paper 2 (Non-Calculator)** *(8 Qs)* · **Paper 4 (Calculator)** *(7 Qs)*

- **E9.1** Introduction to probability
  → **Paper 2 (Non-Calculator)** *(10 Qs)* · **Paper 4 (Calculator)** *(5 Qs)*

- **E9.2** Relative and expected frequencies
  → **Paper 2 (Non-Calculator)** *(6 Qs)* · **Paper 4 (Calculator)** *(9 Qs)*

- **E9.3** Probability of combined events
  → **Paper 2 (Non-Calculator)** *(22 Qs)* · **Paper 4 (Calculator)** *(23 Qs)*

- **E10.1** Classifying statistical data
  → **Paper 2 (Non-Calculator)** *(8 Qs)* · **Paper 4 (Calculator)** *(10 Qs)*

- **E10.2** Interpreting statistical data
  → **Paper 2 (Non-Calculator)** *(8 Qs)* · **Paper 4 (Calculator)** *(10 Qs)*

- **E10.3** Discrete and continuous data
  → **Paper 2 (Non-Calculator)** *(9 Qs)* · **Paper 4 (Calculator)** *(9 Qs)*

- **E10.4** Averages and measures of spread
  → **Paper 2 (Non-Calculator)** *(6 Qs)* · **Paper 4 (Calculator)** *(9 Qs)*

- **E10.5** Averages on a calculator
  → **Paper 4 (Calculator)** *(15 Qs)*

- **E10.6** Statistical charts and diagrams
  → **Paper 2 (Non-Calculator)** *(6 Qs)* · **Paper 4 (Calculator)** *(12 Qs)*

- **E10.7** Scatter diagrams
  → **Paper 2 (Non-Calculator)** *(3 Qs)* · **Paper 4 (Calculator)** *(12 Qs)*

- **E10.8** Cumulative frequency diagrams
  → **Paper 2 (Non-Calculator)** *(4 Qs)* · **Paper 4 (Calculator)** *(13 Qs)*

**1** **(a)** Express 504 as a product of its prime factors.

..... **[2]**

**(b)** Find the smallest positive integer $k$ such that $504k$ is a square number.

$k =$ ..... **[2]**

**2** **(a)** Write 84 as a product of its prime factors.

..... **[2]**

**(b)** Find the lowest common multiple (LCM) of 84 and 126.

..... **[2]**

**3** Here is a list of numbers.

2 9 12 23 36 45 64 81

**(a)** Write down a number from the list which is:

**(i)** a multiple of 12,

..... **[1]**

**(ii)** a cube number,

..... **[1]**

**(iii)** both a prime number and a factor of 46.

..... **[1]**

**(b)** Find the lowest common multiple (LCM) of 36 and 64.

..... **[2]**

**4** Here is a list of numbers.

$$0.25 \quad \sqrt{7} \quad 8 \quad 9 \quad 13 \quad -5 \quad 21 \quad 64$$

**(a)** From the list, write down a prime number.

..... **[1]**

**(b)** From the list, write down an irrational number.

..... **[1]**

**(c)** From the list, write down an integer less than zero.

..... **[1]**

**(d)** From the list, write down the reciprocal of 4.

..... **[1]**

**(e)** From the list, write down a number that is both a square number and a cube number.

..... **[1]**

**5** Here is a list of numbers.

8 15 23 27 36 49 64

**(a)** From the list, write down

**(i)** a prime number,

..... **[1]**

**(ii)** a cube number,

..... **[1]**

**(iii)** a number that is both a square number and a cube number.

..... **[1]**

**(b)** Express 36 as a product of its prime factors.

..... **[2]**

**6** Here is a list of numbers.

2 9 15 16 23 27 64 72

**(a)** From the list, write down

**(i)** a prime number,

..... **[1]**

**(ii)** a cube number,

..... **[1]**

**(iii)** a number that is both a square number and a cube number,

..... **[1]**

**(iv)** a triangle number.

..... **[1]**

**(b)** Express 72 as a product of its prime factors.

..... **[2]**

**7** **(a)** Write 168 as a product of its prime factors.

..... **[2]**

**(b)** Find the highest common factor (HCF) of 168 and 252.

..... **[2]**

**(c)** Find the lowest common multiple (LCM) of 168 and 252.

..... **[2]**

**8** Here is a list of numbers.

$$-5 \qquad -1 \qquad \frac{3}{8} \qquad 8 \qquad 13 \qquad 25 \qquad \sqrt{30} \qquad \pi$$

**(a)** From this list, write down:

**(i)** a prime number,

..... **[1]**

**(ii)** a cube number,

..... **[1]**

**(iii)** an irrational number,

..... **[1]**

**(iv)** an integer that is not a natural number.

..... **[1]**

**(b)** Find the reciprocal of 0.8.
Give your answer as a fraction in its simplest form.

..... **[2]**

**9** Two numbers, $A$ and $B$, are written as products of their prime factors.

$A = 2^3 \times 3^2 \times 5$

$B = 2^2 \times 3^4 \times 7$

**(a)** Find the highest common factor (HCF) of $A$ and $B$.

..... **[2]**

**(b)** Find the lowest common multiple (LCM) of $A$ and $B$.
Give your answer as a product of prime factors.

..... **[2]**

**(c)** The integer $C$ is such that:

- the HCF of $A$ and $C$ is 6
- the LCM of $A$ and $C$ is $2^3 \times 3^2 \times 5 \times 11$

Find the value of $C$.

$C =$ ..... **[3]**

**10** **(a)** Let $x$ be a rational number and $y$ be an irrational number.
State whether $x + y$ is rational or irrational.

..... **[1]**

**(b)** Give an example of an irrational number $y$ to show that $y^2$ can be a rational number.

**[1]**

**(c)** The $n$th term of a sequence is given by the expression

$n^2 + n + 41$

**(i)** Work out the 3rd term of this sequence, and state whether it is a prime number.

..... **[2]**

**(ii)** Explain why the 41st term of this sequence is not a prime number.

..... **[2]**

**11** The number $N$ is written as a product of its prime factors as $N = 2^3 \times 3^x \times 5^2$.

**(a)** The highest common factor (HCF) of $N$ and 540 is 180.
Find the value of $x$.

$x =$ ..... **[3]**

**(b)** The lowest common multiple (LCM) of $N$ and a number $M$ is $2^4 \times 3^3 \times 5^2 \times 7$.
Given that $M = 2^y \times 3^z \times 7$, find the value of $y$ and the value of $z$.

$y =$ ..... , $z =$ ..... **[2]**

**(c)** Find the smallest positive integer $k$ such that $Nk$ is a cube number.

$k =$ ..... **[2]**

**12** **(a)** The highest common factor (HCF) of $x$ and 12 is 4.
The lowest common multiple (LCM) of $x$ and 12 is 120.
Find the value of $x$.

$x =$ ..... **[2]**

**(b)** **(i)** Express 1008 as a product of its prime factors.

..... **[2]**

**(ii)** An integer $N$ is a perfect cube.
The highest common factor (HCF) of $N$ and 1008 is 72.
Find the smallest possible positive value of $N$.

$N =$ ..... **[3]**

**13** Let $m = p^3q$ and $n = pq^3$, where $p$ and $q$ are distinct prime numbers.

**(a)** Express, in terms of $p$ and $q$:

**(i)** the highest common factor (HCF) of $m$ and $n$,

..... **[1]**

**(ii)** the lowest common multiple (LCM) of $m$ and $n$.

..... **[1]**

**(b)** The HCF of $m$ and $n$ is 15.
It is given that $m > n$.
Find the value of $p$ and the value of $q$.

$p =$ ..... , $q =$ ..... **[3]**

**(c)** Show algebraically that the product $m \times n$ is a square number.

**[2]**

**(d)** Using your values of $p$ and $q$, find the smallest positive integer $k$ such that $m \times k$ is a cube number.

$k =$ ..... **[2]**

**14** **(a)** The lowest common multiple (LCM) of two numbers $p$ and $q$ is 1260.
The highest common factor (HCF) of $p$ and $q$ is 14.
Given that $p = 70$, find the value of $q$.

$q =$ ..... **[3]**

**(b)** Consider the two numbers $M$ and $N$ given by:

$M = 2^x \times 3^2 \times 5$

$N = 2^3 \times 3^y \times 7$

The highest common factor (HCF) of $M$ and $N$ is 72.
The lowest common multiple (LCM) of $M$ and $N$ is 10080.
Find the value of $x$ and the value of $y$.

$x =$ ..... , $y =$ ..... **[4]**

**(c)** Given that $n$ is a positive integer, explain why $n^2 + n$ is always an even number.

.....

.....

..... **[2]**

**15** Three numbers $A$, $B$ and $C$ are given as products of their prime factors.

$$A = 2^3 \times 3^2 \times 5 \qquad B = 2^2 \times 3^x \times 5^2 \qquad C = 2^y \times 3 \times 7$$

The highest common factor (HCF) of $A$ and $B$ is 180.
The lowest common multiple (LCM) of $B$ and $C$ is 12600.

Find the value of $x$ and the value of $y$.

$x =$ ..... , $y =$ ..... **[4]**

**16** Two numbers, $A$ and $B$, are given as a product of their prime factors.

$$A = 2^3 \times 3^2 \times 5 \qquad B = 2^2 \times 3^3 \times 7$$

**(a)** Find the highest common factor (HCF) of $A$ and $B$.

..... **[1]**

**(b)** Find the lowest common multiple (LCM) of $A$ and $B$.
Give your answer as a product of its prime factors.

..... **[2]**

**(c)** $A \times k$ is a perfect square.
Find the smallest positive integer value of $k$.

$k =$ ..... **[2]**

**17** Number $A$ and Number $B$ are given as products of their prime factors.

$A = 2^2 \times 3^3 \times 5$

$B = 2^x \times 3 \times 7$

**(a)** The highest common factor (HCF) of $A$ and $B$ is 6.
Find the value of $x$.

$x =$ ..... **[2]**

**(b)** Number $C = 2^2 \times 3^y \times 5^2$.
The lowest common multiple (LCM) of $A$ and $C$ is 8100.
Find the value of $y$.

$y =$ ..... **[3]**

**(c)** Here is a list of numbers.

$$\sqrt{11} \qquad -3.5 \qquad 4\pi \qquad \frac{7}{9} \qquad \sqrt{144} \qquad 0.\dot{4}$$

Write down all the irrational numbers from this list.

..... **[2]**

**18** **(a)** **(i)** Write down a rational number between 0.4 and 0.5.

..... **[1]**

**(ii)** Write down an irrational number between 4 and 5.

..... **[1]**

**(b)** Give an example to show that the sum of two different prime numbers can be a square number.

**[2]**

**(c)** Write down the 6th triangle number.

..... **[1]**

**(d)** Find the highest common factor (HCF) of $24a^3b^2$ and $36a^2b^4$.

..... **[2]**

**19** **(a)** Find the highest common factor (HCF) of 84 and 120.

..... **[2]**

**(b)** Find the lowest common multiple (LCM) of 84 and 120.

..... **[2]**

**(c)** Write down the reciprocal of 0.8 as a fraction in its simplest form.

..... **[2]**

**(d)** State whether $\sqrt{120}$ is rational or irrational.
Give a reason for your answer.

..... **[2]**

**20** **(a)** $x, y$ and $z$ are positive integers.

- $x$ is a prime number.
- $y$ is a cube number greater than 1.
- $z$ is a triangle number.

It is given that $x + y = z$ and $x < 10$.

Find two possible sets of values for $x, y$ and $z$.

**(i)** First set of values.

$x =$ ..... , $y =$ ..... , $z =$ ..... **[2]**

**(ii)** Second set of values.

$x =$ ..... , $y =$ ..... , $z =$ ..... **[2]**

**(b)** $p$ and $q$ are two different irrational numbers.

Write down an example of $p$ and $q$ such that:

**(i)** $p + q$ is a rational number.

$p =$ ..... , $q =$ ..... **[1]**

**(ii)** $p \times q$ is a rational number.

$p =$ ..... , $q =$ ..... **[1]**

**21** **(a)** The number 168 can be written as $2^3 \times 3 \times 7$.
The number $N$ can be written as $2^x \times 3^2 \times 7^y$, where $x$ and $y$ are integers.
The highest common factor (HCF) of 168 and $N$ is 84.
The lowest common multiple (LCM) of 168 and $N$ is 3528.

Find the value of $x$ and the value of $y$.

$x =$ ..... , $y =$ ..... **[3]**

**(b)** Find the exact value of $N$.

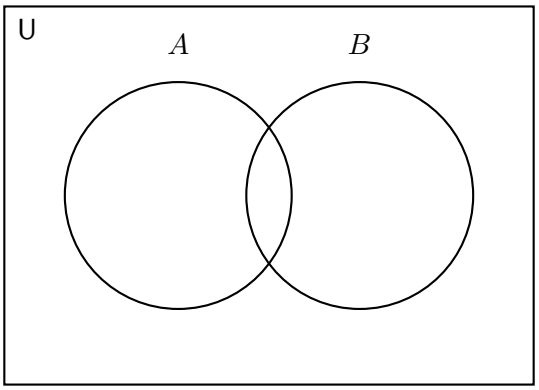
$N =$ ..... **[1]**

**(c)** Two flashing lights, A and B, are turned on at the same time.
Light A flashes every 45 seconds.
Light B flashes every 72 seconds.
They both flash together at 10:15:00.
Work out the next time they flash together.

..... **[3]**

**22** The universal set $\mathsf{U}$ contains all integers from 1 to 10 inclusive.
Set $A$ contains all the even numbers in $\mathsf{U}$.
Set $B$ contains all the prime numbers in $\mathsf{U}$.

**(a)** Complete the Venn diagram to show the elements of these sets.




**[2]**

**(b)** List the elements of $A \cap B$.

..... **[1]**

**(c)** Find $n(A \cup B)'$.

..... **[1]**

**23** $\mathsf{U} = \{x : x \text{ is an integer and } 1 \leq x \leq 10\}$
$A = \{x : x \text{ is a prime number}\}$
$B = \{x : x \text{ is a factor of } 12\}$

**(a)** List the members of set $A$.

..... **[1]**

**(b)** List the members of set $B$.

..... **[1]**

**(c)** Find $n(A \cup B)$.

..... **[1]**

**(d)** Write down the elements of $A \cap B$.

..... **[1]**

**(e)** Write down a number that is in $\mathsf{U}$ but is not in $A \cup B$.

..... **[1]**

**24** The universal set $\mathsf{U}$ and sets $A$ and $B$ are defined as follows:

- $\mathsf{U} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$
- $A = \{2, 3, 5, 7\}$
- $B = \{1, 3, 5, 7, 9\}$

**(a)** List the elements of $A \cap B$.

..... **[1]**

**(b)** Find $n((A \cup B)')$.

..... **[2]**

**(c)** Write down the correct set notation symbol $(\in, \notin, \subset, \cup, \cap)$ to complete the statement:

$2 \ldots A$

**[1]**

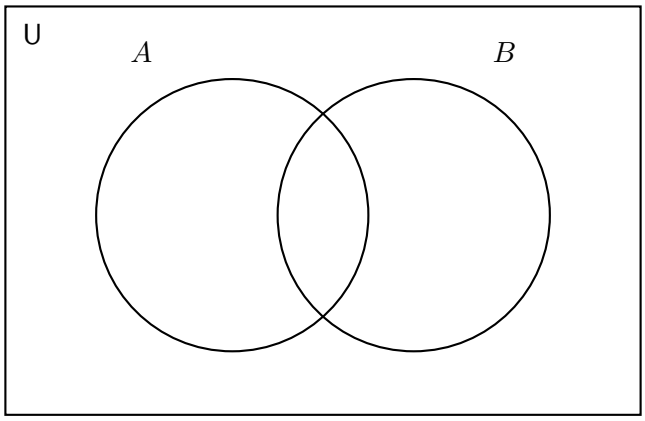
**(d)** Write down the correct set notation symbol to complete the statement:

$\{1, 9\} \ldots B$

**[1]**

**25** Let U be the set of integers from 1 to 10 inclusive.
Let $A$ be the set of prime numbers.
Let $B$ be the set of odd numbers.

**(a)** Complete the Venn diagram to show the elements of these sets.




**[2]**

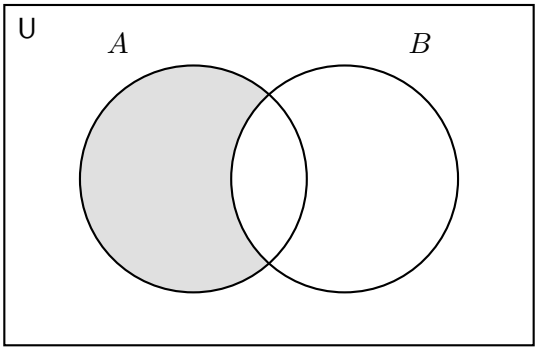
**(b)** List the elements of $A \cap B'$.

..... **[1]**

**(c)** Find $n(A \cup B)$.

..... **[1]**

**(d)** Write down the set notation for the shaded region in the Venn diagram above.




..... **[1]**

**26** The universal set $\mathsf{U}$ and sets $P$ and $F$ are defined as follows:

- $\mathsf{U} = \{x : x \text{ is an integer} 1 \leq x \leq 10\}$
- $P = \{x : x \text{ is a prime number}\}$
- $F = \{x : x \text{ is a factor of } 18\}$

**(a)** Write down the elements of set $F$.

$F = \{$ ..... $\}$ **[1]**

**(b)** List the elements of $P \cap F$.

$P \cap F = \{$ ..... $\}$ **[1]**

**(c)** Find $n((P \cup F)')$.

..... **[2]**

**(d)** Write down the correct set symbol, $\in$ or $\notin$, to make this statement true:

$5 \ldots\ldots F$

**[1]**

**27** The universal set $\mathsf{U}$ and sets $A$ and $B$ are defined as follows:

$\mathsf{U} = \{x : x \text{ is an integer and } 1 \leq x \leq 10\}$

$A = \{x : x \text{ is a prime number}\}$

$B = \{x : x \text{ is a factor of } 10\}$

**(a)** List the elements of set $A$.

..... **[1]**

**(b)** Find $n(A \cup B)$.

..... **[2]**

**(c)** Complete each statement by inserting the correct symbol from $\in, \notin, \subseteq, \cup, \cap$.

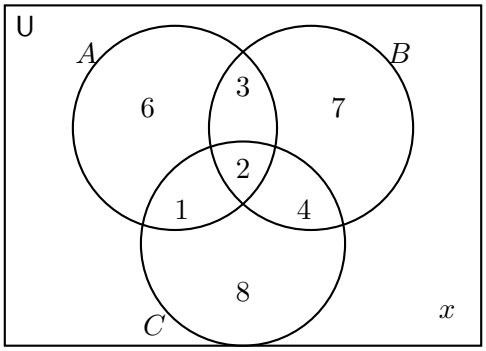
**(i)** $4$ ..... $A$

**[1]**

**(ii)** $A \cap B$ ..... $A$

**[1]**

**28** The Venn diagram shows the number of elements in each region for sets $A$, $B$, and $C$.




**(a)** Given that $n(\mathsf{U}) = 35$, find the value of $x$.

$x =$ ..... **[2]**

**(b)** Find $n(A \cap C')$.

..... **[1]**

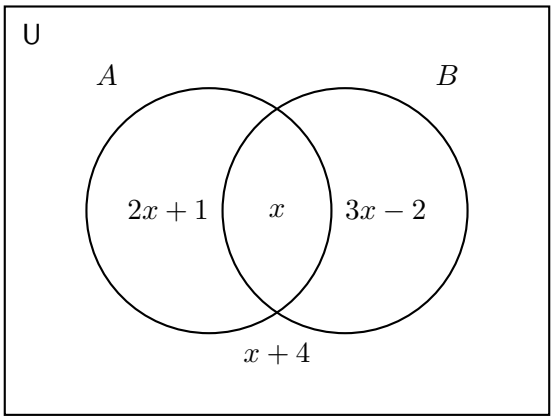
**(c)** Use set notation to describe the set of elements whose number is 4.

..... **[1]**

**(d)** Find $n(A \cup (B \cap C))$.

..... **[2]**

**29** The Venn diagram shows the number of elements in sets $A$ and $B$, in terms of $x$.
$\mathsf{U}$ is the universal set and $n(\mathsf{U}) = 52$.



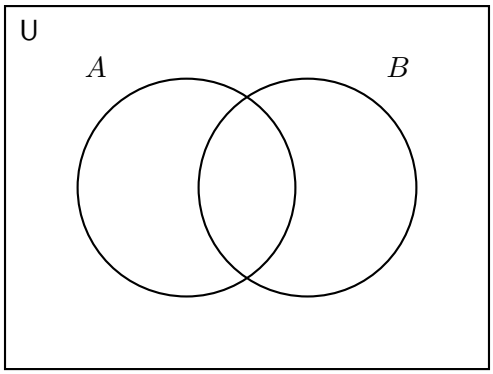

**(a)** Form an equation in $x$ and solve it.

$x =$ ..... **[3]**

**(b)** Find $n(A \cup B')$.

..... **[2]**

**(c)** On the Venn diagram below, shade the region $A \cap B'$.




**[1]**

**30** The universal set $\mathsf{U}$ is the set of all real numbers.
Four sets $P$, $Q$, $R$ and $T$ are defined as follows:

$P = \{x : -2 < x \leq 5\}$

$Q = \{x : -4 \leq x < 3\}$

$R = \{x : x > 1\}$

$T = \{x : x \leq -1 \text{ or } x \geq 2\}$

**(a)** Write down the set $P \cap Q$ as an inequality.

..... **[2]**

**(b)** Find the integer elements of the set $P \cap R'$.

..... **[2]**

**(c)** Let $S = \{x : x \leq k\}$.
Given that $P \cap S = P$, find the range of possible values for $k$.

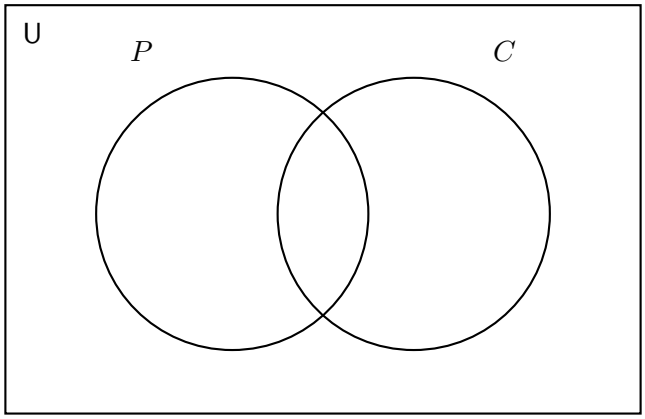
..... **[2]**

**(d)** Write down the set $Q \cap T$ as inequalities.

..... **[2]**

**31** There are 45 students in a class.
Let U be the set of students in this class.
Let $P$ be the set of students who study Physics.
Let $C$ be the set of students who study Chemistry.
$n(P) = 28$
$n(C) = 22$
$n(P \cap C) = x$
$n((P \cup C)') = 4$

**(a)** Complete the Venn diagram in terms of $x$.




**[2]**

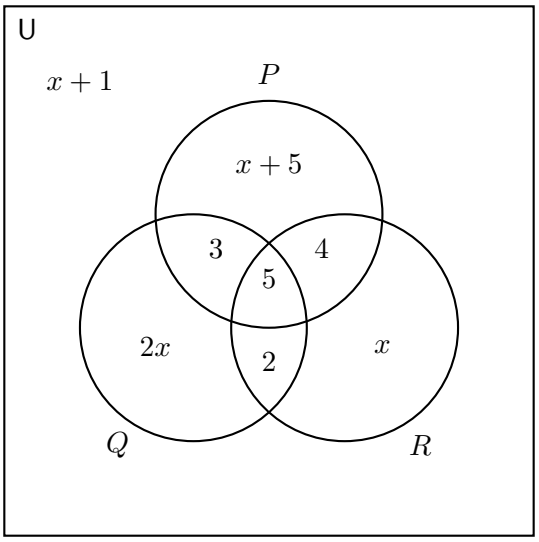
**(b)** Form an equation in $x$ and solve it to find the value of $x$.

$x =$ ..... **[2]**

**(c)** Find $n(P \cap C')$.

..... **[1]**

**32** The Venn diagram shows the number of elements in each region for the sets $P$, $Q$ and $R$.



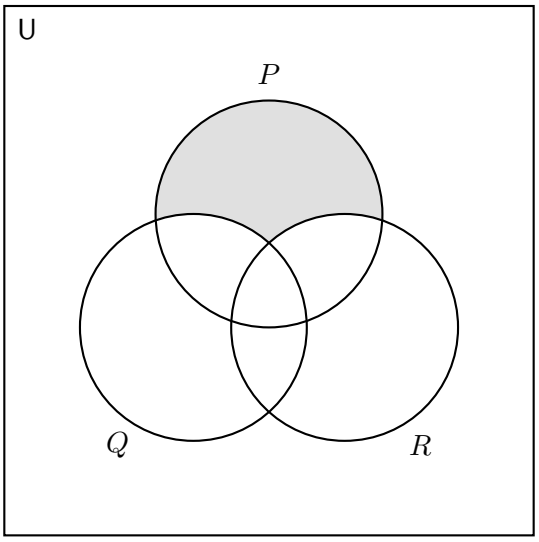

**(a)** The total number of elements in the universal set is 45.
Form an equation in terms of $x$ and solve it to find the value of $x$.

$x =$ ..... **[3]**

**(b)** Find $n(P \cup Q)$.

..... **[1]**

**(c)** On the Venn diagram below, shade the region $P \cap (Q \cup R)'$.

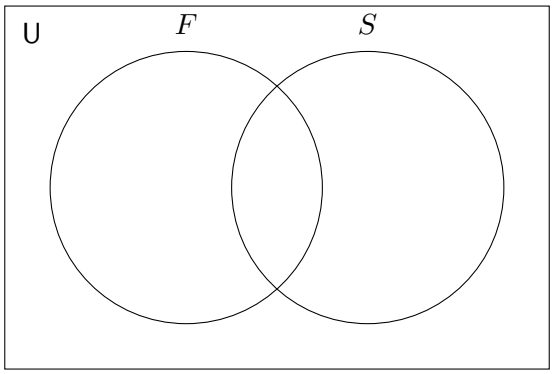



**[1]**

**33** In a class of 40 students:

- 25 study French ($F$)
- 18 study Spanish ($S$)
- 5 study neither French nor Spanish

**(a)** Complete the Venn diagram to show this information.




**[3]**

**(b)** Find $n(F \cap S')$.

..... **[1]**

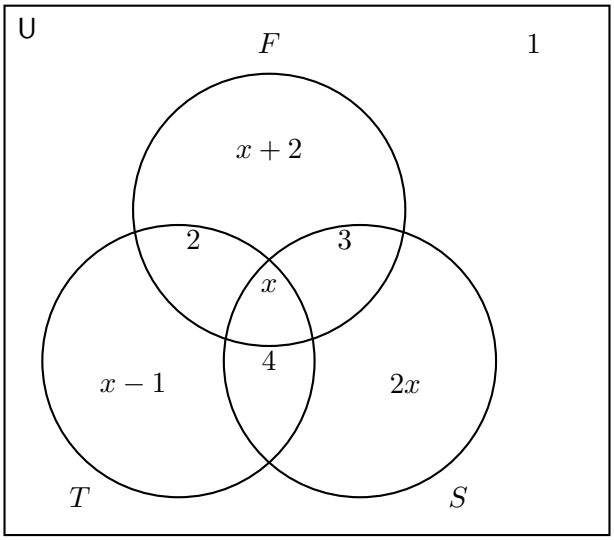
**(c)** A student is chosen at random from the class.
Find the probability that they study exactly one of the two languages.

..... **[2]**

**(d)** Two students are chosen at random from those who study French.
Find the probability that both also study Spanish.

..... **[1]**

**34** A group of 36 students were asked which sports they play.
The sports are Football ($F$), Tennis ($T$) and Swimming ($S$).
The Venn diagram shows the number of students in each region, in terms of $x$.




**(a)** Write down an equation in $x$ and solve it to find the value of $x$.

$x =$ ..... **[3]**

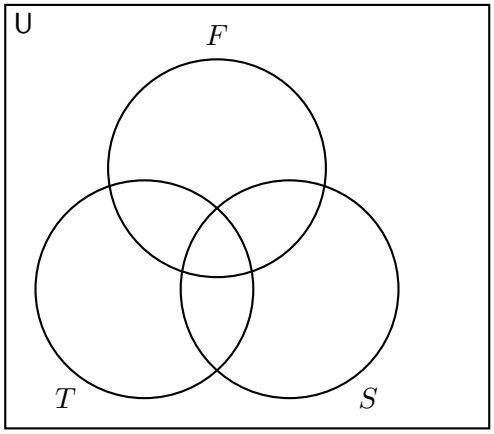
**(b)** Find the number of students who play Football.

..... **[2]**

**(c)** Find $n(T' \cap S)$.

..... **[2]**

**(d)** On the Venn diagram, shade the region $F \cap (T \cup S)'$.




**[1]**

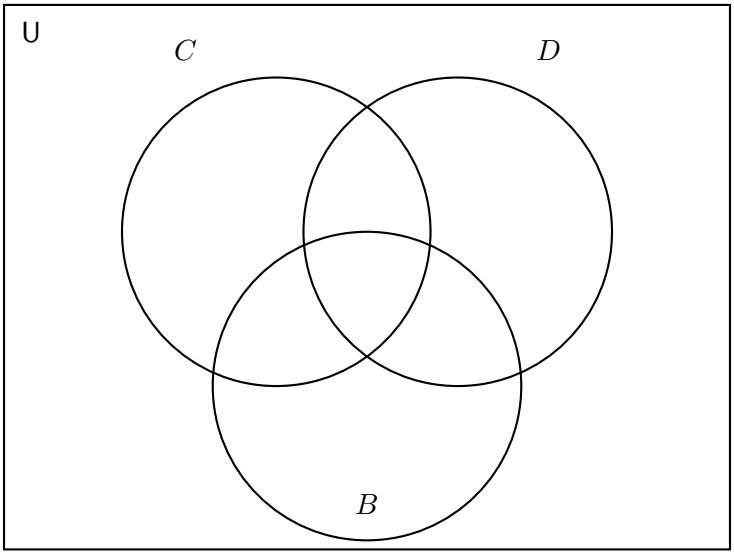
**35** In a survey of 100 people, information was collected about the pets they own.

Let $\mathsf{U}$ be the set of people surveyed.
Let $C$ be the set of people who own a cat.
Let $D$ be the set of people who own a dog.
Let $B$ be the set of people who own a bird.

- 8 people own all three pets.
- 12 people own a cat and a dog but not a bird.
- 7 people own a cat and a bird but not a dog.
- 10 people own a dog and a bird but not a cat.
- 3 people do not own any of these pets.

The number of people who own only a cat is $x$.
The number of people who own only a bird is also $x$.
The number of people who own only a dog is $2x$.

**(a)** Complete the Venn diagram to show this information.




**[3]**

**(b)** Find the value of $x$.

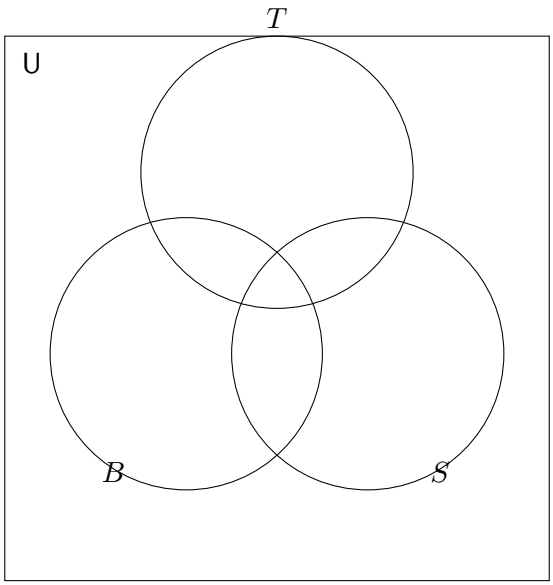
$x =$ ..... **[2]**

**(c)** Find $n((C \cup D) \cap B')$.

..... **[2]**

**36** There are 60 people in a sports club.

- 30 play Tennis ($T$)
- 28 play Badminton ($B$)
- 25 play Squash ($S$)
- 12 play both Tennis and Badminton
- 9 play both Tennis and Squash
- 11 play both Badminton and Squash
- 4 play all three sports




**(a)** Complete the Venn diagram.

**[3]**

**(b)** Find $n(T \cup B \cup S)'$.

..... **[1]**

**(c)** Find $n(T \cap (B \cup S))$.

..... **[1]**

**(d)** Use set notation to describe the region containing the number 9 (the number of people who play Badminton only).

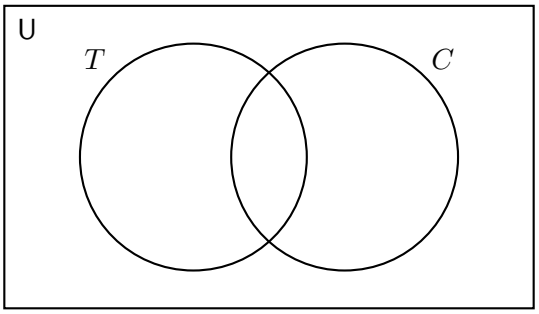
..... **[2]**

**37** In a sports club, the universal set $\mathsf{U}$ represents all the students.
The set $T$ represents students who play Tennis.
The set $C$ represents students who play Cricket.

You are given the following information:

- $n(\mathsf{U}) = 50$
- $n(T) = 3x$
- $n(C) = 4x$
- $n(T \cap C) = x$
- $n((T \cup C)') = 2$

**(a)** Complete the Venn diagram, writing the number of students in each region in terms of $x$ where appropriate.




**[3]**

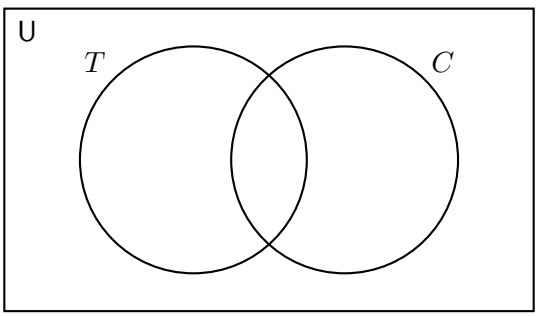
**(b)** Form an equation in $x$ and solve it to find the value of $x$.

$x =$ ..... **[2]**

**(c)** Find the number of students who play Tennis but not Cricket.

..... **[1]**

**(d)** On the Venn diagram below, shade the region $T' \cup C$.

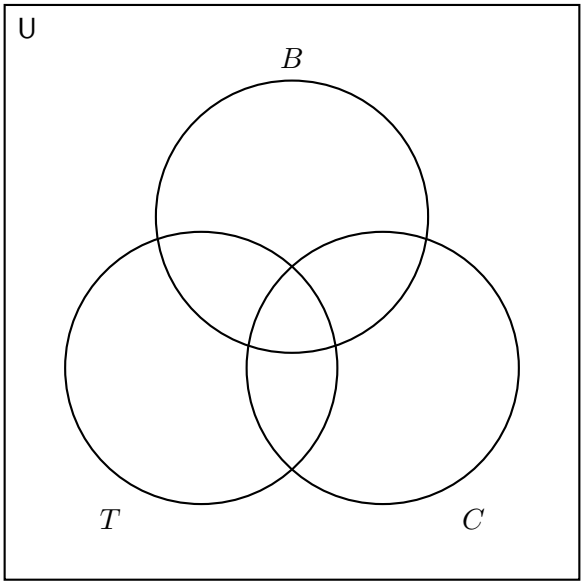



**[2]**

**38** In a survey of 103 people, the following information was recorded about their use of three types of transport: Bus ($B$), Train ($T$), and Car ($C$).

- 15 people use all three types.
- 28 people use both a Train and a Car.
- 22 people use both a Bus and a Car.
- 30 people use both a Bus and a Train.
- $x$ people use only a Bus.
- The number of people who use only a Train is twice the number of people who use only a Bus.
- The number of people who use only a Car is 5 more than the number of people who use only a Bus.
- 8 people do not use any of these types of transport.

**(a)** Complete the Venn diagram in terms of $x$ to show this information.




**[3]**

**(b)** Find the value of $x$.

$x =$ ..... **[2]**

**(c)** Find $n(T \cup C)$.

..... **[1]**

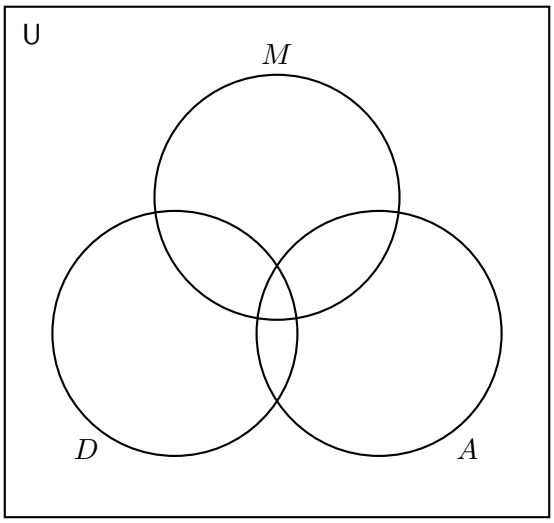
**(d)** A person is chosen at random from those who use a Train.
Find the probability that this person also uses a Car but does not use a Bus.

..... **[2]**

**39** In a year group of 120 students, they can choose to study Music ($M$), Drama ($D$), and Art ($A$).

- 12 students study all three subjects.
- 25 students study Music and Drama.
- 32 students study Drama and Art.
- 28 students study Music and Art.
- 50 students study Music.
- 65 students study Drama.
- $x$ students study Art only.
- 15 students study none of these three subjects.

**(a)** Use this information to complete the Venn diagram.




**[3]**

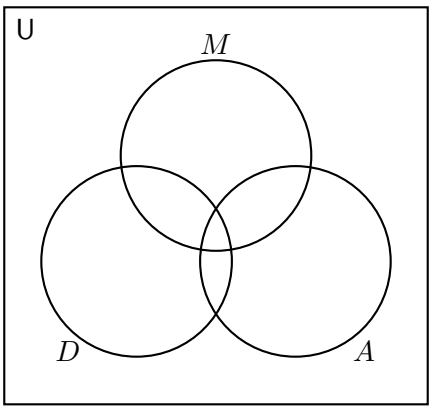
**(b)** Find the value of $x$.

$x =$ ..... **[2]**

**(c)** Find $n(M \cup (D \cap A))$.

..... **[2]**

**(d)** On the Venn diagram below, shade the region $M \cap (D \cup A)'$.




**[1]**

**40** **(a)** Write down the value of $\sqrt{121}$.

..... **[1]**

**(b)** Work out the value of

$$4^3 - \sqrt[3]{27}$$

..... **[2]**

**41** **(a)** Work out the value of

$$2^3 \times \sqrt{81}$$

..... **[2]**

**(b)** Write down the value of $\sqrt[3]{125}$.

..... **[1]**

**42** **(a)** Work out the value of

$$\sqrt{169} + 2^3$$

..... **[2]**

**(b)** Work out the value of

$$4^2 \times \sqrt[3]{125}$$

..... **[2]**

**43** **(a)** Work out the value of

$3^3 - \sqrt{144}$

..... **[2]**

**(b)** Work out the value of

$2^4 \times \sqrt[3]{125}$

..... **[2]**

**44** **(a)** Write down the value of

$\sqrt{196}$

..... **[1]**

**(b)** Work out the value of

$4^3 - \sqrt[3]{125}$

..... **[2]**

**(c)** Calculate the exact value of

$\sqrt{13^2 - 12^2}$

..... **[2]**

**45** **(a)** Work out the value of

$5^2 - \sqrt{121}$

..... **[1]**

**(b)** Work out the value of

$\sqrt[3]{27} \times 2^4$

..... **[2]**

**(c)** Work out the exact value of

$$\frac{6^2 + \sqrt{64}}{\sqrt[3]{125}}$$

..... **[2]**

**46** **(a)** Work out the exact value of

$$\frac{5^2 + \sqrt{144}}{2^3 - \sqrt[3]{1}}$$

..... **[2]**

**(b)** Find the value of $k$ when

$$3^k = \sqrt{81} \times 3^2$$

$k =$ ..... **[2]**

**47** **(a)** Calculate the value of

$$\frac{4^3 + \sqrt{225}}{\sqrt[3]{64}}$$

Give your answer as a decimal.

..... **[3]**

**(b)** Given that $p = 5$ and $q = -2$, work out the value of

$$p^2 - q^3$$

..... **[2]**

**48** **(a)** A cube has a volume of 125 $cm^3$.
Find the length of one edge of the cube.

..... cm **[1]**

**(b)** Work out the total surface area of this cube.

..... $cm^2$ **[2]**

**(c)** Given that $\sqrt{N} = 3^3 - 4^2$, find the value of $N$.

$N =$ ..... **[2]**

**49** **(a)** Work out the exact value of

$$\left(\frac{\sqrt{225}+\sqrt[3]{1000}}{5^2}\right)^{-2}$$

..... **[3]**

**(b)** Solve the equation

$$x^{\frac{2}{3}} = \frac{\sqrt{144}}{3}$$

$x =$ ..... **[3]**

**50** Solve the following equations.

**(a)**

$$\sqrt[3]{3x+4} = 4$$

$x =$ ..... **[3]**

**(b)**

$$x^2 = 4^3 + 6^2$$

$x =$ ..... , or $x =$ ..... **[2]**

**(c)**

$$\sqrt{x} + 2\sqrt{x} = 21$$

$x =$ ..... **[2]**

**51** **(a)** Find the integer $k$ such that

$$\sqrt[3]{k} = 4^2 - 3^2 - 2$$

$k =$ ..... **[2]**

**(b)** Calculate the exact value of

$$\frac{\sqrt{10^2 - 6^2}}{\sqrt[3]{125} - 3^2}$$

..... **[3]**

**(c)** Given that $y > 0$ and $\sqrt[3]{y} + \sqrt[3]{y} = 6$, work out the value of $y^2$.

..... **[3]**

**52** **(a)** Calculate the value of

$$\frac{\sqrt[3]{450} + 12.5^2}{3.4^3 - \sqrt{80}}$$

Write down all the figures on your calculator display.

..... **[2]**

**(b)** Write your answer to part (a) correct to 3 decimal places.

..... **[1]**

**(c)** Find the largest integer $n$ such that $n^4 < 2000$.

$n =$ ..... **[2]**

**53** **(a)** **(i)** Calculate the value of

$$\sqrt[3]{8.5^2 - 4.1^3}$$

Write down your full calculator display.

..... **[2]**

**(ii)** Write your answer to part (a)(i) correct to 3 significant figures.

..... **[1]**

**(b)** Calculate the exact value of

$$\frac{4.8^2}{\sqrt{0.0576}}$$

..... **[2]**

**54** **(a)** Calculate the value of

$$\frac{\sqrt{15.2^2 - 8.9^2}}{3.1^3}$$

..... **[2]**

**(b)** Given that $p = 4.2 \times 10^5$ and $q = 8.1 \times 10^{-3}$, work out the value of $\sqrt[3]{p \times q}$.
Give your answer correct to 3 significant figures.

..... **[3]**

**55** **(a)** Calculate the value of

$$\frac{\sqrt[3]{45.6} + 2.1^4}{3.8^2}$$ Give your answer correct to 3 significant figures.

..... **[2]**

**(b)** A solid cube has a volume of 4913 $cm^3$.
Calculate the total surface area of the cube.

..... $cm^2$ **[3]**

**56** **(a)** Calculate the value of

$$\sqrt[5]{7.59375}$$

..... **[1]**

**(b)** Find the two possible values of $x$ in the equation

$$2x^4 - 17 = 1233$$

$x =$ ..... , $x =$ ..... **[3]**

**(c)** It is given that

$$y = \sqrt{\frac{3a^2 + b}{c}}$$

Work out the value of $y$ when $a = 4.2$, $b = -5.7$ and $c = 1.8$.
Give your answer correct to 2 decimal places.

$y =$ ..... **[3]**

**57** **(a)** Calculate the value of

$$\frac{\sqrt[4]{150000}}{4.5^3 - \sqrt{75}}$$ Give your answer correct to 3 decimal places.

..... **[2]**

**(b)** $x$ is a positive integer such that $\sqrt[3]{x^2} = 16$.
Find the value of $x$.

$x =$ ..... **[3]**

**(c)** The volume of a sphere is equal to the volume of a cube with edge length 12 cm.
Calculate the radius of the sphere.
The volume, $V$, of a sphere with radius $r$ is $V = \frac{4}{3}\pi r^3$.

..... cm **[3]**

**58** **(a)** Write the following in order of size, starting with the smallest.

$$0.65 \qquad \frac{2}{3} \qquad 64\% \qquad \frac{5}{8}$$

..... **[2]**

**(b)** Write 0.85 as a fraction in its simplest form.

..... **[1]**

**59** **(a)** Write $0.\dot{7}$ as a fraction.

..... **[1]**

**(b)** Write $0.\dot{3}\dot{6}$ as a fraction in its simplest form.

..... **[2]**

**60** Work out

$$3\frac{1}{3} \div 2\frac{4}{5} - \frac{5}{14}$$

Give your answer as a fraction in its simplest form.

..... **[3]**

**61** **(a)** Write 45% as a fraction in its simplest form.

..... **[1]**

**(b)** Write $\frac{9}{25}$ as a decimal.

..... **[1]**

**(c)** Write 2.04 as a mixed number in its simplest form.

..... **[2]**

**62** **(a)** Write 0.45 as a fraction in its simplest form.

..... **[1]**

**(b)** Write $\frac{3}{8}$ as a decimal.

..... **[1]**

**(c)** Write these in order of size, starting with the smallest.

$$\frac{5}{12} \qquad 0.4 \qquad 42\% \qquad \frac{3}{7}$$

..... **[2]**

**63** **(a)** Write 0.075 as a fraction in its simplest form.

..... **[2]**

**(b)** Write $3\frac{2}{5}$ as a decimal.

..... **[1]**

**(c)** Write the following in order of size, starting with the smallest.

$$62\% \qquad \frac{5}{8} \qquad 0.6 \qquad \frac{7}{11}$$

..... **[2]**

**64** The length of a rectangle is increased by $33.\dot{3}\%$.
The width of the rectangle is decreased by $x\%$.
The overall area of the rectangle increases by $16.\dot{6}\%$.
Calculate the value of $x$.

$x =$ ..... **[5]**

**65** **(a)** A tank is initially $\frac{1}{5}$ full of water.
When 24 litres of water are added, the tank is exactly 60% full.
Work out the total capacity of the tank.

..... litres **[3]**

**(b)** A bottle contains $x$ millilitres of juice.
30% of the juice is poured out.
Then $\frac{1}{4}$ of the remaining juice is poured out.
The bottle now contains 210 millilitres of juice.
Work out the value of $x$.

$x =$ ..... **[3]**

**66** A bag contains red counters, blue counters and green counters.
The number of red counters is $x$.
The number of blue counters is $2x + 5$.
The number of green counters is $3x - 1$.

**(a)** Exactly 35% of the counters in the bag are blue.
Work out the value of $x$.

$x =$ ..... **[4]**

**(b)** Work out the percentage of the counters in the bag that are green.
Write your answer as a mixed number.

..... % **[3]**

**67** **(a)** Write the recurring decimal $0.2\dot{4}\dot{5}$ as a fraction in its simplest form.
You must show all your working.

..... **[3]**

**(b)** The fraction $\frac{N}{42}$, where $N$ is a positive integer, is in its simplest form.
Explain why the decimal equivalent of $\frac{N}{42}$ is a recurring decimal.

.....

..... **[2]**

**(c)** The recurring decimal $0.\dot{x}\dot{y}$, where $x$ and $y$ are distinct digits, can be written as a fraction $\frac{k}{11}$, where $k$ is an integer.
Find the value of $x + y$.

$x + y =$ ..... **[2]**

**68** **(a)** By converting the recurring decimal to a fraction, show that

$$0.2\dot{7} = \frac{5}{18}$$

**[2]**

**(b)** In a school, $\frac{2}{5}$ of the students are boys and the rest are girls.
45% of the boys play a sport.
30% of the girls play a sport.

**(i)** Work out the percentage of all students in the school who play a sport.

..... % **[3]**

**(ii)** There are 144 students who play a sport.
Work out the total number of students in the school.

..... **[2]**

**69** A train departs a station with $x$ passengers.
At the first stop, $\frac{1}{3}$ of the passengers get off the train and 12 passengers get on.
At the second stop, 25% of the passengers now on the train get off and nobody gets on.
After the second stop, there are 15 fewer passengers on the train than there were when it departed.

**(a)** Write down an expression in terms of $x$, in its simplest form, for the number of passengers on the train after the first stop.

..... **[2]**

**(b)** Form an equation in $x$ and solve it to find the number of passengers on the train when it departed.

$x =$ ..... **[4]**

**(c)** What percentage of the original passengers remained on the train after the second stop? Give your answer as a mixed number.

..... % **[3]**

**70** In a school of 840 students, 35% are in the lower school, $\frac{3}{8}$ are in the middle school, and the rest are in the upper school.

**(a)** Work out the number of students in the middle school.

..... **[1]**

**(b)** Work out the percentage of students in the upper school.

..... % **[2]**

**(c)** 15% of the middle school students play a musical instrument.
Work out the fraction of the total number of students who are middle school students that play a musical instrument.
Give your answer in its simplest form.

..... **[2]**

**71** A tank contains $V$ litres of water.
During the first hour, the tank loses 24% of its water.
During the second hour, the tank loses $\frac{2}{5}$ of its remaining water.
After these two hours, there are 45.6 litres of water left in the tank.

**(a)** Calculate the value of $V$.

$V =$ ..... **[4]**

**(b)** Express the total amount of water lost over the two hours as a fraction of the original volume $V$.
Give your answer in its simplest form.

..... **[2]**

**72** In a school, $\frac{3}{8}$ of the students are in the Junior section, 45% are in the Middle section, and the rest are in the Senior section.
There are 140 students in the Senior section.

**(a)** Work out the total number of students in the school.

..... **[3]**

**(b)** The number of boys in the Junior section is 150% of the number of girls in the Junior section.
Work out the number of girls in the Junior section.

..... **[3]**

**73** **(a)** Write the following in order of size, starting with the smallest.

$$\frac{5}{7} \qquad 71\% \qquad 0.714 \qquad \frac{3}{4}$$

Smallest ..... **[2]**

**(b)** Find the fraction which is exactly halfway between $\frac{2}{5}$ and $\frac{5}{8}$.
Give your answer in its simplest form.

..... **[2]**

**(c)** A fraction is equivalent to $\frac{4}{9}$.
The difference between the denominator and the numerator is 35.
Find the fraction.

..... **[2]**

**74** A company has three departments: Sales, Production, and Administration.

- $\frac{2}{5}$ of the employees are in Sales.
- 28% of the employees are in Production.
- The remaining 24 employees are in Administration.

**(a)** Work out the total number of employees in the company.

..... **[3]**

**(b)** 60% of the Sales employees are women.
$\frac{4}{7}$ of the Production employees are men.
Half of the Administration employees are women.
Work out the percentage of the whole company who are women.

..... % **[4]**

**75** **(a)** A television costs \$450. In a sale, the price is reduced by 18%.
Calculate the sale price of the television.

\$ ..... **[2]**

**(b)** In the same sale, a laptop is sold for \$629 after a 15% reduction.
Calculate the original price of the laptop.

\$ ..... **[3]**

**(c)** The price of a tablet is increased by $\frac{1}{5}$.
Later, this new price is decreased by 20%.
Calculate the overall percentage decrease in the price of the tablet.

..... % **[2]**

**76** Insert one of the symbols $<$, $>$ or $=$ to make each statement correct.

(i) $0.45 \ldots\ldots \dfrac{2}{5}$

(ii) $-3 \ldots\cdots -7$

**[2]**

**77** Write the following in order of size, starting with the smallest.

$$\frac{5}{8} \qquad 0.62 \qquad 61.5\% \qquad \frac{3}{5}$$

..... **[2]**

**78** Write $<, >$ or $=$ to make each statement true.

**(a)** $0.45 \ldots\ldots\ldots \frac{4}{9}$

**[1]**

**(b)** $2^{-3} \ldots\ldots\ldots 0.125$

**[1]**

**(c)** $-1.5 \ldots\ldots \cdots -\frac{7}{5}$

**[1]**

**79** **(a)** Insert one of the symbols $<, =$ or $>$ to make each statement true.

**(i)** $\frac{5}{8} \ldots 0.65$

**[1]**

**(ii)** $72\% \ldots \frac{18}{25}$

**[1]**

**(b)** Write the following in order of size, starting with the smallest.

$$0.55 \qquad \frac{6}{11} \qquad 54\% \qquad \frac{5}{9}$$

..... **[2]**

**80** **(a)** Write the following numbers in order of size, starting with the smallest.

$$\frac{3}{4} \qquad 0.72 \qquad 74\% \qquad \frac{5}{7} \qquad 0.7\dot{4}$$

..... **[2]**

**(b)** Complete each statement by inserting the correct symbol from $=$, $>$, or $<$.

**(i)** $-\frac{5}{8}$ ..... $-0.65$

**[1]**

**(ii)** $(0.3)^2$ ..... $9\%$

**[1]**

**81** **(a)** Insert $<$, $>$ or $=$ to make each statement correct.

**(i)**

$$0.4 \ldots\ldots \frac{2}{5}$$

**[1]**

**(ii)**

$$-7 \ldots\cdots -9$$

**[1]**

**(iii)**

$$2^3 \ldots\ldots \sqrt{25}$$

**[1]**

**(b)** Write these in order of size, starting with the smallest.

$$33\% \qquad \frac{1}{3} \qquad 0.3 \qquad \frac{3}{8}$$

..... **[2]**

**82** **(a)** Given that $0 < p < 1$, write the following expressions in order of size, starting with the smallest.

$$p \qquad p^2 \qquad \sqrt{p} \qquad \frac{1}{p}$$

..... **[2]**

**(b)** Given that $-1 < q < 0$, write $<, >$ or $=$ to make each statement true.

**(i)** $q^3 \ldots\ldots\ldots q$

**[1]**

**(ii)** $\frac{1}{q} \ldots\ldots\ldots q$

**[1]**

**83** **(a)** Given that $0 < x < 1$, write the following expressions in order of size, starting with the smallest.

$$x \qquad x^2 \qquad \frac{1}{x} \qquad \sqrt{x}$$

..... **[2]**

**(b)** Given that $x < -1$, write the following expressions in order of size, starting with the largest.

$$x \qquad x^3 \qquad x^{-2} \qquad \frac{1}{x}$$

..... **[2]**

**84** **(a)** Given that $x = -0.4$, write the following in order of size, starting with the smallest.

$$x \qquad x^2 \qquad x^3 \qquad \frac{1}{x}$$

..... **[2]**

**(b)** Insert the correct symbol ($<, =$, or $>$) to make the statement true.
Explain your reasoning.

$5\sqrt{3} \ldots 4\sqrt{5}$

**[2]**

**(c)** Find the integer $m$ that satisfies the inequality

$$\frac{m}{7} < \frac{5}{9} < \frac{m+1}{7}$$

$m =$ ..... **[3]**

**85** **(a)** It is given that $0 < p < 1$.
Write the following values in order of size, starting with the smallest.

$$p \qquad \sqrt{p} \qquad p^2 \qquad \frac{1}{p}$$

..... **[2]**

**(b)** Write the correct symbol $<$, $>$ or $=$ to complete each logical statement.

**(i)** If $x < 0$, then $x^3 \ldots\ldots x^2$.

**[1]**

**(ii)** If $x > 1$, then $x^{-1} \ldots\ldots 1$.

**[1]**

**(c)** Given $a = 2^{50}$, $b = 3^{30}$ and $c = 5^{20}$, write $a$, $b$ and $c$ in order of size, starting with the smallest.
You must show all your working.

..... **[3]**

**86** Let $x$ be a real number such that $0 < x < 1$.

**(a)** Write the following four expressions in order of size, starting with the smallest.

$$x^2 \qquad \sqrt{x} \qquad \frac{1}{x} \qquad x$$

..... **[2]**

**(b)** Now let $k$ be a negative integer such that $k \leq -2$.
Write the following four expressions in order of size, starting with the smallest.

$$k^3 \qquad \frac{1}{k} \qquad k^2 \qquad -\frac{1}{k^2}$$

..... **[3]**

**(c)** Find the range of values of $n$ for which $n^2 \leq n$ and $n \neq 0$.
Express your answer as an inequality.

..... **[3]**

**87** Write the following in order of size, starting with the smallest.

$$\frac{4}{33} \qquad 0.12 \qquad 12.1\% \qquad \sqrt{0.015}$$

..... **[2]**

**88** Write the following masses in order of size, starting with the largest.

$5.2 \times 10^{-4}$ kg  0.04 g  $4.5 \times 10^{3}$ mg  $6 \times 10^{-7}$ tonnes

..... **[3]**

**89** **(a)** Here are four numerical expressions.

$a = 3.1 \times 10^3$
$b = \sqrt{9.8 \times 10^6}$
$c = 5^5$
$d = 56^2$

Work out the value of each expression and write them in order of size, starting with the largest.

..... **[3]**

**(b)** State the integer $n$ such that

$$n < \sqrt{250} < n + 1$$

$n =$ ..... **[1]**

**90** **(a)** Write these numbers in order of size, starting with the smallest.

$$5^{-3} \qquad 8.5 \times 10^{-3} \qquad 0.00805 \qquad \frac{1}{120}$$

..... **[2]**

**(b)** The mass, $m$ kilograms, of a bag of rice is 12 kg, correct to the nearest whole kilogram. Write down the inequality representing the possible values of $m$.

..... **[2]**

**91** **(a)** Write the following numbers in order of size, starting with the smallest.

$$\frac{7}{8} \qquad 86\% \qquad 0.865 \qquad \frac{5}{6}$$

..... **[2]**

**(b)** Write down a fraction, $f$, such that $\frac{1}{3} < f < \frac{1}{2}$.

..... **[2]**

**92** Consider the five numbers $p$, $q$, $r$, $s$ and $t$ defined below:

$p = 4.8 \times 10^{-3}$

$q = \dfrac{1}{205}$

$r = \sqrt{2.3 \times 10^{-5}}$

$s = 0.485\%$

$t = (0.07)^2$

**(a)** Write the five numbers $p$, $q$, $r$, $s$ and $t$ in order of magnitude, starting with the largest.

..... **[3]**

**(b)** The number $x$ satisfies the inequality $r < x \leq p$.
Write down a possible value for $x$ in standard form.

$x =$ ..... **[1]**

**(c)** Given that $y \neq t$ and $y \geq q$, determine whether the statement $y > s$ is always true, sometimes true, or never true.
Explain your answer.

.....

..... **[2]**

**93** **(a)** Given that $x > 0$ and $y < 0$, insert the correct symbol $<$, $>$ or $=$ to make the statement correct.

$x^2y \ldots\ldots 0$

**[1]**

**(b)** Write these expressions in order of size, starting with the smallest, when $a = 0.5$.

$$a^2 \qquad a \qquad \sqrt{a} \qquad \frac{1}{a}$$

..... **[2]**

**(c)** State whether the following statement is True or False.
Give a reason for your answer.
If $p > q$, then $\frac{1}{p} < \frac{1}{q}$ for all non-zero real numbers $p$ and $q$.

..... **[2]**

**94** Work out the exact value of

$$-3 - 4 \times \left(\frac{1}{2} - \frac{3}{4}\right)^2$$

..... **[3]**

**95** **(a)** The temperature at midnight was $-5°C$.
By noon, the temperature had risen by $14°C$.
Write down the temperature at noon.

..... **[1]**

**(b)** Work out the difference between $-8°C$ and $17°C$.

..... **[1]**

**(c)** Work out the value of

$$24 - 12 \div (3 - 7)$$

..... **[2]**

**96** **(a)** Work out the value of

$$12 - 2 \times (-3) + 4^2$$

..... **[2]**

**(b)** Work out the value of

$$\frac{3}{4} + \frac{1}{6} \times \frac{3}{2}$$

Give your answer in its simplest form.

..... **[2]**

**97** **(a)** Work out the value of

$$12 - 3 \times (-4) + (15 \div -3)$$

..... **[2]**

**(b)** Work out the value of

$$\frac{2}{5} \div \frac{4}{7} - \frac{1}{10}$$

Give your answer as a fraction in its simplest form.

..... **[2]**

**98** **(a)** Work out the value of

$$15 - 3 \times (4 - (-2))$$

..... **[2]**

**(b)** Work out the value of

$$2\frac{1}{4} \div \frac{3}{8} - 1\frac{1}{2}$$

Give your answer in its simplest form.

..... **[3]**

**99** **(a)** Work out the value of

$12 - 4 \times (5 - 9)$

..... **[2]**

**(b)** Work out the value of

$1.2 \div 0.04$

..... **[2]**

**(c)** The temperature at 04:00 was $-5.2°C$.
By 14:00, the temperature had risen by $8.7°C$.
Work out the temperature at 14:00.

..... °C **[1]**

**100** **(a)** Work out the exact value of

$\frac{5}{6} + \frac{2}{3} \times \frac{9}{10}$ Give your answer as a mixed number in its simplest form.

..... **[3]**

**(b)** Work out the exact value of

$2\frac{1}{4} \div \left(1\frac{1}{8} - \frac{3}{4}\right)$

..... **[3]**

**101** **(a)** Calculate the exact value of

$$2\frac{2}{5} \div \left(1\frac{1}{3} - \frac{5}{6}\right)$$

Show all your working and give your answer as a mixed number in its simplest form.

..... **[4]**

**(b)** Work out the exact value of

$$\frac{0.45 \times 1.2}{0.06}$$

..... **[3]**

**102** Solve the equation, giving your answer as a fraction in its simplest form.

$$\frac{x+\frac{1}{3}}{2-0.4} = -1\frac{1}{4}$$

$x =$ ..... **[5]**

**103** Work out the exact value of

$$\frac{2\frac{1}{3} \times 1\frac{2}{7} - \sqrt{\frac{25}{9}}}{\left(\frac{1}{2}\right)^3 \div \frac{3}{16}}$$

..... **[5]**

**104** Without using a calculator, work out

$$\frac{3\frac{2}{3} + 2\frac{1}{4} \times 1\frac{1}{3}}{4\frac{1}{2} - 1\frac{5}{6}}$$

You must show all your working and give your answer as a fraction in its simplest form.

..... **[5]**

**105** **(a)** Work out the value of

$$2\frac{2}{3} \div \left(1\frac{1}{4} - \frac{5}{6}\right)$$

Give your answer as a mixed number in its simplest form.

..... **[4]**

**(b)** A rectangular field has a length of $12\frac{1}{2}$ m and an area of $83\frac{1}{3}$ $m^2$.
Work out the perimeter of the field.
Give your answer as a mixed number in its simplest form.

..... $m$ **[4]**

**106** **(a)** At 06:00, the temperature in a city is $-4.5°C$.
By 14:00, the temperature has risen by $11.2°C$.
By 22:00, the temperature has dropped by $8.1°C$ from the temperature at 14:00.

Work out the temperature at 22:00.

..... **[2]**

**(b)** Use your calculator to work out the value of

$$\frac{4.72^2 - \sqrt[3]{18.5}}{3.1 \times (0.45 + 1.2)}$$

Give your answer correct to 3 significant figures.

..... **[2]**

**107** A water tank has a capacity of 85.5 litres.
It is initially completely empty.
A hose pumps water into the tank at a rate of $\frac{3}{4}$ of a litre per minute.
At the same time, a leak in the tank loses water at a rate of 0.45 litres per minute.

Work out how long it takes to completely fill the tank.
Give your answer in hours and minutes.

hours ..... , minutes ..... **[4]**

**108** The temperature of a liquid is $-4.5°C$.
It is heated so that its temperature rises by $2.8°C$ every minute for 15 minutes.

**(a)** Work out the temperature of the liquid after the first 15 minutes.

..... **[2]**

**(b)** After the 15 minutes, the liquid is cooled so its temperature falls by $1.2°C$ every minute for $m$ minutes.
After the $m$ minutes of cooling, the temperature of the liquid is $9.9°C$.
Work out the value of $m$.

$m =$ ..... **[3]**

**109** A scientific freezer is used to change the temperature of liquid samples.

**(a)** Sample A has an initial temperature of 12.5°C.
It cools at a constant rate of 3.2°C per minute.
Calculate the temperature of Sample A after 8 minutes.

..... **[2]**

**(b)** Sample B has an initial temperature of −4.5°C.
To find its final temperature, the initial temperature is multiplied by −2, and the result is then decreased by 18.3°C.
Calculate the positive difference between the final temperature of Sample B and the final temperature of Sample A.

..... **[4]**

**110** A large container has a total capacity of 1000 litres.

**(a)** The container is initially $\frac{2}{5}$ full of water.
Water is pumped into the container at a constant rate of 18.5 litres per minute for 20 minutes.
Work out the volume of water in the container after these 20 minutes.

..... litres **[3]**

**(b)** Water then leaks from the container at a constant rate of 4.5 litres per minute.
Calculate the time taken for the container to become exactly half full.

..... minutes **[3]**

**(c)** Once the container is half full, the leak is fixed.
A different pump then fills the container completely in 1 hour and 20 minutes.
Calculate the average rate at which this pump fills the container.
Give your answer in litres per minute.

..... litres per minute **[3]**

**111** **(a)** A tank contains 450.5 litres of oil.
The oil is poured into bottles.
Each bottle holds exactly 1.25 litres.
Work out the number of completely filled bottles and the volume of oil left over.

Number of bottles = ..... , Volume left over = ..... **[3]**

**(b)** A machine pumps water at a rate of 1800 litres per minute.
It is used to fill a rectangular pool of length 24.5 m, width 12 m and depth 1.8 m.
($1\ m^3 = 1000$ litres)

Calculate the time taken to fill the pool completely.
Give your answer in hours and minutes.

hours, minutes **[4]**

**(c)** A freezer is set at $-18°C$.
During a power cut, the temperature inside the freezer rises by $1.5°C$ per hour.
The power cut lasts for 6 hours and 20 minutes.
Work out the temperature in the freezer when the power is restored.

..... °C **[3]**

**112** **(a)** Find the value of $7^0$.

..... **[1]**

**(b)** Find the value of $81^{\frac{1}{2}}$.

..... **[1]**

**(c)** Write as a single power of $x$.

$x^5 \times x^{-2}$

..... **[1]**

**113** **(a)** Work out the value of

$125^{\frac{1}{3}}$

..... **[1]**

**(b)** Work out the value of

$6^{-2}$

..... **[1]**

**(c)** Simplify.

$t^3 \times t^5$

..... **[1]**

**114** **(a)** Simplify.

$m^5 \times m^2$

..... **[1]**

**(b)** Find the value of

$5^{-2}$

..... **[1]**

**(c)** Find the value of

$49^{\frac{1}{2}}$

..... **[1]**

**115** **(a)** Find the value of

$3^{-2}$

..... **[1]**

**(b)** Find the value of

$\left(\frac{8}{27}\right)^{\frac{1}{3}}$

..... **[1]**

**(c)** Simplify.

$x^4 \times x^{-2}$

..... **[1]**

**(d)** Simplify.

$(y^3)^4$

..... **[1]**

**116** **(a)** Find the value of

$2^{-3} \times 2^{5}$

..... **[2]**

**(b)** Work out the value of

$81^{\frac{1}{2}}$

..... **[1]**

**(c)** Find the exact value of

$64^{0}$

..... **[1]**

**117** **(a)** Work out the value of

$$16^{-\frac{3}{4}}$$

..... **[2]**

**(b)** Simplify.

$$\frac{12w^6y^3}{3w^{-2}y^5}$$

..... **[3]**

**118** **(a)** Simplify fully

$(3xy^2)^3$

..... **[2]**

**(b)** Find the value of $x$ when

$16^x = 2^{10}$

$x =$ ..... **[3]**

**119** **(a)** Simplify.

$(4x^3y)^2$

..... **[2]**

**(b)** Simplify.

$$\frac{18a^6b^2}{6a^2b^5}$$

..... **[2]**

**(c)** Work out the value of

$27^{-\frac{2}{3}}$

..... **[2]**

**120** **(a)** Find the exact value of each of the following.

**(i)** $64^{\frac{2}{3}}$

..... **[2]**

**(ii)** $2^{-4}$

..... **[1]**

**(b)** **(i)** Simplify fully.

$$\frac{15a^6b^3}{3a^2b^7}$$

..... **[2]**

**(ii)** Simplify fully.

$$(4m^6n^{-2})^{\frac{1}{2}}$$

..... **[2]**

**121** **(a)** Find the exact value of

$16^{-\frac{3}{4}}$

..... **[2]**

**(b)** Simplify fully.

$(5x^3y^{-2})^2$

..... **[2]**

**(c)** Solve the equation.

$2^{3x-1} \times 4^x = 32$

$x =$ ..... **[3]**

**122** **(a)** Work out the exact value of

$$125^{-\frac{2}{3}}$$

..... **[3]**

**(b)** Simplify

$$\left(\frac{16a^8}{81b^{12}}\right)^{\frac{3}{4}}$$

..... **[3]**

**123** **(a)** Simplify.

$$\left(\frac{25x^4}{16y^8}\right)^{-\frac{1}{2}}$$

..... **[3]**

**(b)** Solve the equation.

$$2^{3x-1} \times 4^{x+2} = 8^{x-3}$$

$x =$ ..... **[4]**

**124** **(a)** Simplify fully.

$$\left(\frac{16x^8}{y^4}\right)^{-\frac{1}{2}}$$

..... **[3]**

**(b)** Solve the equation.

$$4^x - 2^{2x+1} + 8 = 0$$

$x =$ ..... **[4]**

**125** **(a)** Simplify fully.

$$\left(\frac{16x^{12}}{81y^{8}}\right)^{-\frac{3}{4}}$$

..... **[3]**

**(b)** Find the value of $x$ when

$$8^{x} = 4^{x+3}$$

$x =$ ..... **[3]**

**(c)** Solve the equation.

$$5 \times 25^{k} = 125$$

$k =$ ..... **[3]**

**126** **(a)** Find the value of $k$ when $5^k = \dfrac{1}{\sqrt{125}}$.

$k =$ ..... **[3]**

**(b)** Simplify fully.

$$\left(\frac{16x^{12}}{81y^8}\right)^{-\frac{3}{4}}$$

..... **[3]**

**(c)** Solve the equation.

$$2^{2x-1} \times 4^x = 32$$

$x =$ ..... **[3]**

**127** **(a)** Write 0.00407 in standard form.

..... **[1]**

**(b)** Work out $(3 \times 10^4) \times (5 \times 10^6)$.
Give your answer in standard form.

..... **[2]**

**128** **(a)** Write 450000 in standard form.

..... **[1]**

**(b)** Write $3.08 \times 10^{-4}$ as an ordinary number.

..... **[1]**

**(c)** Work out.

$(2 \times 10^3) \times (4 \times 10^5)(0)$

*Give your answer in standard form.*

..... **[1]**

**129** **(a)** Write 452000 in standard form.

..... **[1]**

**(b)** Write $6.07 \times 10^{-3}$ as an ordinary number.

..... **[1]**

**(c)** Work out $(4 \times 10^4) \times (3 \times 10^5)$.
Give your answer in standard form.

..... **[2]**

**130** **(a)** Write 0.00714 in standard form.

..... **[1]**

**(b)** Write $5.02 \times 10^4$ as an ordinary number.

..... **[1]**

**(c)** Work out the value of

$$(3 \times 10^5) \times (6 \times 10^4)$$

Give your answer in standard form.

..... **[2]**

**131** Consider the following numerical values.

**(a)** Convert the following numbers as instructed.

**(i)** Write 0.00034 in standard form.

..... **[1]**

**(ii)** Write $5.2 \times 10^4$ as an ordinary number.

..... **[1]**

**(b)** Work out, giving your answer in standard form:

$(2 \times 10^3) \times (4 \times 10^5)$

..... **[2]**

**132** **(a)** Write 0.000452 in standard form.

..... **[1]**

**(b)** Write $3.71 \times 10^5$ as an ordinary number.

..... **[1]**

**(c)** Calculate, giving your answer in standard form.

$(2 \times 10^4) \times (4 \times 10^3)$

..... **[2]**

**133** The mass of the Earth is $5.97 \times 10^{24}$ kg.
The mass of the Moon is $7.35 \times 10^{22}$ kg.

**(a)** Calculate the difference between the mass of the Earth and the mass of the Moon.
Give your answer in standard form.

..... *kg* **[2]**

**(b)** The mass of the Sun is $1.99 \times 10^{30}$ kg.
Find the value of $k$ when

mass of the Sun = k ×mass of the Earth
Give your answer to 3 significant figures.

$k =$ ..... **[3]**

**134** **(a)** Work out $(8.5 \times 10^6) - (4 \times 10^5)$.
Give your answer in standard form.

..... **[2]**

**(b)** It is given that $p = 2 \times 10^9$ and $q = 5 \times 10^4$.
Work out the value of $\sqrt{\frac{p}{q}}$.
Give your answer in standard form.

..... **[3]**

**135** **(a)** Work out, giving your answer in standard form.

$$\frac{3.6 \times 10^{-4}}{8 \times 10^{-7}}$$

..... **[3]**

**(b)** Work out, giving your answer in standard form.

$$(5 \times 10^{8}) + (7 \times 10^{7})$$

..... **[2]**

**(c)** Given that $x = 2 \times 10^{m}$ and $y = 8 \times 10^{n}$, where $m$ and $n$ are integers.
Find an expression for $x^2y$ in standard form, in terms of $m$ and $n$.

..... **[2]**

**136** **(a)** Work out the exact value of

$$\frac{5.4 \times 10^5 + 6 \times 10^4}{2 \times 10^{-3}}$$

Give your answer in standard form.

..... **[3]**

**(b)** Evaluate $\sqrt{2.5 \times 10^9}$.
Give your answer in standard form.

..... **[2]**

**(c)** Given that $p = 2 \times 10^x$ and $q = 8 \times 10^y$, find an expression for $p^2 \div q$.
Give your answer in standard form in terms of $x$ and $y$.

..... **[3]**

**137** $p = 5 \times 10^m$ and $q = 8 \times 10^n$, where $m$ and $n$ are integers.

**(a)** Find an expression, in terms of $m$ and $n$, for $p \times q$.
Give your answer in standard form.

..... **[2]**

**(b)** Find an expression, in terms of $m$ and $n$, for $\dfrac{p^2}{q}$.
Give your answer in standard form.

..... **[3]**

**(c)** Given that $p + q = 5.08 \times 10^m$, work out the exact value of $m - n$.

$m - n =$ ..... **[3]**

**138** The distance from the Earth to the Sun is $1.5 \times 10^8$ km.

**(a)** Light travels at a speed of $3.0 \times 10^5$ km/s.
Calculate the time taken for light to travel from the Sun to the Earth.
Give your answer in seconds.

..... seconds **[2]**

**(b)** A spacecraft travels at a constant speed of $2.5 \times 10^4$ km/h.
Work out the time it takes the spacecraft to travel from the Earth to the Sun.
Give your answer in hours, in standard form.

..... hours **[2]**

**139** The mass of Venus is $4.87 \times 10^{24}$ kg.
The mass of the Sun is $1.99 \times 10^{30}$ kg.

**(a)** The mass of Earth is 1.23 times the mass of Venus.
Calculate the mass of Earth.
Give your answer in standard form correct to 3 significant figures.

..... kg **[2]**

**(b)** Calculate how many times greater the mass of the Sun is than the mass of Venus.
Give your answer in standard form correct to 3 significant figures.

..... **[2]**

**140** The mass of the Sun is $1.99 \times 10^{30}$ kg.
The mass of Jupiter is $1.90 \times 10^{27}$ kg.

**(a)** Calculate how many times greater the mass of the Sun is than the mass of Jupiter.
Give your answer in standard form correct to 3 significant figures.

..... **[3]**

**(b)** A comet has a mass of $4.5 \times 10^{13}$ kg.
Calculate the mass of the comet in grams.
Give your answer in standard form.

..... $g$ **[2]**

**141** **(a)** Calculate, giving your answer in standard form.

$3.4 \times 10^7 + 2.1 \times 10^6$

..... **[2]**

**(b)** A computer can do a calculation in $4 \times 10^{-9}$ seconds.
Work out how many of these calculations the computer can do in 5 minutes.
Give your answer in standard form.

..... **[3]**

**142** The mass of the Earth is approximately $5.97 \times 10^{24}$ kg.
The mass of the Moon is approximately $7.35 \times 10^{22}$ kg.

**(a)** Calculate the total mass of the Earth and the Moon.
Give your answer in standard form.

..... kg **[2]**

**(b)** Calculate how many times greater the mass of the Earth is than the mass of the Moon.
Give your answer correct to the nearest integer.

..... **[2]**

**(c)** The mass of a planet is $p \times 10^{25}$ kg.
The mass of this planet is 300 times the mass of the Earth.
Work out the value of $p$, giving your answer correct to 3 significant figures.

$p =$ ..... **[2]**

**143** **(a)** The mass of Earth is $5.97 \times 10^{24}$ kg and the mass of the Moon is $7.35 \times 10^{22}$ kg.
Calculate the difference between the mass of Earth and the mass of the Moon.
Give your answer in standard form.

..... kg **[2]**

**(b)** The distance from the Sun to Earth is $1.50 \times 10^{8}$ km.
The speed of light is $3.00 \times 10^{5}$ km/s.

Calculate the time it takes for light to travel from the Sun to Earth.
Give your answer in standard form.

..... $s$ **[2]**

**(c)** The Earth is modelled as a sphere with radius $r = 6.37 \times 10^{6}$ m.
Calculate the surface area of the Earth.
Give your answer in standard form, correct to 3 significant figures.
The surface area, $A$, of a sphere with radius $r$ is $A = 4\pi r^2$.

..... $m^2$ **[3]**

**144** A large cylindrical water tank has a base radius of $4.5 \times 10^2$ cm and a height of $1.2 \times 10^3$ cm.

**(a)** Calculate the volume of the water tank in $cm^3$.
Give your answer in standard form correct to 4 significant figures.

..... $cm^3$ **[3]**

**(b)** The empty tank is being filled with water at a constant rate of $1.5 \times 10^{-4}$ $m^3$ per second.
Note that $1\ m^3 = 1 \times 10^6\ cm^3$.

Calculate the time taken to fill the tank completely.
Give your answer in hours, written in standard form correct to 3 significant figures.

..... hours **[4]**

**145** **(a)** Write 23.496 correct to 2 decimal places.

..... **[1]**

**(b)** Write 0.040719 correct to 3 significant figures.

..... **[1]**

**146** Consider the number 47.5381.

**(a)** Write 47.5381 correct to:

**(i)** 2 decimal places.

..... **[1]**

**(ii)** 2 significant figures.

..... **[1]**

**(iii)** the nearest integer.

..... **[1]**

**147** Write the following numbers correct to the specified degree of accuracy.

**(a)** 782.51 correct to the nearest ten.

..... **[1]**

**(b)** 0.03561 correct to 2 significant figures.

..... **[1]**

**(c)** 4.098 correct to 2 decimal places.

..... **[1]**

**148** **(a)** Write 48.975 correct to 2 decimal places.

..... **[1]**

**(b)** Write 0.003058 correct to 2 significant figures.

..... **[1]**

**(c)** By writing each number correct to 1 significant figure, estimate the value of

$$\frac{7.84 \times 31.6}{0.385}$$

..... **[2]**

**149** **(a)** Round the following numbers to the specified degree of accuracy.

**(i)** 34.567 correct to 2 decimal places.

..... **[1]**

**(ii)** 0.070381 correct to 2 significant figures.

..... **[1]**

**(b)** By writing each number correct to 1 significant figure, estimate the value of

$$\frac{61.3 \times 3.92}{0.485}$$

..... **[2]**

**150** **(a)** Write 0.04758 correct to 2 significant figures.

..... **[1]**

**(b)** Write 384.6 correct to the nearest 10.

..... **[1]**

**(c)** By writing each number correct to 1 significant figure, estimate the value of

$$\frac{58.7 \times 4.12}{19.6 - 2.87}$$

..... **[2]**

**151** **(a)** Write 203.498 correct to

**(i)** 2 decimal places.

..... **[1]**

**(ii)** 3 significant figures.

..... **[1]**

**(iii)** the nearest ten.

..... **[1]**

**(b)** By writing each number correct to 1 significant figure, estimate the value of

$$\frac{6.89 \times 41.2}{0.513}$$

..... **[2]**

**152** By writing each number correct to 1 significant figure, estimate the value of

$$\frac{58.4 \times 3.09}{0.187}$$

..... **[3]**

**153** **(a)** By writing each number correct to 1 significant figure, estimate the value of

$$\frac{7.84 \times 31.6}{0.381}$$

..... **[2]**

**(b)** By writing each number correct to 1 significant figure, estimate the value of

$$\frac{\sqrt{98.6} + 19.3}{0.513^2}$$

..... **[3]**

**154** **(a)** By writing each number correct to 1 significant figure, estimate the value of

$$\frac{\sqrt{98.5} \times 4.12^2}{0.187}$$

..... **[3]**

**(b)** A rectangular field has length 489 m and width 215 m.

Estimate the area of the field, in square metres, by rounding each length to 1 significant figure.

..... $m^2$ **[2]**

**155** By writing each number correct to 1 significant figure, estimate the value of

$$\sqrt{\frac{398.2 \times 0.0514}{0.187}}$$

..... **[4]**

**156** **(a)** By writing each number correct to 1 significant figure, estimate the value of

$$\frac{\sqrt{3.87 \times 10^6}}{0.0524 \times 21.3}$$

..... **[3]**

**(b)** Is your estimate in part (a) greater than or less than the exact value of the expression? Give a reason for your answer.

.....

..... **[2]**

**157** **(a)** By writing each number correct to 1 significant figure, estimate the value of

$$\frac{\sqrt[3]{8.25} \times 19.3^2}{0.485}$$

..... **[3]**

**(b)** The number of bacteria in an experiment is given by the formula

$$N = 4.12 \times 10^5 \times (1.95)^t$$

where $t$ is the time in hours.
By writing each base number correct to 1 significant figure, estimate the number of bacteria when $t = 3$.
Give your answer in standard form.

$N \approx$ ..... **[3]**

**158** **(a)** By writing each number correct to 1 significant figure, estimate the value of

$$\frac{\sqrt{98.5} \times 4.1^3}{0.192 \times 48.7}$$

*You must show all your working.*

..... **[3]**

**(b)** Given that $y = \dfrac{ax}{b-c}$, find an estimate for the value of $y$ when
$a = 3.92$, $x = 0.0514$, $b = 1.08$ and $c = 0.812$.
Write all numbers correct to 1 significant figure in your calculation.

$y =$ ..... **[3]**

**159** **(a)** By writing each number correct to 1 significant figure, estimate the value of

$$\frac{38.7 \times 5.12^2}{\sqrt[3]{996}}$$

..... **[3]**

**(b)** A solid metal cylinder has radius 3.89 cm and height 11.2 cm.

By writing each dimension correct to 1 significant figure, estimate the volume of the cylinder.
Take the value of $\pi$ to be 3.

..... $cm^3$ **[3]**

**(c)** A train travels a distance of 4210 km at an average speed of 585 km/h.

By writing both the distance and the speed correct to 1 significant figure, estimate the time taken for the journey.
Give your answer in hours and minutes.

hours, minutes **[3]**

**160** A solid metal cone has a base radius of 5.42 cm and a perpendicular height of 12.8 cm.

**(a)** Calculate the volume of the cone.
Write your answer correct to 1 decimal place.

..... $cm^3$ **[2]**

**(b)** The cone is melted down to make identical small spheres of radius 0.85 cm.
Calculate the maximum number of full spheres that can be made.

..... **[3]**

**161** A rectangular field has length 125 m and width 74 m.

**(a)** Calculate the area of the field.
Give your answer correct to 2 significant figures.

..... $m^2$ **[2]**

**(b)** A different field is a square of side $x$ metres.
The area of this square field is equal to the area of the rectangular field.
Calculate the value of $x$.
Give your answer correct to 1 decimal place.

$x =$ ..... **[3]**

**162** A solid metal cone has radius $r = 5.82$ cm and height $h = 11.4$ cm.

**(a)** Calculate the volume of the cone.
Give your answer correct to 3 significant figures.
The volume, $V$, of a cone with radius $r$ and height $h$ is $V = \frac{1}{3}\pi r^2 h$.

..... $cm^3$ **[2]**

**(b)** By writing each of the values 5.82, 11.4 and $\pi$ correct to 1 significant figure, estimate the volume of the cone.
Show your working.

..... $cm^3$ **[2]**

**(c)** A worker packs these cones into boxes.
Each box holds exactly 12 cones.
The factory produces 850 cones a day.
Calculate the maximum number of boxes that can be completely filled each day.

..... **[2]**

**163** **(a)** Calculate the value of

$$\frac{4.2^3 + \sqrt{18.5}}{0.75 \times 1.2}$$

Give your answer correct to 3 significant figures.

..... **[2]**

**(b)** A spherical water tank has a radius of 1.85 metres.

**(i)** Calculate the volume of the tank.
Give your answer correct to the nearest cubic metre.
The volume, $V$, of a sphere with radius $r$ is $V = \frac{4}{3}\pi r^3$.

..... $m^3$ **[2]**

**(ii)** The tank is initially empty.
Water is pumped into the tank at a rate of 12.5 litres per minute.
Calculate the time required to fill the tank completely.
Give your answer in hours and minutes, correct to the nearest minute.

hours, minutes **[3]**

**164** **(a)** Calculate the value of

$$p = \frac{28.7^2 - 11.2^2}{\sqrt{98.5}}$$

**(i)** Write down all the figures on your calculator display.

$p =$ ..... **[1]**

**(ii)** Write your answer to part (a)(i) correct to 3 significant figures.

$p =$ ..... **[1]**

**(b)** By writing each number in the original expression correct to 1 significant figure, estimate the value of $p$.

$p \approx$ ..... **[2]**

**165** A solid metal cylinder has a base radius of 4.15 cm and a height of 19.8 cm. The metal has a density of 8.13 grams per cubic centimetre.

**(a)** By writing each number correct to 1 significant figure, estimate the mass of the cylinder in grams.
Use $\pi \approx 3$ for your estimation.
Show your working clearly.

..... $g$ **[3]**

**(b)** Calculate the actual mass of the cylinder.
Give your answer correct to 3 significant figures.

..... $g$ **[3]**

**166** **(a)** Write 45 : 75 : 105 in its simplest form.

..... **[1]**

**(b)** Alice, Bob and Charlie share \$240 in the ratio of their ages. Alice is 8 years old, Bob is 10 years old and Charlie is 12 years old.
Work out how much Bob receives.

% ..... **[2]**

**167** **(a)** Write the ratio 1.5 hours to 45 minutes in its simplest form.

..... **[1]**

**(b)** Alex, Ben and Carl share \$240 in the ratio 2 : 3 : 5.
Calculate how much more Carl receives than Alex.

% ..... **[2]**

**168** **(a)** Write the ratio 450 g to 1.2 kg in its simplest form.

..... **[2]**

**(b)** Anna, Ben and Chloe share \$420 in the ratio 3 : 4 : 5.
Calculate the amount Ben receives.

\$ ..... **[2]**

**169** Alex, Ben and Charlie share \$840 in the ratio 2 : 3 : 5.

**(a)** Calculate the amount of money Ben receives.

\$ ..... **[2]**

**(b)** Express Charlie's share as a fraction of the total amount, giving your answer in its simplest form.

..... **[2]**

**170** Alex, Ben and Chloe share \$120 in the ratio 3 : 4 : 5.

**(a)** Show that Chloe receives \$50.

**[2]**

**(b)** Work out the amount Alex receives.

\$ ..... **[2]**

**(c)** Write down the ratio of Ben's share to Chloe's share in its simplest form.

..... **[1]**

**171** **(a)** Write the ratio 45 minutes to 2 hours in its simplest form.

..... **[2]**

**(b)** A piece of wood of length 3.6 m is cut into three pieces in the ratio 2 : 3 : 7.
Calculate the length, in cm, of the longest piece.

..... cm **[3]**

**172** A sports club has $m$ male members and $f$ female members.
When 30 new male members join the club, the ratio of male members to female members becomes 3 : 1.
When, instead, 20 new female members join the original members, the ratio of male members to female members becomes 1 : 1.

**(a)** Write down two simultaneous equations in $m$ and $f$.

.....

..... **[2]**

**(b)** Solve the equations to find the value of $m$ and the value of $f$.

$m =$ ..... , $f =$ ..... **[3]**

**(c)** Find the original ratio of male members to female members in its simplest form.

..... **[1]**

**173** A bag contains red balls, blue balls and green balls.
The ratio of red balls to blue balls is 3 : 4.
The ratio of blue balls to green balls is 5 : 2.

**(a)** Find the ratio of red balls to blue balls to green balls in its simplest form.

..... **[2]**

**(b)** There is a total of 129 balls in the bag.
Calculate the number of green balls.

..... **[2]**

**(c)** $y$ blue balls and $y$ green balls are removed from the bag.
The new ratio of blue balls to green balls is 3 : 1.
Work out the value of $y$.

$y =$ ..... **[3]**

**174** A map is drawn to a scale of 1 : 50000.

**(a)** The length of a road on the map is 4.5 cm.
Calculate the actual length of the road in kilometres.

..... km **[2]**

**(b)** A lake has an actual area of 14.2 $km^2$.
Calculate the area of the lake on the map in $cm^2$.

..... $cm^2$ **[3]**

**175** A map is drawn to a scale of 1 : 250 000.

**(a)** The distance between two towns on the map is 8.4 cm. Calculate the actual distance between the two towns in kilometres.

..... km **[3]**

**(b)** A forest has an actual area of 35 square kilometres.
Calculate the area of the forest on the map in square centimetres.

..... $cm^2$ **[3]**

**176** A map is drawn to a scale of 1 : 250 000.

**(a)** The distance between two cities on the map is 16.4 cm.
Calculate the actual distance between the two cities in kilometres.

..... km **[2]**

**(b)** The actual area of a lake is 35 $km^2$.
Calculate the area of the lake on the map in square centimetres.

..... $cm^2$ **[3]**

**(c)** Another map is drawn such that a length of 5 cm represents an actual distance of 20 km.
Write the scale of this map in the form 1 : $n$.

1 : ..... **[1]**

**177** **(a)** A map has a scale of 1 : 50000.
The actual distance between two towns is 14.5 km.
Calculate the distance, in cm, between the two towns on the map.

..... cm **[2]**

**(b)** An area of a forest on the map is 12.4 $cm^2$.
Calculate the actual area of the forest in $km^2$.

..... $km^2$ **[2]**

**(c)** A recipe for 8 people requires 240 g of flour and 150 g of butter.
Marco wants to make the recipe for 12 people.
He has 400 g of flour and 200 g of butter.
Determine if Marco has enough ingredients.
Show your working.

.....

..... **[3]**

**178** A map is drawn to a scale of $1 : 250\,000$.

**(a)** The distance between two cities on the map is 16.8 cm.
Calculate the actual distance between the cities, in km.

..... km **[2]**

**(b)** A lake has an actual area of 56 $km^2$.
Calculate the area of the lake on the map, in $cm^2$.

..... $cm^2$ **[2]**

**(c)** On a different map of the same region, the area of the same lake is 14 $cm^2$.
Find the scale of this new map in the form $1 : n$.

$1 :$ ..... **[3]**

**179** **(a)** The ratio of adults to children in a theater is 5 : 3.
There are 84 more adults than children.
Calculate the total number of people in the theater.

..... **[3]**

**(b)** A map is drawn to a scale of 1 : 25000.

**(i)** Two towns are 8.5 cm apart on the map.
Calculate the actual distance between the towns in kilometres.

..... $km$ **[2]**

**(ii)** A forest has an actual area of 3.5 $km^2$.
Calculate the area of the forest on the map in square centimetres.

..... $cm^2$ **[3]**

**180** In a factory, the number of male workers to female workers is in the ratio 5 : 3.
The ratio of male workers who work full-time to male workers who work part-time is 3 : 2.
The ratio of female workers who work full-time to female workers who work part-time is 4 : 1.

**(a)** Find the ratio of the total number of full-time workers to the total number of part-time workers in the factory.
Give your answer in its simplest form.

..... **[4]**

**(b)** There are 42 more full-time male workers than part-time female workers.
Calculate the total number of workers in the factory.

..... **[3]**

**181** In a factory, the number of workers on the day shift and the number of workers on the night shift are in the ratio 5 : 3.

**(a)** There are 42 more workers on the day shift than on the night shift.
Calculate the total number of workers in the factory.

..... **[3]**

**(b)** The ratio of male workers to female workers on the day shift is 3 : 4.
The ratio of male workers to female workers on the night shift is 2 : 7.
Calculate the overall ratio of male workers to female workers in the factory.
Give your answer in its simplest form.

..... **[4]**

**182** **(a)** In a club, the ratio of the number of boys to the number of girls is 3 : 5.
When 4 boys leave and 8 girls join the club, the new ratio of boys to girls becomes 1 : 3.
Calculate the original number of boys in the club.

..... **[4]**

**(b)** The ratio of the mass of object $A$ to object $B$ is 3 : 4.
The ratio of the mass of object $B$ to object $C$ is 5 : 6.
The mass of object $C$ is 45 kg more than the mass of object $A$.
Calculate the total mass of the three objects.

..... kg **[3]**

**183** Three partners, Alex, Ben and Chris, share a total profit of $\$P$. The ratio of Alex's share to Ben's share is 3 : 4. The ratio of Ben's share to Chris's share is 5 : 6.

**(a)** Find the ratio of Alex's share to Ben's share to Chris's share in its simplest form.

..... **[2]**

**(b)** Chris receives \$16 200 more than Alex. Calculate the value of $P$.

\$ ..... **[3]**

**(c)** Alex invests his share of the profit in a bank account that pays compound interest at a rate of 4.2% per year.
Calculate the value of Alex's investment at the end of 5 years.
Give your answer correct to the nearest dollar.

\$ ..... **[3]**

**184** A train departs at 08 : 15 and arrives at its destination at 09 : 45.
The train travels a distance of 120 km.
Calculate the average speed of the train.

..... km/h **[2]**

**185** A train travels 102 km in 1 hour 12 minutes.
Work out its average speed.

..... km/h **[3]**

**186** A car travels 120 km in 1 hour and 30 minutes.
Calculate the average speed of the car in km/h.

..... km/h **[3]**

**187** **(a)** A car travels a distance of 108 km in 1 hour 30 minutes.
Calculate its average speed in km/h.

..... km/h **[2]**

**(b)** Change your answer in part (a) into m/s.

..... $m/s$ **[2]**

**188** **(a)** A train travels a distance of 120 km in 1 hour and 30 minutes.
Calculate the average speed of the train.

..... km/h **[2]**

**(b)** A car uses 8 litres of fuel for every 100 km travelled.
Work out how much fuel the car uses to travel 350 km.

..... litres **[2]**

**189** **(a)** A train travels a distance of 180 km in 2 hours and 15 minutes.
Calculate the average speed of the train in km/h.

..... km/h **[2]**

**(b)** An empty tank is filled with water at a constant rate of 1.5 litres per second.
Work out the time it takes to fill the tank with 450 litres of water.
Give your answer in minutes.

..... minutes **[2]**

**190** A cyclist travels 30 km at an average speed of $v$ km/h.
He then travels a further 40 km at an average speed of $(v + 5)$ km/h.
The total time taken for the entire journey is 4 hours.

**(a)** Write down an equation in terms of $v$ and show that it simplifies to $2v^2 - 25v - 75 = 0$.

**[3]**

**(b)** Solve the equation $2v^2 - 25v - 75 = 0$ to find the possible values of $v$.

$v =$ ..... , or $v =$ ..... **[3]**

**(c)** Find the time taken for the first 30 km of the journey.

..... hours **[1]**

**191** A train travels a distance of 385 km.
The journey takes 4 hours 25 minutes.
Calculate the average speed of the train in km/h.
Give your answer correct to 1 decimal place.

..... $km/h$ **[3]**

**192** Water flows from a pipe into a rectangular tank at a rate of 45 litres per minute.

**(a)** Calculate the volume of water, in cubic centimetres ($cm^3$), that flows into the tank in 1 second.

..... $cm^3$ **[2]**

**(b)** The tank has a base measuring 1.5 m by 80 cm. Calculate the rate at which the water level is rising in the tank, giving your answer in cm/s.

..... $cm/s$ **[3]**

**193** A car uses fuel at an average rate of 6.4 litres per 100 km.

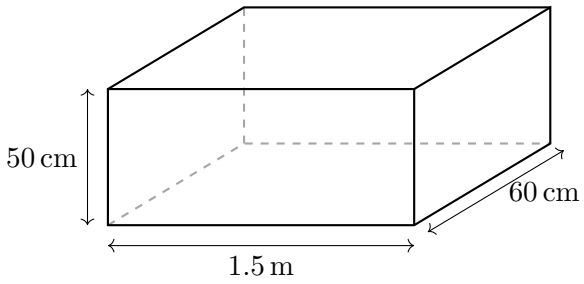
**(a)** Calculate the amount of fuel used for a journey of 350 km.

..... litres **[2]**

**(b)** The cost of fuel is 1.35 euros per litre.
The exchange rate is \$1 = 0.85 euros. Calculate the cost of the fuel for this journey in dollars (\$).
Give your answer correct to the nearest cent.

\$ ..... **[3]**

**194** A rectangular water tank measures 1.5 m by 60 cm by 50 cm.




**(a)** Calculate the volume of the tank in litres.

..... litres **[2]**

**(b)** Water is pumped into the empty tank at a constant rate of 24 litres per minute.
Calculate the time taken to fill the tank completely.
Give your answer in minutes and seconds.

minutes ..... , seconds ..... **[3]**

**195** A swimming pool has a volume of 52 $m^3$.

**(a)** Water flows into the empty pool from a pipe at a rate of 1.6 litres per second.
Calculate the time taken to completely fill the pool.
Give your answer in hours and minutes, correct to the nearest minute.

hours, minutes **[3]**

**(b)** The cost of the water is \$2.15 per cubic metre. The exchange rate is \$1 = 0.84 euros.
Calculate the total cost of the water used to fill the pool in euros.
Give your answer correct to 2 decimal places.

euros ..... **[3]**

**196** **(a)** A car travels 325 km.
It uses fuel at a rate of 8.4 litres per 100 km.
Fuel costs \$1.35 per litre.
Calculate the total cost of the fuel used for this journey.

\$ ..... **[3]**

**(b)** On a different journey, the car travels at an average speed of 75 km/h.
The journey takes 4 hours and 24 minutes.
Work out the distance travelled.

..... km **[3]**

**197** Water flows from a pipe into a rectangular tank at a rate of 2.5 litres per second.
The tank has a length of 1.8 m, a width of 1.5 m, and a height of 1.2 m.
The tank is initially empty.

**(a)** Calculate the capacity of the tank in litres.

..... litres **[2]**

**(b)** Work out how long it takes to fill the tank completely.
Give your answer in minutes.

..... minutes **[2]**

**(c)** Another pipe empties the full tank at a rate of $x$ litres per minute.
It takes 45 minutes to empty the tank completely.
Calculate the value of $x$.

$x =$ ..... **[2]**

**198** **(a)** Water flows from a pipe into an empty tank at a rate of 0.35 litres per second.
The tank is a cuboid with length 2.5 m, width 1.4 m and height 1.2 m.

Calculate the time taken to fill the tank completely.
Give your answer in hours and minutes.

$h$ ..... , min ..... **[4]**

**(b)** A mobile phone costs \$850 in the USA and €780 in Germany. The exchange rate is \$1 = €0.84.

Calculate the difference in price between the two countries.
Give your answer in dollars (\$).

\$ ..... **[3]**

**199** A car travels from City A to City B, a distance of $d$ km, at an average speed of $v$ km/h.
The car then travels from City B to City C, a distance of $2d$ km, at an average speed of $(v + 15)$ km/h.
The total time taken for the entire journey from City A to City C is $T$ hours.

**(a)** Show that the total time $T$ can be written as

$$T = \frac{3dv + 15d}{v(v + 15)}$$

**[3]**

**(b)** Given that $d = 120$ and $T = 4.5$,

Form an equation in $v$ and show that it simplifies to

$$3v^2 - 195v - 1200 = 0$$

**[4]**

**200** A cyclist travels $x$ km at a speed of 12 km/h. She then travels a further $(x + 5)$ km at a speed of 15 km/h. The total time taken for the whole journey is 2 hours 35 minutes.

**(a)** Work out the value of $x$.

$x =$ ..... **[4]**

**(b)** Calculate her average speed for the whole journey.
Give your answer correct to 1 decimal place.

..... km/h **[3]**

**201** An aeroplane flies from City A to City B, a distance of 2400 km.

**(a)** The aeroplane flies at an average speed of $v$ km/h.
Write down an expression, in terms of $v$, for the time taken in hours.

..... hours **[1]**

**(b)** The aeroplane returns from City B to City A.
The average speed on the return journey is $(v + 40)$ km/h.
The total time taken for the two flights is 10 hours and 30 minutes.
Write down an equation in $v$ and show that it simplifies to

$$7v^2 - 2920v - 192000 = 0$$

**[4]**

**(c)** Solve the equation $7v^2 - 2920v - 192000 = 0$ to find the average speed of the aeroplane from City A to City B.
Give your answer correct to the nearest integer.

..... km/h **[2]**

**202** A cyclist travels from Town A to Town B, a distance of 48 km.
The cyclist travels the first 30 km at an average speed of $x$ km/h. They travel the remaining distance at an average speed of $(x - 4)$ km/h. The total time taken for the journey is 2 hours and 30 minutes.

**(a)** Show that $5x^2 - 116x + 240 = 0$.

**[4]**

**(b)** Solve the equation $5x^2 - 116x + 240 = 0$ to find the possible values of $x$.
Show all your working and give your answers correct to 2 decimal places.

$x =$ ..... **[3]**

**203** A man runs 15 km at an average speed of $x$ km/h.
He then walks 12 km at an average speed of $(x - 4)$ km/h.
The total time taken for the journey is 3.5 hours.

**(a)** Show that $x$ satisfies the equation

$$7x^2 - 82x + 120 = 0$$

**[3]**

**(b)** **(i)** Solve the equation

$$7x^2 - 82x + 120 = 0$$

$x =$ ..... , or $x =$ ..... **[3]**

**(ii)** Find the time taken for the man to run the 15 km.

..... hours **[2]**

**204** A car travels 150 km at an average speed of $x$ km/h.

**(a)** Write down an expression, in terms of $x$, for the time taken in hours.

..... hours **[1]**

**(b)** The car then travels a further 90 km at an average speed of $(x + 15)$ km/h.
The total time for the whole journey is 3 hours.
Show that $x^2 - 65x - 750 = 0$.

**[4]**

**(c)** Solve the equation $x^2 - 65x - 750 = 0$ to find the initial average speed of the car.

$x =$ ..... km/h **[3]**

**205** A shopkeeper buys a television for \$320. He sells it for \$400.
Calculate his percentage profit.

..... % **[2]**

**206** **(a)** A shop has a sale.
All items are sold with a 15% discount.
Calculate the sale price of a coat originally costing \$80.

\$ ..... **[2]**

**(b)** A car was bought for \$4000 and sold for \$3200.
Work out the percentage loss.

..... % **[2]**

**207** **(a)** Write 24 as a percentage of 60.

..... % **[2]**

**(b)** A coat has a normal price of \$80. In a sale, the price is reduced by 15%.
Work out the sale price of the coat.

\$ ..... **[2]**

**208** **(a)** Find 15% of \$320.

\$ ..... **[2]**

**(b)** In a sale, the original price of a shirt is reduced by 20%.
The original price of the shirt is \$45.
Work out the sale price.

\$ ..... **[3]**

**209** A shop sells a jacket for \$120. During a sale, the shop offers a 15% discount on the price of the jacket.

**(a)** Calculate the sale price of the jacket.

\$ ..... **[2]**

**(b)** The original price of \$120 gives the shop a profit of 20% on the cost price.
Calculate the cost price of the jacket.

\$ ..... **[3]**

**210** **(a)** Work out 15% of \$240.

\$ ..... **[2]**

**(b)** A shirt costs \$40. In a sale, the price is reduced to \$32.
Calculate the percentage reduction in the price.

..... % **[2]**

**(c)** The price of a pair of shoes is increased by 20% to \$60.
Calculate the original price of the shoes.

\$ ..... **[2]**

**211** **(a)** A shop reduces its prices by 20% in a sale.
The original price of a book is \$15.
Calculate the sale price of the book.

\$ ..... **[2]**

**(b)** The sale price of a jacket in the same shop is \$56.
Calculate the original price of the jacket.

\$ ..... **[3]**

**(c)** A different shop buys a watch for \$30 and sells it for \$42.
Calculate the percentage profit.

..... % **[2]**

**212** **(a)** Alice invests \$4500 at a rate of 3.2% per year compound interest.
Calculate the total value of her investment at the end of 5 years.

\$ ..... **[3]**

**(b)** A shopkeeper buys a television for \$320. He sells it for \$408.
Calculate his percentage profit.

..... % **[2]**

**213** **(a)** In 2021, the population of a town was 45 000.
In 2022, the population increased by 4.2%.
Calculate the population in 2022.

..... **[2]**

**(b)** In a sale, the price of a television is reduced by 24%.
The sale price is \$342.
Calculate the original price of the television.

\$ ..... **[3]**

**214** Amira invests \$4500 in a savings account that pays compound interest at a rate of 2.4% per year.

**(a)** Calculate the total amount in Amira's account at the end of 5 years.

\$ ..... **[3]**

**(b)** Ben invests \$4500 in a different savings account that pays simple interest at a rate of $r$% per year. At the end of 5 years, Ben has the same total amount in his account as Amira has in hers.

Calculate the value of $r$.

$r =$ ..... **[3]**

**215** **(a)** Anja invests \$8000 in an account paying compound interest at a rate of 3.5% per year.

Calculate the total value of her investment at the end of 4 years.
Give your answer correct to the nearest cent.

\$ ..... **[3]**

**(b)** The population of a town decreases by 4% each year.
In 2026, the population is 45 000.
Calculate the population of the town in 2023.
Give your answer correct to the nearest integer.

..... **[3]**

**216** **(a)** Amira invests \$4500 in a savings account paying compound interest at a rate of 2.4% per year.
Calculate the total value of her investment at the end of 5 years.

\$ ..... **[3]**

**(b)** In a sale, the price of a computer is reduced by 18%.
The sale price is \$557.60.
Calculate the original price of the computer.

\$ ..... **[3]**

**217** **(a)** An investment of \$8000 earns compound interest at a rate of 3.5% per year. Calculate the value of the investment at the end of 4 years.
Give your answer correct to the nearest cent.

\$ ..... **[3]**

**(b)** A car is bought for \$24000. Its value decreases by 12% each year. Calculate the number of complete years it takes for the value of the car to fall below \$10000.

..... years **[3]**

**(c)** In 2022, the price of a house was \$150000. This was a 25% increase on its price in 2017. Calculate the price of the house in 2017.

\$ ..... **[3]**

**218** **(a)** Amira invests \$4000 in a bank account paying compound interest at a rate of 2.5% per year.
Calculate the total interest earned by Amira at the end of 5 years.

\$ ..... **[3]**

**(b)** Boris invests \$4000 in a different bank account paying simple interest at a rate of 3.2% per year.
Calculate the number of years it takes for the total value of Boris's investment to become \$4896.

..... years **[3]**

**(c)** Diana invests a sum of money in a bank account paying compound interest at a rate of 1.8% per year.
At the end of 6 years, her investment is worth \$6676.70.
Calculate the amount Diana originally invested, giving your answer correct to the nearest dollar.

\$ ..... **[3]**

**219** **(a)** The population of a city increases by 4.5% each year.
The population is now $1.25 \times 10^6$.
Calculate the expected population in 10 years' time.
Give your answer in standard form, correct to 3 significant figures.

..... **[3]**

**(b)** In a sale, a washing machine is sold for \$299.25. This is a 14% reduction on the original price.
Calculate the original price.

\$ ..... **[2]**

**(c)** A car depreciates in value by 15% in the first year and by 10% in each subsequent year.
The value of the car after 3 years is \$17212.50.
Calculate the original value of the car.

\$ ..... **[2]**

**220** The population of a city increases by $x\%$ each year.
In 2015, the population was 1 250 000.
In 2018, the population was 1 367 631.

**(a)** Show that the value of $x$ is approximately 3.

**[3]**

**(b)** Assume the population continues to increase by exactly 3% each year.
Calculate the year in which the population will first exceed double its 2015 value.

..... **[4]**

**221** **(a)** A shopkeeper buys a television for $\$p$. He sells it for \$960, making a percentage profit of 20% on his cost price.
Calculate the value of $p$.

$p = \$ \ldots\ldots\ldots\ldots\ldots$ **[2]**

**(b)** The shopkeeper buys another television for $\$q$. He wants to make a 15% profit on his cost price after allowing a 10% discount on the marked price.

Write down an expression, in its simplest form, for the marked price in terms of $q$.

$\$ \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots$ **[3]**

**(c)** An investment of \$500 earns compound interest at a rate of $r$% per year. At the end of 10 years, the value of the investment is \$740.12.
Work out the value of $r$.

$r = \ldots\ldots\ldots\ldots\ldots$ **[3]**

**222** **(a)** Ali invests \$5000 at a rate of 2.5% per year compound interest. Calculate the value of his investment at the end of 3 years.
Give your answer correct to the nearest cent.

\$ ..... **[2]**

**(b)** Beatrice invests a sum of money at a rate of $r$% per year compound interest.
After 5 years, the value of her investment has increased by 30%.
Calculate the value of $r$.

$r =$ ..... **[3]**

**(c)** The value of a car depreciates by 12% each year.
After 3 years, the value of the car is \$17 036.80.
Calculate the original value of the car.

\$ ..... **[3]**

**223** **(a)** A car's value depreciates by 15% in the first year.
In each subsequent year, its value depreciates by 8%.
After exactly 3 years, the value of the car is \$17 986.
Calculate the original value of the car.

\$ ..... **[4]**

**(b)** Calculate the overall percentage decrease in the car's value over the 3 years.
Give your answer correct to 1 decimal place.

..... % **[2]**

**(c)** The population of a city is $P$.
The population increases by $r$% each year.
After 12 years, the population has exactly doubled.
Calculate the value of $r$.

$r =$ ..... **[3]**

**224** **(a)** A car depreciates in value by 15% each year.
After 3 years, the value of the car is \$12 282.50.
Calculate the original value of the car.

\$ ..... **[3]**

**(b)** An antique clock was bought for \$15 000. Its value increases exponentially at a rate of $x$% per year. After 5 years, its value is \$21 500.
Calculate the value of $x$, giving your answer correct to 1 decimal place.

$x =$ ..... **[3]**

**(c)** A house is bought for \$250 000. Its value increases by 4% in the first year, decreases by 2% in the second year, and increases by 5% in the third year. Calculate the overall percentage increase in the value of the house over the 3 years.
Give your answer correct to 2 decimal places.

..... % **[4]**

**225** **(a)** A population of bacteria increases by $r\%$ every hour.
The population doubles in 6 hours.
Calculate the value of $r$, giving your answer correct to 3 significant figures.

$r =$ ..... **[3]**

**(b)** A merchant buys goods for $\$X$. He sets the selling price such that he makes a 40% profit on the cost price. During a sale, he gives a 15% discount on this selling price. The final sale price is \$142.80.
Calculate the value of $X$.

$X =$ ..... **[3]**

**(c)** An amount of $\$P$ is invested at a rate of $x\%$ per year simple interest. The total simple interest earned in 5 years is \$250.
The same amount $\$P$ is invested at a rate of $x\%$ per year compound interest. The total compound interest earned in 2 years is \$102.50.
Find the value of $P$ and the value of $x$.

$P =$ \$ ..... , x = ..... **[5]**

**226** Calculate the value of

$$\frac{\sqrt{15.2 - 3.4^2}}{4.1 \times 0.25}$$

Write down your full calculator display.

..... **[2]**

**227** Work out the value of

$$\frac{\sqrt[3]{25.6 - 3.2^2}}{1.8 \times 0.45}$$

**(a)** Write down all the figures on your calculator display.

..... **[2]**

**(b)** Write your answer to part (a) correct to 3 decimal places.

..... **[1]**

**228** Consider the following calculation.

**(a)** Work out the value of

$$\frac{\sqrt{4.1^2 + 5.2}}{2.3^3 - 1.4}$$

Write down all the figures on your calculator display.

..... **[2]**

**(b)** Give your answer to part (a) correct to 3 significant figures.

..... **[1]**

**229** **(a)** Work out the value of

$$\frac{\sqrt{15.2^2 - 8.5^2}}{3.2 \times 1.4}$$

Write down your full calculator display.

..... **[2]**

**(b)** Write your answer to part (a) correct to 3 significant figures.

..... **[1]**

**(c)** Calculate

$$\frac{4.5 \times 10^5}{1.2 \times 10^{-3}}$$

Give your answer in standard form.

..... **[1]**

**230** **(a)** Consider the following calculation.

Work out the value of

$$\frac{8.3^2 - 4.1}{\sqrt{5.6 + 2.1^3}}$$

Write down all the figures on your calculator display.

..... **[2]**

**(b)** Give your answer to part (a) correct to 3 significant figures.

..... **[1]**

**(c)** A computer program runs for 6.45 hours.
Write this time in hours and minutes.

hours ..... , minutes ..... **[1]**

**231** **(a)** Work out the value of

$$\frac{4.5^2 - \sqrt{18.2}}{3.1 + 0.8^3}$$

..... **[2]**

**(b)** Calculate the value of

$$2\sin(48^\circ) - \cos(35^\circ)$$

Give your answer correct to 3 decimal places.

..... **[2]**

**232** **(a)** A shopkeeper calculates the cost of an item using the expression

$$\frac{17.3 + 8.14}{5.3}$$

Work out the cost.
Give your answer as a money value in dollars, formatted correctly.

$ ..... **[2]**

**(b)** A movie lasts for 2 hours and 18 minutes.
Write this time as a decimal number of hours.

..... hours **[2]**

**233** **(a)** Use your calculator to find the value of

$$3.5^4 + \sqrt[3]{87.6}$$

Give your answer correct to 2 decimal places.

..... **[2]**

**(b)** Calculate the value of $x$ when

$$x = \frac{2 \times \sin(43°)}{\cos(18°)}$$

Give your answer correct to 3 significant figures.

$x =$ ..... **[2]**

**234** A student uses a calculator to solve three different problems.

**(a)** The answer to a problem about money is displayed on the calculator as 8.4.
Write this amount appropriately in dollars and cents.

..... **[1]**

**(b)** The answer to a problem about time is displayed on the calculator as 5.15 hours.
Write this time appropriately in hours and minutes.

hours ..... , minutes ..... **[1]**

**(c)** A car travels 280 km at an average speed of 85 km/h. Calculate the time taken for this journey.
Give your answer in hours and minutes, correct to the nearest minute.

hours ..... , minutes ..... **[3]**

**235** **(a)** Calculate the value of

$$\frac{3.5 + 1.2^2}{\sqrt{4.1 - 2.8}}$$

**(i)** Write down all the figures on your calculator display.

..... **[2]**

**(ii)** Write your answer to part (a)(i) correct to 3 significant figures.

..... **[1]**

**(b)** A calculator displays a journey time as 5.35 hours.
Convert this time to hours and minutes.

$h$ ..... , min ..... **[2]**

**236** **(a)** A circular swimming pool has a radius of 4.2 metres.
Calculate the area of the pool.
Give your answer correct to the nearest square metre.

..... $m^2$ **[2]**

**(b)** A train journey takes 4 hours and 23 minutes.
The average speed of the train is 78 km/h.

Calculate the distance travelled by the train.
Give your answer correct to 1 decimal place.

..... km **[4]**

**237** A plumber charges a fixed fee of \$45 plus an hourly rate of \$32.50.

**(a)** The plumber works on a job for 3 hours and 40 minutes.
Calculate the total cost of this job.
Give your answer to the nearest cent.

\$ ..... **[3]**

**(b)** On another job, the total cost charged by the plumber is \$230.25.
Calculate the time the plumber worked, giving your answer in hours and minutes.

hours, minutes **[3]**

**238** A metal sphere has a radius of 4.85 cm.

**(a)** Calculate the volume of the sphere.
The volume, $V$, of a sphere with radius $r$ is $V = \frac{4}{3}\pi r^3$.
Give your answer correct to 1 decimal place.

..... $cm^3$ **[2]**

**(b)** The metal sphere is melted down and made into identical small cones.
Each cone has a radius of 1.2 cm and a height of 3.5 cm.
The volume, $V$, of a cone with radius $r$ and height $h$ is $V = \frac{1}{3}\pi r^2 h$.
Calculate the maximum number of full cones that can be made.

..... **[4]**

**(c)** Calculate the percentage of metal wasted from the original sphere.

..... % **[1]**

**239** Amir works part-time at a factory.

**(a)** In one week, Amir works for 24 hours and 40 minutes.
He is paid \$18.60 per hour.
Calculate his total pay for the week.
Give your answer in the correct format for money.

\$ ..... **[3]**

**(b)** The following month, Amir's total pay increases by 4.5%.
Amir uses his calculator to work out his new total pay.
The calculator displays '479.446'.
Write down Amir's new total pay correct to the nearest cent.

\$ ..... **[1]**

**(c)** One of the factory machines produces a widget every 3.4 minutes.
Calculate the total time it takes the machine to produce 150 widgets.
Give your answer in hours and minutes.

hours ..... , minutes ..... **[3]**

**240** A water tank is in the shape of a sphere with radius 1.4 m.
The tank is completely full of water.

**(a)** Work out the volume of water in the tank.
Write down all the figures on your calculator display.

..... $m^3$ **[2]**

**(b)** Water leaks from the tank at a constant rate of 0.035 cubic metres per minute.
Calculate the time taken for the tank to become completely empty.
Do not round any intermediate values during your working.
Give your answer in hours and minutes, correct to the nearest minute.

hours ..... , minutes ..... **[4]**

**241** **(a)** Calculate the exact value of

$$\left(3.2 \times 10^{5}\right) \times \left(4.5 \times 10^{-2}\right)$$

Give your answer in standard form.

..... **[2]**

**(b)** Work out the value of

$$\frac{1}{\sqrt{2\pi}} \times e^{-0.5^2}$$

Give your answer correct to 4 decimal places.

..... **[2]**

**(c)** A formula is given by $T = 2\pi\sqrt{\frac{l}{g}}$.

Find the value of $l$ when $T = 1.5$ and $g = 9.81$.

Give your answer correct to 2 decimal places.

$l =$ ..... **[2]**

**242** **(a)** Calculate

$$\frac{3.4 \times 10^5 + 2.7 \times 10^4}{8.1 \times 10^{-3}}$$

Give your answer in standard form.

..... **[3]**

**(b)** A solid metal cone has a base radius of $r$ cm and a height of $h$ cm.
$r = 5.2$ correct to 1 decimal place.
$h = 12.4$ correct to 1 decimal place.
Calculate the upper bound for the volume of the cone.
The volume, $V$, of a cone with radius $r$ and height $h$ is $V = \frac{1}{3}\pi r^2 h$.

..... $cm^3$ **[3]**

**243** The formula for the period, $T$ seconds, of a simple pendulum is given by

$$T = 2\pi\sqrt{\frac{l}{g}}$$

where $l$ is the length of the pendulum in metres and $g$ is the acceleration due to gravity, 9.81 m/s$^2$.

**(a)** A pendulum has a length of 85 cm.
Calculate the period $T$.
Give your answer correct to 3 decimal places.

$T =$ ..... seconds **[2]**

**(b)** Rearrange the formula to make $l$ the subject.

$l =$ ..... **[2]**

**(c)** A different pendulum has a period of exactly 2.5 seconds.
Calculate the length of this pendulum in centimetres.
Give your answer correct to 1 decimal place.

..... cm **[3]**

**244** The volume, $V$, of a spherical cap is given by the formula

$$V = \frac{1}{3}\pi h^2(3R - h)$$

where $R$ is the radius of the sphere and $h$ is the height of the cap.

**(a)** Calculate $V$ when $R = 8.45$ cm and $h = 3.2$ cm. Give your answer correct to 1 decimal place.

$V =$ ..... $cm^3$ **[2]**

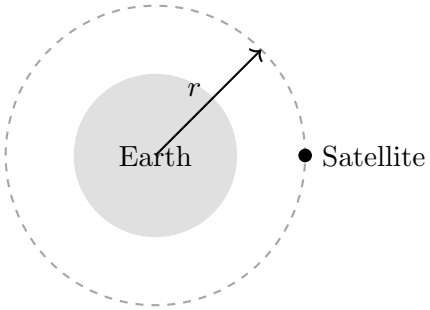
**(b)** Rearrange the formula to make $R$ the subject.

$R =$ ..... **[3]**

**(c)** A different spherical cap has volume $V = 500$ cm$^3$ and height $h = 5.6$ cm. Calculate the radius, $R$, of this sphere.
Give your answer correct to 3 significant figures.

$R =$ ..... cm **[2]**

**245** A satellite is in a circular orbit around the Earth.




**(a)** The speed, $v$ m/s, of the satellite is given by the formula

$$v = \sqrt{\frac{GM}{r}}$$

where:

- $G = 6.67 \times 10^{-11}$
- $M = 5.97 \times 10^{24}$
- $r = 6.80 \times 10^{6}$

Work out the value of $v$.
Give your answer correct to 3 significant figures.

$v =$ ..... **[3]**

**(b)** The time, $T$ seconds, taken for the satellite to complete one full orbit is given by

$$T = \frac{2\pi r}{v}$$

Using the exact value of $v$ from your calculator, calculate the time taken for one full orbit.
Give your answer in hours, minutes and seconds, correct to the nearest second.

hours ..... , minutes ..... , seconds ..... **[4]**

**246** The volume, $V$, of a spherical cap is given by the formula

$$V = \frac{1}{3}\pi h^2(3R - h)$$

**(a)** Calculate the volume $V$ when $R = 8.74$ and $h = 2.31$.
Give your answer correct to 1 decimal place.

$V =$ ..... **[3]**

**(b)** A student calculates $V$ by first rounding the values of $R$ and $h$ to 1 decimal place before substituting them into the formula.
Calculate the difference between this incorrect answer and the exact volume.
Give your difference correct to 2 significant figures.

Difference = ..... **[4]**

**247** A train journey from Portsmouth to London is shown in the timetable.

| Station | Time |
|---|---|
| Portsmouth | 08 45 |
| London | 11 20 |

**(a)** Work out the journey time in hours and minutes.

hours ..... , minutes ..... **[1]**

**(b)** The return journey takes exactly the same amount of time.
The train leaves London at 3.35 p.m.

Write down the time it arrives in Portsmouth using the 24-hour clock.

..... **[2]**

**248** A train timetable shows that a train leaves Paris at 09 55 and arrives in Lyon at 12 10.

**(a)** Work out the time taken for this journey in hours and minutes.

hours ..... , minutes ..... **[1]**

**(b)** Another train leaves Paris at 16 45.
This journey takes 2 hours and 35 minutes.
Work out the time this train arrives in Lyon.

..... **[2]**

**249** A flight departs from London to Dubai at 21 : 45 local time.

The flight time is 6 hours and 50 minutes.
The local time in Dubai is 4 hours ahead of the local time in London.
Work out the local time in Dubai when the flight arrives.

..... **[3]**

**250** A flight leaves Dubai at 14 30 local time on Friday and arrives in London at 19 15 local time on the same day.
The local time in Dubai is 4 hours ahead of the local time in London.

**(a)** Work out the duration of the flight in hours and minutes.

hours, minutes **[2]**

**(b)** A second flight leaves Dubai at 22 45 local time on Friday.
The duration of this flight is 7 hours and 20 minutes.
Work out the local time and the day it arrives in London.

Time = ..... , Day = ..... **[2]**

**251** **(a)** A train departs from Paris at 22:50.
The journey to Rome takes 14 hours and 35 minutes.
Work out the arrival time in Rome using the 24-hour clock.

..... **[2]**

**(b)** Convert 3.4 hours into hours and minutes.

$h$ ..... , min ..... **[2]**

**252** The timetable shows part of a train journey.

| Station | Arrive | Depart |
|---|---|---|
| Alphaville | – | 09 45 |
| Betatown | 11 20 | 11 35 |
| Gammacity | 14 15 | – |

**(a)** Calculate how long the journey from Alphaville to Betatown takes.

hours ..... , minutes ..... **[2]**

**(b)** A later train leaves Alphaville at 14 55.
The journey time to Gammacity is exactly 4 hours and 10 minutes.
At what time does this train arrive at Gammacity?

..... **[2]**

**253** **(a)** A movie starts at 19 35 and lasts for 115 minutes.
Work out the time the movie ends in the 24-hour clock.

..... **[2]**

**(b)** A plane leaves New York at 21 40 local time.
The flight to Paris takes 7 hours and 15 minutes.
The local time in Paris is 6 hours ahead of New York.
Find the local time in Paris when the plane arrives.

..... **[3]**

**254** A flight departs from Dubai at 14 30 local time and arrives in Sydney.
The flight time is 14 hours and 15 minutes.
The local time in Sydney is 7 hours ahead of the local time in Dubai.
Work out the local time in Sydney when the flight arrives.

..... **[3]**

**255** Light travels at $3 \times 10^5$ km/s.

**(a)** Calculate the distance light travels in one year (365 days).
Give your answer in standard form.

..... km **[2]**

**(b)** The distance from Earth to a star is $4.5 \times 10^{13}$ km.
Calculate the time taken for light to travel from the star to Earth.
Give your answer in years, correct to 3 significant figures.

..... years **[2]**

**256** A train travels from Station A to Station B. The total distance is 420 km.
The train departs Station A at 08 : 15 and arrives at Station B at 11 : 03.

**(a)** Calculate the time taken for the journey, in hours and minutes.

hours, minutes **[2]**

**(b)** Calculate the average speed of the train in km/h.

..... km/h **[3]**

**257** Dubai is in the time zone UTC+4 and London is in the time zone UTC+0.

**(a)** An airplane departs Dubai at 02 15 local time on Tuesday and flies to London.
The flight takes 7 hours and 45 minutes.

Find the local time and day in London when the flight arrives.

Time ..... , Day ..... **[3]**

**(b)** An astronaut spends exactly 1.5 years in space.
Given that 1 year = 365 days, calculate the total number of minutes the astronaut spends in space.

Give your answer in standard form.

..... minutes **[3]**

**258** **(a)** A flight from Tokyo to London takes 14 hours and 15 minutes.
Tokyo is 9 hours ahead of London.
The flight arrives in London at 16:30 local time on Tuesday.
Calculate the departure time from Tokyo in local time.

..... **[3]**

**(b)** A machine completes one cycle every 1 minute and 45 seconds.
Calculate the maximum number of full cycles it can complete in 8 hours.

..... **[3]**

**259** A flight departs from London and travels to Dubai.
The local time in Dubai is 4 hours ahead of the local time in London.

**(a)** The flight leaves London at 13 45 local time.
It arrives in Dubai at 00 15 local time the next day.
Calculate the total duration of the flight.

hours ..... , minutes ..... **[3]**

**(b)** The total distance of the flight is 5486 km.
Using your answer to part (a), calculate the average speed of the airplane in km/h.

..... km/h **[3]**

**260** **(a)** A marathon runner completes a 42.2 km race at an average speed of 12.5 km/h.

Work out the time taken to complete the race.
Give your answer in hours, minutes and seconds, correct to the nearest second.

hours = ..... , minutes = ..... , seconds = ..... **[4]**

**(b)** Another runner finishes the race at 13 04.
She started the race at 08 35.
Work out how long she took to complete the race, in hours and minutes.

hours = ..... , minutes = ..... **[2]**

**261** A communications satellite orbits the Earth.
It completes one full orbit every 94 minutes and 30 seconds.

**(a)** Work out the number of full orbits the satellite completes in a 30-day month.

..... **[3]**

**(b)** The satellite passes directly over an observatory at 05 50 on 1st March.
Find the time and date when it completes its 100th orbit after this point.

Time ..... , Date ..... **[4]**

**262** A factory machine produces items continuously, 24 hours a day.

**(a)** The machine produces items at a constant rate of 80 items per minute.
Calculate the time it takes to produce exactly 1 000 000 items.
Give your answer in days, hours and minutes.

days ..... , hours ..... , minutes ..... **[3]**

**(b)** A company in Tokyo orders a batch of 1 000 000 items.
The factory is located in New York.
The local time in Tokyo is 14 hours ahead of the local time in New York.
The machine starts producing this batch at 08 00 local New York time on Monday.
Calculate the local Tokyo time and day when the batch is finished.

Day ..... , Time ..... **[4]**

**263** A ship travels a distance of 920 km at an average speed of 24 km/h.

**(a)** Work out the time taken for the journey.
Give your answer in hours and minutes.

hours ..... , minutes ..... **[3]**

**(b)** The ship left the port on Wednesday at 17 50.
Determine the day and the time it arrives at its destination.

Day ..... , Time ..... **[4]**

**264** An aircraft flies from City A to City B, a distance of 6600 km.
The local time at City B is 5 hours behind the local time at City A.
The aircraft leaves City A at 21 40 local time on Tuesday.
The average speed of the aircraft is 800 km/h.

**(a)** Calculate the time of the flight in hours and minutes.

hours, minutes **[2]**

**(b)** Find the local time and the day when the aircraft arrives at City B.

Time = ..... , Day = ..... **[2]**

**(c)** The aircraft stays at City B for 14 hours and 20 minutes before flying back to City A.
The return flight takes 7 hours and 45 minutes.
Find the local time and the day when the aircraft arrives back at City A.

Time = ..... , Day = ..... **[3]**

**265** A cyclist travels 35 km at an average speed of $x$ km/h.
She then travels a further 42 km at an average speed of $(x - 5)$ km/h.

**(a)** Write down an expression, in terms of $x$, for the total time taken in hours.

..... **[1]**

**(b)** The total time for the journey is 4 hours and 33 minutes.
Show that $13x^2 - 285x + 500 = 0$.

**[4]**

**(c)** Solve the equation to find the value of $x$.

$x =$ ..... **[3]**

**266** Consider the following applications of time, speed, and dates.

**(a)** A satellite completes one orbit of the Earth in exactly 92 minutes.
Calculate the number of full orbits it completes in exactly one non-leap year (365 days).

..... orbits **[4]**

**(b)** A plane flies from City P to City Q. The distance is 11 440 km and the average speed is 880 km/h.

**(i)** Calculate the flight time.

..... hours **[1]**

**(ii)** City P is in the time zone UTC−4.
City Q is in the time zone UTC+9.
The plane departs from City P on Monday at 22 : 45 local time.
Find the local day and time of arrival in City Q.

Day = ..... , Time = ..... **[3]**

**267** **(a)** A satellite takes 92 minutes and 45 seconds to complete one orbit of the Earth.
Work out the total time taken to complete 15 orbits.
Give your answer in hours, minutes and seconds.

$h$ ..... , min ..... , $s$ ..... **[2]**

**(b)** A flight departs City A at 09:45 local time.
The flight time is 8 hours and 20 minutes.
It arrives in City B at 02:05 local time the next day.
City B is $h$ hours ahead of City A.

Find the value of $h$.

$h =$ ..... **[3]**

**(c)** Train P leaves Station $X$ at 08:00 and travels to Station $Y$ at a constant speed of 90 km/h.
Train Q leaves Station $Y$ at 08:30 and travels to Station $X$ at a constant speed of 120 km/h.
The distance between $X$ and $Y$ is 360 km.
Work out the exact time when the two trains pass each other.

..... **[4]**

**268** Hassan travels from Paris to Dubai.
He changes €400 into United Arab Emirates Dirhams (AED).
The exchange rate is €1 = 4.20 AED.

**(a)** Calculate how many AED he receives.

..... AED **[1]**

**(b)** Hassan spends 1050 AED in Dubai.
He changes his remaining AED back into euros.
The new exchange rate is €1 = 4.50 AED.

Calculate how many euros he receives.

€ ..... **[2]**

**269** John has \$400. He buys a new camera that costs 255 Euros (). The exchange rate is \$1 = 0.85.
Work out how much money John has left in dollars.

\$ ..... **[3]**

**270** Aisha buys 3 pens costing \$1.20 each and 4 notebooks costing \$2.50 each.
She pays with a \$20 note.
Work out how much change she receives.

% ..... **[3]**

**271** The exchange rate between pounds (£) and dollars ($) is £1 = $1.50.
Anna changes £300 into dollars.
She spends $210 on a tablet.
She changes the remaining dollars back into pounds at the same exchange rate.

Work out how many pounds she receives.

£ ..... **[3]**

**272** The exchange rate between US dollars ($) and euros (€) is $1 = 0.85.

**(a)** Liam changes $400 into euros.
Work out how many euros he receives.

€ ..... **[2]**

**(b)** A jacket costs 102 in Paris.
Work out the cost of this jacket in US dollars.

$ ..... **[2]**

**273** **(a)** Adele buys 4 pens costing \$2.35 each and 3 notebooks costing \$4.50 each.
She pays with a \$50 note.
Work out how much change she receives.

% ..... **[3]**

**(b)** The exchange rate between euros () and US dollars (\$) is 1 = \$1.20.
Change 45 into US dollars.

% ..... **[1]**

**274** Amira goes on holiday to Japan.
She changes \$850 into Japanese Yen (JPY). The exchange rate is \$1 = 148 JPY.

**(a)** Calculate how many JPY she receives.

..... JPY **[2]**

**(b)** In Japan, she spends 45 000 JPY on a camera and 31 500 JPY on food.
At the end of the holiday, she changes her remaining JPY back to dollars (\$). The new exchange rate is \$1 = 152 JPY.
Calculate how many dollars she receives.
Give your answer correct to the nearest cent.

\$ ..... **[3]**

**275** A car dealer buys a car in Japan for 1 250 000 JPY.
The exchange rate is £1 = 145 JPY.

**(a)** Calculate the cost of the car in pounds (£).

£ ..... **[2]**

**(b)** The dealer pays shipping costs of £600.
He then sells the car in the UK for £12 500.

Calculate his percentage profit on the total cost of buying and shipping the car.

..... % **[3]**

**276** **(a)** The exchange rate between euros () and US dollars ($) is 1 = $1.14.
Convert 450 into US dollars.

% ..... **[2]**

**(b)** The exchange rate between British pounds (£) and US dollars ($) is £1 = $1.32.
Calculate the exchange rate between British pounds and euros.
Give your answer in the form £1 = $x$, correct to 2 decimal places.

£1 = ..... **[3]**

**277** A shop sells laptops in London and in New York.
The same model of laptop costs £680 in London and $850 in New York.

**(a)** The exchange rate is £1 = $1.32.

Calculate the difference in the cost of the laptop in London and the laptop in New York. Give your answer in dollars ($).

$ ..... **[3]**

**(b)** A customer buys the laptop in New York for $850. They must also pay a local tax of 4% on the cost of the laptop. The customer pays the total amount in euros (€). The exchange rate is $1 = €0.85.

Calculate the total amount the customer pays in euros.

€ ..... **[3]**

**278** **(a)** Tomas changes £450 into euros ().
The exchange rate is £1 = 1.18.
He spends 285 on his holiday.
He changes the remaining money back into pounds.
The new exchange rate is £1 = 1.22.
Calculate the amount of pounds he receives.
Give your answer correct to the nearest penny.

£ ..... **[4]**

**(b)** A television is priced at $850. In a sale, the price is reduced by 15%. A customer has a voucher that gives a further 5% discount off the sale price.
Calculate the final price paid for the television.

% ..... **[2]**

**279** A travel agency displays the following currency exchange rates:

£1 = \$1.28

£1 = CHF 1.14

**(a)** Elena changes \$704 into Swiss francs (CHF).
Calculate the amount she receives in Swiss francs.

CHF ..... **[3]**

**(b)** A laptop costs £850 in London and CHF 995 in Zurich.
Calculate the difference in price between London and Zurich.
Give your answer in US dollars (\$), correct to 2 decimal places.

\$ ..... **[3]**

**280** Amira travels from the UK to Japan.
She changes 850 into Japanese Yen (JPY).
The bank charges a 2% fee on the amount exchanged.
The exchange rate is 1 = 180 JPY.

**(a)** Calculate the amount of JPY she receives.

..... JPY **[3]**

**(b)** While in Japan, she spends 115000 JPY.
She changes her remaining JPY back to pounds ().
The new exchange rate is 1 = 184 JPY.
The bank charges a fixed fee of 500 JPY for this transaction.
Calculate the amount of pounds she receives.

..... **[4]**

**281** Carlos buys a car in the USA for \$17 400. He pays a shipping fee of \$1 200.

**(a)** The exchange rate is £1 = \$1.24. Calculate the total cost of the car and shipping in pounds (£).

£ ..... **[2]**

**(b)** Carlos sells the car in the UK for £17 100.
Calculate his percentage profit on his total cost.

..... % **[3]**

**(c)** Carlos changes his profit of £2 100 into euros.
The exchange rate is £1 = 1.16 euros.
The bank charges a 2% commission fee for the exchange.
Calculate the amount he receives in euros.

..... euros **[3]**

**282** A trader buys 1000 items for $y$ each.
He pays a shipping fee of 250.
He sells all 1000 items in the USA for \$4.20 each.

The exchange rate is 1 = \$1.20.
The trader makes a profit of 40% on his total cost in Euros.

Calculate the value of $y$.

$y =$ ..... **[5]**

**283** **(a)** Carlos changes \$800 into euros () when the exchange rate is \$1 = 0.85.
He spends 420 on holiday.
He changes the remaining euros back into dollars when the exchange rate is \$1 = 0.82.
Calculate the amount of dollars he receives.
Give your answer correct to the nearest cent.

% ..... **[4]**

**(b)** In a sale, a shop reduces all prices by 15%.
The sale price of a jacket is \$61.20.
Calculate the original price of the jacket.

% ..... **[2]**

**284** A shop sells two brands of coffee, Brand A and Brand B.
Brand A costs \$12.50 per 500 grams. Brand B costs \$8.40 per 300 grams.

**(a)** Which brand is better value for money?
You must show all your working.

..... **[3]**

**(b)** The shop owner mixes $x$ grams of Brand A with $y$ grams of Brand B to make a new blend.
She makes 2 kg of the new blend.
The total cost of the coffee in the new blend is \$54.10.

Work out the value of $x$ and the value of $y$.

$x =$ ..... , $y =$ ..... **[4]**

**285** Marcus plans a business trip from Singapore to Tokyo.
Before leaving, he exchanges SGD 2500 into Japanese yen (JPY) at a bank that charges a 1.5% commission on the original amount before conversion.
The exchange rate is SGD 1 = JPY 112.

**(a)** Calculate the amount of Japanese yen Marcus receives.

JPY ..... **[3]**

**(b)** During his trip, Marcus spends JPY 198 400.
When he returns to Singapore, he converts his remaining Japanese yen back into Singapore dollars at an exchange rate of SGD 1 = JPY 115, with no commission charged.
Calculate how much he receives in Singapore dollars, correct to the nearest cent.

SGD ..... **[3]**

**(c)** If Marcus had exchanged the JPY 77 400 back at the original rate of SGD 1 = JPY 112 with no commission, how much more in Singapore dollars would he have received?
Give your answer correct to 2 decimal places.

SGD ..... **[2]**

**286** In Year 1, the exchange rate between dollars ($) and euros (EUR) was $\$1 = x$ EUR.
In Year 2, the exchange rate changed to $\$1 = (x - 0.05)$ EUR.

In Year 1, a laptop cost 400 EUR. In Year 2, the same model laptop cost 450 EUR.

The cost of the laptop in dollars increased by $100 from Year 1 to Year 2.

**(a)** Write down an expression in terms of $x$ for the cost of the laptop in dollars in Year 1.

$\$$ ..... **[1]**

**(b)** Form an equation in $x$ and show that it simplifies to $20x^2 - 11x - 4 = 0$.

**[4]**

**(c)** Solve the equation to find the value of $x$.

$x =$ ..... **[3]**

**287** **(a)** Car A costs \$24 000 in the USA. Car B costs €21 500 in Germany. The exchange rate is \$1 = €$x$.
The difference in the cost of the two cars is exactly €740.
Car B is more expensive than Car A.

Find the value of $x$.

$x =$ ..... **[3]**

**(b)** A customer changes a sum of dollars (\$) into British pounds (£). The exchange rate is £1 = \$1.28.
The bank deducts a 2.5% commission from the initial sum in dollars.
The remaining dollars are then changed into pounds.
The customer receives £780.

Calculate the initial sum of dollars the customer wanted to change.

\$ ..... **[3]**

**(c)** A company earns revenue in dollars (\$) which is then converted into euros (€). In 2022, the exchange rate was \$1 = €0.828.
In 2023, the company's revenue in dollars decreased by 8%.
However, the company's revenue in euros was exactly the same in 2023 as it was in 2022.

Calculate the exchange rate in 2023, giving your answer in the form \$1 = €$k$.

$k =$ ..... **[3]**

**288** A shop sells laptops.
In 2022, the price of a laptop was \$840. In 2023, the price increased by 15%. In 2024, the price decreased by $r$% from the 2023 price. The price of the laptop in 2024 was \$821.10.

**(a)** Calculate the value of $r$.

$r =$ ..... **[3]**

**(b)** The shop buys 450 of these laptops from a supplier in China.
The total cost is 2 430 000 Chinese Yuan (CNY).
The exchange rate is \$1 = 7.2 CNY.

Calculate the profit the shop makes on each laptop when sold at the 2022 price of \$840.

\$ ..... **[3]**

**(c)** A customer in the UK buys one of these laptops in 2022 for \$840. They pay with a credit card that charges a 2.5% foreign transaction fee. The exchange rate is £1 = \$1.32.
Calculate the total amount the customer pays in pounds (£).
Give your answer correct to the nearest penny.

£ ..... **[3]**

**289** A population of bacteria increases exponentially at a constant rate of $p\%$ per hour.
At 09 : 00, the population is 8 000.
At 12 : 00 on the same day, the population is 27 000.

**(a)** **(a)** Work out the value of $p$.

$p =$ ..... **[3]**

**(b)** Work out the population of the bacteria at 14 : 00 on the same day.

..... **[2]**

**290** **(a)** The value of a painting increases exponentially by $r\%$ each year.
After 2 years, the total percentage increase in the value of the painting is 44%.
Work out the value of $r$.

$r =$ ..... **[3]**

**(b)** The value of a rare stamp increases exponentially.
Its value doubles every 15 years.
The current value of the stamp is \$80.

Work out the value of the stamp after 45 years.

% ..... **[2]**

**291** A new car is bought for \$24 000. The value of the car depreciates exponentially at a rate of 8% per year.

Calculate the value of the car after 5 years.
Give your answer correct to the nearest dollar.

\$ ..... **[3]**

**292** The population of a city is 125 000.
The population increases exponentially at a rate of 1.8% per year.
Calculate the population of the city after 8 years.
Give your answer correct to the nearest integer.

..... **[3]**

**293** A new car is bought for \$26000. The value of the car depreciates exponentially at a rate of 15% each year.

Work out the value of the car after 4 years.
Give your answer correct to the nearest dollar.

% ..... **[3]**

**294** A rare stamp is bought for \$450. Its value increases exponentially at a rate of 6.5% per year. Calculate the value of the stamp after 12 years.

\$ ..... **[3]**

**295** A machine is bought for \$32 000. It depreciates exponentially at a rate of 12% each year.

Calculate the value of the machine after 5 years.
Give your answer correct to the nearest dollar.

\$ ..... **[3]**

**296** The population of a town is 65000.
The population increases exponentially at a rate of 2.8% per year.

**(a)** Calculate the expected population of the town after 5 years.
Give your answer correct to the nearest integer.

..... **[2]**

**(b)** Calculate the overall percentage increase in the population over the 5 years.
Give your answer correct to 1 decimal place.

..... % **[2]**

**297** The population of a town at the start of 2020 was 42 000.
The population increases at a constant rate of 2.5% each year.

**(a)** Calculate the population of the town at the start of 2026.
Give your answer correct to the nearest hundred.

..... **[2]**

**(b)** Calculate the percentage increase in the population from the start of 2020 to the start of 2026.
Give your answer correct to 1 decimal place.

..... % **[2]**

**298** A population of bacteria in a dish grows exponentially.

**(a)** Initially, there are 1500 bacteria in the dish.
The population increases by 8% every hour.
Work out the population of bacteria after 12 hours.
Give your answer correct to the nearest whole number.

..... **[2]**

**(b)** Work out the least number of complete hours it will take for the initial population of 1500 bacteria to be greater than 3000.

..... hours **[3]**

**299** The population of a town was 85 000 on 1 January 2010.
The population increases exponentially at a constant rate of $r\%$ per year.
On 1 January 2022, the population was 112 000.

**(a)** **(a)** Calculate the value of $r$, correct to 3 significant figures.

$r =$ ..... **[3]**

**(b)** Assuming the rate of increase remains the same, calculate the expected population on 1 January 2030.
Give your answer correct to the nearest thousand.

..... **[2]**

**300** A car is purchased for \$28 000. Its value depreciates exponentially at a rate of $r\%$ each year. After 4 years, its value is \$16 150.

**(a)** Calculate the value of $r$.
Give your answer correct to 3 significant figures.

$r =$ ..... **[3]**

**(b)** The value of the car continues to depreciate at the same rate.
Calculate the value of the car after a further 3 years.
Give your answer correct to the nearest dollar.

\$ ..... **[2]**

**301** A machine costs \$45000. Its value depreciates exponentially at a rate of $x\%$ per year. After 4 years, the value of the machine is \$26000.

**(a)** Calculate the value of $x$.
Give your answer correct to 3 significant figures.

$x =$ ..... **[3]**

**(b)** The machine will be replaced when its value first falls below \$10000.
Find the number of complete years before the machine is replaced.

..... years **[3]**

**302** **(a)** The population of a city is 125 000.
It grows exponentially at a rate of 2.4% per year.

Calculate the expected population after 10 years.
Give your answer correct to the nearest hundred.

..... **[3]**

**(b)** The population of another city was 85 000 in the year 2015.
By 2023, the population had decreased exponentially to 68 000.

Calculate the annual percentage decrease.

..... % **[3]**

**303** A delivery van is bought new for \$36 000. Its value depreciates exponentially by 14% each year.

**(a)** Calculate the value of the van after 4 years.
Give your answer correct to the nearest dollar.

\$ ..... **[2]**

**(b)** Calculate the number of complete years it takes for the value of the van to first fall below \$10 000.

..... years **[4]**

**304** Consider the depreciation of two different vehicles over time.

**(a)** A car is purchased for \$28 500. Its value depreciates exponentially at a rate of 12% per year. Calculate the value of the car after 5 years, giving your answer correct to the nearest dollar.

\$ ..... **[3]**

**(b)** A van, initially purchased for \$35 000, depreciates exponentially to a value of \$15 200 after 4 years.
Calculate the annual percentage rate of depreciation for this van.

..... % **[4]**

**305** The mass, $M$ grams, of a radioactive substance decays exponentially at a rate of $r\%$ per day.
After 2 days, the mass is 80 grams.
After 5 days, the mass is 40.96 grams.

**(a)** Calculate the value of $r$.

$r =$ ..... **[3]**

**(b)** Find the initial mass of the substance.

..... grams **[2]**

**(c)** Calculate the number of complete days from the start when the mass is first less than 10 grams.

..... days **[3]**

**306** A radioactive substance decays exponentially.

**(a)** The mass of the substance decreases by 12% every day.
The initial mass is 500 grams.

Calculate the mass remaining after 1 week (7 days).

..... $g$ **[2]**

**(b)** After $d$ complete days, the mass of the substance falls below 50 grams.

Find the smallest integer value of $d$.

$d =$ ..... **[3]**

**(c)** A different radioactive substance decays exponentially such that its mass halves every 15 days.

Calculate the percentage decrease in its mass each day.

..... % **[3]**

**307** The mass, $M$ grams, of a radioactive substance remaining after $t$ hours is modelled by the formula

$$M = M_0 \times k^{-t}$$

where $M_0$ and $k$ are positive constants.
At $t = 2$, the mass is 256 grams.
At $t = 5$, the mass is 108 grams.

**(a)** Show that $k = \frac{4}{3}$.

**[3]**

**(b)** Find the value of $M_0$.

$M_0 =$ ..... **[2]**

**(c)** Calculate the value of $t$ when the mass of the substance has decreased to 10% of its initial mass $M_0$.
Give your answer correct to 2 decimal places.

$t =$ ..... **[3]**

**308** Alex and Ben each invest \$4000 for 10 years. Alex invests his \$4000 in an account paying compound interest at a rate of 4.5% per year.
Ben invests his \$4000 in an account paying compound interest at a rate of $r$% per year.

**(a)** Calculate the total amount in Alex's account after 10 years.

\$ ..... **[2]**

**(b)** After 10 years, Ben has \$500 more than Alex.
Calculate the value of $r$.

$r =$ ..... **[4]**

**(c)** A different investment of \$2500 grows exponentially. It doubles in value every 8 years.
Calculate its value after 20 years.

\$ ..... **[2]**

**309** The population of a species of insect increases exponentially.
Initially, there are 2500 insects.
After 6 weeks, there are 9100 insects.

**(a)** Calculate the percentage increase in the population per week.

..... % **[4]**

**(b)** Calculate the insect population after 10 weeks, giving your answer correct to the nearest 100.

..... **[2]**

**(c)** Find the number of complete weeks it takes for the initial population to increase to over 50 000.

..... weeks **[3]**

**310** Simplify

$$\sqrt{75} - \sqrt{27} + \sqrt{12}$$

..... **[2]**

**311** Simplify.

$\sqrt{75} + \sqrt{48} - \sqrt{27}$

..... **[2]**

**312** Simplify fully.

$\sqrt{72} + \sqrt{50} - \sqrt{18}$

..... **[2]**

**313** Simplify

$$\sqrt{75} - \sqrt{27} + \sqrt{12}$$

..... **[2]**

**314** Simplify.

$\sqrt{75} + \sqrt{48} - \sqrt{27}$

..... **[2]**

**315** **(a)** Simplify.

$\sqrt{45}$

..... **[1]**

**(b)** Simplify.

$\sqrt{20} + \sqrt{80}$

..... **[2]**

**316** Simplify

$$\sqrt{75} + \sqrt{27} - \sqrt{12}$$

..... **[3]**

**317** Simplify the following expression, leaving your answer in the form $k\sqrt{3}$.

$\sqrt{48} - \sqrt{12}$

..... **[3]**

**318** **(a)** Simplify.

$$\sqrt{75} + \sqrt{12}$$

..... **[2]**

**(b)** Rationalise the denominator.

$$\frac{14}{\sqrt{7}}$$

..... **[2]**

**319** **(a)** Simplify

$$\sqrt{75} - \sqrt{12}$$

..... **[2]**

**(b)** Rationalise the denominator and simplify

$$\frac{18}{\sqrt{6}}$$

..... **[2]**

**320** **(a)** Simplify.

$\sqrt{75}$

..... **[2]**

**(b)** Simplify.

$2\sqrt{18} - \sqrt{8}$

..... **[2]**

**321** **(a)** **(i)** Simplify.

$$\sqrt{45} + \sqrt{20}$$

..... **[2]**

**(ii)** Rationalise the denominator.

$$\frac{18}{\sqrt{3}}$$

..... **[2]**

**322** **(a)** Simplify.

$\sqrt{45}$

..... **[1]**

**(b)** Work out the value of the following expression, giving your answer in the form $k\sqrt{2}$.

$\sqrt{72} + \sqrt{50} - \sqrt{18}$

..... **[2]**

**(c)** Rationalise the denominator.

$$\frac{21}{\sqrt{7}}$$

..... **[2]**

**323** **(a)** Expand and simplify.

$$(4 - \sqrt{3})(2 + \sqrt{3})$$

..... **[2]**

**(b)** Rationalise the denominator and simplify.

$$\frac{21}{\sqrt{7}}$$

..... **[2]**

**324** **(a)** Expand and simplify

$$(3 - \sqrt{5})^2$$

..... **[2]**

**(b)** Rationalise the denominator.

$$\frac{14}{\sqrt{7}}$$

..... **[2]**

**325** **(a)** Expand and simplify.

$$(3 - 2\sqrt{5})(4 + \sqrt{5})$$

..... **[2]**

**(b)** Rationalise the denominator and simplify.

$$\frac{14}{\sqrt{7}}$$

..... **[2]**

**326 (a) (a)** Expand and simplify

$$(3 - 2\sqrt{5})(4 + \sqrt{5})$$

..... **[2]**

**(b)** Rationalise the denominator.

$$\frac{15}{\sqrt{3}}$$

..... **[2]**

**327** **(a)** Expand and simplify

$$(3+\sqrt{2})(4-\sqrt{2})$$

..... **[2]**

**(b)** Rationalise the denominator.

$$\frac{22}{5-\sqrt{3}}$$

..... **[3]**

**328** **(a)** Expand and simplify.

$(3 + \sqrt{2})^2$

..... **[2]**

**(b)** Rationalise the denominator and simplify.

$$\frac{4}{3 - \sqrt{5}}$$

..... **[3]**

**329** **(a)** Expand and simplify.

$$(2\sqrt{5}-3)(2\sqrt{5}+3)$$

..... **[2]**

**(b)** Rationalise the denominator.

$$\frac{14}{3-\sqrt{2}}$$

..... **[3]**

**330** **(a)** Expand and simplify.

$$(3\sqrt{2}-1)(2\sqrt{2}+5)$$

..... **[2]**

**(b)** Rationalise the denominator and simplify.

$$\frac{26}{4-\sqrt{3}}$$

..... **[3]**

**331** **(a)** Expand and simplify

$$(3-\sqrt{5})^2$$

..... **[2]**

**(b)** Write $\dfrac{11}{4-\sqrt{5}}$ in the form $a+b\sqrt{5}$, where $a$ and $b$ are integers.

..... **[3]**

**332** **(a)** Rationalise the denominator and simplify fully.

$$\frac{15}{\sqrt{5}}$$

..... **[2]**

**(b)** Expand and simplify.

$$(2+\sqrt{3})(5-\sqrt{3})$$

..... **[3]**

**333** **(a)** Expand and simplify.

$$(3+\sqrt{5})(2-\sqrt{5})$$

..... **[2]**

**(b)** Rationalise the denominator and simplify.

$$\frac{14}{\sqrt{7}}$$

..... **[3]**

**334** **(a)** Expand and simplify.

$$(5 - 2\sqrt{7})(3 + \sqrt{7})$$

..... **[2]**

**(b)** Rationalise the denominator and simplify fully.

$$\frac{22}{5 - \sqrt{3}}$$

..... **[3]**

**335** **(a)** Expand and simplify.

$(4 - 3\sqrt{2})(5 + 2\sqrt{2})$

..... **[2]**

**(b)** Rationalise the denominator and simplify.

$$\frac{11}{4 - \sqrt{5}}$$

..... **[3]**

**(c)** Solve the equation.

$x\sqrt{27} - 8 = x\sqrt{3} + 10$

Give your answer in the form $k\sqrt{3}$ where $k$ is an integer.

$x =$ ..... **[3]**

**336** A rectangle has an area of $5 + \sqrt{7}$ cm$^2$ and a width of $3 - \sqrt{7}$ cm.
Calculate the length of the rectangle.
Give your answer in the form $a + b\sqrt{7}$ where $a$ and $b$ are integers.

..... *cm* **[4]**

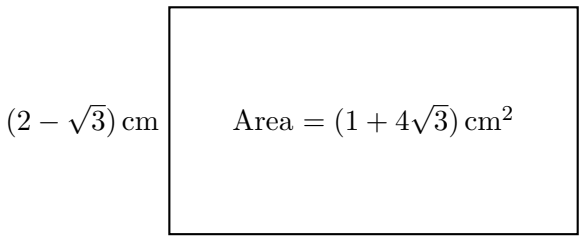
**337** Solve the equation.

$$x\sqrt{3} - 2 = x + \sqrt{12}$$

Give your answer in the form $a + b\sqrt{3}$, where $a$ and $b$ are integers.

$x =$ ..... **[5]**

**338** The diagram shows a rectangle with width $(2 - \sqrt{3})$ cm and length $(a + b\sqrt{3})$ cm, where $a$ and $b$ are integers.
The area of the rectangle is $(1 + 4\sqrt{3})\,\text{cm}^2$.




NOT TO SCALE

Work out the value of $a$ and the value of $b$.

$a =$ ..... , $b =$ ..... **[5]**

**339** **(a)** Rationalise the denominator and simplify.

$$\frac{5+\sqrt{3}}{2-\sqrt{3}}$$

..... **[3]**

**(b)** Solve the equation.
Give your answer in the form $a+b\sqrt{2}$, where $a$ and $b$ are integers.

$x(3-\sqrt{2})=14$

$x =$ ..... **[3]**

**340** **(a)** Express $\dfrac{5+\sqrt{3}}{2-\sqrt{3}}$ in the form $a+b\sqrt{3}$, where $a$ and $b$ are integers.

..... **[3]**

**(b)** A rectangle has an area of $12+4\sqrt{5}$ square centimetres and a width of $3+\sqrt{5}$ centimetres. Calculate the exact length of the rectangle, giving your answer in its simplest form.

Length = ..... cm **[4]**

**341** **(a)** Rationalise the denominator and simplify.

$$\frac{22}{5-\sqrt{3}}$$

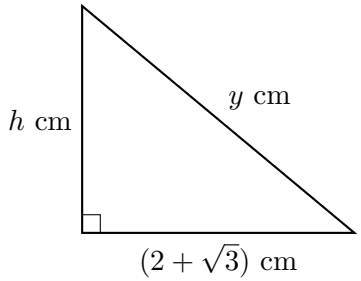
..... **[3]**

**(b)** Solve the equation.

$x\sqrt{2} = x + \sqrt{8}$ Give your answer in the form $a + b\sqrt{2}$, where $a$ and $b$ are integers.

$x =$ ..... **[4]**

**342** A right-angled triangle has a base of $(2 + \sqrt{3})$ cm and an area of $(5 + 3\sqrt{3})$ cm$^2$.




**(a)** Show that the exact height, $h$, is $(2 + 2\sqrt{3})$ cm.

**[4]**

**(b)** The hypotenuse of the triangle is $y$ cm.
Work out the exact value of $y^2$.
Give your answer in the form $a + b\sqrt{3}$, where $a$ and $b$ are integers.

$y^2 =$ ..... **[3]**

**343** **(a)** A rectangle has an area of $(5 + 2\sqrt{3})$ cm$^2$ and a length of $(2 + \sqrt{3})$ cm.

Work out the width of the rectangle in the form $a + b\sqrt{3}$, where $a$ and $b$ are integers.

..... cm **[3]**

**(b)** Solve the equation.

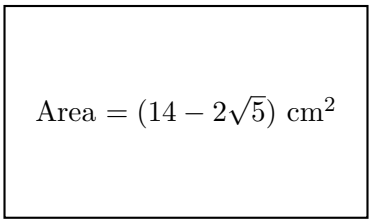
$$x\sqrt{5} - 2 = 2x + \sqrt{20}$$

Give your answer in the form $c + d\sqrt{5}$, where $c$ and $d$ are integers.

Show all your working.

$x =$ ..... **[4]**

**344** A rectangle has length $(3 + \sqrt{5})$ cm and area $(14 - 2\sqrt{5})$ cm$^2$.




**(a)** Calculate the width of the rectangle.
Give your answer in the form $a + b\sqrt{5}$, where $a$ and $b$ are integers.

..... cm **[4]**

**(b)** The length of the diagonal of the rectangle is $d$ cm.
Show that $d^2 = 308 - 124\sqrt{5}$.

**[3]**

**345** **(a)** Rationalise the denominator and simplify fully.

$$\frac{6}{3-\sqrt{3}}$$

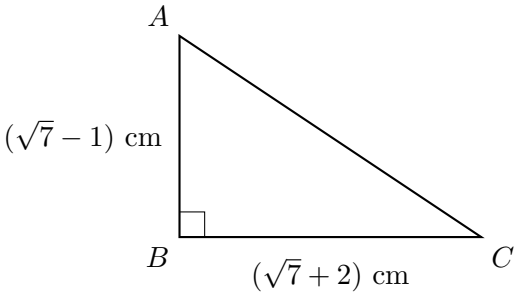
..... **[4]**

**(b)** Write the following expression in the form $a\sqrt{2}+b\sqrt{3}$, where $a$ and $b$ are integers.

$$\frac{\sqrt{24}+\sqrt{54}}{\sqrt{3}}-\frac{10}{\sqrt{2}}$$

..... **[3]**

**346** The diagram shows a right-angled triangle $ABC$.
$AB = (\sqrt{7} - 1)$ cm, $BC = (\sqrt{7} + 2)$ cm and angle $ABC = 90°$.




**(a)** Calculate the area of triangle $ABC$.
Give your answer in the form $\dfrac{p + q\sqrt{7}}{2}$, where $p$ and $q$ are integers.

Area = ..... $cm^2$ **[3]**

**(b)** Find the length of $AC$ in the form $\sqrt{a + b\sqrt{7}}$, where $a$ and $b$ are integers.

$AC$ = ..... cm **[4]**

**347** A right-angled triangle has a base of length $(3+\sqrt{2})$ cm and an area of $(5+2\sqrt{2})$ cm$^2$.

**(a)** Show that the height of the triangle is $\dfrac{2(5+2\sqrt{2})}{3+\sqrt{2}}$ cm.

**[2]**

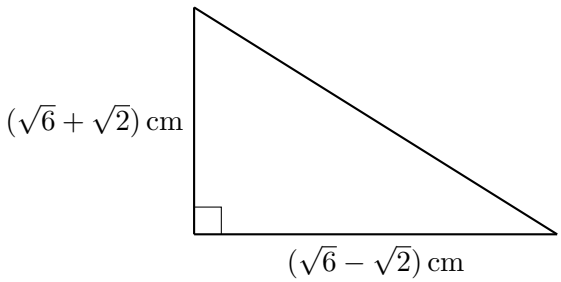
**(b)** Rationalise the denominator of $\dfrac{2(5+2\sqrt{2})}{3+\sqrt{2}}$ to express the height in the form $\dfrac{a+b\sqrt{2}}{c}$, where $a$, $b$ and $c$ are integers.

..... cm **[4]**

**(c)** Given instead that another expression for a length is $\dfrac{1}{\sqrt{7}-2}$, write this length in the form $\dfrac{\sqrt{7}+c}{d}$ where $c$ and $d$ are integers.

..... **[2]**

**348** The diagram shows a right-angled triangle.




**(a)** Calculate the area of the triangle.

..... $cm^2$ **[3]**

**(b)** Show that the length of the hypotenuse is 4 cm.

**[4]**

**(c)** Another right-angled triangle is mathematically similar to this triangle.
The area of the similar triangle is 18 $cm^2$.
Calculate the length of its hypotenuse.

..... cm **[3]**

**349** Work out the value of the expression

$$3x^2 - 5xy$$

when $x = -4$ and $y = 2$.

..... **[2]**

**350** Work out the value of

$$5x^2 - 3y$$

when $x = -2$ and $y = 4$.

..... **[2]**

**351** Find the value of $3a - 2b$ when $a = 4$ and $b = -5$.

..... **[2]**

**352** Work out the value of

$$3x^2 - 4y$$

when $x = -3$ and $y = -5$.

..... **[2]**

**353** Work out the value of

$$\frac{3x - y^2}{2z}$$

when $x = 5$, $y = -4$ and $z = 0.5$.

..... [**2**]

**354** Given the formula $v = u + at$.

Find the value of $v$ when $u = 5$, $a = -2$, and $t = 3$.

$v =$ ..... **[3]**

**355** **(a)** Work out the value of $4x - 3y$ when $x = -2$ and $y = 5$.

..... **[2]**

**(b)** Work out the value of $\dfrac{p^2 - 7}{2q}$ when $p = 5$ and $q = -3$.

..... **[2]**

**356** **(a)** Work out the value of $3a - 5b$ when $a = -2$ and $b = 4$.

..... **[2]**

**(b)** Work out the value of $x^2 + 4x$ when $x = -3$.

..... **[2]**

**357** Consider the formula

$$P = 3a^2 - 2ab + 5b$$

**(a)** Calculate the value of $P$ when $a = 4$ and $b = -3$.

$P =$ ..... **[2]**

**(b)** Calculate the value of $b$ when $P = 19$ and $a = -2$.

$b =$ ..... **[2]**

**358** **(a)** Work out the value of $5x - 2y$ when $x = 3$ and $y = -4$.

..... **[2]**

**(b)** Evaluate

$$\frac{a^2 - b^3}{2c}$$

when $a = -4$, $b = 2$ and $c = 3$.

..... **[2]**

**359 (a) (a)** The cost of hiring a plumber is calculated using the formula

$C = 40 + 25h$ where $C$ is the cost in dollars and $h$ is the number of hours worked.
Work out the cost of hiring a plumber for 3.5 hours.

$C = \$ \ldots\ldots\ldots\ldots\ldots$ **[2]**

**(b)** Given that $v = u + at$, find the value of $v$ when $u = -12$, $a = 9.8$ and $t = 3$.

$v = \ldots\ldots\ldots\ldots\ldots$ **[3]**

**360** **(a)** A plumber charges a fixed fee of \$40 plus \$25 for each hour he works.
Write down an expression, in terms of $h$, for the total cost, in dollars, for $h$ hours of work.

\$ ..... **[1]**

**(b)** Given the formula

$$P = \frac{k^3 - m^2}{\sqrt{n}}$$

Work out the exact value of $P$ when $k = -2$, $m = -3$ and $n = \frac{1}{16}$.

$P =$ ..... **[3]**

**361** Find the exact value of

$$W = \frac{p^{-2} + q^3}{\sqrt[3]{r}}$$

when $p = \frac{1}{2}$, $q = -2$ and $r = -\frac{1}{8}$.

$W =$ ..... **[4]**

**362** The distance, $d$, between two points $(x_1, y_1)$ and $(x_2, y_2)$ is given by the formula

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

**(a)** Work out the value of $d$ when $x_1 = 4$, $y_1 = -3$, $x_2 = -8$, and $y_2 = 2$.

$d =$ ..... **[2]**

**(b)** Given that $d = 17$, $x_1 = k$, $y_1 = 8$, $x_2 = 3$, and $y_2 = -7$, work out the two possible values of $k$.

$k =$ ..... , $k =$ ..... **[4]**

**363** The $n$th term of a sequence is given by the expression

$$T_n = an^2 + bn + c$$

where $a$, $b$, and $c$ are constants.
The first three terms of the sequence are $T_1 = 4$, $T_2 = 13$, and $T_3 = 28$.

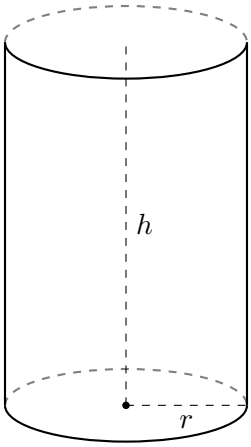
**(a)** By substituting $n = 1$, $n = 2$, and $n = 3$ into the expression, find the values of $a$, $b$, and $c$.

$a =$ ..... , $b =$ ..... , $c =$ ..... **[5]**

**(b)** Find the value of $k$ such that $T_{k+1} - T_k = 57$.

$k =$ ..... **[2]**

**364** A cylindrical container of radius $r$ and height $h$ is open at the top.
The total exterior surface area of the container, $A$, consists of the curved surface area and the circular base.




**(a)** Show that the total surface area $A$ can be written as

$$A = \pi r(2h + r)$$

**[2]**

**(b)** **(i)** Calculate the exact surface area $A$ when $r = 4$ and $h = 6$.
Give your answer in terms of $\pi$.

$A =$ ..... **[2]**

**(ii)** The volume of the cylinder is $V$.
The formula for the volume is $V = \pi r^2 h$.
Express $h$ in terms of $V$, $\pi$ and $r$, and substitute this into the expression for $A$ to show that

$$A = \frac{2V}{r} + \pi r^2$$

**[3]**

**365** Consider the expression $P = \dfrac{2x^2 - 3y}{x - y}$.

**(a)** Calculate the value of $P$ when $x = -2$ and $y = 4$.

$P =$ ..... **[3]**

**(b)** Rearrange the formula to make $y$ the subject.

$y =$ ..... **[4]**

**366** The mathematical operation $*$ is defined for all non-zero numbers $a$ and $b$ as

$$a * b = \frac{a^2 - b^2}{ab}$$

**(a)** Evaluate the following, giving each answer as a fraction in its simplest form.

**(i)** $6 * (-2)$

..... **[2]**

**(ii)** $\frac{1}{2} * 4$

..... **[3]**

**(b)** Given that $x * 2 = \frac{3}{2}$, find the positive value of $x$.

$x =$ ..... **[3]**

**367** A plumber charges a fixed call-out fee of $\$C$ and an hourly rate of $\$R$.

**(a)** Write down an expression for the total cost, $T$ dollars, for a job that takes $h$ hours.

$T =$ ..... **[1]**

**(b)** The call-out fee is \$45 and the hourly rate is \$32.50.
Work out the total cost for a job that takes 3.5 hours.

\$ ..... **[2]**

**368** A formula is given by

$$P = \frac{\sqrt{x - y}}{z^3}$$

Calculate the value of $P$ when $x = 12.5$, $y = 3.84$, and $z = -1.2$.
Give your answer correct to 3 significant figures.

$P =$ ..... **[3]**

**369** A solid shape has a volume, $V$, given by the formula

$$V = \pi r^2 h + \frac{2}{3}\pi r^3$$

Work out the value of $V$ when $r = 4.5$ and $h = 8.2$.
Give your answer correct to 3 significant figures.

$V =$ ..... **[3]**

**370** **(a)** A plumber charges a fixed fee of \$45 plus \$32 for each hour worked.
Write down a formula for the total charge, $C$ dollars, when the plumber works for $h$ hours.

$C =$ ..... **[1]**

**(b)** Work out the total charge when the plumber works for 3.5 hours.

\$ ..... **[1]**

**(c)** Work out the value of

$$x^3 - 4xy$$

when $x = -3$ and $y = 2.5$.

..... **[2]**

**371** The volume, $V$, of a cone with radius $r$ and height $h$ is given by the formula

$$V = \frac{1}{3}\pi r^2 h$$

**(a)** Calculate the value of $V$ when $r = 6.4$ and $h = 12.6$.
Give your answer correct to 1 decimal place.

$V =$ ..... **[2]**

**(b)** Work out the value of $r$ when $V = 800$ and $h = 14$.
Give your answer correct to 3 significant figures.

$r =$ ..... **[3]**

**372** The cost, $C$ dollars, of hiring a campervan for $d$ days and travelling $m$ kilometres is given by the formula

$$C = 85d + 0.24m$$

**(a)** Calculate the cost of hiring the campervan for 7 days and travelling 450 kilometres.

$ ..... **[2]**

**(b)** A family hires the campervan for 14 days.
The total cost is $1475.60.
Work out the number of kilometres they travelled.

..... km **[3]**

**373** The cost, $C$ dollars, of hiring a plumber for $h$ hours is given by the formula:

$C = 45 + 35h$

**(a)** Calculate the cost of hiring the plumber for 3.5 hours.

$C =$ ..... **[2]**

**(b)** The plumber charges a total of \$325 for a job.
Calculate the number of hours the plumber worked.

$h =$ ..... hours **[3]**

**374** The total surface area, $A$ cm$^2$, of a solid cone with base radius $r$ cm and vertical height $h$ cm is given by the formula

$$A = \pi r \left( r + \sqrt{r^2 + h^2} \right)$$

**(a)** Calculate the value of $A$ when $r = 5$ and $h = 12$.
Give your answer correct to 3 significant figures.

$A =$ ..... **[2]**

**(b)** Another solid cone has base radius $r = 6$ and total surface area $A = 96\pi$.
Calculate the vertical height, $h$, of this cone.

$h =$ ..... **[4]**

**375** The volume, $V$, of a spherical cap of height $h$ and base radius $a$ is given by the formula

$$V = \frac{1}{6}\pi h(3a^2 + h^2)$$

**(a)** Calculate the volume of a spherical cap with height $h = 4.2$ cm and base radius $a = 6.5$ cm. Give your answer correct to 1 decimal place.

$V =$ ..... cm$^3$ **[2]**

**(b)** Rearrange the formula to make $a^2$ the subject.

$a^2 =$ ..... **[3]**

**(c)** Find the base radius, $a$, of a spherical cap with volume $V = 150$ cm$^3$ and height $h = 3$ cm. Give your answer correct to 3 significant figures.

$a =$ ..... cm **[1]**

**376** The value of $y$ is given by the formula

$$y = \frac{a - b^2}{c - \sqrt{d}}$$

Work out the value of $y$ when $a = -4.5$, $b = -3.2$, $c = 1.8$ and $d = 0.64$.

$y =$ ..... **[3]**

**377** A formula used in physics is

$$s = ut + \frac{1}{2}at^2$$

**(a)** **(a)** Calculate the value of $s$ when $u = 5.2$, $t = 4.5$ and $a = -9.8$.

$s =$ ..... **[2]**

**(b)** Calculate the positive value of $t$ when $s = -9.8$, $u = 4.9$ and $a = -9.8$.

$t =$ ..... **[3]**

**378** The volume, $V$, and surface area, $A$, of a solid are given by the formulas below.

**(a)** The formula for the volume is

$$V = \frac{1}{3}\pi h(r^2 + rR + R^2)$$

Calculate the value of $V$ when $h = 12.5$, $r = 4.2$, and $R = 7.8$.
Give your answer correct to 3 significant figures.

$V =$ ..... **[3]**

**(b)** The formula for the surface area is

$$A = \pi(R + r)\sqrt{(R - r)^2 + h^2} + \pi r^2 + \pi R^2$$

Calculate the value of $A$ when $h = 12.5$, $r = 4.2$, and $R = 7.8$.
Give your answer correct to 1 decimal place.

$A =$ ..... **[3]**

**379** Expand and simplify.

$(3x - 2)(2x + 5)$

..... **[2]**

**380** **(a)** Expand and simplify.

$3(2x - 1) - 2(x + 4)$

..... **[2]**

**(b)** Factorise fully.

$15x^2y - 20xy^2$

..... **[2]**

**381** **(a)** Expand and simplify.

$4(2a - 3b) - 3(a - 5b)$

..... **[2]**

**(b)** Factorise completely.

$12x^2y - 18xy^2$

..... **[2]**

**382** **(a)** Expand and simplify.

$4(2x - 3y) - 3(x - 5y)$

..... **[2]**

**(b)** Factorise completely.

$14a^2b - 21ab^2$

..... **[2]**

**383** **(a)** Expand and simplify

$$4(3x - 2) - 2(x - 5)$$

..... **[2]**

**(b)** Factorise fully

$$15a^2b - 10ab^2$$

..... **[2]**

**384** **(a)** Expand and simplify.

$3(2x - 5) - 2(x + 4)$

..... **[2]**

**(b)** Factorise completely.

$8x^2y - 12xy^2$

..... **[2]**

**385** **(a)** Simplify

$$4x^2 - 7xy + 3 - 9x^2 + 2xy - 8$$

..... **[2]**

**(b)** Factorise fully

$$14a^3b - 21a^2b^2$$

..... **[2]**

**386** **(a)** Expand and simplify

$$3(2x - 5) - 4(x - 2)$$

..... **[2]**

**(b)** Factorise fully

$$12x^2y - 18xy^2$$

..... **[2]**

**387** **(a)** Simplify.

$3a - 5b + 2a + 8b$

..... **[2]**

**(b)** Factorise completely.

$12x^2y - 18xy^2$

..... **[2]**

**388** **(a)** Expand and simplify.

$3(2x - 5) - 4(x - 3)$

..... **[2]**

**(b)** Factorise fully.

$14x^2y + 21xy^2$

..... **[2]**

**389** **(a)** Factorise completely.

$12x^2 - 27y^2$

..... **[2]**

**(b)** Factorise completely.

$2ax - 6ay + bx - 3by$

..... **[2]**

**390** **(a)** Expand and simplify.

$(2x - 3)(x + 5)$

..... **[2]**

**(b)** Factorise.

$y^2 - 16x^2$

..... **[1]**

**(c)** Factorise.

$p^2 - 4pq - 5q^2$

..... **[2]**

**391** **(a)** Expand and simplify.

$(2x - 3)(x + 5)$

..... **[2]**

**(b)** Factorise.

$ac - 3bc + 2ad - 6bd$

..... **[2]**

**(c)** Factorise.

$16x^2 - 25y^2$

..... **[1]**

**392** **(a)** Expand and simplify.

$(2x - 3)(x + 4)$

..... **[2]**

**(b)** Factorise completely.

$ax - bx - 3ay + 3by$

..... **[2]**

**(c)** Factorise.

$49x^2 - 16y^2$

..... **[2]**

**393** **(a)** Factorise.

$x^2 - 8x + 15$

..... **[2]**

**(b)** Factorise.

$2x^2 - 5x - 12$

..... **[2]**

**(c)** Factorise.

$ax - 3a + 2bx - 6b$

..... **[2]**

**394** **(a)** Expand and simplify

$$(3x-2)(x+5)-(2x-3)^2$$

..... **[3]**

**(b)** Factorise completely:

**(i)** $3x^3-12xy^2$

..... **[2]**

**(ii)** $6x^2-11x-10$

..... **[2]**

**395** **(a)** Expand and simplify.

$(2x - 5)(3x + 1)$

..... **[2]**

**(b)** Factorise.

$x^2 - 7x - 18$

..... **[2]**

**(c)** Factorise completely.

$18x^2 - 50y^2$

..... **[3]**

**396** **(a)** Expand and simplify.

$(x-2)(2x+3)(x-5)$

..... **[3]**

**(b)** Factorise completely.

$12x^3 + 17x^2 - 5x$

..... **[3]**

**397** **(a)** Factorise completely.

$18x^2 - 50y^2$

..... **[3]**

**(b)** Factorise completely.

$3x^3 + 10x^2 - 8x$

..... **[3]**

**398** **(a)** Expand and simplify.

$$(x-2)(x+3)(2x-1)$$

..... **[3]**

**(b)** Factorise completely.

**(i)**

$$6x^2 - 11x - 10$$

..... **[2]**

**(ii)**

$$ax - ay + 3cx - 3cy$$

..... **[2]**

**399** **(a)** Factorise

$$cx - 5dx + 3cy - 15dy$$

..... **[2]**

**(b)** Factorise fully

$$2x^3 - 18x$$

..... **[2]**

**(c)** Factorise fully

$$12x^3 - 14x^2 - 10x$$

..... **[3]**

**400** **(a)** Expand and simplify.

$(x-4)(2x+1)(x+3)$

..... **[3]**

**(b)** Factorise completely.

$2ax - 3by + 2ay - 3bx$

..... **[2]**

**(c)** Factorise.

$6x^2 + 7x - 20$

..... **[2]**

**401** **(a)** Expand and simplify

$$(2x-1)(x+3)(3x-2)$$

..... **[3]**

**(b)** Factorise completely

$$4ax - 6ay - 6bx + 9by$$

..... **[2]**

**(c)** Given that $x^2 - y^2 = 45$ and $x - y = 5$, find the value of $(x+y)^2$.

..... **[3]**

**402** **(a)** Factorise

$$6ax - 2ay - 3bx + by$$

..... **[2]**

**(b)** Factorise completely.

**(i)**

$$50x^2 - 8y^2$$

..... **[3]**

**(ii)**

$$2x^3 + 5x^2 - 12x$$

..... **[3]**

**403** **(a)** Expand and simplify

$$(x-3)(2x+1)(x+4)$$

..... **[3]**

**(b)** Factorise

$$2x^2 - 7x - 15$$

..... **[2]**

**404** **(a)** Expand and simplify.

$(2x - 3)(x + 5)$

..... **[2]**

**(b)** Factorise.

$x^2 - 7x - 18$

..... **[2]**

**(c)** Factorise completely.

$4x^2 - 25y^2$

..... **[2]**

**405** **(a)** Expand and simplify

$$(x-4)(2x+3)(x+2)$$

..... **[3]**

**(b)** Factorise

$$16x^2 - 49y^2$$

..... **[2]**

**(c)** Factorise

$$3x^2 - 10x - 8$$

..... **[2]**

**406** **(a)** Expand and simplify.

$(x-4)(2x+3)(x+1)$

..... **[3]**

**(b)** Factorise completely.

$6x^2 - 11x - 10$

..... **[2]**

**(c)** Factorise completely.

$x^3 + 3x^2 - 4x - 12$

..... **[3]**

**407** **(a)** Expand and simplify.

$(x-2)(2x+1)(x+3)$

..... **[3]**

**(b)** Factorise.

$6x^2 - 11x - 10$

..... **[2]**

**(c)** Factorise completely.

$3x^3 + 5x^2 - 2x$

..... **[3]**

**408** **(a)** Expand and simplify.

$(x-2)(x+3)(2x-1)$

..... **[3]**

**(b)** Factorise.

$6x^2 - 11x - 10$

..... **[2]**

**(c)** Factorise completely.

$2x^3 - 18x$

..... **[3]**

**409** Write as a single fraction in its simplest form.

$$\frac{2x}{5} - \frac{x-1}{3}$$

..... **[2]**

**410** Write as a single fraction in its simplest form.

$$\frac{2x-1}{3}+\frac{x+4}{5}$$

..... **[2]**

**411** Write as a single fraction in its simplest form.

$$\frac{3x}{4} - \frac{x-2}{5}$$

..... **[2]**

**412** Write as a single fraction in its simplest form.

$$\frac{5}{x+2} + \frac{2}{3x-1}$$

..... **[3]**

**413** Write as a single fraction in its simplest form.

$$\frac{x+3}{4} - \frac{2x-1}{5}$$

..... **[3]**

**414** Write as a single fraction in its simplest form.

$$\frac{3x}{4} - \frac{x-2}{3}$$

..... **[3]**

**415** Write as a single fraction in its simplest form.

$$\frac{2x}{3} - \frac{x-4}{5}$$

..... **[3]**

**416** **(a)** Write as a single fraction in its simplest form.

$$\frac{2x}{3} + \frac{x-1}{4}$$

..... **[2]**

**(b)** Simplify fully.

$$\frac{15x^3y^2}{5xy^4}$$

..... **[2]**

**417** **(a)** Write as a single fraction in its simplest form

$$\frac{3x}{4} - \frac{2(x-3)}{5}$$

..... **[2]**

**(b)** Simplify

$$\frac{6a^3b}{5c^2} \div \frac{9a^2}{10bc}$$

..... **[2]**

**418** **(a)** Simplify fully.

$$\frac{3x}{4} + \frac{2x}{5}$$

..... **[2]**

**(b)** Simplify fully.

$$\frac{5a^2b}{12} \div \frac{10ab^2}{9}$$

..... **[2]**

**419** Simplify fully.

$$\frac{x^2 - 16}{x^2 - 5x + 4}$$

..... **[3]**

**420** Simplify fully.

$$\frac{3x^2 - 12}{x^2 + 3x - 10}$$

..... **[4]**

**421** **(a)** Write as a single fraction in its simplest form.

$$\frac{3}{x-2} - \frac{2}{x+5}$$

..... **[3]**

**(b)** Simplify fully.

$$\frac{x^2 - 16}{x^2 - 2x - 8}$$

..... **[3]**

**422** **(a)** Simplify fully.

$$\frac{x^2 - 9}{2x^2 + 5x - 3}$$

..... **[3]**

**(b)** Write as a single fraction in its simplest form.

$$\frac{4}{x - 2} - \frac{3}{x + 1}$$

..... **[3]**

**423** **(a)** Write as a single fraction in its simplest form

$$\frac{4}{x-3} - \frac{3}{x+2}$$

..... **[3]**

**(b)** Simplify

$$\frac{x^2-9}{x^2+5x+6}$$

..... **[3]**

**424** Simplify fully.

$$\frac{2x^2 + 5x - 12}{4x^2 - 9}$$

..... **[4]**

**425** Simplify fully.

$$\frac{3x^2 - 5x - 2}{x^2 - 4} \div \frac{3x + 1}{x^2 - x - 2}$$

..... **[6]**

**426** **(a)** Simplify fully

$$\frac{2x^2 - 5x - 3}{4x^2 - 1} \div \frac{x^2 - 9}{2x^2 + 5x - 3}$$

..... **[4]**

**(b)** Write as a single fraction in its simplest form

$$\frac{3}{x^2 - 4} - \frac{2}{x^2 + 3x + 2}$$

..... **[3]**

**427** **(a)** Simplify fully.

$$\frac{2x^2 - 5x - 3}{4x^2 - 9}$$

..... **[3]**

**(b)** Write as a single fraction in its simplest form.

$$\frac{3}{x-2} + \frac{4}{x+1} - \frac{2x}{x^2 - x - 2}$$

..... **[4]**

**428** Simplify fully.

$$\frac{x^2 - 9}{2x^2 + 5x - 3}$$

..... **[3]**

**429** Write as a single fraction in its simplest form.

$$\frac{5}{x-2} - \frac{3}{2x+1}$$

..... **[3]**

**430** Simplify fully.

$$\frac{2x^2 - 9x + 9}{2x^2 - 18}$$

..... **[4]**

**431** Simplify fully.

$$\frac{2x^2 - 8}{x^2 + x - 6}$$

..... **[5]**

**432** **(a)** Write as a single fraction in its simplest form.

$$\frac{x+3}{2x-1} - \frac{x-2}{x+4}$$

..... **[3]**

**(b)** Simplify fully.

$$\frac{x^2-16}{x^2+4x}$$

..... **[2]**

**433** Simplify fully.

$$\frac{x^2 - 25}{x^2 - 3x} \div \frac{x^2 + 6x + 5}{2x - 6}$$

..... **[4]**

**434** **(a)** Simplify fully.

$$\frac{2x^2 - 8x}{x^2 - 16} \div \frac{6x}{3x + 12}$$

..... **[4]**

**(b)** Write as a single fraction in its simplest form.

$$\frac{5}{x - 3} + \frac{2x + 1}{x^2 - 9}$$

..... **[3]**

**435** **(a)** Write as a single fraction in its simplest form.

$$\frac{4}{x-3} - \frac{3}{2x-1}$$

..... **[3]**

**(b)** Simplify fully.

$$\frac{x^2-25}{2x^2-9x-5} \div \frac{x+5}{4x+2}$$

..... **[4]**

**436** **(a)** Write as a single fraction in its simplest form.

$$\frac{3}{x-2} - \frac{x+5}{x^2-4}$$

..... **[3]**

**(b)** Simplify fully.

$$\frac{4x^2-25}{2x^2-5x} \times \frac{6x^3}{4x^2+10x}$$

..... **[4]**

**437** **(a)** Write as a single fraction in its simplest form.

$$\frac{5}{x-2}+\frac{4}{2x+3}$$

..... **[3]**

**(b)** Simplify fully.

$$\frac{2x^2-5x-12}{4x^2-9}\div\frac{x-4}{2x-3}$$

..... **[4]**

**438** **(a)** Simplify fully.

$$\frac{2x^2 + 5x - 3}{4x^2 - 1}$$

..... **[4]**

**(b)** Express as a single fraction in its simplest form.

$$\frac{5}{x - 3} + \frac{x + 2}{x^2 - 4x + 3}$$

..... **[4]**

**439** Simplify fully.

$4x^3y^{-2} \times 3x^5y^6$

..... **[2]**

**440** **(a)** Simplify.

$$4x^2 \times 3x^{-5}$$

..... **[1]**

**(b)** Simplify.

$$\left(\frac{8x^9}{27}\right)^{\frac{1}{3}}$$

..... **[2]**

**441** **(a)** Simplify.

$$4x^3 \times 5x^{-2}$$

..... **[1]**

**(b)** Simplify.

$$\left(\frac{2x^4}{y^3}\right)^2$$

..... **[2]**

**(c)** Solve the equation.

$$3^p = \frac{1}{27}$$

$p =$ ..... **[1]**

**442** **(a)** Simplify.

$$4x^3y^{-2} \times 3x^5y^6$$

..... **[2]**

**(b)** Find the exact value of

$$16^{-\frac{1}{4}}$$

..... **[2]**

**443** **(a)** Simplify.

$5x^4 \times 3x^{-2}$

..... **[2]**

**(b)** Find the value of

$16^{\frac{3}{4}}$

..... **[2]**

**444** **(a)** Simplify.

$y^5 \times y^{-2}$

..... **[1]**

**(b)** Work out the value of

$5^{-2}$

..... **[1]**

**(c)** Simplify.

$(2x^4)^3$

..... **[2]**

**445** **(a)** Simplify

$$4x^{-3} \times \frac{3}{2}x^{\frac{1}{2}}$$

..... **[2]**

**(b)** Simplify

$$\left(\frac{3y^4}{2}\right)^3$$

..... **[2]**

**446** **(a)** Find the value of $x$ when

$$2^x = \frac{1}{8}$$

$x =$ ..... **[1]**

**(b)** Simplify

$$(4y^6)^{\frac{1}{2}}$$

..... **[2]**

**(c)** Work out the exact value of

$$16^{-\frac{3}{4}}$$

..... **[1]**

**447** **(a)** Simplify.

$$w^4 \times w^7$$

..... **[1]**

**(b)** Simplify.

$$(3x^3)^2$$

..... **[2]**

**(c)** Simplify.

$$\frac{15y^8}{5y^{-2}}$$

..... **[2]**

**448** **(a)** Simplify.

$p^4 \times p^7$

..... **[1]**

**(b)** Simplify.

$(3q^3)^2$

..... **[2]**

**(c)** Work out the value of $27^{\frac{2}{3}}$.

..... **[2]**

**449** **(a)** Simplify fully

$$\frac{12x^5y^3}{3x^2y^{-1}}$$

..... **[2]**

**(b)** Solve the equation

$$27^{x-1} = 9^{2x+1}$$

$x =$ ..... **[3]**

**450** **(a)** Work out the exact value of

$$32^{-\frac{2}{5}}$$

..... **[2]**

**(b)** Solve the equation.

$$5^{x+2} = 125^{x}$$

$x =$ ..... **[3]**

**451** **(a)** Work out the exact value of

$$125^{\frac{2}{3}}$$

..... **[2]**

**(b)** Simplify fully.

$$\frac{15x^4y^{-3}}{5x^{-2}y^5}$$

..... **[3]**

**452** **(a)** Solve the equation.

$$16^x = 2$$

$x =$ ..... **[1]**

**(b)** Work out the exact value of

$$125^{\frac{2}{3}} \times 5^{-2}$$

..... **[2]**

**(c)** Solve the equation.

$$5^{2x-1} = 125^x$$

$x =$ ..... **[3]**

**453** **(a)** Simplify fully.

$$4x^{-3} \times \frac{3}{2}x^{\frac{1}{2}}$$

..... **[2]**

**(b)** Solve the equation.

$$3^{2x+1} = 27$$

$x =$ ..... **[2]**

**(c)** Simplify fully.

$$\left(\frac{8a^6}{b^3}\right)^{\frac{1}{3}}$$

..... **[2]**

**454** **(a)** Simplify fully.

$$\left(\frac{2x^5}{3y^2}\right)^3$$

..... **[3]**

**(b)** Solve the equation.

$$9^x = 27^{x+1}$$

$x =$ ..... **[3]**

**455** **(a)** Work out the value of

$$\left(\frac{27}{64}\right)^{-\frac{2}{3}}$$

..... **[2]**

**(b)** Solve the equation

$$8^{2x-1} = 32^{x+2}$$

$x =$ ..... **[4]**

**456** **(a)** Simplify fully.

$$\left(\frac{27x^9}{64y^{12}}\right)^{-\frac{1}{3}}$$

..... **[3]**

**(b)** Solve the equation.

$$2^{3x-1} = 32^x$$

$x =$ ..... **[3]**

**457** **(a)** Simplify fully.

$$\left(\frac{16x^{12}}{y^8}\right)^{\frac{3}{4}}$$

..... **[3]**

**(b)** Solve the equation.

$$2^{x+3} = 8^x$$

$x =$ ..... **[3]**

**458** **(a)** Solve the equation.

$$9^{2x} = 27^{x+1}$$

$x =$ ..... **[3]**

**(b)** Simplify fully.

$$\left(\frac{16a^8}{81b^{12}}\right)^{-\frac{3}{4}}$$

..... **[3]**

**459** **(a)** Solve the equation.

$$16^{2x-1} = 8^{x+2}$$

$x =$ ..... **[3]**

**(b)** Simplify fully.

$$\left(\frac{81x^{12}}{16y^8}\right)^{-\frac{3}{4}}$$

..... **[3]**

**460** **(a)** Simplify fully.

$$\frac{16x^{\frac{5}{2}}y^{-2}}{2x^{\frac{1}{2}}y^{3}} \div \frac{4x^{3}}{y^{4}}$$

..... **[3]**

**(b)** Solve the equation.

$$5^{2x-1} \times 25^{x+2} = 125^{x}$$

$x =$ ..... **[4]**

**461** **(a)** Simplify fully.

$$\frac{4x^{-\frac{1}{2}} \times 3x^2}{6x^{\frac{1}{3}}}$$

..... **[3]**

**(b)** Solve the equation.

$$9^{2x-1} \times 3^{x+1} = 27^x$$

$x =$ ..... **[4]**

**462** **(a)** Simplify fully

$$\frac{4x^{\frac{3}{2}} \times (9x^{-4})^{\frac{1}{2}}}{6x^{-\frac{3}{2}}}$$

..... **[4]**

**(b)** Given that $\dfrac{3^{2k-1}}{27^k} = \sqrt{3}$, find the value of $k$.

$k =$ ..... **[3]**

**463** **(a)** Simplify fully.

$$\frac{15x^{\frac{2}{3}} \times 2x^{-\frac{1}{6}}}{5x^{\frac{3}{2}}}$$

..... **[3]**

**(b)** Solve the equation.

$$x^{-\frac{2}{3}} = \frac{4}{9}$$

$x =$ ..... **[3]**

**(c)** Solve the equation.

$$9^x \times 3^{2x-1} = 27^{x+1}$$

$x =$ ..... **[4]**

**464** **(a)** Simplify.

$$5x^2y^3 \times 2x^4y^{-1}$$

..... **[2]**

**(b)** Simplify.

$$\left(\frac{27a^6}{b^9}\right)^{\frac{1}{3}}$$

..... **[2]**

**(c)** Evaluate.

$$16^{-\frac{1}{2}}$$

..... **[1]**

**465** **(a)** Simplify fully.

$$\left(\frac{16x^{12}}{y^8}\right)^{\frac{3}{4}}$$

..... **[3]**

**(b)** Solve the equation.

$$4^{2x-1} = 8^{x+2}$$

$x =$ ..... **[3]**

**466** **(a)** Solve the equation.

$$4^{2x-1} = 32^x$$

$x =$ ..... **[3]**

**(b)** Simplify.

$$\frac{12x^5y^{-3}}{3x^2y}$$

..... **[2]**

**(c)** Solve the equation.

$$x^{\frac{3}{2}} = 64$$

$x =$ ..... **[2]**

**467** **(a)** Simplify fully.

$$\frac{24x^{\frac{3}{2}}y^{-2}}{6x^{-\frac{1}{2}}y}$$

..... **[2]**

**(b)** Solve the equation.

$$9^x \times 27^{x-1} = 3\sqrt{3}$$

$x =$ ..... **[4]**

**468** **(a)** Solve the equation.

$$4^x \times 8^{2x-1} = 16$$

$x =$ ..... **[3]**

**(b)** Simplify fully.

$$\frac{12x^{\frac{3}{2}} \times 2x^{-\frac{1}{2}}}{8x^{-3}}$$

..... **[3]**

**469** **(a)** Simplify

$$\left(\frac{8x^9}{27y^6}\right)^{-\frac{2}{3}}$$

..... **[3]**

**(b)** Find the values of $a$ and $b$ given that

$$\frac{5^{a-2} \times 25^b}{125^{a+1}} = 5^4$$

$a =$ ..... , $b =$ ..... **[4]**

**470** **(a)** Simplify fully.

$$\frac{12x^{\frac{3}{2}}y^{-2}}{4x^{-\frac{1}{2}}y^{4}}$$

..... **[3]**

**(b)** Solve the equation.

$$4^{2x-1} \times 8^{x+1} = 32^{x}$$

$x =$ ..... **[4]**

**471** **(a)** Simplify fully.

$$\frac{18a^3b^{-2}}{45a^{-1}b^4} \times \left(\frac{a^2b}{2}\right)^{-2}$$

..... **[4]**

**(b)** Solve the simultaneous equations.

$$9^x \times 3^y = 2187$$
$$\frac{8^x}{4^y} = 128$$

$x =$ ..... , $y =$ ..... **[5]**

**472** Solve the equation.

$$\frac{3x-1}{4} = \frac{x+5}{3}$$

$x =$ ..... **[3]**

**473** Solve the equation.

$$\frac{2x-3}{4}+\frac{x+1}{3}=2$$

$x =$ ..... **[3]**

**474** Solve the simultaneous equations.

$2x - y = 7$

$3x + 2y = 14$

$x =$ ..... , $y =$ ..... **[3]**

**475** Solve the simultaneous equations.
You must show all your working.

$4x + 3y = 5$

$2x - 5y = 22$

$x =$ ..... , $y =$ ..... **[3]**

**476** **(a)** Solve the equation.

$$4(x - 3) = 2x + 10$$

$x =$ ..... **[2]**

**(b)** Rearrange the formula to make $x$ the subject.

$$y = \frac{2x - 1}{5}$$

$x =$ ..... **[2]**

**477** **(a)** Solve the equation.

$3(2x - 1) = 5x + 8$

$x =$ ..... **[2]**

**(b)** Rearrange the formula to make $x$ the subject.

$y = 4x - 7$

$x =$ ..... **[2]**

**478** **(a)** Solve the equation.

$$3(2x - 5) = 4(x + 1)$$

$x =$ ..... **[3]**

**(b)** Make $h$ the subject of the formula.

$$V = \frac{1}{3}\pi r^2 h$$

$h =$ ..... **[2]**

**479** **(a)** Solve the equation.

$$3(2x - 1) = 5x + 8$$

$x =$ ..... **[2]**

**(b)** Make $x$ the subject of the formula.

$$y = \frac{2x + 3}{5}$$

$x =$ ..... **[3]**

**480** **(a)** Solve the equation

$$4(2x - 3) = 5x + 9$$

$x =$ ..... **[2]**

**(b)** Solve the simultaneous equations.

$$3x + 2y = 16$$
$$5x - 3y = 14$$

$x =$ ..... , $y =$ ..... **[3]**

**481** **(a)** Solve the equation.

$3(2x - 1) = 4x + 11$

$x =$ ..... **[2]**

**(b)** Solve the simultaneous equations.

$3x + 2y = 16$

$x - y = 2$

$x =$ ..... , $y =$ ..... **[3]**

**482** **(a)** Solve

$$5x - 3 = 2(x + 6)$$

$x =$ ..... **[2]**

**(b)** Solve

$$\frac{x+2}{3} + \frac{2x-1}{4} = 3$$

$x =$ ..... **[3]**

**483** **(a)** Solve the equation.

$$3(2x - 5) = 4x + 7$$

$x =$ ..... **[3]**

**(b)** Rearrange the formula to make $x$ the subject.

$$y = \frac{2x + 3}{5}$$

$x =$ ..... **[2]**

**484** **(a)** Solve the equation.

$$4(2x - 1) = 3x + 11$$

$x =$ ..... **[3]**

**(b)** Make $x$ the subject of the formula.

$$y = \frac{3x - 5}{2}$$

$x =$ ..... **[3]**

**485** Consider the following algebraic tasks.

**(a)** Solve the equation.

$$5(x-2) = 3x + 14$$

$x =$ ..... **[3]**

**(b)** Rearrange the formula to make $b$ the subject.

$$A = \frac{h(a+b)}{2}$$

$b =$ ..... **[3]**

**486** **(a)** Solve the equation.

$$3(2x - 5) = 4x + 7$$

$x =$ ..... **[3]**

**(b)** Solve the simultaneous equations.
You must show all your working.

$$2x + 3y = 13 \qquad 5x - 2y = 4$$

$x =$ ..... , $y =$ ..... **[3]**

**487** **(a)** Solve.

$$4(x - 3) = 2x + 7$$

$x =$ ..... **[3]**

**(b)** Solve the simultaneous equations.
You must show all your working.

$$5x - 2y = 16$$

$$3x + 4y = 7$$

$x =$ ..... , $y =$ ..... **[3]**

**488** Rearrange the formula to make $x$ the subject.

$$y = \frac{3x - 2}{x + 4}$$

$x =$ ..... **[4]**

**489** Rearrange the formula to make $x$ the subject.

$$y = \frac{2x + 3}{x - 5}$$

$x =$ ..... **[4]**

**490** Rearrange the formula to make $x$ the subject.

$$y = \frac{3x - 5}{2x + 1}$$

$x =$ ..... **[4]**

**491** **(a)** Solve the simultaneous equations.
You must show all your working.

$3x + 2y = 12$

$5x - 3y = 1$

$x =$ ..... , $y =$ ..... **[3]**

**(b)** Solve the equation by factorisation.

$2x^2 - 5x - 12 = 0$

$x =$ ..... , $x =$ ..... **[3]**

**492** **(a)** Solve the simultaneous equations.
You must show all your working.

$2x + 3y = 13$
$5x - 2y = 4$

$x =$ ..... , $y =$ ..... **[3]**

**(b)** Make $x$ the subject of the formula.

$$y = \frac{3x - 1}{x + 2}$$

$x =$ ..... **[4]**

**493** **(a)** Rearrange the formula to make $x$ the subject.

$$y = \frac{3x - 2}{x + 4}$$

$x =$ ..... **[4]**

**(b)** Solve the equation.

$$\frac{5}{x - 2} = \frac{2}{x + 1}$$

$x =$ ..... **[3]**

**494** **(a)** Solve the simultaneous equations.
You must show all your working.

$3x + 2y = 12$

$5x - 3y = 1$

$x =$ ..... , $y =$ ..... **[4]**

**(b)** Solve the equation by factorisation.

$2x^2 - 7x - 15 = 0$

$x =$ ..... , or $x =$ ..... **[3]**

**495** **(a)** Solve the simultaneous equations.
You must show all your working.

$3x - 2y = 19$

$4x + y = 18$

$x =$ ..... , $y =$ ..... **[3]**

**(b)** Rearrange the formula to make $p$ the subject.

$$T = \frac{3p + 4}{p - 2}$$

$p =$ ..... **[4]**

**496** **(a)** Solve the simultaneous equations.
You must show all your working.

$2x - y = 9$ $3x + 2y = 10$

$x =$ ..... , $y =$ ..... **[4]**

**(b)** Solve the equation.

$$\frac{2x}{3} - \frac{x-4}{4} = \frac{5}{2}$$

$x =$ ..... **[4]**

**497** Solve the following equations.

**(a)** Solve the simultaneous equations.
You must show all your working.

$4x + 3y = 5$

$2x - 5y = 9$

$x =$ ..... , $y =$ ..... **[4]**

**(b)** Solve the equation by factorisation.

$3x^2 - 10x - 8 = 0$

$x =$ ..... or $x =$ **[4]**

**498** Solve the equation.

$$\frac{2}{x-1} + \frac{1}{x-2} = 1$$

Give your answers in the form $a \pm \sqrt{b}$, where $a$ and $b$ are integers.
You must show all your working.

$x =$ ..... **[6]**

**499** Consider the equation

$$\frac{4}{x-1} - \frac{3}{2x+1} = 2$$

**(a)** Show that this equation simplifies to $4x^2 - 7x - 9 = 0$.

**[4]**

**(b)** Solve the equation $4x^2 - 7x - 9 = 0$.
Give your answers in the form $\dfrac{p \pm \sqrt{q}}{8}$, where $p$ and $q$ are integers.

$x =$ ..... **[3]**

**500** **(a)** Make $t$ the subject of the formula

$$y = \sqrt{\frac{3t-5}{t+2}}$$

$t =$ ..... **[3]**

**(b)** Use your graphic display calculator to solve the equation

$$x^3 - 4x = 3$$

$x =$ ..... **[3]**

**501** **(a)** Tickets for a school show cost \$12 for an adult and \$8 for a child.
For one performance, a total of 150 tickets were sold and the total amount received was \$1480.
Let $x$ be the number of adult tickets sold and $y$ be the number of child tickets sold.

Write down a pair of simultaneous equations in $x$ and $y$ and solve them to find the value of $x$ and the value of $y$.

$x =$ ..... , $y =$ ..... **[4]**

**(b)** Make $m$ the subject of the formula.

$$T = 2\pi\sqrt{\frac{m}{k}}$$

$m =$ ..... **[3]**

**502** **(a)** Rearrange the formula to make $x$ the subject.

$$p = \frac{3x - 1}{x + 4}$$

$x =$ ..... **[4]**

**(b)** Solve the equation by factorising.

$$2x^2 - 5x - 12 = 0$$

$x =$ ..... , $x =$ ..... **[3]**

**503** **(a)** A rectangle has length $(3x + 2)$ cm and width $(x - 1)$ cm.
The perimeter of the rectangle is 42 cm.

Work out the value of $x$.

$x =$ ..... **[3]**

**(b)** Make $x$ the subject of the formula.

$$p = \sqrt{\frac{x-3}{x}}$$

$x =$ ..... **[4]**

**504** **(a)** Make $x$ the subject of the formula.

$$y = \frac{2x + 5}{x - 3}$$

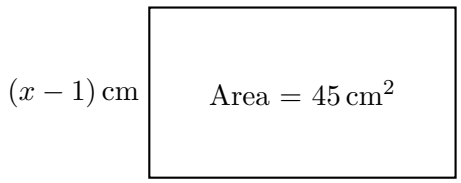
$x =$ ..... **[3]**

**(b)** Solve the equation.
Show all your working and give your answers correct to 2 decimal places.

$$\frac{x + 1}{3} - \frac{2}{x - 2} = 1$$

$x =$ ..... , $x =$ ..... **[6]**

**505** The diagram shows a rectangle with length $(2x + 3)$ cm and width $(x - 1)$ cm.
The area of the rectangle is $45\,\text{cm}^2$.




NOT TO SCALE

**(a)** Show that $2x^2 + x - 48 = 0$.

**[2]**

**(b)** Solve the equation $2x^2 + x - 48 = 0$.
You must show all your working and give your answers correct to 2 decimal places.

$x =$ ..... , $x =$ ..... **[4]**

**(c)** Make $x$ the subject of the formula.

$$y = \frac{ax + 5}{x - 3}$$

$x =$ ..... **[4]**

**506** Solve the equation.
Give your answers correct to 2 decimal places.

$$\frac{3}{x+2} + \frac{1}{x-1} = 2$$

$x =$ ..... , $x =$ ..... **[6]**

**507** Solve the equation.
Show all your working and give your answers correct to 2 decimal places.

$$\frac{3}{x-2} + \frac{4}{2x+1} = 2$$

$x =$ ..... , $x =$ ..... **[6]**

**508** Solve the equation.
Show all your working and give your answers correct to 2 decimal places.

$$\frac{5}{x-2} - \frac{3}{2x+1} = 2$$

$x =$ ..... , $x =$ ..... **[7]**

**509** **(a)** Show that $\dfrac{3}{x-1} - \dfrac{2}{x+3} = 1$ simplifies to $x^2 + x - 14 = 0$.

**[4]**

**(b)** Solve the equation $x^2 + x - 14 = 0$.
Show all your working and give your answers correct to 2 decimal places.

$x =$ ..... , $x =$ ..... **[3]**

**510** **(a)** A car travels 120 km at an average speed of $x$ km/h.
A train travels the same distance at an average speed of $(x + 10)$ km/h.
The time taken by the car is 36 minutes more than the time taken by the train.

**(i)** Show that $x^2 + 10x - 2000 = 0$.

**[3]**

**(ii)** Solve the equation $x^2 + 10x - 2000 = 0$.

$x =$ ..... or $x =$ ..... **[2]**

**(b)** Solve the equation.

$3x^2 - 7x - 5 = 0$ Give your answers correct to 2 decimal places.
You must show all your working.

$x =$ ..... or $x =$ ..... **[3]**

**511** **(a)** A train travels 120 km at an average speed of $x$ km/h.
It then travels a further 150 km at an average speed of $(x + 10)$ km/h.
The total time for the journey is 4 hours.
Write down an equation in $x$ and show that it simplifies to $2x^2 - 115x - 600 = 0$.

**[4]**

**(b)** Solve the equation $2x^2 - 115x - 600 = 0$ to find the value of $x$.
Give your answer correct to 1 decimal place.

$x =$ ..... **[2]**

**(c)** Use your graphic display calculator to solve the equation.

$$2^x - 3 = \frac{5}{x}$$

Give your answers correct to 3 significant figures.

$x =$ ..... , $x =$ ..... **[2]**

**512** A cyclist travels 40 km at an average speed of $x$ km/h.

**(a)** Write down an expression, in terms of $x$, for the time taken in hours.

..... hours **[1]**

**(b)** A car travels the same 40 km journey at an average speed of $(x + 20)$ km/h.
The car takes 45 minutes less than the cyclist.
Form an equation in $x$ and show that it simplifies to $3x^2 + 60x - 3200 = 0$.

**[4]**

**(c)** Solve the equation.

$$3x^2 + 60x - 3200 = 0$$

Give your answers correct to 2 decimal places.

$x =$ ..... , or $x =$ ..... **[3]**

**513** **(a)** Solve the equation.

$$\frac{5}{x+2} + \frac{2}{x-3} = 1$$

Show all your working and give your answers correct to 2 decimal places.

$x =$ ..... , $x =$ ..... **[5]**

**(b)** Make $x$ the subject of the formula.

$$y = \frac{2x-3}{5x+1}$$

$x =$ ..... **[4]**

**514** **(a)** Solve the equation.

Give your answers in the exact form $\frac{p \pm \sqrt{q}}{2}$.

$$\frac{x}{x+2} = \frac{3}{x-6}$$

$x =$ ..... **[5]**

**(b)** Rearrange the formula to make $x$ the subject.

$$y = \frac{2x-5}{3x+1}$$

$x =$ ..... **[4]**

**515** **(a)** Solve the equation.

$$\frac{3}{x-1} + \frac{2}{2x+1} = 2$$

Show all your working and give your answers correct to 2 decimal places.

$x =$ ..... , or $x =$ ..... **[5]**

**(b)** The sum of the squares of two consecutive positive even numbers is 340.

Write down an equation in $x$ and solve it to find the two numbers.

The numbers are ..... , and ..... **[4]**

**516** A bus travels 100 km at an average speed of $x$ km/h.
A car travels 100 km at an average speed of $(x + 15)$ km/h.
The car takes 30 minutes less than the bus.

**(a)** Write down an equation in $x$ and show that it simplifies to $x^2 + 15x - 3000 = 0$.

**[3]**

**(b)** Solve the equation $x^2 + 15x - 3000 = 0$.
You must show all your working and give your answers correct to 2 decimal places.

$x =$ ..... , $x =$ ..... **[4]**

**(c)** Work out the time taken by the car to travel the 100 km.
Give your answer in hours and minutes, correct to the nearest minute.

hours, minutes **[3]**

**517** **(a)** Make $x$ the subject of the formula.

$$y = \frac{3x - 2}{x + 4}$$

$x =$ ..... **[4]**

**(b)** Solve the equation.
Show all your working and give your answers correct to 2 decimal places.

$$\frac{5}{x - 2} - \frac{3}{2x + 1} = 2$$

$x =$ ..... , or $x =$ ..... **[6]**

**518** Consider the following algebraic equations and formulas.

**(a)** Rearrange the formula to make $x$ the subject.

$$y = \frac{3x - 1}{x + 2}$$

$x =$ ..... **[4]**

**(b)** Solve the equation.
Give your answers correct to 2 decimal places.
Show all your working.

$$\frac{2}{x - 1} + \frac{3}{x + 2} = 1$$

$x =$ ..... or $x =$ **[6]**

**519** A train travels 150 km at an average speed of $x$ km/h.

**(a)** Write down an expression, in terms of $x$, for the time taken in hours.

..... hours **[1]**

**(b)** A bus travels the same distance of 150 km at an average speed of $(x - 15)$ km/h.
The bus takes 30 minutes longer than the train.
Show that $x$ satisfies the equation $x^2 - 15x - 4500 = 0$.

**[4]**

**(c)** Solve the equation $x^2 - 15x - 4500 = 0$ to find the average speed of the train.

..... km/h **[3]**

**(d)** Rearrange the formula to make $q$ the subject.

$$P = \frac{2q + 3}{q - 1}$$

$q =$ ..... **[4]**

**520** Solve the inequality.

$3x - 5 < 7x + 3$

..... **[2]**

**521** **(a)** Solve the inequality.

$$3(x-2) > 5x + 4$$

..... **[2]**

**(b)** Write down the largest integer that satisfies the inequality in part (a).

..... **[1]**

**522** Solve the inequality.

$4(3x - 1) - 5x \geq 2(x + 8)$

..... **[3]**

**523** **(a)** Solve the inequality

$$3(x-4) < 5x - 2$$

..... **[2]**

**(b)** Write down the smallest integer value of $x$ that satisfies this inequality.

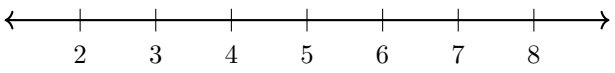
..... **[1]**

**524** **(a)** Solve the inequality.

$$3(2x - 1) > 4x + 7$$

..... **[2]**

**(b)** On the number line, represent the solution to the inequality in part (a).




**[1]**

**(c)** Write down the smallest integer value of $x$ that satisfies this inequality.

..... **[1]**

**525** **(a)** Solve the inequality.

$$-5 < 3x + 1 \leq 13$$

..... **[2]**

**(b)** Write down all the integer values of $x$ that satisfy this inequality.

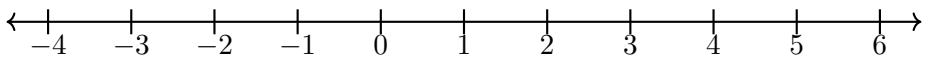
..... **[2]**

**526** **(a)** Solve the inequality.

$$-5 < 2x - 1 \leq 7$$

..... **[2]**

**(b)** Show your solution to part (a) on the number line.




**[2]**

**527** **(a)** Solve the inequality.

$4x - 3 \leq 17$

..... **[2]**

**(b)** Write down the largest integer $x$ that satisfies the inequality.

$3(x + 2) < 20$

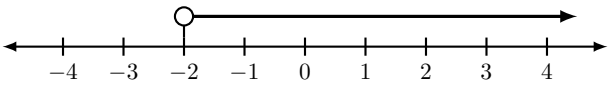
$x =$ ..... **[2]**

**528** **(a)** Solve the inequality.

$$4x - 3 < 2x + 9$$

..... **[2]**

**(b)** Write down the inequality shown on the number line.



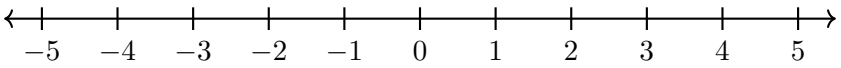

..... **[2]**

**529** **(a)** Solve the inequality.

$$-5 < 2x - 1 \leq 7$$

..... **[3]**

**(b)** Represent your answer to part (a) on the number line provided.



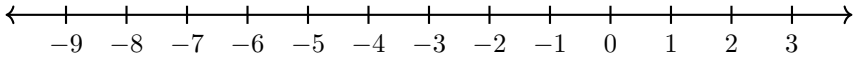

**[2]**

**530** **(a)** Solve the inequality.

$$3(x-2) \leq 5x+8$$

..... **[3]**

**(b)** Represent the solution to part (a) on the number line.



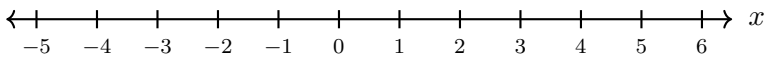

**[2]**

**531** **(a)** Solve the inequality

$$-5 < 2x - 1 \leq 7$$

..... **[2]**

**(b)** Represent your solution to part (a) on the number line.




**[2]**

**(c)** Write down the smallest integer value of $x$ that satisfies the inequality $-5 < 2x - 1 \leq 7$.

..... **[1]**

**532** **(a)** Solve the inequality.

$3x - 2 > 5x + 7$

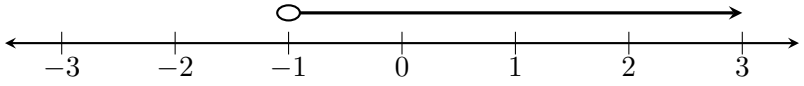
..... **[2]**

**(b)** Write down all the integer values of $n$ that satisfy the inequality.

$-5 < 2n - 1 \leq 3$

..... **[3]**

**(c)** An inequality is shown on the number line above.
Write down the inequality.




..... **[1]**

**533** **(a)** Solve the inequality.

$$4x - 3 \leq 7x + 12$$

..... **[2]**

**(b)** Solve the inequality.

$$-8 < 2x - 1 \leq 7$$

..... **[2]**

**(c)** Write down the largest integer and the smallest integer that satisfy the inequality in part (b).

Largest = ..... , Smallest = ..... **[2]**

**534** **(a)** Solve the inequality.

$2 - 3x < 11$

..... **[2]**

**(b)** Solve the inequality.

$4(2x - 1) \leq 5x + 17$

..... **[3]**

**(c)** $n$ is an integer such that $2 - 3n < 11$ and $4(2n - 1) \leq 5n + 17$.
Write down all the possible values of $n$.

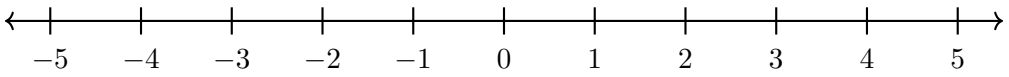
..... **[2]**

**535** **(a)** Solve the inequality.

$3x - 5 \leq 7x + 3$

..... **[2]**

**(b)** On the number line, show the solution to the inequality in part (a).



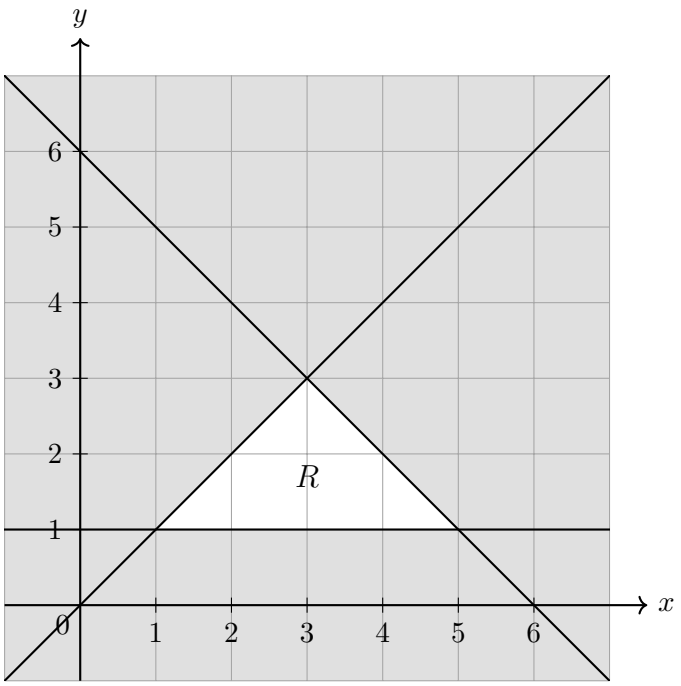

**[2]**

**(c)** List all the integers that satisfy the inequality.

$-2 < 3x - 5 \leq 4$

..... **[3]**

**536** The unshaded region $R$ in the diagram is defined by three inequalities.



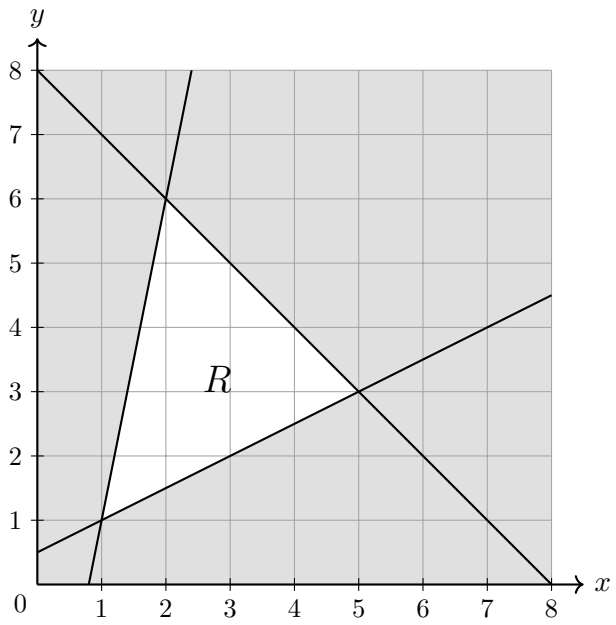

Write down these three inequalities.

.....

.....

..... **[3]**

**537** The diagram shows a region $R$, which is unshaded.
The unwanted regions are shaded.



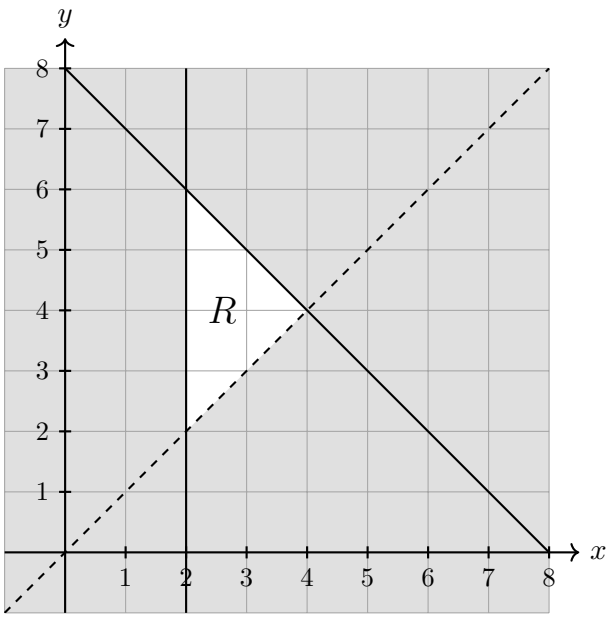

Write down the three inequalities that define region $R$.

.....

.....

..... **[3]**

**538** The diagram shows a region $R$ defined by three inequalities.
The unwanted regions are shaded.

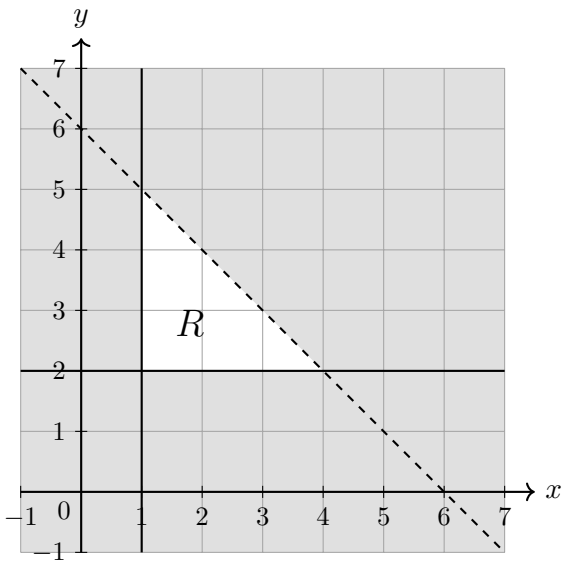



Write down the three inequalities that define the unshaded region $R$.

.....

..... **[3]**

**539** The unshaded region $R$ is defined by three inequalities.
The boundaries of the region are shown on the coordinate grid.




Write down the three inequalities that define the unshaded region $R$.

.....

.....

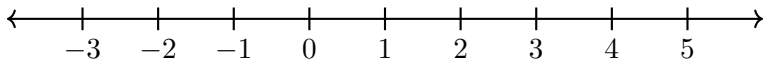
..... **[3]**

**540** **(a)** Solve the inequality.

$$-4 < 3x + 2 \leq 14$$

..... **[2]**

**(b)** Represent your answer to part (a) on the number line.



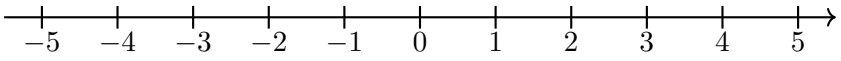

**[2]**

**541** **(a)** Solve the inequality.

$$-7 < 2x - 1 \leq 5$$

..... **[3]**

**(b)** Represent your answer to part (a) on the number line.




**[2]**

**542** **(a)** Solve the inequality.

$$-5 \leq 2x - 1 < 6$$

**(a)** Solve the inequality.

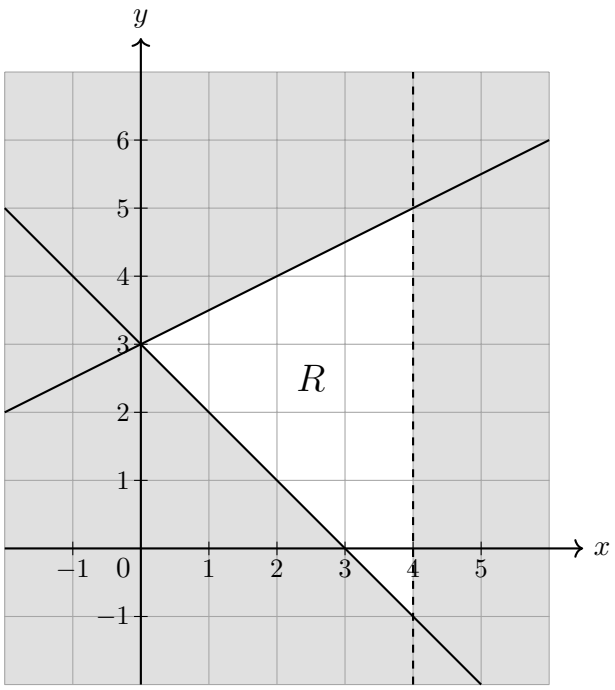
$$-5 \leq 2x - 1 < 6$$

..... **[3]**

**(b)** Write down all the integer values of $x$ that satisfy this inequality.

..... **[2]**

**543** The unshaded region $R$ is defined by three inequalities.




**(a)** Write down these three inequalities.

.....

.....

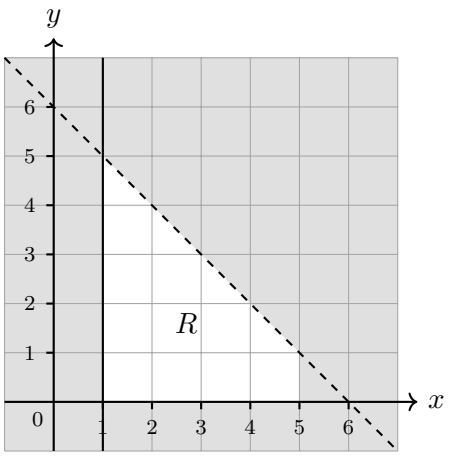
..... **[3]**

**(b)** The point $(a, b)$, where $a$ and $b$ are integers, lies in the region $R$.
It is given that $a = 3$.

Find all possible values of $b$.

..... **[3]**

**544** The unshaded region $R$ in the grid is defined by three inequalities.
One of these inequalities is $x \geq 1$.




**(a)** Write down the other two inequalities that define region $R$.

.....

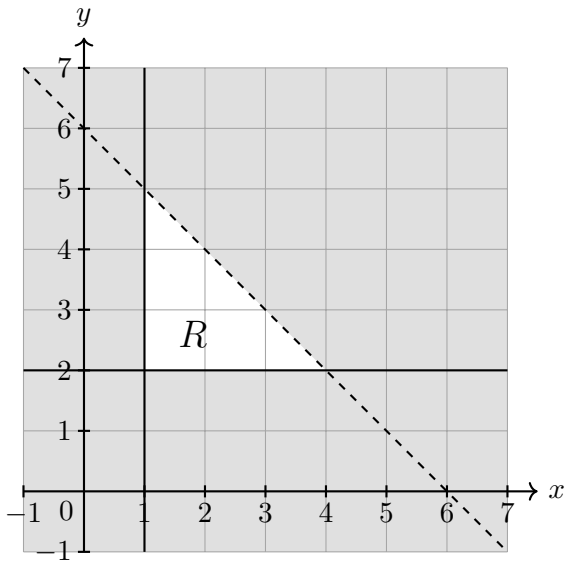
.....

..... **[4]**

**(b)** Find the maximum value of $2x + 3y$ for points with integer coordinates $(x, y)$ inside region $R$.

..... **[2]**

**545** The diagram shows a region $R$, which is unshaded.
The region is bounded by three lines.
One of the lines is broken.



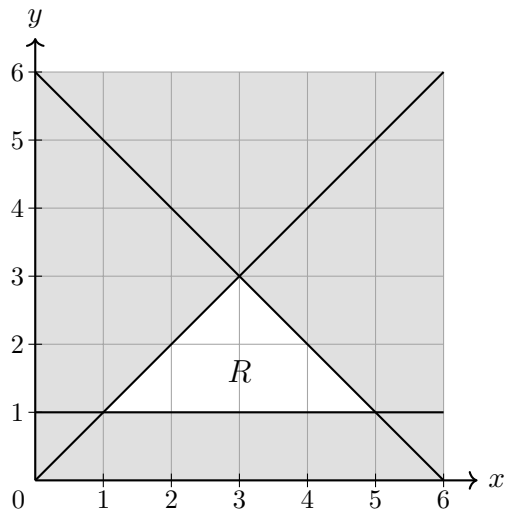

Write down the three inequalities that define the region $R$.

.....

.....

..... **[3]**

**546** The diagram shows a grid with three straight lines and an unshaded region labeled $R$.



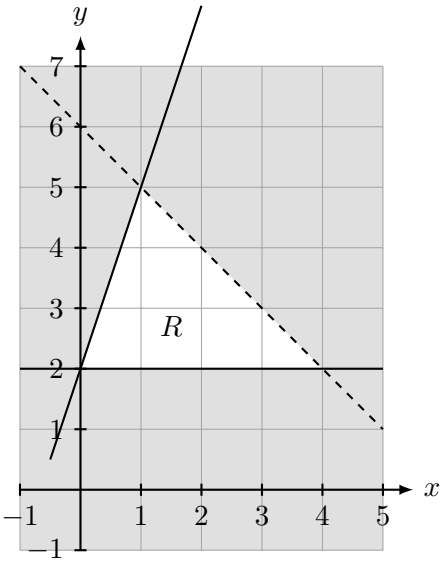

**(a)** Write down the three inequalities that define the unshaded region $R$.

**[3]**

**(b)** Find the maximum value of $3x + 2y$ for integers $x$ and $y$ that lie in the region $R$.

..... **[2]**

**547** A region $R$ is defined by three inequalities.




The region $R$, unshaded, is defined by three inequalities.
Write down these three inequalities.

.....

.....

..... **[6]**

**548** **(a)** Solve the inequality.

$$\frac{3x-2}{5} - \frac{2x-5}{3} \leq 2$$

..... **[4]**

**(b)** A region $R$ is defined by the three inequalities $y > x$, $y < 4$, and $x + y > 2$.
Write down all the points in region $R$ that have integer coordinates.

..... **[3]**

**549** **(a)** Solve the inequality.

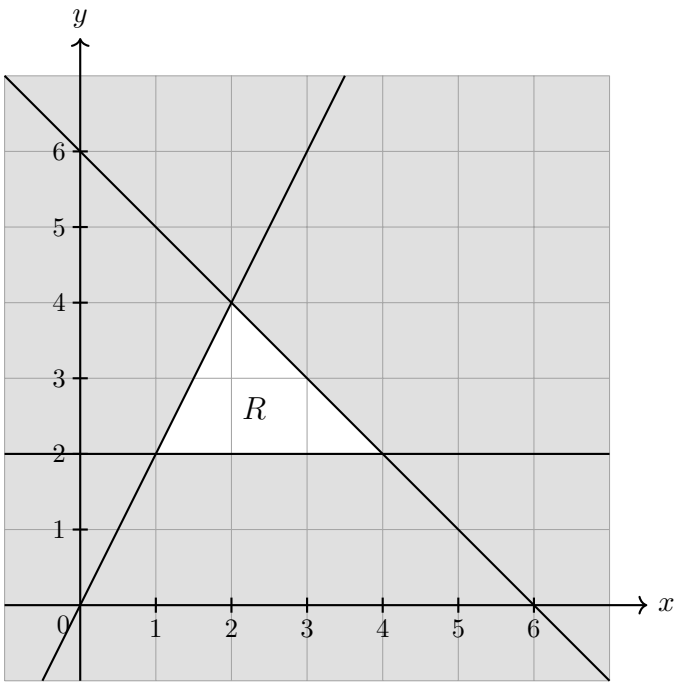
$$5x - 3 \leq 2x + 12$$

..... **[2]**

**(b)** $x$ is a positive integer.
Write down all the possible values of $x$.

..... **[2]**

**550** The diagram shows a region $R$ unshaded.



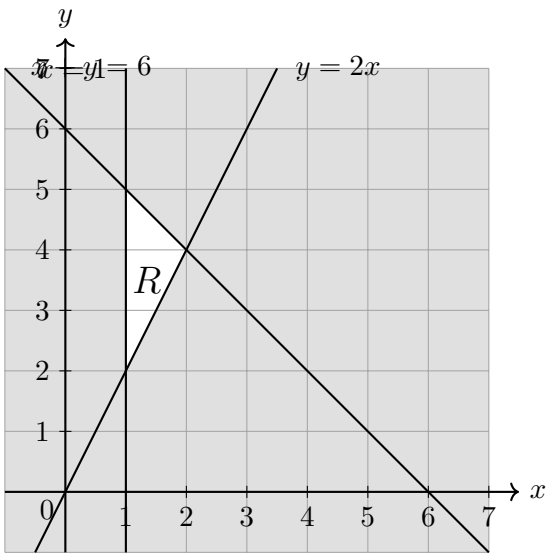

Write down the three inequalities that define the region $R$.

.....

.....

..... **[3]**

**551** The diagram shows three straight lines: $x = 1$, $y = 2x$, and $x + y = 6$.
The unshaded region $R$ is bounded by these lines.




Write down the three inequalities that define the unshaded region $R$.

.....

.....

..... **[3]**

**552** **(a)** Solve the inequality.

$$-7 < 3x + 2 \leq 11$$

..... **[3]**

**(b)** Write down the largest prime number that satisfies this inequality.

..... **[1]**

**553** **(a)** Solve the double inequality.

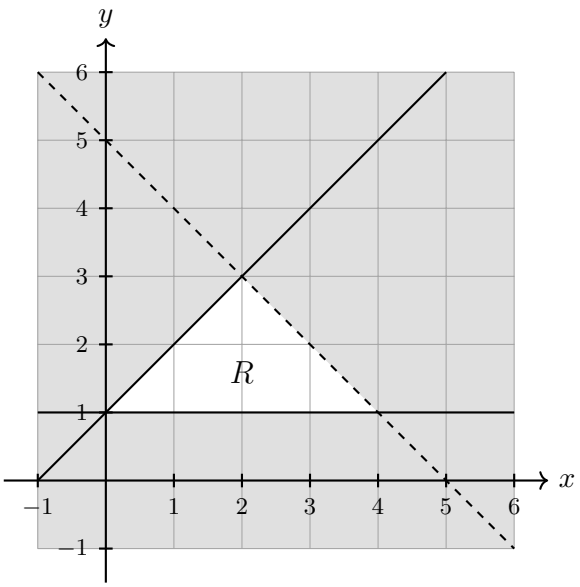
$$-7 \leq 3x - 1 < 11$$

..... **[3]**

**(b)** Write down all the integer values of $x$ that satisfy the inequality in part (a).

..... **[2]**

**554** The diagram shows a region $R$ which is unshaded.
The unwanted regions are shaded.




**(a)** Write down the three inequalities that define region $R$.

.....

.....

..... **[4]**

**(b)** Find the point $(x, y)$, with integer coordinates, inside region $R$ which maximizes the value of $2y - x$.

( ( ..... , ..... ) ) **[2]**

**555** **(a)** Solve the inequality.

$$-5 < 2x + 1 \leq 11$$

..... **[3]**

**(b)** Write down all the integers that satisfy the inequality.

$$-2.5 \leq y < 1$$

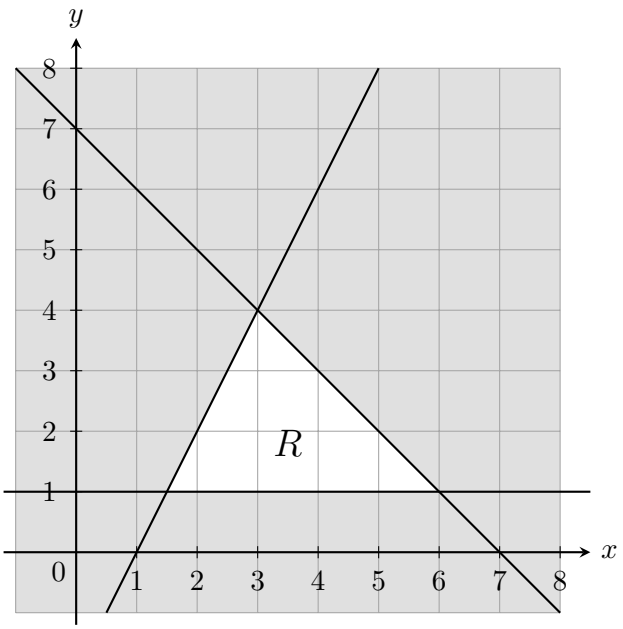
..... **[2]**

**(c)** Use your graphic display calculator to solve the inequality.

$$2x^2 - 7x + 3 > 0$$

..... **[2]**

**556** The diagram shows a grid with three straight lines.




**(a)** The unshaded region $R$ is defined by three inequalities.
The boundaries are included in the region $R$.
Write down the three inequalities.

.....

.....

..... **[5]**

**(b)** A point $(x, y)$ lies in the region $R$.
Both $x$ and $y$ are integers.
Find the maximum value of $2x + y$.

..... **[2]**

**557** **(a)** Solve the inequality.

$$-5 < 2x - 1 \leq 7$$

..... **[2]**

**(b)** Write down all the integers that satisfy the inequality in part (a).

..... **[2]**

**(c)** A rectangular garden has length $(3x + 2)$ metres and width $x$ metres.
The perimeter of the garden is strictly less than 40 metres.

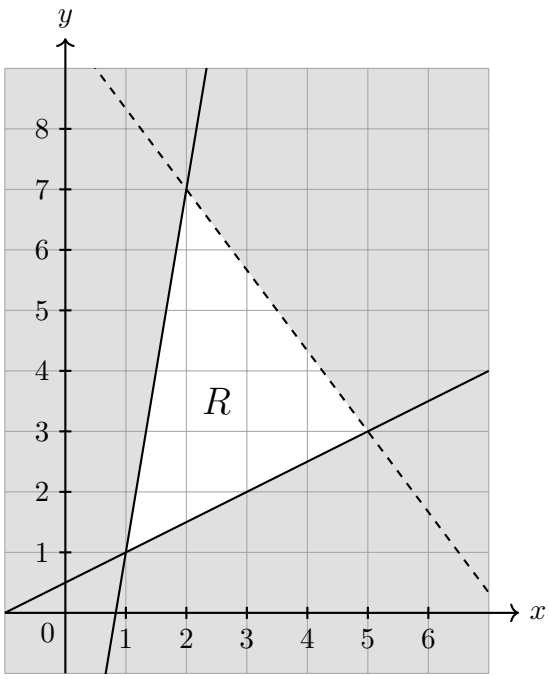
**(i)** Write down an inequality in terms of $x$ and show that it simplifies to $x < 4.5$.

**[2]**

**(ii)** Given that $x$ is a positive integer, find the maximum possible area of the garden.

Area = ..... $m^2$ **[2]**

**558** The diagram shows a coordinate grid.
The unshaded region $R$ is bounded by three lines.




**(a)** Write down the three inequalities that define the region $R$.

.....

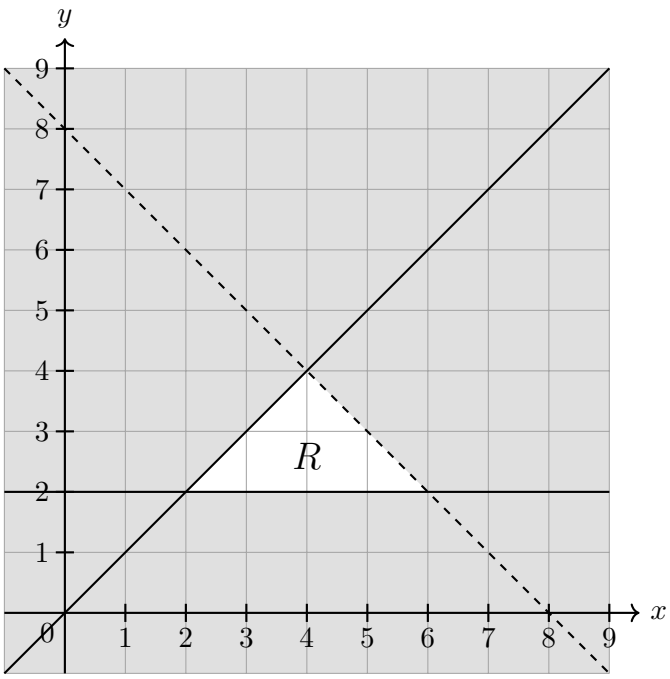
.....

..... **[3]**

**(b)** A point $(x, y)$ in region $R$ has integer coordinates.
Find the maximum possible value of $x + y$.

..... **[2]**

**559** The diagram shows a grid with three lines drawn.
The unwanted regions are shaded grey.




**(a)** Write down the three inequalities that define the unshaded region $R$.

.....

.....

..... **[3]**

**(b)** Find the coordinates of the point $(x, y)$ in region $R$ where $x$ and $y$ are integers, such that the value of $2x + 3y$ is a maximum.

( ( ..... , ..... ) ) **[2]**

**560** A factory produces two types of bicycles: Mountain bikes and City bikes.
In a week, the factory produces $x$ Mountain bikes and $y$ City bikes.

- The factory produces at least 15 Mountain bikes.
- The factory can produce at most 40 bicycles in a week.
- The number of City bikes produced is at least half the number of Mountain bikes produced.

**(a)** Write down three inequalities, in terms of $x$ and $y$, to show this information.

.....

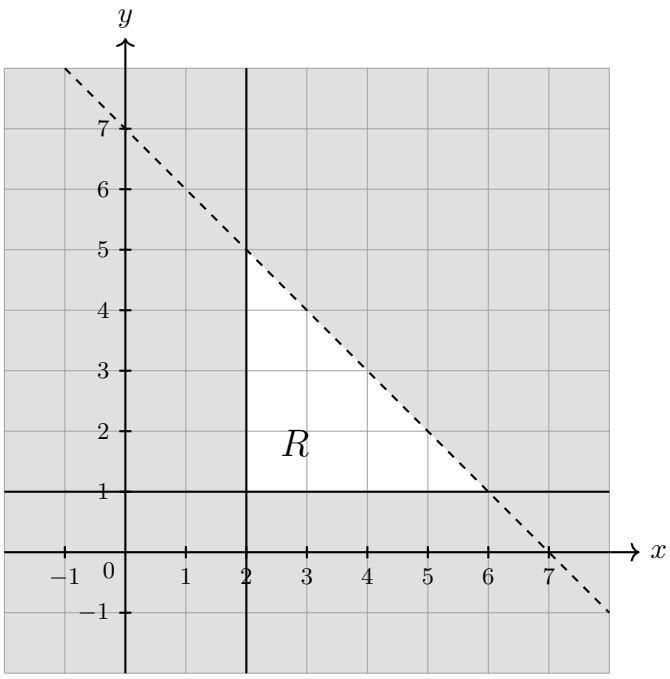
.....

..... **[3]**

**(b)** The profit on each Mountain bike is \$400 and the profit on each City bike is \$150.
Work out the maximum possible profit the factory can make in a week.

\$ ..... **[3]**

**561** The unshaded region $R$ in the diagram is defined by three inequalities.




**(a)** Write down the three inequalities that define the unshaded region $R$.

.....

.....

..... **[3]**

**(b)** Use a graphic display calculator to solve the inequality.

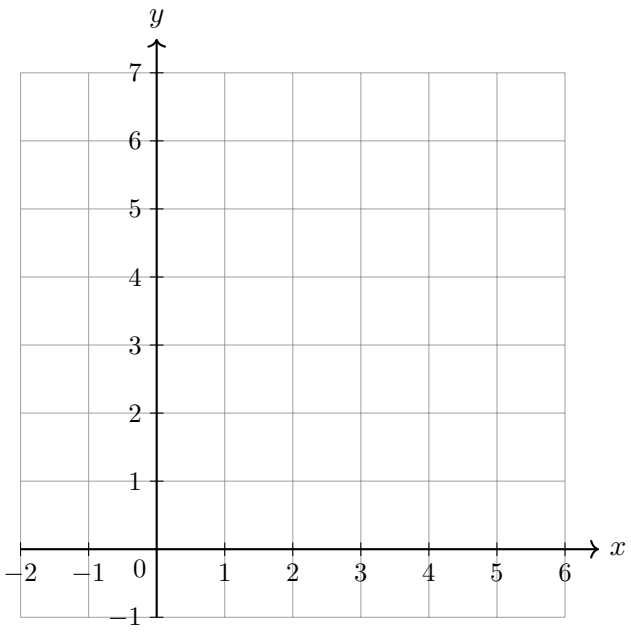
$x^2 - 4x - 2 \leq 0$

Give your answer correct to 2 decimal places.

..... **[3]**

**562** **(a)** On the grid, draw the lines $y = 2$, $y = x + 2$, and $2x + y = 8$.
Shade the unwanted regions to show the region $R$ defined by the inequalities:

- $y \geq 2$
- $y \leq x + 2$
- $2x + y \leq 8$




**[4]**

**(b)** Find the largest value of $3x + 2y$ for points $(x, y)$ in the region $R$ where both $x$ and $y$ are integers.

..... **[2]**

**563** A rectangular garden has length $(2x + 5)$ metres and width $(x - 1)$ metres, where $x > 1$.
The perimeter of the garden is less than 50 metres and the area of the garden is at least 42 $m^2$.

**(a)** Use the information about the perimeter to show that $x < 7$.

**[2]**

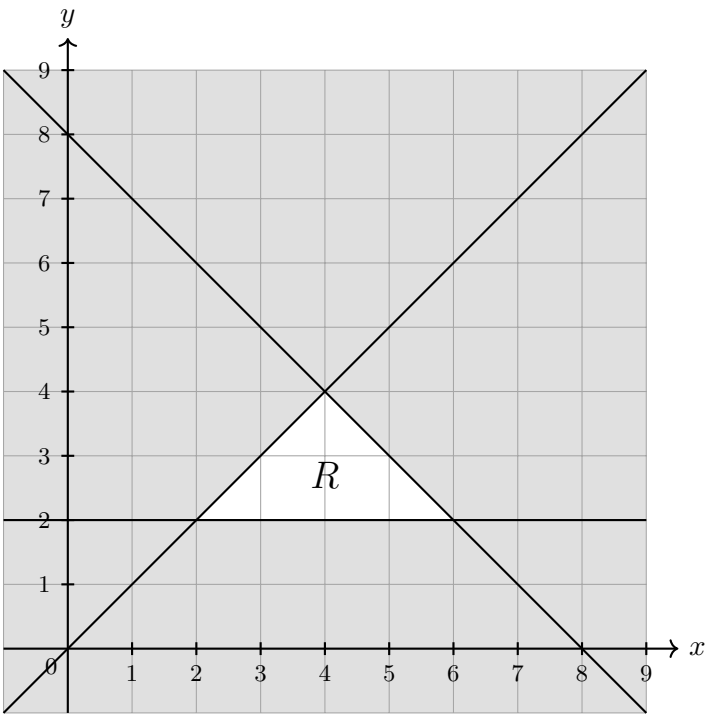
**(b)** Use the information about the area to show that $2x^2 + 3x - 47 \geq 0$.

**[2]**

**(c)** Use your graphic display calculator to solve the inequality $2x^2 + 3x - 47 \geq 0$ for $x > 1$, and hence state the range of possible values of $x$.
Give your boundary values correct to 2 decimal places where appropriate.

..... **[3]**

**564** The diagram shows three straight lines on a coordinate grid.




**(a)** The region $R$ is unshaded.
Write down the three inequalities that define the region $R$.

.....

.....

..... **[3]**

**(b)** Find the maximum value of $2x + 3y$ given that $x$ and $y$ are integers that satisfy these three inequalities.

..... **[3]**

**(c)** A point is chosen at random from the points with integer coordinates in region $R$.
Work out the probability that $x = y$.

..... **[2]**

**565** **(a)** Solve the inequality.

$$-5 < 2x - 1 \leq 7$$

..... **[2]**

**(b)** Find the smallest integer $x$ that satisfies the inequality.

$$\frac{2x-5}{3} - \frac{x-3}{4} > 1$$

$x =$ ..... **[3]**

**(c)** A taxi company, City Cabs, charges a fixed fare of \$3.20 plus \$1.50 per kilometre.
Another company, Metro Taxis, charges a fixed fare of \$5.00 plus \$1.10 per kilometre.
Construct and solve an inequality to find the minimum whole number of kilometres a passenger must travel for City Cabs to be more expensive than Metro Taxis.

..... kilometres **[3]**

**566** Tickets for a community show cost \$5 for children and \$10 for adults.
The maximum number of tickets that can be sold is 500.
The organizers must sell at least 150 adult tickets.
The total amount of money raised from ticket sales must be at least \$3000.

Let $x$ be the number of children's tickets sold and $y$ be the number of adult tickets sold.

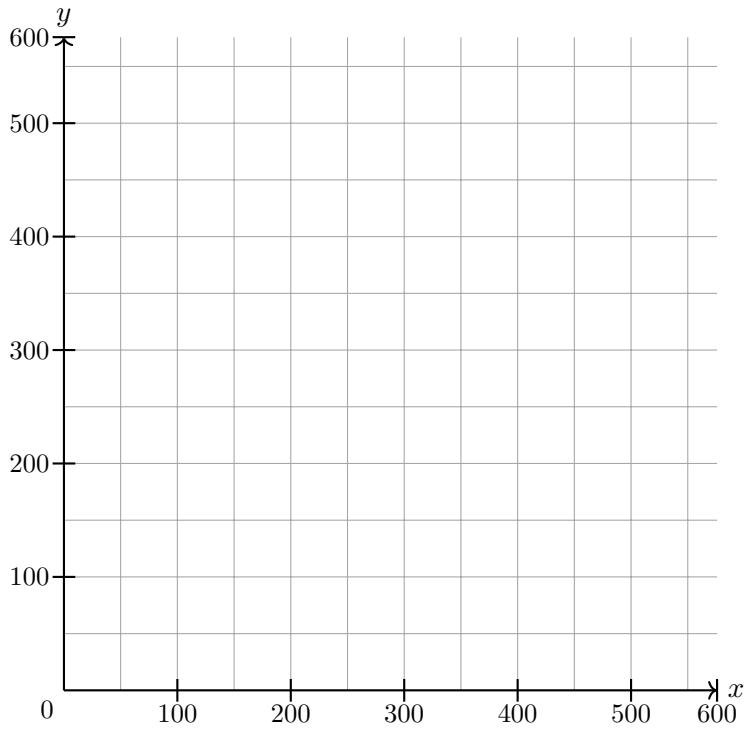
**(a)** Write down three inequalities in $x$ and $y$ to show this information.

.....

.....

..... **[3]**

**(b)** On the grid, draw three lines to show these inequalities.
Shade the unwanted regions to leave the region representing the possible number of tickets sold.




**[4]**

**(c)** The organizers want to maximize the number of children's tickets sold.
Find this maximum number of children's tickets and the corresponding number of adult tickets.

Children's tickets = ..... , Adult tickets = ..... **[2]**

**567** A carpenter makes $x$ tables and $y$ chairs each week.
The total number of tables and chairs made each week must not exceed 60.
The number of tables made must be at least 10.
The number of chairs made must be at least half the number of tables made.

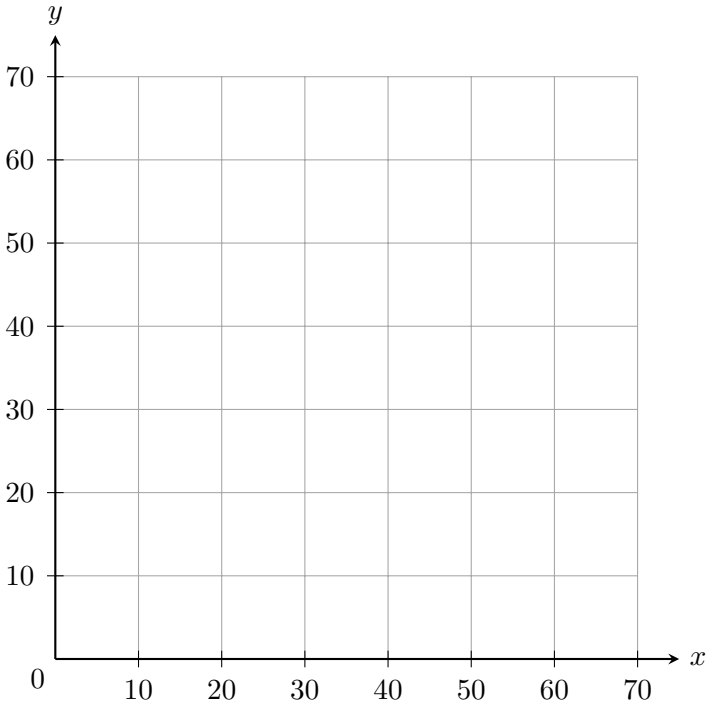
**(a)** Write down three inequalities, in terms of $x$ and $y$, to show this information.

.....

.....

..... **[3]**

**(b)** On the grid, draw three straight lines and shade the unwanted regions to show the region which satisfies these inequalities.
Label the region $R$.




**[4]**

**(c)** The profit on each table is \$40 and the profit on each chair is \$25.
Work out the maximum possible weekly profit.

\$ ..... **[2]**

**568** A clothing company manufactures two types of shirts: Standard and Premium.
In one day, they make $x$ Standard shirts and $y$ Premium shirts.
They can make a maximum of 60 shirts in total per day.
Each Standard shirt requires 1 metre of fabric and each Premium shirt requires 2 metres of fabric.
The company has a maximum of 80 metres of fabric available per day.
They must make at least 10 Premium shirts per day.

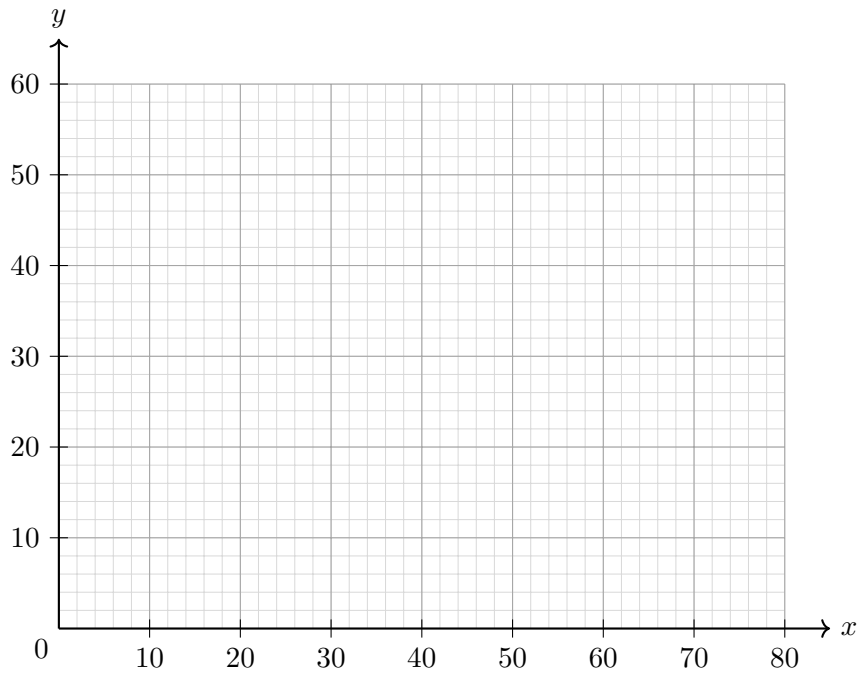
**(a)** Write down three inequalities in $x$ and $y$, other than $x \geq 0$, to represent this information.

.....

.....

..... **[3]**

**(b)** On the grid provided, draw three lines to show these inequalities.
Shade the unwanted regions to leave the region that satisfies the inequalities unshaded.




**[4]**

**(c)** The profit on each Standard shirt is \$5 and the profit on each Premium shirt is \$8.
Find the maximum possible profit the company can make in one day.

$ ..... **[2]**

**569** **(a)** Find the exact range of values of $x$ for which

$$\frac{2x-3}{4} < \frac{x+5}{3} \leq \frac{3x-1}{2}$$

..... **[5]**

**(b)** A mobile phone company offers two plans.
Plan A costs \$12 per month plus \$0.07 per minute of calls.
Plan B costs \$25 per month for unlimited calls.

Let $x$ be the number of minutes of calls in a month.

**(i)** Write down an inequality in terms of $x$ to show the condition when Plan B is strictly cheaper than Plan A.

..... **[1]**

**(ii)** Solve your inequality to find the minimum number of whole minutes of calls required for Plan B to be cheaper than Plan A.

..... minutes **[3]**

**570** **(a)** A factory makes $x$ Mountain bicycles and $y$ Road bicycles each day.

- The factory makes at most 60 bicycles in total each day.
- It makes at least 10 Road bicycles.
- It makes at least 15 Mountain bicycles.
- The number of Road bicycles is at most twice the number of Mountain bicycles.

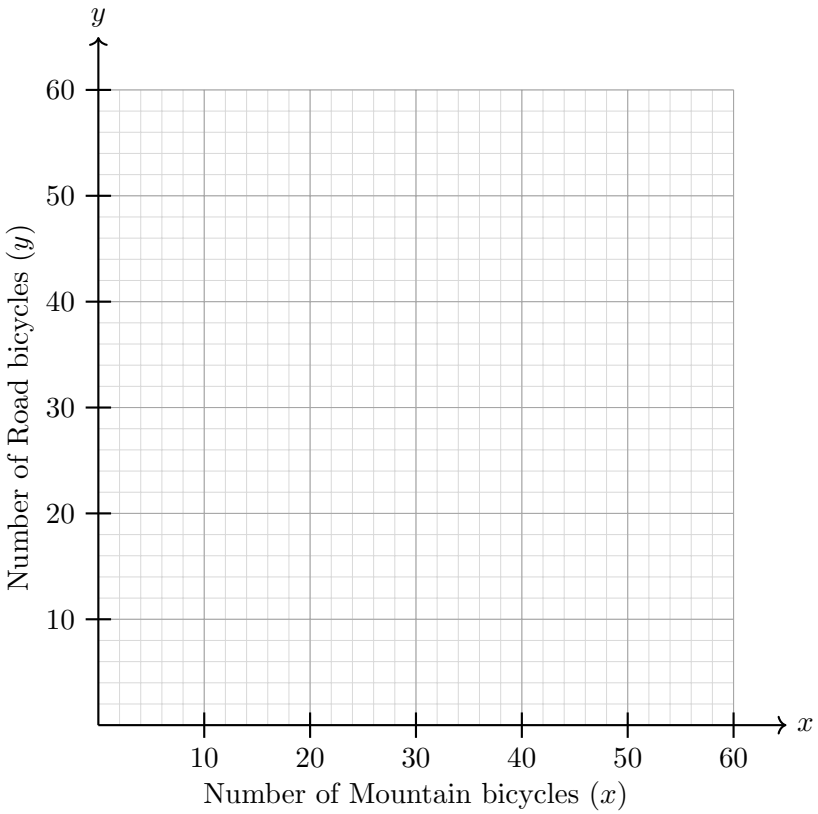
Write down four inequalities, in terms of $x$ and $y$, to show this information.

.....

.....

..... **[4]**

**(b)** On the grid, draw four lines to show these inequalities.
By shading the unwanted regions, indicate the region $R$ which satisfies all four inequalities.




**[5]**

**(c)** The profit on each Mountain bicycle is \$30 and the profit on each Road bicycle is \$50.
Find the maximum possible profit the factory can make in one day, and the number of each type of bicycle it must make to achieve this profit.

Maximum profit = \$ .....

Mountain bicycles = .....

Road bicycles = ..... **[3]**

**571** Here are the first five terms of a number sequence.

7 11 15 19 23

**(a)** Write down the next term in this sequence.

..... **[1]**

**(b)** Find an expression, in terms of $n$, for the $n$th term of this sequence.

..... **[2]**

**572** Here are the first four terms of a sequence.

5 11 17 23

**(a)** Find the next term of this sequence.

..... **[1]**

**(b)** Find an expression for the $n$th term of this sequence.

..... **[2]**

**(c)** Explain why 150 is not a term in this sequence.

..... **[1]**

**573** Here are the first four terms of a sequence.

5 12 19 26

**(a)** Write down the next term in the sequence.

..... **[1]**

**(b)** Find an expression for the $n$th term of the sequence.

..... **[2]**

**(c)** Explain why 150 is not a term in this sequence.

..... **[1]**

**574** Here are the first four terms of a sequence.

5 11 17 23

**(a)** Find the next term in this sequence.

..... **[1]**

**(b)** Find an expression for the $n$th term of this sequence.

..... **[2]**

**(c)** Explain why 150 is not a term in this sequence.

..... **[1]**

**575** **(a)** Find the $n$th term of the sequence.

$$5 \quad 9 \quad 13 \quad 17 \quad 21$$

..... **[2]**

**(b)** Find the $n$th term of the sequence.

$$2 \quad 6 \quad 18 \quad 54 \quad 162$$

..... **[2]**

**576** The first four terms of a sequence are

7 11 15 19

**(a)** Write down the next two terms of the sequence.

..... **[1]**

**(b)** Find an expression for the $n$th term of this sequence.

$n$th term = ..... **[2]**

**(c)** Explain why 102 is not a term in this sequence.

..... **[2]**

**577** **(a)** Sequence $A$ is a linear sequence.

$$5 \quad 9 \quad 13 \quad 17 \quad \ldots$$

**(i)** Find the next two terms of sequence $A$.

..... **[1]**

**(ii)** Find the $n$th term of sequence $A$.

..... **[2]**

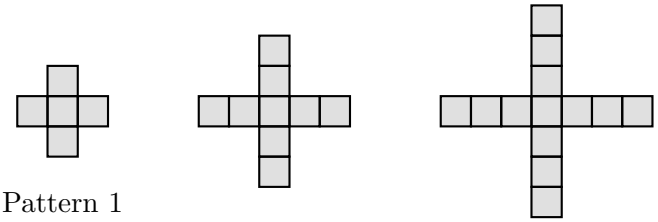
**(b)** Sequence $B$ is a geometric sequence.

$$2 \quad 6 \quad 18 \quad 54 \quad \ldots$$

Find the $n$th term of sequence $B$.

..... **[2]**

**578** The diagram shows a sequence of patterns made from small squares.



Pattern 1 Pattern 2 Pattern 3

**(a)** Find the number of squares in Pattern 4.

..... **[1]**

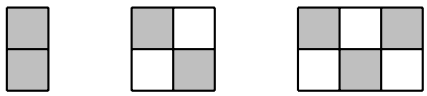
**(b)** Find an expression, in terms of $n$, for the number of squares in Pattern $n$.

..... **[2]**

**(c)** Which pattern has exactly 141 squares?

Pattern ..... **[2]**

**579** The diagram shows the first three patterns in a sequence made from congruent grey and white squares.



Pattern 1 Pattern 2 Pattern 3

**(a)** Complete the table for Pattern 4 and Pattern 5.

| Pattern number ($n$) | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Number of grey squares | 2 | 2 | 3 | | |
| Number of white squares | 0 | 2 | 3 | | |
| Total number of squares | 2 | 4 | 6 | | |

**[2]**

**(b)** Write down an expression, in terms of $n$, for the total number of squares in Pattern $n$.

..... **[1]**

**(c)** Pattern $k$ has a total of 94 squares.
Find the value of $k$.

$k =$ ..... **[2]**

**580** The first four terms of a sequence are

$$-1 \qquad 4 \qquad 13 \qquad 26$$

**(a)** Find an expression for the $n$th term of the sequence.

..... **[3]**

**(b)** Find the value of $n$ when the $n$th term is 433.

$n =$ ..... **[3]**

**581** The first five terms of a quadratic sequence $T$ are shown below.

$$5 \quad 12 \quad 23 \quad 38 \quad 57$$

**(a)** Write down the next term in sequence $T$.

..... **[1]**

**(b)** Find an expression, in terms of $n$, for $T_n$, the $n$th term of sequence $T$.

$T_n =$ ..... **[3]**

**(c)** Sequence $U$ has $n$th term $U_n = 3 \times 2^{n-1}$.
Find the value of $T_6 - U_6$.

..... **[2]**

**582** The first five terms of a sequence, $T_n$, are given below.

$$4 \qquad 14 \qquad 34 \qquad 68 \qquad 120$$

**(a)** By finding the first, second and third differences, show that the $n$th term of the sequence is a cubic expression in $n$.

**[2]**

**(b)** Find an expression for $T_n$, the $n$th term of the sequence.

$T_n =$ ..... **[5]**

**583** Here are the first five terms of a sequence.

3 9 17 27 39

Find an expression, in terms of $n$, for the $n$th term of this sequence.

..... **[3]**

**584** Here are the first five terms of a sequence.

4 11 22 37 56

**(a)** Work out the next term of the sequence.

..... **[1]**

**(b)** Find an expression for the $n$th term of the sequence.

..... **[3]**

**585** The $n$th term of a sequence is $u_n = an^2 + bn$.

**(a)** Given that $u_1 = 5$ and $u_2 = 14$, write down two simultaneous equations in $a$ and $b$.

.....

..... **[2]**

**(b)** Solve these simultaneous equations to find the value of $a$ and the value of $b$.

$a =$ ..... , $b =$ ..... **[3]**

**586** **(a)** The $n$th term of a sequence is $n^2 - 3n + 5$.
Work out the first three terms of this sequence.

1st term = ..... , 2nd term = ..... , 3rd term = .....
**[2]**

**(b)** The first four terms of another sequence are

2 9 20 35

Find an expression for the $n$th term of this sequence.

..... **[3]**

**587** The first four terms of a sequence, $T$, are

2 9 22 41

**(a)** Find an expression for the $n$th term, $T_n$.

$T_n =$ ..... **[3]**

**(b)** A second sequence is given by the formula $U_n = 2^n + 3$.
Work out the value of $U_5$.

$U_5 =$ ..... **[1]**

**(c)** Find the value of $n$ for which $U_n = 131$.

$n =$ ..... **[2]**

**588** The first four terms of a sequence are

$$-1 \qquad 2 \qquad 9 \qquad 20$$

**(a)** Find the $n$th term of this sequence.

..... **[3]**

**(b)** The $n$th term of another sequence is given by $3^n - 2$.
Find the value of the term in this sequence that is closest to 1000.

..... **[3]**

**589** The first five terms of a sequence are

2 9 20 35 54 ...

**(a)** Find the next term.

..... **[1]**

**(b)** Find the $n$th term of the sequence.

..... **[3]**

**(c)** The $k$th term of the sequence is 464.
Find the value of $k$.

$k =$ ..... **[3]**

**590** Consider the following three sequences, $P$, $Q$ and $R$.
Sequence $P$:

$$2 \quad 9 \quad 28 \quad 65 \quad 126 \quad \ldots$$

Sequence $Q$:

$$3 \quad 8 \quad 15 \quad 24 \quad 35 \quad \ldots$$

Sequence $R$:

$$\frac{2}{3} \quad \frac{9}{8} \quad \frac{28}{15} \quad \frac{65}{24} \quad \frac{126}{35} \quad \ldots$$

**(a)** Find the $n$th term of sequence $P$.

..... **[2]**

**(b)** Find the $n$th term of sequence $Q$.

..... **[2]**

**(c)** Write down the $n$th term of sequence $R$.

..... **[1]**

**(d)** Find the 12th term of sequence $R$, giving your answer as a fraction in its simplest form.

..... **[2]**

**591** Here are the first five terms of a sequence.

2 9 28 65 126

**(a)** Find the next term of this sequence.

..... **[1]**

**(b)** Find an expression for the $n$th term of this sequence.

..... **[3]**

**(c)** The $k$th term of the sequence is 8001.
Find the value of $k$.

$k =$ ..... **[3]**

**592** The table shows the first five terms of sequences $A$, $B$ and $C$.

| $n$ | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Sequence $A$ | 2 | 5 | 10 | 17 | 26 |
| Sequence $B$ | 1 | 8 | 27 | 64 | 125 |
| Sequence $C$ | $-1$ | 3 | 17 | 47 | 99 |

**(a)** Find the $n$th term of sequence $A$.

..... **[2]**

**(b)** Find the $n$th term of sequence $B$.

..... **[1]**

**(c)** Use your answers to parts (a) and (b) to find an expression, in terms of $n$, for the $n$th term of sequence $C$.
Give your answer in its simplest form.

..... **[2]**

**(d)** Determine which term in sequence $A$ has a value of 10001.

$n =$ ..... **[2]**

**593** **(a)** The first five terms of a sequence are

$$-1 \quad 8 \quad 33 \quad 80 \quad 155$$

Use the method of differences to find an expression for the $n$th term of this sequence.

..... **[4]**

**(b)** An exponential sequence has $n$th term $P_n = 5 \times 3^{n-1}$.
The $k$th term of this sequence is 3645.
Work out the value of $k$.

$k =$ ..... **[2]**

**(c)** The $n$th term of sequence $T$ is $T_n = \dfrac{2n-1}{3n+2}$.
Write down the value that $T_n$ approaches as $n$ becomes very large.

..... **[1]**

**594** The first five terms of a cubic sequence $A$ are shown below.

$$2 \quad 10 \quad 30 \quad 68 \quad 130$$

**(a)** Use the method of differences to find an expression, in terms of $n$, for $A_n$, the $n$th term of sequence $A$.

$A_n =$ ..... **[4]**

**(b)** Sequence $B$ is defined by $B_n = A_{n+1} - A_n$.
Show that $B_n = 3n^2 + 3n + 2$.

**[2]**

**(c)** Find the value of $n$ for which $B_n = 272$.

$n =$ ..... **[2]**

**595** **(a)** A sequence has the following first four terms:

1 15 53 127

Use the method of differences to find an expression for the $n$th term.

$n$th term = ..... **[4]**

**(b)** Another sequence, $V$, has its $n$th term defined by

$$V_n = \frac{n^2 + 1}{2^n}$$

**(i)** Find the values of $V_1$, $V_2$ and $V_3$.
Give each fraction in its simplest form.

$V_1$ = ..... , $V_2$ = ..... , $V_3$ = ..... **[2]**

**(ii)** Work out the exact value of $V_5 - V_4$.

..... **[3]**

**596** **(a)** The first five terms of a sequence are

4 5 14 37 80

Use the method of differences to find an expression for the $n$th term.

..... **[4]**

**(b)** The $n$th term of a different sequence is given by $T_n = a \times 2^n + b$.
The first term of this sequence is 10 and the third term is 34.

Find the values of $a$ and $b$.

$a =$ ..... , $b =$ ..... **[4]**

**(c)** Calculate the 12th term of the sequence described in part (b).

..... **[2]**

**597** The first five terms of three sequences, A, B and C, are shown below.

| | | | | | |
|---|---|---|---|---|---|
| Sequence A: | 5 | 9 | 13 | 17 | 21 |
| Sequence B: | 2 | 6 | 20 | 50 | 102 |
| Sequence C: | $\frac{5}{2}$ | $\frac{3}{2}$ | $\frac{13}{20}$ | $\frac{17}{50}$ | $\frac{7}{34}$ |

**(a)** Find the $n$th term of sequence A.

..... **[2]**

**(b)** **(i)** Work out the 6th term of sequence B.

..... **[1]**

**(ii)** Find an expression for the $n$th term of sequence B.

..... **[4]**

**(c)** **(i)** Work out the 6th term of sequence C, giving your answer as a fraction in its simplest form.

..... **[1]**

**(ii)** Find an expression for the $n$th term of sequence C.

..... **[2]**

**598** $y$ is directly proportional to the square of $x$.
When $x = 3$, $y = 36$.

**(a)** Find an equation connecting $y$ and $x$.

$y =$ ..... **[2]**

**(b)** Find the value of $y$ when $x = 5$.

$y =$ ..... **[1]**

**599** $y$ is directly proportional to $(x + 2)$.
When $x = 3$, $y = 20$.

Find the value of $y$ when $x = 7$.

$y =$ ..... **[3]**

**600** $y$ is directly proportional to the square of $x$.
When $x = 3$, $y = 36$.

**(a)** Find an equation connecting $y$ and $x$.

$y =$ ..... **[2]**

**(b)** Find the possible values of $x$ when $y = 64$.

$x =$ ..... **[2]**

**601** $y$ is directly proportional to $(x+2)^2$.
When $x = 3$, $y = 75$.

**(a)** Find an equation for $y$ in terms of $x$.

$y =$ ..... **[2]**

**(b)** Calculate the value of $y$ when $x = 6$.

$y =$ ..... **[2]**

**602** $y$ is directly proportional to $x$.
When $x = 5$, $y = 15$.

**(a)** Find an equation connecting $y$ and $x$.

..... **[2]**

**(b)** Work out the value of $y$ when $x = 12$.

$y =$ ..... **[2]**

**603** $y$ is directly proportional to $(x+2)^3$.
When $x = 1$, $y = 135$.

**(a)** Find an equation for $y$ in terms of $x$.

$y =$ ..... **[2]**

**(b)** Find the value of $y$ when $x = 0$.

$y =$ ..... **[1]**

**(c)** Find the value of $x$ when $y = 320$.

$x =$ ..... **[2]**

**604** The variable $y$ is directly proportional to the square of $x$.
When $x = 3$, $y = 36$.

**(a)** Find an equation connecting $y$ and $x$.

$y =$ ..... **[2]**

**(b)** Find the value of $y$ when $x = 5$.

$y =$ ..... **[1]**

**(c)** Find the values of $x$ when $y = 196$.

$x =$ ..... **[2]**

**605** $y$ is directly proportional to the cube of $x$.
When $x = 2$, $y = 24$.

**(a)** Find an equation connecting $y$ and $x$.

$y =$ ..... **[2]**

**(b)** Work out the value of $y$ when $x = 5$.

$y =$ ..... **[1]**

**(c)** Work out the value of $x$ when $y = 81$.

$x =$ ..... **[2]**

**606** The table shows some corresponding positive values of $p$ and $q$.

| $p$ | 2 | 4 | 10 |
|---|---|---|---|
| $q$ | 100 | 25 | 4 |

**(a)** Determine the algebraic equation connecting $q$ and $p$.

$q =$ ..... **[2]**

**(b)** Find the value of $q$ when $p = 5$.

$q =$ ..... **[1]**

**(c)** Find the value of $p$ when $q = 1$.

$p =$ ..... **[2]**

**607** The resistance, $R$ ohms, of a uniform cylindrical wire is directly proportional to its length, $L$ metres, and inversely proportional to the square of its diameter, $d$ millimetres.
When $L = 12$ and $d = 2$, $R = 9$.

**(a)** Find a formula for $R$ in terms of $L$ and $d$.

$R =$ ..... **[3]**

**(b)** A second wire made of the same material has length 40 metres and resistance 7.5 ohms.
Calculate the diameter, $d$, of this second wire.

$d =$ ..... mm **[2]**

**(c)** The length of a wire is increased by 25% and its diameter is halved.
Find the percentage increase in its resistance.

..... % **[2]**

**608** $P$ is directly proportional to the square root of $Q$.
$Q$ is inversely proportional to $R^3$.
When $Q = 9$, $P = 12$.
When $R = 2$, $Q = 2$.

**(a)** Find an equation for $P$ in terms of $R$.

$P =$ ..... **[5]**

**(b)** Calculate the value of $P$ when $R = 4$.

$P =$ ..... **[2]**

**609** $A$ is directly proportional to the square of $B$.
$B$ is inversely proportional to $(C + 2)$.
Assume that $B > 0$ and $(C + 2) > 0$.
When $B = 4$, $A = 80$.
When $C = 2$, $B = 3$.

**(a)** Find the value of $A$ when $C = 4$.

$A =$ ..... **[4]**

**(b)** Find an expression for $C$ in terms of $A$.

$C =$ ..... **[4]**

**610** The table shows corresponding values of two positive variables $p$ and $q$.

| $p$ | 4 | 9 | 25 | 64 |
|---|---|---|---|---|
| $q$ | 18 | 12 | 7.2 | 4.5 |

**(a)** State which variation model best fits the data in the table, writing your answer using the proportional symbol $\propto$.

..... **[2]**

**(b)** Find an equation connecting $p$ and $q$.

$q =$ ..... **[2]**

**(c)** Calculate the value of $p$ when $q = 2.4$.

$p =$ ..... **[2]**

**611** $y$ is inversely proportional to the square of $x$.
When $x = 4$, $y = 3$.

**(a)** Find an equation connecting $y$ and $x$.

..... **[3]**

**(b)** Work out the possible values of $x$ when $y = 0.75$.

$x =$ ..... **[3]**

**612** $P$ is inversely proportional to the square of $Q$.
When $Q = 4$, $P = 1.5$.

**(a)** Find an equation connecting $P$ and $Q$.

$P =$ ..... **[3]**

**(b)** Find the value of $P$ when $Q = 10$.

$P =$ ..... **[1]**

**(c)** Find the positive value of $Q$ when $P = 24$.

$Q =$ ..... **[2]**

**613** The quantity $P$ is inversely proportional to the square root of $Q$.
When $Q = 9$, $P = 12$.

**(a)** **(i)** Find the value of $P$ when $Q = 144$.

$P =$ ..... **[3]**

**(ii)** Find the value of $Q$ when $P = 1.5$.

$Q =$ ..... **[2]**

**(b)** Calculate the percentage decrease in $P$ when the value of $Q$ is multiplied by 4.

..... % **[2]**

**614** $M$ is inversely proportional to the square root of $N$.
When $N = 16$, $M = 15$.

**(a)** Find an equation connecting $M$ and $N$.

$M =$ ..... **[2]**

**(b)** Find the value of $M$ when $N = 100$.

$M =$ ..... **[2]**

**(c)** Find the value of $N$ when $M = 1.2$.

$N =$ ..... **[3]**

**615** The table shows some values of two variables, $x$ and $y$.
The relationship between $x$ and $y$ can be modelled by a variation equation.

| $x$ | 2 | 4 | 6 |
|---|---|---|---|
| $y$ | 36 | 9 | 4 |

**(a)** Use the data to identify the correct variation model and find an equation connecting $y$ and $x$.

$y =$ ..... **[3]**

**(b)** Find the positive value of $x$ when $y = 2.25$.

$x =$ ..... **[2]**

**(c)** When $x$ is multiplied by 3, $y$ is multiplied by a constant $p$.
Find the value of $p$.

$p =$ ..... **[2]**

**616** $p$ is inversely proportional to the square root of $q$.
When $q = 144$, $p = 5$.

**(a)** Find an equation connecting $p$ and $q$.

..... **[2]**

**(b)** Find the value of $q$ when $p = 15$.

$q =$ ..... **[2]**

**(c)** The value of $q$ is increased by 300%.
Find the percentage decrease in $p$.

..... % **[3]**

**617** $P$ is inversely proportional to the square root of $Q$.
When $Q = 36$, $P = 4$.

**(a)** Find an equation for $P$ in terms of $Q$.

$P =$ ..... **[2]**

**(b)** Calculate the value of $P$ when $Q = 144$.

$P =$ ..... **[1]**

**(c)** Calculate the value of $Q$ when $P = 1.5$.

$Q =$ ..... **[2]**

**(d)** The value of $Q$ is increased by 300%.
Calculate the percentage decrease in the value of $P$.

..... % **[3]**

**618** **(a)** Some corresponding values of $x$ and $y$ are given below.

$$x = 2,\ y = 75 \qquad x = 4,\ y = 18.75 \qquad x = 10,\ y = 3$$

Identify which of the following models best fits the data:

$$y \propto x \qquad y \propto x^2 \qquad y \propto \frac{1}{x} \qquad y \propto \frac{1}{x^2}$$

Hence, find the equation for $y$ in terms of $x$.

$y =$ ..... **[3]**

**(b)** $P$ is directly proportional to the cube of $Q$.
When $Q$ is increased by 50%, work out the percentage increase in $P$.

..... % **[3]**

**619** **(a)** $y$ is directly proportional to the cube of $x$.

When $x$ increases by 20%, find the percentage increase in $y$.

..... % **[3]**

**(b)** The table shows some corresponding values of $v$ and $A$.

Determine the best variation model connecting $A$ and $v$ from the following list:

$$A \propto v \qquad A \propto v^2 \qquad A \propto \sqrt{v} \qquad A \propto \frac{1}{v}$$

You must show working to justify your decision.

| $v$ | 2 | 4 | 10 |
|---|---|---|---|
| $A$ | 12 | 48 | 300 |

..... **[3]**

**620** **(a)** The table shows some values of $x$ and $y$.

| $x$ | 2 | 4 | 5 | 8 |
|---|---|---|---|---|
| $y$ | 32 | 8 | 5.12 | 2 |

**(i)** Identify the variation model and write down the equation for $y$ in terms of $x$.

$y =$ ..... **[3]**

**(ii)** Calculate the value of $y$ when $x = 10$.

$y =$ ..... **[1]**

**(b)** $A$ is directly proportional to the cube of $B$.
When $B$ is increased by $x\%$, the value of $A$ increases by 72.8%.
Find the value of $x$.

$x =$ ..... **[3]**

**621** **(a)** $y$ is inversely proportional to the square root of $x$.

When $x$ is increased by 44%, find the percentage change in $y$.

..... % **[4]**

**(b)** $P$ is directly proportional to the cube of $Q$.
$Q$ is inversely proportional to $R$.
When $R = 2$, $P = 10$.

Find the value of $P$ when $R = 4$.

$P =$ ..... **[3]**

**622** The table shows some corresponding values of $x$ and $y$.

| $x$ | 2 | 4 | 8 |
|---|---|---|---|
| $y$ | 100 | 25 | 6.25 |

**(a)** Determine the relationship between $x$ and $y$ and find an equation connecting them.

..... **[3]**

**(b)** $z$ is directly proportional to $x$.
When $x = 4$, $z = 10$.

**(i)** Find an equation for $x$ in terms of $z$.

$x =$ ..... **[2]**

**(ii)** Hence find an equation for $y$ in terms of $z$.
Give your answer in its simplest form.

$y =$ ..... **[3]**

**623** The table shows some corresponding values of $x$ and $y$.

| $x$ | 2 | 4 | 10 |
|---|---|---|---|
| $y$ | 40 | 5 | 0.32 |

**(a)** Determine the proportional relationship between $y$ and $x$ and find an equation connecting them.

$y =$ ..... **[3]**

**(b)** Find the value of $x$ when $y = 2560$.

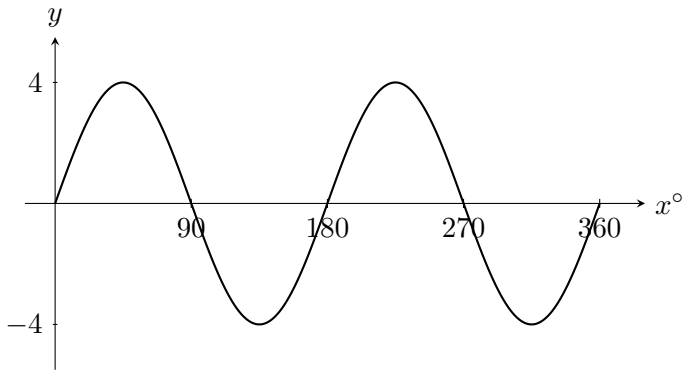
$x =$ ..... **[2]**

**(c)** The variable $x$ is inversely proportional to $w$.
When $w = 5$, $x = 2$.

Find an equation connecting $y$ and $w$.
Give your answer in its simplest form.

$y =$ ..... **[4]**

**624** The diagram shows the graph of $y = a\sin(bx)$ for $0° \leq x \leq 360°$.



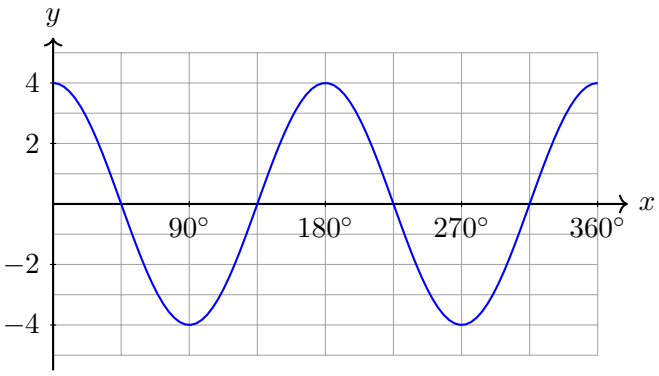

**(a)** Find the value of $a$.

$a =$ ..... **[1]**

**(b)** Find the value of $b$.

$b =$ ..... **[1]**

**625** The diagram shows a sketch of the graph of $y = a\cos(bx)$ for $0° \leq x \leq 360°$.



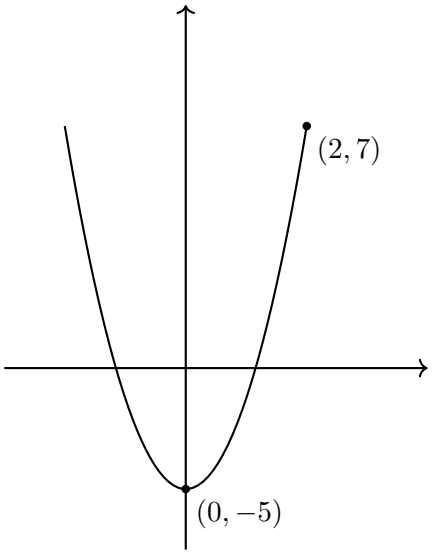

**(a)** Write down the value of $a$.

$a =$ ..... **[1]**

**(b)** Find the value of $b$.

$b =$ ..... **[2]**

**626** The diagram shows a sketch of the graph of the function $f(x) = ax^2 + c$.
The graph passes through the points $(0, -5)$ and $(2, 7)$.



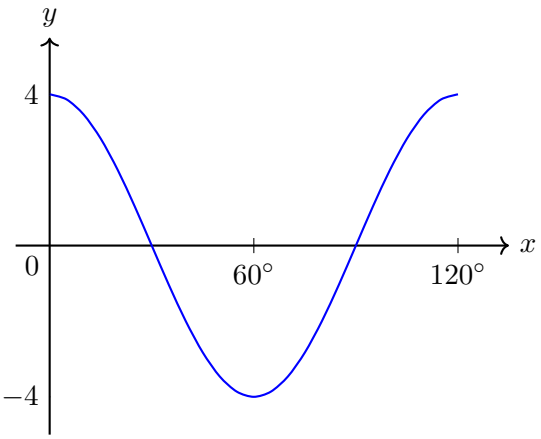

**(a)** Write down the value of $c$.

$c =$ ..... **[1]**

**(b)** Find the value of $a$.

$a =$ ..... **[2]**

**627** The diagram shows the sketch of the graph of $y = a\cos(bx)$ for $0° \leq x \leq 120°$.



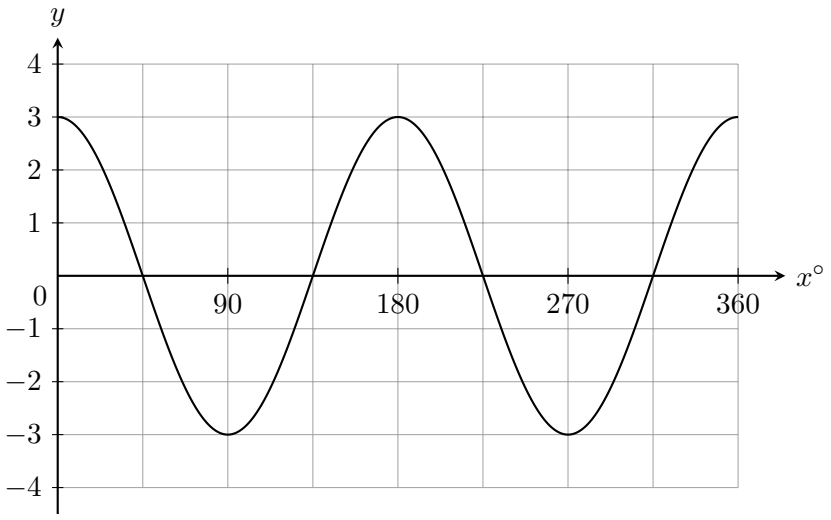

**(a)** Write down the amplitude of the graph.

..... **[1]**

**(b)** Find the value of $a$ and the value of $b$.

$a =$ ..... , $b =$ ..... **[2]**

**628** The diagram shows the graph of the trigonometric function $f(x) = a\cos(bx)$ for $0° \leq x \leq 360°$.



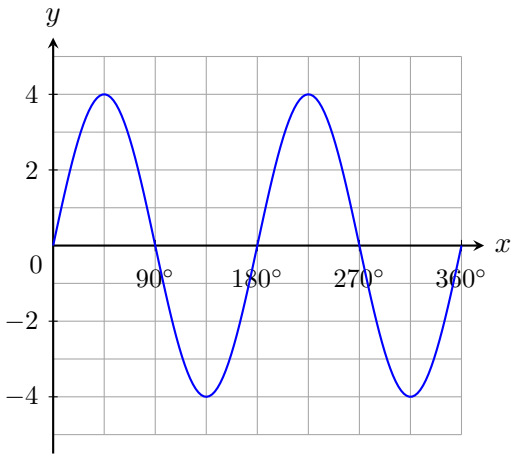

**(a)** Write down the value of $a$ and the value of $b$.

$a =$ ..... , $b =$ ..... **[2]**

**(b)** Write down the coordinates of the first minimum point where $x > 0$.

( ..... , ..... ) **[1]**

**629** The diagram shows the graph of $y = a\sin(bx)$ for $0° \leq x \leq 360°$.



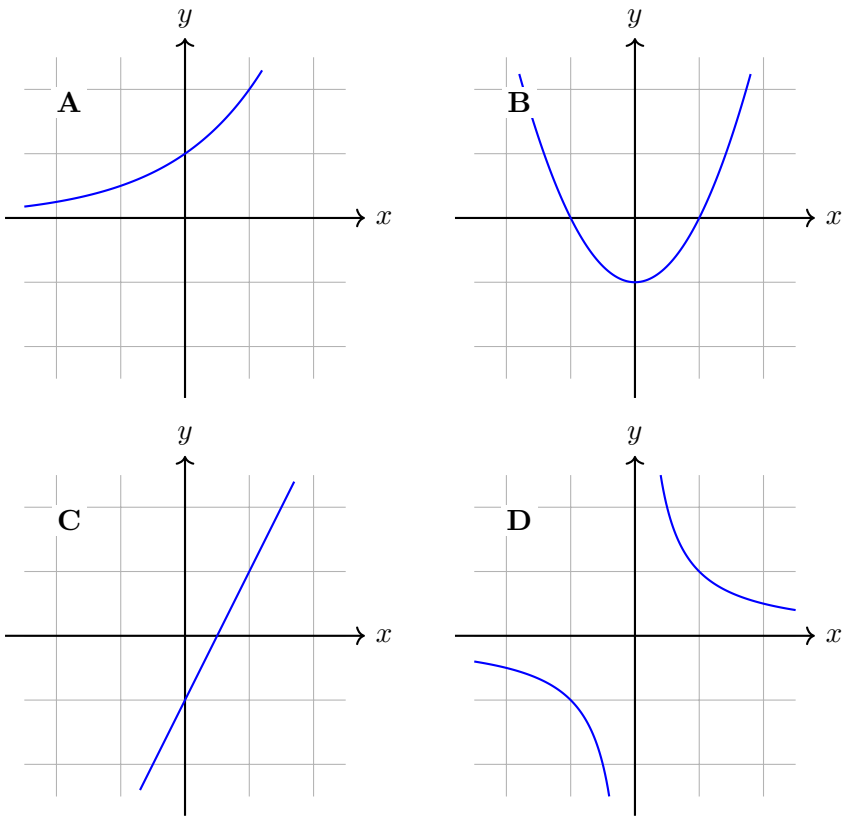

**(a)** Find the value of $a$ and the value of $b$.

$a =$ ..... , $b =$ ..... **[2]**

**(b)** Write down the period of the function.

..... ° **[1]**

**630** Four graphs are shown below, labelled A, B, C, and D.




**(a)** Write down the letter of the graph that represents each function.

**(i)**

$$y = x^2 - 1$$

..... **[1]**

**(ii)**

$$y = 2^x$$

..... **[1]**

**(iii)**

$$y = \frac{1}{x}$$

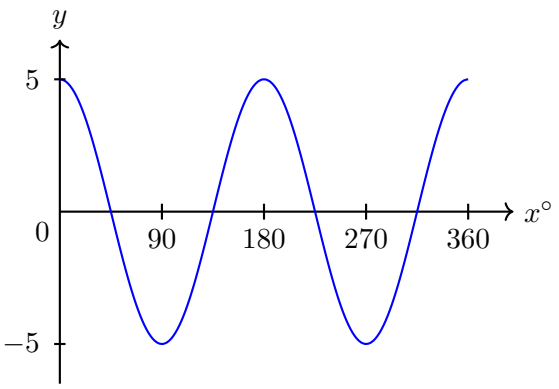
..... **[1]**

**(iv)**

$$y = 2x - 1$$

..... **[1]**

**631** The diagram shows the graph of $y = a\cos(bx)$ for $0° \leq x \leq 360°$.




**(a)** Write down the amplitude of the graph.

..... **[1]**

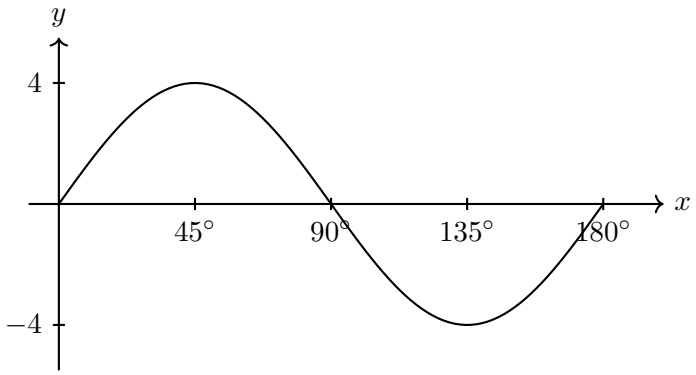
**(b)** Write down the period of the graph.

..... ° **[1]**

**(c)** Find the value of $a$ and the value of $b$.

$a =$ ..... , $b =$ ..... **[2]**

**632** The diagram shows the graph of $y = a\sin(bx)$ for $0° \leq x \leq 180°$.




**(a)** State the amplitude.

..... **[1]**

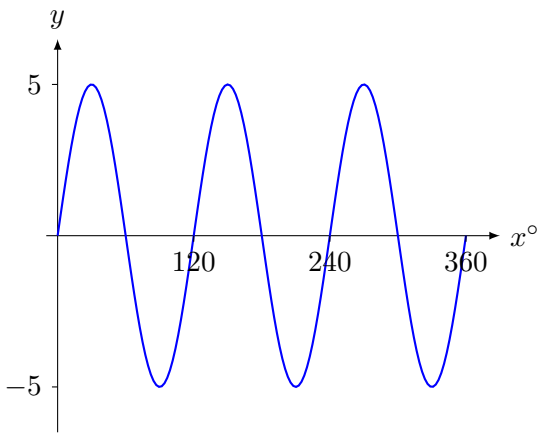
**(b)** State the period of the function.

..... ° **[1]**

**(c)** Write down the value of $a$ and the value of $b$.

$a =$ ..... , $b =$ ..... **[2]**

**633** The diagram shows the graph of $y = a\sin(bx)$ for $0° \leq x \leq 360°$.



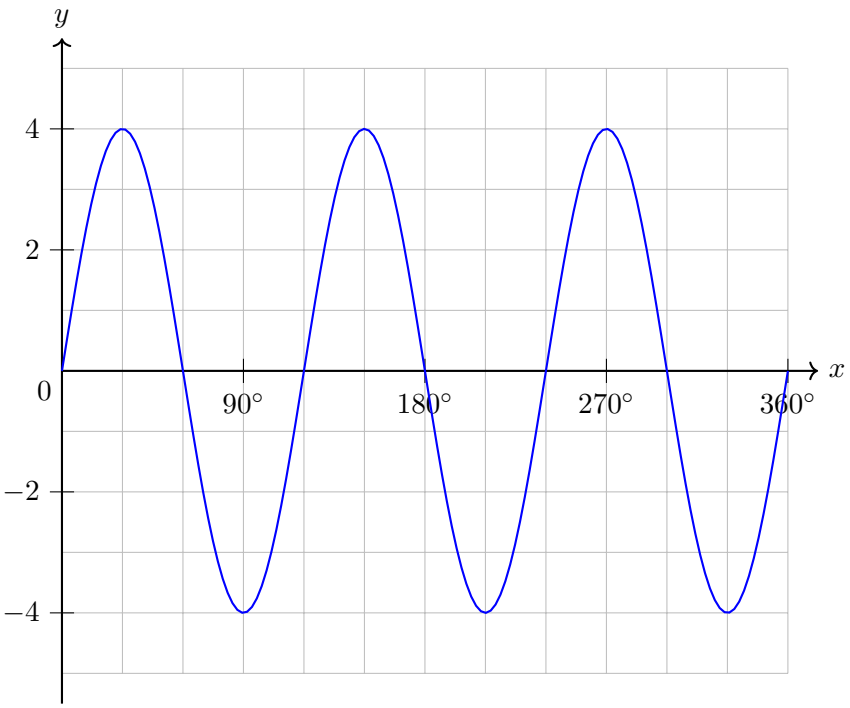

**(a)** Find the value of $a$ and the value of $b$.

$a =$ ..... , $b =$ ..... **[2]**

**(b)** The graph of $y = kx^3$ passes through the point $(2, -24)$.
Find the value of $k$.

$k =$ ..... **[2]**

**634** The diagram shows the graph of $y = a\sin(bx)$ for $0° \leq x \leq 360°$.
The values of $a$ and $b$ are positive integers.




**(a)** Find the value of $a$ and the value of $b$.

$a =$ ..... , $b =$ ..... **[2]**

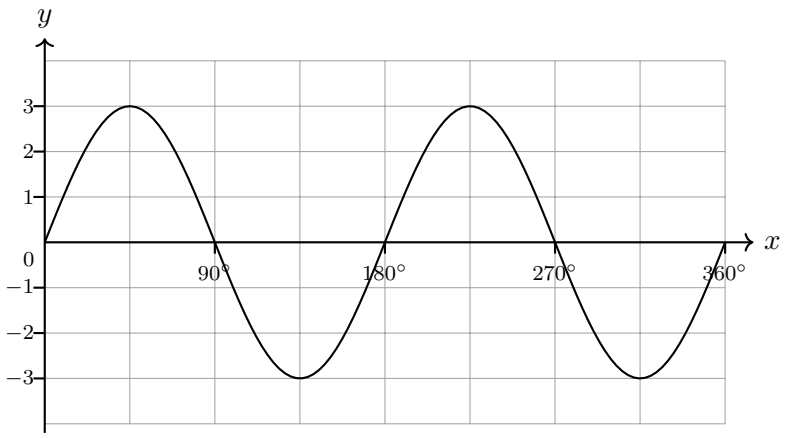
**(b)** Write down the period of the function.

..... ° **[1]**

**(c)** State the number of solutions to the equation $a\sin(bx) = 2$ in the interval $0° \leq x \leq 360°$.

..... **[1]**

**635** The diagram shows the graph of the trigonometric function $y = a\sin(bx)$ for $0° \leq x \leq 360°$, where $a$ and $b$ are positive constants.




**(a)** Write down the amplitude and the period of the function.

Amplitude = ..... , Period = ..... **[2]**

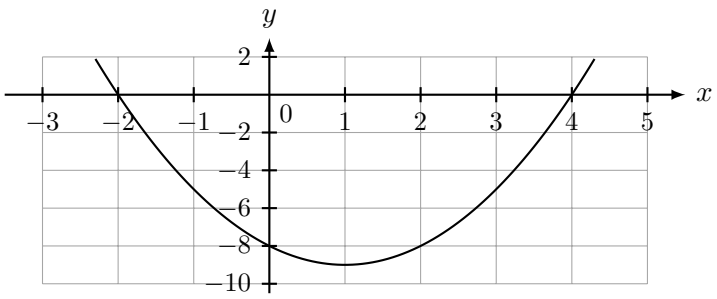
**(b)** Determine the values of $a$ and $b$.

$a$ = ..... , $b$ = ..... **[2]**

**(c)** Use the graph to state the number of solutions to the equation $a\sin(bx) = -2$ for $0° \leq x \leq 360°$.

..... **[1]**

**636** The diagram shows the graph of the function $y = x^2 + bx + c$.




**(a)** Write down the value of $c$.

$c =$ ..... **[1]**

**(b)** Find the value of $b$.

$b =$ ..... **[2]**

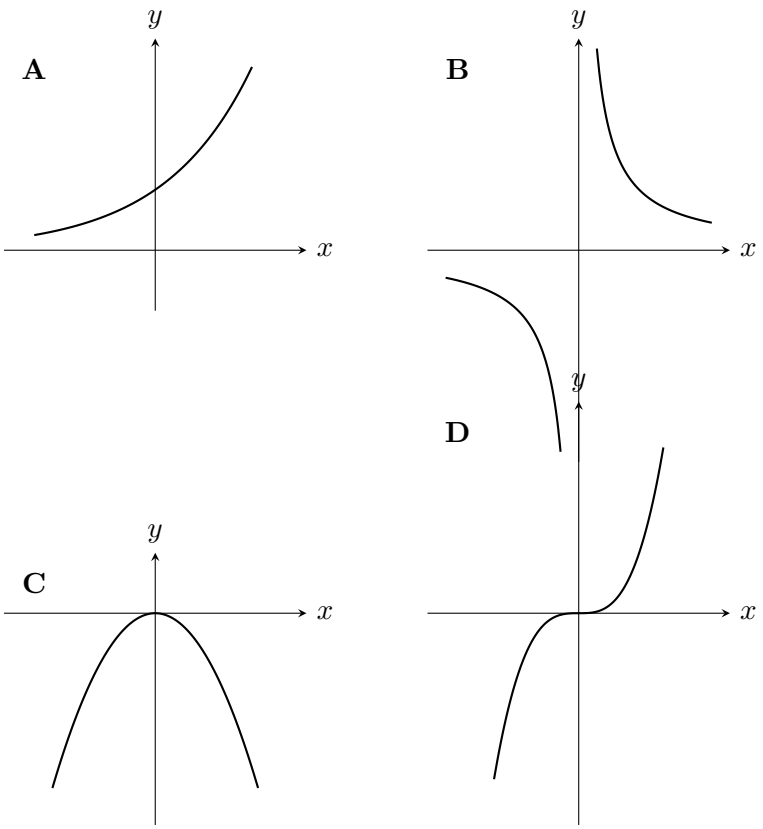
**(c)** Write down the coordinates of the minimum point of the graph.

( ..... , ..... ) **[2]**

**637** Six equations are shown below.

$$y = 2^x \qquad y = x^2 \qquad y = -x^2 \qquad y = x^3 \qquad y = \frac{1}{x} \qquad y = \cos x$$

The diagrams show four of these graphs, labelled A, B, C and D.

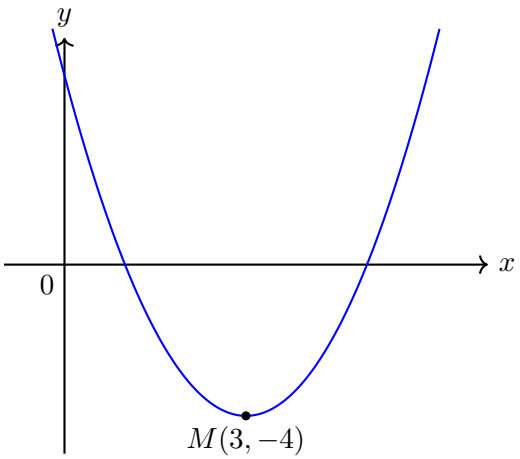



Write down the correct equation for each graph.

$A =$ ..... , $B =$ ..... , $C =$ ..... , $D =$ .....

**[4]**

**638** The diagram shows a sketch of the graph of $y = x^2 + bx + c$.
The minimum point on the graph is $M(3, -4)$.



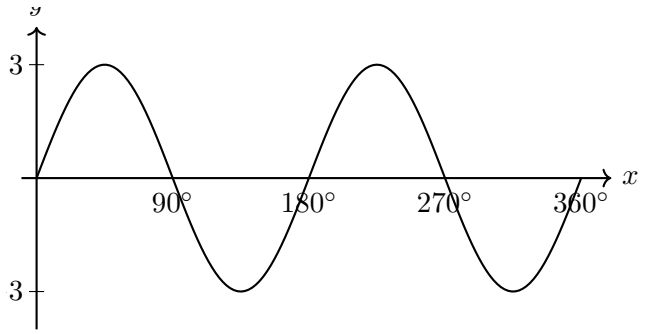

**(a)** Work out the value of $b$ and the value of $c$.

$b =$ ..... , $c =$ ..... **[3]**

**(b)** Find the coordinates of the points where the graph crosses the $x$-axis.

..... **[2]**

**639** The diagram shows a sketch of the graph of $f(x) = a\sin(bx)$ for $0° \leq x \leq 360°$.




**(a)** Write down the value of $a$.

$a =$ ..... **[1]**

**(b)** Write down the value of $b$.

$b =$ ..... **[1]**

**(c)** Solve the equation $f(x) = 1.5$ for $0° \leq x \leq 180°$.

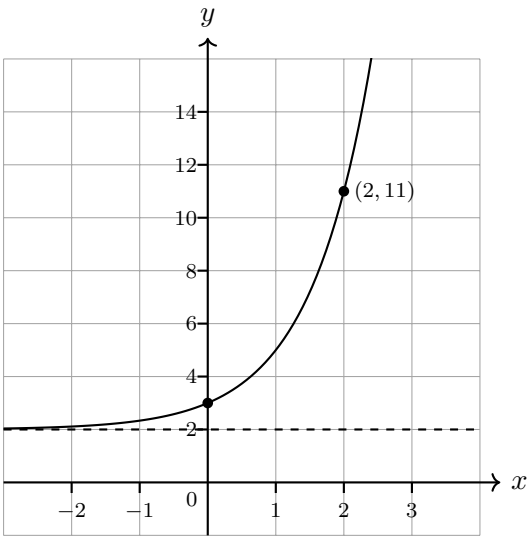
$x =$ ..... **[3]**

**640** The graph of the quadratic function $f(x) = ax^2 + bx + c$ has a line of symmetry at $x = 2$.
The graph passes through the points $(0, -5)$ and $(2, 3)$.

Find the values of $a$, $b$ and $c$.

$a =$ ..... , $b =$ ..... , $c =$ ..... **[5]**

**641** The diagram shows the graph of $f(x) = a^x + c$, where $a > 0$ and $c$ are constants.
The curve passes through the points $(0, 3)$ and $(2, 11)$, and has a horizontal asymptote at $y = 2$.




**(a)** Find the values of $a$ and $c$.

$a =$ ..... , $c =$ ..... **[3]**

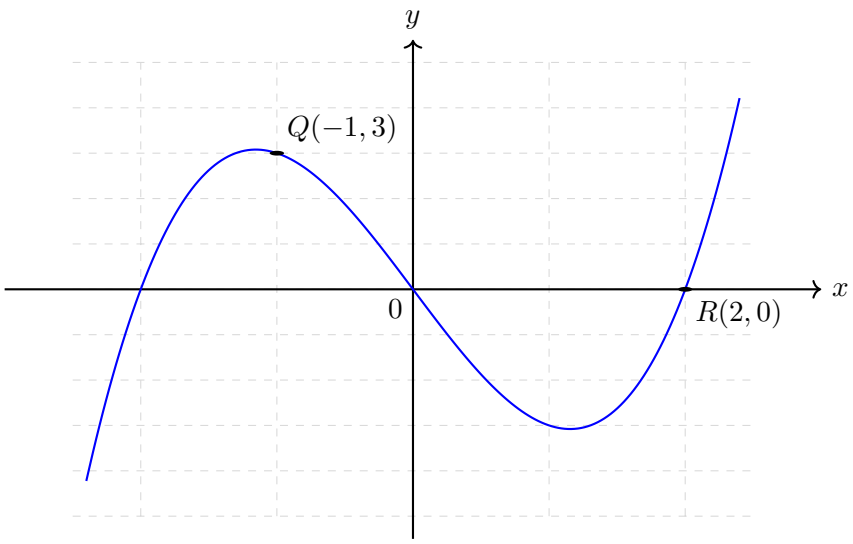
**(b)** Find the value of $f(-1)$, giving your answer as a fraction.

$f(-1) =$ ..... **[1]**

**(c)** State the range of the function $f(x)$.

..... **[2]**

**642** The diagram shows a sketch of the curve with equation $y = x^3 + ax^2 + bx$.
The curve passes through the points $Q(-1, 3)$ and $R(2, 0)$.




**(a)** Use the coordinates of $Q$ and $R$ to form two simultaneous equations in terms of $a$ and $b$.

.....

..... **[2]**

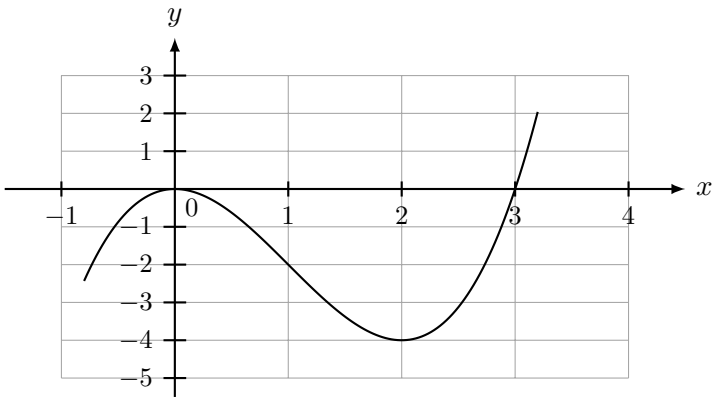
**(b)** Solve your equations to find the value of $a$ and the value of $b$.

$a =$ ..... , $b =$ ..... **[2]**

**(c)** Find the coordinates of the other two points where the curve crosses the $x$-axis.

..... **[3]**

**643** The diagram shows the graph of $y = f(x)$ where $f(x) = ax^3 + bx^2$.
The graph passes through the origin $(0, 0)$, the point $(3, 0)$ and has a local minimum point at $(2, -4)$.




**(a)** Show that $3a + b = 0$.

**[1]**

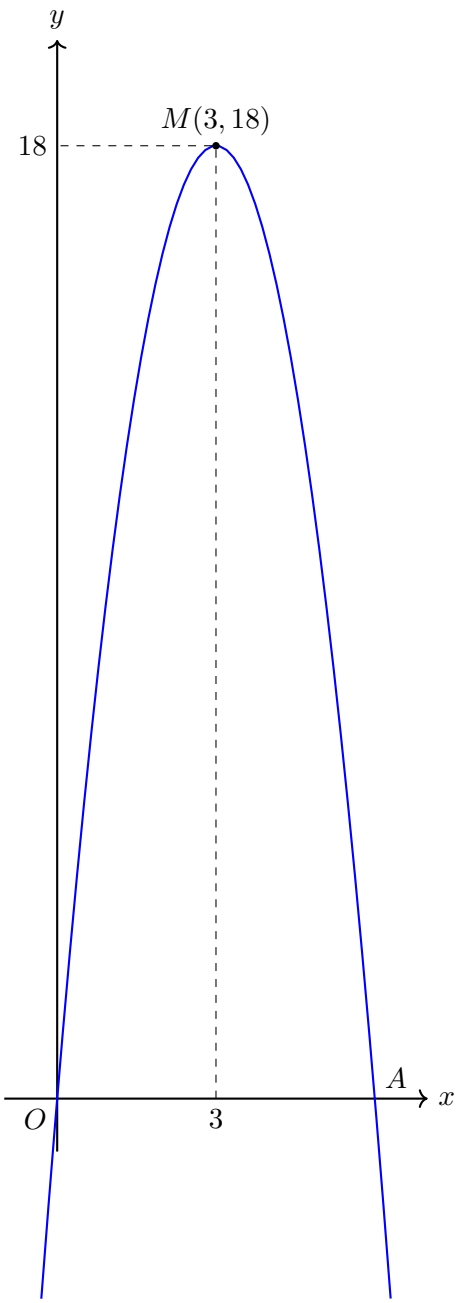
**(b)** Find the value of $a$ and the value of $b$.

$a =$ ..... , $b =$ ..... **[3]**

**(c)** Write down the range of values of $k$ for which the equation $f(x) = k$ has exactly one solution.

..... **[2]**

**644** The diagram shows a sketch of the graph of $y = f(x)$, where $f(x) = ax^2 + bx + c$.
The curve passes through the origin $O$ and intersects the $x$-axis at the point $A$.
The point $M(3, 18)$ is the maximum point of the curve.




**(a)** Write down the value of $c$.

$c =$ ..... **[1]**

**(b)** Write down the coordinates of $A$.

( ..... , ..... ) **[1]**

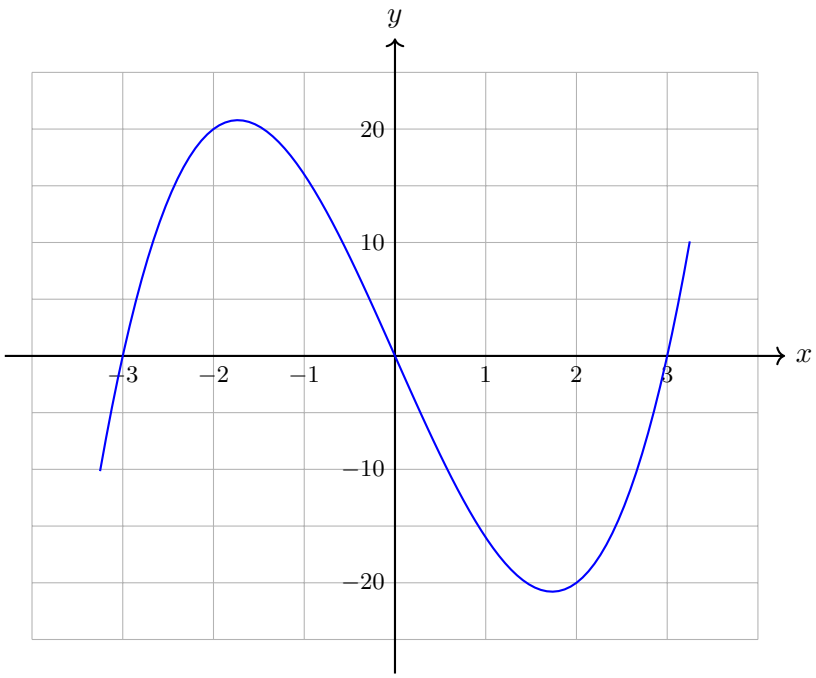
**(c)** Find the value of $a$ and the value of $b$.

$a =$ ..... , $b =$ ..... **[3]**

**(d)** Find the range of values of $k$ for which the equation $f(x) = k$ has two distinct positive real roots.

..... **[2]**

**645** The diagram shows the graph of the cubic function $y = ax^3 + bx^2 + cx + d$.




**(a)** Write down the value of $d$.

$d =$ ..... **[1]**

**(b)** The graph intersects the $x$-axis at $x = -3$ and $x = 3$.
Show that $b = 0$ and $9a + c = 0$.

**[3]**

**(c)** The graph passes through the point $(1, -16)$.
Find the value of $a$ and the value of $c$.

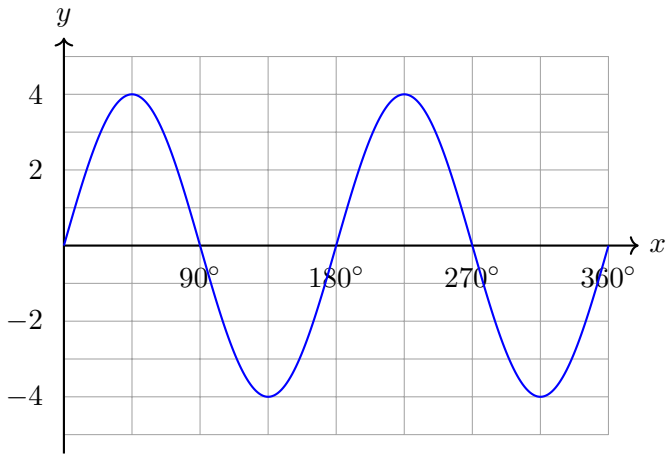
$a =$ ..... , $c =$ ..... **[3]**

**(d)** A different function $g(x) = px^2 + qx + r$ has a minimum point at $(0, -5)$ and passes through the point $(2, 7)$.

Find the values of $p$, $q$ and $r$.

$p =$ ..... , $q =$ ..... , $r =$ ..... **[3]**

**646** The diagram shows the graph of $y = a\sin(bx)$ for $0° \leq x \leq 360°$.




**(a)** Write down the value of $a$.

$a =$ ..... **[1]**

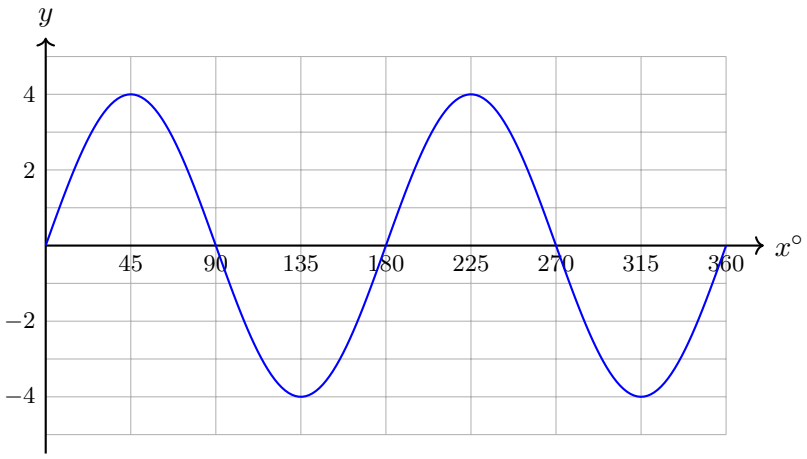
**(b)** Find the value of $b$.

$b =$ ..... **[1]**

**(c)** Calculate the value of $y$ when $x = 30°$.
Give your answer correct to 3 significant figures.

$y =$ ..... **[2]**

**647** The diagram shows the graph of $y = a\sin(bx)$ for $0° \leq x \leq 360°$.




**(a)** Find the value of $a$ and the value of $b$.

$a =$ ..... , $b =$ ..... **[2]**

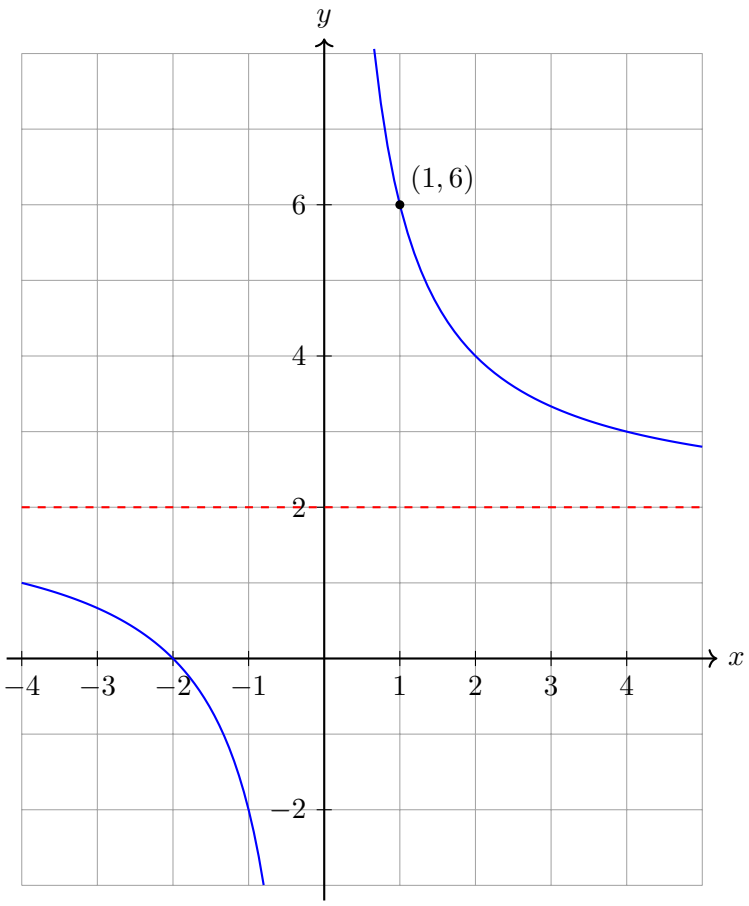
**(b)** The graph of $y = p^x$ passes through the point $(3, 125)$.
Find the value of $p$.

$p =$ ..... **[1]**

**(c)** The graph of $y = \dfrac{k}{x}$ passes through the point $(0.5, 8)$.
Find the value of $k$.

$k =$ ..... **[1]**

**648** The diagram shows the graph of $y = \frac{a}{x} + b$.
The equation of the horizontal asymptote is $y = 2$.
The graph passes through the point $(1, 6)$.




**(a)** Write down the value of $b$.

$b =$ ..... **[1]**

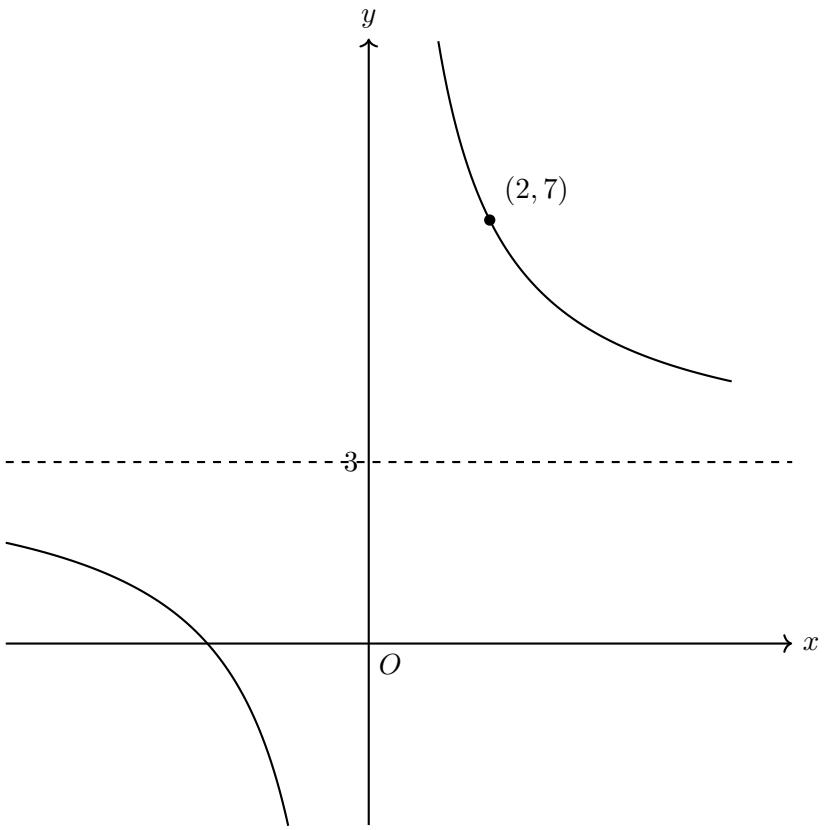
**(b)** Find the value of $a$.

$a =$ ..... **[2]**

**(c)** The graph of $y = k^x$, where $k > 0$, also passes through the point $(2, 4)$, which lies on the curve $y = \frac{a}{x} + b$.
Find the value of $k$.

$k =$ ..... **[2]**

**649** The diagram shows a sketch of the graph of $f(x) = \dfrac{a}{x} + b$, where $xe0$.
The dashed line is a horizontal asymptote.
The graph passes through the point $(2, 7)$.




**(a)** Write down the value of $b$.

$b =$ ..... **[1]**

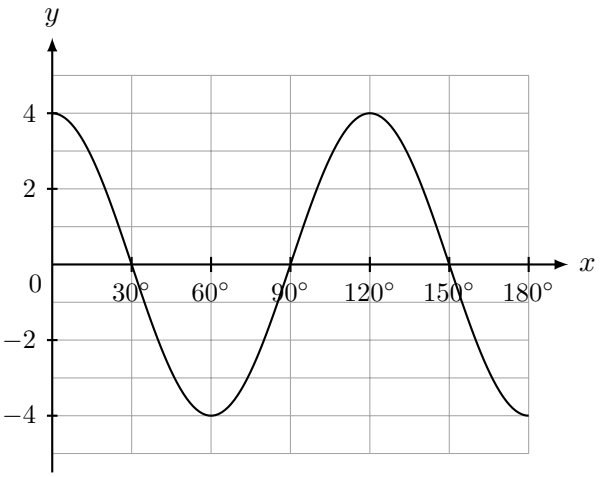
**(b)** Find the value of $a$.

$a =$ ..... **[2]**

**(c)** Find the coordinates of the point where the graph crosses the $x$-axis.

( ..... , ..... ) **[2]**

**650** The diagram shows a sketch of the graph of $f(x) = a\cos(bx)$ for $0° \leq x \leq 180°$.




**(a)** Write down the value of $a$.

$a =$ ..... **[1]**

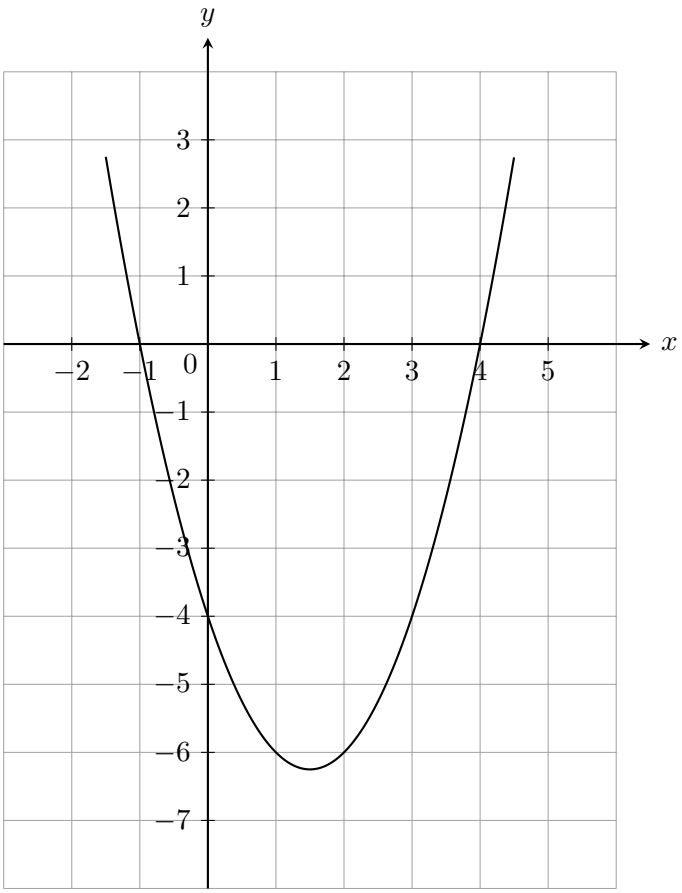
**(b)** Find the value of $b$.

$b =$ ..... **[2]**

**(c)** Determine the number of solutions to the equation $a\cos(bx) = 1.5$ in the interval $0° \leq x \leq 360°$.

..... **[2]**

**651** The diagram shows the graph of a quadratic function $f(x) = ax^2 + bx + c$.
The graph crosses the $y$-axis at $(0, -4)$ and the $x$-axis at $(-1, 0)$ and $(4, 0)$.




**(a)** Write down the value of $c$.

$c =$ ..... **[1]**

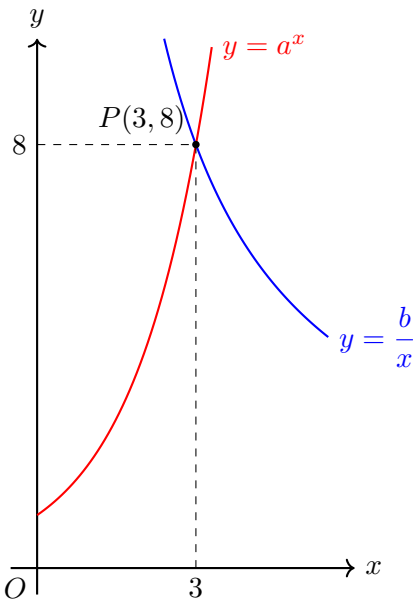
**(b)** Determine the value of $a$ and the value of $b$.

$a =$ ..... , $b =$ ..... **[3]**

**(c)** Calculate the exact coordinates of the minimum point of the graph.

( ..... , ..... ) **[2]**

**652** The diagram shows a sketch of the graphs of $y = a^x$ and $y = \dfrac{b}{x}$ for $x > 0$.
The graphs intersect at the point $P(3, 8)$.




**(a)** Find the value of $a$ and the value of $b$.

$a =$ ..... , $b =$ ..... **[3]**

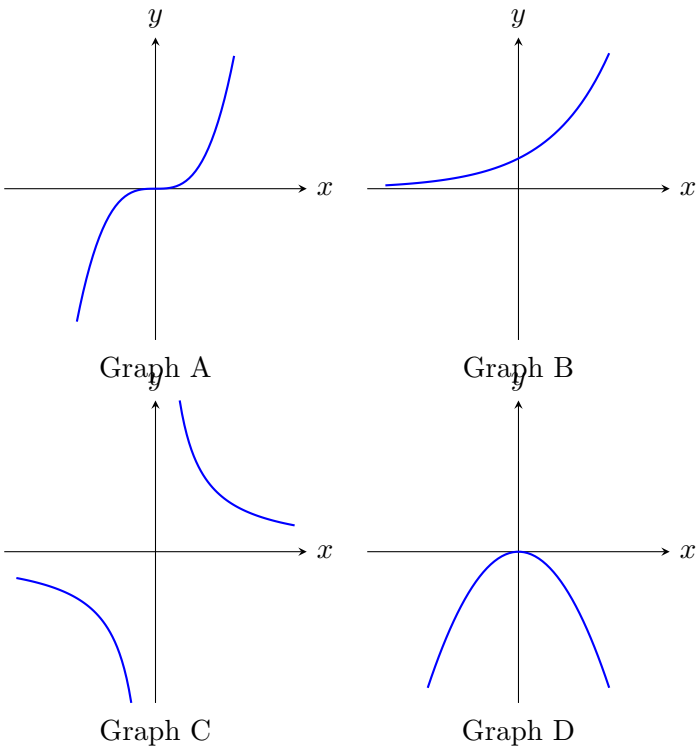
**(b)** The graph of $y = a^x$ intersects the $y$-axis at the point $Q$.
Write down the coordinates of $Q$.

( ..... , ..... ) **[1]**

**(c)** Calculate the gradient of the straight line joining $Q$ and $P$.

Gradient = ..... **[2]**

**653** Four graphs A, B, C and D are shown below.




**(a)** Write down the letter of the graph which could represent the function

**(i)** $f(x) = \dfrac{a}{x}$ where $a > 0$

..... **[1]**

**(ii)** $f(x) = a^x$ where $a > 1$

..... **[1]**

**(iii)** $f(x) = -ax^2$ where $a > 0$

..... **[1]**

**(b)** The equation of Graph B is $y = p^x$.
The graph passes through the point $(3, 125)$.
Find the value of $p$.

$p =$ ..... **[2]**

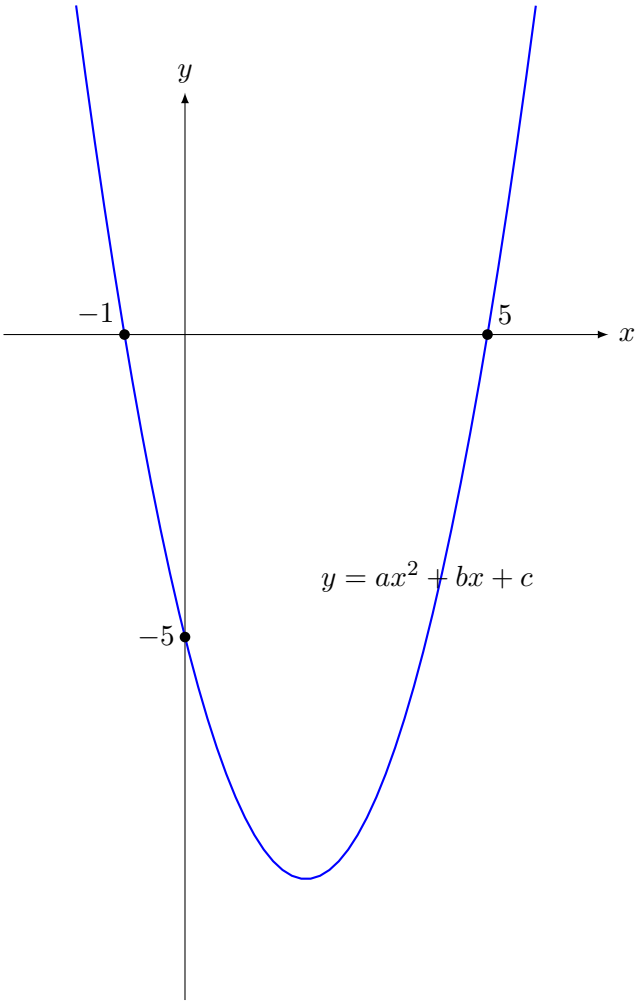
**(c)** The equation of Graph C is $y = \dfrac{q}{x}$.

The graph passes through the point $(0.5, 8)$.

Find the value of $q$.

$q =$ ..... **[2]**

**654** The diagram shows the graph of $y = ax^2 + bx + c$.
The graph intersects the $x$-axis at $(-1, 0)$ and $(5, 0)$, and the $y$-axis at $(0, -5)$.




**(a)** Find the values of $a$, $b$ and $c$.

$a =$ ..... , $b =$ ..... , $c =$ ..... **[3]**

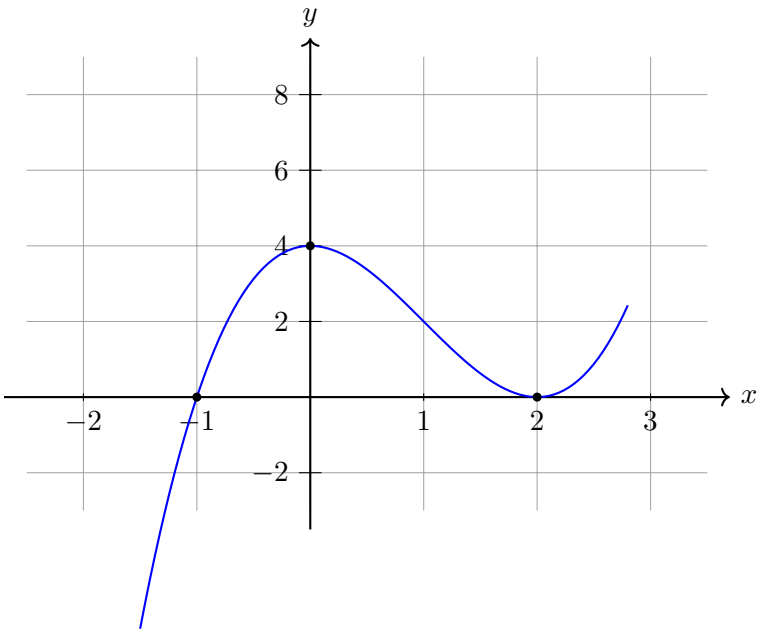
**(b)** Write down the coordinates of the minimum point of the graph.

( ..... , ..... ) **[2]**

**(c)** The graph of $y = p \times q^x$ passes through the points $(0, 2)$ and $(2, 18)$, where $q > 0$.
Find the value of $p$ and the value of $q$.

$p =$ ..... , $q =$ ..... **[3]**

**655** The diagram shows a sketch of the curve $f(x) = x^3 + ax^2 + bx + c$.
The curve crosses the $y$-axis at $(0, 4)$.
It crosses the $x$-axis at $(-1, 0)$ and touches the $x$-axis at $(2, 0)$.




**(a)** Write down the value of $c$.

$c =$ ..... **[1]**

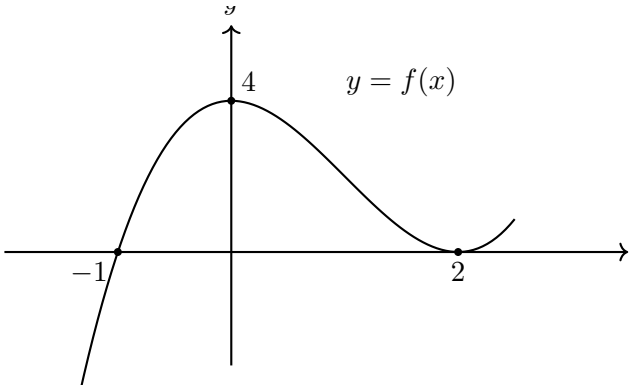
**(b)** By using the $x$-intercepts, find the value of $a$ and the value of $b$.

$a =$ ..... , $b =$ ..... **[4]**

**(c)** Use the graph to state the number of solutions to the equation $f(x) = 3$.

..... **[1]**

**656** The diagram shows a sketch of the graph of $f(x) = x^3 + bx^2 + cx + d$.
The curve intersects the $y$-axis at 4 and the $x$-axis at $-1$ and 2.
The point $(2, 0)$ is a local minimum.




**(a)** Write down the value of $d$.

$d =$ ..... **[1]**

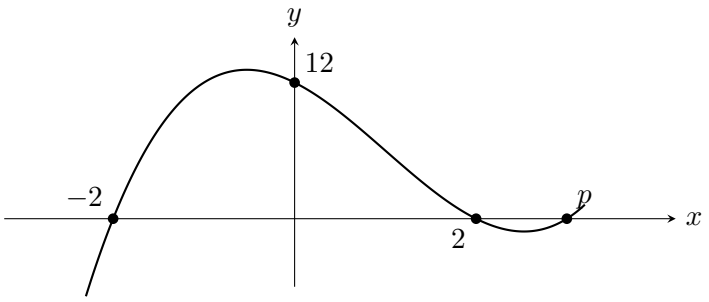
**(b)** Find the values of $b$ and $c$.

$b =$ ..... , $c =$ ..... **[4]**

**(c)** Write down the number of solutions to the equation $f(x) = -3$.

Number of solutions = ..... **[1]**

**657** The diagram shows the graph of $y = x^3 + bx^2 + cx + d$.
The graph crosses the axes at $(-2, 0)$, $(2, 0)$, $(p, 0)$ and $(0, 12)$.




**(a)** Write down the value of $d$.

$d =$ ..... **[1]**

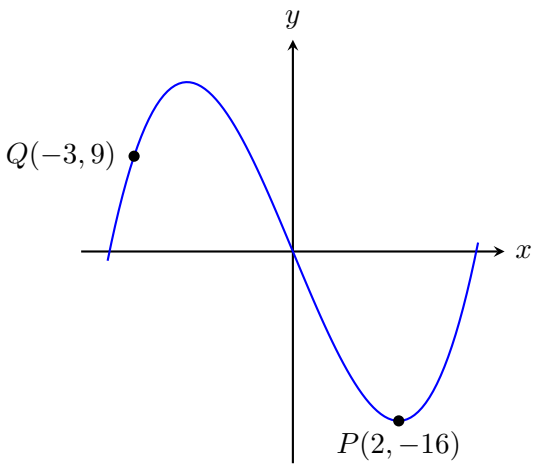
**(b)** Find the values of $b$ and $c$.

$b =$ ..... , $c =$ ..... **[4]**

**(c)** Find the value of $p$.

$p =$ ..... **[2]**

**658** The diagram shows a sketch of the graph of $f(x) = ax^3 + bx$.
The curve passes through the points $P(2, -16)$ and $Q(-3, 9)$.




**(a)** Find the value of $a$ and the value of $b$.

$a =$ ..... , $b =$ ..... **[3]**

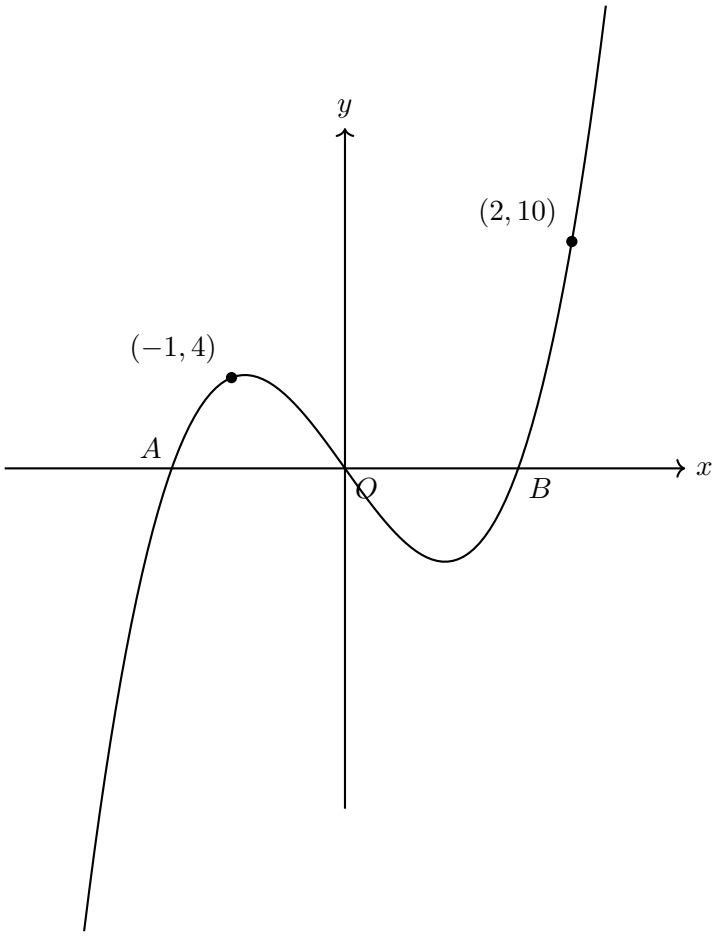
**(b)** Find the coordinates of the local maximum point of the graph.

( ..... , ..... ) **[2]**

**(c)** The equation $f(x) = k$ has exactly one solution.
Write down the range of values for $k$.

..... **[2]**

**659** The diagram shows a sketch of the curve $y = ax^3 + bx$.
The curve passes through the origin $O$, and the points $(2, 10)$ and $(-1, 4)$.
The curve crosses the $x$-axis at the points $A$, $O$, and $B$.



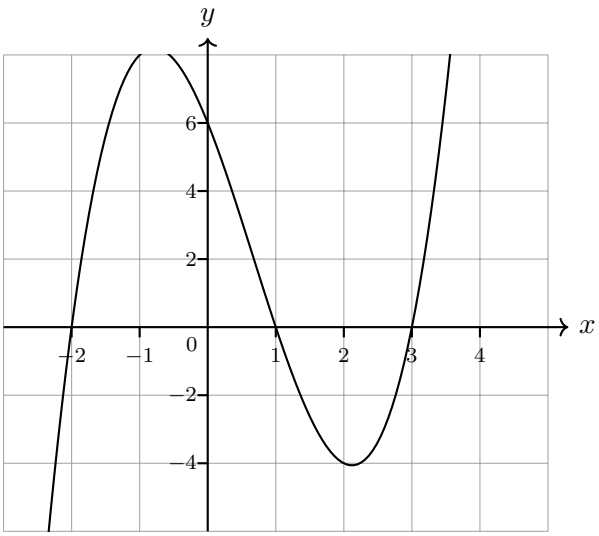

**(a)** Find the value of $a$ and the value of $b$.

$a =$ ..... , $b =$ ..... **[4]**

**(b)** Calculate the exact length of the line segment $AB$.

..... **[3]**

**660** The diagram shows the graph of the cubic function $g(x) = x^3 + bx^2 + cx + d$, where $b$, $c$, and $d$ are integers.
The curve crosses the $x$-axis at $(-2, 0)$, $(1, 0)$, and $(3, 0)$.




**(a)** Find the values of $b$, $c$, and $d$.

$b =$ ..... , $c =$ ..... , $d =$ ..... **[3]**

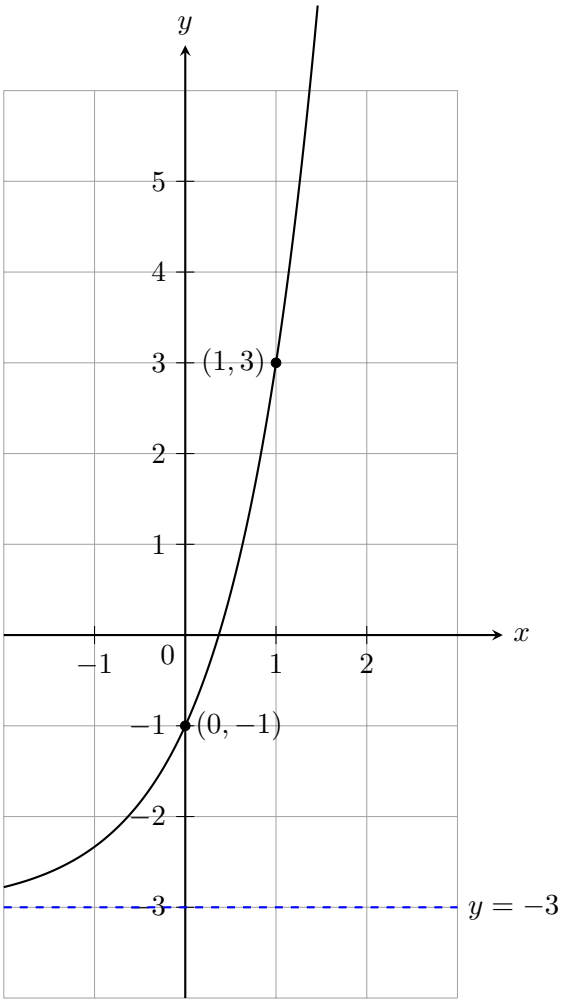
**(b)** Write down the coordinates of the $y$-intercept of the graph of $y = g(x)$.

( ..... , ..... ) **[1]**

**(c)** Use your graphic display calculator to find the $x$-coordinates of the local maximum and local minimum points on the graph of $y = g(x)$.
Give your answers correct to 2 decimal places.

Local maximum $x =$ ..... , Local minimum $x =$ ..... **[3]**

**661** The diagram shows the graph of the exponential function $f(x) = a \cdot b^x + c$.
The horizontal line $y = -3$ is an asymptote to the graph.
The graph passes through the points $(0, -1)$ and $(1, 3)$.




**(a)** State the value of $c$.

$c =$ ..... **[1]**

**(b)** Work out the value of $a$ and the value of $b$.

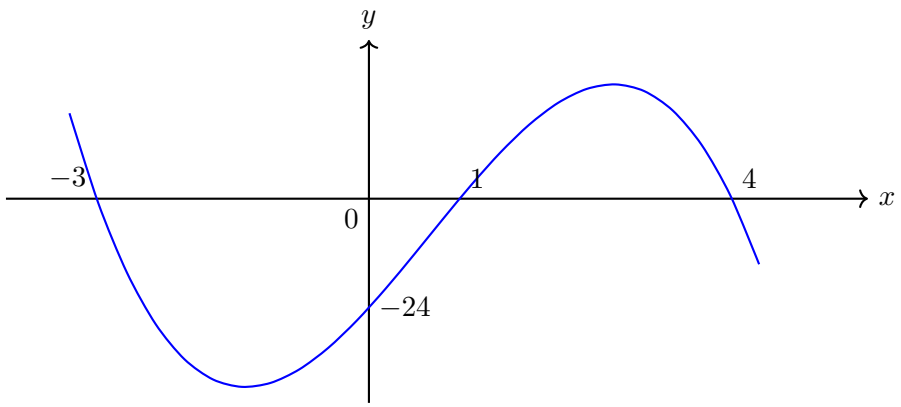
$a =$ ..... , $b =$ ..... **[4]**

**(c)** The graph of $g(x) = \dfrac{k}{x}$ intersects the graph of $f(x)$ at the point where $x = 1$.

Find the value of $k$.

$k =$ ..... **[2]**

**662** The diagram shows the graph of a cubic function $f(x) = ax^3 + bx^2 + cx + d$.
The graph crosses the $x$-axis at $(-3, 0)$, $(1, 0)$ and $(4, 0)$.
It crosses the $y$-axis at $(0, -24)$.




**(a)** Find the values of $a$, $b$, $c$ and $d$.

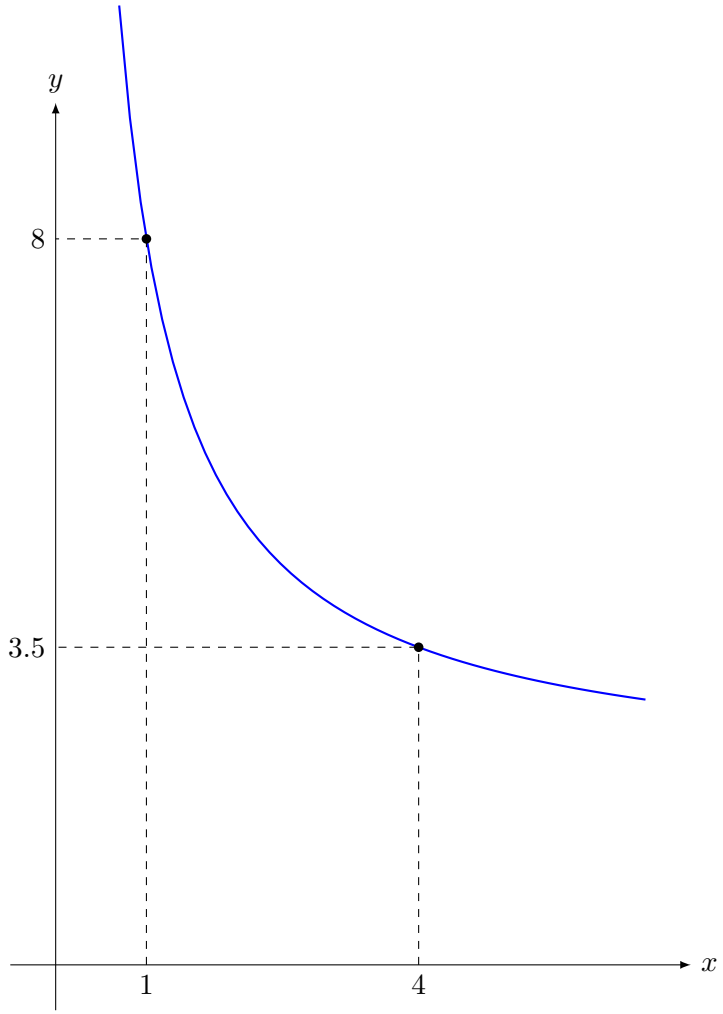
$a =$ ..... , $b =$ ..... , $c =$ ..... , $d =$ ..... **[6]**

**(b)** The graph of $y = f(x)$ is translated by the vector $\begin{pmatrix} 0 \\ 10 \end{pmatrix}$ to give the graph of $y = g(x)$.

Find the coordinates of the $y$-intercept of the graph of $y = g(x)$.

( ..... , ..... ) **[2]**

**663** The diagram shows part of the graph of $y = \frac{a}{x} + b$ for $x > 0$.
The graph passes through the points $(1, 8)$ and $(4, 3.5)$.




**(a)** Find the value of $a$ and the value of $b$.

$a =$ ..... , $b =$ ..... **[3]**

**(b)** Write down the equation of the horizontal asymptote to the graph.

..... **[1]**

**(c)** The line $y = x - 3$ intersects the graph of $y = \frac{6}{x} + 2$ at the point $P$.

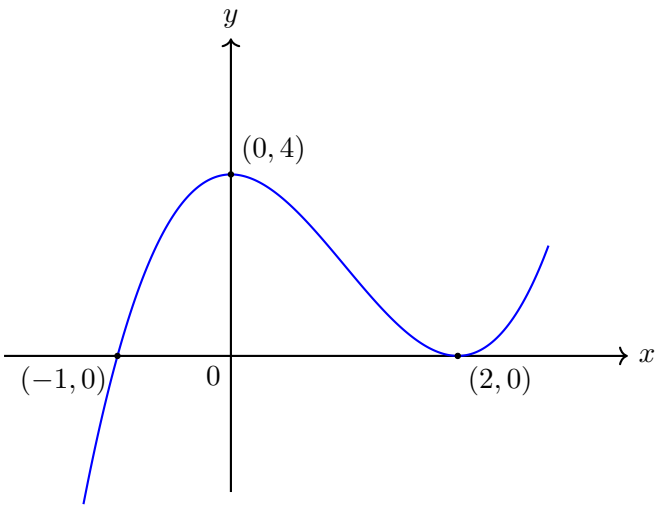
**(i)** Show that the $x$-coordinate of $P$ satisfies the equation $x^2 - 5x - 6 = 0$.

**[2]**

**(ii)** Find the coordinates of $P$.

( ..... , ..... ) **[2]**

**664** The diagram shows a sketch of the graph of $f(x) = x^3 + bx^2 + cx + 4$.
The graph crosses the $x$-axis at $(-1, 0)$ and touches the $x$-axis at $(2, 0)$.
The graph crosses the $y$-axis at $(0, 4)$.




**(a)** **(i)** Use the point $(-1, 0)$ to show that $b - c = -3$.

**[2]**

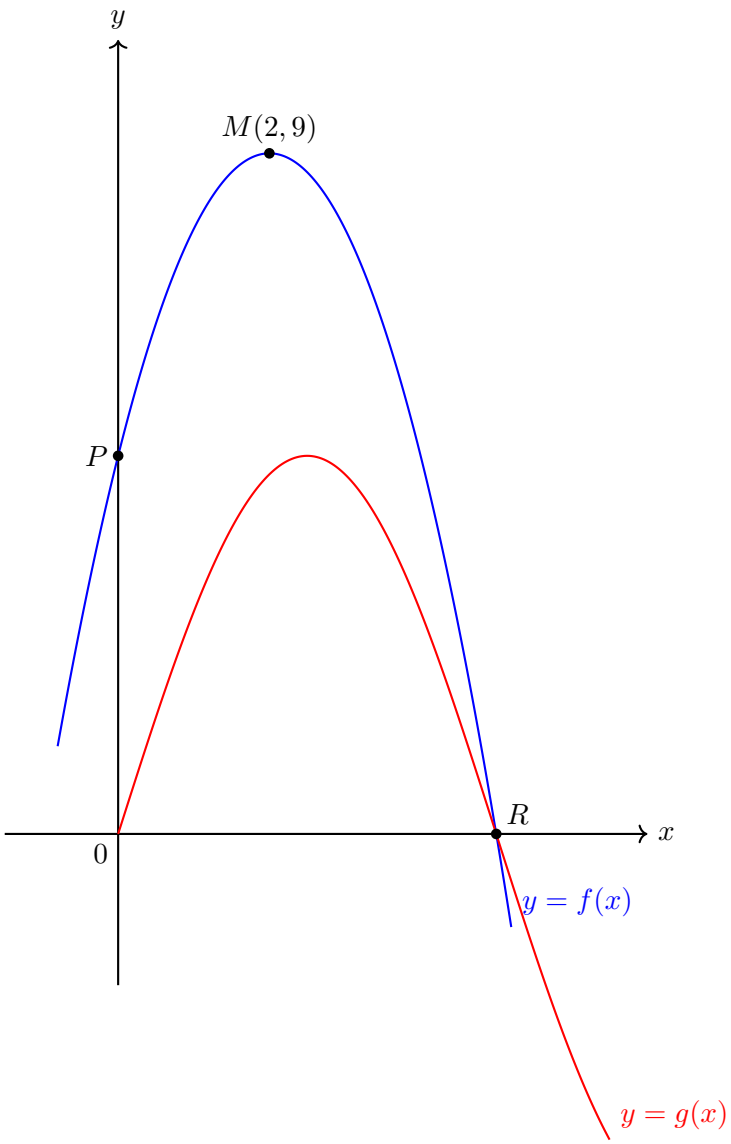
**(ii)** Find the values of $b$ and $c$.

$b =$ ..... , $c =$ ..... **[4]**

**(b)** The graph of $g(x) = p \cdot q^x$ passes through the points $(0, 2)$ and $(3, 54)$.
Given that $q > 0$, work out the values of $p$ and $q$.

$p =$ ..... , $q =$ ..... **[3]**

**665** The diagram shows the graphs of $y = f(x)$ and $y = g(x)$ for $x \geq 0$.
It is given that $f(x) = px^2 + qx + r$.
The graph of $y = f(x)$ has a maximum point at $M(2, 9)$ and intersects the $y$-axis at $P(0, 5)$.




**(a)** Find the values of $p, q$, and $r$.

$p =$ ..... , $q =$ ..... , $r =$ ..... **[4]**

**(b)** The graph of $y = f(x)$ intersects the positive $x$-axis at the point $R$.
Find the coordinates of $R$.

( ..... , ..... ) **[2]**

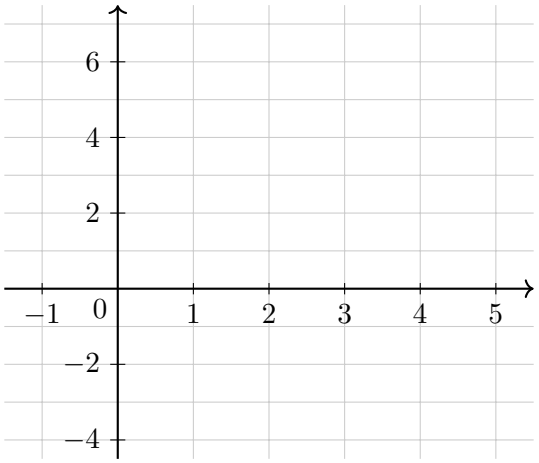
**(c)** It is given that $g(x) = a\sin(bx^\circ)$, where $a > 0$ and $b > 0$.
The graph of $y = g(x)$ has its first positive $x$-intercept at the point $R$.
The amplitude of $g(x)$ is equal to the $y$-coordinate of $P$.

Find the value of $a$ and the value of $b$.

$a =$ ..... , $b =$ ..... **[3]**

**666** Consider the function $f(x) = x^2 - 4x + 1$ for $-1 \leq x \leq 5$.

**(a)** Use your graphic display calculator to sketch the graph of $y = f(x)$ on the axes provided.




**[2]**

**(b)** Write down the coordinates of the vertex of the graph.

( ..... , ..... ) **[1]**

**(c)** Find the zeros of $f(x)$.
Give your answers correct to 2 decimal places.

$x =$ ..... , $x =$ ..... **[2]**

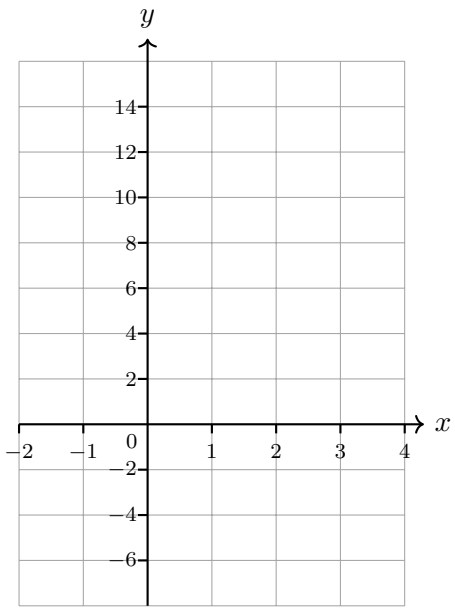
**667** Consider the function $f(x) = 2x^2 - 5x - 3$ for $-2 \leq x \leq 4$.

**(a)** Use your graphic display calculator to complete the table of values for $f(x)$.

| $x$ | $-2$ | $-1$ | $0$ | $1$ | $2$ | $3$ | $4$ |
|---|---|---|---|---|---|---|---|
| $f(x)$ | $15$ | | $-3$ | | $-5$ | | $9$ |

**[2]**

**(b)** On the grid, plot the points from your table and draw a smooth curve to show the graph of $y = f(x)$ for $-2 \leq x \leq 4$.



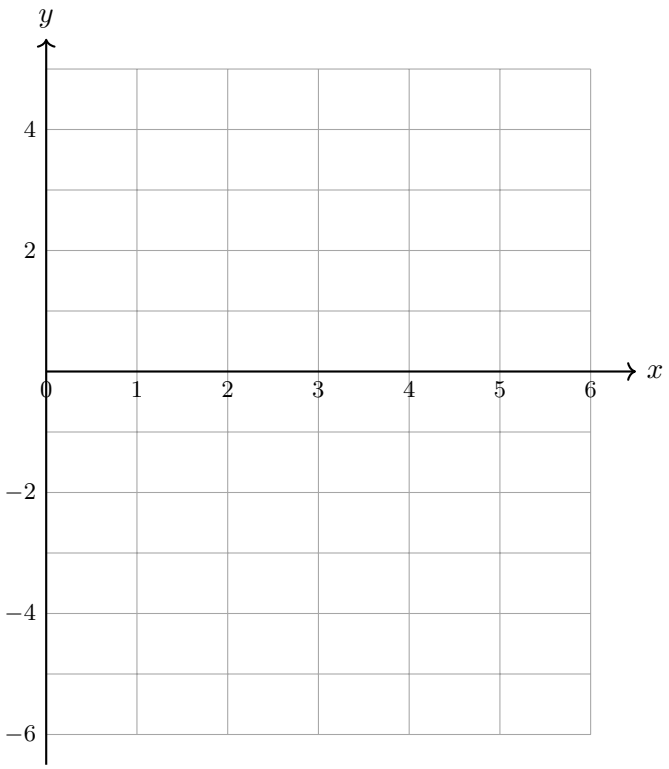

**[2]**

**(c)** Use your graphic display calculator to find the coordinates of the vertex of the graph of $y = f(x)$.

( ..... , ..... ) **[1]**

**668** Let $f(x) = -x^2 + 6x - 5$.

**(a)** On the axes provided, sketch the graph of $y = f(x)$ for $0 \leq x \leq 6$.




**[2]**

**(b)** Find the zeros of $f(x)$.

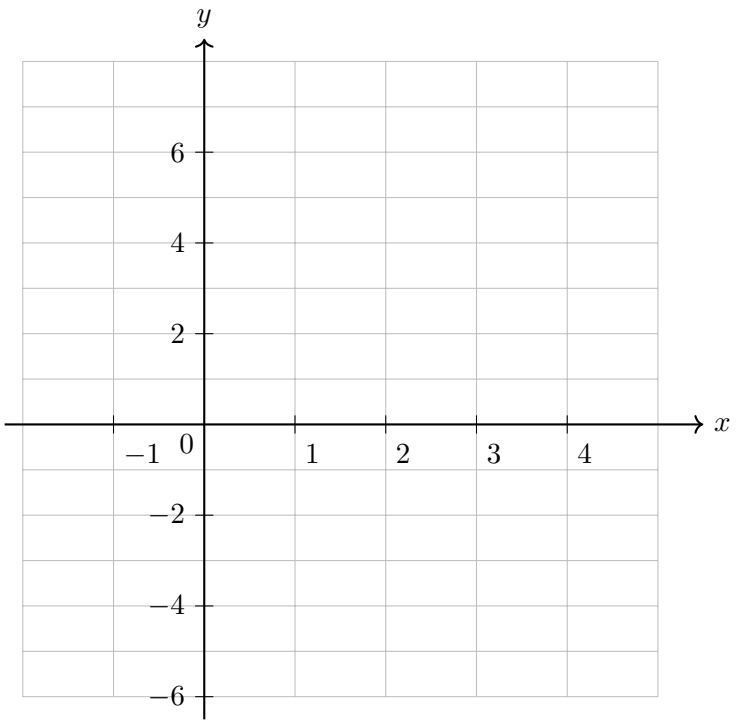
$x =$ ..... , $x =$ ..... **[2]**

**(c)** Find the coordinates of the vertex of the graph.

( ..... , ..... ) **[2]**

**669** Let $f(x) = -2x^2 + 5x + 3$.

**(a)** On the axes, sketch the graph of $y = f(x)$ for $-1.5 \leq x \leq 4$.




**[2]**

**(b)** Find the coordinates of the local maximum.

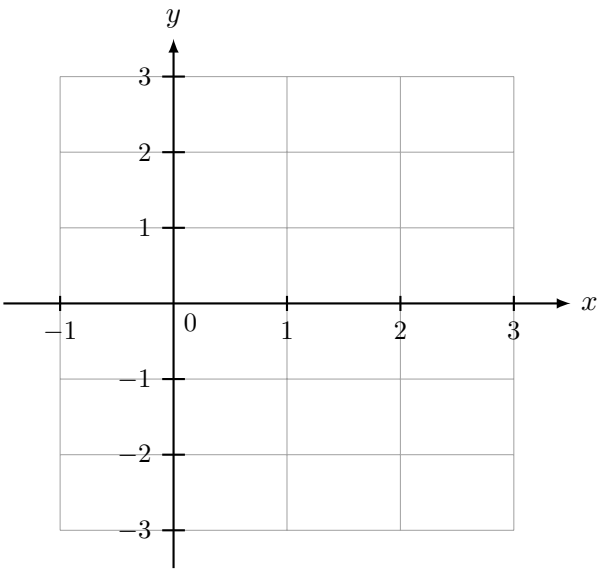
( ..... , ..... ) **[2]**

**(c)** Find the zeros of $f(x)$.

$x =$ ..... , $x =$ ..... **[2]**

**670** Consider the function $f(x) = x^3 - 3x^2 + 2$ for $-1 \leq x \leq 3$.

**(a)** On the grid, sketch the graph of $y = f(x)$.




**[2]**

**(b)** Find the zeros of $f(x)$.

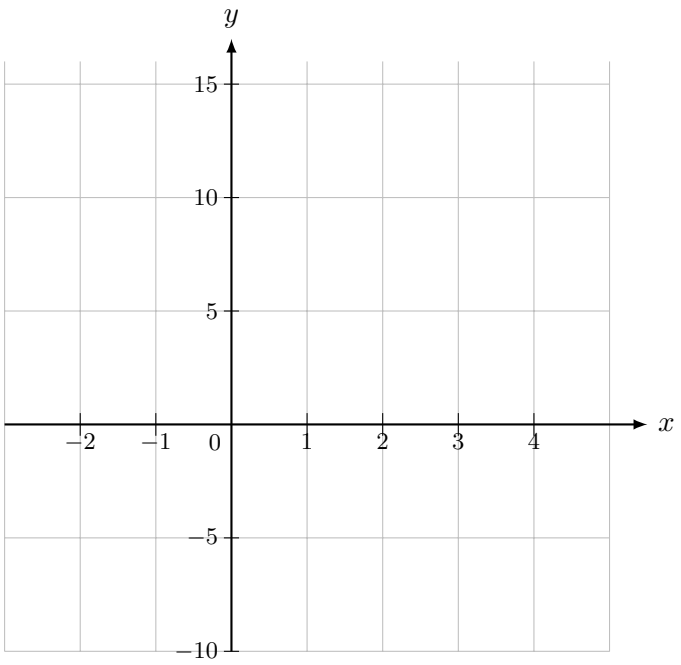
$x =$ ..... **[2]**

**(c)** Find the coordinates of the local minimum point of the graph.

( ..... , ..... ) **[2]**

**671** A function is given by $f(x) = 2x^2 - 5x - 3$ for the domain $-2 \leq x \leq 4$.

**(a)** On the axes, sketch the graph of $y = f(x)$.




**[2]**

**(b)** Write down the zeros of $f(x)$.

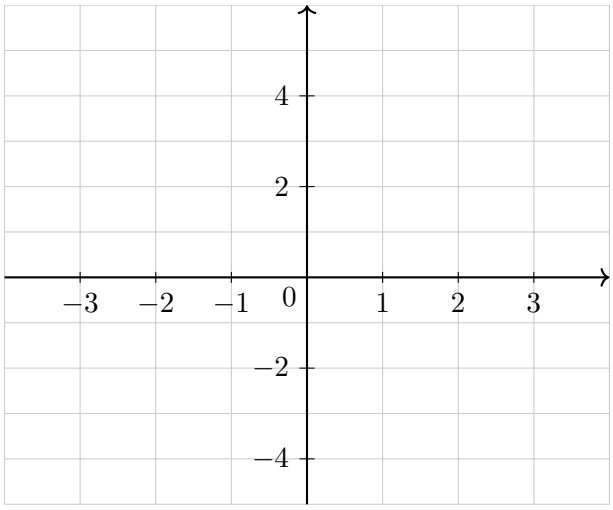
$x =$ ..... , $x =$ ..... **[2]**

**(c)** Write down the coordinates of the local minimum.

( ..... , ..... ) **[2]**

**672** Consider the function $f(x) = \dfrac{x^3}{4} - 2x + 1$ for $-3.5 \leq x \leq 3.5$.

**(a)** On the grid, sketch the graph of $y = f(x)$.




**[2]**

**(b)** Find the coordinates of the local maximum point and the local minimum point.
Give your answers correct to 2 decimal places.

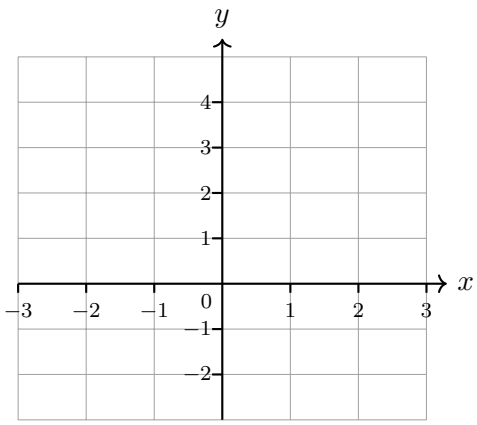
Local maximum = ..... , Local minimum = ..... **[2]**

**(c)** Write down the range of values of $k$ for which the equation $f(x) = k$ has three distinct real solutions.

..... **[2]**

**673** Consider the functions $f(x) = \dfrac{4}{x^2+1}$ and $g(x) = 2x - 1$ for $-3 \leq x \leq 3$.

**(a)** Use your graphic display calculator to sketch the graphs of $y = f(x)$ and $y = g(x)$ on the grid.




**[3]**

**(b)** Find the coordinates of the point of intersection of the graphs of $y = f(x)$ and $y = g(x)$. Give your answers correct to 3 significant figures.

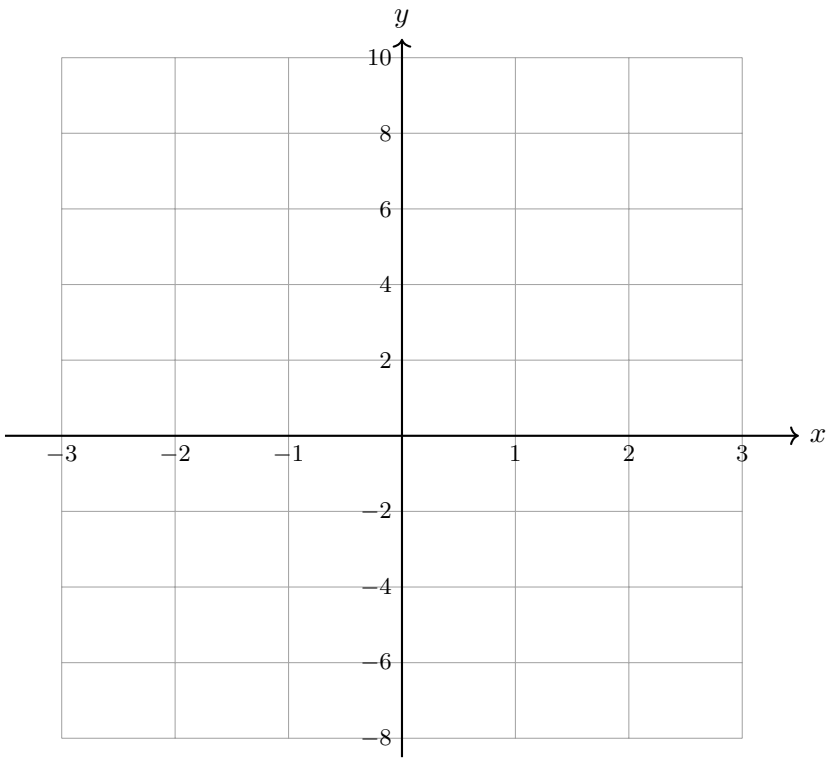
( ..... , ..... ) **[2]**

**(c)** Write down the equation of the horizontal asymptote of the graph of $y = f(x)$.

..... **[1]**

**674** Let $f(x) = x^3 - 3x + 1$.

**(a)** On the axes provided, sketch the graph of $y = f(x)$ for $-2.5 \leq x \leq 2.5$.




**[2]**

**(b)** Find the coordinates of the local maximum point.

( ..... , ..... ) **[2]**

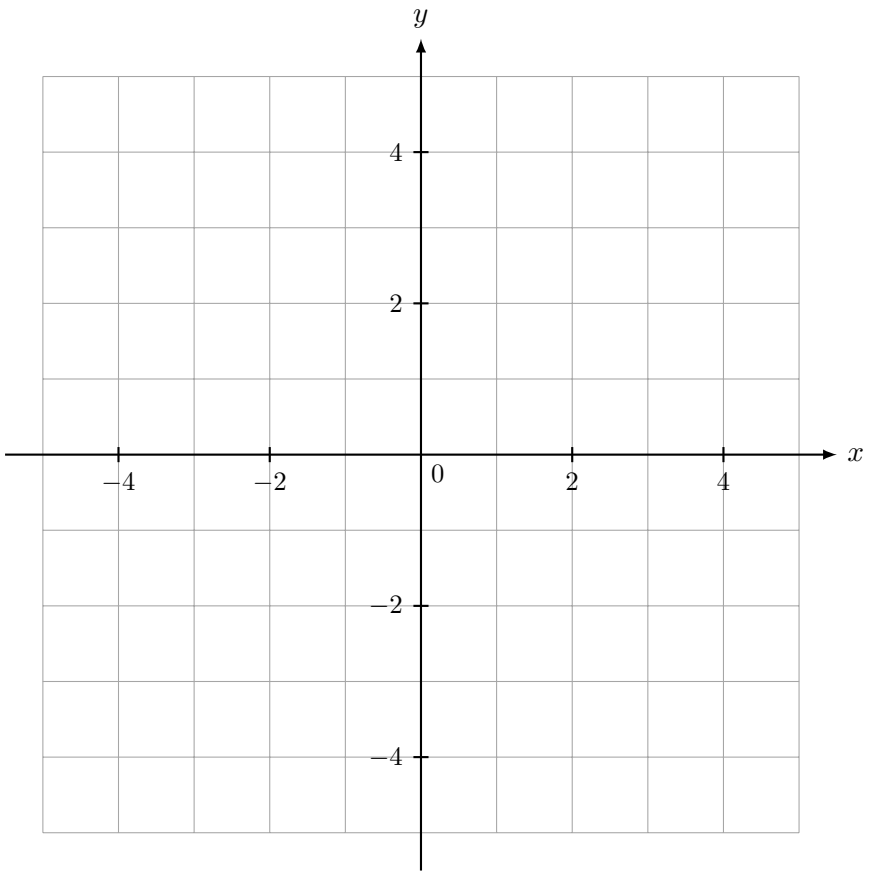
**(c)** On the same axes, sketch the graph of $y = 2x - 1$. **[1]**

**(d)** Find the $x$-coordinates of the points of intersection of the two graphs.

$x =$ ..... , $x =$ ..... , $x =$ ..... **[2]**

**675** Let $f(x) = \dfrac{2}{x+1}$ and $g(x) = x - 2$ for $-5 \leq x \leq 5$.

**(a)** On the grid, sketch the graphs of $y = f(x)$ and $y = g(x)$.




**[3]**

**(b)** Write down the equation of the vertical asymptote of the graph of $y = f(x)$.

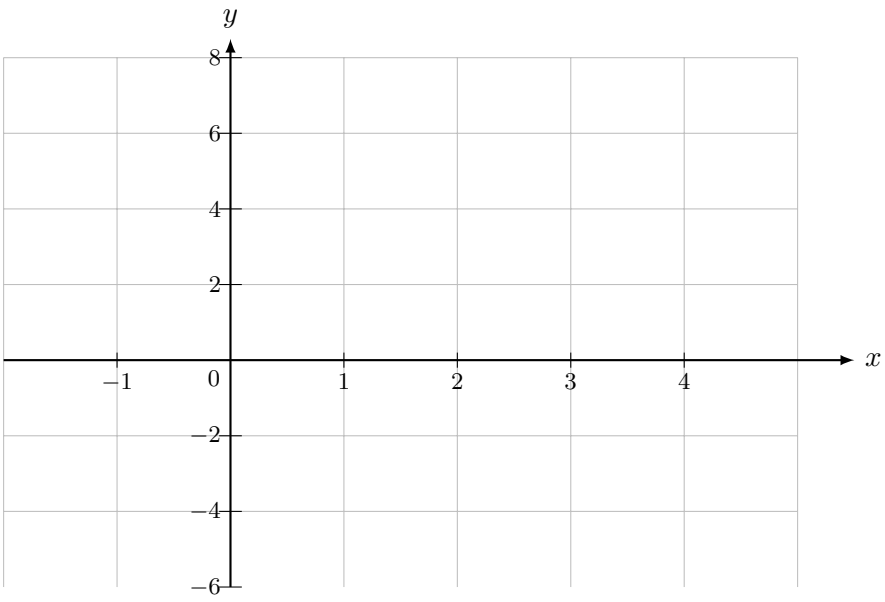
..... **[1]**

**(c)** Find the coordinates of the points of intersection of the two graphs.

Point 1 = ..... , Point 2 = ..... **[3]**

**676** Let $f(x) = x^3 - 4x^2 + x + 2$ for the domain $-1 \leq x \leq 4$.

**(a)** On the axes, sketch the graph of $y = f(x)$.




**[2]**

**(b)** Write down the $x$-coordinates of the points where $f(x) = 0$.

$x =$ ..... , $x =$ ..... , $x =$ ..... **[3]**

**(c)** Write down the coordinates of the local minimum.

( ..... , ..... ) **[2]**

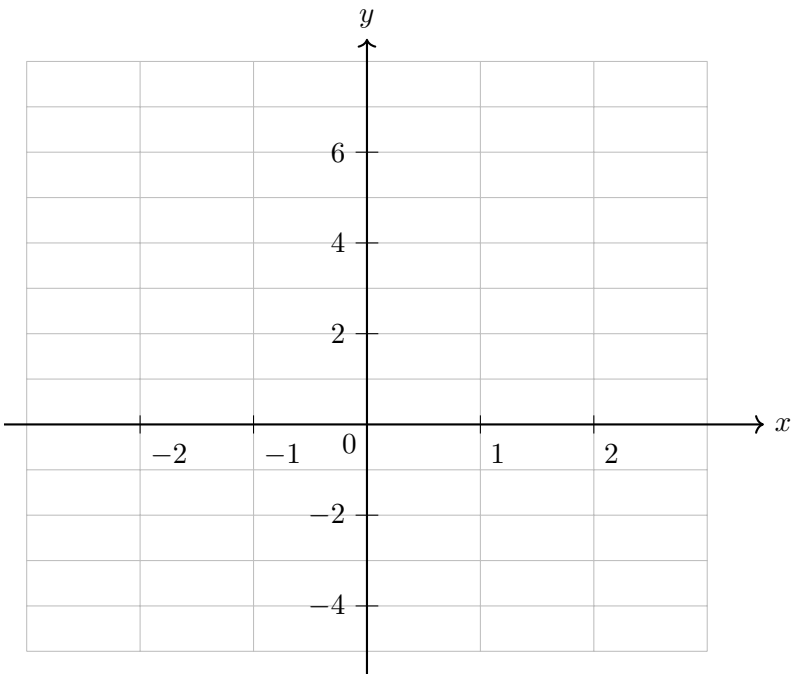
**(d)** On the same axes, sketch the graph of $g(x) = -2x + 1$. **[1]**

**(e)** Find the $x$-coordinates of the points of intersection of the graphs of $f(x)$ and $g(x)$.

$x =$ ..... , $x =$ ..... , $x =$ ..... **[3]**

**677** Let $f(x) = x^3 - 3x + 2$.

**(a)** On the axes, sketch the graph of $y = f(x)$ for $-2.5 \leq x \leq 2.5$.




**[3]**

**(b)** Find the coordinates of the local maximum.

( ..... , ..... ) **[2]**

**(c)** Find the zeros of $f(x)$.

$x =$ ..... , $x =$ ..... **[2]**

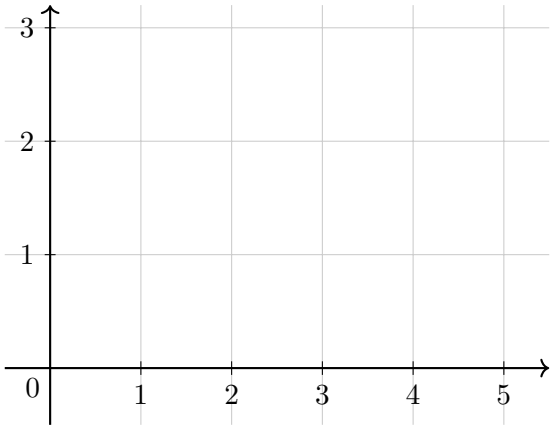
**(d)** On the same diagram, sketch the graph of $y = 2x + 1$. **[2]**

**(e)** Find the $x$-coordinates of the points of intersection of $y = f(x)$ and $y = 2x + 1$.

..... **[3]**

**678** Let $f(x) = 3x \cdot 2^{-x}$ and $g(x) = 0.5x + 0.2$ for $0 \leq x \leq 5$.

**(a)** On the axes provided, sketch the graph of $y = f(x)$ and the graph of $y = g(x)$ for $0 \leq x \leq 5$.




**[3]**

**(b)** Find the coordinates of the local maximum of $y = f(x)$.
Give your answer correct to 2 decimal places.

( ..... , ..... ) **[2]**

**(c)** Solve the inequality $f(x) > g(x)$ for $0 \leq x \leq 5$.
Give your answer correct to 2 decimal places.

..... **[2]**

**679** Consider the function $h(x) = x^3 - 3x^2 - 2x + 5$ for $-2 \leq x \leq 4$.

**(a)** Use your graphic display calculator to find the zeros of $h(x)$.
Give your answers correct to 2 decimal places.

$x =$ ..... **[3]**

**(b)** Find the coordinates of the local maximum point and the local minimum point of the graph of $y = h(x)$.
Give your answers correct to 2 decimal places.

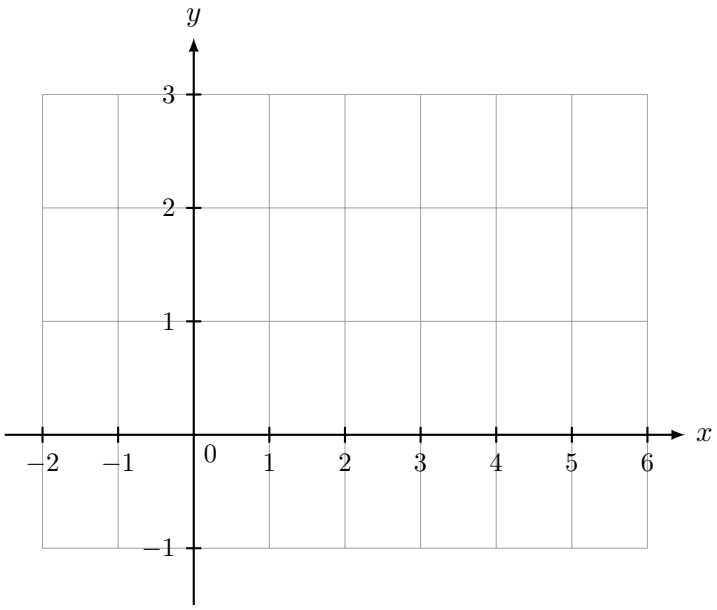
Local maximum = ..... , Local minimum = ..... **[2]**

**(c)** The equation $h(x) = k$ has three distinct real solutions.
Write down the range of possible values of $k$, giving your boundary values correct to 2 decimal places.

..... **[2]**

**680** The function $f(x)$ is defined as $f(x) = \dfrac{x^2}{2^x}$ for the domain $-1 \leq x \leq 5$.

**(a)** On the grid, sketch the graph of $y = f(x)$.




**[3]**

**(b)** Find the coordinates of the local maximum point.

( ..... , ..... ) **[2]**

**(c)** Find the $x$-coordinates of the points where $f(x) = 0.5$.

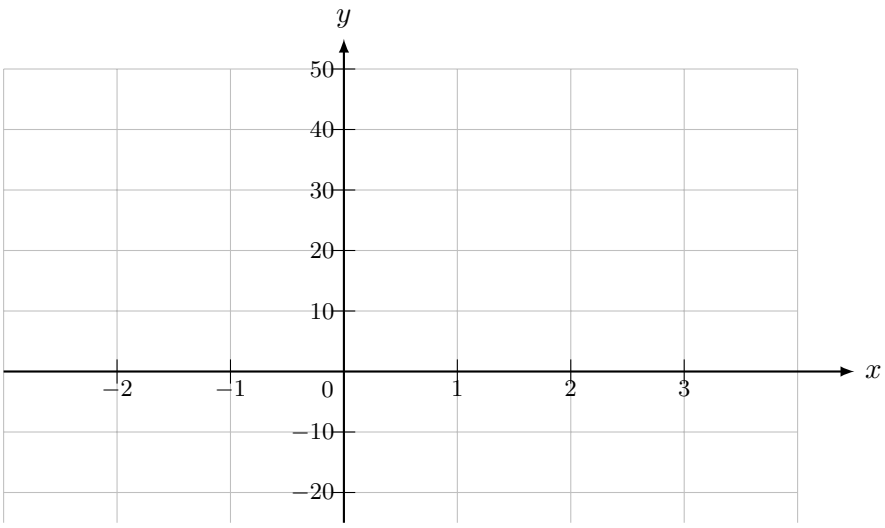
$x =$ ..... **[2]**

**(d)** Write down the range of values of $k$ for which the equation $f(x) = k$ has exactly 3 different solutions.

..... **[2]**

**681** The function $f(x)$ is defined as $f(x) = 3x^4 - 8x^3 - 6x^2 + 24x$ for the domain $-2 \leq x \leq 3$.

**(a)** On the axes provided, sketch the graph of $y = f(x)$.




**[3]**

**(b)** Write down the coordinates of the two local minima.

.....

..... **[2]**

**(c)** Write down the coordinates of the local maximum.

( ..... , ..... ) **[1]**

**(d)** The equation $f(x) = k$ has exactly four distinct solutions.
Write down the range of values for $k$.

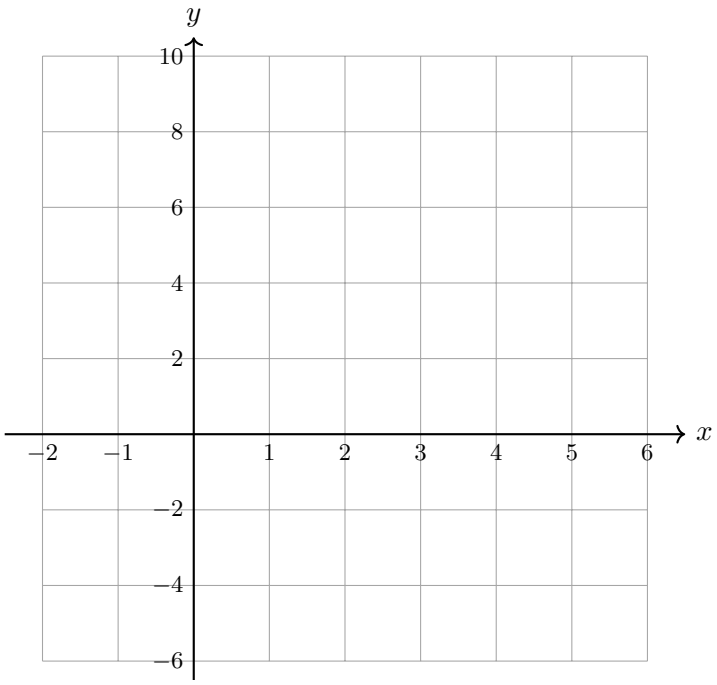
..... **[2]**

**(e)** The graph of $y = f(x)$ is translated by the vector $\begin{pmatrix} 0 \\ 5 \end{pmatrix}$.
Write down the coordinates of the local maximum of the translated graph.

( ..... , ..... ) **[1]**

**682** Let $f(x) = \dfrac{x^2 - 2x + 4}{x - 2}$.

**(a)** On the axes provided, sketch the graph of $y = f(x)$ for $-2 \leq x \leq 6, x \neq 2$.




**[3]**

**(b)** Write down the equation of the vertical asymptote.

..... **[1]**

**(c)** Find the coordinates of the local maximum point.

( ..... , ..... ) **[2]**

**(d)** Find the coordinates of the local minimum point.

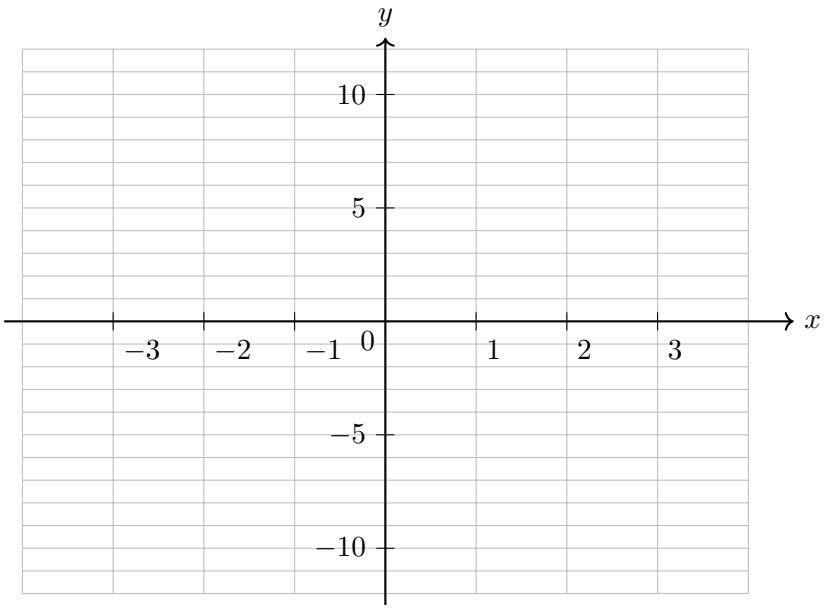
( ..... , ..... ) **[2]**

**(e)** The equation $f(x) = k$ has no real solutions.
Write down the range of values of $k$.

..... **[2]**

**683** Let $f(x) = \dfrac{x^3 + 2}{x}$.

**(a)** On the axes, sketch the graph of $y = f(x)$ for $-3 \leq x \leq 3$.




**[3]**

**(b)** Find the coordinates of the local minimum.

( ..... , ..... ) **[2]**

**(c)** Find the zero of $f(x)$.

$x =$ ..... **[1]**

**(d)** Write down the equation of the vertical asymptote.

..... **[1]**

**(e)** Solve the equation $f(x) = 4$.

..... **[3]**

**684** Functions $f$ and $g$ are defined as follows.

$f(x) = 2x - 5$, $g(x) = x^2 + 1$

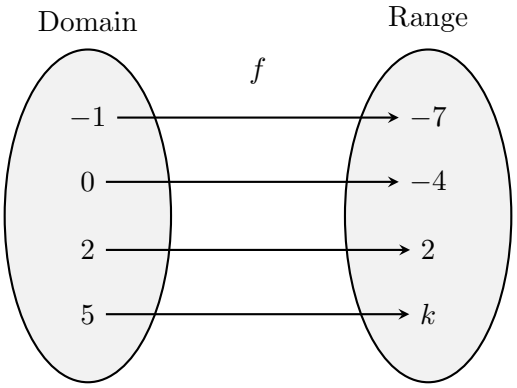
**(a)** Find the value of $f(g(3))$.

..... **[2]**

**(b)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

**685** The mapping diagram shows the function $f(x) = 3x - 4$ for the domain $\{-1, 0, 2, 5\}$.




**(a)** Work out the value of $k$.

$k =$ ..... **[1]**

**(b)** Write down the range of $f$.

..... **[1]**

**(c)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

**686** It is given that $f(x) = 3x - 2$ and $g(x) = \dfrac{10}{x}$.

**(a)** Find $f(4)$.

..... **[1]**

**(b)** Find $x$ when $g(x) = 2$.

$x =$ ..... **[1]**

**(c)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

**687** The functions $f$ and $g$ are defined as $f(x) = 3x - 2$ and $g(x) = x^2 + 4$.

**(a)** Work out

**(i)**

hspace*0.8cm $f(-4)$

..... **[1]**

**(ii)**

hspace*0.8cm $gf(2)$

..... **[2]**

**(b)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

**688** $f(x) = 4x - 5$
$g(x) = \dfrac{x}{3} + 2$

**(a)** Work out $f(-2)$.

..... **[1]**

**(b)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

**(c)** Find $fg(x)$ in its simplest form.

$fg(x) =$ ..... **[2]**

**689** The functions $f$ and $g$ are defined as $f(x) = 5x - 2$, $g(x) = x^2 + 3$.

**(a)** Find the value of $f(4)$.

..... **[1]**

**(b)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

**(c)** Find $fg(2)$.

..... **[2]**

**690** The functions $f$ and $g$ are defined as $f(x) = 4x - 5$, $g(x) = \dfrac{12}{x}$.

**(a)** Work out the value of $f(3)$.

..... **[1]**

**(b)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

**(c)** Work out $fg(2)$.

..... **[2]**

**691** Consider the functions $f(x) = 3x - 2$ and $g(x) = x^2 + 4$.

**(a)** Find $f(4)$.

..... **[1]**

**(b)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

**(c)** Find $gf(2)$.

..... **[2]**

**692** Consider the functions $f(x) = 4x - 7$ and $g(x) = 2x^2 + 1$.

**(a)** Find $f(-3)$.

..... **[1]**

**(b)** Find $gf(2)$.

..... **[2]**

**(c)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

**693** Let $f(x) = 5 - 2x$, $g(x) = x^2 - 3$.

**(a)** Work out $f(-4)$.

..... **[1]**

**(b)** Work out $gf(2)$.

..... **[2]**

**(c)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

**694** $f(x) = 4x - 3$, $g(x) = 2 - 5x$

**(a)** Work out $f(2)$.

..... **[1]**

**(b)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

**(c)** Work out $gf(1)$.

..... **[2]**

**695** Let $f(x) = 4x - 3$ and $g(x) = 2x^2 + 1$.

**(a)** Work out $fg(3)$.

..... **[2]**

**(b)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

**(c)** Solve the equation

$$f(x) = 17$$

$x =$ ..... **[2]**

**696** Two functions, $f$ and $g$, are defined as follows:

$$f(x) = 3x - 2 \qquad g(x) = x^2 + 1$$

**(a)** Work out $f(4)$.

..... **[1]**

**(b)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

**(c)** Find $gf(x)$ in its simplest form.

$gf(x) =$ ..... **[3]**

**697** Consider the functions $f(x) = \dfrac{5}{x-3}$ for $xe3$, and $g(x) = 2x + 1$.

**(a)** Find $fg(x)$, giving your answer as a single fraction in its simplest form.

$fg(x) =$ ..... **[2]**

**(b)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

**(c)** Solve the equation $gf(x) = 6$.

$x =$ ..... **[2]**

**698** The functions $f$ and $g$ are defined as $f(x) = \dfrac{3x+2}{x-1}$, $x \neq 1$, $g(x) = x - 2$.

**(a)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[3]**

**(b)** Solve the equation.

$f(x) = g(x)$

$x =$ ..... , $x =$ ..... **[4]**

**699** Consider the functions $h(x) = \dfrac{12}{x-1}$ and $k(x) = 2x + 3$.

**(a)** State the value of $x$ which must be excluded from the domain of $h(x)$.

$x =$ ..... **[1]**

**(b)** Find $k^{-1}(x)$.

$k^{-1}(x) =$ ..... **[2]**

**(c)** Find $hk(x)$, giving your answer in its simplest form.

$hk(x) =$ ..... **[2]**

**(d)** Solve the equation.

$h(x) = k(2)$

$x =$ ..... **[3]**

**700** Let $f(x) = \dfrac{4}{2x-3}$ for $x \neq 1.5$, and $g(x) = x + 5$.

**(a)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[3]**

**(b)** Solve the equation $f(x) = g(-3)$.

$x =$ ..... **[3]**

**(c)** Express $gf(x)$ as a single fraction in its simplest form.

$gf(x) =$ ..... **[3]**

**701** The functions $f$, $g$ and $h$ are defined as $f(x) = x^2 - 2$, $g(x) = \dfrac{x-3}{2}$, $h(x) = 5 - 3x$.

**(a)** Find $h^{-1}(x)$.

$h^{-1}(x) =$ ..... **[2]**

**(b)** Solve the equation

$f(x) = 34$

$x =$ ..... , $x =$ ..... **[2]**

**(c)** Find $hg(x)$ in its simplest form.

..... **[3]**

**(d)** Work out $fh(2)$.

..... **[2]**

**702** The functions $f$ and $g$ are defined as $f(x) = \dfrac{x+3}{x-1}$, where $x \neq 1$, and $g(x) = 2x - 5$.

**(a)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[3]**

**(b)** Show that $ff(x) = x$.

**[3]**

**(c)** Solve the equation $g(x^2) = f(3)$.

$x =$ ..... **[4]**

**703** $f(x) = \dfrac{3x+2}{x-1}$, $g(x) = \dfrac{2}{x+1}$

**(a)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[3]**

**(b)** Find $fg(x)$, giving your answer as a single fraction in its simplest form.

..... **[3]**

**(c)** Solve the equation.

$f(x) = f^{-1}(x)$

Give your answers in the form $a \pm \sqrt{b}$, where $a$ and $b$ are integers.

$x =$ ..... **[4]**

**704** The functions $f$ and $g$ are defined as follows:

$f(x) = \dfrac{3x+5}{x-1}, x \neq 1$, and $g(x) = x^2 + 1$

**(a)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[4]**

**(b)** Solve the equation $f(x) = x$.

$x =$ ..... , $x =$ ..... **[4]**

**(c)** Find $gf(2)$.

$gf(2) =$ ..... **[2]**

**705** $f(x) = \dfrac{4}{x-1}$ for $xe1$, $g(x) = 2x + 3$, $h(x) = x^2 - 4$.

**(a)** Find an expression, in its simplest form, for

**(i)** $fg(x)$,

$fg(x) =$ ..... **[2]**

**(ii)** $fh(x) - 1$ as a single fraction.

..... **[3]**

**(b)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[3]**

**(c)** Solve the equation $gf(x) = 5$.

$x =$ ..... **[3]**

**706** The functions $h$ and $k$ are defined as $h(x) = \dfrac{2x+1}{3}$ and $k(x) = 5 - x$.

**(a)** Find $h^{-1}(x)$.

$h^{-1}(x) =$ ..... **[2]**

**(b)** Find an expression for $hk(x)$ in its simplest form.

$hk(x) =$ ..... **[2]**

**(c)** Solve the equation $h(x) = k(x)$.

$x =$ ..... **[3]**

**707** Functions $h$ and $k$ are defined as follows.

$h(x) = \dfrac{4}{x-3}$ for $x \neq 3$

$k(x) = 2x + 1$

**(a)** Find $k(h(5))$.

..... **[2]**

**(b)** Find $h^{-1}(x)$.

$h^{-1}(x) =$ ..... **[2]**

**(c)** Solve the equation $h(k(x)) = 2$.

$x =$ ..... **[3]**

**708** The functions $f$, $g$ and $h$ are defined as follows:
$f(x) = 2x + 3$, $g(x) = x^2 - 4$, $h(x) = 5^x$

**(a)** Find $h(-1)$ as a fraction in its simplest form.

..... **[1]**

**(b)** Find $gf(x)$ in its simplest form.

$gf(x) =$ ..... **[3]**

**(c)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

**(d)** Find the value of $x$ when $h(x) = 125$.

$x =$ ..... **[1]**

**709** Two functions, $f$ and $g$, are defined as follows:

$$f(x) = \frac{5}{x-2} \qquad g(x) = x + 2$$

**(a)** State the value of $x$ that must be excluded from the domain of $f(x)$.

$x =$ ..... **[1]**

**(b)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[3]**

**(c)** Solve the equation $f(x) = g(x)$.

$x =$ ..... , $x =$ ..... **[4]**

**710** $f(x) = 2x - 3$, $g(x) = x^2 - x - 7$

**(a)** Work out $g(-2)$.

..... **[2]**

**(b)** Find $gf(x)$ in its simplest form.

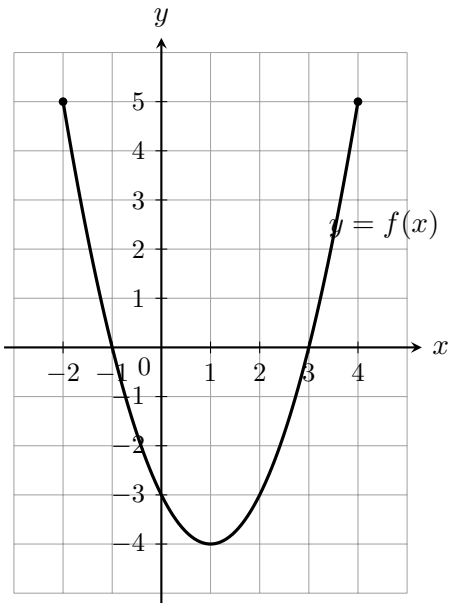
..... **[3]**

**(c)** Solve the equation.

$f(x) = g(x)$

$x =$ ..... , $x =$ ..... **[3]**

**711** The diagram shows the graph of $y = f(x)$ for $-2 \leq x \leq 4$, where $f(x) = x^2 - 2x - 3$.
The function $g$ is defined by $g(x) = 2x + 1$.




**(a)** Use the graph to write down the range of $f$ for $-2 \leq x \leq 4$.

..... **[2]**

**(b)** Find $fg(3)$.

$fg(3) =$ ..... **[2]**

**(c)** Find $gf(x)$, giving your answer in the form $ax^2 + bx + c$.

$gf(x) =$ ..... **[2]**

**(d)** Solve the equation $g^{-1}(x) = 6$.

$x =$ ..... **[2]**

**712** Let $f(x) = \dfrac{2x+1}{3}$ and $g(x) = \dfrac{5}{x-2}$, where $x \neq 2$.

**(a)** Find $gf(x)$.
Give your answer as a single fraction in its simplest form.

..... **[3]**

**(b)** Find $g^{-1}(x)$.

$g^{-1}(x) =$ ..... **[3]**

**(c)** Solve the equation

$$f(x) = g(3)$$

$x =$ ..... **[3]**

**713** $f(x) = 2x - 1$
$g(x) = x^2 - 4$
$h(x) = \dfrac{2x+1}{x-3}, \quad x \neq 3$

**(a)** Work out $g(-3)$.

..... **[1]**

**(b)** Find $h^{-1}(x)$.

$h^{-1}(x) =$ ..... **[3]**

**(c)** Find $gf(x)$ in the form $ax^2 + bx + c$.

$gf(x) =$ ..... **[3]**

**(d)** Solve the equation $f(x) = g(x)$.

$x =$ ..... , $x =$ ..... **[3]**

**714** Consider the functions $f(x) = \dfrac{2x+3}{x-1}$ for $xe1$, and $h(x) = x^2 - 4$ for $x \geq 0$.

Show that $ff(x) = x$ for all $xe1$.

**[3]**

**715** The functions $f$, $g$ and $h$ are defined as $f(x) = 4 - 3x$, $g(x) = \dfrac{5}{x-2}$, $x \neq 2$, $h(x) = x^2 - x - 7$.

**(a)** **(i)** Find the value of $f(0.5)$.

..... **[1]**

**(ii)** Find $gf(x)$ in its simplest form.

$gf(x) =$ ..... **[2]**

**(b)** Find the range of $h(x)$.

..... **[2]**

**(c)** Solve the equation.

$h(x) = 2x - 1$ Give your answers correct to 2 decimal places.

$x =$ ..... , $x =$ ..... **[4]**

**716** Functions $f$, $g$ and $h$ are defined as follows.

$f(x) = 3x - 2$

$g(x) = x^2 - 4x$

$h(x) = \dfrac{2x+1}{x-1}$ for $x \neq 1$

**(a)** Find $h^{-1}(x)$.

$h^{-1}(x) =$ ..... **[3]**

**(b)** Find an expression for $g(f(x))$ in the form $ax^2 + bx + c$.

..... **[3]**

**(c)** Solve the equation $f(g(x)) = x$.
Give your answers correct to 2 decimal places.
You must show all your working.

$x =$ ..... , x =

..... **[4]**

**717** Let $f(x) = 3x + 2$, $g(x) = x^2 - x$ and $h(x) = 5^x$.

**(a)** Find $fg(x)$.
Give your answer in the form $ax^2 + bx + c$.

..... **[2]**

**(b)** Find $gf(x)$.
Give your answer in the form $ax^2 + bx + c$.

..... **[3]**

**(c)** Solve the equation

$$gf(x) = 0$$

$x =$ ..... , $x =$ ..... **[3]**

**(d)** Find the value of $x$ when $h(x) = \dfrac{1}{125}$.

$x =$ ..... **[2]**

**718** $f(x) = \dfrac{3}{x+2}, \quad x \neq -2$
$g(x) = 5x - 4$

**(a)** Find $g^{-1}(x)$.

$g^{-1}(x) =$ ..... **[2]**

**(b)** Find $fg(x)$ as a single fraction in its simplest form.

$fg(x) =$ ..... **[2]**

**(c)** Find $ff(x)$ as a single fraction in its simplest form.

$ff(x) =$ ..... **[3]**

**(d)** Solve the equation $f(x) = g(x)$.
Show all your working.

$x =$ ..... , $x =$ ..... **[4]**

**719** Let $f(x) = \dfrac{2x+5}{x-3}$ for $x \neq 3$, and $g(x) = 3x - 1$.

**(a)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[3]**

**(b)** Express $fg(x)$ as a single fraction in its simplest form.

$fg(x) =$ ..... **[3]**

**(c)** Solve the equation $gf(x) = x$.
Show all your working and give your answers correct to 2 decimal places.

$x =$ ..... , $x =$ ..... **[5]**

**720** Consider the functions $f(x) = \dfrac{3}{x+2}$, $g(x) = (3x+5)^2$ and $h(x) = \dfrac{3x+5}{x-2}$.

**(a)** Find $h^{-1}(x)$.

$h^{-1}(x) =$ ..... **[3]**

**(b)** Find $fg(x)$.
Give your answer as a single fraction in its simplest form.

$fg(x) =$ ..... **[3]**

**(c)** Solve the equation.

$h(x) = x$

Show all your working and give your answers correct to 2 decimal places.

$x =$ ..... , or $x =$ ..... **[5]**

**721** The functions $f$ and $g$ are defined as:

$$f(x) = \frac{2x+3}{x-1} \qquad g(x) = x^2 - 4x$$

**(a)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[4]**

**(b)** Find $ff(x)$, giving your answer as a single fraction in its simplest form.

$ff(x) =$ ..... **[4]**

**(c)** Solve the equation $g(x) = f(2)$.
Show all your working and give your answers correct to 2 decimal places.

$x =$ ..... , $x =$ ..... **[4]**

**722** The functions $f$ and $g$ are defined as $f(x) = \dfrac{x+5}{x-2}$, $g(x) = x^2 + 3x$.

**(a)** State the value of $x$ for which $f(x)$ is undefined.

$x =$ ..... **[1]**

**(b)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[4]**

**(c)** Solve the equation

$g(x) = 10$

$x =$ ..... , $x =$ ..... **[3]**

**(d)** Solve the equation

$f(x) = x$

Show all your working and give your answers correct to 2 decimal places.

$x =$ ..... , $x =$ ..... **[4]**

**723** A quadratic function has $x$-intercepts at $x = -3$ and $x = 4$.
The coefficient of $x^2$ is 1.

Write down the equation of this function in the form $y = ax^2 + bx + c$.

$y =$ ..... **[2]**

**724** A quadratic function has $x$-intercepts at $x = 4$ and $x = -2$.
The coefficient of $x^2$ is 1.

Find the equation of the quadratic function in the form $y = x^2 + bx + c$.

$y =$ ..... **[2]**

**725** A quadratic function $f(x)$ has $x$-intercepts at $x = -4$ and $x = 3$.
The coefficient of $x^2$ is 1.

Find the equation of the function in the form $f(x) = x^2 + bx + c$.

$f(x) =$ ..... **[2]**

**726** A quadratic function has $x$-intercepts at $x = -4$ and $x = 7$.
The coefficient of $x^2$ is 1.
Write down the equation of the function.
Give your answer in the form $y = x^2 + bx + c$.

$y =$ ..... **[2]**

**727** A quadratic function has the equation $y = x^2 + bx + c$.
The vertex of the graph of this function is $(4, -3)$.

Find the value of $b$ and the value of $c$.

$b =$ ..... , $c =$ ..... **[3]**

**728** A quadratic function is given by

$$y = x^2 + bx + c$$

The vertex of the graph of this function is $(3, -4)$.

Find the value of $b$ and the value of $c$.

$b =$ ..... , $c =$ ..... **[3]**

**729** A quadratic function has a minimum point at $(3, -4)$ and passes through the point $(5, 0)$.
The coefficient of $x^2$ is 1.

Find the equation of the function in the form $y = x^2 + bx + c$.

$y =$ ..... **[3]**

**730** The graph of a quadratic function, $y = x^2 + bx + c$, crosses the $x$-axis at $(-4, 0)$ and $(2, 0)$.

**(a)** Work out the value of $b$ and the value of $c$.

$b =$ ..... , $c =$ ..... **[2]**

**(b)** Find the coordinates of the vertex of the graph.

( ..... , ..... ) **[2]**

**731** The graph of a quadratic function $f(x) = x^2 + bx + c$ crosses the $x$-axis at $(-3, 0)$ and $(5, 0)$.

**(a)** Find the values of $b$ and $c$.

$b =$ ..... , $c =$ ..... **[3]**

**(b)** Write down the coordinates of the $y$-intercept of the graph of $y = f(x)$.

( ..... , ..... ) **[1]**

**732** A quadratic function $y = f(x)$ has its vertex at $(3, -4)$.
The coefficient of $x^2$ is 1.

**(a)** Find the equation of the function in the form $y = x^2 + bx + c$.

$y =$ ..... **[2]**

**(b)** Find the coordinates of the points where the graph of $y = f(x)$ crosses the $x$-axis.

..... **[2]**

**733** A quadratic function is given by the equation $y = (x - h)^2 + k$.
The vertex of the graph of this function is $(3, -2)$.

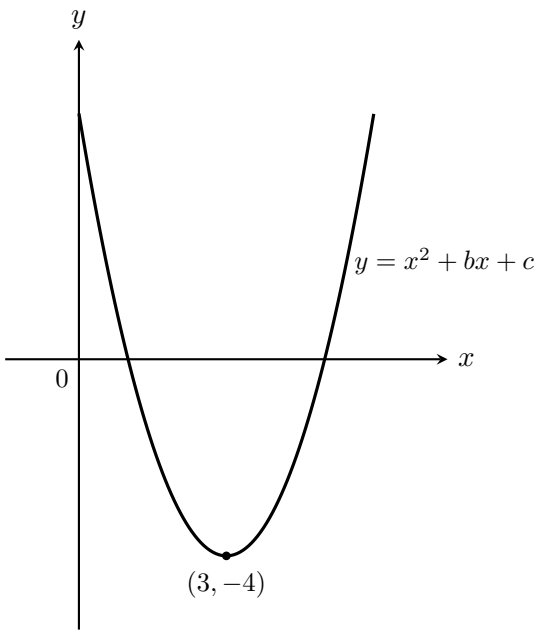
**(a)** Write down the value of $h$ and the value of $k$.

$h =$ ..... , $k =$ ..... **[2]**

**(b)** The graph passes through the point $(5, p)$.
Work out the value of $p$.

$p =$ ..... **[2]**

**734** The diagram shows a sketch of the quadratic curve $y = x^2 + bx + c$.
The vertex of the curve is at $(3, -4)$.




**(a)** Write down the equation of the curve in the form $y = (x - h)^2 + k$.

$y =$ ..... **[2]**

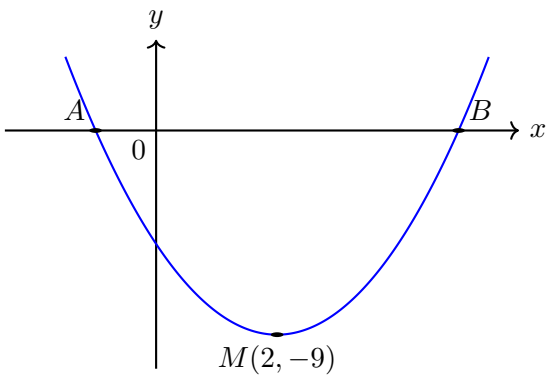
**(b)** Find the values of $b$ and $c$.

$b =$ ..... , $c =$ ..... **[2]**

**(c)** Write down the coordinates of the point where the curve crosses the $y$-axis.

( ..... , ..... ) **[1]**

**735** The diagram shows the graph of a quadratic function $y = x^2 + px + q$.
The graph has a minimum point (vertex) at $M(2, -9)$.




**(a)** Find the value of $p$ and the value of $q$.

$p =$ ..... , $q =$ ..... **[3]**

**(b)** The graph crosses the $x$-axis at the points $A$ and $B$.
Find the coordinates of $A$ and the coordinates of $B$.

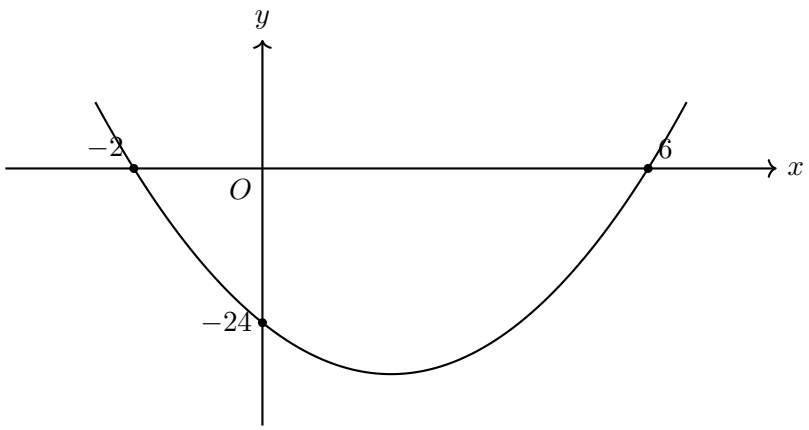
$A =$ ..... , $B =$ ..... **[3]**

**736** The graph of a quadratic function $y = f(x)$ intersects the $x$-axis at $(-2, 0)$ and $(5, 0)$.
The graph intersects the $y$-axis at $(0, -20)$.

Work out the function $f(x)$, giving your answer in the form $ax^2 + bx + c$.

$f(x) =$ ..... **[4]**

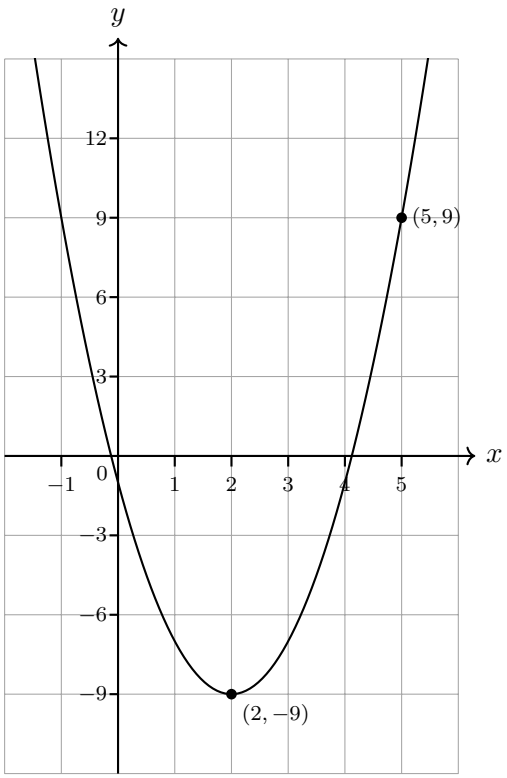
**737** The diagram shows a sketch of the quadratic graph $y = f(x)$.
The graph crosses the $x$-axis at $(-2, 0)$ and $(6, 0)$.
It crosses the $y$-axis at $(0, -24)$.




Find the expression for $f(x)$ in the form $ax^2 + bx + c$.

$f(x) =$ ..... **[4]**

**738** The diagram shows the graph of a quadratic function $g(x)$.
The vertex of the parabola is at $(2, -9)$ and the curve passes through the point $(5, 9)$.




**(a)** Find the equation of the quadratic function in the form $g(x) = a(x - h)^2 + k$.

$g(x) =$ ..... **[3]**

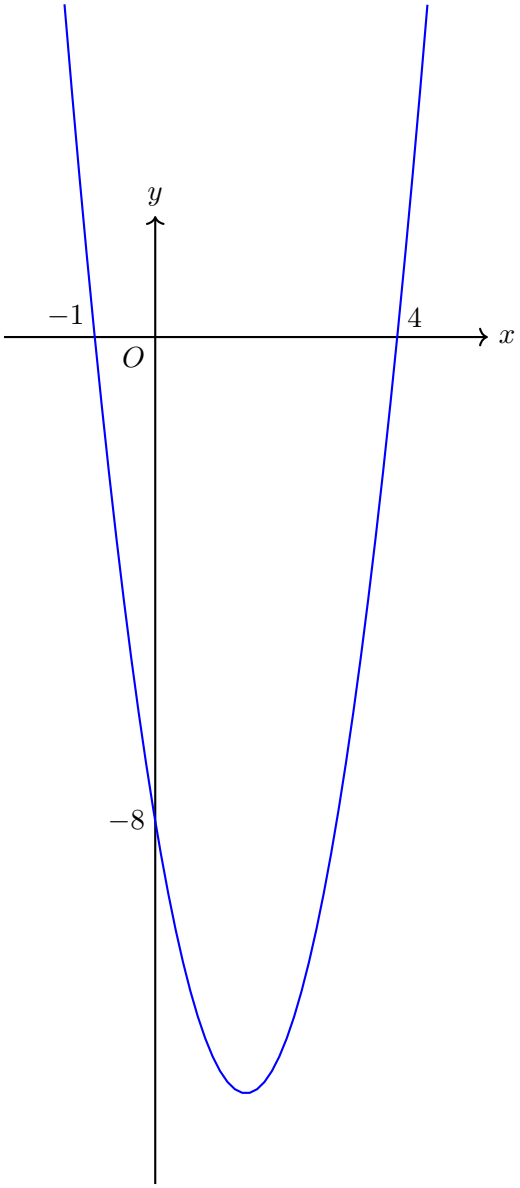
**(b)** Express $g(x)$ in the form $ax^2 + bx + c$.

$g(x) =$ ..... **[2]**

**(c)** Write down the equation of the line of symmetry of the graph of $y = g(x)$.

..... **[1]**

**739** The diagram shows a sketch of the quadratic function $y = f(x)$.
The graph crosses the $x$-axis at $(-1, 0)$ and $(4, 0)$.
The graph crosses the $y$-axis at $(0, -8)$.



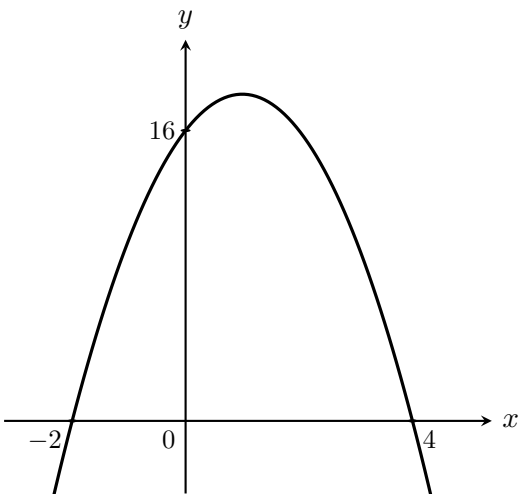

**(a)** Find the equation of the quadratic function in the form $y = ax^2 + bx + c$.

$y =$ ..... **[3]**

**(b)** Find the coordinates of the minimum point of the graph.

( ..... , ..... ) **[3]**

**740** The diagram shows the graph of a quadratic function $y = ax^2 + bx + c$.
The curve crosses the $x$-axis at $(-2, 0)$ and $(4, 0)$, and crosses the $y$-axis at $(0, 16)$.




**(a)** Find the equation of the curve in the form $y = ax^2 + bx + c$.

$y =$ ..... **[4]**

**(b)** Find the coordinates of the turning point of the curve.

( ..... , ..... ) **[2]**

**(c)** Write down the equation of the line of symmetry of the curve.

..... **[1]**

**741** A quadratic function is given by $f(x) = a(x - h)^2 + k$.
The graph of $y = f(x)$ has a maximum point at $(-1, 8)$ and passes through the point $(2, -10)$.

**(a)** Write down the value of $h$ and the value of $k$.

$h =$ ..... , $k =$ ..... **[2]**

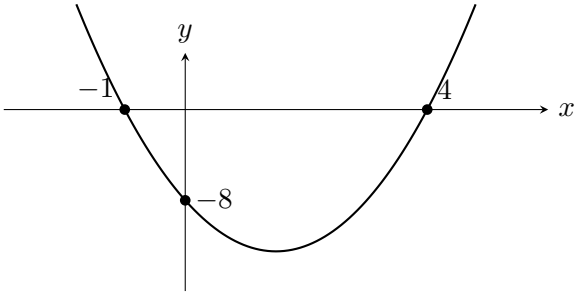
**(b)** Find the value of $a$.

$a =$ ..... **[3]**

**(c)** Write $f(x)$ in the form $ax^2 + bx + c$.

$f(x) =$ ..... **[3]**

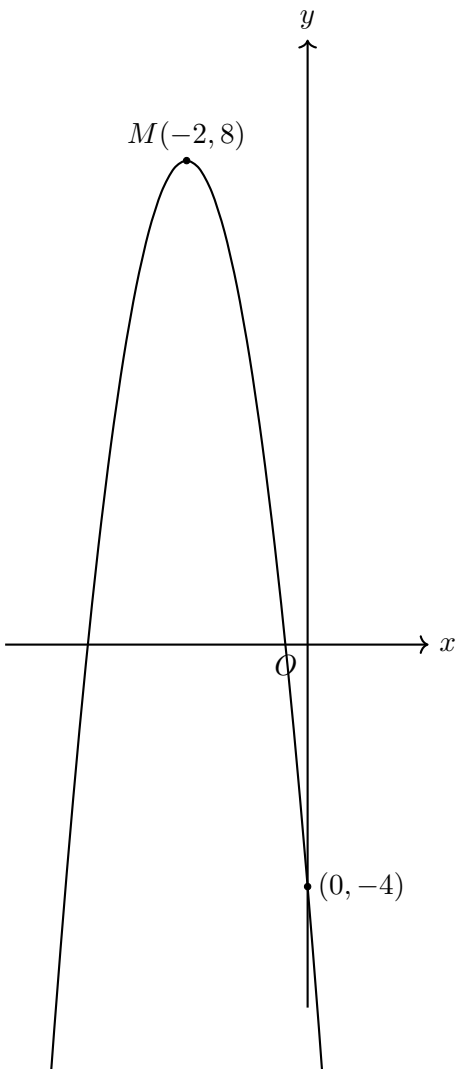
**742** The diagram shows the graph of a quadratic function $y = f(x)$.
The graph crosses the $x$-axis at $x = -1$ and $x = 4$.
The graph crosses the $y$-axis at $(0, -8)$.




Find the equation of the quadratic function in the form $f(x) = ax^2 + bx + c$.

$f(x) =$ ..... **[4]**

**743** The diagram shows a sketch of the curve $y = f(x)$ where $f(x)$ is a quadratic function.
The maximum point on the curve is $M(-2, 8)$ and the curve crosses the $y$-axis at $(0, -4)$.




**(a)** Find the equation of the curve.
Give your answer in the form $f(x) = ax^2 + bx + c$.

$f(x) =$ ..... **[4]**

**(b)** Solve $f(x) = 0$ to find the exact $x$-coordinates of the points where the curve crosses the $x$-axis.

Give your answers in the form $\frac{p \pm q\sqrt{6}}{r}$, where $p$, $q$ and $r$ are integers.

$x =$ ..... **[3]**

**744** A quadratic curve has its vertex at $(3, -5)$ and passes through the point $(1, 7)$.
Find the equation of the curve in the form $y = a(x - h)^2 + k$.

$y =$ ..... **[3]**

**745** The graph of a quadratic function has its vertex at $(-2, 5)$ and passes through the point $(1, -13)$. Find the equation of the function in the form $y = a(x - h)^2 + k$.

$y =$ ..... **[3]**

**746** The graph of a quadratic function has its vertex at $(2, 7)$ and passes through the point $(4, -5)$.

**(a)** Find the equation of the function in the form $y = a(x - h)^2 + k$.

$y =$ ..... **[3]**

**(b)** Find the $y$-intercept of the graph.

..... **[1]**

**747** A quadratic curve crosses the $x$-axis at $(-1, 0)$ and $(5, 0)$.
The curve also passes through the point $(2, -18)$.
Find the equation of the curve in the form $y = ax^2 + bx + c$.

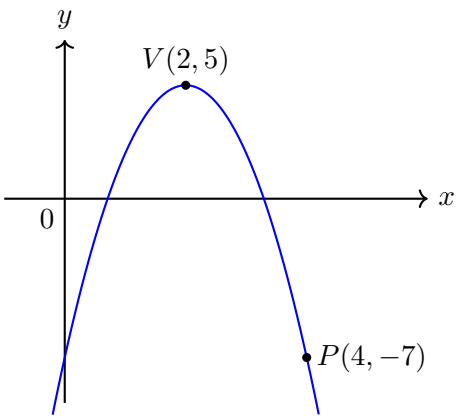
$y =$ ..... **[4]**

**748** A quadratic curve has a vertex at the point $(2, -5)$ and passes through the point $(4, 7)$.

Find the equation of the curve.
Give your answer in the form $y = ax^2 + bx + c$.

$y =$ ..... **[4]**

**749** The diagram shows a sketch of the graph of a quadratic function $y = a(x - h)^2 + k$.
The vertex of the graph is $V(2, 5)$.
The graph passes through the point $P(4, -7)$.




**(a)** Write down the value of $h$ and the value of $k$.

$h =$ ..... , $k =$ ..... **[2]**

**(b)** Work out the value of $a$.

$a =$ ..... **[2]**

**(c)** Find the coordinates of the $y$-intercept of the graph.

( ..... , ..... ) **[2]**

**750** The graph of a quadratic function $f(x)$ intersects the $x$-axis at $x = -2$ and $x = 5$.
The graph passes through the point $(3, 10)$.

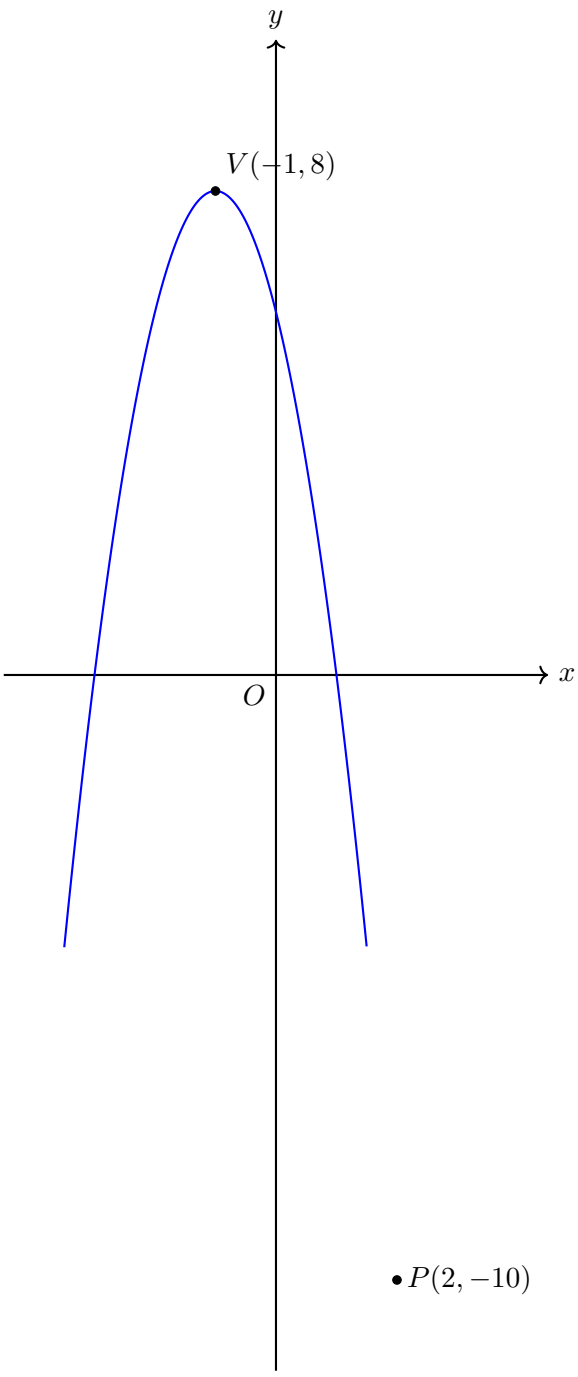
**(a)** Find an expression for $f(x)$ in the form $ax^2 + bx + c$.

$f(x) =$ ..... **[4]**

**(b)** Find the coordinates of the vertex of the graph of $y = f(x)$.

( ..... , ..... ) **[2]**

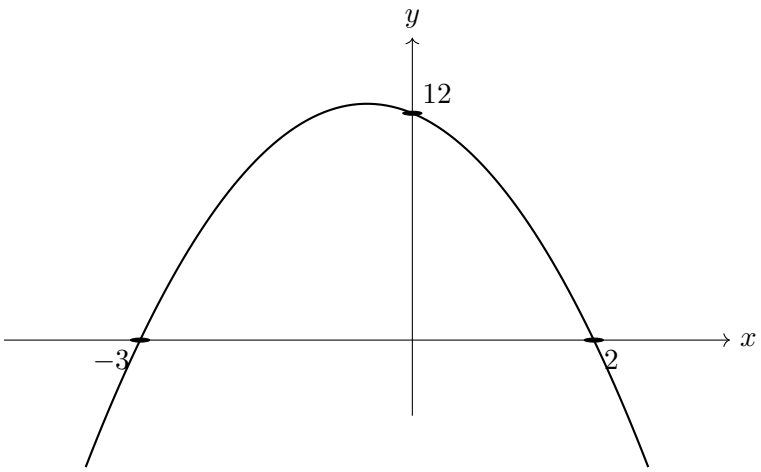
**751** The diagram shows a sketch of the graph of $y = ax^2 + bx + c$.
The vertex of the graph is $V(-1, 8)$ and it passes through the point $P(2, -10)$.




Work out the values of $a$, $b$ and $c$.

$a =$ ..... , $b =$ ..... , $c =$ ..... **[5]**

**752** The diagram shows the sketch of a quadratic function.
The curve crosses the $x$-axis at $-3$ and 2, and intersects the $y$-axis at 12.



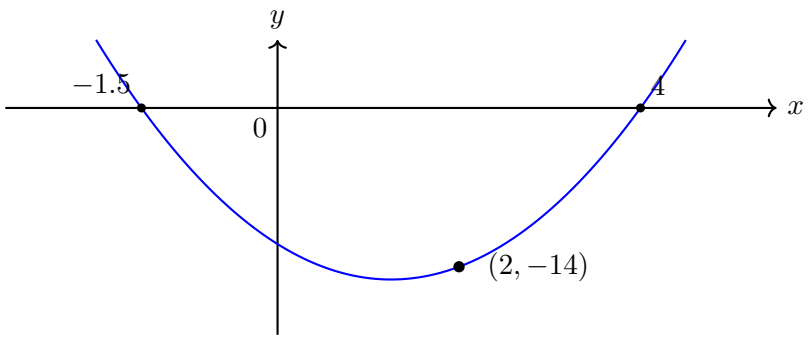

**(a)** Find the equation of the quadratic function.
Give your answer in the form $y = ax^2 + bx + c$.

$y =$ ..... **[4]**

**(b)** Calculate the coordinates of the vertex of this curve.

( ( ..... , ..... ) ) **[2]**

**753** The diagram shows a sketch of the quadratic function $y = f(x)$.
The curve crosses the $x$-axis at $x = -1.5$ and $x = 4$.
It passes through the point $(2, -14)$.




**(a)** Find the equation of the quadratic function in the form $f(x) = ax^2 + bx + c$.

$f(x) =$ ..... **[4]**

**(b)** Find the exact coordinates of the turning point of the graph.

( ..... , ..... ) **[3]**

**754** A quadratic function is given by $f(x) = a(x - h)^2 + k$.
The maximum point of the graph of $y = f(x)$ is $(-4, 20)$.
The graph intersects the $x$-axis at $x = -2$ and $x = p$.

**(a) (i)** Write down the value of $h$ and the value of $k$.

$h =$ ..... , $k =$ ..... **[2]**

**(ii)** Find the value of $a$.

$a =$ ..... **[2]**

**(iii)** Find the value of $p$.

$p =$ ..... **[1]**

**(iv)** Express $f(x)$ in the form $ax^2 + bx + c$.

$f(x) =$ ..... **[2]**

**755** A quadratic function $h(x) = ax^2 + bx + c$ has $x$-intercepts at $(-2, 0)$ and $(4, 0)$, and passes through the point $(1, 13.5)$.

**(a)** Find the values of $a$, $b$, and $c$.

$a =$ ..... , $b =$ ..... , $c =$ ..... **[4]**

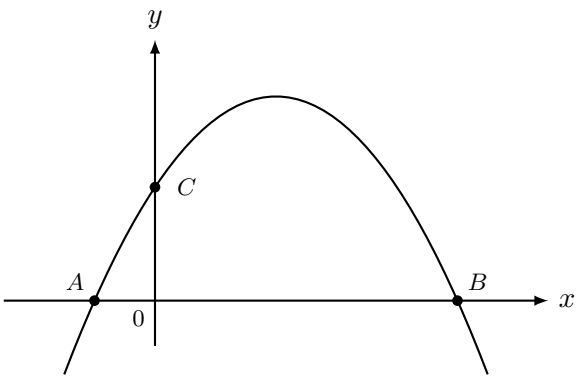
**(b)** Hence find the coordinates of the vertex of the graph of $y = h(x)$.

( ..... , ..... ) **[2]**

**(c)** State the range of $h(x)$.

..... **[1]**

**756** The graph of a quadratic function $y = f(x)$ intersects the $x$-axis at the points $A(-1, 0)$ and $B(5, 0)$. The graph intersects the $y$-axis at the point $C(0, 10)$.




**(a)** Find the equation of the function $f(x)$.
Give your answer in the form $f(x) = ax^2 + bx + c$.

$f(x) =$ ..... **[4]**

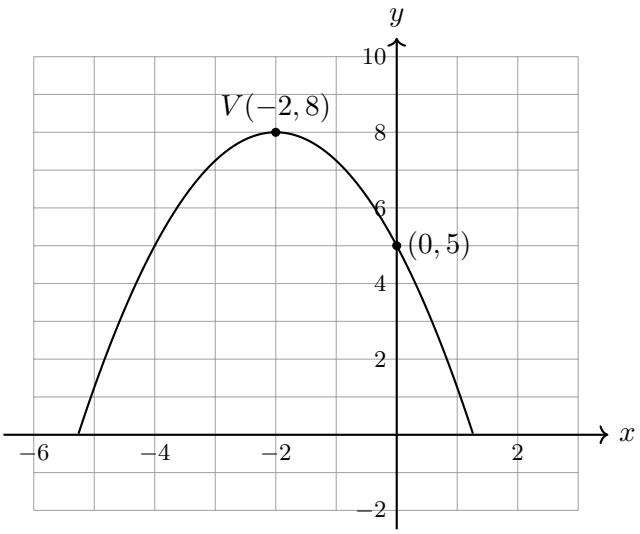
**(b)** Find the coordinates of the turning point (vertex) of the graph.

( ..... , ..... ) **[2]**

**(c)** The graph of $y = f(x)$ is translated by the vector $\begin{pmatrix} 2 \\ -3 \end{pmatrix}$ to give the graph of $y = g(x)$.
Find the equation of $g(x)$ in the form $g(x) = a(x - h)^2 + k$.

$g(x) =$ ..... **[2]**

**757** The diagram shows the graph of a quadratic function $y = f(x)$.
The vertex of the graph is at $V(-2, 8)$ and it intersects the $y$-axis at $(0, 5)$.




**(a)** Find the equation of the function $f(x)$ in the form $f(x) = a(x - h)^2 + k$.

$f(x) =$ ..... **[3]**

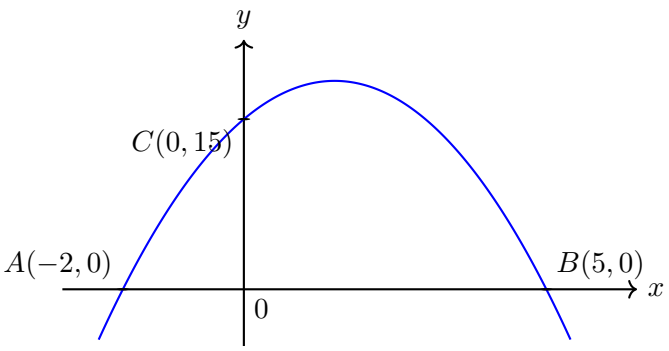
**(b)** Write $f(x)$ in the form $ax^2 + bx + c$.

$f(x) =$ ..... **[2]**

**(c)** Calculate the exact $x$-coordinates of the points where the graph intersects the $x$-axis.
Give your answers correct to 3 significant figures.

$x =$ ..... , $x =$ ..... **[3]**

**758** The diagram shows a sketch of the quadratic curve $y = ax^2 + bx + c$.
The curve crosses the $x$-axis at $A(-2, 0)$ and $B(5, 0)$, and crosses the $y$-axis at $C(0, 15)$.




**(a)** Find the values of $a$, $b$, and $c$.

$a =$ ..... , $b =$ ..... , $c =$ ..... **[5]**

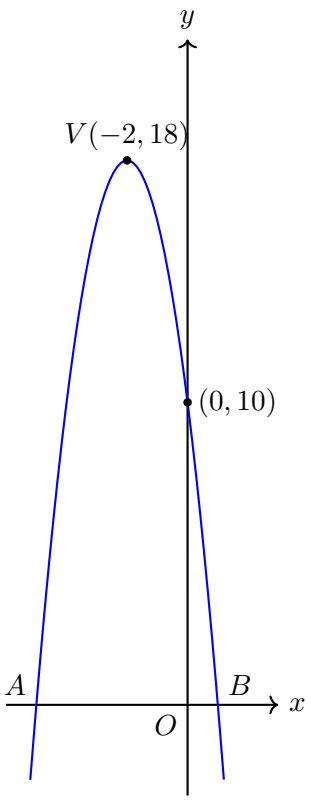
**(b)** Find the coordinates of the vertex of the curve.

( ..... , ..... ) **[3]**

**(c)** The equation $ax^2 + bx + c = k$ has no real roots.
Write down the inequality for $k$.

..... **[1]**

**759** The diagram shows a sketch of the graph of a quadratic function $y = f(x)$.
The maximum point of the graph is $V(-2, 18)$.
The graph crosses the $y$-axis at $(0, 10)$.




**(a)** Find the equation of the quadratic function in the form $y = ax^2 + bx + c$.

$y =$ ..... **[3]**

**(b)** The graph crosses the $x$-axis at $A$ and $B$.
Find the length of the line segment $AB$.

$AB =$ ..... **[3]**

**(c)** The line $y = -2x + k$ is a tangent to the curve $y = f(x)$.

Find the value of $k$.

$k =$ ..... **[4]**

**760** A quadratic function $f(x) = a(x-h)^2 + k$ has its vertex at $(4, -5)$ and passes through the point $(1, 8.5)$.

**(a)** Find the values of $a$, $h$, and $k$.

$a =$ ..... , $h =$ ..... , $k =$ ..... **[3]**

**(b)** Solve the equation $f(x) = 0$, giving your answers correct to 2 decimal places.

$x =$ ..... , $x =$ ..... **[3]**

**(c)** The line $y = 2x - 7$ intersects the curve $y = f(x)$ at two points, $P$ and $Q$.
Find the $x$-coordinates of $P$ and $Q$, giving your answers correct to 2 decimal places.

$x =$ ..... , $x =$ ..... **[3]**

**(d)** The graph of $y = f(x)$ is translated by the vector $\begin{pmatrix} -3 \\ 9 \end{pmatrix}$ to give the graph of $y = g(x)$.
Write down the expression for $g(x)$ in the form $p(x-q)^2 + r$.

$g(x) =$ ..... **[2]**

**761** The quadratic function $f(x) = ax^2 + bx + c$ has a minimum turning point at $(1, -8)$.
The graph of $y = f(x)$ crosses the $x$-axis at $x = -1$ and $x = 3$.

**(a)** Show that $f(x) = 2x^2 - 4x - 6$.

**[3]**

**(b)** Solve the equation.

$f(x) = 10$

$x =$ ..... , $x =$ ..... **[3]**

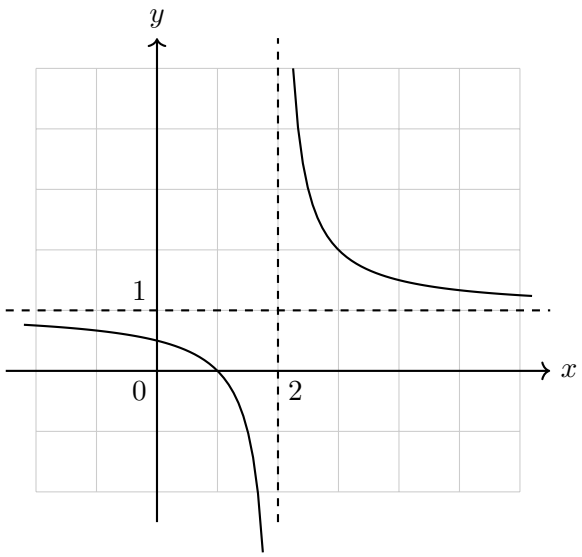
**(c)** A straight line has equation $y = mx - 8$.
The line is a tangent to the graph of $y = f(x)$.
Find the possible values of $m$.

$m =$ ..... , $m =$ ..... **[4]**

**(d)** The graph of $y = f(x)$ is translated by the vector $\begin{pmatrix} 0 \\ k \end{pmatrix}$.
The new graph has no $x$-intercepts.
Write down the inequality for $k$.

..... **[2]**

**762** The diagram shows the sketch of the graph of a function $y = f(x)$.
The dashed lines are the asymptotes to the curve.




**(a)** Write down the equation of the vertical asymptote.

..... **[1]**

**(b)** Write down the equation of the horizontal asymptote.

..... **[1]**

**763** Let $f(x) = \dfrac{7}{x-4} + 3$.

**(a)** Write down the equation of the vertical asymptote of the graph of $y = f(x)$.

..... **[1]**

**(b)** Write down the equation of the horizontal asymptote of the graph of $y = f(x)$.

..... **[1]**

**764** The function $f$ is given by $f(x) = \dfrac{5}{x+3} - 2$.

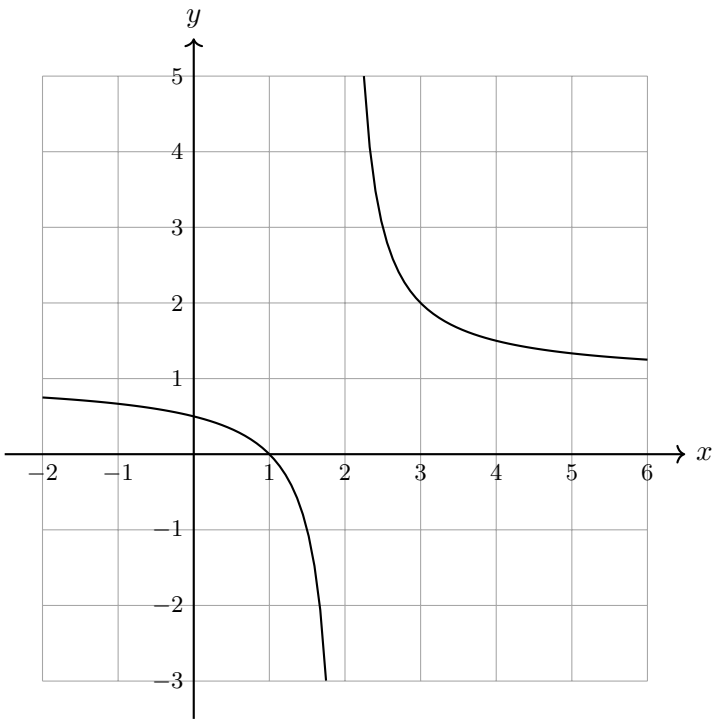
**(a)** Write down the equation of the vertical asymptote of the graph of $y = f(x)$.

$x =$ ..... **[1]**

**(b)** Write down the equation of the horizontal asymptote of the graph of $y = f(x)$.

$y =$ ..... **[1]**

**765** The diagram shows the graph of a function $y = f(x)$.
The graph has two asymptotes, each parallel to one of the coordinate axes.
The equations of the asymptotes involve only integer values.




**(a)** Write down the equation of the vertical asymptote of the graph of $y = f(x)$.

..... **[1]**

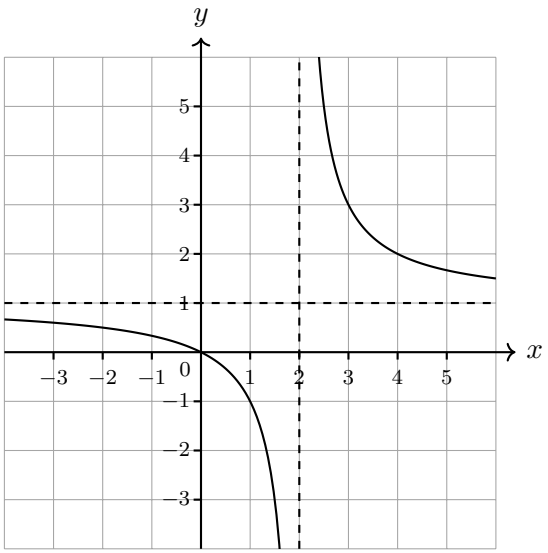
**(b)** Write down the equation of the horizontal asymptote of the graph of $y = f(x)$.

..... **[1]**

**(c)** The graph of $y = f(x)$ is reflected in the $y$-axis to form the graph of $y = g(x)$.
Write down the equation of the vertical asymptote of the graph of $y = g(x)$.

..... **[1]**

**766** The diagram shows the graph of a rational function $y = f(x)$.




**(a)** Write down the equation of the vertical asymptote of the graph of $y = f(x)$.

..... **[1]**

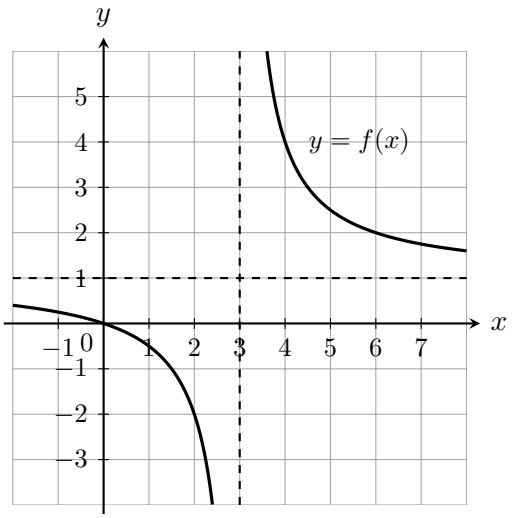
**(b)** Write down the equation of the horizontal asymptote of the graph of $y = f(x)$.

..... **[1]**

**(c)** Use the graph to find the value of $f(0)$ and the solution to the equation $f(x) = 0$.

$f(0) =$ ..... , $x =$ ..... **[2]**

**767** The diagram shows the graph of $y = f(x)$ and its two asymptotes, which are shown as dashed lines.




**(a)** Write down the equation of the vertical asymptote of the graph.

..... **[1]**

**(b)** Write down the equation of the horizontal asymptote of the graph.

..... **[1]**

**(c)** Use the graph to write down

**(i)** the value of $f(4)$,

$f(4) =$ ..... **[1]**

**(ii)** the value of $x$ for which $f(x) = -2$.

$x =$ ..... **[1]**

**768** A function is defined as $f(x) = \dfrac{4}{x-1} + 3$.

**(a)** Write down the equation of the

**(i)** vertical asymptote,

..... **[1]**

**(ii)** horizontal asymptote.

..... **[1]**

**(b)** Find the coordinates of the point where the graph of $y = f(x)$ crosses the $x$-axis.

( ( ..... , ..... ) ) **[2]**

**(c)** Find the coordinates of the point where the graph of $y = f(x)$ crosses the $y$-axis.

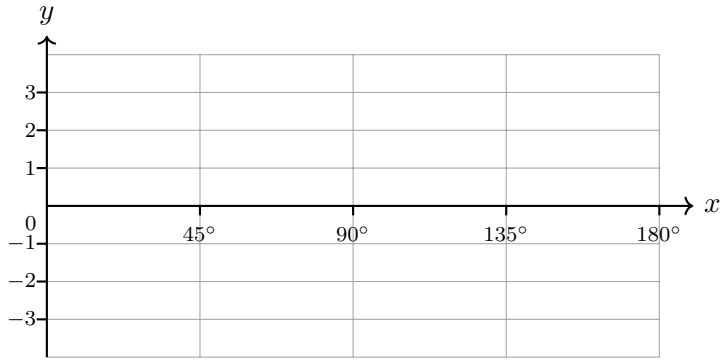
( ( ..... , ..... ) ) **[1]**

**769** Consider the trigonometric function $g(x) = \tan(2x)$ for $0° \leq x \leq 180°$.

**(a)** Write down the equations of the two vertical asymptotes of the graph of $y = g(x)$ for $0° \leq x \leq 180°$.

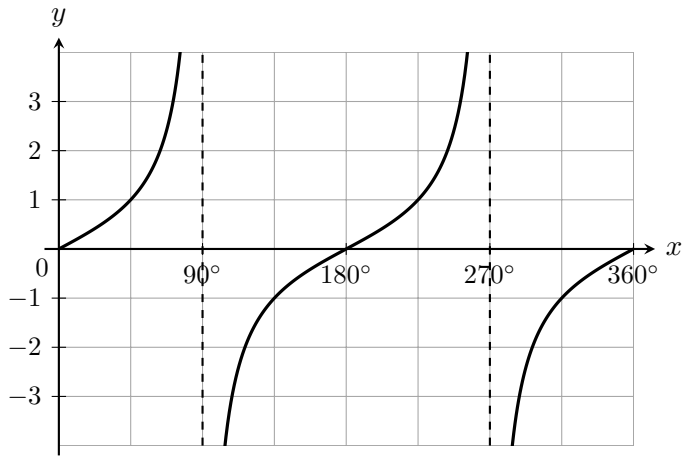
Asymptote 1: ..... , Asymptote 2: ..... **[2]**

**(b)** On the grid, sketch the graph of $y = \tan(2x)$ for $0° \leq x \leq 180°$, drawing the asymptotes as dashed lines.




**[3]**

**770** The diagram shows the graph of $f(x) = \tan x$ for $0° \leq x \leq 360°$.




**(a)** Write down the equations of the two asymptotes of the graph of $f(x) = \tan x$ for $0° \leq x \leq 360°$.

$x =$ ..... , $x =$ ..... **[2]**

**(b)** Explain how the graph shows that $\tan 90°$ is undefined.

..... **[1]**

**(c)** The graph of $y = \tan x$ is translated by the vector $\begin{pmatrix} 35° \\ 0 \end{pmatrix}$.
Write down the equations of the two asymptotes of the translated graph for $0° \leq x \leq 360°$.

$x =$ ..... , $x =$ ..... **[2]**

**(d)** Given that $\tan 45° = 1$, use the graph to write down the other solution to $\tan x = 1$ for $0° \leq x \leq 360°$.

$x =$ ..... ° **[1]**

**771** Consider the functions $g(x) = \tan x$ and $h(x) = \dfrac{4x-1}{2x+5}$.

**(a)** Write down the equations of the two asymptotes of the graph of $y = g(x)$ for $0° \leq x \leq 360°$.

$x =$ ..... , $x =$ ..... **[2]**

**(b)** Write down the equations of the vertical and horizontal asymptotes of the graph of $y = h(x)$.

Vertical asymptote $x =$ ..... , Horizontal asymptote $y =$ ..... **[2]**

**(c)** Solve the equation

$h(x) = 0$

$x =$ ..... **[2]**

**772** A function is defined as $f(x) = \dfrac{3}{x+2} + 1$.

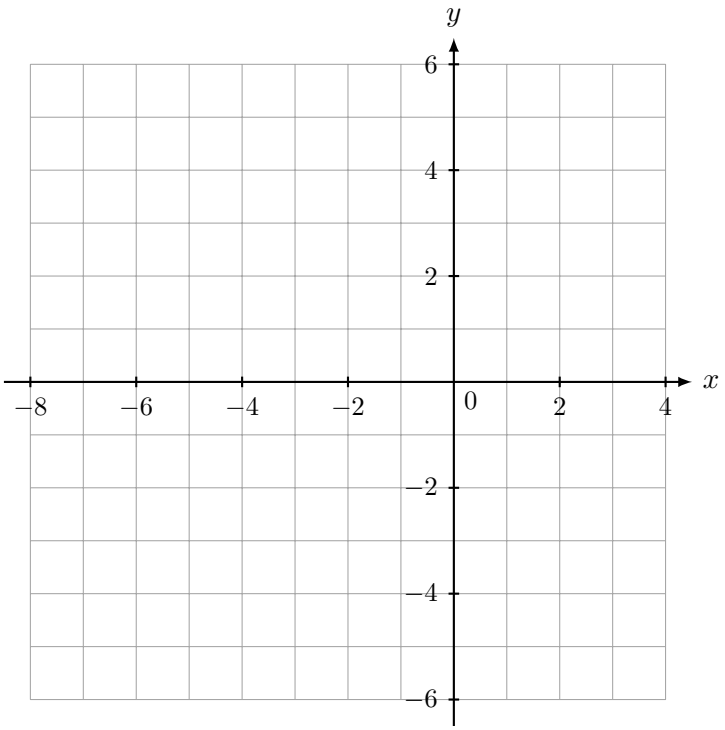
**(a)** Write down the equations of the two asymptotes of the graph of $y = f(x)$.

Vertical asymptote: ..... , Horizontal asymptote: ..... **[2]**

**(b)** Find the coordinates of the points where the graph of $y = f(x)$ crosses the $x$-axis and the $y$-axis.

$x$-axis at ..... , $y$-axis at ..... **[2]**

**(c)** On the grid, sketch the graph of $y = f(x)$.




**[3]**

**773** Let $f(x) = \dfrac{3}{x+2} - 4$.

**(a)** Write down the equation of the vertical asymptote of the graph of $y = f(x)$.

..... **[1]**

**(b)** Write down the equation of the horizontal asymptote of the graph of $y = f(x)$.

..... **[1]**

**(c)** Find the coordinates of the points where the graph of $y = f(x)$ intersects the coordinate axes.

$x$-intercept = ..... , $y$-intercept = ..... **[5]**

**(d)** The graph of $y = \tan x$ has an infinite number of vertical asymptotes.
Write down the equations of the two asymptotes in the interval $0° \leq x \leq 360°$.

$x =$ ..... , $x =$ ..... **[2]**

**774** A function is given by $f(x) = \dfrac{a}{x-b} + c$, where $a$, $b$ and $c$ are constants.
The graph of $y = f(x)$ has a vertical asymptote at $x = 2$ and a horizontal asymptote at $y = -1$.
The graph intersects the $x$-axis at the point $(5, 0)$.

**(a)** Write down the value of $b$ and the value of $c$.

$b =$ ..... , $c =$ ..... **[2]**

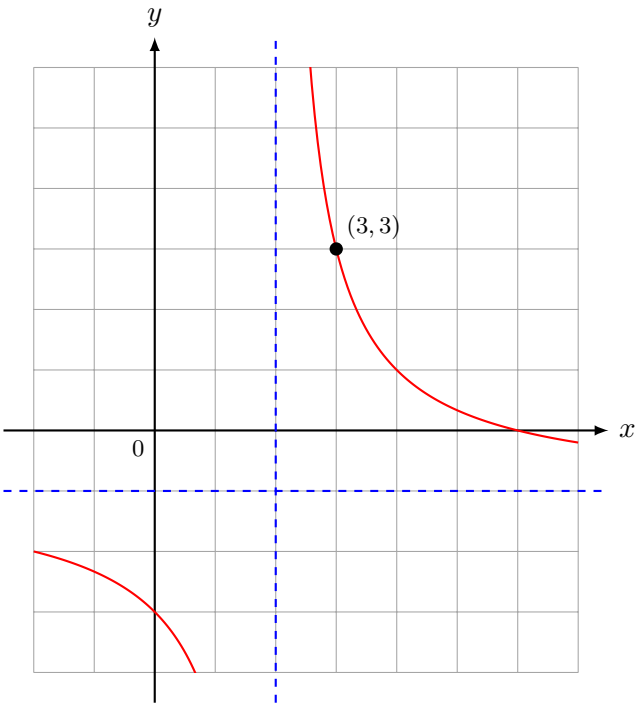
**(b)** Work out the value of $a$.

$a =$ ..... **[2]**

**(c)** The graph of $y = f(x)$ intersects the straight line $y = -x + 5$ at two points.
Find the $x$-coordinates of these two points.

$x =$ ..... , $x =$ ..... **[2]**

**775** The diagram shows the graph of $y = f(x)$ where $f(x) = \dfrac{p}{x - q} + r$.
The graph has asymptotes at $x = 2$ and $y = -1$.
The graph passes through the point $(3, 3)$.




**(a)** Find the values of $q$ and $r$.

$q =$ ..... , $r =$ ..... **[2]**

**(b)** Find the value of $p$.

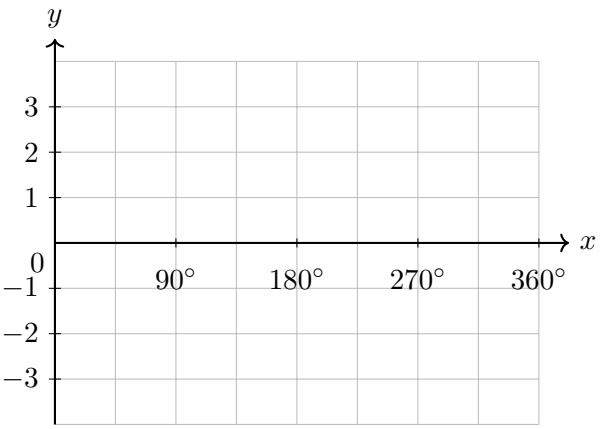
$p =$ ..... **[2]**

**(c)** Find the coordinates of the point where the graph of $y = f(x)$ crosses the $x$-axis.

( ..... , ..... ) **[3]**

**776** The function $g(x) = \tan x$ is defined for $0° \leq x \leq 360°$.

**(a)** On the grid, sketch the graph of $y = g(x)$.




**[2]**

**(b)** Write down the equations of the asymptotes of the graph of $y = g(x)$ in this domain.

**[2]**

**(c)** Solve the equation $2\tan x + 3 = 0$ for $0° \leq x \leq 360°$.
Give your answers correct to 1 decimal place.

$x =$ ..... , $x =$ ..... **[3]**

**777** Consider the function $h(x) = \dfrac{2x^2 - 5x - 3}{x - 2}$ for $-4 \leq x \leq 6$, $xe2$.

**(a)** Use your graphic display calculator to find the zeros of $h(x)$.

$x =$ ..... , $x =$ ..... **[2]**

**(b)** Write down the equation of the vertical asymptote of the graph of $y = h(x)$.

..... **[1]**

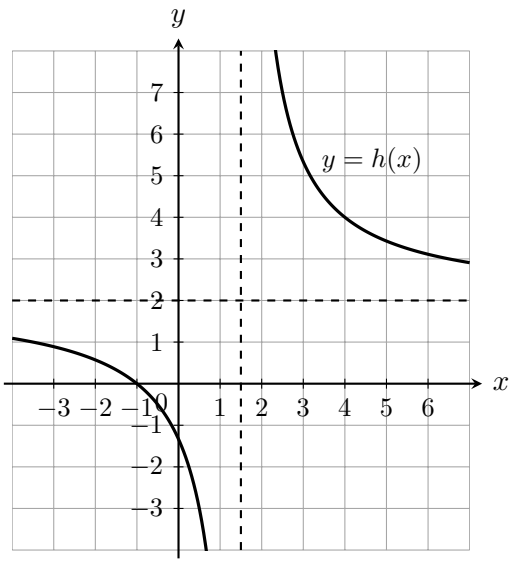
**(c)** Explain why the graph of $y = h(x)$ does not have a horizontal asymptote.

..... **[2]**

**(d)** The equation $h(x) = k$ has no real solutions.
Find the range of possible values of $k$, giving your boundary values correct to 2 decimal places.

..... **[2]**

**778** The diagram shows the graph of $y = h(x)$, where $h(x) = \dfrac{5}{x - 1.5} + 2$ for $x \in 1.5$.
The dashed lines show the asymptotes of the curve.




**(a)** From the graph, write down the equation of

**(i)** the vertical asymptote,

..... **[1]**

**(ii)** the horizontal asymptote.

..... **[1]**

**(b)** Explain why the equation $h(x) = 2$ has no solution.

..... **[1]**

**(c)** The graph of $y = h(x)$ is reflected in the $y$-axis.
Write down the equations of the vertical and horizontal asymptotes of the reflected graph.

Vertical asymptote: ..... , Horizontal asymptote: ..... **[2]**

**(d)** Solve the equation $h(x) = 3.4$, giving your answer correct to 3 significant figures.

$x =$ ..... **[3]**

**779** The function $f(x)$ is given by $f(x) = \dfrac{p}{x-q} + r$, where $p, q$, and $r$ are constants.
The graph of $y = f(x)$ has a vertical asymptote at $x = 5$ and a horizontal asymptote at $y = -3$.
The graph passes through the point $(7, -2)$.

**(a)** Write down the value of $q$ and the value of $r$.

$q =$ ..... , $r =$ ..... **[2]**

**(b)** Find the value of $p$.

$p =$ ..... **[3]**

**(c)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[4]**

**(d)** Write down the equations of the asymptotes of the graph of $y = f^{-1}(x)$.

$x =$ ..... , $y =$ ..... **[2]**

**780** A function is defined as $y = f(x)$.

**(a)** Describe fully the single transformation that maps the graph of $y = f(x)$ onto the graph of $y = f(x - 4)$.

.....

..... **[2]**

**(b)** The graph of $y = f(x)$ is translated by the vector $\begin{pmatrix} 0 \\ 7 \end{pmatrix}$ to give the graph of $y = g(x)$.
Write down the expression for $g(x)$ in terms of $f(x)$.

$g(x) =$ ..... **[1]**

**781** **(a)** The graph of $y = x^2$ is translated by the vector $\begin{pmatrix} 0 \\ -5 \end{pmatrix}$ to map onto the graph of $y = g(x)$. Write down $g(x)$ in terms of $x$.

$g(x) =$ ..... **[1]**

**(b)** The graph of $y = x^2$ is translated by the vector $\begin{pmatrix} 3 \\ 0 \end{pmatrix}$ to map onto the graph of $y = h(x)$. Write down $h(x)$ in terms of $x$.

$h(x) =$ ..... **[1]**

**(c)** Describe fully the single transformation that maps the graph of $y = x^2$ onto the graph of $y = (x + 4)^2 - 1$.

.....

.....

..... **[2]**

**782** A function is defined as $f(x) = 2x - 5$.

**(a)** The graph of $y = f(x)$ is translated by the vector $\begin{pmatrix} 0 \\ 7 \end{pmatrix}$ to give the graph of $y = g(x)$.
Find $g(x)$ in terms of $x$, giving your answer in its simplest form.

$g(x) =$ ..... **[2]**

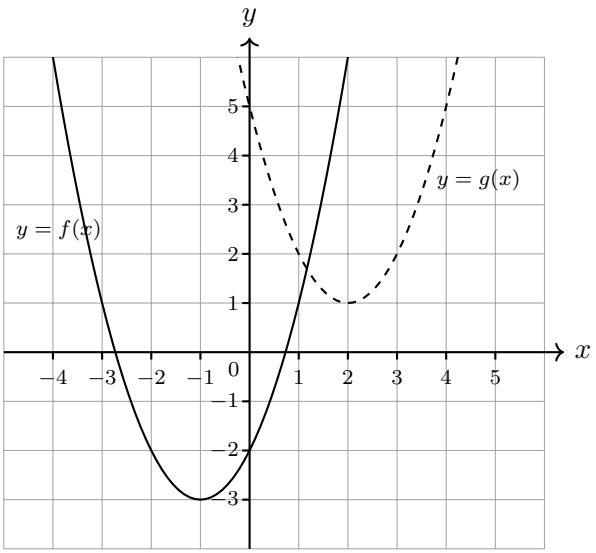
**(b)** Describe fully the single transformation that maps the graph of $y = f(x)$ onto the graph of $y = f(x - 4)$.

.....

.....

..... **[2]**

**783** The diagram shows the graph of $y = f(x)$ and the graph of $y = g(x)$.




**(a)** Describe fully the single transformation that maps the graph of $y = f(x)$ onto the graph of $y = g(x)$.

.....

.....

..... **[2]**

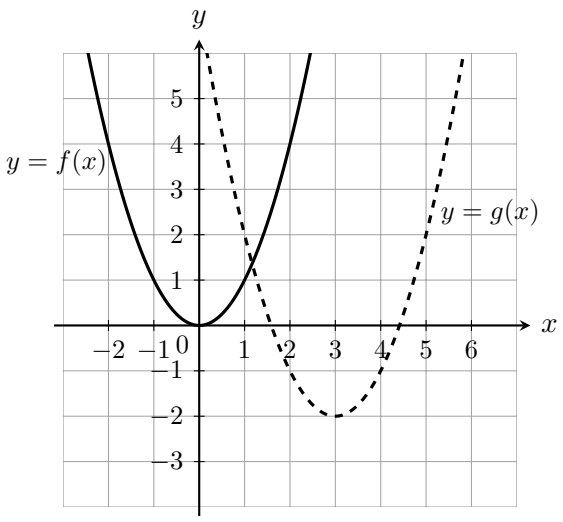
**(b)** Write $g(x)$ in terms of $f(x)$.

$g(x) =$ ..... **[2]**

**(c)** The graph of $y = f(x)$ is translated by the vector $\begin{pmatrix} -2 \\ 0 \end{pmatrix}$.
Write down the coordinates of the vertex of the resulting graph.

( ..... , ..... ) **[1]**

**784** The diagram shows the graphs of $y = f(x)$ and $y = g(x)$ on a coordinate grid.




**(a)** Write down the column vector that describes the translation mapping the graph of $y = f(x)$ onto the graph of $y = g(x)$.

$$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$$ **[2]**

**(b)** Given that $f(x) = x^2$, write down $g(x)$ in the form $(x - a)^2 + b$.

$g(x) =$ ..... **[2]**

**(c)** Write down the coordinates of the vertex of the graph of $y = f(x + 5) - 1$.

( ..... , ..... ) **[2]**

**785** The graph of $y = f(x)$ has a local maximum at the point $P(-2, 5)$ and a local minimum at the point $Q(3, -1)$.

**(a)** Write down the coordinates of the local maximum of the graph of $y = f(x) - 4$.

( ..... , ..... ) **[1]**

**(b)** Write down the coordinates of the local minimum of the graph of $y = f(x - 2)$.

( ..... , ..... ) **[1]**

**(c)** The graph of $y = f(x)$ is translated by the vector $\begin{pmatrix} a \\ b \end{pmatrix}$ such that the image of the local maximum $P$ is at the origin $(0, 0)$.

**(i)** Find the value of $a$ and the value of $b$.

$a =$ ..... , $b =$ ..... **[2]**

**(ii)** Write down the coordinates of the image of the local minimum $Q$ under the same translation.

( ..... , ..... ) **[1]**

**786** Consider the function $f(x) = x^2 - 4x + 1$.

**(a)** The graph of $y = f(x)$ is translated by the vector $\begin{pmatrix} 2 \\ -5 \end{pmatrix}$ to give the graph of $y = g(x)$.
Find $g(x)$ in the form $ax^2 + bx + c$.

$g(x) =$ ..... **[3]**

**(b)** The graph of $y = f(x)$ is mapped onto the graph of $y = h(x)$, where $h(x) = x^2 + 2x - 2$.
Find the translation vector that maps the graph of $y = f(x)$ onto the graph of $y = h(x)$.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[3]**

**787** Let $f(x) = 3x^2$.

**(a)** The graph of $y = f(x)$ is translated by the vector $\begin{pmatrix} -2 \\ 0 \end{pmatrix}$ to give the graph of $y = g(x)$.
Find $g(x)$ in terms of $x$.
Give your answer in the form $ax^2 + bx + c$.

$g(x) =$ ..... **[3]**

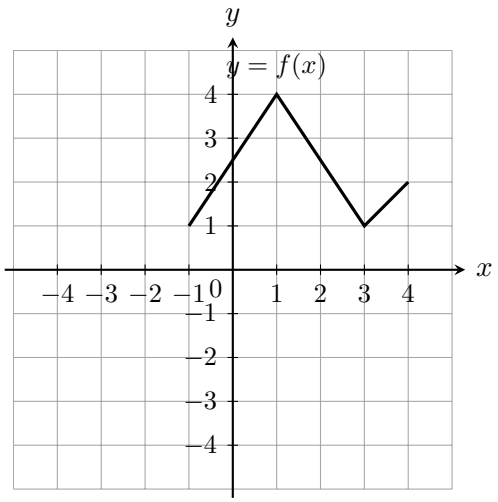
**(b)** The graph of $y = g(x)$ is translated by the vector $\begin{pmatrix} 0 \\ -8 \end{pmatrix}$ to give the graph of $y = h(x)$.
Write down the coordinates of the minimum point of the graph of $y = h(x)$.

( ..... , ..... ) **[2]**

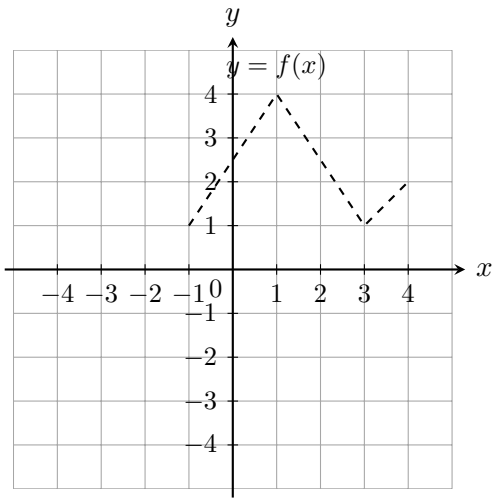
**(c)** Find the $y$-intercept of the graph of $y = h(x)$.

..... **[1]**

**788** The diagram shows the graph of $y = f(x)$ on a coordinate grid.




**(a)** On the grid below, draw the graph of $y = f(x + 2) - 3$.




**[2]**

**(b)** The graph of $y = f(x)$ is mapped onto the graph of $y = h(x)$ by a single transformation.
The point $(1, 4)$ on $y = f(x)$ is mapped onto $(1, -1)$ on $y = h(x)$, and the point $(3, 1)$ is mapped onto $(3, -4)$.
Describe fully the single transformation that maps the graph of $y = f(x)$ onto the graph of $y = h(x)$.

.....

.....

..... **[2]**

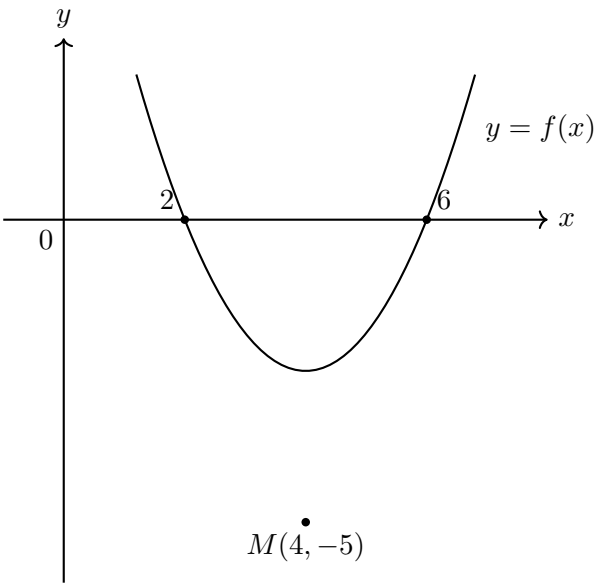
**(c)** Write down $h(x)$ in terms of $f(x)$.

$h(x) =$ ..... **[1]**

**(d)** Find the coordinates of the image of the point $(4, 2)$ on $y = f(x)$ when it is mapped onto the graph of $y = f(x - 3) + 4$.

( ..... , ..... ) **[2]**

**789** The diagram shows a sketch of the graph of $y = f(x)$.
The curve intersects the $x$-axis at $(2, 0)$ and $(6, 0)$ and has a minimum point at $M(4, -5)$.




**(a)** A transformation maps the graph of $y = f(x)$ onto the graph of $y = f(x) + 4$.
Write down the coordinates of the image of point $M$.

( ( ..... , ..... ) ) **[1]**

**(b)** The graph of $y = f(x)$ is mapped onto the graph of $y = g(x)$ by a translation with vector $\begin{pmatrix} -3 \\ 2 \end{pmatrix}$.
Write down the expression for $g(x)$ in terms of $f(x)$.

$g(x) =$ ..... **[2]**

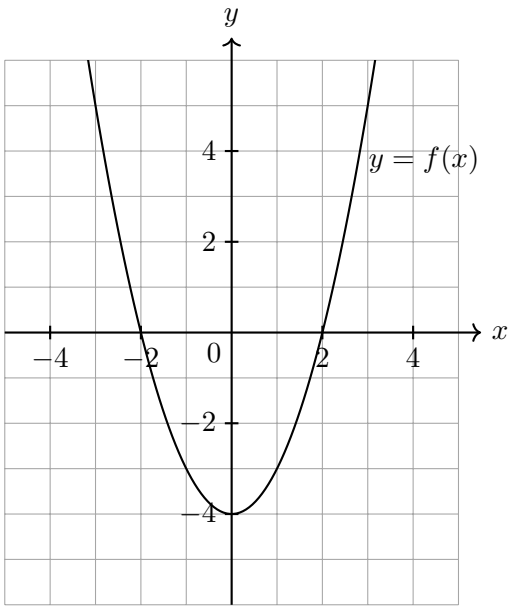
**(c)** The graph of $y = f(x - k)$ passes through the origin $(0, 0)$.
Find the two possible values of the constant $k$.

$k =$ ..... , $k =$ ..... **[2]**

**(d)** The function $h(x)$ is such that the graph of $y = h(x)$ has a minimum point at $(7, -5)$.
Find $h(x)$ in terms of $f(x)$.

$h(x) =$ ..... **[1]**

**790** The diagram shows the graph of $y = f(x)$, where $f(x) = x^2 - 4$.



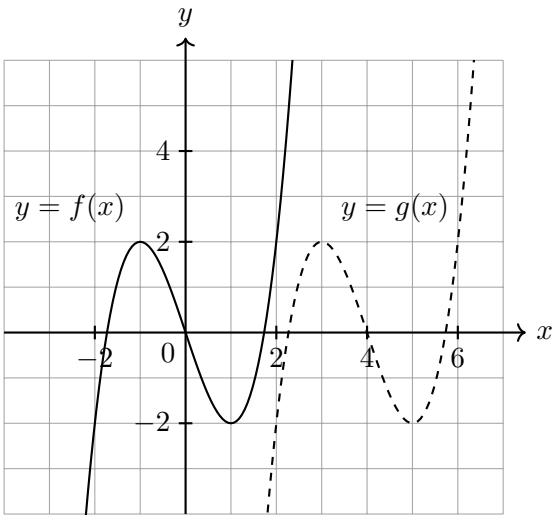

**(a)** The graph of $y = f(x)$ is translated by the vector $\begin{pmatrix} 0 \\ 5 \end{pmatrix}$ to give the graph of $y = g(x)$.
Write down the expression for $g(x)$ in terms of $x$, simplifying your answer.

$g(x) =$ ..... **[2]**

**(b)** Write down the coordinates of the turning point of the graph of $y = f(x + 3)$.

( ..... , ..... ) **[2]**

**791** The diagram shows the graph of $y = f(x)$ and its image $y = g(x)$ after a single transformation.




**(a)** Describe fully the single transformation that maps the graph of $y = f(x)$ onto the graph of $y = g(x)$.

.....

.....

..... **[2]**

**(b)** Given that $f(x) = x^3 - 3x$, find $g(x)$ in terms of $x$, giving your answer in expanded and simplified form.

$g(x) =$ ..... **[4]**

**792** A function is given as $f(x) = x^2 - 2x - 3$.

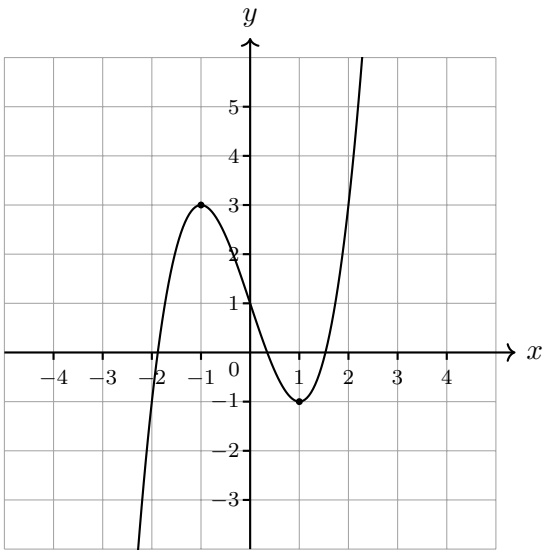
**(a)** The graph of $y = f(x)$ is translated by the vector $\begin{pmatrix} -4 \\ 0 \end{pmatrix}$ to form the graph of $y = g(x)$.
Find $g(x)$ in the form $ax^2 + bx + c$.

$g(x) =$ ..... **[4]**

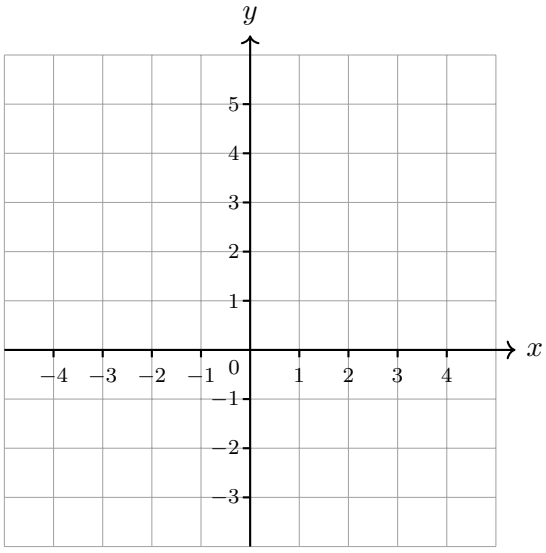
**(b)** The graph of $y = f(x)$ is mapped onto the graph of $y = h(x)$ by a translation with vector $\begin{pmatrix} 0 \\ -5 \end{pmatrix}$.
Solve the equation $h(x) = 0$.

$x =$ ..... , $x =$ ..... **[3]**

**793** The diagram shows the graph of $y = f(x)$ for $-4 \leq x \leq 4$.
The curve has a local maximum at $(-1, 3)$, a local minimum at $(1, -1)$, and passes through the origin $(0, 1)$.




**(a)** On the grid, sketch the graph of $y = f(x - 2) + 1$.




**[3]**

**(b)** Given that $f(x) = x^3 - 3x + 1$, find the coordinates of the local minimum point on the graph of $y = f(x + k) - 2$, in terms of $k$.

( ..... , ..... ) **[2]**

**(c)** The graph of $y = f(x) + p$ has exactly two $x$-intercepts.
Write down the two possible values of $p$.

$p =$ ..... , $p =$ ..... **[2]**

**794** The function $f$ is defined by $f(x) = 2x^2 - 8x + 11$.

**(a)** Write $f(x)$ in the form $a(x + h)^2 + k$.

$f(x) =$ ..... **[3]**

**(b)** The graph of $y = f(x)$ is translated to give the graph of $y = g(x)$, where $g(x) = 2x^2 + 4x - 3$. Find the translation vector that maps the graph of $y = f(x)$ onto the graph of $y = g(x)$.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[4]**

**795** The functions $f(x)$ and $g(x)$ are defined as $f(x) = x^2 - 6x + 14$ and $g(x) = \dfrac{3}{x+2} - 5$.

**(a)** By writing $f(x)$ in the form $(x-a)^2 + b$, find the column vector of the translation that maps the graph of $y = x^2$ onto the graph of $y = f(x)$.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[3]**

**(b)** The graph of $y = g(x)$ is translated by the vector $\begin{pmatrix} 4 \\ 2 \end{pmatrix}$ to map onto the graph of $y = h(x)$.
Find $h(x)$ in its simplest form.

$h(x) =$ ..... **[3]**

**(c)** Write down the equations of the vertical and horizontal asymptotes of the graph of $y = h(x)$.

$x =$ ..... , $y =$ ..... **[2]**

**796** A function is defined as $f(x) = \dfrac{20}{x}$.

**(a)** The graph of $y = f(x)$ is mapped onto the graph of $y = g(x)$ by a translation with vector $\begin{pmatrix} 0 \\ -4 \end{pmatrix}$.

**(i)** Write down an expression for $g(x)$.

$g(x) =$ ..... **[1]**

**(ii)** Solve the equation $g(x) = 0$.

$x =$ ..... **[2]**

**(b)** The graph of $y = f(x)$ is translated by the vector $\begin{pmatrix} k \\ 0 \end{pmatrix}$ to give the graph of $y = h(x)$.
The graph of $y = h(x)$ passes through the point $(8, 5)$.
Work out the value of $k$.

$k =$ ..... **[3]**

**(c)** Let $p(x) = f(x - 3) + 7$.
Write down the equations of the vertical and horizontal asymptotes of the graph of $y = p(x)$.

Vertical asymptote: ..... , Horizontal asymptote: ..... **[2]**

**797** The function $f$ is defined by $f(x) = 2x^2 - 5x + 1$.

**(a)** The graph of $y = f(x)$ is translated by the vector $\begin{pmatrix} -2 \\ -6 \end{pmatrix}$ to give the graph of $y = g(x)$.

**(i)** Write down $g(x)$ in terms of $f(x)$.

$g(x) =$ ..... **[1]**

**(ii)** Find $g(x)$ in the form $ax^2 + bx + c$.

$g(x) =$ ..... **[3]**

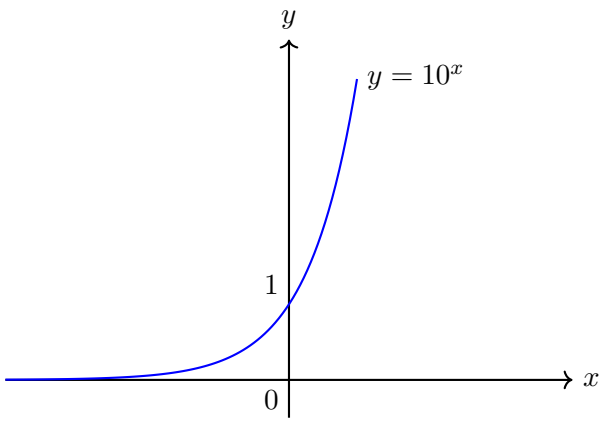
**(b)** The graph of $y = f(x)$ is mapped onto the graph of $y = h(x)$ by a single translation, where $h(x) = 2x^2 - 13x + 18$.
Find the column vector of this translation.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[4]**

**(c)** Find the coordinates of the point of intersection of the graphs $y = f(x)$ and $y = g(x)$.

( ..... , ..... ) **[3]**

**798** The diagram shows a sketch of the function $f(x) = 10^x$.




**(a)** Write down the inverse function, $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[1]**

**(b)** Work out the exact value of $f^{-1}\left(\frac{1}{100}\right)$.

..... **[1]**

**(c)** Write down the exact solution to the equation $f(x) = 45$.

$x =$ ..... **[1]**

**799** **(a)** Write $10^3 = 1000$ as a logarithmic equation.

..... **[1]**

**(b)** Find the exact value of $x$ when $\log x = -2$.

$x =$ ..... **[1]**

**(c)** Solve $5^x = 12$, giving your answer in the form $\dfrac{\log a}{\log b}$.

$x =$ ..... **[1]**

**800** The functions $f$ and $g$ are defined as $f(x) = 10^x$ and $g(x) = \log x$.

**(a)** Find the value of $g(100)$.

..... **[1]**

**(b)** Find the value of $x$ when $f(x) = 0.001$.

$x =$ ..... **[1]**

**(c)** Write down the expression for $g(f(x))$ in its simplest form.

$g(f(x)) =$ ..... **[1]**

**801** **(a)** Work out the value of

$\log 10000$

..... **[1]**

**(b)** Write the following equation in the form $\log_a b = c$.

$5^3 = 125$

..... **[1]**

**(c)** Solve the equation

$\log_2 x = 6$

$x =$ ..... **[1]**

**802** **(a)** Find the value of $x$ when $\log_2 x = 5$.

$x =$ ..... **[1]**

**(b)** Find the value of $y$ when $10^y = 0.01$.

$y =$ ..... **[1]**

**(c)** Write the equation $P = k^t$ in logarithmic form.

..... **[1]**

**803** **(a)** Write the following as a logarithmic equation.

$5^3 = 125$

..... **[1]**

**(b)** Find the value of $\log 1000$.

..... **[1]**

**(c)** Solve the equation.

$\log_2 x = 5$

$x =$ ..... **[2]**

**804** **(a)** Write the equation $10^5 = 100000$ in the form $\log a = b$.

..... **[1]**

**(b)** Find the value of $x$ when $\log x = 3$.

$x =$ ..... **[2]**

**(c)** The function $f(x)$ is defined as $f(x) = 10^x$.
Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[1]**

**805** All logarithms in this question are in the base shown, or base
10
10 where no base is written.

**(a)** Write down the value of

**(i)** log
1000
log1000,

..... **[1]**

**(ii)** log
2
32
log
2
32.

..... **[1]**

**(b)** Given that
log
3
=
4
log
3
x=4, find the value of
x.

=
x= ..... **[1]**

**(c)** The function
f is defined by
(
)
=
10
f(x)=10
x
.
Write down
1
(

)
f
1
(x).

1
(
)
=
f
1
(x)= ..... **[1]**

**806** **(a)** Find the value of $x$ in each of the following equations.

**(i)** $10^x = 100000$

$x =$ ..... **[1]**

**(ii)** $\log x = -2$

$x =$ ..... **[1]**

**(iii)** $\log_2 64 = x$

$x =$ ..... **[1]**

**(b)** Given that $f(x) = 10^x$, write down an expression for $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[1]**

**807** **(a)** Write down the value of

$\log 1000$

..... **[1]**

**(b)** Solve the equation.

$\log x = -2$

$x =$ ..... **[2]**

**(c)** Write the equation $5^x = 125$ in the form $x = \log_a b$.

$x =$ ..... **[1]**

**808** Answer the following independent questions on logarithmic and exponential equations.

**(a)** Write the exponential equation $10^4 = 10000$ in logarithmic form.

..... **[1]**

**(b)** Find the exact value of $x$ when $\log_5 x = 3$.

$x =$ ..... **[2]**

**(c)** The solution to the equation $7^x = 25$ can be written in the form $x = \dfrac{\log b}{\log a}$.
Write down the value of $a$ and the value of $b$.

$a =$ ..... , $b =$ ..... **[2]**

**809** Consider the function $f(x) = \log_3 x$ for $x > 0$.

**(a)** Find the value of $f(81)$.

..... **[1]**

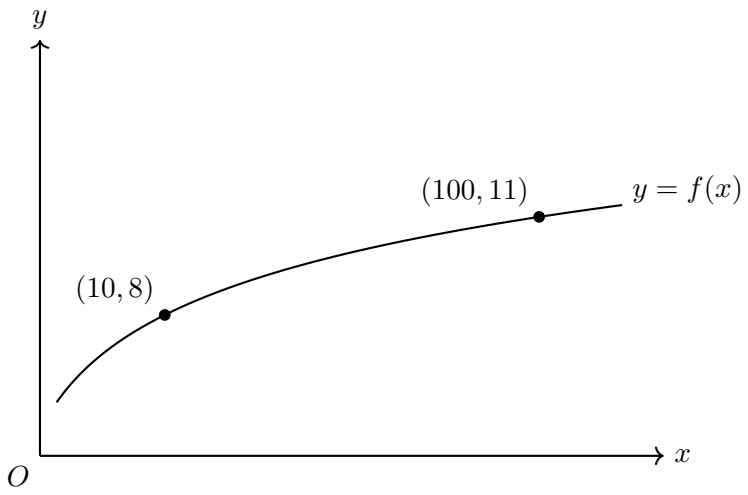
**(b)** Find the value of $f\left(\frac{1}{9}\right)$.

..... **[2]**

**(c)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

**810** The diagram shows a sketch of the graph of $y = f(x)$ for $x > 0$.
The function is defined as $f(x) = p \log x + q$, where $\log x$ denotes the logarithm to base 10.
The graph passes through the points $(10, 8)$ and $(100, 11)$.




**(a)** **(a)** Find the value of $p$ and the value of $q$.

$p =$ ..... , $q =$ ..... **[3]**

**(b)** Solve the equation $f(x) = 23$.

$x =$ ..... **[2]**

**(c)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

**811** The logarithmic function is the inverse of the exponential function.

**(a)** Write the equation $y = 6^x$ in logarithmic form.

**(a)** Write the equation $y = 6^x$ in logarithmic form.

$x =$ ..... **[1]**

**(b)** Solve the equation.

$3^x = 20$ Give your answer correct to 3 significant figures.

$x =$ ..... **[2]**

**812** A population of insects is growing exponentially.
The number of insects, $N$, after $t$ days is given by

$$N = 250 \times 1.08^t$$

**(a)** Work out the number of insects after 10 days.
Give your answer to the nearest integer.

..... **[2]**

**(b)** Calculate the number of days it takes for the population to reach 1000.
Give your answer correct to 1 decimal place.

..... days **[3]**

**813** The value of a machine, $V$ dollars, depreciates exponentially according to the formula

$$V = 15000 \times 0.8^t$$

where $t$ is the number of years after it is purchased.

**(a)** Calculate the value of the machine after 4 years.

$ ..... **[2]**

**(b)** Find the number of years it takes for the value of the machine to decrease to $7500.
Give your answer correct to 3 significant figures.

..... years **[3]**

**814** A machine is purchased for \$25000. Its value depreciates exponentially by 12% each year.

**(a)** Calculate the value of the machine after 4 years.
Give your answer correct to the nearest dollar.

% ..... **[2]**

**(b)** Find the number of complete years it takes for the value of the machine to fall below \$10000.
You must show all your working.

..... years **[3]**

**815** **(a)** Solve the equation.

$3^{2x} = 50$ Give your answer correct to 3 significant figures.

$x =$ ..... **[3]**

**(b)** Make $x$ the subject of the formula.

$y = 10^{x-4}$

$x =$ ..... **[2]**

**816** A population of bacteria is modelled by the formula

$P = 5000 \times 1.12^t$ where $P$ is the population and $t$ is the time in hours after the start of an experiment.

**(a)** **(a)** Write down the initial population of the bacteria.

..... **[1]**

**(b)** Find the time it takes for the population to reach 20 000.
Give your answer correct to 1 decimal place.

$t =$ ..... hours **[4]**

**817** An investment of \$8000 earns compound interest at a rate of 3.5% per year. The value of the investment after $n$ years is \$$V$.

**(a)** Write down an equation for $V$ in terms of $n$.

$V =$ ..... **[1]**

**(b)** Calculate the minimum number of complete years it takes for the value of the investment to exceed \$12000.
Show all your working.

..... years **[3]**

**(c)** Solve the equation.

$2^x = 55$ Give your answer correct to 3 significant figures.

$x =$ ..... **[2]**

**818** **(a)** Solve the equation $3^x = 20$.
Give your answer correct to 3 significant figures.

$x =$ ..... **[3]**

**(b)** The mass, $M$ grams, of a radioactive substance after $t$ days is given by the formula
$M = 500 \times 0.85^t$ Calculate the value of $t$ when $M = 100$.
Give your answer correct to 3 significant figures.

$t =$ ..... **[3]**

**819** The value, $V$ dollars, of a painting $t$ years after it was bought is given by the formula

$$V = 4500 \times 1.08^t$$

**(a)** Calculate the value of the painting 5 years after it was bought.
Give your answer correct to the nearest dollar.

..... **[2]**

**(b)** Calculate the time it takes for the value of the painting to reach \$10 000.
Give your answer in years, correct to 2 decimal places.

..... years **[4]**

**820** The value, $\$V$, of a vintage motorcycle $t$ years after it was purchased is modelled by the formula

$$V = 4500 \times 1.08^t$$

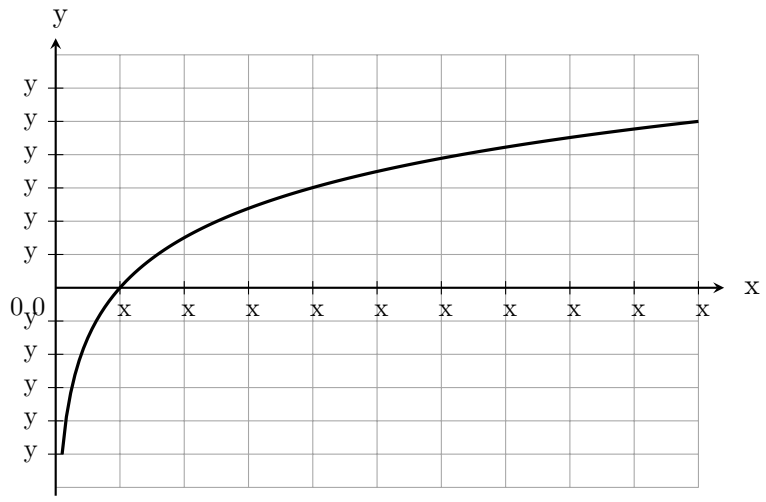
**(a)** Calculate the value of the motorcycle after 5 years.

$\$$ ..... **[2]**

**(b)** Calculate the number of years it takes for the value of the motorcycle to reach \$10 000. Give your answer correct to 1 decimal place.

..... years **[4]**

**821** The diagram shows the graph of
=
log
y=logx for
0.1
10
0.1x10.




**(a)** Use the graph to estimate the value of
log
6
log6.

..... **[1]**

**(b)** Use the graph to solve the equation
log
=
0.5
logx=0.5.

=
x= ..... **[1]**

**(c)** Solve the equation
3
+
1
=
20
3
x+1
=20, giving your answer correct to

3
3 significant figures.

=
x= . . . . . **[3]**

**(d)** Solve the equation
log
(
2
3
)
=
1.4
log(2x3)=1.4, giving your answer correct to
3
3 significant figures.

=
x= . . . . . **[2]**

**822** The mass, $M$ grams, of a radioactive substance after $t$ days is given by the formula $M = 400 \times 0.88^t$.

**(a)** Calculate the mass of the substance after 5 days.
Give your answer correct to 1 decimal place.

..... grams **[2]**

**(b)** Find the number of days it takes for the mass of the substance to reduce to exactly half of its initial value.

..... days **[3]**

**(c)** Another radioactive substance has an initial mass of 800 grams.
After 10 days, its mass is 300 grams.
It decays according to the formula $M = 800 \times k^t$.
Find the value of $k$, giving your answer correct to 3 significant figures.

$k =$ ..... **[3]**

**823** An amount of \$8000 is invested in an account that pays compound interest at a rate of 4.5% per year. The value of the investment, $V$ dollars, after $t$ years is given by the formula

$$V = 8000 \times 1.045^t$$

**(a)** Calculate the value of the investment after 5 years.

\$ ..... **[2]**

**(b)** Use logarithms to calculate the number of complete years it takes for the value of the investment to first exceed \$15000.

..... years **[3]**

**(c)** Solve the equation.

$7^{x-1} = 250$ Give your answer correct to 3 significant figures.

$x =$ ..... **[3]**

**824** A population of insects grows exponentially.
The number of insects, $N$, after $t$ weeks is given by the formula

$$N = 800 \times 2^t$$

**(a)** Work out the number of insects after 4 weeks.

..... **[1]**

**(b)** The number of insects reaches 102 400.
Work out the exact value of $t$.

$t =$ ..... **[2]**

**(c)** Another population of insects can be modelled by the formula $P = 1500 \times 3^t$.
Find the value of $t$ when this population reaches 20 000.
Give your answer correct to 3 significant figures.

$t =$ ..... **[3]**

**(d)** A sum of \$2000 is invested at an annual compound interest rate of 5%. The total value, $V$, of the investment after $n$ years is given by

$$V = 2000 \times 1.05^n$$

Calculate the minimum number of complete years required for the investment to double in value.

..... years **[3]**

**825** A function $f$ is defined as $f(x) = 4 \times 10^{2x} - 7$.

**(a)** Find $f^{-1}(x)$.

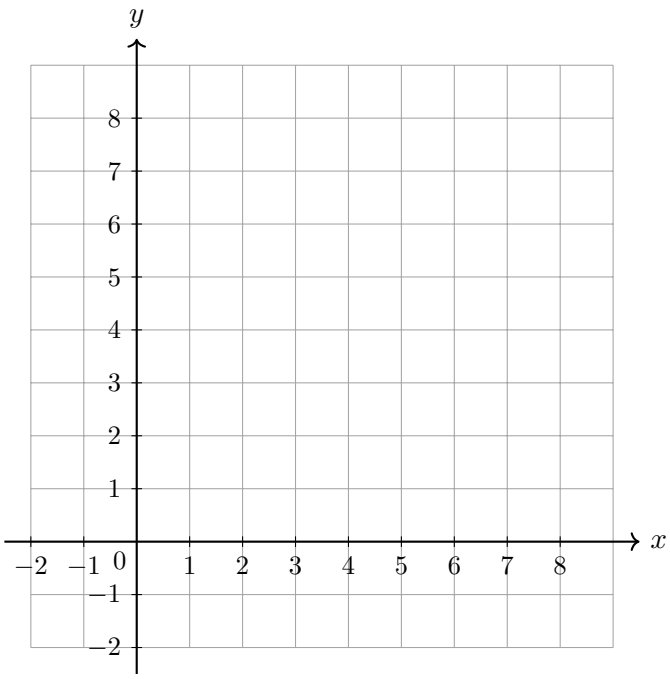
$f^{-1}(x) =$ ..... **[3]**

**(b)** Solve the equation $f(x) = 53$.
Give your answer correct to 3 significant figures.

$x =$ ..... **[3]**

**826** A function is defined as $f(x) = 2^x$.

**(a)** On the grid, sketch the graph of $y = f(x)$ for $-2 \leq x \leq 3$.




**[2]**

**(b)** Write down the equation of the inverse function, $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[1]**

**(c)** On the same grid in part (a), sketch the graph of $y = f^{-1}(x)$ for $0 < x \leq 8$. **[2]**

**(d)** Solve the equation $2^x = 12$.
Give your answer correct to 3 significant figures.

$x =$ ..... **[2]**

**827** **(a)** The mass, $M$ grams, of a radioactive substance after $t$ days is given by $M = 250 \times 0.92^t$.

**(i)** Write down the initial mass of the substance.

..... grams **[1]**

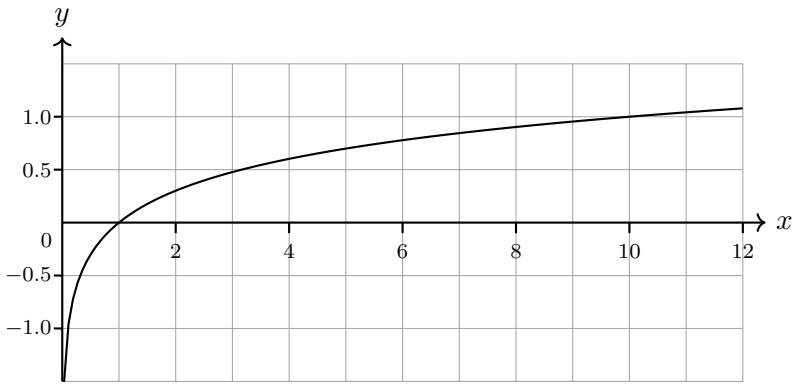
**(ii)** Find the number of days it takes for the mass to reduce to 50 grams.
Give your answer correct to 3 significant figures.

..... days **[3]**

**(b)** An amount of \$800 is invested at a rate of 6% per year compound interest. Calculate the number of complete years it takes for the investment to first exceed \$2000.

..... years **[3]**

**828** The diagram shows the graph of $y = \log x$ for $0 < x \leq 12$, where $\log x$ denotes $\log_{10} x$.




**(a)** Write down the coordinates of the $x$-intercept and the equation of the vertical asymptote of the graph of $y = \log x$.

$x$-intercept = ..... , Asymptote: ..... **[2]**

**(b)** Use the graph to estimate the solution to the equation $\log x = 0.7$.

$x =$ ..... **[1]**

**(c)** Solve algebraically the equation

$$3^{2x-1} = 5^{x+2}$$

Give your answer correct to 3 significant figures.

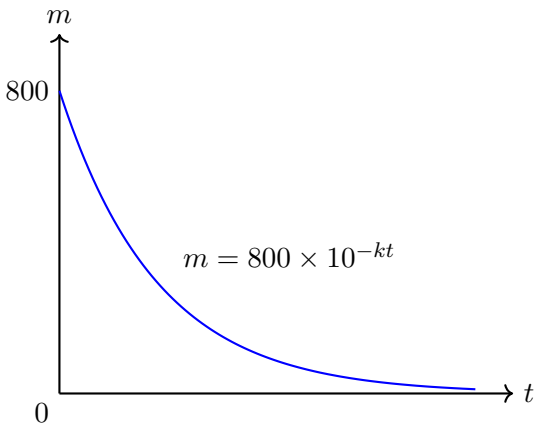
$x =$ ..... **[4]**

**829** The mass, $m$ grams, of a radioactive isotope decays according to the formula

$m = 800 \times 10^{-kt}$ where $t$ is the time in years and $k$ is a constant.

The diagram shows a sketch of the decay curve.

After 15 years, the mass is 250 grams.




**(a)** Show that $k = 0.0337$ correct to 3 significant figures.

**[3]**

**(b)** Find the number of years it takes for the mass to become 50 grams.

..... years **[3]**

**(c)** Make $t$ the subject of the formula $m = 800 \times 10^{-kt}$.

$t =$ ..... **[2]**

**830** Let $f(x) = 5 \times 10^x$ and $g(x) = \log x$.

**(a)** Find the value of $f(g(100))$.

..... **[2]**

**(b)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

**(c)** Solve the equation.

$f(x) = 200$ Give your answer correct to 3 significant figures.

$x =$ ..... **[2]**

**(d)** Find $g(f(x))$ in the form $x + k$, where $k$ is a constant.

..... **[2]**

**831** A function is defined as $f(x) = 5 \times 3^x - 2$.

**(a)** Find the value of $f(-2)$, giving your answer as a fraction in its simplest form.

..... **[2]**

**(b)** Find $f^{-1}(x)$.

Give your answer in terms of base 10 logarithms.

$f^{-1}(x) =$ ..... **[3]**

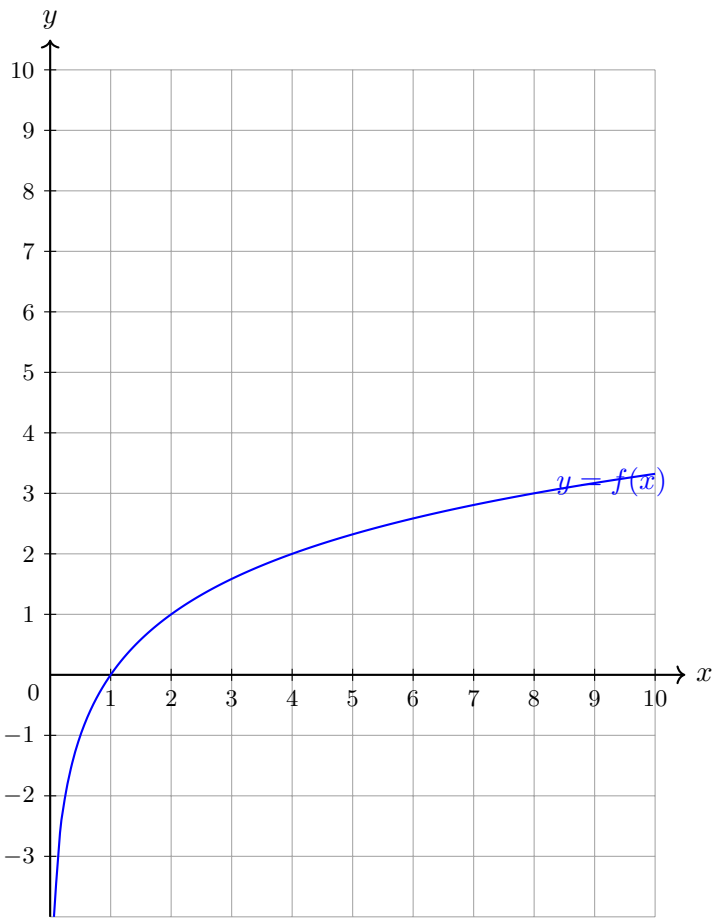
**(c)** Solve the equation.

$f(x) = 500$

Give your answer correct to 3 significant figures.

$x =$ ..... **[3]**

**832** The diagram shows the graph of $y = f(x)$.




**(a)** The graph of $y = f(x)$ passes through the point $(8, 3)$.
Given that $f(x) = \log_a x$, show that $a = 2$.

**[2]**

**(b)** **(i)** On the grid, sketch the graph of $y = f^{-1}(x)$. **[2]**

**(ii)** Write down the equation of the asymptote of the graph of $y = f^{-1}(x)$.

..... **[1]**

**(c)** By drawing a suitable straight line on the grid, find an approximate solution to the equation $\log_2 x + x = 8$.

$x =$ ..... **[3]**

**833** The function $f(x)$ is defined as $f(x) = 4 \times 10^{0.5x}$.

**(a)** Find the value of $f(3)$.
Give your answer correct to 3 significant figures.

..... **[2]**

**(b)** Find $f^{-1}(x)$.

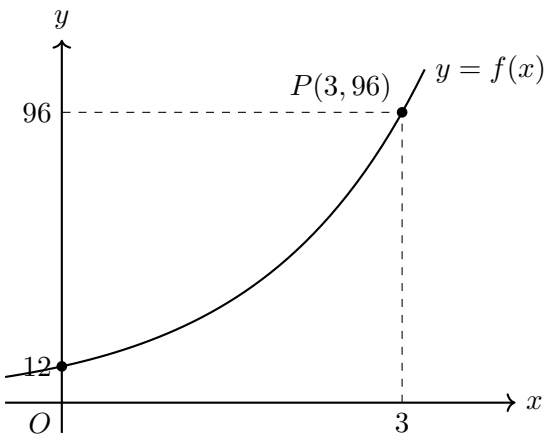
$f^{-1}(x) =$ ..... **[3]**

**(c)** Solve the equation

$$f(x) = 200$$

$x =$ ..... **[3]**

**834** The diagram shows the graph of $y = f(x)$ where $f(x) = a \times b^x$.
The graph intersects the $y$-axis at the point $(0, 12)$ and passes through the point $P(3, 96)$.




**(a)** Find the value of $a$ and the value of $b$.

$a =$ ..... , $b =$ ..... **[3]**

**(b)** Find $f^{-1}(x)$.
Give your answer in terms of base 10 logarithms.

$f^{-1}(x) =$ ..... **[3]**

**(c)** Solve the equation $f(x) = 500$.
Give your answer correct to 3 significant figures.

$x =$ ..... **[3]**

**835** The value, \$V, *of a machine t years after it is purchased is modelled by the formula* $V = 24000 \times 0.85^t$.

**(a)** Calculate the value of the machine after 4 years, giving your answer correct to the nearest dollar.

\$ ..... **[2]**

**(b)** Find the number of years it takes for the value of the machine to fall to \$9000.
Give your answer correct to 3 significant figures.

$t =$ ..... years **[3]**

**(c)** A different asset increases in value by 6.8% each year.
Find the minimum whole number of years it takes for the value of this asset to double.

..... years **[3]**

**(d)** The function $g$ is defined by $g(x) = 5 \times 2^x + 3$.
Find $g^{-1}(x)$.

$g^{-1}(x) =$ ..... **[3]**

**836** The function $f(x) = A \times b^x$, where $b > 0$, passes through the points $(0, 7)$ and $(3, 56)$.

**(a)** Find the value of $A$ and the value of $b$.

$A =$ ..... , $b =$ ..... **[3]**

**(b)** Find the inverse function, $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[3]**

**(c)** The equation $f(x) = 600$ is equivalent to $2^x = k$.

**(i)** Work out the exact value of $k$ as a fraction.

$k =$ ..... **[1]**

**(ii)** Solve the equation $f(x) = 600$.
Give your answer correct to 3 decimal places.

$x =$ ..... **[2]**

**(d)** A different function $g(x)$ is defined as $g(x) = \log_c(x)$.
The graph of $y = g(x)$ passes through the point $(81, 4)$.

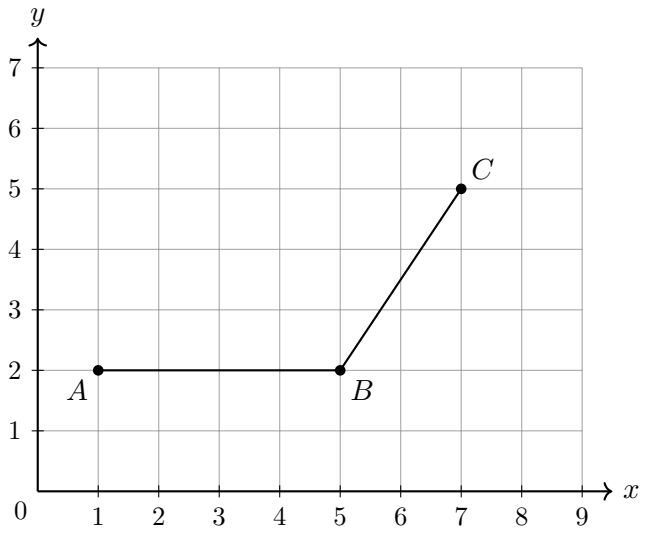
**(i)** Find the value of $c$.

$c =$ ..... **[2]**

**(ii)** Solve the equation $g(x) = -2$.

$x =$ ..... **[1]**

**837** The diagram shows three vertices, $A(1, 2)$, $B(5, 2)$ and $C(7, 5)$, of a parallelogram $ABCD$.




**(a)** Find the coordinates of vertex $D$.

( ( ..... , ..... ) ) **[2]**

**(b)** Calculate the area of the parallelogram $ABCD$.

..... **[2]**

**838** Points $P(-4, 2)$ and $Q(6, 2)$ are plotted on a Cartesian grid.

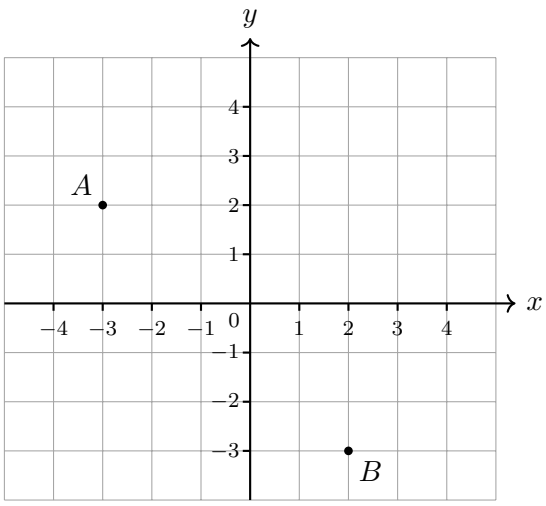
**(a)** Write down the coordinates of the midpoint of the line segment $PQ$.

( ..... , ..... ) **[1]**

**(b)** Point $R$ lies on the positive $y$-axis.
The area of triangle $PQR$ is 35 square units.
Find the coordinates of $R$.

( ..... , ..... ) **[3]**

**839** The diagram shows points $A$ and $B$ plotted on a coordinate grid.




**(a)** Write down the coordinates of point $A$ and point $B$.

$A =$ ..... , $B =$ ..... **[2]**

**(b)** Point $C$ is the reflection of point $A$ in the $y$-axis.
Write down the coordinates of point $C$.

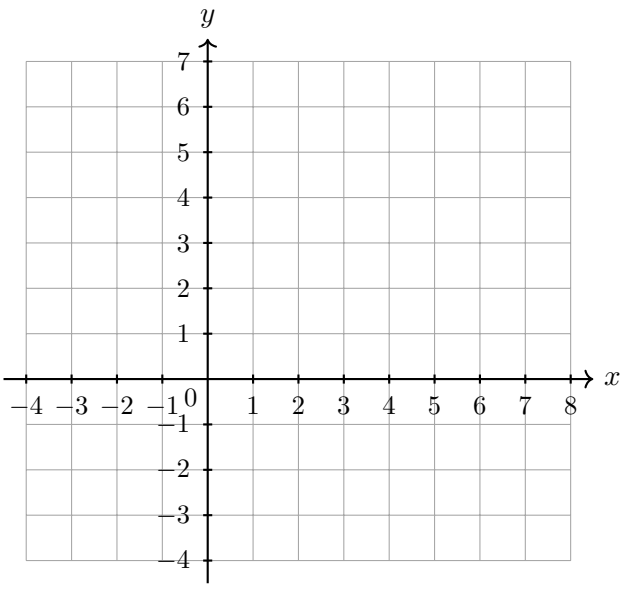
( ..... , ..... ) **[1]**

**(c)** Point $D$ is plotted on the grid so that $ABCD$ is a rectangle.
Find the coordinates of point $D$ and calculate the area of rectangle $ABCD$.

$D =$ ..... , Area = ..... **[2]**

**840** The points $A(-1, 3)$, $B(2, -1)$ and $C(6, 1)$ are three vertices of a parallelogram $ABCD$.

**(a)** On the grid, plot the points $A$, $B$ and $C$.




**[1]**

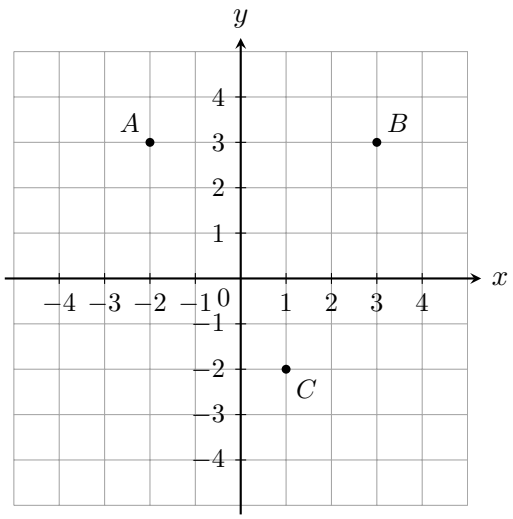
**(b)** Find the coordinates of the fourth vertex, $D$.

( ..... , ..... ) **[2]**

**(c)** Write down the coordinates of the centre of the parallelogram.

( ..... , ..... ) **[2]**

**841** The diagram shows three points, $A$, $B$, and $C$, plotted on a Cartesian coordinate grid.




**(a)** Write down the coordinates of point $A$ and point $C$.

$A =$ ..... , $C =$ ..... **[2]**

**(b)** Write down the coordinates of point $D$ such that $ABCD$ is a parallelogram.

$D =$ ( ..... , ..... ) **[2]**

**(c)** Calculate the area of parallelogram $ABCD$.

..... square units **[2]**

**842** Three vertices of a parallelogram $PQRS$ are $P(-2, 1)$, $Q(4, 3)$, and $R(6, -2)$.

**(a)** Find the coordinates of vertex $S$.

( ..... , ..... ) **[2]**

**(b)** The diagonals $PR$ and $QS$ intersect at point $M$.
Find the coordinates of $M$.

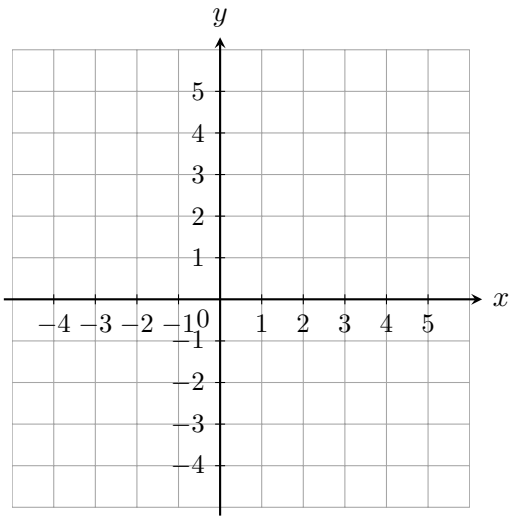
( ..... , ..... ) **[2]**

**(c)** Calculate the area of parallelogram $PQRS$.

Area = ..... **[2]**

**843** In a Cartesian coordinate plane, a kite $PQRS$ has vertices $P(1, 5)$, $Q(4, 2)$, $R(1, -4)$, and $S(-2, 2)$.

**(a)** Plot the points $P(1, 5)$, $Q(4, 2)$, $R(1, -4)$, and $S(-2, 2)$ on the grid below and join them in order to form the kite $PQRS$.




**[2]**

**(b)** Write down the coordinates of the point where the diagonals of kite $PQRS$ intersect.

( ..... , ..... ) **[1]**

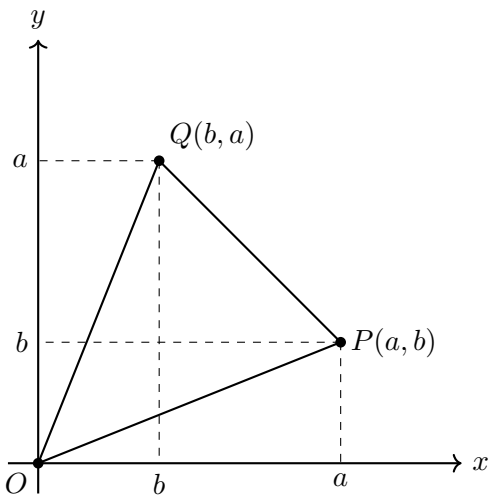
**(c)** Calculate the area of kite $PQRS$.

..... square units **[2]**

**(d)** Point $P$ is reflected in the $x$-axis to give point $P'$, and point $Q$ is reflected in the $y$-axis to give point $Q'$.
Write down the coordinates of $P'$ and $Q'$.

$P' =$ ..... , $Q' =$ ..... **[2]**

**844** The points $P(a, b)$ and $Q(b, a)$ are in the first quadrant, with $a > b$.
The origin is denoted by $O(0, 0)$.




**(a)** By drawing a suitable rectangle or otherwise, show that the area of triangle $OPQ$ is $\frac{1}{2}(a^2 - b^2)$.

**[3]**

**(b)** The area of triangle $OPQ$ is 24 and the length of $PQ$ is $4\sqrt{2}$.
Find the values of $a$ and $b$.

$a =$ ..... , $b =$ ..... **[5]**

**845** A rectangle has its sides parallel to the coordinate axes.
Two opposite vertices of the rectangle are $A(-2, -4)$ and $C(x, y)$, where $x > 0$ and $y > 0$.
The area of the rectangle is 54 square units and its perimeter is 30 units.

**(a)** Show that $x + y = 9$.

**[2]**

**(b)** Form an equation in $x$ and solve it to find the two possible pairs of coordinates for $C$.

**[4]**

**(c)** For the case where $x > y$, write down the coordinates of the other two vertices of the rectangle.

**[2]**

**846** Points $A$ and $B$ have coordinates $(3k, -k)$ and $(k+4, -5k)$ respectively, where $k$ is a constant.

**(a)** The midpoint of the line segment $AB$ lies on the $y$-axis.
Find the value of $k$.

$k =$ ..... **[2]**

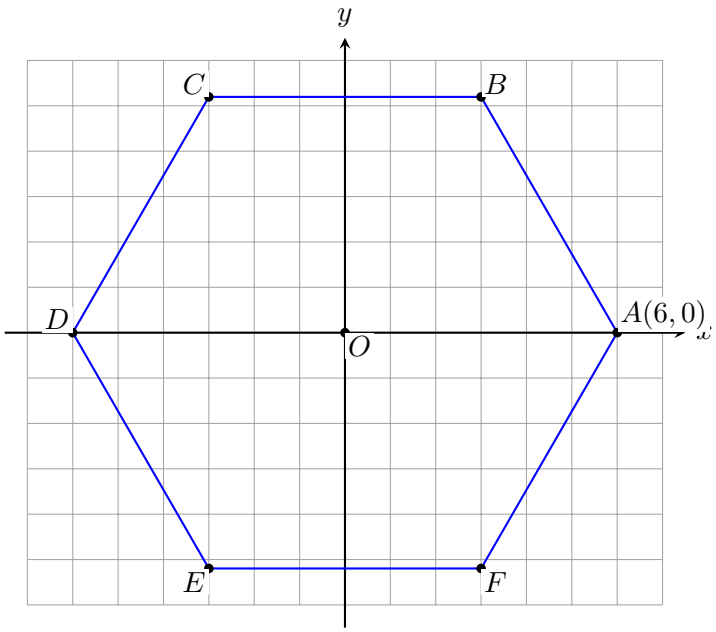
**(b)** Using your value of $k$ from part (a), find the exact length of $AB$.
Give your answer in the form $a\sqrt{b}$ where $b$ is a prime number.

..... **[3]**

**(c)** A point $C$ lies on the $x$-axis.
The distance $AC$ is equal to the distance $BC$.
Find the coordinates of $C$.

( ..... , ..... ) **[4]**

**847** A regular hexagon $ABCDEF$ has its center at the origin $O(0, 0)$.
The point $A$ has coordinates $(6, 0)$.




**(a)** Write down the coordinates of point $D$.

( ..... , ..... ) **[1]**

**(b)** Calculate the coordinates of point $B$.
Give your values correct to 2 decimal places.

( ..... , ..... ) **[3]**

**(c)** Point $P$ lies on the line segment $AD$ such that $AP : PD = 1 : 3$.
Find the coordinates of $P$.

( ..... , ..... ) **[2]**

**848** A straight line has the equation $2y + 3x = 12$.
The line intersects the $y$-axis at $A$ and the $x$-axis at $B$.
The point $O$ is the origin $(0, 0)$.

**(a)** Find the coordinates of $A$ and $B$.

$A =$ ..... , $B =$ ..... **[2]**

**(b)** Point $C$ is such that $OACB$ is a rectangle.
Write down the coordinates of $C$.

( ( ..... , ..... ) ) **[1]**

**(c)** Point $P$ lies on the line segment $AB$ and has an $x$-coordinate of 2.
Find the coordinates of $P$.

( ( ..... , ..... ) ) **[2]**

**(d)** Find the area of triangle $OPB$.

..... **[2]**

**849** A straight line $L$ has the equation $3x + 2y = 18$.

**(a)** The line $L$ crosses the $x$-axis at point $A$ and the $y$-axis at point $B$.
Find the coordinates of $A$ and $B$.

$A =$

$B =$ **[2]**

**(b)** Point $P$ lies on the line segment $AB$ such that the ratio $AP : PB = 2 : 1$.
Find the coordinates of $P$.

( ..... , ..... ) **[3]**

**(c)** Does the point $(4, 3)$ lie on the line $L$?
Show how you decide.

**[2]**

**850** Point $A$ has coordinates $(2k-1, k+3)$ and point $B$ has coordinates $(5-k, 3k-5)$, where $k$ is a constant.

**(a)** Given that point $A$ lies in the second quadrant and both of its coordinates are integers, write down the possible values of $k$.

$k =$ ..... **[2]**

**(b)** In another case, point $A$ lies on the line $y = 2x - 4$.
Find the value of $k$ and write down the coordinates of point $B$.

$k =$ ..... , $B =$ ..... **[3]**

**(c)** Find the value of $k$ for which the midpoint of $AB$ lies on the $y$-axis.

$k =$ ..... **[2]**

**851** A rectangle $ABCD$ has vertices $A(-3,-1)$, $B(5,-1)$, $C(5,k)$, and $D(h,k)$, where $k>0$.

**(a)** State the value of $h$ and the quadrant in which point $D$ lies.

$h =$ ..... , Quadrant: ..... **[2]**

**(b)** Given that the area of rectangle $ABCD$ is 52 square units, calculate the value of $k$.

$k =$ ..... **[3]**

**(c)** Point $M$ is the centre of rectangle $ABCD$.
Find the coordinates of $M$.

$M =$ ( ..... , ..... ) **[2]**

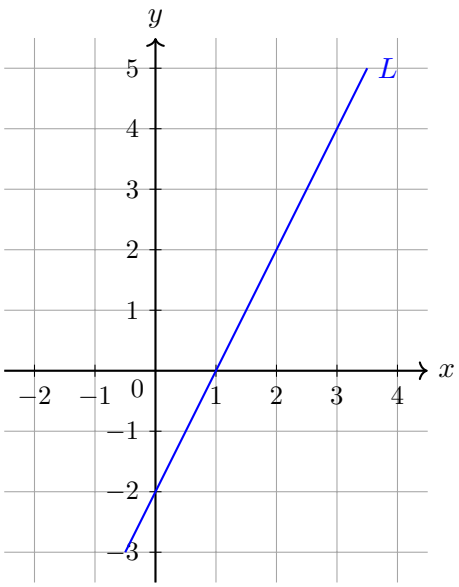
**(d)** Point $E(p,q)$ lies on the $y$-axis such that triangle $ABE$ has an area of 18 square units and $q>0$.
Find the coordinates of $E$.

$E =$ ( ..... , ..... ) **[3]**

**852** Points $A$ and $B$ have coordinates $(2, -3)$ and $(6, 5)$ respectively.
Work out the gradient of the line $AB$.

..... **[2]**

**853** The diagram shows a straight line $L$ drawn on a coordinate grid.




**(a)** Find the gradient of line $L$.

..... **[2]**

**(b)** Write down the coordinates of the $y$-intercept of line $L$.

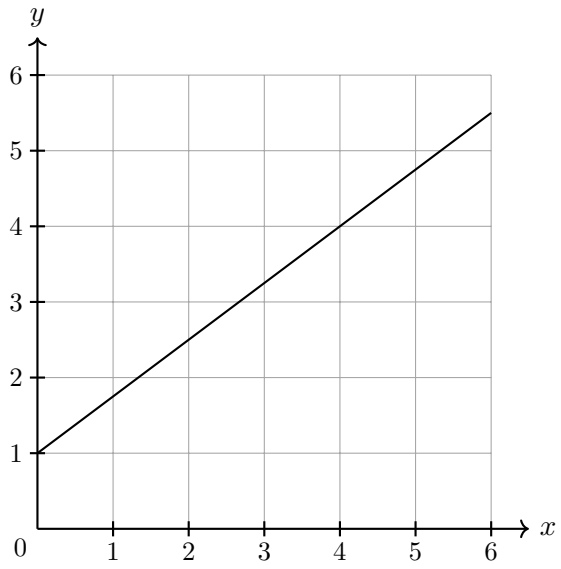
( ..... , ..... ) **[1]**

**854** The points $A$ and $B$ have coordinates $(-2, 8)$ and $(4, -4)$ respectively.

**(a)** Work out the gradient of the line $AB$.

..... **[2]**

**(b)** A straight line $L$ is drawn on the grid.
Find the gradient of line $L$.




..... **[2]**

**855** The points $A(-4, 7)$ and $B(2, -5)$ lie on a straight line.

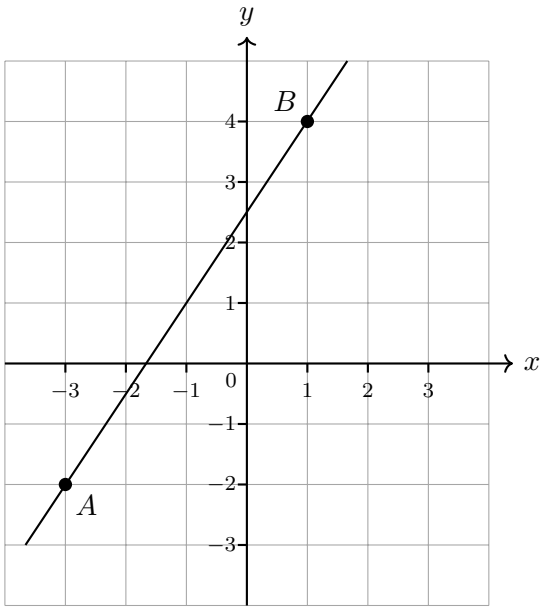
**(a)** Calculate the gradient of the line $AB$.

..... **[2]**

**(b)** Point $C$ has coordinates $(0, c)$.
The gradient of the line $AC$ is equal to the gradient of the line $AB$.
Find the value of $c$.

$c =$ ..... **[2]**

**856** **(a)** The diagram shows a straight line passing through points $A$ and $B$.
Work out the gradient of the line $AB$.

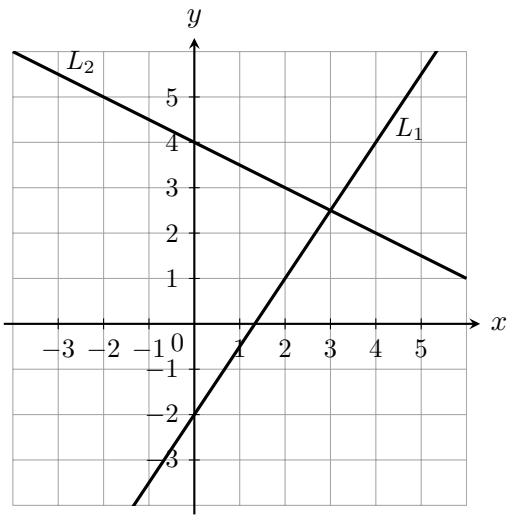



Gradient = ..... **[2]**

**(b)** Point $C$ has coordinates $(-2, 5)$ and point $D$ has coordinates $(4, -7)$.
Calculate the gradient of the line $CD$.

Gradient = ..... **[2]**

**857** The diagram shows two straight lines, $L_1$ and $L_2$, drawn on a coordinate grid.




**(a)** Find the gradient of line $L_1$.

..... **[2]**

**(b)** Find the gradient of line $L_2$.

..... **[2]**

**(c)** A straight line $L_3$ passes through the points $P(-3, 7)$ and $Q(5, -9)$.
Calculate the gradient of line $L_3$.

..... **[2]**

**858** **(a)** A straight line passes through the points $P(2, k)$ and $Q(5, 3k - 1)$.

**(i)** Find an expression for the gradient of the line $PQ$ in terms of $k$.
Give your answer in its simplest form.

Gradient = ..... **[2]**

**(ii)** The gradient of the line $PQ$ is 4.
Find the value of $k$.

$k$ = ..... **[2]**

**859** The points $A(-2, 5)$, $B(4, k)$, and $C(10, -1)$ lie in a coordinate plane.

**(a)** Given that the gradient of the line segment $AB$ is $-\frac{1}{2}$, calculate the value of $k$.

$k =$ ..... **[3]**

**(b)** Using your value of $k$ from part (a), show that the points $A$, $B$, and $C$ lie on the same straight line.

**[2]**

**(c)** Point $D$ has coordinates $(p, 3p - 1)$.
Find the value of $p$ for which the line $AD$ has a gradient of 2.

$p =$ ..... **[3]**

**860** **(a)** Point $P$ has coordinates $(2a, 3a)$ and point $Q$ has coordinates $(5a, a + 4)$.

Find an expression for the gradient of the line $PQ$ in terms of $a$.
Give your answer in its simplest form.

..... **[2]**

**(b)** The line $PQ$ is perpendicular to a line with gradient 3.
Work out the value of $a$.

$a =$ ..... **[3]**

**(c)** The points $A(-1, 2)$, $B(3, 8)$ and $C(c, 23)$ lie on a straight line.
Work out the value of $c$.

$c =$ ..... **[3]**

**861** The line $L_1$ passes through the points $P(-2, 3a)$ and $Q(4, a + 6)$.

**(a)** Find an expression for the gradient of $L_1$ in terms of $a$.
Give your answer in its simplest form.

..... **[3]**

**(b)** The line $L_2$ has equation $y = 3x - 5$.
Given that $L_1$ is parallel to $L_2$, find the value of $a$.

$a =$ ..... **[2]**

**(c)** Using your value of $a$, find the coordinates of the midpoint of $PQ$.

( ..... , ..... ) **[3]**

**862** The points $A(1, 2)$, $B(5, k)$ and $C(k, 2k)$ are three distinct collinear points, where $k$ is a constant.

**(a)** Write down an expression, in terms of $k$, for the gradient of the line segment $AB$.

..... **[1]**

**(b)** By finding an expression for the gradient of $BC$, show that $k^2 - 11k + 10 = 0$.

**[3]**

**(c)** Solve the equation to find the two possible values of $k$.

$k =$ ..... , k =

..... **[2]**

**(d)** Calculate the exact gradient of the line passing through point $B$ and the origin $(0, 0)$.

..... **[2]**

**863** The points $A$, $B$ and $C$ have coordinates $(1, 3)$, $(4, 9)$ and $(8, y)$ respectively.

**(a)** The points $A$, $B$ and $C$ lie on the same straight line.
Work out the value of $y$.

$y =$ ..... **[2]**

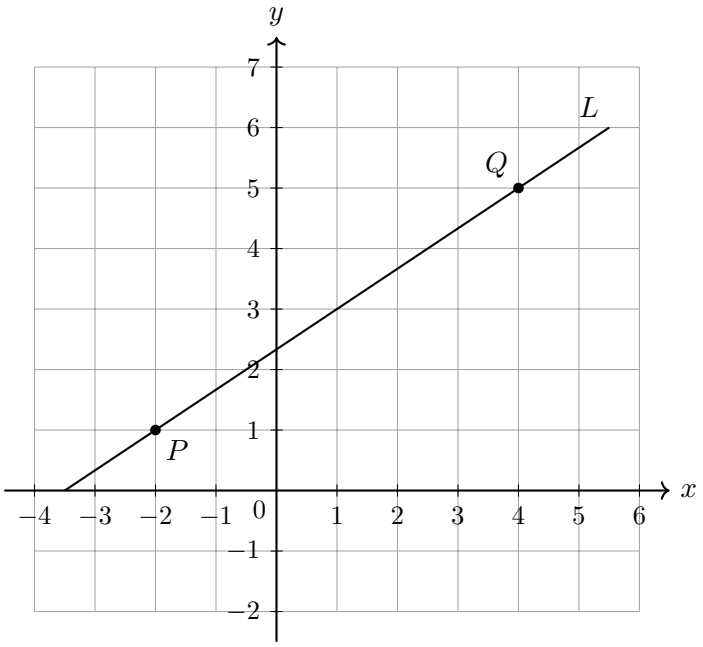
**(b)** A different point $D$ has coordinates $(p, p^2 - p - 3)$.
The gradient of the line $AD$ is 2.
Work out the possible values of $p$.

$p =$ ..... , $p =$ ..... **[4]**

**(c)** A line $L$ has gradient $m$.
The line $L$ passes through the points $(3, m)$ and $(m + 1, 12)$.
Find the two possible values of $m$.

$m =$ ..... , $m =$ ..... **[4]**

**864** **(a)** The diagram shows a straight line $L$ passing through the points $P(-2, 1)$ and $Q(4, 5)$.
Find the gradient of line $L$.




..... **[2]**

**(b)** A different line passes through the points $(3, k)$ and $(8, -2)$.
The gradient of this line is $-1.4$.
Calculate the value of $k$.

$k =$ ..... **[3]**

**865** Point $A$ has coordinates $(-3, 5)$ and point $B$ has coordinates $(4, -9)$.

**(a)** Calculate the gradient of the line $AB$.

..... **[2]**

**(b)** Point $C$ has coordinates $(k, 1)$.
The gradient of the line $AC$ is equal to the gradient of the line $AB$.
Find the value of $k$.

$k =$ ..... **[3]**

**866** Point $P$ has coordinates $(k-3, 2k)$ and point $Q$ has coordinates $(2, 5)$.

**(a)** The gradient of the line $PQ$ is $-3$.
Work out the value of $k$.

$k =$ ..... **[3]**

**(b)** Using your value of $k$ from part (a), find the coordinates of the midpoint of $PQ$.

( ..... , ..... ) **[2]**

**867** A straight line passes through the points $P(3k, 4)$ and $Q(k-2, 8)$.
The gradient of this line is $-\frac{1}{2}$.

**(a)** Find the value of $k$.

$k =$ ..... **[3]**

**(b)** Write down the coordinates of $P$ and $Q$.

$P =$ ..... , $Q =$ ..... **[1]**

**(c)** Another point $R$ lies on the $y$-axis such that the gradient of the line passing through $P$ and $R$ is 1.
Find the coordinates of $R$.

( ..... , ..... ) **[3]**

**868** Points $A(-1, -4)$ and $B(3, 8)$ lie on a straight line.

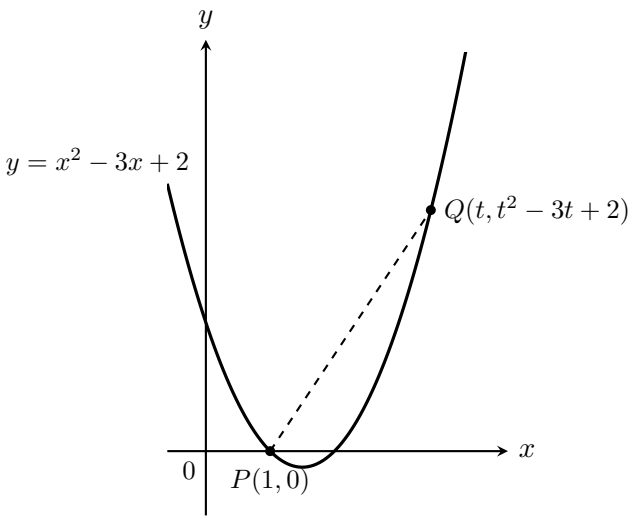
**(a)** Point $C(p, 23)$ also lies on the line passing through $A$ and $B$.
Find the value of $p$.

$p =$ ..... **[3]**

**(b)** Point $D$ has coordinates $(q, 2q - 1)$.
The gradient of the line $AD$ is equal to $-\frac{1}{3}$.
Find the value of $q$.

$q =$ ..... **[3]**

**869** The diagram shows a sketch of the curve $y = x^2 - 3x + 2$ and two points, $P(1, 0)$ and $Q(t, t^2 - 3t + 2)$, on the curve, where $t > 1$.




**(a)** Show that the gradient of the chord $PQ$ simplifies to $t - 2$.

**[3]**

**(b)** Find the coordinates of point $Q$ when the gradient of the chord $PQ$ is 2.8.

$Q = (\ldots\ldots\ ,\ \ldots\ldots)$ **[3]**

**(c)** Point $R$ has coordinates $(3.2, -1.5)$.
Find the value of $t$, where $t > 1$, for which the gradient of $QR$ is equal to the gradient of $PQ$.
Give your answer correct to 3 significant figures.

$t =$ ..... **[4]**

**870** The points $A$ and $B$ have coordinates $(2, -3)$ and $(8, 5)$ respectively.

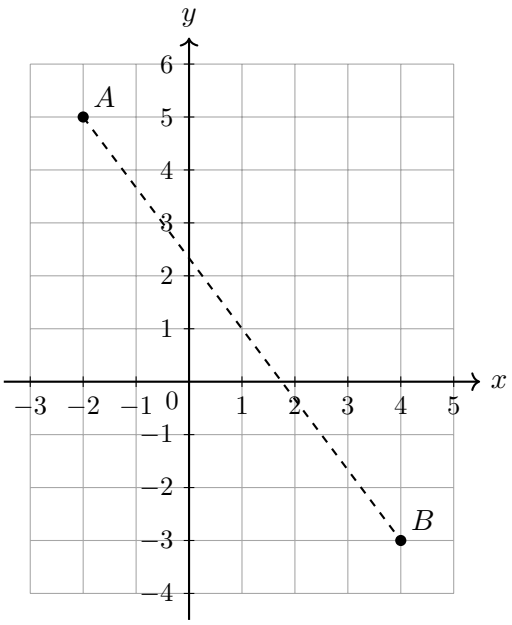
**(a)** Find the coordinates of the midpoint of $AB$.

( ..... , ..... ) **[2]**

**(b)** Calculate the length of the line segment $AB$.

$AB =$ ..... **[2]**

**871** The points $A(-2, 5)$ and $B(4, -3)$ are shown on the coordinate grid.




**(a)** Find the coordinates of the midpoint of the line segment $AB$.

( ( ..... , ..... ) ) **[2]**

**(b)** Calculate the exact length of the line segment $AB$.

..... **[3]**

**872** Point $A$ has coordinates $(1, 4)$ and point $B$ has coordinates $(7, -4)$.

**(a)** Calculate the coordinates of the midpoint of the line segment $AB$.

( ..... , ..... ) **[2]**

**(b)** Calculate the length of the line segment $AB$.

..... **[3]**

**873** The points $A$ and $B$ have coordinates $(2, -3)$ and $(8, 5)$ respectively.

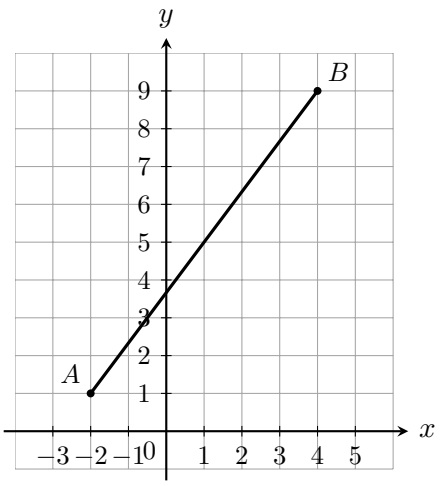
**(a)** Find the coordinates of the midpoint of the line segment $AB$.

( ..... , ..... ) **[2]**

**(b)** Calculate the exact length of the line segment $AB$.

..... **[3]**

**874** The diagram shows a line segment $AB$ plotted on a Cartesian coordinate grid.




**(a)** Write down the coordinates of point $A$ and point $B$.

$A =$ ..... , $B =$ ..... **[2]**

**(b)** Find the coordinates of the midpoint $M$ of the line segment $AB$.

$M =$ ( ..... , ..... ) **[2]**

**(c)** Calculate the length of the line segment $AB$.

..... units **[2]**

**875** The point $M(4, 1)$ is the midpoint of the line segment joining $P(-2, 7)$ and $Q$.

**(a)** Find the coordinates of $Q$.

( ..... , ..... ) **[2]**

**(b)** The point $R$ has coordinates $(k, 10)$, where $k$ is a constant.
The length of $PR$ is 5.
Find the two possible values of $k$.

$k =$ ..... , $k =$ ..... **[3]**

**876** A circle has its centre at the origin, $O(0, 0)$.
The point $P(-3, 4)$ lies on the circumference of this circle.

**(a)** Work out the radius of the circle.

..... **[2]**

**(b)** The line segment $PQ$ is a diameter of the circle.
Find the coordinates of $Q$.

( ..... , ..... ) **[2]**

**(c)** The point $R$ has coordinates $(k, \sqrt{11})$.
$R$ also lies on the circumference of the circle, and $k$ is a negative number.
Work out the exact value of $k$.

$k =$ ..... **[3]**

**877** The line segment $PQ$ has midpoint $M(3, -1)$.
The coordinates of point $P$ are $(-2, 4)$.

**(a)** Find the coordinates of point $Q$.

$Q = (\ldots\ldots\ldots\ ,\ \ldots\ldots\ldots)$ **[2]**

**(b)** Calculate the exact length of $PQ$, giving your answer in the form $k\sqrt{2}$, where $k$ is an integer.

$PQ = \ldots\ldots\ldots\ldots\ldots$ **[3]**

**(c)** Point $R$ lies on the $y$-axis such that $PR = QR$.
Find the coordinates of $R$.

$R = (\ldots\ldots\ldots\ ,\ \ldots\ldots\ldots)$ **[4]**

**878** The points $A(-2,-1)$, $B(4,3)$ and $C(8,-3)$ form a triangle.
$M$ is the midpoint of $AB$ and $N$ is the midpoint of $AC$.

**(a)** Find the coordinates of $M$ and the coordinates of $N$.

$M =$ ..... , $N =$ ..... **[2]**

**(b)** Show that the length of $MN$ is exactly half the length of $BC$.

**[4]**

**879** Points $A$ and $B$ have coordinates $(3p, p-2)$ and $(p, 2p+1)$ respectively, where $p$ is a constant.

**(a)** The midpoint of $AB$ lies on the line $y = x + 1$.
Find the value of $p$.

$p =$ ..... **[3]**

**(b)** Given instead that the length of the line segment $AB$ is 6, show that $5p^2 + 6p - 27 = 0$.

**[3]**

**(c)** Solve the equation $5p^2 + 6p - 27 = 0$ to find the two possible values of $p$.

$p =$ ..... , $p =$ ..... **[3]**

**880** A circle has diameter $AB$, where $A$ is the point $(-1, 2)$ and $B$ is the point $(5, 8)$.

**(a)** Find the coordinates of the centre of the circle, $C$.

( ..... , ..... ) **[2]**

**(b)** Find the exact radius of the circle.
Give your answer in the form $a\sqrt{b}$, where $a$ and $b$ are positive integers and $b$ is as small as possible.

..... **[3]**

**(c)** Point $D$ is such that the midpoint of $CD$ is $M\left(\frac{5}{2}, \frac{3}{2}\right)$.

Find the exact length of $AD$, giving your answer in the form $c\sqrt{d}$, where $c$ and $d$ are integers and $d$ is as small as possible.

..... **[5]**

**881** The points $A$ and $B$ have coordinates $(-2, 5)$ and $(4, -3)$ respectively.

**(a)** Calculate the exact length of the line segment $AB$.

..... **[3]**

**(b)** Find the coordinates of the midpoint of $AB$.

( ..... , ..... ) **[2]**

**882** Point $P$ has coordinates $(3, k)$ and point $Q$ has coordinates $(7, 5)$.
The length of the line segment $PQ$ is 5 units.

**(a)** Find the two possible values of $k$.

$k =$ ..... , $k =$ ..... **[3]**

**(b)** $M$ is the midpoint of $PQ$.
For the case where $k < 5$, find the coordinates of $M$.

( ( ..... , ..... ) ) **[2]**

**883** The midpoint of the line segment $PQ$ is $M(1, -2)$.
The coordinates of $P$ are $(-3, 5)$.

**(a)** Find the coordinates of $Q$.

( ..... , ..... ) **[2]**

**(b)** Calculate the length of the line segment $PQ$.
Give your answer correct to 3 significant figures.

..... **[3]**

**884** Point $P$ has coordinates $(-3, 5)$ and point $Q$ has coordinates $(k, -1)$.
The length of the line segment $PQ$ is 10 units.

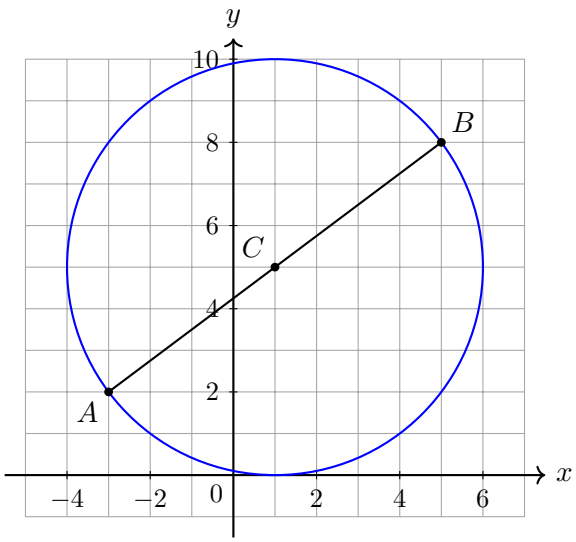
**(a)** Find the two possible values of $k$.

$k =$ ..... , $k =$ ..... **[4]**

**(b)** Given that $k > 0$, point $M$ is the midpoint of $PQ$.
Find the coordinates of $M$.

( ..... , ..... ) **[2]**

**885** The points $A(-3, 2)$ and $B(5, 8)$ are the endpoints of a diameter of a circle.




**(a)** Find the coordinates of the centre of the circle, $C$.

( ( ..... , ..... ) ) **[2]**

**(b)** Calculate the area of the circle.
Give your answer correct to 1 decimal place.

..... **[3]**

**(c)** The point $D(k, 8)$ lies on the circumference of the circle.
Find the possible values of $k$.

$k =$ ..... , $k =$ ..... **[3]**

**886** The points $A(2p, p)$ and $B(3p, 4p)$ are joined by a straight line, where $p > 0$.

**(a)** The length of the line segment $AB$ is $5\sqrt{10}$.
Find the value of $p$.

$p =$ ..... **[4]**

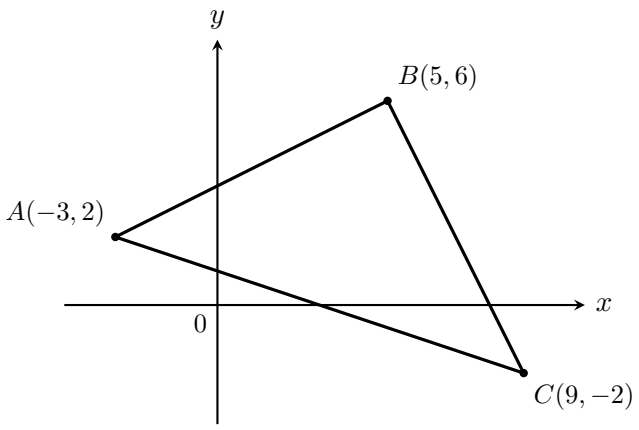
**(b)** Point $C$ lies on the line segment $AB$ such that the ratio $AC : CB = 2 : 3$.
Using your value of $p$ from part (a), find the coordinates of $C$.

( ..... , ..... ) **[3]**

**(c)** The point $D(x, y)$ is such that $C$ is the midpoint of $BD$.
Work out the coordinates of $D$.

( ..... , ..... ) **[2]**

**887** The diagram shows a sketch of triangle $ABC$ with vertices $A(-3, 2)$, $B(5, 6)$, and $C(9, -2)$.




**(a)** Show that triangle $ABC$ is isosceles.

**[4]**

**(b)** $M$ is the midpoint of $AC$.

**(i)** Find the coordinates of $M$.

$M = (\ldots\ldots\ ,\ \ldots\ldots)$ **[2]**

**(ii)** Calculate the length of $BM$, giving your answer correct to 3 significant figures.

$BM = \ldots\ldots\ldots\ldots\ldots$ **[2]**

**(c)** Calculate the area of triangle $ABC$.

..... square units **[2]**

**(d)** Point $D$ is such that $ABCD$ is a square.
Find the coordinates of $D$.

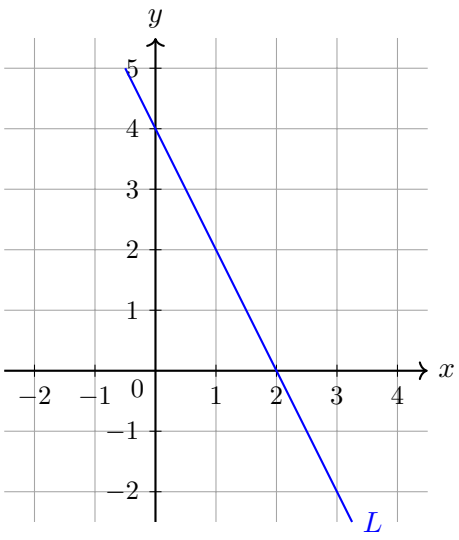
$D = ($ ..... , ..... $)$ **[2]**

**888** Find the gradient and the $y$-intercept of the line with equation

$3x + 5y = 20$

Gradient = ..... , $y$-intercept = ..... **[3]**

**889** The diagram shows a straight line $L$ drawn on a coordinate grid.




**(a)** Find the equation of line $L$.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

**(b)** Write the equation of line $L$ in the form $ax + by = c$, where $a$, $b$ and $c$ are integers.

..... **[1]**

**890** A straight line has the equation $4x + 5y = 20$.

**(a)** Find the gradient and the $y$-intercept of this line.

Gradient = ..... , $y$-intercept = ..... **[3]**

**(b)** A different line is parallel to the $y$-axis and passes through the point $(3, -8)$.
Write down the equation of this line.

..... **[1]**

**891** **(a)** Find the gradient and the $y$-intercept of the line $3x + 5y = 15$.

Gradient = ..... , $y$-intercept = ..... **[3]**

**(b)** Find the equation of the line parallel to $3x + 5y = 15$ that passes through the point $(0, -4)$. Give your answer in the form $y = mx + c$.

$y =$ ..... **[2]**

**892** A straight line $L$ has the equation $3x - 4y = 12$.

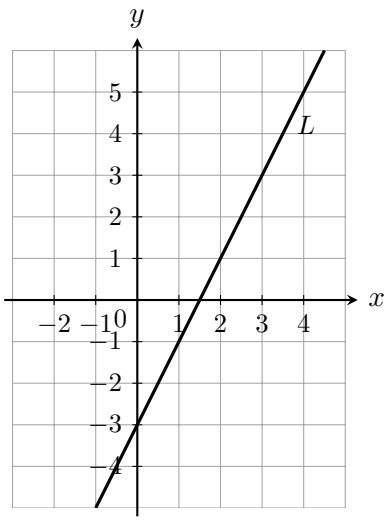
**(a)** Find the gradient and the $y$-intercept of line $L$.

Gradient = ..... , $y$-intercept = ..... **[3]**

**(b)** Another line $M$ is parallel to $L$ and passes through the point $(0, 5)$.
Write down the equation of line $M$ in the form $y = mx + c$.

$y =$ ..... **[2]**

**893** The diagram shows a straight line $L$ drawn on a coordinate grid.




**(a)** Find the equation of line $L$ in the form $y = mx + c$.

$y =$ ..... **[3]**

**(b)** A vertical line $M$ passes through the point $(-2, 4)$.
Write down the equation of line $M$.

..... **[1]**

**(c)** Find the coordinates of the point where line $L$ and line $M$ intersect.

( ..... , ..... ) **[2]**

**894** Line $L_1$ has equation $4x + 5y = 20$.

**(a)** Find the gradient and the $y$-intercept of line $L_1$.

Gradient = ..... , $y$-intercept = ..... **[3]**

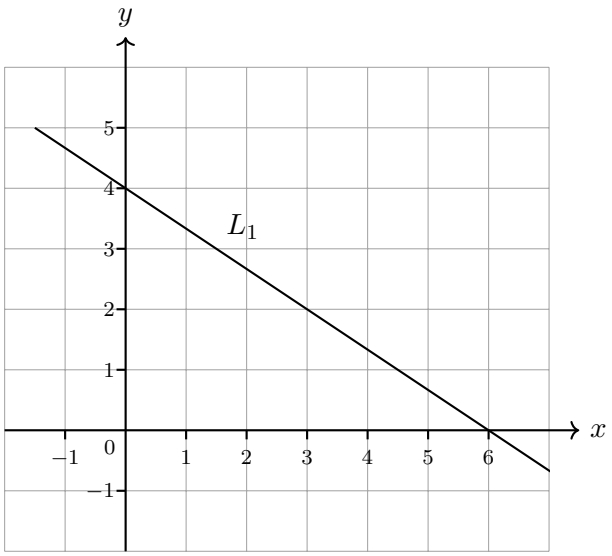
**(b)** Find the coordinates of the point where line $L_1$ crosses the $x$-axis.

( ..... , ..... ) **[2]**

**(c)** Line $L_2$ passes through the points $(-3, -5)$ and $(3, 7)$.
Find the equation of line $L_2$ in the form $ax + by = c$, where $a$, $b$, and $c$ are integers.

..... **[3]**

**895** The diagram shows a straight line $L_1$ drawn on a coordinate grid.




**(a)** Find the equation of line $L_1$.
Give your answer in the form $ax + by = c$, where $a$, $b$ and $c$ are integers.

..... **[3]**

**(b)** The line $L_2$ has equation $\frac{x}{3} - \frac{y}{2} = 1$.

**(i)** Write the equation of $L_2$ in the form $y = mx + c$.

$y =$ ..... **[2]**

**(ii)** Find the coordinates of the point of intersection of $L_1$ and $L_2$.

( ..... , ..... ) **[2]**

**896** $A$ is the point $(p, -1)$ and $B$ is the point $(5, 2p - 1)$.
The gradient of the line $AB$ is 3.

**(a)** Find the value of $p$.

$p =$ ..... **[3]**

**(b)** Find the equation of the line $AB$.
Give your answer in the form $y = mx + c$.

..... **[2]**

**(c)** The line $AB$ intersects the $x$-axis at the point $C$ and the $y$-axis at the point $D$.
Work out the area of triangle $COD$, where $O$ is the origin.

..... **[3]**

**897** A straight line $L$ passes through the points $A(-2, 3k)$ and $B(4, k)$, where $k$ is a non-zero constant.

**(a)** Find the gradient of line $L$ in terms of $k$.
Give your answer in its simplest form.

..... **[2]**

**(b)** Find the equation of line $L$ in the form $ax + by = c$, where $a$, $b$, and $c$ are expressions in terms of $k$.

..... **[3]**

**(c)** Line $L$ also passes through the point $(1, 10)$.
Find the value of $k$.

$k =$ ..... **[2]**

**(d)** Hence, write down the coordinates of the $x$-intercept of line $L$.

( ..... , ..... ) **[2]**

**898** The line $L$ passes through the points $A(2a, a+4)$ and $B(a-1, 3a+2)$.
The gradient of line $L$ is $-4$.

**(a)** Find the value of $a$.

$a =$ ..... **[3]**

**(b)** Find the equation of line $L$.
Give your answer in the form $px + qy = r$, where $p$, $q$ and $r$ are integers and $p > 0$.

..... **[3]**

**(c)** Line $L$ intersects the line $y = 3x - 2$ at the point $M$.
Find the coordinates of $M$.

( ..... , ..... ) **[3]**

**899** $A$ is the point $(-3, 1)$ and $B$ is the point $(5, 5)$.

**(a)** Find the equation of the perpendicular bisector of the line segment $AB$.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[4]**

**(b)** The perpendicular bisector of $AB$ intersects the line $x - 2y = 5$ at the point $P$.
Find the coordinates of $P$.

( ..... , ..... ) **[3]**

**(c)** Find the exact length of $AP$.

..... **[3]**

**900** The line $L$ passes through the points $P(-3, 10)$ and $Q(5, -2)$.

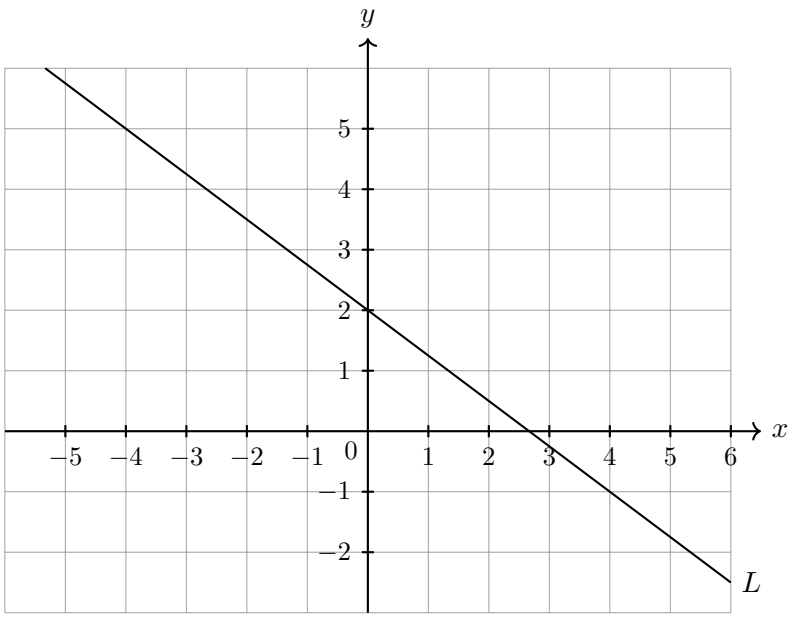
**(a)** Find the equation of line $L$.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

**(b)** Line $M$ has the equation $2y + 3x = 15$.
Find the coordinates of the point where line $M$ intersects the $x$-axis.

( ..... , ..... ) **[2]**

**901** The diagram shows a line $L$ drawn on a coordinate grid.




**(a)** Work out the gradient of line $L$.

..... **[2]**

**(b)** Find the equation of line $L$.
Give your answer in the form $ax + by = c$, where $a$, $b$ and $c$ are integers.

..... **[3]**

**902** Point $A$ has coordinates $(-4, 3)$ and point $B$ has coordinates $(6, -2)$.

**(a)** Find the equation of the line passing through $A$ and $B$.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

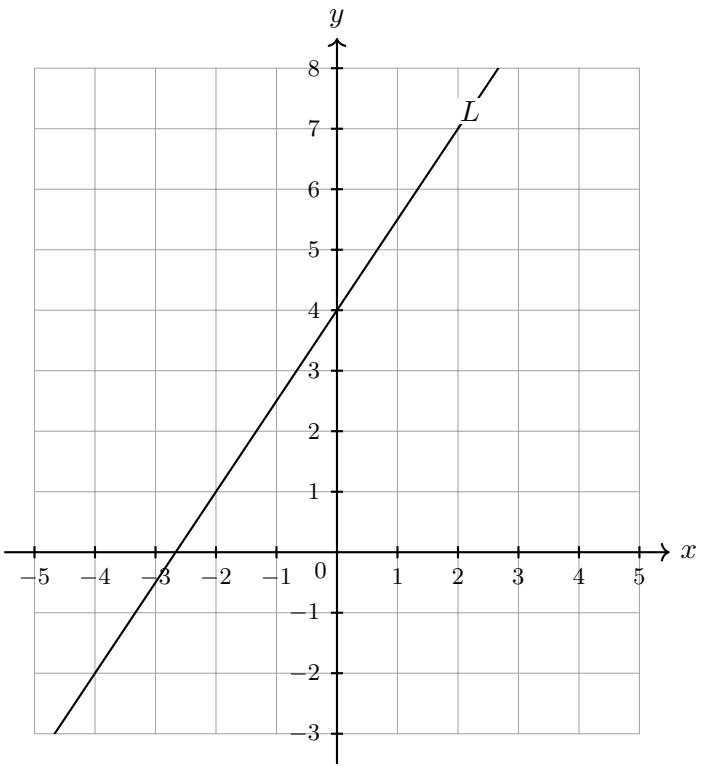
**(b)** The line $3x - 5y = 15$ intersects the $x$-axis at point $C$ and the $y$-axis at point $D$.
Find the coordinates of $C$ and $D$.

$C =$ ..... , $D =$ ..... **[2]**

**(c)** Write down the equation of the line that is parallel to the $y$-axis and passes through point $A$.

..... **[1]**

**903** The diagram shows a straight line $L$.



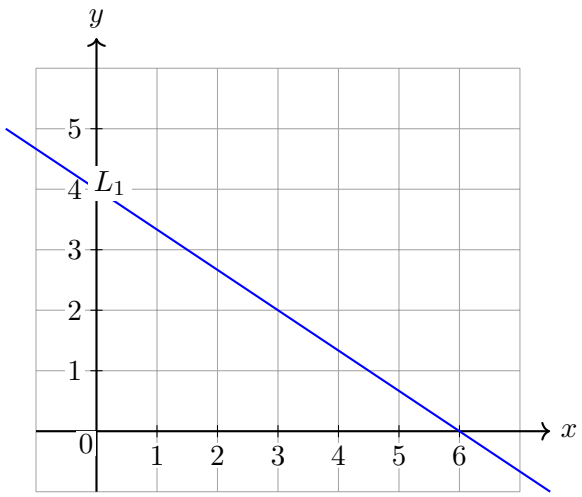

**(a)** Find the equation of line $L$.
Give your answer in the form $ax + by = c$, where $a$, $b$, and $c$ are integers.

..... **[3]**

**(b)** Line $M$ is perpendicular to line $L$ and passes through the point $(9, -1)$.
Find the equation of line $M$ in the form $y = mx + c$.

$y =$ ..... **[3]**

**904** The diagram shows a straight line $L_1$ drawn on a coordinate grid.



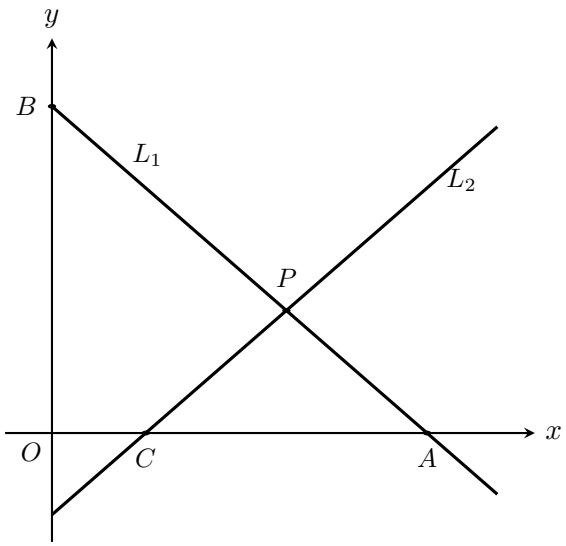

**(a)** Find the equation of line $L_1$ in the form $y = mx + c$.

$y =$ ..... **[3]**

**(b)** Line $L_2$ passes through the point $(2, 7)$ and the midpoint of the segment of $L_1$ that lies between the axes.
Find the equation of $L_2$ in the form $ax + by = c$, where $a$, $b$, and $c$ are integers.

..... **[4]**

**905** The diagram shows a sketch of two straight lines, $L_1$ and $L_2$, which intersect at the point $P$.
Line $L_1$ has equation $3x + 2y = 24$ and crosses the $x$-axis at $A$ and the $y$-axis at $B$.
Line $L_2$ has equation $y = 1.5x - 3$ and crosses the $x$-axis at $C$.




**(a)** Write down the coordinates of $A$, $B$, and $C$.

$A =$ ..... , $B =$ ..... , $C =$ ..... **[3]**

**(b)** Find the coordinates of the point of intersection $P$.

$P =$ ( ..... , ..... ) **[3]**

**(c)** Calculate the area of quadrilateral $OCPB$.

..... square units **[3]**

**(d)** A third line $L_3$ has equation $kx - 4y = 14$ and also passes through point $P$.
Calculate the value of $k$.

$k =$ ..... **[2]**

**906** Find the equation of the line that is parallel to $y = -2x + 7$ and passes through the point $(3, 4)$. Give your answer in the form $y = mx + c$.

$y =$ ..... **[2]**

**907** Line $L$ has the equation $4x + 2y = 9$.
Find the equation of the line parallel to $L$ that passes through the point $(3, -1)$.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

**908** Line $L$ has the equation $y = 6 - 3x$.

Find the equation of the line parallel to $L$ that passes through the point $(-2, 5)$.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

**909** Line $L_1$ has the equation $y = 3x - 7$.

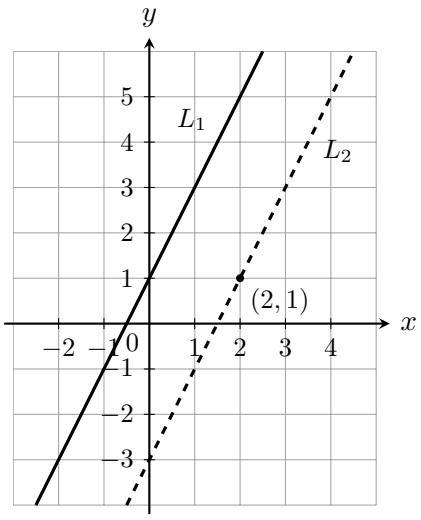
**(a)** Write down the gradient of a line that is parallel to $L_1$.

Gradient = ..... **[1]**

**(b)** Line $L_2$ is parallel to $L_1$ and passes through the point $(2, -5)$.
Find the equation of line $L_2$.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

**910** The diagram shows two parallel straight lines, $L_1$ and $L_2$, on a coordinate grid.
Line $L_1$ has equation $y = 2x + 1$ and line $L_2$ passes through the point $(2, 1)$.




**(a)** Write down the gradient of line $L_1$.

..... **[1]**

**(b)** Find the equation of line $L_2$ in the form $y = mx + c$.

$y =$ ..... **[2]**

**(c)** A third line $L_3$ is parallel to $L_1$ and passes through the point $(-1, 5)$.
Find the equation of line $L_3$ in the form $y = mx + c$.

$y =$ ..... **[2]**

**911** A straight line $L_1$ has the equation $2y - 5x = 8$.

**(a)** Find the gradient of line $L_1$.

..... **[2]**

**(b)** Line $L_2$ is parallel to $L_1$ and passes through the point $(0, -3)$.
Write down the equation of $L_2$.

$y =$ ..... **[2]**

**(c)** Line $L_3$ is parallel to $L_1$ and passes through the point $(4, 1)$.
Find the equation of $L_3$ in the form $ax + by = c$, where $a$, $b$, and $c$ are integers.

..... **[3]**

**912** The points $A$ and $B$ have coordinates $(-3, 5)$ and $(5, -1)$ respectively.
Line $L$ is parallel to the line segment $AB$ and passes through the point $(4, -2)$.

**(a)** Find the gradient of the line segment $AB$.

..... **[2]**

**(b)** Find the equation of line $L$.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

**913** Line $A$ has the equation $3x + 2y = 14$.

**(a)** Find the gradient of line $A$.

Gradient = ..... **[2]**

**(b)** Line $B$ is parallel to line $A$ and passes through the point $(-4, 9)$.
Find the equation of line $B$.
Give your answer in the form $ax + by = c$, where $a$, $b$ and $c$ are integers.

..... **[4]**

**914** Line $L_1$ has equation $3x + 4y = 12$.

**(a)** Find the gradient of line $L_1$.

..... **[2]**

**(b)** Line $L_2$ is parallel to line $L_1$ and passes through the point $(4, 3)$.
Find the equation of line $L_2$ in the form $ax + by = c$, where $a$, $b$, and $c$ are integers.

..... **[3]**

**(c)** Line $L_2$ crosses the $x$-axis at $A$ and the $y$-axis at $B$.
Find the coordinates of the midpoint of $AB$.

( ..... , ..... ) **[3]**

**915** Line $P$ has equation $4x - ky = 8$.
Line $Q$ has equation $y = (k - 3)x + 2$.
The lines $P$ and $Q$ are parallel.

**(a)** Find the two possible values of the constant $k$.

$k =$ ..... **[4]**

**(b)** Given that line $P$ has a positive gradient,

**(i)** find the equation of line $P$ in the form $y = mx + c$.

$y =$ ..... **[2]**

**(ii)** A line $R$ is parallel to line $P$ and passes through the point $(-5, 7)$.
Find the equation of line $R$ in the form $ax + by = c$, where $a$, $b$, and $c$ are integers.

..... **[2]**

**916** Line $M$ has the equation $5x - 2y = 15$.
Point $P$ has coordinates $(-4, 1)$.
Find the equation of the line parallel to $M$ that passes through $P$.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

**917** $A$ is the point $(-3, 2)$ and $B$ is the point $(5, 6)$.

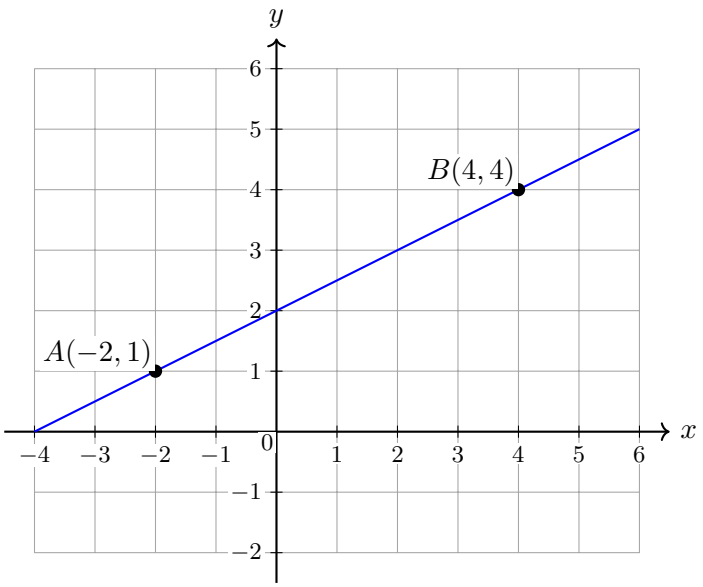
**(a)** Calculate the gradient of the line $AB$.

..... **[2]**

**(b)** Line $L$ is parallel to $AB$ and passes through the point $(4, -1)$.
Find the equation of line $L$, giving your answer in the form $ax + by = c$, where $a$, $b$ and $c$ are integers.

..... **[3]**

**918** The diagram shows the points $A(-2, 1)$ and $B(4, 4)$ joined by a straight line.




**(a)** Calculate the gradient of the line $AB$.

..... **[2]**

**(b)** Line $M$ passes through the point $(2, -3)$ and is parallel to the line $AB$.
Find the equation of line $M$ in the form $y = mx + c$.

$y =$ ..... **[3]**

**(c)** Line $M$ intersects the $x$-axis at point $P$ and the $y$-axis at point $Q$.
Calculate the length of the line segment $PQ$.
Give your answer correct to 3 significant figures.

..... **[4]**

**919** Points $A(-1, 5)$ and $B(3, -3)$ lie on line $L_1$.
Line $L_2$ is parallel to $L_1$ and passes through the point $C(2, 7)$.
Line $L_2$ crosses the $x$-axis at $D$ and the $y$-axis at $E$.

**(a)** Find the equation of line $L_2$.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[4]**

**(b)** Calculate the exact area of triangle $ODE$, where $O$ is the origin $(0, 0)$.

..... $units^2$ **[3]**

**920** The points $P(-2, 3)$, $Q(4, 15)$ and $R(k, 2k + 1)$ lie on a coordinate plane, where $k$ is a constant.

**(a)** Line $L$ passes through $R$ and is parallel to the line segment $PQ$.
Given that the $y$-intercept of line $L$ is $-5$, find the value of $k$.

$k =$ ..... **[4]**

**(b)** Find the equation of the line passing through $Q(4, 15)$ that is parallel to the line $5x - 2y + 9 = 0$.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

**921** Line $L_1$ has the equation $3x - 4y = 12$.
Line $L_2$ is parallel to $L_1$.

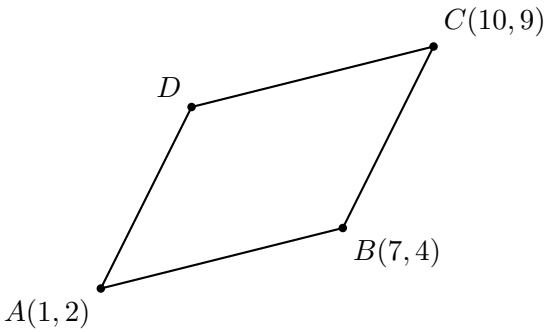
Line $L_2$ crosses the $x$-axis at point $A$ and the $y$-axis at point $B$.
The length of the line segment $AB$ is 15.

Find the two possible equations of $L_2$.
Give your answers in the form $y = mx + c$.

.....

..... **[7]**

**922** The points $A(1, 2)$, $B(7, 4)$, and $C(10, 9)$ are three vertices of a parallelogram $ABCD$.
Line $CD$ is parallel to line $AB$ and line $AD$ is parallel to line $BC$.




*Not to scale*

**(a)** Find the equation of the line $CD$.

..... **[3]**

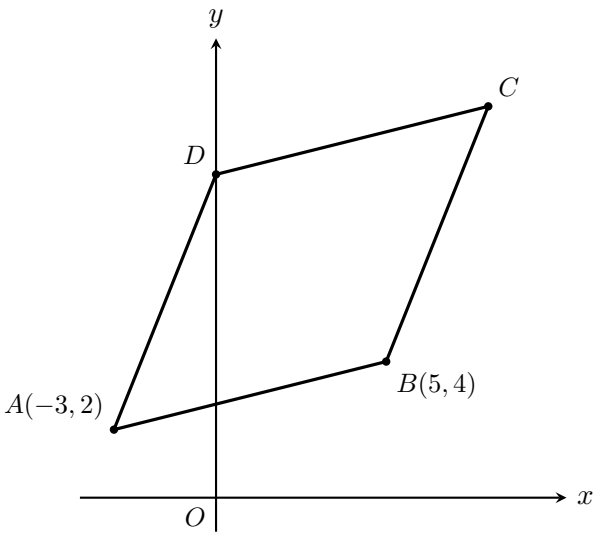
**(b)** Find the equation of the line $AD$.

..... **[3]**

**(c)** Work out the coordinates of the vertex $D$.

( ..... , ..... ) **[3]**

**923** The diagram shows a sketch of a parallelogram $ABCD$ on a coordinate grid.
Vertex $A$ has coordinates $(-3, 2)$ and vertex $B$ has coordinates $(5, 4)$.
The line passing through $B$ and $C$ has equation $5x - 2y = 17$.




**(a)** Find the equation of the line passing through $A$ and $D$ in the form $y = mx + c$.

$y =$ ..... **[3]**

**(b)** Vertex $D$ lies on the $y$-axis.
Write down the coordinates of $D$ and find the coordinates of vertex $C$.

$D =$ ..... , $C =$ ..... **[3]**

**(c)** Line $L$ is parallel to $AB$ and passes through the midpoint of $BC$.
Find the equation of line $L$ in the form $ax + by = c$, where $a$, $b$, and $c$ are integers.

..... **[4]**

**(d)** Calculate the length of diagonal $AC$, giving your answer correct to 3 significant figures.

$AC =$ ..... **[2]**

**924** Line $L$ has the equation $5y + 2x = 15$.
Find the gradient of a line that is perpendicular to $L$.

..... **[2]**

**925** Find the equation of the line that is perpendicular to $y = 3x - 5$ and passes through the point $(6, 2)$. Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

**926** Line $L_1$ has the equation $y = 5 - 2x$.
Line $L_2$ is perpendicular to $L_1$ and passes through the point $(4, 1)$.

**(a)** Find the gradient of line $L_1$.

..... **[1]**

**(b)** Find the equation of line $L_2$.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

**927** Line $L_1$ has the equation $4y = 3x - 8$.

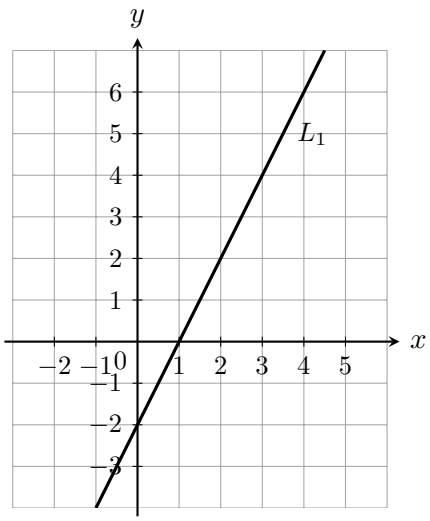
**(a)** Find the gradient of a line that is perpendicular to $L_1$.

Gradient = ..... **[2]**

**(b)** Line $L_2$ is perpendicular to $L_1$ and passes through the point $(6, -5)$.
Find the equation of line $L_2$ in the form $y = mx + c$.

$y =$ ..... **[2]**

**928** The diagram shows a straight line $L_1$ drawn on a coordinate grid.




**(a)** Write down the gradient of line $L_1$.

..... **[1]**

**(b)** Find the gradient of a line that is perpendicular to $L_1$.

..... **[1]**

**(c)** Line $L_2$ is perpendicular to $L_1$ and passes through the point $(4, 3)$.
Find the equation of line $L_2$ in the form $y = mx + c$.

$y =$ ..... **[3]**

**929** A straight line $L$ has the equation $2y = 5x + 8$.

**(a)** Find the gradient of line $L$.

..... **[2]**

**(b)** Find the gradient of a line that is perpendicular to line $L$.

..... **[1]**

**(c)** A line $M$ is perpendicular to line $L$ and passes through the point $(10, -3)$.
Find the equation of line $M$ in the form $y = mx + c$.

$y =$ ..... **[3]**

**930** Point $A$ has coordinates $(-3, 8)$ and point $B$ has coordinates $(9, -2)$.

**(a)** Find the coordinates of the midpoint of $AB$ and the gradient of $AB$.

Midpoint = ..... , Gradient = ..... **[2]**

**(b)** Find the equation of the perpendicular bisector of $AB$.
Give your answer in the form $ax + by = c$, where $a$, $b$ and $c$ are integers.

..... **[3]**

**931** $A$ is the point $(-4, 7)$ and $B$ is the point $(2, -1)$.

Find the equation of the perpendicular bisector of the line segment $AB$.
Give your answer in the form $ax + by = c$, where $a$, $b$ and $c$ are integers.

..... **[5]**

**932** Line $L$ has equation $2x + 3y = 12$.
Point $A$ has coordinates $(-3, 8)$ and point $B$ has coordinates $(9, -2)$.

**(a)** Find the gradient of a line that is perpendicular to line $L$.

..... **[2]**

**(b)** For the line segment joining $A(-3, 8)$ and $B(9, -2)$,

**(i)** find the coordinates of the midpoint of $AB$,

( ..... , ..... ) **[2]**

**(ii)** find the equation of the perpendicular bisector of $AB$, giving your answer in the form $ax + by = c$, where $a$, $b$, and $c$ are integers.

..... **[4]**

**933** $A$ is the point $(-3, 8)$ and $B$ is the point $(9, -2)$.

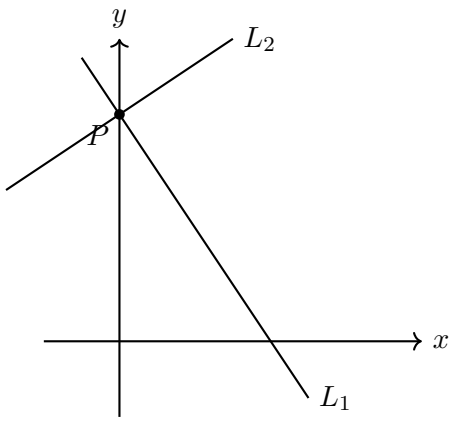
**(a)** Find the coordinates of the midpoint of $AB$.

( ..... , ..... ) **[2]**

**(b)** Find the equation of the perpendicular bisector of $AB$.
Give your answer in the form $ax + by = c$, where $a$, $b$ and $c$ are integers.

..... **[4]**

**934** Line $L_1$ has the equation $3x + 2y = 12$.
Line $L_2$ is perpendicular to $L_1$ and intersects $L_1$ at a point $P$ on the $y$-axis.




*Not to scale*

**(a)** Find the coordinates of point $P$.

( ..... , ..... ) **[1]**

**(b)** Find the equation of line $L_2$.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

**(c)** Point $Q$ lies on $L_2$ and has an $x$-coordinate of 9.
Point $R$ lies on $L_1$ and has an $x$-coordinate of 4.
Work out the area of triangle $PQR$.

..... **[5]**

**935** Line $L_1$ has equation $kx + 2y = 12$.
Line $L_2$ passes through the points $A(0, 5)$ and $B(2, k + 2)$.
The constant $k$ is a non-zero integer.

**(a)** Find the gradient of line $L_1$ in terms of $k$.

..... **[1]**

**(b)** Find the gradient of line $L_2$ in terms of $k$.

..... **[2]**

**(c)** Given that $L_1$ and $L_2$ are perpendicular, find the two possible values of $k$.

$k =$ ..... **[4]**

**(d)** Given that the gradient of line $L_1$ is positive, find the equation of line $L_2$ in the form $ax + by = c$, where $a$, $b$, and $c$ are integers.

..... **[3]**

**936** The equation of line $P$ is $3x - 4y = 12$.
Line $Q$ is perpendicular to line $P$ and passes through the point $(6, -2)$.

Find the equation of line $Q$.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[4]**

**937** The points $A$ and $B$ have coordinates $(-2, 4)$ and $(6, 0)$ respectively.

**(a)** Find the coordinates of the midpoint of $AB$.

( ..... , ..... ) **[2]**

**(b)** Find the equation of the perpendicular bisector of the line segment $AB$.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[4]**

**938** The coordinates of point $A$ are $(-4, 5)$ and the coordinates of point $B$ are $(2, -7)$.

**(a)** Find the coordinates of the midpoint of the line segment $AB$.

( ..... , ..... ) **[2]**

**(b)** Calculate the gradient of the line $AB$.

..... **[2]**

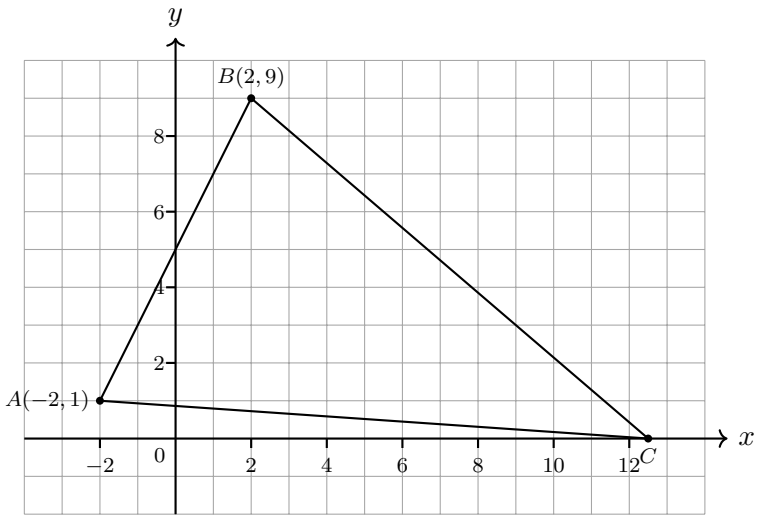
**(c)** Find the equation of the perpendicular bisector of the line segment $AB$.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

**(d)** The perpendicular bisector intersects the $x$-axis at point $P$.
Find the coordinates of $P$.

( ..... , ..... ) **[2]**

**939** The diagram shows a right-angled triangle $ABC$ with $\angle ABC = 90°$.
The coordinates of $A$ and $B$ are $(-2, 1)$ and $(2, 9)$ respectively, and point $C$ lies on the $x$-axis.




**(a)** Find the equation of the line $BC$.

$y =$ ..... **[3]**

**(b)** Find the coordinates of point $C$.

( ..... , ..... ) **[2]**

**(c)** Calculate the area of triangle $ABC$.

Area = ..... **[2]**

**940** The line $L_1$ has equation $2y + 3x = 12$.
The line $L_2$ is perpendicular to $L_1$ and intersects $L_1$ at the point $P$ on the $y$-axis.

**(a)** Find the coordinates of $P$.

( ..... , ..... ) **[2]**

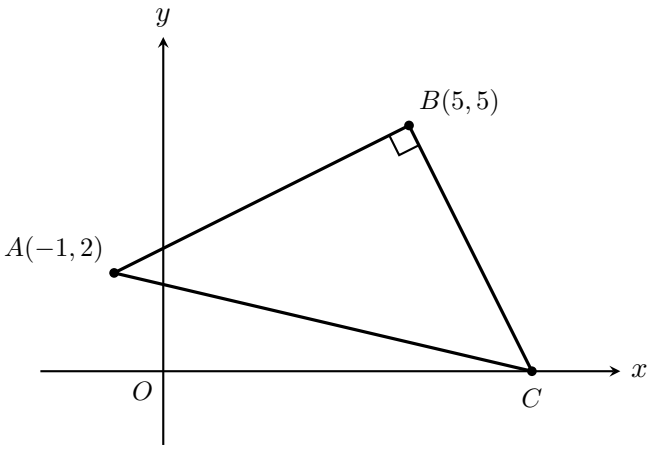
**(b)** Find the equation of line $L_2$.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

**(c)** Line $L_2$ crosses the $x$-axis at the point $Q$.
Calculate the exact length of $PQ$.

..... **[4]**

**941** The diagram shows a sketch of triangle $ABC$ with vertices $A(-1, 2)$ and $B(5, 5)$.
Angle $ABC = 90°$ and vertex $C$ lies on the $x$-axis.




**(a)** Find the equation of the line passing through $B$ and $C$ in the form $y = mx + c$.

$y =$ ..... **[3]**

**(b)** Find the coordinates of vertex $C$.

$C =$ ( ..... , ..... ) **[2]**

**(c)** Calculate the area of triangle $ABC$.

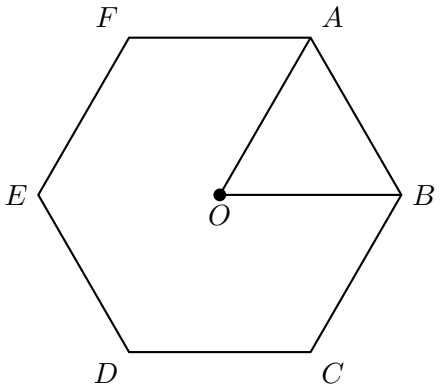
..... square units **[3]**

**(d)** The perpendicular bisector of $AC$ intersects the $y$-axis at point $P$.

Find the coordinates of $P$, giving the $y$-coordinate correct to 3 significant figures.

$P = (\ldots\ldots\ldots\ ,\ \ldots\ldots\ldots)$ **[4]**

**942** The diagram shows a regular polygon $ABCDEF$ with centre $O$.




**(a)** Write down the mathematical name of the polygon $ABCDEF$.

..... **[1]**

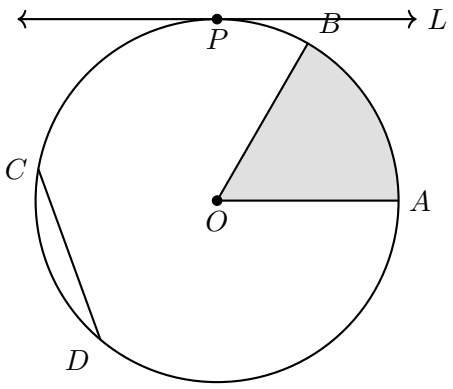
**(b)** Write down the mathematical name for the angle $AFE$.

..... **[1]**

**(c)** Write down the mathematical name for the triangle $AOB$.

..... **[1]**

**943** The diagram shows a circle with centre $O$.
The straight line $L$ touches the circle at the point $P$.




**(a)** Write down the mathematical name for the straight line $CD$.

..... **[1]**

**(b)** Write down the mathematical name for the straight line $L$.

..... **[1]**

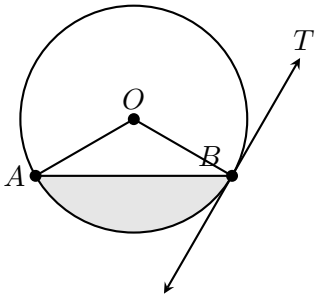
**(c)** Write down the mathematical name for the shaded region $OAB$.

..... **[1]**

**(d)** The angle $AOB$ is $60°$.
Write down the mathematical name for this type of angle.

..... **[1]**

**944** The diagram shows a circle, centre $O$.
$A$ and $B$ are points on the circumference of the circle.
The line passing through $T$ touches the circle at $B$.




**(a)** Write down the geometrical name of the line $AB$.

..... **[1]**

**(b)** Write down the geometrical name of the line $BT$.

..... **[1]**

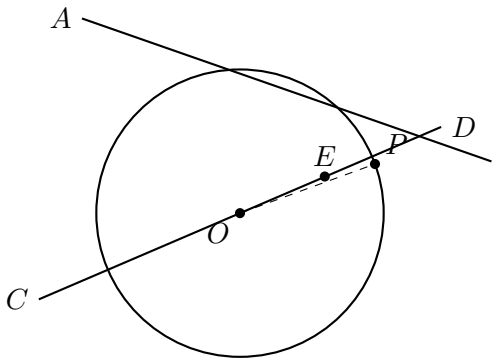
**(c)** Write down the geometrical name of the shaded region.

..... **[1]**

**(d)** Write down the mathematical name of triangle $OAB$.

..... **[1]**

**945** The diagram shows a circle with centre $O$.
The lines $AB$ and $CD$ intersect at $E$.
$AB$ touches the circle at $P$, and $CD$ passes through the centre $O$.




**(a)** Write down the mathematical name for the line segment $OP$.

..... **[1]**

**(b)** Write down the mathematical name for the line $AB$.

..... **[1]**

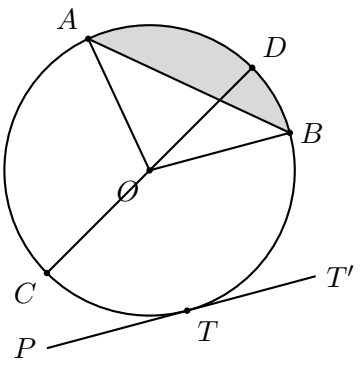
**(c)** Write down the mathematical name for the line segment $CD$.

..... **[1]**

**(d)** State the size of angle $OPT$.

..... ° **[1]**

**946** The diagram shows a circle with centre $O$ and points $A$, $B$, $C$ and $D$ on its circumference.
The line $PT$ is a tangent to the circle at $T$, and the shaded region is bounded by the chord $AB$ and the minor arc $AB$.




**(a)** Write down the geometrical name for:

**(i)** the line segment $CD$,

..... **[1]**

**(ii)** the shaded region bounded by chord $AB$ and minor arc $AB$,

..... **[1]**

**(iii)** the region bounded by radii $OA$, $OB$ and the major arc $AB$.

..... **[1]**

**(b)** State the type of angle formed by the reflex angle $\angle AOB$ if the acute angle $\angle AOB = 100°$.

..... **[1]**

**947** **(a)** **(i)** A polygon has interior angles that sum to 1080°.

Work out the number of sides of this polygon.

..... **[2]**

**(ii)** Write down the mathematical name of this polygon.

..... **[1]**

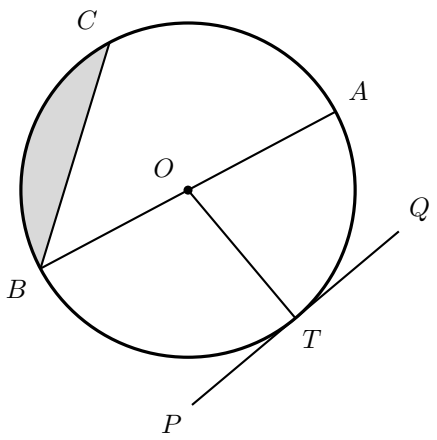
**(b)** Write down the mathematical name of a quadrilateral with all four sides equal in length and no right angles.

..... **[1]**

**(c)** Write down the mathematical name of a 3D solid that has a uniform cross-section and two parallel circular faces.

..... **[1]**

**948** The diagram shows a circle, centre $O$, with diameter $AB$.
The points $A$, $B$, $C$, and $T$ lie on the circumference of the circle.
The straight line $PQ$ touches the circle at $T$.




**(a)** Write down the mathematical name for

**(i)** the straight line $PQ$,

..... **[1]**

**(ii)** the straight line $BC$,

..... **[1]**

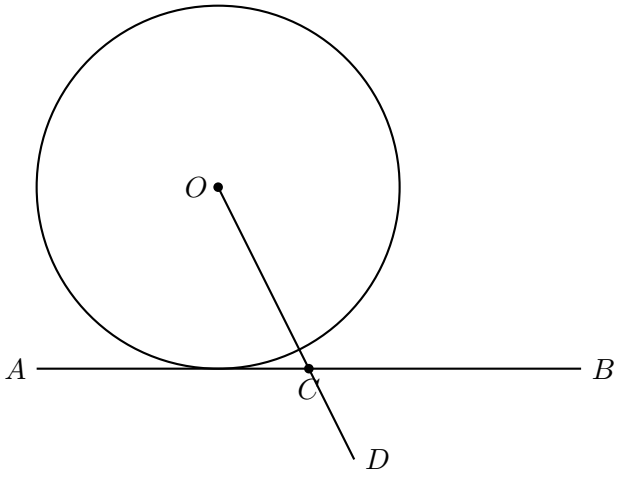
**(iii)** the shaded region bounded by $BC$ and the minor arc $BC$.

..... **[1]**

**(b)** State the geometrical word that describes the relationship between radius $OT$ and line $PQ$, and write down the size of angle $OTQ$.

Relationship: ..... , Angle $OTQ =$ ..... **[2]**

**949** The diagram shows a circle with centre $O$.
The line $AB$ touches the circle at the point $C$.
The line $OD$ passes through $C$.




**(a)** Write down the mathematical name for the straight line $AB$.

..... **[1]**

**(b)** Write down the mathematical name for the straight line $OC$.

..... **[1]**

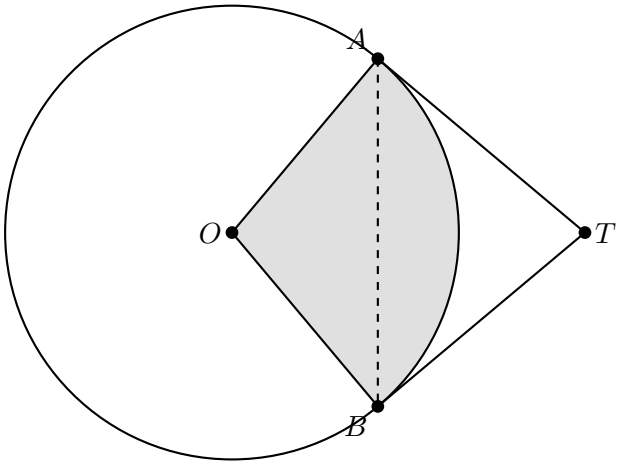
**(c)** Write down the size of angle $OCB$.

Angle $OCB =$ ..... ° **[1]**

**(d)** The length of $OC$ is 5 cm.
Calculate the circumference of the circle.
Give your answer in terms of $\pi$.

..... cm **[2]**

**950** Points $A$ and $B$ lie on a circle, centre $O$.
The lines $TA$ and $TB$ are tangents to the circle.
The region enclosed by the arc $AB$ and the radii $OA$ and $OB$ is shaded.




**(a)** Write down the mathematical name for the quadrilateral $OATB$.

..... **[1]**

**(b)** Write down the mathematical name for the shaded region.

..... **[1]**

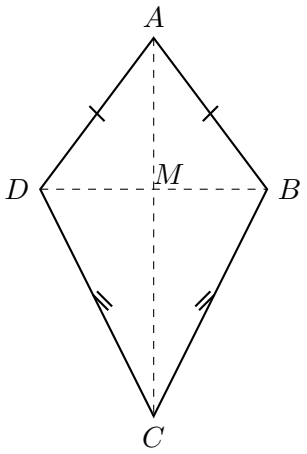
**(c)** Write down the mathematical name for the straight line $AB$.

..... **[1]**

**(d)** Triangle $OAT$ and triangle $OBT$ are exactly the same size and shape.
Write down the mathematical term for two identical shapes.

..... **[1]**

**951** The diagram shows a quadrilateral $ABCD$.
$AB = AD$ and $CB = CD$.
The diagonals $AC$ and $BD$ intersect at $M$.




**(a)** Write down the mathematical name of the quadrilateral $ABCD$.

..... **[1]**

**(b)** Write down the mathematical term that describes the relationship between the line $AC$ and the line $BD$.

..... **[1]**

**(c)** Triangle $ABM$ and Triangle $ADM$ are identical in shape and size.
Write down the mathematical word for this.

..... **[1]**

**(d)** The angle $ABC$ is $110°$.
Write down the mathematical name for this type of angle.

..... **[1]**

**(e)** Write down the mathematical name for the line segment $AC$ in relation to the line segment $BD$.

..... **[1]**

**952** A regular polygon has $n$ sides.
Each interior angle of this polygon is 156°.

**(a)** Work out the size of one exterior angle of this polygon.

..... ° **[1]**

**(b)** Find the value of $n$.

$n =$ ..... **[2]**

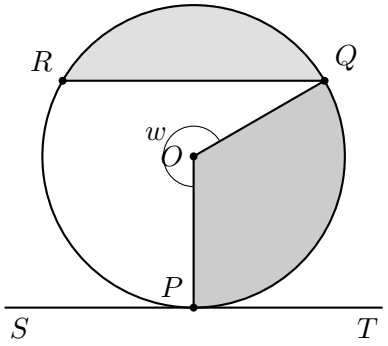
**(c)** Write down the mathematical name for a regular polygon with 10 sides.

..... **[1]**

**(d)** State whether a regular 15-sided polygon has an acute, obtuse, or reflex interior angle.

..... **[1]**

**953** The diagram shows a circle, centre $O$.
$P$, $Q$ and $R$ are points on the circumference.
The straight line $ST$ touches the circle at $P$.




**(a)** Write down the mathematical name of the line $ST$.

..... **[1]**

**(b)** Write down the mathematical name of the line segment $QR$.

..... **[1]**

**(c)** Write down the mathematical name of the shaded region bounded by the line $QR$ and the minor arc $QR$.

..... **[1]**

**(d)** Write down the mathematical name of the shaded region bounded by the radii $OP$ and $OQ$ and the minor arc $PQ$.

..... **[1]**

**(e)** The angle marked $w$ is greater than $180°$ and less than $360°$.
Write down the geometrical term for this type of angle.

..... **[1]**

**954** Quadrilateral $PQRS$ has the following geometrical properties:

- Only one pair of opposite sides are parallel.
- The non-parallel sides are equal in length.
- It has one line of symmetry and no rotational symmetry.

**(a)** Write down the full geometrical name of quadrilateral $PQRS$.

..... **[1]**

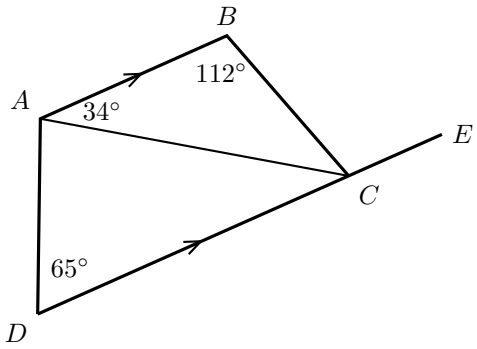
**(b)** A regular polygon has an exterior angle of $36°$.
State the geometrical name of this regular polygon.

..... **[2]**

**(c)** A solid prism has a regular decagon as its cross-section.
Write down the number of faces and the number of edges of this prism.

Faces = ..... , Edges = ..... **[2]**

**955** In the diagram, $ABCD$ is a quadrilateral with $AB$ parallel to $DC$.
The side $DC$ is extended to $E$, and $AC$ is a diagonal.
Angle $ADC = 65°$, angle $BAC = 34°$, and angle $ABC = 112°$.




**(a)** Write down the special mathematical name for quadrilateral $ABCD$.

..... **[1]**

**(b)** For vertex $B$,

**(i)** write down the geometrical term that describes the interior angle $ABC$,

..... **[1]**

**(ii)** calculate the size of the reflex angle at vertex $B$.

..... ° **[1]**

**(c)** Calculate the size of

**(i)** angle $ACD$,

Angle $ACD =$ ..... ° **[1]**

**(ii)** the exterior angle $BCE$.

Angle $BCE =$ ..... ° **[2]**

**956** **(a)** The interior angle of a regular polygon is 144°.

**(i)** Work out the number of sides of this polygon.

..... **[2]**

**(ii)** Write down the geometrical name of this polygon.

..... **[1]**

**(b)** A hexagon has interior angles of $x°$, $2x°$, $(x + 30)°$, $(3x + 60)°$, 110° and 100°.

**(i)** Calculate the value of $x$.

$x =$ ..... **[3]**

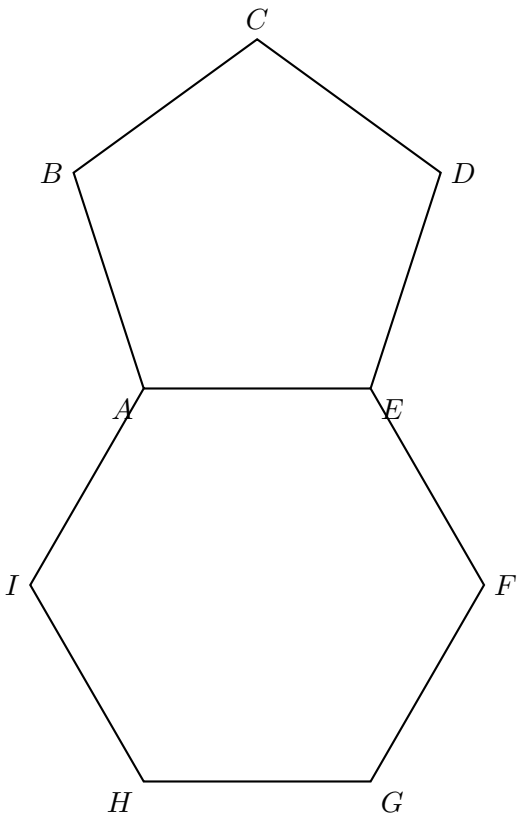
**(ii)** Write down the geometrical name for an angle of $(x + 30)°$.

..... **[1]**

**(iii)** Write down the geometrical name for an angle of $(3x + 60)°$.

..... **[1]**

**957** The diagram shows a regular pentagon $ABCDE$ and a regular hexagon $AEFGHI$ joined along the edge $AE$.




**(a)** Calculate the size of an interior angle of the regular pentagon $ABCDE$.

..... ° **[2]**

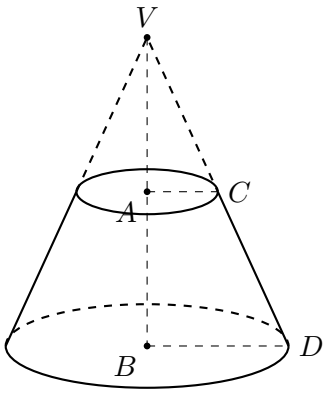
**(b)** Calculate the size of an interior angle of the regular hexagon $AEFGHI$.

..... ° **[2]**

**(c)** Calculate the size of the reflex angle $BAI$.

..... ° **[3]**

**958** The diagram shows a solid frustum formed by removing a smaller right circular cone from the top of a larger right circular cone.
The plane of the circular top face is parallel to the plane of the circular base.




**(a)** State the geometrical relationship between the original cone with vertex $V$ and the removed cone with vertex $V$.

..... **[1]**

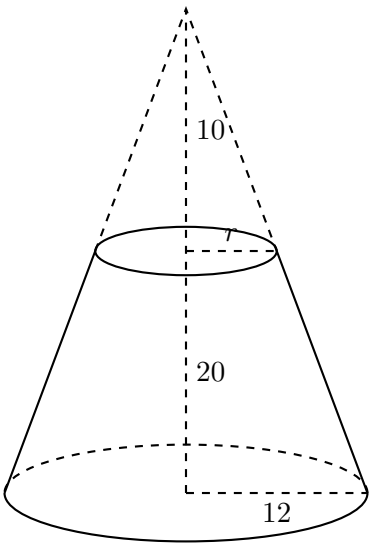
**(b)** Explain why $\triangle VAC$ is a right-angled triangle and why $AC$ is parallel to $BD$.

..... **[2]**

**(c)** Given that $AC = 4.5$ cm, $BD = 7.5$ cm and the perpendicular height of the frustum $AB = 6$ cm, calculate the perpendicular height $VA$ of the removed cone.

$VA =$ ..... cm **[3]**

**959** The diagram shows a large solid cone with a smaller solid cone removed from the top.
The cut is made parallel to the base of the large cone.
All dimensions are in centimetres.




**(a)** Write down the mathematical name of the solid with height 20.

..... **[1]**

**(b)** The small cone is geometrically related to the original large cone.

**(i)** Write down the mathematical word that describes the relationship between the small cone and the large cone.

..... **[1]**

**(ii)** Work out the value of the radius, $r$.

$r =$ ..... **[2]**

**(iii)** Calculate the slant edge of the solid with height 20.

..... *cm* **[3]**

**960** A map shows three towns $A$, $B$ and $C$.
Town $B$ is 45 km from $A$ on a bearing of 062°.
Town $C$ is 45 km from $A$ on a bearing of 138°.

**(a)** Write down the mathematical name of triangle $ABC$.

..... **[1]**

**(b)** Work out the bearing of $A$ from $C$.

..... ° **[2]**

**(c)** **(i)** Point $M$ is the midpoint of the straight line $BC$.
Write down the mathematical name of the line $AM$ in relation to the line $BC$.

..... **[1]**

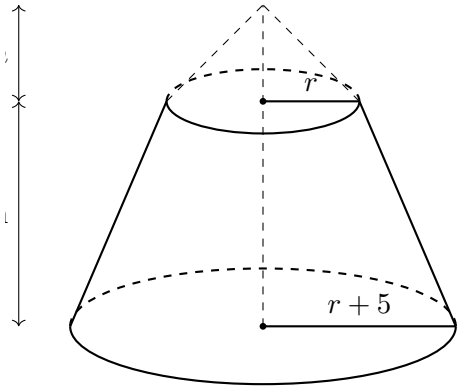
**(ii)** Calculate the bearing of $M$ from $A$.

..... ° **[2]**

**(d)** A fourth town $D$ is located such that the line $AB$ is parallel to the line $DC$ and the line $AD$ is not parallel to the line $BC$.
Write down the mathematical name of the quadrilateral $ABCD$.

..... **[1]**

**961** The diagram shows a solid frustum obtained by cutting a smaller right circular cone of height $h$ cm from a larger right circular cone of height $(h + 12)$ cm.
The radius of the top circular face is $r$ cm and the radius of the bottom circular base is $(r + 5)$ cm.
The two cones are mathematically similar.




**(a)** Given that the scale factor relating the linear dimensions of the smaller cone to the larger cone is $\frac{3}{5}$, show that $h = 18$.

**[3]**

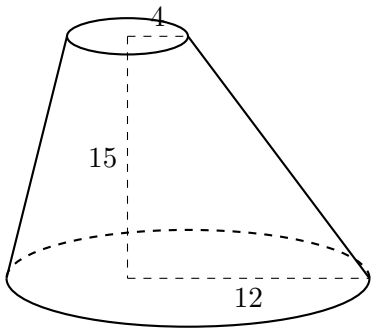
**(b)** Find the value of $r$.

$r =$ ..... **[2]**

**(c)** The ratio of the volume of the smaller cone to the volume of the frustum is $27 : k$.
Find the value of $k$.

$k =$ ..... **[2]**

**962** The diagram shows a frustum of a cone.
The radius of the circular base is 12 cm and the radius of the circular top is 4 cm.
The vertical height of the frustum is 15 cm.




**(a)** Calculate the volume of the frustum.
The volume, $V$, of a cone with radius $r$ and height $h$ is $V = \frac{1}{3}\pi r^2 h$.

..... $cm^3$ **[5]**

**(b)** A different shape is geometrically similar to the frustum.
The ratio of the surface area of this shape to the surface area of the frustum is 9 : 25.
Calculate the vertical height of this shape.

..... cm **[3]**

**963** A regular polygon has $n$ sides.

**(a)** The interior angle of this regular polygon is 4 times the size of its exterior angle.
Calculate the value of $n$ and write down the mathematical name of this polygon.

$n =$ ..... , Name = ..... **[4]**

**(b)** A solid uniform prism has a cross-section in the shape of this regular polygon.
Write down the number of faces, edges, and vertices of this prism.

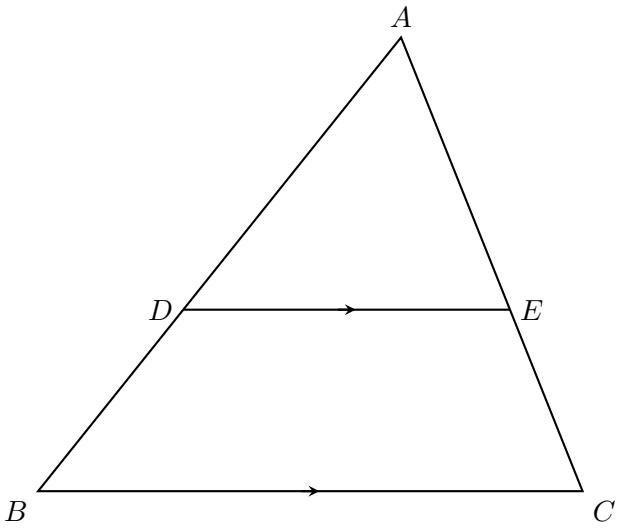
Faces = ..... , Edges = ..... , Vertices = ..... **[3]**

**(c)** Two mathematically similar cones have base areas in the ratio 9 : 16.
The volume of the smaller cone is $108\pi$ cm$^3$.

Calculate the volume of the larger cone.
Give your answer in terms of $\pi$.

..... $cm^3$ **[3]**

**964** The diagram shows triangle $ABC$.
$D$ is a point on $AB$ and $E$ is a point on $AC$ such that $DE$ is parallel to $BC$.
$AD = 6$ cm, $DB = 4$ cm, $AE = 7.5$ cm and $BC = 15$ cm.




**(a)** Explain why triangle $ADE$ is similar to triangle $ABC$.

**[2]**

**(b)** Calculate the length of $DE$.

..... cm **[2]**

**(c)** Calculate the length of $EC$.

..... cm **[2]**

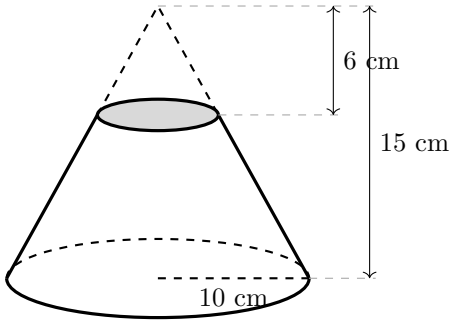
**(d)** State the geometrical name of the quadrilateral $DBCE$.

..... **[1]**

**(e)** The area of triangle $ADE$ is 18 $cm^2$.
Calculate the area of the quadrilateral $DBCE$.

..... $cm^2$ **[3]**

**965** The diagram shows a solid formed by cutting a smaller right cone of height 6 cm from the top of a larger right cone of height 15 cm and base radius 10 cm. The cut is made along a plane parallel to the circular base.




**(a)** Write down the mathematical name for

**(i)** the remaining solid after the top cone is removed,

..... **[1]**

**(ii)** a solid pyramid that has 4 triangular faces, 6 edges, and 4 vertices.

..... **[1]**

**(b)** Explain why the removed cone is similar to the original large cone, and write down the scale factor that maps the original large cone onto the removed cone.

Reason: ..... , Scale factor = ..... **[2]**

**(c)** Calculate the radius of the upper circular face of the remaining solid.

..... cm **[2]**

**(d)** A regular polygon has an exterior angle of 36°.

**(i)** Write down the mathematical name of this regular polygon.

..... **[2]**

**(ii)** A right prism has this regular polygon as its cross-section.
State the total number of faces and the total number of edges of this prism.

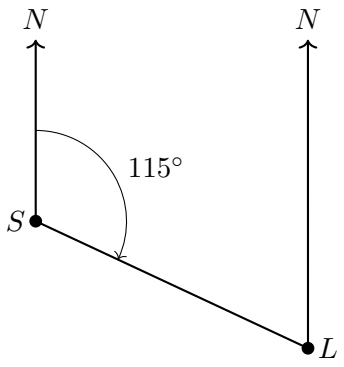
Faces = ..... , Edges = ..... **[2]**

**966** The bearing of point $Y$ from point $X$ is 048°.

Work out the bearing of point $X$ from point $Y$.

..... ° **[2]**

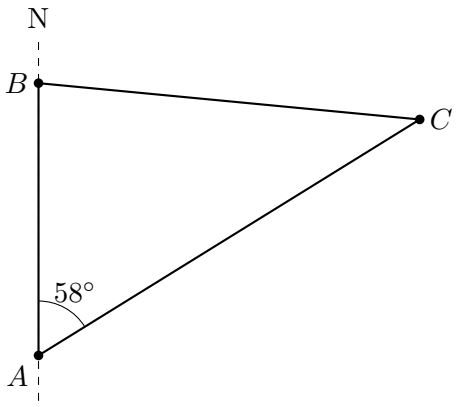
**967** The diagram shows the positions of a ship, $S$, and a lighthouse, $L$.




The bearing of the lighthouse from the ship is 115°.
Work out the bearing of the ship from the lighthouse.

..... ° **[2]**

**968** The diagram shows the positions of three towns, $A$, $B$ and $C$.
$B$ is due north of $A$ and the bearing of $C$ from $A$ is 058°.




**(a)** Find the bearing of $A$ from $C$.

..... ° **[2]**

**(b)** Given that angle $ABC = 84°$, find the bearing of $C$ from $B$.

..... ° **[1]**

**969** The positions of three towns, $A$, $B$ and $C$, are marked on a map.

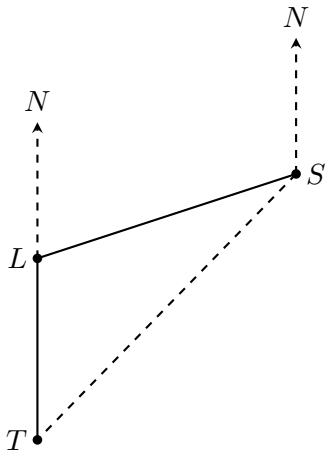
**(a)** The bearing of town $B$ from town $A$ is 068°.
Work out the bearing of town $A$ from town $B$.

..... ° **[2]**

**(b)** Town $C$ is due west of town $B$.
Write down the bearing of town $C$ from town $B$.

..... ° **[1]**

**970** The diagram shows the positions of a lighthouse $L$ and two ships $S$ and $T$.
The bearing of ship $S$ from the lighthouse $L$ is 072°.




**(a)** Calculate the three-figure bearing of the lighthouse $L$ from ship $S$.

..... ° **[2]**

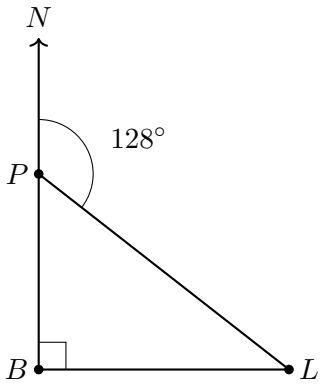
**(b)** Ship $T$ is due South of the lighthouse $L$.
Calculate the size of angle $SLT$.

$\angle SLT =$ ..... ° **[1]**

**(c)** The bearing of ship $T$ from ship $S$ is 215°.
Calculate the size of angle $LST$.

$\angle LST =$ ..... ° **[2]**

**971** The bearing of a lighthouse $L$ from a port $P$ is 128°.
A boat $B$ is positioned such that it is due South of $P$ and due West of $L$.




**(a)** Find the bearing of $P$ from $L$.

..... ° **[2]**

**(b)** Work out the size of angle $BPL$.

..... ° **[1]**

**(c)** Work out the size of angle $PLB$.

..... ° **[2]**

**972** The bearing of a lighthouse, $L$, from a boat, $B$, is 148°.

**(a)** Work out the bearing of $B$ from $L$.

..... ° **[2]**

**(b)** A port, $P$, is due north of $L$ and due east of $B$.

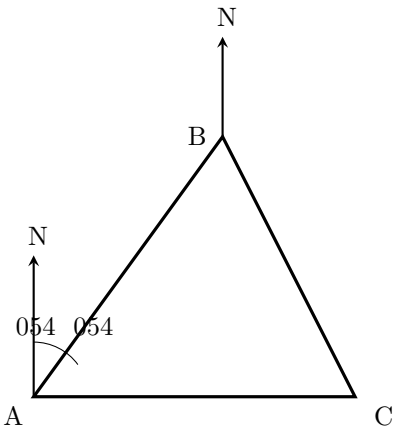
**(i)** Write down the bearing of $B$ from $P$.

..... ° **[1]**

**(ii)** Work out the size of angle $LBP$.

..... ° **[2]**

**973** The diagram shows the positions of three lighthouses,
A,
B, and
C. Lighthouse
C is due east of lighthouse
A, and the bearing of
B from
A is
054
054
.




**(a)** Write down the three-figure bearing of
C from
A.

.....

**[1]**

**(b)** Calculate the three-figure bearing of
A from
B.

.....

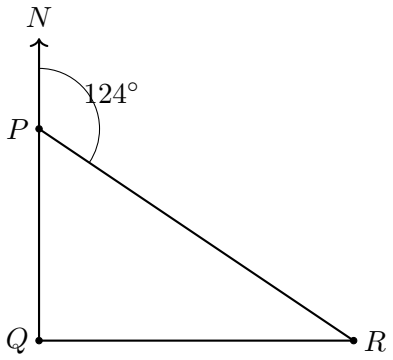
**[2]**

**(c)** Given that
=
AB=AC, calculate the three-figure bearing of
C from
B.

.....

**[2]**

**974** The diagram shows the positions of three points, $P$, $Q$ and $R$.
Point $Q$ is due south of point $P$.
Point $R$ is due east of point $Q$.
The bearing of point $R$ from point $P$ is 124°.



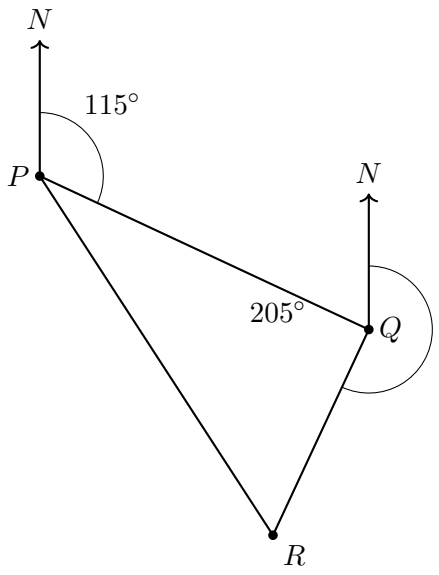

**(a)** Find the bearing of point $P$ from point $R$.

..... ° **[2]**

**(b)** Calculate the angle $PRQ$.

Angle $PRQ$ = ..... ° **[2]**

**975** The diagram shows the positions of three points $P$, $Q$ and $R$.
The bearing of $Q$ from $P$ is 115°.
The bearing of $R$ from $Q$ is 205°.



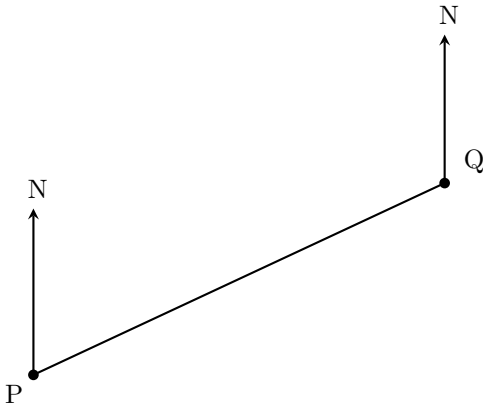

**(a)** Show that angle $PQR = 90°$.

**[2]**

**(b)** The distance $PQ$ is 15 km and the distance $QR$ is 8 km.
Calculate the distance $PR$.

..... km **[3]**

**976** The scale drawing shows the positions of two towns,
P and
Q, on a map.
The scale is
1
1 cm represents
5
5 km.




**(a)** Measure the three-figure bearing of
Q from
P.

.....

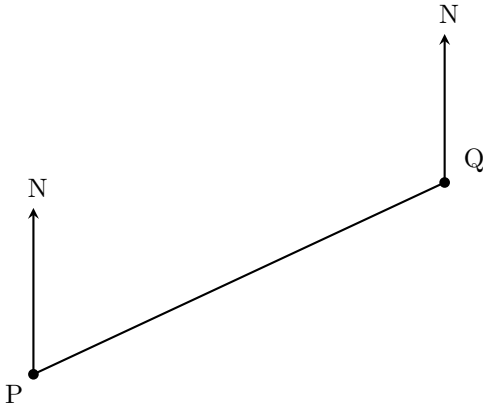
**[1]**

**(b)** Work out the actual distance, in kilometres, from town
P to town
Q.

..... km **[2]**

**(c)** A third town,
R, is
25
25 km from
Q on a bearing of
140
140
.
Using a ruler and protractor, mark and label the position of town

R on the scale drawing below.




**[2]**

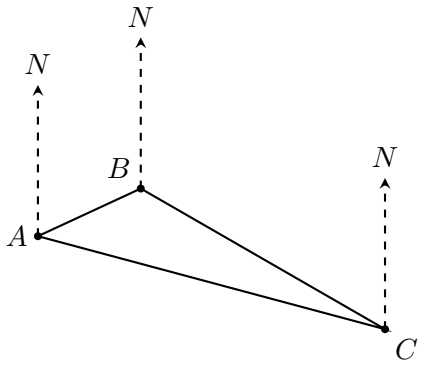
**(d)** Without measuring, calculate the three-figure bearing of P from Q.

.....

**[2]**

**977** The diagram shows the positions of three points, $A$, $B$, and $C$.

- The bearing of $B$ from $A$ is 065°.
- The bearing of $C$ from $B$ is 120°.
- The bearing of $A$ from $C$ is 285°.




**(a)** Calculate the size of angle $ABC$.

$\angle ABC =$ ..... ° **[2]**

**(b)** Calculate the size of angle $BCA$.

$\angle BCA =$ ..... ° **[2]**

**(c)** Point $D$ lies on the straight line $AC$ such that triangle $ABD$ is isosceles with $AB = BD$.

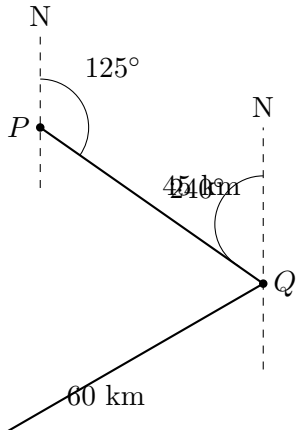
**(i)** Calculate the three-figure bearing of $C$ from $A$.

..... ° **[2]**

**(ii)** Calculate the size of angle $ABD$.

$\angle ABD =$ ..... ° **[2]**

**978** A ship sails from port $P$ on a bearing of 125° for 45 km to point $Q$.
It then sails from $Q$ on a bearing of 240° for 60 km to point $R$.



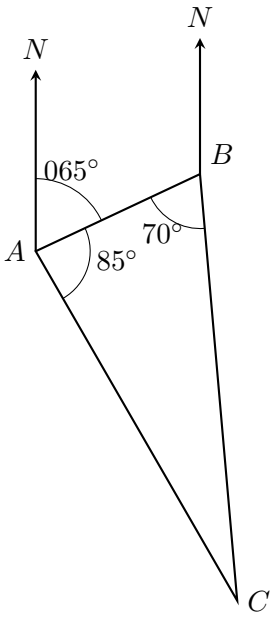

**(a)** Show that angle $PQR = 65°$.

**[2]**

**(b)** Calculate the distance $PR$.

$PR =$ ..... km **[3]**

**979** The diagram shows the positions of three towns, $A$, $B$ and $C$.
The bearing of $B$ from $A$ is 065°.
Angle $CAB = 85°$ and angle $ABC = 70°$.




**(a)** Calculate the bearing of $C$ from $A$.

..... ° **[1]**

**(b)** Calculate the bearing of $A$ from $B$.

..... ° **[2]**

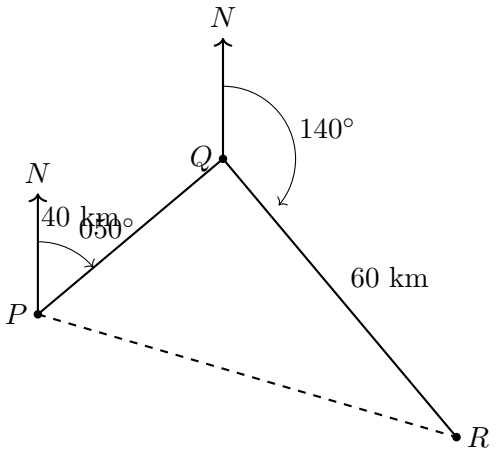
**(c)** Calculate the bearing of $C$ from $B$.

..... ° **[2]**

**(d)** Calculate angle $ACB$.

..... ° **[1]**

**980** A ship sails 40 km from a port $P$ to point $Q$ on a bearing of 050°.
It then sails 60 km from $Q$ to point $R$ on a bearing of 140°.




**(a)** Show that the angle $PQR$ is 90°.

**[2]**

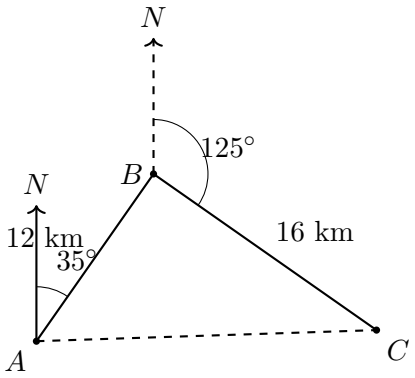
**(b)** Calculate the distance $PR$.

..... km **[2]**

**(c)** Calculate the bearing of $R$ from $P$.

..... ° **[3]**

**981** A ship leaves a port $A$ and travels 12 km on a bearing of 035° to point $B$.
From $B$, the ship travels 16 km on a bearing of 125° to point $C$.




**(a)** Show that angle $ABC = 90°$.

**[2]**

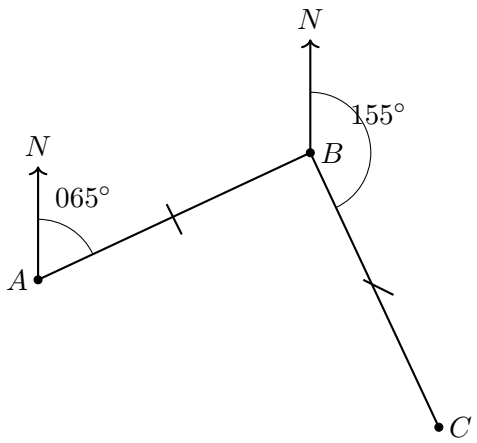
**(b)** Calculate the direct distance $AC$.

$AC =$ ..... km **[3]**

**(c)** Calculate the bearing of $C$ from $A$.

..... ° **[3]**

**982** Point $B$ is on a bearing of 065° from point $A$.
Point $C$ is on a bearing of 155° from point $B$.
The distance $AB = BC$.




**(a)** Show that angle $ABC = 90°$.

**[2]**

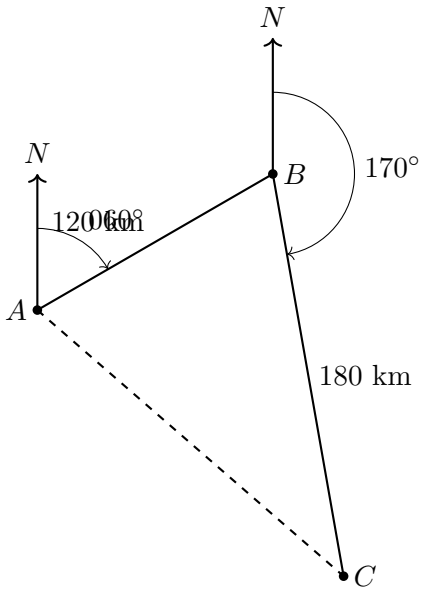
**(b)** Work out the bearing of $A$ from $C$.

..... ° **[3]**

**(c)** Point $D$ is such that the quadrilateral $ABCD$ is a square.
Find the bearing of $D$ from $A$.

..... ° **[2]**

**983** A helicopter flies from base $A$ to base $B$, a distance of 120 km on a bearing of 060°.
It then flies from base $B$ to base $C$, a distance of 180 km on a bearing of 170°.



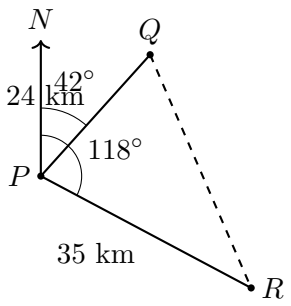

**(a)** Calculate the distance $AC$.

..... km **[4]**

**(b)** Calculate the bearing of $C$ from $A$.

..... ° **[3]**

**984** Three towns are located at points $P$, $Q$, and $R$.
Town $Q$ is 24 km from $P$ on a bearing of 042°.
Town $R$ is 35 km from $P$ on a bearing of 118°.



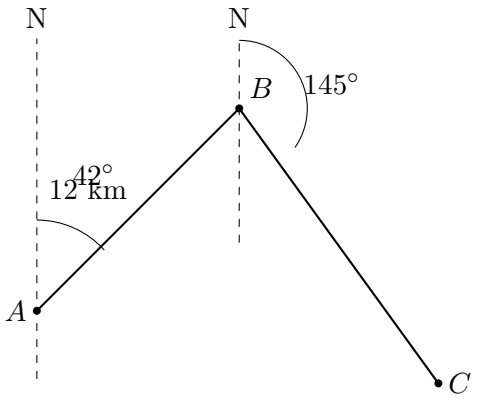

**(a)** Calculate the distance $QR$.

$QR =$ ..... km **[3]**

**(b)** Calculate the bearing of $R$ from $Q$.

..... ° **[4]**

**985** Three lighthouses $A$, $B$ and $C$ are situated such that $B$ is 12 km from $A$ on a bearing of 042°. $C$ is 18 km from $A$ and the bearing of $C$ from $B$ is 145°.



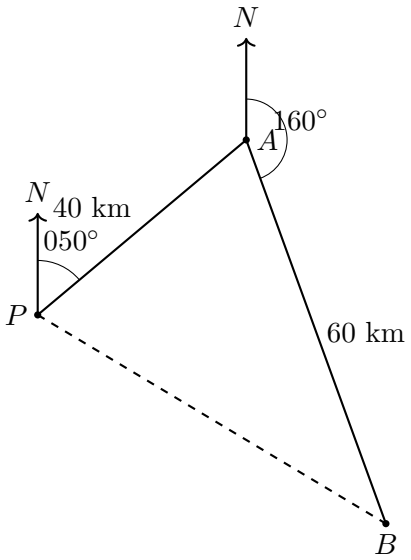

**(a)** Calculate the bearing of $C$ from $A$.

..... ° **[5]**

**(b)** A boat sails from $A$ in a straight line such that it is always equidistant from $B$ and $C$. Calculate the bearing on which the boat sails.

..... ° **[2]**

**986** A ship leaves port $P$ and sails 40 km on a bearing of 050° to reach point $A$.
It then sails 60 km on a bearing of 160° to reach point $B$.



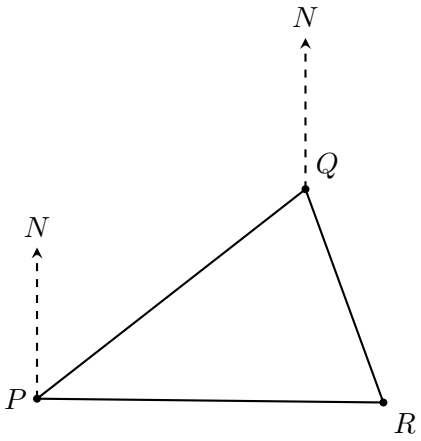

**(a)** Calculate the distance $PB$.

..... km **[4]**

**(b)** Calculate the bearing of $B$ from $P$.

..... ° **[4]**

**987** The points $P$, $Q$, and $R$ represent the positions of three towns.
Town $Q$ is 45 km from town $P$ on a bearing of 052°.
Town $R$ is 30 km from town $Q$ on a bearing of 160°.




**(a)** Show that the angle $PQR$ is 72°.

**[2]**

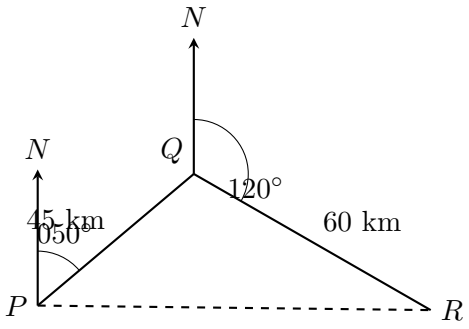
**(b)** Calculate the distance $PR$.

$PR =$ ..... km **[3]**

**(c)** Calculate the three-figure bearing of town $R$ from town $P$.

..... ° **[4]**

**988** A ship sails 45 km from Port $P$ on a bearing of 050° to Point $Q$.
It then sails 60 km on a bearing of 120° to Point $R$.




**(a)** Calculate the distance $PR$.

..... km **[4]**

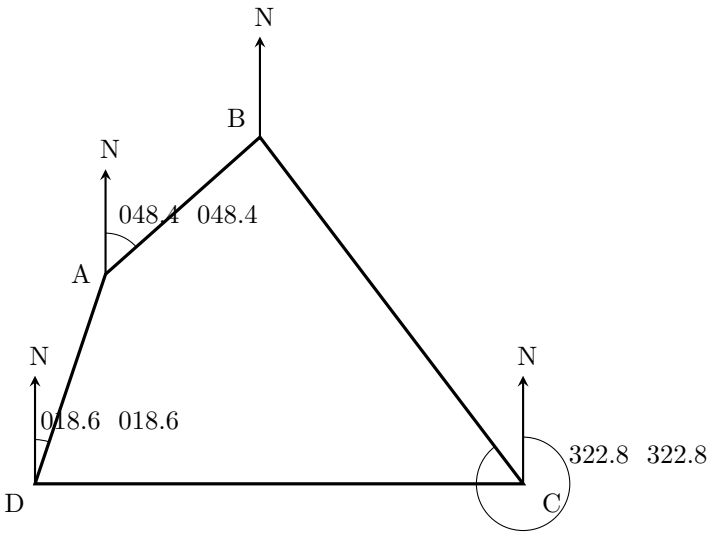
**(b)** Calculate the bearing of $R$ from $P$.

..... ° **[3]**

**(c)** The ship sails directly back from $R$ to $P$.
Calculate the bearing of $P$ from $R$.

..... ° **[2]**

**989** The diagram shows the positions of four radar stations,
A,
B,
C, and
D, on flat ground.
Station
C is due east of station
D. The bearing of
A from
D is
018.6
018.6
, the bearing of
B from
A is
048.4
048.4
, and the bearing of
B from
C is
322.8
322.8
.




**(a)** Write down the three-figure bearing of
D from
C.

.....

**[1]**

**(b)** Calculate the three-figure bearing of
D from
A.

. . . . . .

**[2]**

**(c)** Calculate the size of

**(i)** angle DAB,

*Angle*

=

DAB= . . . . . .

**[2]**

**(ii)** angle ABC.

*Angle*

=

ABC= . . . . . .

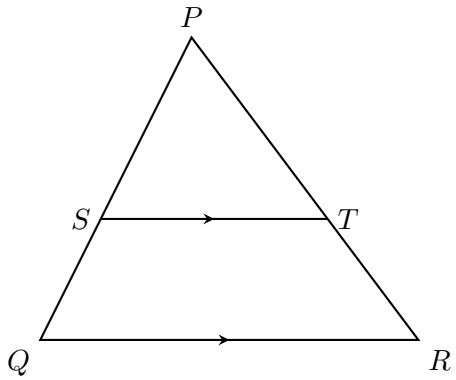
**[3]**

**(d)** Calculate the three-figure bearing of C from B.

. . . . . .

**[2]**

**990** The diagram shows triangle $PQR$.
Point $S$ lies on $PQ$ and point $T$ lies on $PR$.
The line $ST$ is parallel to the line $QR$.

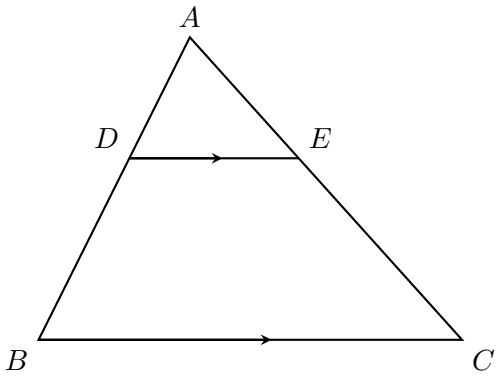



NOT TO SCALE

Given that $PS = 6$ cm, $SQ = 3$ cm and $ST = 8$ cm, calculate the length of $QR$.

$QR =$ ..... cm **[2]**

**991** The diagram shows triangle $ABC$ with a line $DE$ drawn parallel to $BC$.
$AD = 4$ cm, $DB = 6$ cm, and $DE = 5$ cm.



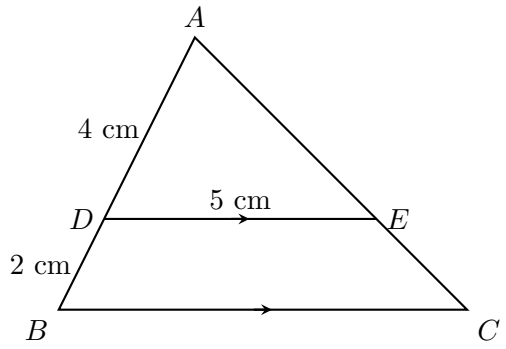

**(a)** Calculate the length of $BC$.

$BC =$ ..... $cm$ **[2]**

**(b)** The area of triangle $ADE$ is 16 cm$^2$.
Calculate the area of triangle $ABC$.

Area = ..... $cm^2$ **[2]**

**992** The diagram shows triangle $ABC$.
$D$ is a point on $AB$ and $E$ is a point on $AC$ such that $DE$ is parallel to $BC$.
$AD = 4$ cm, $DB = 2$ cm and $DE = 5$ cm.



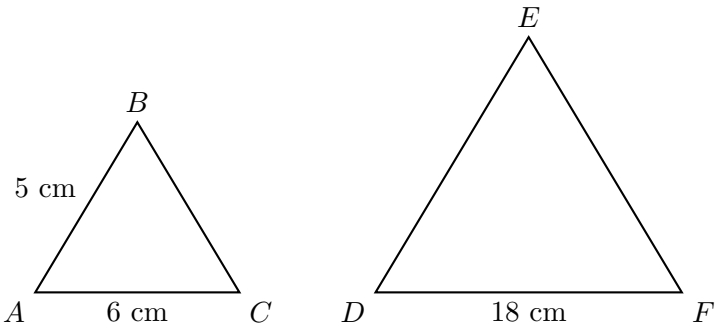

**(a)** Calculate the length of $BC$.

..... *cm* **[2]**

**(b)** The area of triangle $ADE$ is 8 cm$^2$.
Work out the area of the trapezium $DBCE$.

..... *cm*$^2$ **[2]**

**993** Triangle $ABC$ and triangle $DEF$ are mathematically similar.




NOT TO SCALE

**(a)** Calculate the length $DE$.

$DE =$ ..... $cm$ **[2]**

**(b)** The area of triangle $DEF$ is 72 cm$^2$.
Calculate the area of triangle $ABC$.

..... $cm^2$ **[2]**

**994** Two mathematically similar solid cylinders are shown as Cylinder $A$ and Cylinder $B$.
Cylinder $A$ has a height of 6 cm.
Cylinder $B$ has a height of 15 cm.

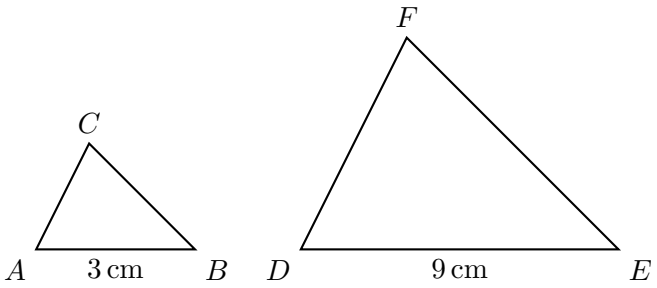
**(a)** The volume of Cylinder $A$ is 48 $cm^3$.
Calculate the volume of Cylinder $B$.

..... $cm^3$ **[2]**

**(b)** The curved surface area of Cylinder $B$ is 200 $cm^2$.
Calculate the curved surface area of Cylinder $A$.

..... $cm^2$ **[2]**

**995** Triangle $ABC$ and triangle $DEF$ are mathematically similar.
$AB = 3$ cm and $DE = 9$ cm.




NOT TO SCALE

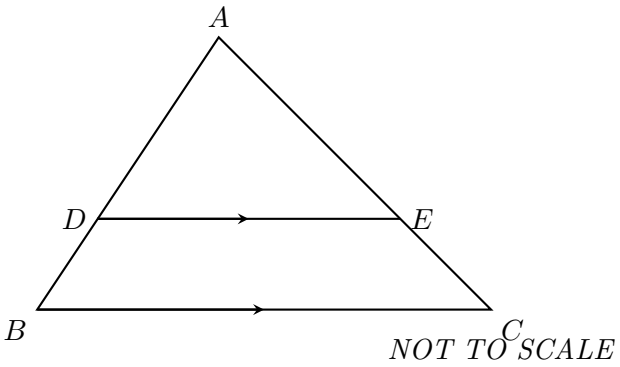
**(a)** The length of $BC$ is 5 cm.
Calculate the length of $EF$.

..... cm **[2]**

**(b)** The area of triangle $DEF$ is 54 cm$^2$.
Calculate the area of triangle $ABC$.

..... $cm^2$ **[2]**

**996** In the diagram, $ADE$ is a triangle and $DE$ is parallel to $BC$.
$AD = 4$ cm, $DB = 2$ cm and $DE = 5$ cm.



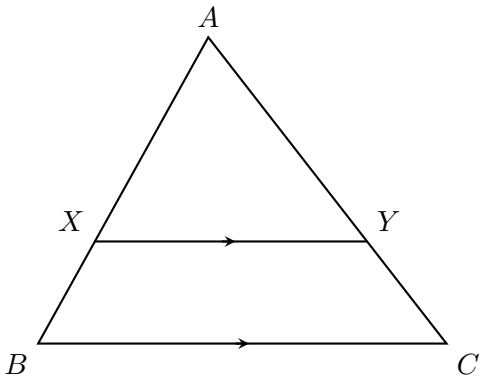

**(a)** Calculate the length of $BC$.

$BC =$ ..... cm **[2]**

**(b)** The area of triangle $ADE$ is 8 cm$^2$.
Calculate the area of the trapezium $DBCE$.

Area = ..... $cm^2$ **[3]**

**997** In the diagram, $ABC$ is a triangle and the line $XY$ is parallel to $BC$.
$AX = 4$ cm, $XB = 2$ cm and $XY = 6$ cm.



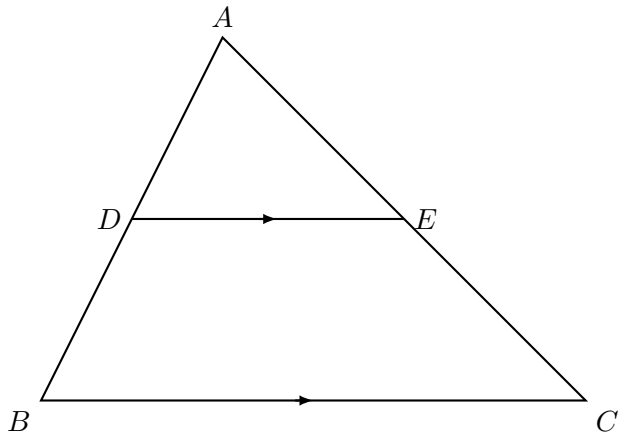

**(a)** Calculate the length of $BC$.

$BC =$ ..... cm **[2]**

**(b)** The area of triangle $AXY$ is 32 cm$^2$.
Calculate the area of the trapezium $XBCY$.

Area = ..... $cm^2$ **[3]**

**998** The diagram shows a triangle $ABC$.
$D$ is a point on $AB$ and $E$ is a point on $AC$ such that $DE$ is parallel to $BC$.
$AD = 4$ cm, $DB = 2$ cm and $DE = 6$ cm.




**(a)** Calculate the length of $BC$.

..... cm **[2]**

**(b)** The area of triangle $ADE$ is 12 cm$^2$.
Calculate the area of the trapezium $DBCE$.

..... $cm^2$ **[3]**

**999** Two mathematically similar bottles have heights of 10 cm and 15 cm.

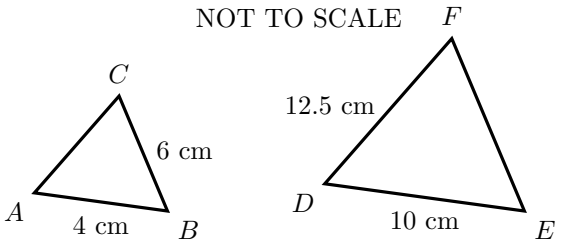
**(a)** The surface area of the smaller bottle is 120 $cm^2$.
Work out the surface area of the larger bottle.

..... $cm^2$ **[3]**

**(b)** The volume of the larger bottle is 540 $cm^3$.
Work out the volume of the smaller bottle.

..... $cm^3$ **[3]**

**1000** The diagram shows two mathematically similar triangles, $ABC$ and $DEF$.
$AB = 4$ cm, $BC = 6$ cm, $DE = 10$ cm, and $DF = 12.5$ cm.




**(a)** Calculate the length of $EF$.

$EF =$ ..... cm **[2]**

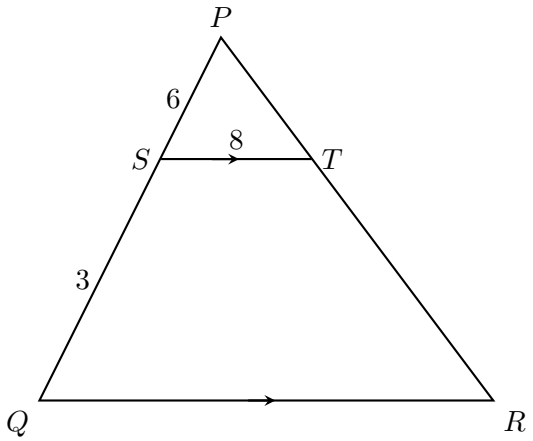
**(b)** Calculate the length of $AC$.

$AC =$ ..... cm **[2]**

**(c)** Given that the area of triangle $ABC$ is 10 cm$^2$, calculate the area of triangle $DEF$.

..... $cm^2$ **[2]**

**1001** In the diagram, the line $ST$ is parallel to $QR$.
$PS = 6$ cm, $SQ = 3$ cm and $ST = 8$ cm.
The length of $PR$ is 12 cm.




**(a)** Calculate the length of $QR$.

..... cm **[2]**

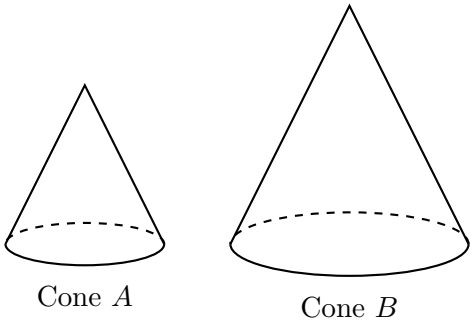
**(b)** Calculate the length of $PT$.

..... cm **[2]**

**(c)** The area of triangle $PST$ is 20 cm$^2$.
Calculate the area of the trapezium $SQRT$.

..... $cm^2$ **[3]**

**1002** Cone $A$ and Cone $B$ are mathematically similar.
The total surface area of Cone $A$ is 20 cm$^2$ and the total surface area of Cone $B$ is 45 cm$^2$.



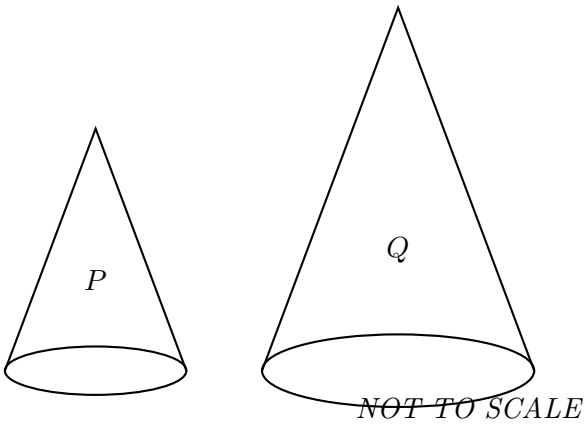

**(a)** The volume of Cone $A$ is 16 cm$^3$.
Calculate the volume of Cone $B$.

..... $cm^3$ **[3]**

**(b)** The height of Cone $B$ is 12 cm.
Calculate the height of Cone $A$.

..... cm **[2]**

**1003** The diagram shows two similar cones, $P$ and $Q$.
Cone $P$ has a volume of 400 cm$^3$ and a height of 12 cm.
Cone $Q$ has a volume of 1350 cm$^3$.



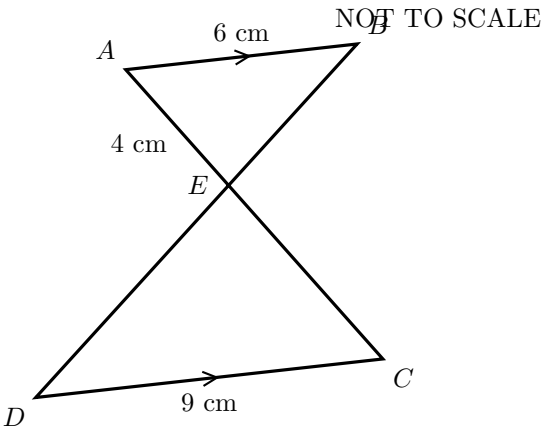

**(a)** Calculate the height of cone $Q$.

Height = ..... cm **[3]**

**(b)** The curved surface area of cone $Q$ is 360 cm$^2$.
Calculate the curved surface area of cone $P$.

Surface area = ..... $cm^2$ **[2]**

**1004** In the diagram, the straight lines $AC$ and $BD$ intersect at $E$, and $AB$ is parallel to $DC$.
$AB = 6$ cm, $DC = 9$ cm, $AE = 4$ cm, and $BD = 12.5$ cm.




**(a)** Explain why triangle $ABE$ is similar to triangle $CDE$.

.....

.....

..... **[3]**

**(b)** Calculate the length of

**(i)** $EC$,

$EC =$ ..... cm **[2]**

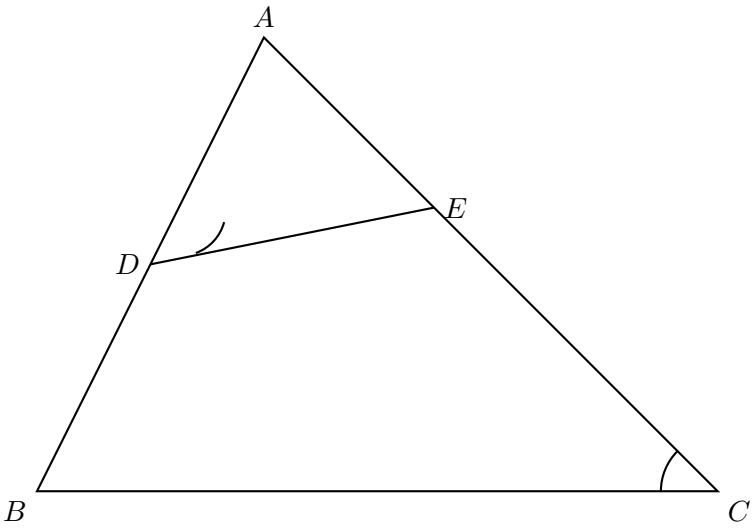
**(ii)** $BE$.

$BE =$ ..... cm **[2]**

**(c)** Given that the area of triangle $ABE$ is 12 cm$^2$, calculate the area of triangle $CDE$.

..... $cm^2$ **[2]**

**1005** The diagram shows a triangle $ABC$.
The point $D$ lies on $AB$ and the point $E$ lies on $AC$.
The angle $ADE$ is equal to the angle $ACB$.




**(a)** Explain why triangle $ADE$ is mathematically similar to triangle $ACB$.

**[3]**

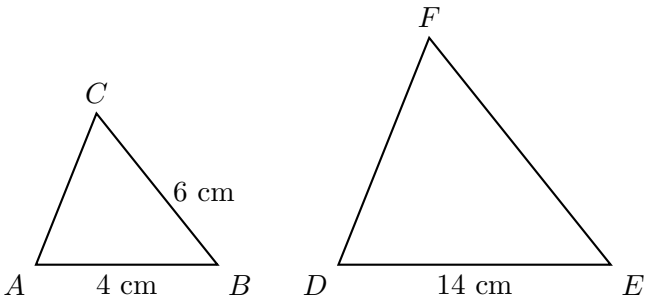
**(b)** It is given that $AD = x$ cm, $DB = (x + 1)$ cm, $AE = 4$ cm and $EC = 5$ cm.
Find the value of $x$.

$x =$ ..... **[4]**

**(c)** Find the ratio of the area of triangle $ADE$ to the area of the quadrilateral $DBCE$.
Give your answer in its simplest form.

. . . . . **[2]**

**1006** Triangles $ABC$ and $DEF$ are mathematically similar.
In the diagram, $AB = 4$ cm, $BC = 6$ cm and $DE = 14$ cm.



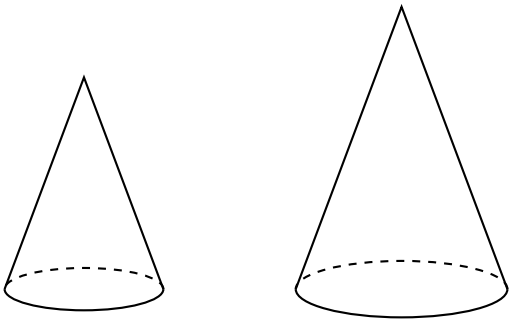

**(a)** Calculate the length of $EF$.

..... cm **[2]**

**(b)** The area of triangle $ABC$ is 8.5 cm$^2$.
Calculate the area of triangle $DEF$.

..... $cm^2$ **[2]**

**1007** Cone $A$ and Cone $B$ are mathematically similar.
The surface area of Cone $A$ is 450 cm$^2$ and the surface area of Cone $B$ is 800 cm$^2$.



Cone $A$ Cone $B$

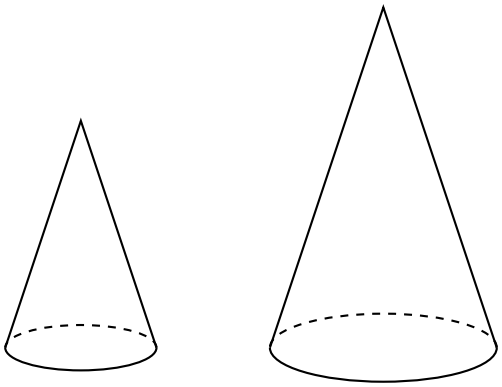
**(a)** The volume of Cone $A$ is 1350 cm$^3$.
Calculate the volume of Cone $B$.

Volume = ..... $cm^3$ **[3]**

**(b)** The height of Cone $B$ is 48 cm.
Calculate the height of Cone $A$.

Height = ..... $cm$ **[2]**

**1008** Two solid cones, $A$ and $B$, are mathematically similar.



Cone $A$ Cone $B$

**(a)** Cone $A$ has a volume of 150 cm$^3$ and a height of 12 cm.
Cone $B$ has a volume of 1200 cm$^3$.
Calculate the height of cone $B$.

..... $cm$ **[3]**

**(b)** The surface area of cone $B$ is 440 cm$^2$.
Calculate the surface area of cone $A$.

..... $cm^2$ **[2]**

**1009** Two mathematically similar containers, $A$ and $B$, have volumes of 320 $cm^3$ and 1080 $cm^3$ respectively.

**(a)** Container $A$ has a height of 18 cm.

Calculate the height of container $B$.

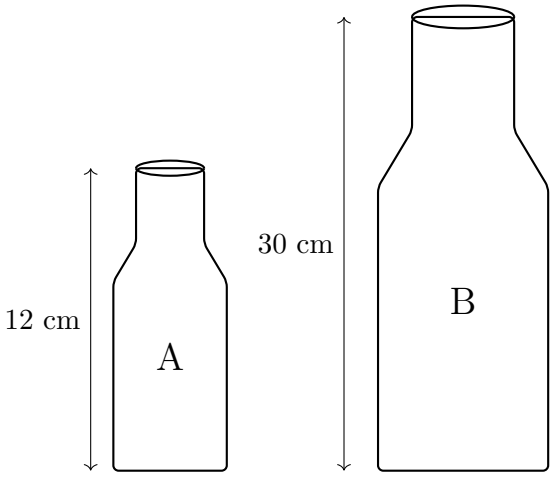
Height = ..... cm **[3]**

**(b)** Container $B$ has a surface area of 630 $cm^2$.

Calculate the surface area of container $A$.

Surface area = ..... $cm^2$ **[3]**

**1010** Two mathematically similar containers, A and B, are shown in the diagram.
Container A has a height of 12 cm and container B has a height of 30 cm.




NOT TO SCALE

**(a)** Container A has a volume of 320 $cm^3$.
Calculate the volume of container B.

Volume = ..... $cm^3$ **[2]**

**(b)** Container B has a surface area of 1125 $cm^2$.
Calculate the surface area of container A.

Surface area = ..... $cm^2$ **[2]**

**(c)** The containers are completely filled with liquid.
The cost of filling container A is \$4.80.
Assuming the cost is directly proportional to the volume, calculate the cost of filling container B.

..... **[2]**

**1011** Two mathematically similar solids, $P$ and $Q$, have volumes of 320 $cm^3$ and 8640 $cm^3$ respectively.

**(a)** The height of solid $P$ is 12 cm.
Calculate the height of solid $Q$.

..... cm **[3]**

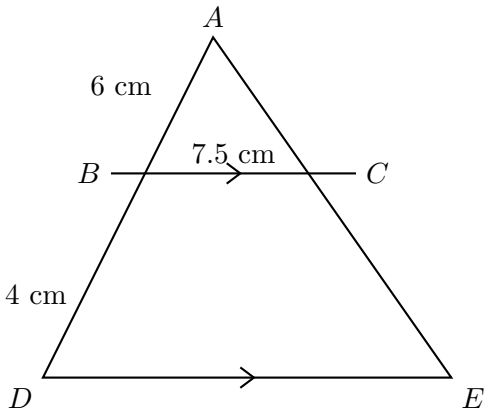
**(b)** The surface area of solid $Q$ is 2250 $cm^2$.
Calculate the surface area of solid $P$.

..... $cm^2$ **[2]**

**(c)** Solid $P$ is made of a metal with a density of 8.5 $g/cm^3$.
Calculate the mass of solid $P$ in kilograms.

..... kg **[2]**

**1012** The diagram shows triangle $ADE$ with a line $BC$ drawn parallel to $DE$.
$AB = 6$ cm, $BD = 4$ cm and $BC = 7.5$ cm.




NOT TO SCALE

**(a)** Explain why triangle $ABC$ is mathematically similar to triangle $ADE$.

**[2]**

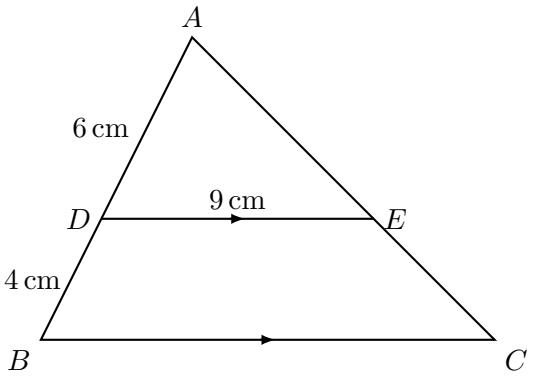
**(b)** Calculate the length of $DE$.

$DE =$ ..... $cm$ **[2]**

**(c)** The area of triangle $ABC$ is 18 cm$^2$.
Calculate the area of the quadrilateral $BCED$.

..... $cm^2$ **[3]**

**1013** In the diagram, $DE$ is parallel to $BC$.
$AD = 6$ cm, $DB = 4$ cm and $DE = 9$ cm.




NOT TO SCALE

**(a)** Explain why triangle $ADE$ is similar to triangle $ABC$.

.....

.....

..... **[2]**

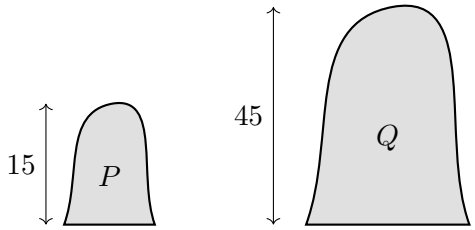
**(b)** Calculate the length of $BC$.

$BC =$ ..... cm **[3]**

**(c)** The area of triangle $ADE$ is 27 $cm^2$.
Calculate the area of the trapezium $DBCE$.

..... $cm^2$ **[3]**

**1014** The diagram shows two mathematically similar solid statues, $P$ and $Q$.
Statue $P$ has a height of 15 cm and a mass of 500 g.
Statue $Q$ has a height of 45 cm.
Assume that both statues are made from the same uniform material.




**(a)** Calculate the mass of statue $Q$.
Give your answer in kilograms.

..... kg **[3]**

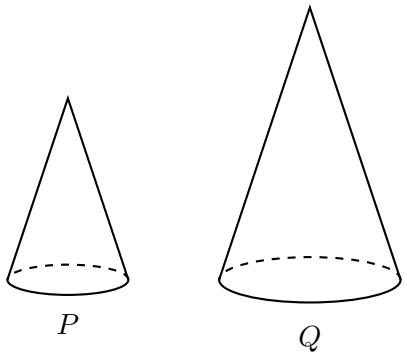
**(b)** A third similar statue, $R$, has a mass of 4 kg.
Calculate the height of statue $R$.

..... cm **[3]**

**(c)** The surface area of statue $Q$ is 2430 $cm^2$.
Calculate the surface area of statue $P$.

..... $cm^2$ **[2]**

**1015** The diagram shows two geometrically similar solid metal cones, $P$ and $Q$.
The volume of cone $P$ is 320 $cm^3$ and the volume of cone $Q$ is 1080 $cm^3$.




**(a)** The height of cone $P$ is 12 cm.
Calculate the height of cone $Q$.

..... cm **[3]**

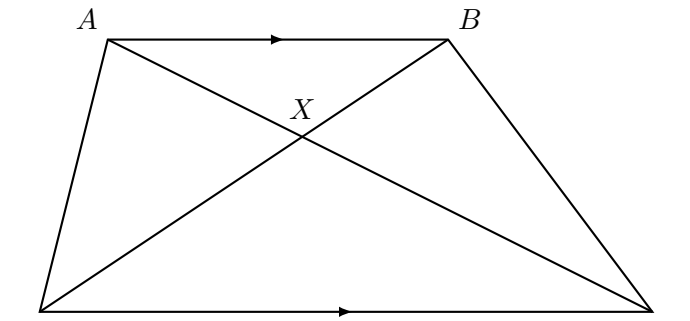
**(b)** The surface area of cone $Q$ is 450 $cm^2$.
Calculate the surface area of cone $P$.

..... $cm^2$ **[2]**

**(c)** Both cones are melted down to make a single new solid cone $R$, which is geometrically similar to $P$ and $Q$.
Calculate the height of cone $R$.
Give your answer correct to 3 significant figures.

..... cm **[3]**

**1016** The diagram shows a trapezium $ABCD$ with $AB$ parallel to $DC$.
The diagonals $AC$ and $BD$ intersect at point $X$.
$AB = 9$ cm, $DC = 15$ cm and $BX = 6$ cm.
The area of triangle $ABX$ is 27 $cm^2$.




NOT TO SCALE

**(a)** Explain why triangle $ABX$ is similar to triangle $CDX$.

**[2]**

**(b)** Calculate the length of $BD$.

..... $cm$ **[3]**

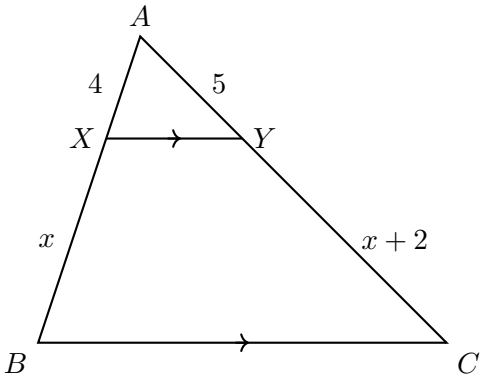
**(c)** Calculate the area of triangle $CDX$.

..... $cm^2$ **[2]**

**(d)** Calculate the area of triangle $BCX$.

..... $cm^2$ **[2]**

**1017** In the diagram, the line $XY$ is parallel to $BC$.
$AX = 4$ cm, $XB = x$ cm, $AY = 5$ cm and $YC = (x + 2)$ cm.




**(a)** Calculate the value of $x$.

$x =$ ..... **[3]**

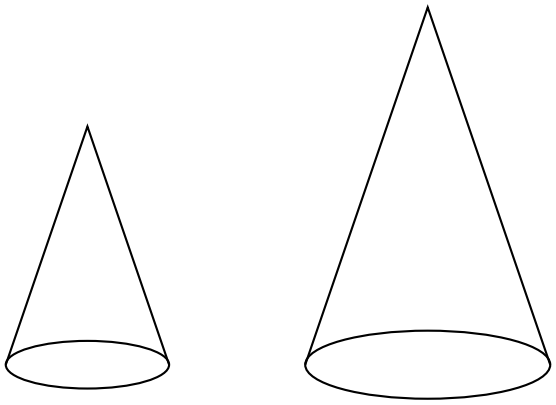
**(b)** $XY = 3.5$ cm.
Calculate the length of $BC$.

..... cm **[2]**

**(c)** The area of triangle $AXY$ is $A$ cm$^2$.
Find an expression for the area of the trapezium $XBCY$ in terms of $A$.
Give your answer in its simplest form.

..... **[2]**

**1018** Cone $A$ and Cone $B$ are mathematically similar solids.




NOT TO SCALE

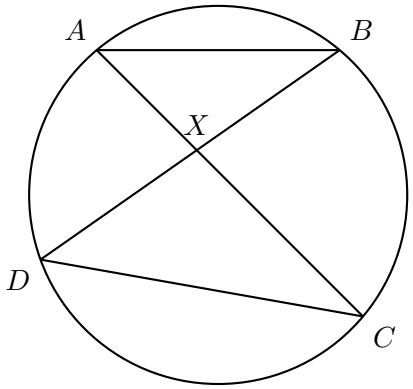
**(a)** The volume of Cone $A$ is 108 cm$^3$ and the volume of Cone $B$ is 500 cm$^3$.
The sum of the surface areas of Cone $A$ and Cone $B$ is 340 cm$^2$.
Calculate the surface area of Cone $B$.

..... $cm^2$ **[4]**

**(b)** Cone $B$ has a total vertical height of 20 cm.
A smaller cone, mathematically similar to Cone $B$, is removed from the top of Cone $B$ to leave a frustum.
The volume of the frustum is $\frac{7}{8}$ of the volume of Cone $B$.
Calculate the vertical height of the removed small cone.

..... $cm$ **[3]**

**1019** The diagram shows a circle with points $A$, $B$, $C$ and $D$ on its circumference.
The chords $AC$ and $BD$ intersect at $X$.




**(a)** Show that triangle $AXB$ is similar to triangle $DXC$.
Give a geometric reason for each statement you make.

**[3]**

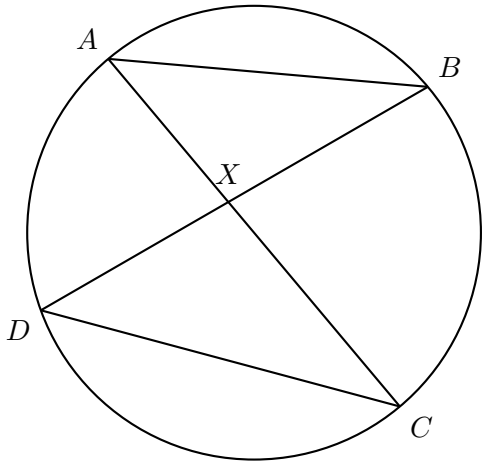
**(b)** $AX = 4$ cm, $BX = 5$ cm, $CX = 7.5$ cm and $AB = 6$ cm.
Calculate the length of $DX$ and the length of $DC$.

$DX =$ ..... , $DC =$ ..... **[3]**

**(c)** The area of triangle $AXB$ is 11.8 cm$^2$.
Calculate the area of triangle $DXC$.

..... $cm^2$ **[2]**

**1020** Points $A$, $B$, $C$, and $D$ lie on the circumference of a circle.
The straight lines $AC$ and $BD$ intersect at the point $X$.




**(a)** Show that triangle $AXB$ is similar to triangle $DXC$, giving geometrical reasons for your statements.

**[3]**

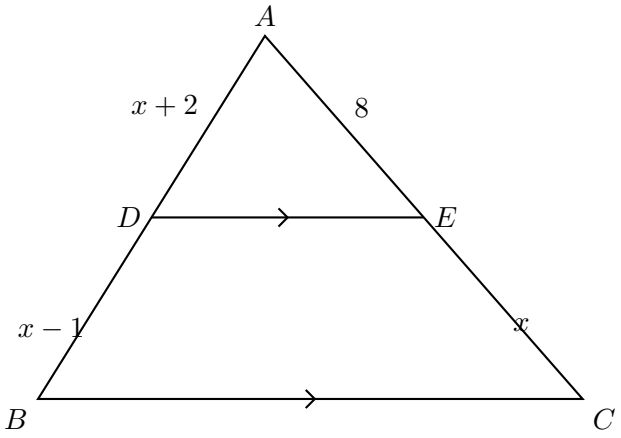
**(b)** $AX = (x + 2)$ cm, $CX = (x - 2)$ cm, $BX = (x - 3)$ cm, and $DX = (x + 4)$ cm.
Calculate the value of $x$.

$x =$ ..... **[3]**

**(c)** The area of triangle $DXC$ is 45 cm$^2$.
Calculate the area of triangle $AXB$.

Area = ..... $cm^2$ **[2]**

**1021** In the diagram, $DE$ is parallel to $BC$.
$AD = (x + 2)$ cm, $DB = (x - 1)$ cm, $AE = 8$ cm and $EC = x$ cm.




NOT TO SCALE

**(a)** Show that $x^2 - 6x + 8 = 0$.

**[3]**

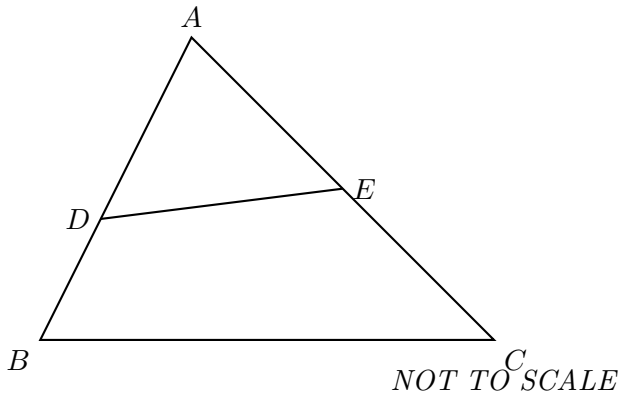
**(b)** Solve the equation $x^2 - 6x + 8 = 0$.

$x =$ ..... , $x =$ ..... **[2]**

**(c)** Given that the area of triangle $ADE$ is 24 cm$^2$ and $x = 4$, calculate the area of the trapezium $DBCE$.

..... $cm^2$ **[3]**

**1022** In the diagram, $D$ is a point on $AB$ and $E$ is a point on $AC$.
Angle $AED$ = Angle $ABC$.
$AD = 6$ cm, $AE = 5$ cm and $DB = 9$ cm.




**(a)** Show that triangle $ADE$ is similar to triangle $ACB$.
Give reasons for your statements.

**[3]**

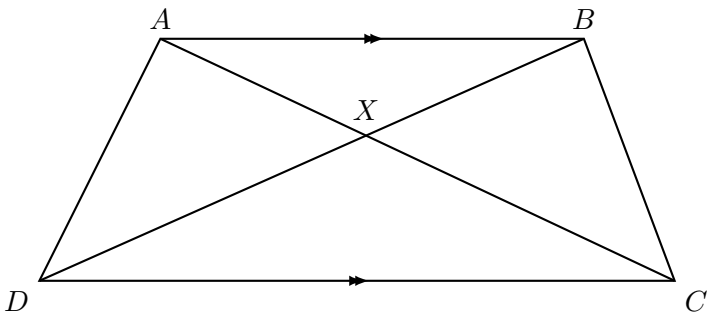
**(b)** Calculate the length of $EC$.

$EC =$ ..... cm **[3]**

**(c)** Find the ratio of the area of triangle $ADE$ to the area of quadrilateral $BCED$ in its simplest form.

..... **[2]**

**1023** The diagram shows a trapezium $ABCD$ with $AB$ parallel to $DC$.
The diagonals $AC$ and $BD$ intersect at $X$.
$AB = 7$ cm and $DC = 10.5$ cm.




**(a)** Explain why triangle $ABX$ is similar to triangle $CDX$.

**[2]**

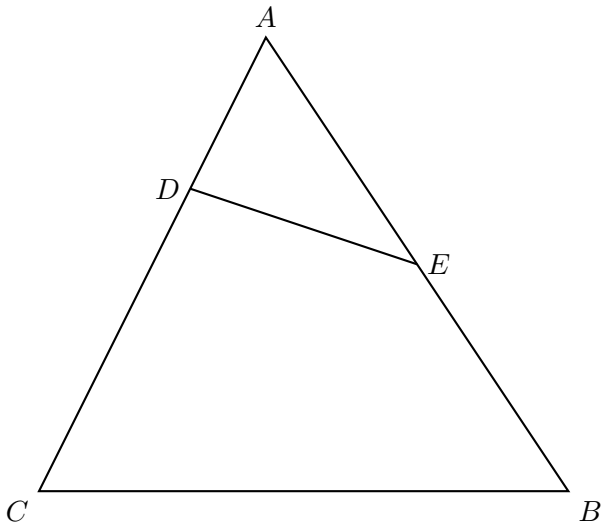
**(b)** $AX = 4$ cm.
Calculate the length of $AC$.

$AC =$ ..... cm **[3]**

**(c)** The area of triangle $ABX$ is 16 cm$^2$.
Calculate the area of the trapezium $ABCD$.

..... $cm^2$ **[4]**

**1024** The diagram shows triangle $ABC$.
$D$ is a point on $AC$ and $E$ is a point on $AB$.
Angle $ABC$ = Angle $AED$.




NOT TO SCALE

**(a)** Explain why triangle $ABC$ is similar to triangle $AED$.

.....

.....

..... **[2]**

**(b)** Given that $AC = 15$ cm and $AD = 6$ cm, write down the scale factor of the enlargement that maps triangle $AED$ onto triangle $ABC$.

..... **[1]**

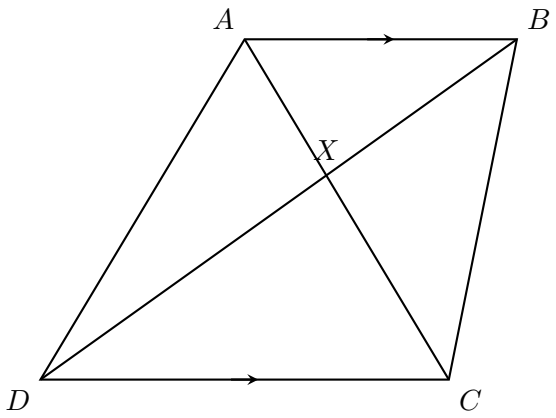
**(c)** $AE = x$ cm and $AB = (2x + 2.5)$ cm.
Use similarity to set up an equation and find the value of $x$.

$x =$ ..... **[3]**

**(d)** The area of the quadrilateral $BCDE$ is 42 cm$^2$.
Calculate the area of triangle $ADE$.

..... $cm^2$ **[3]**

**1025** The diagram shows a trapezium $ABCD$ with $AB$ parallel to $DC$.
The diagonals $AC$ and $BD$ intersect at $X$.




**(a)** Explain, giving geometric reasons, why triangle $ABX$ is similar to triangle $CDX$.

**[3]**

**(b)** $AX = 4$ cm and $CX = 6$ cm.
$AB = x$ cm and $DC = (2x - 3)$ cm.

Work out the value of $x$.

$x =$ ..... **[3]**

**(c)** The area of triangle $ABX$ is 24 cm$^2$.
Calculate the area of the trapezium $ABCD$.

Area = ..... $cm^2$ **[4]**

**1026** Two solid statues, A and B, are mathematically similar.

**(a)** Statue A has a height of 12 cm and a volume of 800 $cm^3$.
Statue B has a volume of 2700 $cm^3$.
Calculate the height of Statue B.

..... cm **[3]**

**(b)** The cost to paint a statue is directly proportional to its surface area.
It costs \$18 to paint Statue A.
Calculate the cost to paint Statue B.

\$ ..... **[3]**

**(c)** A third mathematically similar statue, C, has a surface area that is 44% greater than the surface area of Statue A.
Calculate the volume of Statue C.

..... $cm^3$ **[4]**

**1027** Solid Cone $A$ has a base radius of 10 cm and a perpendicular height of 24 cm.

**(a)** Calculate the total surface area of Cone $A$.
Give your answer in terms of $\pi$.

..... $cm^2$ **[3]**

**(b)** Cone $B$ is mathematically similar to Cone $A$.
The volume of Cone $B$ is $100\pi$ $cm^3$.
Calculate the total surface area of Cone $B$.
Give your answer in terms of $\pi$.

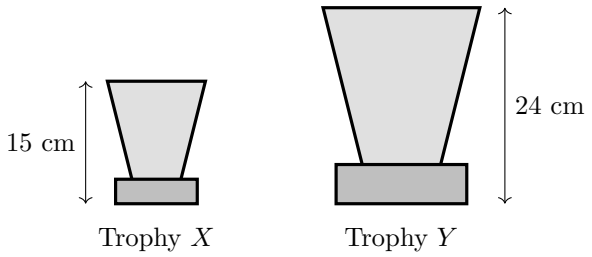
..... $cm^2$ **[4]**

**(c)** A frustum is made by cutting a smaller cone, which is geometrically identical to Cone $B$, from the top of Cone $A$.
This frustum is melted down and recast completely into a solid sphere.
Calculate the radius of this sphere.

..... $cm$ **[4]**

**1028** The diagram shows two mathematically similar solid metal trophies, Trophy $X$ and Trophy $Y$.
Trophy $X$ has a height of 15 cm and a total surface area of 320 $cm^2$.
Trophy $Y$ has a height of 24 cm and a volume of 2048 $cm^3$.

NOT TO SCALE




**(a)** Calculate the total surface area of Trophy $Y$.

..... $cm^2$ **[2]**

**(b)** Calculate the volume of Trophy $X$.

..... $cm^3$ **[3]**

**(c)** A third trophy, Trophy $Z$, is mathematically similar to Trophy $X$ and has a volume of 1372 $cm^3$.
Calculate the height of Trophy $Z$.

..... cm **[3]**

**(d)** Trophy $Y$ is melted down to make as many complete copies of Trophy $X$ as possible.
The remaining metal is used to make a solid sphere of radius $r$ cm. Calculate the value of $r$, giving your answer correct to 3 significant figures.

$r =$ ..... **[4]**

**1029** Answer the following questions about the symmetry of two-dimensional shapes.

**(a)** Write down the number of lines of symmetry of a regular nonagon (a polygon with 9 sides).

..... **[1]**

**(b)** Write down the order of rotational symmetry of a parallelogram.

..... **[1]**

**(c)** Write down the mathematical name of a triangle that has exactly 1 line of symmetry and rotational symmetry of order 1.

..... **[1]**

**1030** Answer the following questions about 2D symmetry.

**(a)** Write down the order of rotational symmetry of a regular decagon.

..... **[1]**

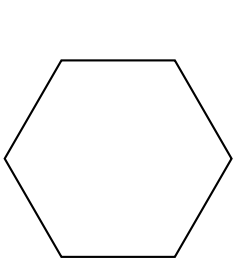
**(b)** A quadrilateral has exactly two lines of symmetry and rotational symmetry of order 2.
State the mathematical name of this quadrilateral.

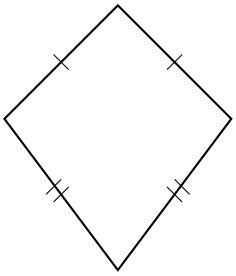
..... **[1]**

**(c)** A shape has rotational symmetry of order 5.
Calculate the smallest angle of rotation that maps the shape onto itself.

..... $^\circ$ **[1]**

**1031** The diagram shows a regular hexagon and a kite.





**(a)** For the regular hexagon, write down:

**(i)** the number of lines of symmetry,

..... **[1]**

**(ii)** the order of rotational symmetry.

..... **[1]**

**(b)** For the kite, write down:

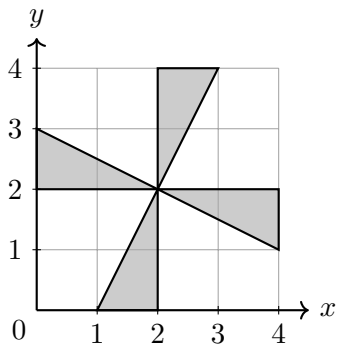
**(i)** the number of lines of symmetry,

..... **[1]**

**(ii)** the order of rotational symmetry.

..... **[1]**

**1032** The diagram shows a shaded shape on a coordinate grid.



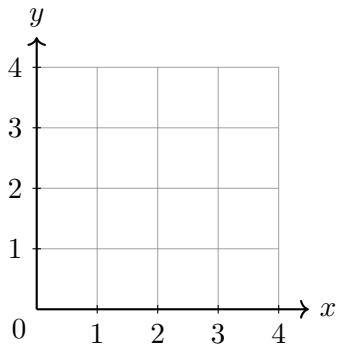

**(a)** Write down the order of rotational symmetry of the shaded shape.

..... **[1]**

**(b)** Write down the number of lines of symmetry of the shaded shape.

..... **[1]**

**(c)** On the grid below, draw the image of the shape after a reflection in the line $x = 2$.




**[2]**

**1033** Write down the symmetry properties for the following geometric shapes.

**(a)** A regular polygon has 15 sides.

Write down:

**(i)** the number of lines of symmetry,

..... **[1]**

**(ii)** the order of rotational symmetry.

..... **[1]**

**(b)** A quadrilateral has exactly two lines of symmetry and rotational symmetry of order 2.

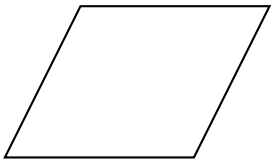
Write down the mathematical name of this quadrilateral.

..... **[1]**

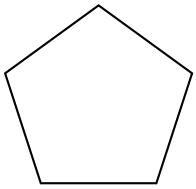
**(c)** Write down the number of planes of symmetry of a solid right pyramid with a regular pentagonal base.

..... **[1]**

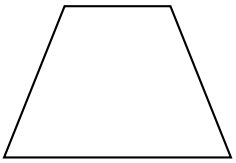
**1034** Three 2D shapes are shown below.



Parallelogram



Regular Pentagon



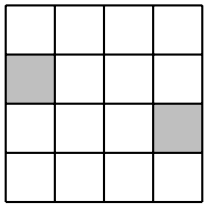
Isosceles Trapezium

Complete the table to show the number of lines of symmetry and the order of rotational symmetry for each shape.

| | | |
|---|---|---|
| | | |
| | | |
| | | |

**[6]**

**1035** The diagram shows a $4 \times 4$ grid with two squares shaded.



**(a)** For the diagram above, write down

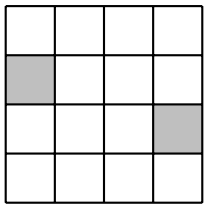
**(i)** the number of lines of symmetry,

..... **[1]**

**(ii)** the order of rotational symmetry.

..... **[1]**

**(b)** On the grid below, shade two more squares so that the completed pattern has exactly 2 lines of symmetry.

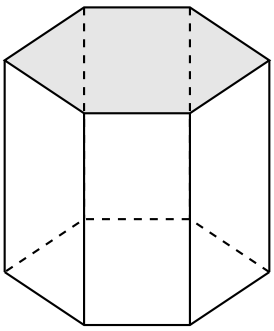


**[2]**

**(c)** A regular polygon has rotational symmetry of order 12.
Write down the number of lines of symmetry of this regular polygon and calculate the size of each exterior angle.

Lines of symmetry = ..... , Exterior angle = ..... **[2]**

**1036** The diagram shows a solid regular hexagonal prism.



**(a)** Write down the number of planes of symmetry of this regular hexagonal prism.

..... **[2]**

**(b)** Write down the order of rotational symmetry of the prism about the vertical axis passing through the centres of its hexagonal faces.

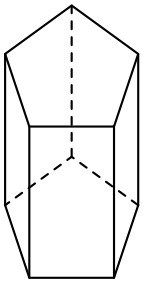
..... **[1]**

**(c)** A different solid is a right square-based pyramid.

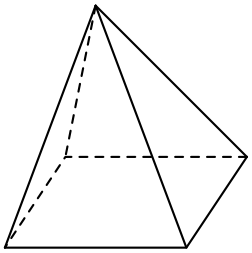
Write down the number of planes of symmetry of this pyramid.

..... **[1]**

**1037** Solid $A$ is a right regular pentagonal prism.
Solid $B$ is a right square-based pyramid.



Solid $A$



Solid $B$

**(a)** State the number of planes of symmetry for:

**(i)** Solid $A$,

..... **[1]**

**(ii)** Solid $B$,

..... **[1]**

**(iii)** a solid cylinder.

..... **[1]**

**(b)** A regular polygon has exactly 12 lines of symmetry.
Calculate the size of one of its interior angles.

..... ° **[2]**

**1038** Consider the symmetry properties of the following 3D solid shapes.

**(a)** A right square-based pyramid has a base of side length 8 cm and perpendicular height 10 cm.
Write down the number of planes of symmetry for this pyramid.

..... **[1]**

**(b)** Write down the total number of planes of symmetry of a solid regular pentagonal prism.

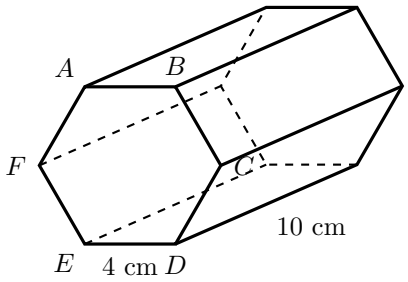
..... **[2]**

**(c)** A right circular cone has base radius $r$ and slant height $l$.
Describe fully its axis of symmetry.

..... **[2]**

**1039** The diagram shows a right prism whose cross-section is a regular hexagon $ABCDEF$ of side length 4 cm. The length of the prism is 10 cm.

NOT TO SCALE




**(a)** For this regular hexagonal prism, write down

**(i)** the total number of planes of symmetry,

..... **[2]**

**(ii)** the order of rotational symmetry about the axis passing through the centres of the two hexagonal faces.

..... **[1]**

**(b)** A solid right pyramid has a regular pentagon as its base.
Write down the number of planes of symmetry and the order of rotational symmetry of this pyramid.

Planes of symmetry = ..... , Order of rotational symmetry = .....
**[2]**

**(c)** A solid right circular cone has base radius 6 cm and vertical height 8 cm. State the number of planes of symmetry of the cone, and calculate the total surface area of the cone, in $cm^2$, in terms of $\pi$.

Planes of symmetry = ..... , Total surface area = ..... **[3]**

**1040** The points $A(1, 6)$ and $B(7, 4)$ are two vertices of a quadrilateral $ABCD$.
The point $M(3, 2)$ is the centre of rotational symmetry of $ABCD$.
The quadrilateral $ABCD$ has rotational symmetry of order 4.

**(a)** Find the coordinates of $C$.

( ..... , ..... ) **[2]**

**(b)** The line $AC$ is a line of symmetry of the quadrilateral.

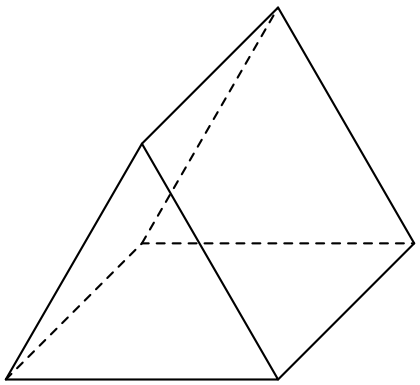
Find the equation of the line $AC$.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

**(c)** Calculate the exact area of the quadrilateral $ABCD$.

..... **[3]**

**1041** The diagram shows a solid right prism.
The cross-section of the prism is an equilateral triangle.



**(a)** State the number of planes of symmetry of this solid prism.

..... **[2]**

**(b)** State the order of rotational symmetry of the prism about the axis passing through the centroids of the two equilateral triangular faces.

..... **[1]**

**(c)** A different solid is a right regular hexagonal prism.

State the number of planes of symmetry of this hexagonal prism.

..... **[2]**

**1042** **(a)** A regular polygon has an exterior angle of 15°.

**(i)** Calculate the number of sides of this polygon.

..... **[2]**

**(ii)** Write down the number of lines of symmetry of this polygon.

..... **[1]**

**(b)** Write down the mathematical name of a quadrilateral that has exactly 1 line of symmetry and rotational symmetry of order 1.

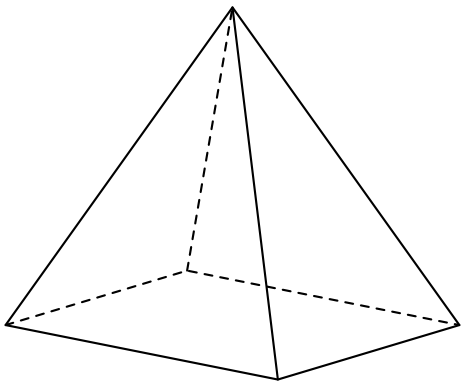
..... **[1]**

**(c)** A solid cone has a base radius of 5 cm and a perpendicular height of 12 cm.

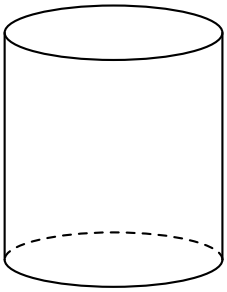
State the number of planes of symmetry of this cone.

..... **[1]**

**1043** The diagram shows a solid square-based pyramid and a solid cylinder.



Square-based pyramid



Cylinder

**(a)** State the number of planes of symmetry for each solid.

**(i)** The solid square-based pyramid.

..... **[1]**

**(ii)** The solid cylinder.

..... **[1]**

**(b)** A solid square-based pyramid has a base side length of 6 cm.
The pyramid has exactly 4 planes of symmetry and its volume is 48 $cm^3$.
Calculate the length of a sloping edge of this pyramid.

..... cm **[5]**

**1044** A rhombus $ABCD$ has its centre at the origin $(0, 0)$.

**(a)** The line of symmetry passing through vertices $A$ and $C$ has the equation $y = 3x$.
Find the equation of the line of symmetry passing through vertices $B$ and $D$.

$y =$ ..... **[2]**

**(b)** Write down the order of rotational symmetry of the rhombus $ABCD$.

..... **[1]**

**(c)** A different shape is a regular polygon with $n$ sides.
The number of lines of symmetry of this polygon is equal to $n$.
The interior angle of this polygon is $165°$.

Calculate the value of $n$.

$n =$ ..... **[3]**

**1045** **(a)** A solid wooden block is in the shape of a right prism.
The cross-section of the prism is a regular octagon.

**(i)** Write down the order of rotational symmetry of the cross-section.

..... **[1]**

**(ii)** State the total number of planes of symmetry of this right prism.

..... **[2]**

**(b)** A different solid is a right pyramid with a regular polygon of $n$ sides as its base.
The pyramid has exactly 12 planes of symmetry.

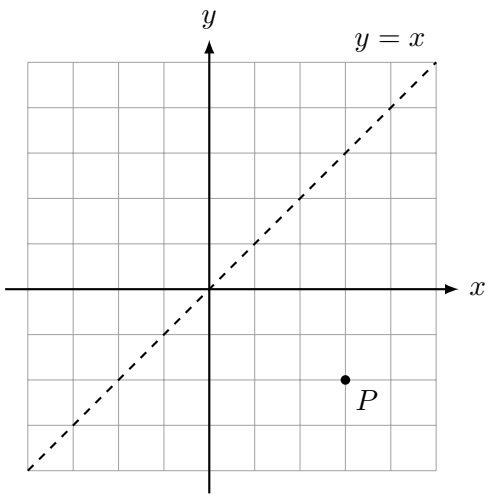
**(i)** Write down the value of $n$.

$n =$ ..... **[1]**

**(ii)** Calculate the sum of the interior angles of the base of this pyramid.

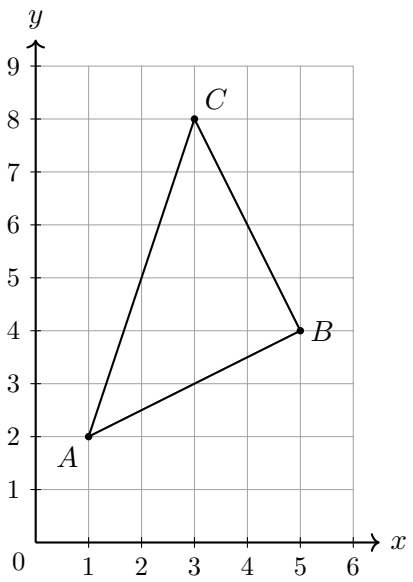
..... ° **[2]**

**(c)** A shape $S$ is drawn on a coordinate grid.
The shape $S$ has a line of symmetry with equation $y = x$.
One vertex of the shape is at the point $P(3, -2)$.
Write down the coordinates of the image of $P$ after a reflection in this line of symmetry.




( ..... , ..... ) **[2]**

**1046** Triangle $ABC$ has vertices at $A(1, 2)$, $B(5, 4)$ and $C(3, 8)$.




**(a)** Show that triangle $ABC$ is an isosceles triangle.

**[3]**

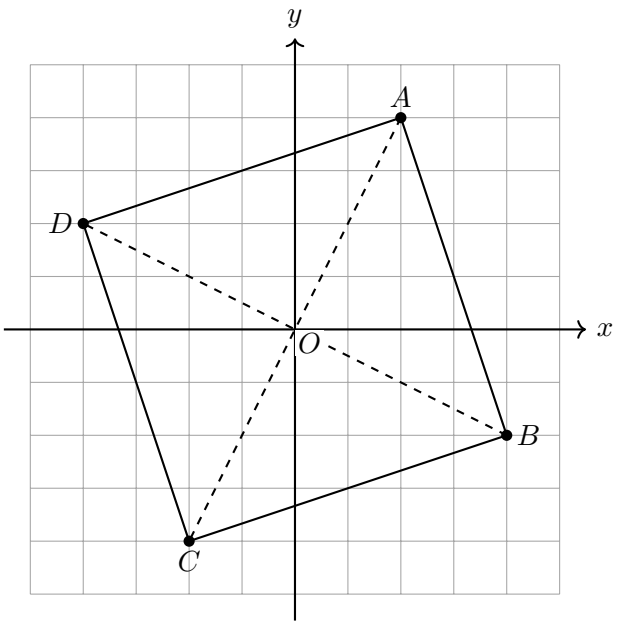
**(b)** Triangle $ABC$ has exactly one line of symmetry.
Find the equation of this line of symmetry.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[4]**

**(c)** A different shape is a regular polygon with $n$ sides.
It has exactly 15 lines of symmetry.
Calculate the size of one interior angle of this polygon.

..... ° **[2]**

**1047** The diagram shows a rhombus $ABCD$.
The diagonals $AC$ and $BD$ intersect at the origin $O(0, 0)$.
The line of symmetry passing through $A$ and $C$ has the equation $y = 2x$.




**(a)** Write down the order of rotational symmetry of the rhombus $ABCD$.

..... **[1]**

**(b)** Find the equation of the line of symmetry passing through $B$ and $D$.

$y =$ ..... **[2]**

**(c)** The coordinates of $A$ are $(k, 2k)$, where $k > 0$.
The area of the rhombus is 40 square units, and the length of the diagonal $BD$ is $4\sqrt{5}$ units.

**(i)** Show that the length of the diagonal $AC$ is $4\sqrt{5}$ units.

**[2]**

**(ii)** Calculate the coordinates of $A$.

$A = (\ldots\ldots\ ,\ \ldots\ldots)$ **[2]**

**(iii)** Calculate the coordinates of $B$.

$B = (\ldots\ldots\ ,\ \ldots\ldots)$ **[3]**

**1048** A right solid prism has a uniform cross-section which is a regular polygon with $n$ sides.
The prism has exactly 7 planes of symmetry.
The length of the prism is not equal to its width.

**(a)** Explain why $n = 6$.

**[2]**

**(b)** The longest diagonal of the hexagonal cross-section is 12 cm.
The volume of the prism is 450 $cm^3$.

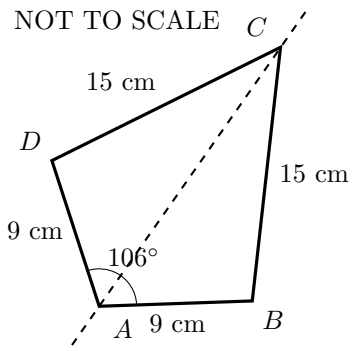
**(i)** Calculate the length of the prism.

..... cm **[4]**

**(ii)** Calculate the total surface area of the prism.

..... $cm^2$ **[4]**

**1049** The diagram shows a quadrilateral $ABCD$ that has exactly one line of symmetry, which passes through vertices $A$ and $C$.
$AB = AD = 9$ cm, $BC = CD = 15$ cm, and angle $BAD = 106°$.




**(a)** Write down the special mathematical name for quadrilateral $ABCD$ and state its order of rotational symmetry.

Name: ..... , Order of rotational symmetry = ..... **[2]**

**(b)** The diagonals $AC$ and $BD$ intersect at $M$.
Using the symmetry of $ABCD$, explain why angle $AMB = 90°$.

..... **[1]**

**(c)** Calculate

**(i)** the length of diagonal $BD$, giving your answer correct to 3 significant figures,

$BD =$ ..... cm **[3]**

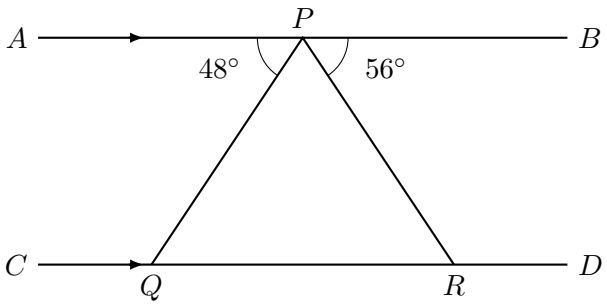
**(ii)** the size of angle $BCD$, giving your answer correct to 1 decimal place.

Angle $BCD =$ ..... ° **[3]**

**(d)** A right prism has quadrilateral $ABCD$ as its uniform cross-section.
Write down the total number of planes of symmetry of this prism.

..... **[2]**

**1050** In the diagram, the line $AB$ is parallel to the line $CD$.
$P$ is a point on $AB$.
$Q$ and $R$ are points on $CD$.
The line $PQ$ and the line $PR$ are drawn.
Angle $APQ = 48°$ and angle $BPR = 56°$.




NOT TO SCALE

**(a)** Write down the value of angle $PQR$.

Angle $PQR =$ ..... ° **[1]**

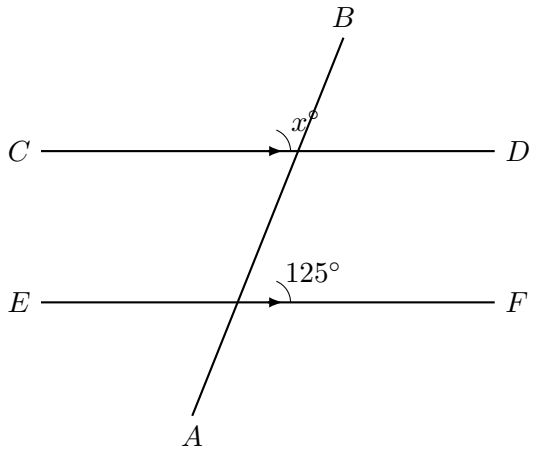
**(b)** Write down the value of angle $PRQ$.

Angle $PRQ =$ ..... ° **[1]**

**(c)** Work out the value of angle $QPR$.

Angle $QPR =$ ..... ° **[1]**

**1051** The diagram shows a straight line $AB$ intersecting two parallel lines, $CD$ and $EF$.




**(a)** Find the value of $x$.

$x =$ ..... **[1]**

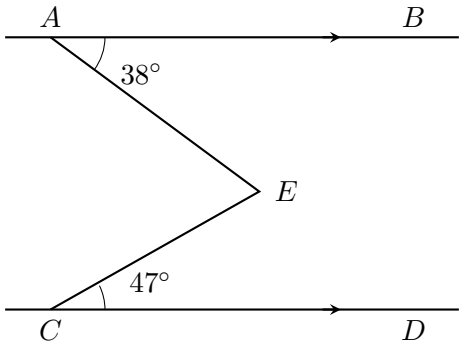
**(b)** Give a geometric reason for your answer to part (a).

..... **[1]**

**(c)** Calculate the interior angle of a regular decagon (10-sided polygon).

..... ° **[1]**

**1052** In the diagram, the line $AB$ is parallel to the line $CD$.
The point $E$ lies between the two parallel lines.




NOT TO SCALE

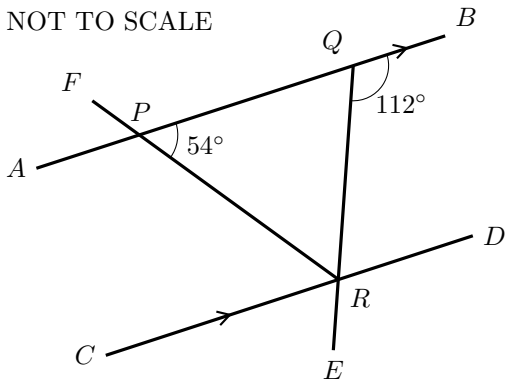
**(a)** Calculate the size of angle $AEC$.

Angle $AEC$ = ..... ° **[2]**

**(b)** A regular polygon has an exterior angle of 24°.
Work out the number of sides of this polygon.

..... **[2]**

**1053** In the diagram, $AB$ is parallel to $CD$.
The straight line $FPR$ intersects $AB$ at $P$ and $CD$ at $R$.
The straight line $QRE$ intersects $AB$ at $Q$ and $CD$ at $R$.
Angle $QPR = 54°$ and angle $BQR = 112°$.




**(a)** Write down the size of angle $PRC$ and give a geometrical reason for your answer.

Angle $PRC =$ ..... , Reason: ..... **[2]**

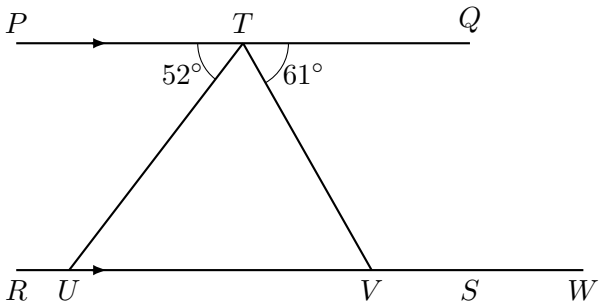
**(b)** Calculate the size of angle $PRQ$.

Angle $PRQ =$ ..... ° **[2]**

**(c)** Write down the size of angle $CRE$.

Angle $CRE =$ ..... ° **[1]**

**1054** In the diagram, the line $PQ$ is parallel to the line $RS$.
The point $T$ lies on $PQ$, and the points $U$ and $V$ lie on $RS$.
The line $RS$ is extended to a point $W$.
Angle $PTU = 52°$ and angle $QTV = 61°$.




**(a)** Calculate angle $TUV$.

Angle $TUV$ = ..... ° **[1]**

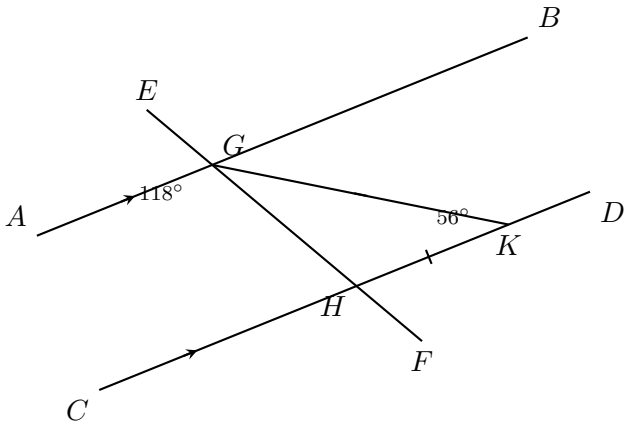
**(b)** Calculate angle $UTV$.

Angle $UTV$ = ..... ° **[2]**

**(c)** Calculate angle $TVW$.

Angle $TVW$ = ..... ° **[2]**

**1055** In the diagram, $AB$ is parallel to $CD$.
The straight line $EF$ intersects $AB$ at $G$ and $CD$ at $H$.
Point $K$ lies on $CD$ such that $GK = HK$, $\angle AGH = 118°$ and $\angle GKH = 56°$.




**(a)** Find the size of angle $GHD$, giving a geometrical reason for your answer.

$\angle GHD =$ ..... ° **[2]**

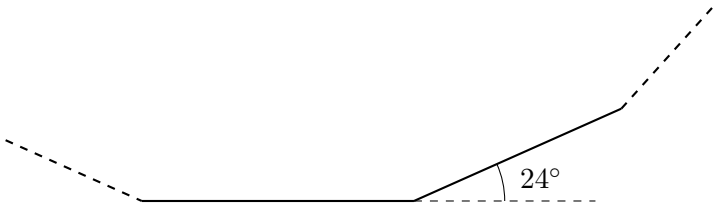
**(b)** Calculate the size of angle $GHK$.

$\angle GHK =$ ..... ° **[1]**

**(c)** Calculate the size of angle $BGK$.

$\angle BGK =$ ..... ° **[2]**

**1056** The diagram shows part of a regular polygon with $n$ sides.
The exterior angle of this polygon is 24°.




**(a)** Calculate the value of $n$.

$n =$ ..... **[2]**

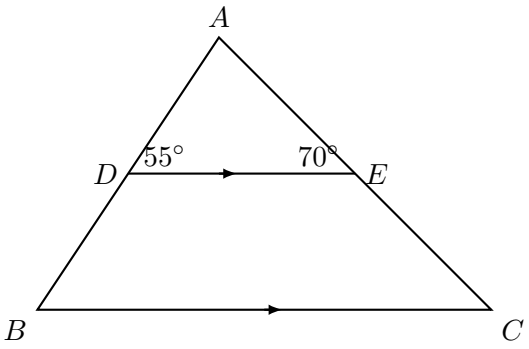
**(b)** Calculate the sum of the interior angles of this polygon.

..... ° **[2]**

**(c)** A different regular polygon has an interior angle of 165°.
Work out the number of sides of this polygon.

..... **[2]**

**1057** The diagram shows a triangle $ABC$ with a line $DE$ drawn parallel to $BC$.
$D$ lies on $AB$ and $E$ lies on $AC$.
Angle $ADE = 55°$ and angle $AED = 70°$.




**(a)** Calculate angle $DAE$.

Angle $DAE =$ ..... ° **[2]**

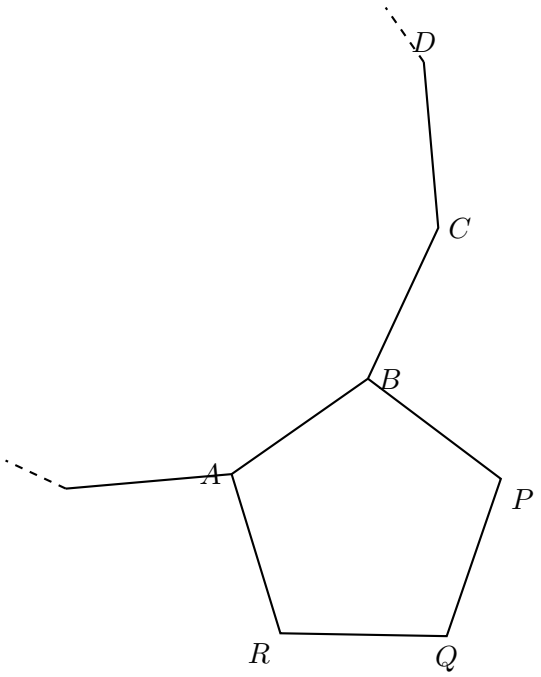
**(b)** Calculate angle $ABC$.

Angle $ABC =$ ..... ° **[1]**

**(c)** Explain why triangle $ADE$ is an isosceles triangle.

**[2]**

**1058** The diagram shows part of a regular $n$-sided polygon $ABCDE\ldots$ and a regular pentagon $ABPQR$ sharing the common side $AB$.




**(a)** Calculate the size of one interior angle of the regular pentagon $ABPQR$.

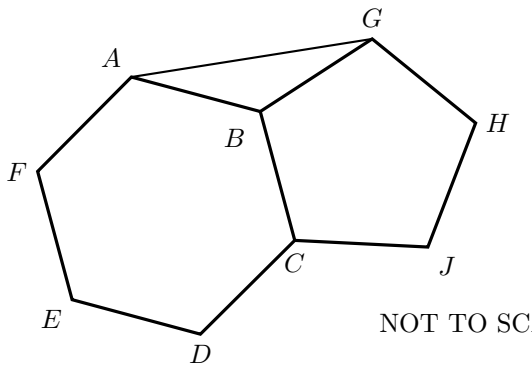
..... ° **[2]**

**(b)** The reflex angle $CBP$ is 102°.

Calculate the value of $n$, the number of sides of the regular polygon $ABCDE\ldots$.

$n =$ ..... **[4]**

**1059** The diagram shows a regular hexagon $ABCDEF$ and a regular pentagon $BCGHJ$ sharing a common side $BC$.
Vertices $A$ and $G$ are joined by a straight line.




**(a)** Calculate the size of

**(i)** interior angle $ABC$ of the regular hexagon,

Angle $ABC$ = ..... ° **[2]**

**(ii)** interior angle $CBG$ of the regular pentagon.

Angle $CBG$ = ..... ° **[1]**

**(b)** Calculate the size of angle $BAG$.

Angle $BAG$ = ..... ° **[3]**

**(c)** An irregular quadrilateral has interior angles $(2x + 10)°$, $(3x - 15)°$, $(x + 25)°$, and $100°$.
Calculate the size of the largest interior angle of this quadrilateral.

..... ° **[3]**

**1060** Answer the following questions about polygons and angles.

**(a)** An irregular polygon has $n$ sides.
Three of the interior angles are each $140°$.
The remaining $(n - 3)$ interior angles are each $165°$.

Work out the value of $n$.

$n =$ ..... **[4]**

**(b)** In triangle $ABC$, the angles are $\angle A = a°$, $\angle B = b°$ and $\angle C = c°$.
It is given that $c - b = a$.

Prove that $ABC$ is a right-angled triangle.

**[2]**

**(c)** Polygon $P$ is a regular polygon with 12 sides.
Polygon $Q$ is a regular polygon with $k$ sides.
The exterior angle of Polygon $Q$ is half the size of the exterior angle of Polygon $P$.
Polygon $P$ and Polygon $Q$ share a common edge $AB$.

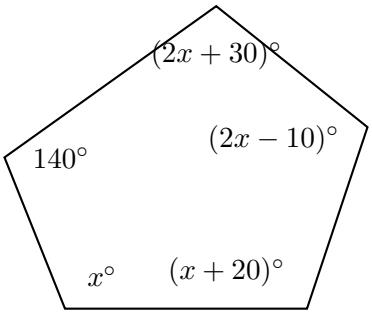
**(i)** Find the value of $k$.

$k =$ ..... **[2]**

**(ii)** Work out the size of the angle at vertex $B$ that lies outside both polygons.

..... ° **[2]**

**1061** The diagram shows an irregular pentagon.
The interior angles of the pentagon are $x°$, $(x + 20)°$, $(2x - 10)°$, $(2x + 30)°$ and $140°$.




NOT TO SCALE

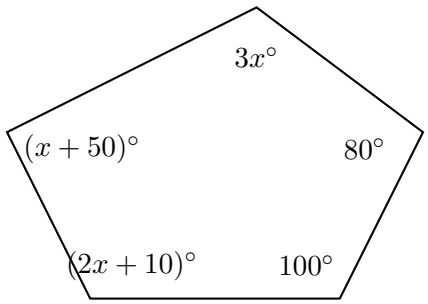
**(a)** Calculate the value of $x$.

$x =$ ..... **[3]**

**(b)** Find the size of the largest interior angle of the pentagon.

..... ° **[2]**

**1062** The diagram shows an irregular pentagon.




NOT TO SCALE

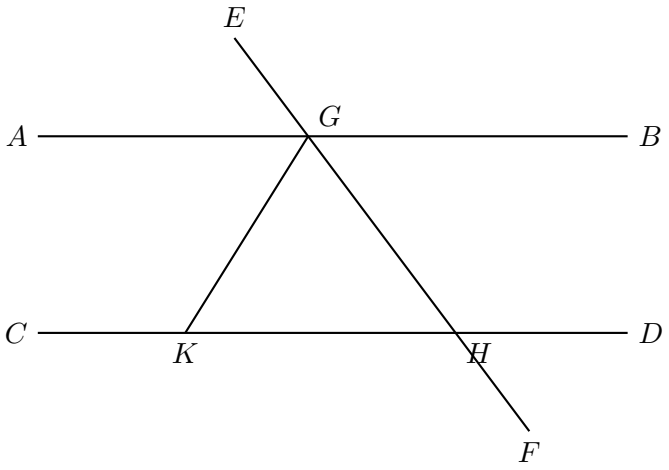
**(a)** Work out the value of $x$.

$x =$ ..... **[3]**

**(b)** Another regular polygon has an interior angle that is 4 times the size of its exterior angle.
Calculate the number of sides of this polygon.

..... **[3]**

**1063** In the diagram, the line $AB$ is parallel to the line $CD$.
The transversal $EF$ intersects $AB$ at $G$ and $CD$ at $H$.
The point $K$ lies on $CD$.




**(a)** Angle $EGB = (5x + 10)°$ and angle $GHD = (7x - 30)°$.
Find the value of $x$.
Give a geometrical reason for your working.

$x =$ ..... , Reason: ..... **[3]**

**(b)** Angle $HGK = 35°$.
Calculate angle $GKH$.

..... ° **[3]**

**1064** **(a)** This question is about the interior and exterior angles of polygons.

**(i)** The interior angles of a hexagon are $x°$, $2x°$, $(x + 30)°$, $(2x + 10)°$, $(3x - 20)°$, and $160°$.
Work out the value of $x$.

$x =$ ..... **[3]**

**(ii)** A regular polygon has an interior angle that is 6.5 times its exterior angle.
Calculate the number of sides of this polygon.

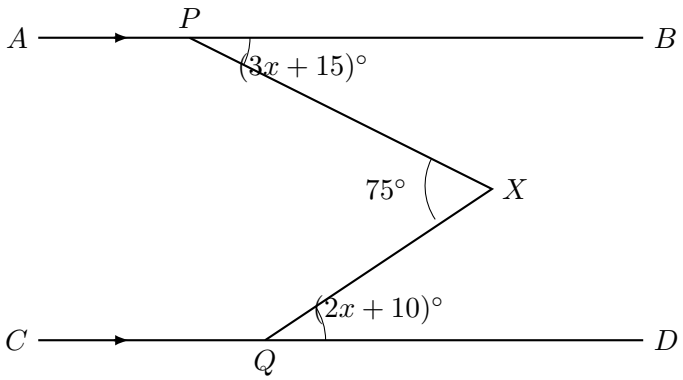
..... **[3]**

**(iii)** Two regular polygons, $P$ and $Q$, have $m$ sides and $n$ sides respectively.
The sum of their exterior angles is $54°$.
The difference between their interior angles is $18°$.
Find the value of $m$ and the value of $n$.

$m =$ ..... , $n =$ ..... **[5]**

**1065** **(a)** In the diagram, the line $AB$ is parallel to the line $CD$.
$X$ is a point between $AB$ and $CD$.
The point $P$ is on $AB$ and the point $Q$ is on $CD$.
Angle $BPX = (3x + 15)°$, angle $DQX = (2x + 10)°$ and angle $PXQ = 75°$.

Find the value of $x$.




NOT TO SCALE

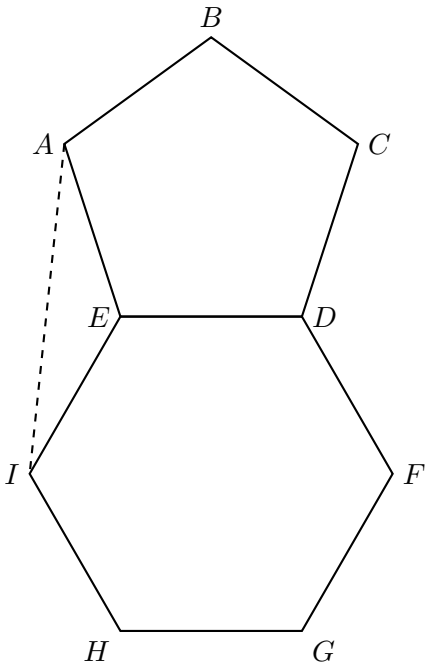
$x =$ ..... **[3]**

**(b)** Polygon $P$ is a regular polygon with $n$ sides.
Polygon $Q$ is a regular polygon with $3n$ sides.
The size of an interior angle of polygon $Q$ is 16° greater than the size of an interior angle of polygon $P$.

Calculate the value of $n$.

$n =$ ..... **[4]**

**1066** The diagram shows a regular pentagon $ABCDE$ and a regular hexagon $EDFGHI$ sharing a common edge $ED$.




**(a)** Calculate angle $CDE$.

Angle $CDE =$ ..... ° **[2]**

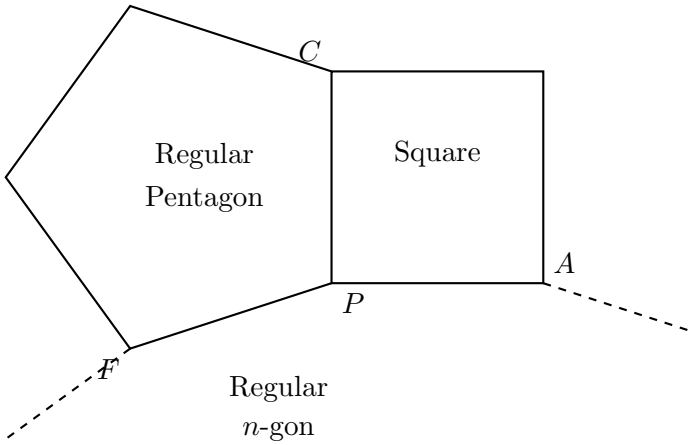
**(b)** Calculate angle $AEI$.

Angle $AEI =$ ..... ° **[2]**

**(c)** Calculate angle $EAI$.

Angle $EAI =$ ..... ° **[3]**

**1067** The diagram shows three regular polygons meeting at the point $P$.
The polygons are a square, a regular pentagon, and a regular polygon with $n$ sides.
The square shares the side $PC$ with the pentagon and the side $PA$ with the $n$-gon.
The pentagon shares the side $PF$ with the $n$-gon.
They meet with no gaps and do not overlap.




**(a)** Calculate the interior angle of a regular pentagon.

..... ° **[2]**

**(b)** Find the value of $n$.

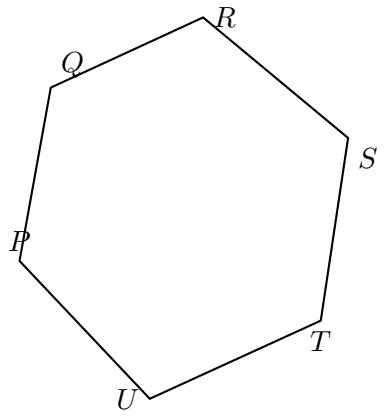
$n =$ ..... **[3]**

**(c)** The vertices $A$ and $F$ of the $n$-gon are joined by a straight line.
Calculate angle $PFA$.

..... ° **[2]**

**1068** The diagram shows an irregular hexagon $PQRSTU$.
The interior angles of the hexagon are:

- $\angle P = (2x + 15)^\circ$
- $\angle Q = (3x - 20)^\circ$
- $\angle R = (2x + 35)^\circ$
- $\angle S = (4x - 45)^\circ$
- $\angle T = (x + 50)^\circ$
- $\angle U = (3x + 10)^\circ$



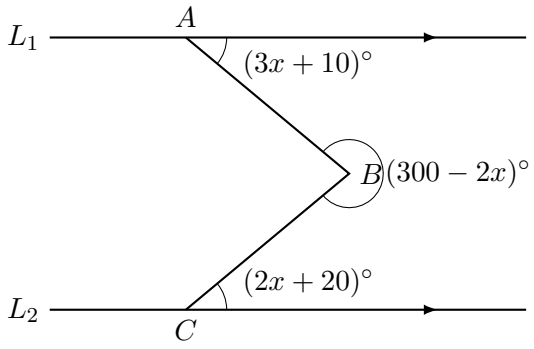

**(a)** Calculate the value of $x$.

$x =$ ..... **[4]**

**(b)** Determine whether $PQ$ is parallel to $TS$, showing clear geometrical working to support your answer.

..... **[3]**

**1069** The diagram shows two parallel lines, $L_1$ and $L_2$.
The point $A$ lies on $L_1$ and the point $C$ lies on $L_2$.
The point $B$ lies between the two parallel lines.
The acute angle between $L_1$ and $AB$ is $(3x + 10)°$.
The acute angle between $L_2$ and $CB$ is $(2x + 20)°$.
The reflex angle $ABC$ is $(300 - 2x)°$.




**(a)** Show that $x = 10$.

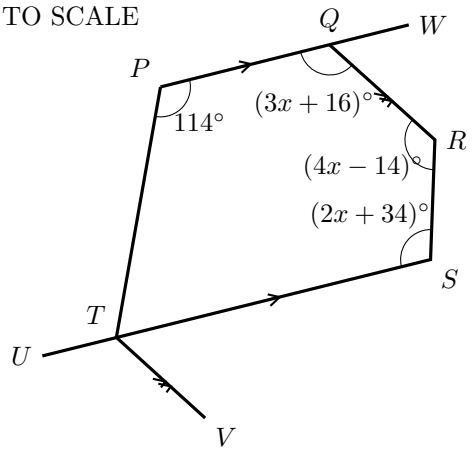
**[4]**

**(b)** The points $A$, $B$, and $C$ form a triangle $ABC$.
Given that $AB = BC$, calculate the angle $BCA$.

Angle $BCA =$ ..... ° **[3]**

**1070** The diagram shows an irregular pentagon $PQRST$.
The side $PQ$ is extended to $W$ and the side $ST$ is extended to $U$.
$PW$ is parallel to $US$, and $QR$ is parallel to $TV$.
Angle $QPT = 114°$, angle $PQR = (3x + 16)°$, angle $QRS = (4x - 14)°$, and angle $RST = (2x + 34)°$.

NOT TO SCALE




**(a)** Find the size of angle $PTS$ and give a geometrical reason for your answer.

Angle $PTS =$ ..... , Reason: ..... **[2]**

**(b)** Show that $x = 36$.

**[3]**

**(c)** Calculate the size of

**(i)** angle $PTU$,

Angle $PTU =$ ..... ° **[1]**

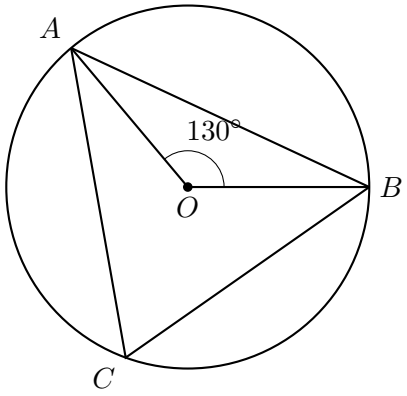
**(ii)** angle $STV$.

Angle $STV =$ ..... ° **[2]**

**(d)** A regular polygon has $n$ sides.
Each interior angle of this regular polygon is 11 times the size of each exterior angle.
Calculate the value of $n$.

$n =$ ..... **[3]**

**1071** Points $A$, $B$ and $C$ lie on a circle, centre $O$.
Angle $AOB = 130°$.



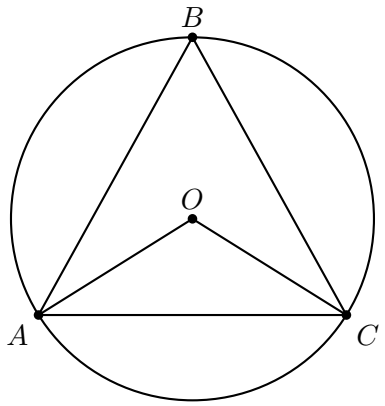

**(a)** Find the size of angle $ACB$.

..... ° **[1]**

**(b)** Find the size of angle $OAB$.

..... ° **[2]**

**1072** $A$, $B$ and $C$ are points on a circle, center $O$.
Angle $ABC = 58°$.



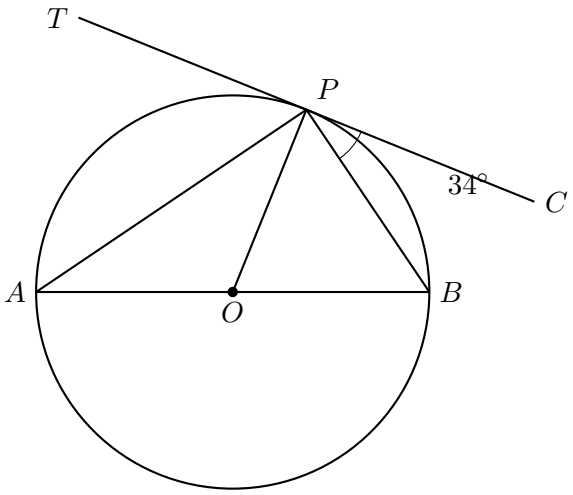

**(a)** Work out the obtuse angle $AOC$.

Angle $AOC$ = ..... ° **[1]**

**(b)** Work out angle $OAC$.

Angle $OAC$ = ..... ° **[2]**

**1073** The diagram shows a circle, centre $O$.
$A$, $B$ and $P$ are points on the circumference.
$AB$ is a diameter of the circle.
The line $TC$ is a tangent to the circle at the point $P$.
Angle $BPC = 34°$.




NOT TO SCALE

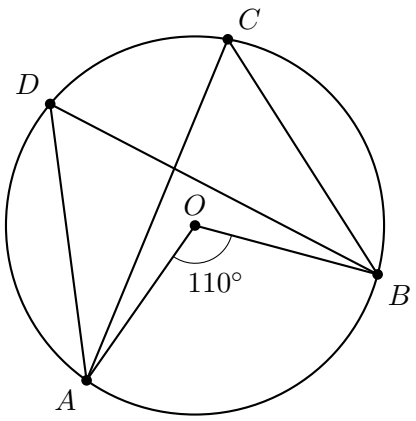
**(a)** Write down the value of angle $PAB$.

Angle $PAB =$ ..... ° **[1]**

**(b)** Work out the value of angle $POB$.

Angle $POB =$ ..... ° **[2]**

**1074** $A$, $B$, $C$ and $D$ are points on a circle, centre $O$.
Angle $AOB = 110°$.




NOT TO SCALE

**(a)** Find the size of angle $ACB$.

Angle $ACB =$ ..... ° **[1]**

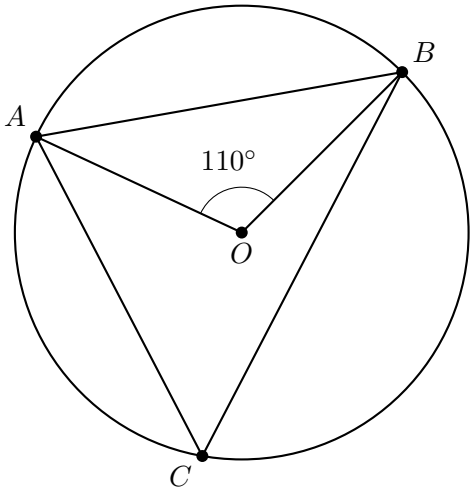
**(b)** Find the size of angle $ADB$.

Angle $ADB =$ ..... ° **[1]**

**(c)** State the geometrical reason for your answer to part (b).

..... **[1]**

**1075** The diagram shows a circle, centre $O$.
$A$, $B$ and $C$ are points on the circumference.
Angle $AOB = 110°$.



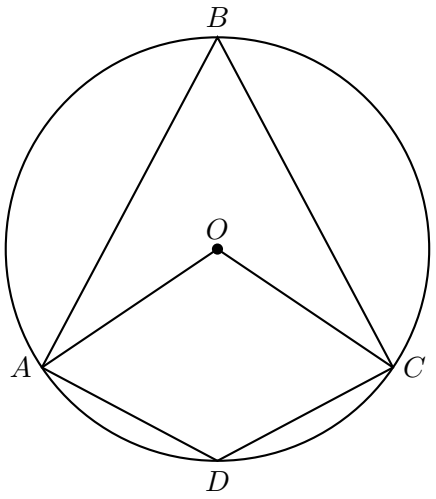

**(a)** **(i)** Calculate the size of angle $ACB$.

Angle $ACB$ = ..... ° **[1]**

**(ii)** Calculate the size of angle $OAB$.

Angle $OAB$ = ..... ° **[2]**

**1076** $A$, $B$, $C$ and $D$ are points on a circle, center $O$.
Angle $ABC = 56°$.




**(a)** Calculate the obtuse angle $AOC$.

..... ° **[1]**

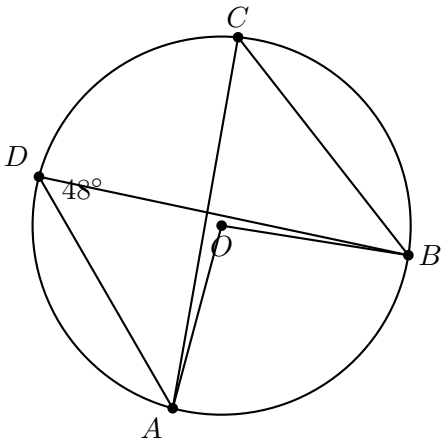
**(b)** Calculate angle $ADC$.

..... ° **[1]**

**(c)** State the geometric reason for your answer to part (b).

..... **[1]**

**1077** $A$, $B$, $C$ and $D$ are points on a circle, centre $O$.
Angle $ADB = 48°$.




**(a)** **(i)** Find angle $ACB$.

Angle $ACB =$ ..... ° **[1]**

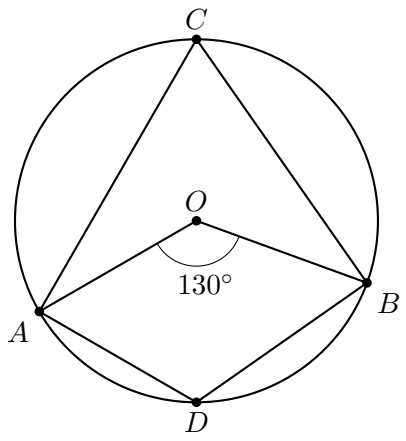
**(ii)** Give a geometrical reason for your answer to part (a)(i).

..... **[1]**

**(b)** Find angle $AOB$.

Angle $AOB =$ ..... ° **[1]**

**1078** Points $A$, $B$, $C$ and $D$ lie on the circumference of a circle, center $O$.
Angle $AOB = 130°$.



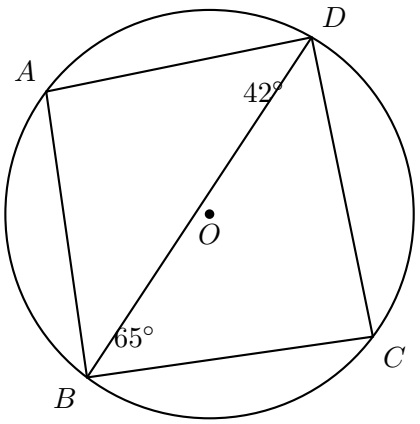

**(a)** Calculate angle $ACB$.

..... ° **[1]**

**(b)** Calculate angle $ADB$.

..... ° **[2]**

**1079** $A$, $B$, $C$ and $D$ are points on the circumference of a circle, centre $O$.
Angle $BDC = 42°$ and angle $CBD = 65°$.




**(a)** Find angle $BAC$.

Angle $BAC =$ ..... ° **[1]**

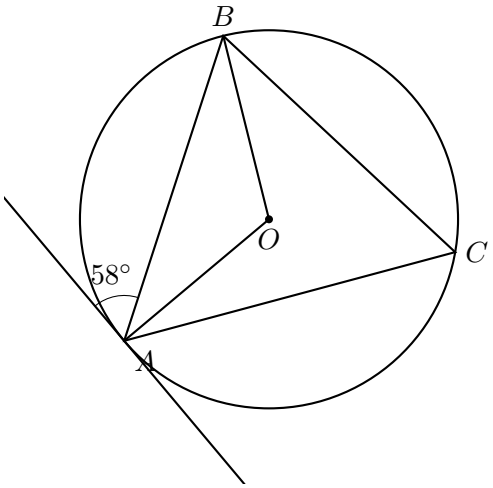
**(b)** Find angle $BCD$.

Angle $BCD =$ ..... ° **[1]**

**(c)** Find angle $BAD$.

Angle $BAD =$ ..... ° **[2]**

**1080** The points $A, B$, and $C$ lie on a circle, centre $O$.
The line $SAT$ is a tangent to the circle at $A$.
Angle $SAB = 58°$.




**(a)** Find angle $ACB$ and state a geometrical reason for your answer.

**(i)** Angle $ACB$

..... ° **[1]**

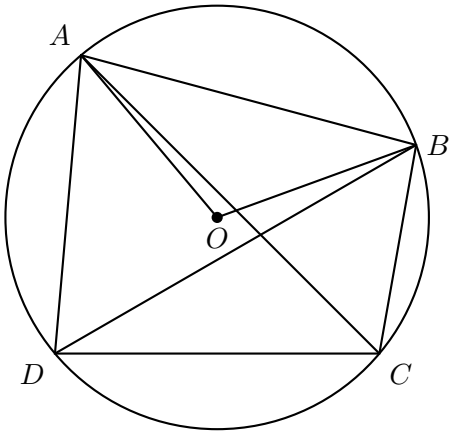
**(ii)** Reason

..... **[1]**

**(b)** Find angle $OBA$.

..... ° **[2]**

**1081** The diagram shows a circle with centre $O$.
The points $A$, $B$, $C$ and $D$ lie on the circumference of the circle.
Angle $AOB = 110°$ and angle $BDC = 30°$.




**(a)** Calculate angle $ACB$.

..... ° **[1]**

**(b)** State the geometrical reason for your answer to part (a).

..... **[1]**

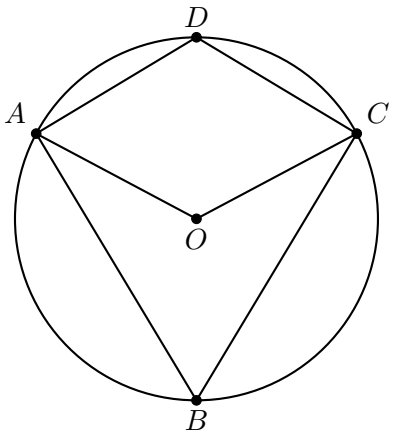
**(c)** Calculate angle $BAC$.

..... ° **[1]**

**(d)** Calculate angle $OAB$.

..... ° **[2]**

**1082** $A$, $B$, $C$ and $D$ are points on the circumference of a circle, centre $O$.
Angle $AOC = 124°$.




**(a)** Calculate angle $ABC$.

..... ° **[2]**

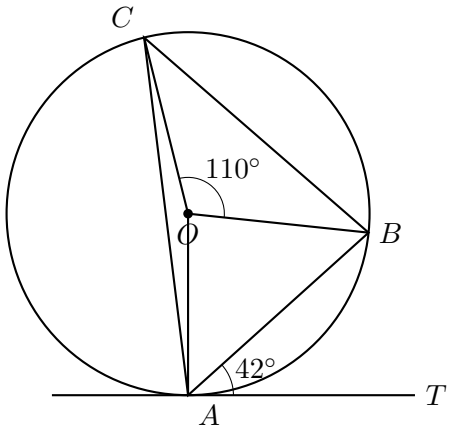
**(b)** Calculate angle $ADC$.

..... ° **[2]**

**(c)** Calculate angle $OAC$.

..... ° **[2]**

**1083** $A$, $B$ and $C$ are points on a circle, centre $O$.
$TA$ is a tangent to the circle at $A$.
$C$ is a point on the major arc $AB$.
Angle $TAB = 42°$ and angle $BOC = 110°$.




**(a)** Write down the value of

**(i)** angle $BCA$,

Angle $BCA =$ ..... ° **[1]**

**(ii)** angle $BAC$.

Angle $BAC =$ ..... ° **[1]**

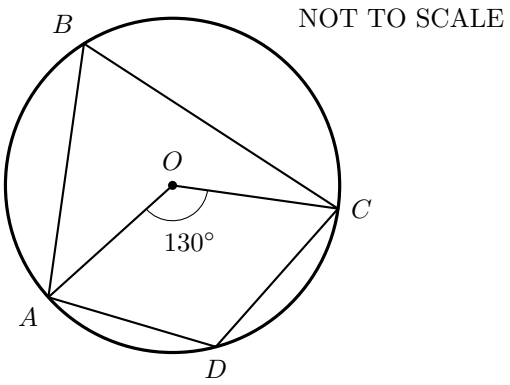
**(b)** Calculate angle $OAB$.

Angle $OAB =$ ..... ° **[2]**

**(c)** Calculate angle $ABC$.

Angle $ABC =$ ..... ° **[2]**

**1084** The diagram shows a circle, centre $O$, with points $A$, $B$, $C$, and $D$ lying on the circumference. Angle $AOC = 130°$ and angle $OAB = 22°$.




**(a)** Find the size of angle $ABC$ and give a geometrical reason for your answer.

Angle $ABC =$ ..... , Reason: ..... **[2]**

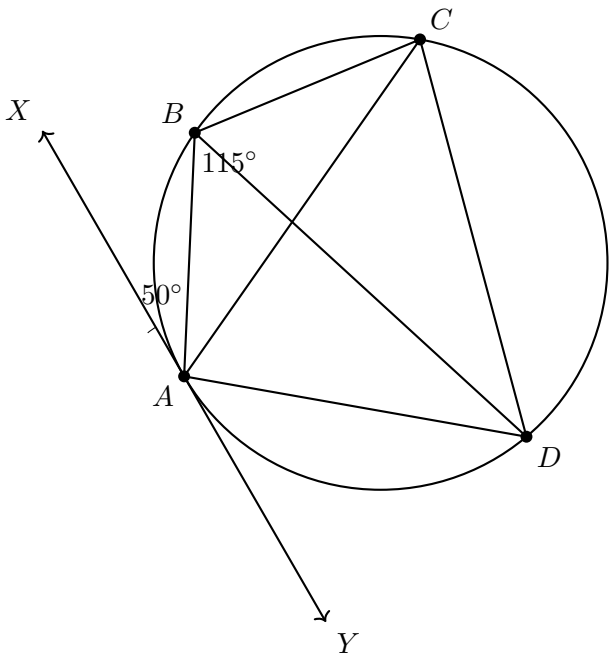
**(b)** Find the size of angle $ADC$ and give a geometrical reason for your answer.

Angle $ADC =$ ..... , Reason: ..... **[2]**

**(c)** Calculate the size of angle $OCB$.

Angle $OCB =$ ..... ° **[2]**

**1085** The diagram shows a cyclic quadrilateral $ABCD$.
The line $XY$ is a tangent to the circle at $A$.
Angle $XAD = 50°$ and angle $ABC = 115°$.




**(a)** **(i)** Write down the size of angle $ACD$.

Angle $ACD =$ ..... ° **[1]**

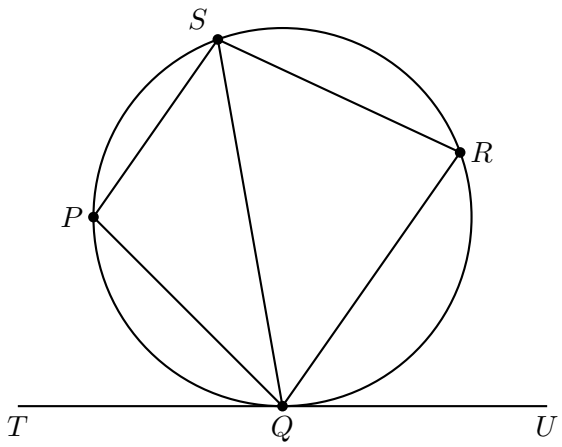
**(ii)** Calculate the size of angle $ADC$.

Angle $ADC =$ ..... ° **[1]**

**(iii)** Calculate the size of angle $CAD$.

Angle $CAD =$ ..... ° **[2]**

**1086** $P$, $Q$, $R$ and $S$ are points on a circle.
$TQU$ is a tangent to the circle at $Q$.
Angle $RQU = 55°$ and Angle $QPS = 100°$.




NOT TO SCALE

**(a)** Find the size of angle $QSR$.

Angle $QSR =$ ..... ° **[1]**

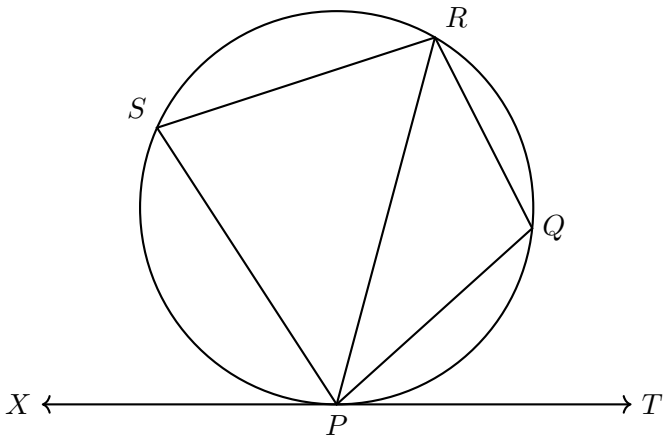
**(b)** Find the size of angle $QRS$.

Angle $QRS =$ ..... ° **[2]**

**(c)** Calculate the size of angle $SQR$.

Angle $SQR =$ ..... ° **[2]**

**1087** Points $P$, $Q$, $R$ and $S$ lie on a circle.
The straight line $XPT$ is a tangent to the circle at $P$.
Angle $QPT = 42°$, angle $PQR = 105°$ and angle $SPR = 48°$.




**(a)** Find the size of angle $PRQ$.

..... ° **[1]**

**(b)** Find the size of angle $PSR$.

..... ° **[1]**

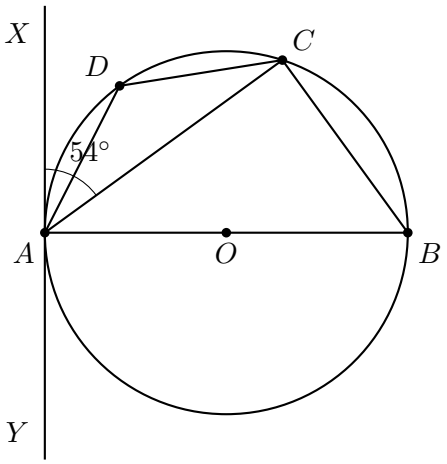
**(c)** Find the size of angle $PRS$.

..... ° **[2]**

**(d)** Find the size of angle $XPS$.

..... ° **[1]**

**1088** In the diagram, $AB$ is a diameter of the circle, center $O$.
The line $XAY$ is a tangent to the circle at $A$.
$C$ and $D$ are points on the circle.
Angle $XAC = 54°$.




**(a)** **(i)** Write down the size of angle $ACB$.

..... ° **[1]**

**(ii)** Give a geometric reason for your answer.

..... **[1]**

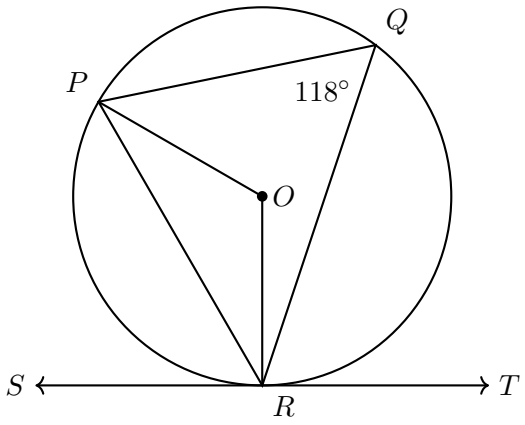
**(b)** Find angle $ABC$.

..... ° **[1]**

**(c)** Find angle $ADC$.

..... ° **[2]**

**1089** $P$, $Q$ and $R$ are points on a circle, centre $O$.
$ST$ is a tangent to the circle at $R$.
Angle $PQR = 118°$.




**(a)** Find reflex angle $POR$.

Reflex angle $POR =$ ..... ° **[2]**

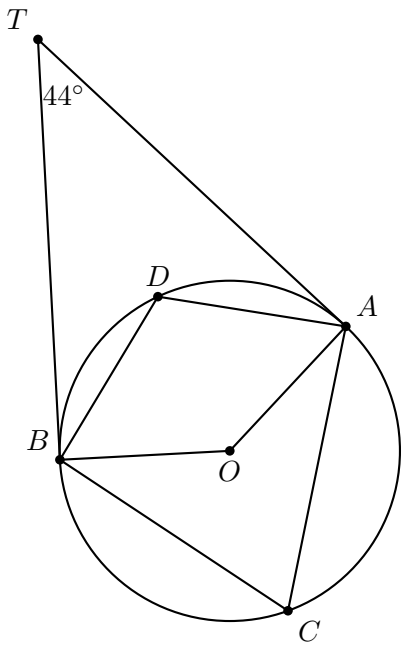
**(b)** Find obtuse angle $POR$.

Obtuse angle $POR =$ ..... ° **[1]**

**(c)** Find angle $PRS$.

Angle $PRS =$ ..... ° **[2]**

**1090** $A$, $B$, $C$ and $D$ are points on a circle, centre $O$.
$TA$ and $TB$ are tangents to the circle at $A$ and $B$.
Angle $ATB = 44°$.




**(a)** Calculate angle $AOB$.

Angle $AOB =$ ..... ° **[2]**

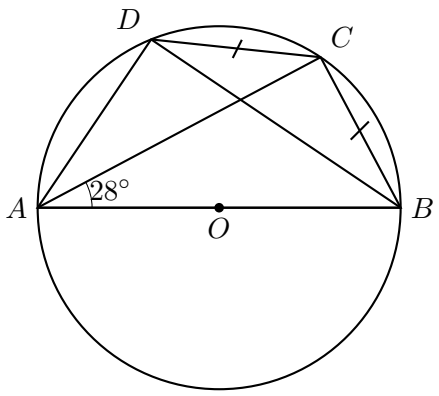
**(b)** Calculate angle $ACB$.

Angle $ACB =$ ..... ° **[1]**

**(c)** Calculate angle $ADB$.

Angle $ADB =$ ..... ° **[2]**

**1091** $A, B, C$ and $D$ are points on a circle.
$AB$ is a diameter of the circle.
$CD = CB$ and angle $CAB = 28°$.




**(a) (i)** Find angle $CBA$.

Angle $CBA =$ ..... ° **[1]**

**(ii)** Give a geometrical reason for your answer.

..... **[1]**

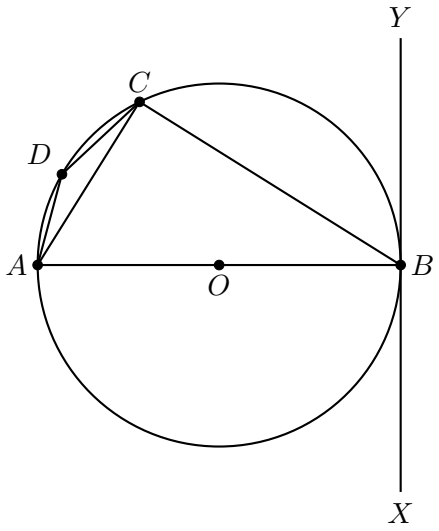
**(b)** Find angle $CAD$.

Angle $CAD =$ ..... ° **[1]**

**(c)** Find angle $ADC$.

Angle $ADC =$ ..... ° **[2]**

**1092** $A$, $B$, $C$ and $D$ lie on a circle, centre $O$.
$AB$ is a diameter of the circle.
The line $XY$ is a tangent to the circle at $B$.
Angle $CBY = 58°$.




**(a)** **(i)** Find angle $BAC$.

..... ° **[1]**

**(ii)** State the circle theorem that gives this angle directly from angle $CBY$.

..... **[1]**

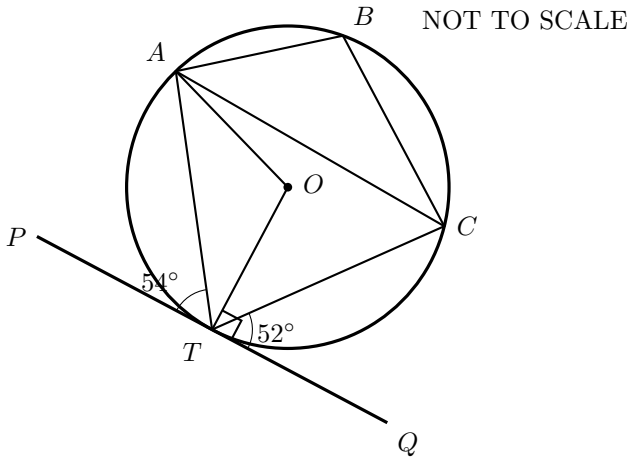
**(b)** Calculate angle $ABC$.

..... ° **[3]**

**(c)** Calculate angle $ADC$.

..... ° **[2]**

**1093** The diagram shows a circle, centre $O$, with points $A$, $B$, $C$, and $T$ lying on the circumference.
The straight line $PTQ$ is a tangent to the circle at $T$.
Angle $PTA = 54°$, angle $QTC = 52°$, and angle $BCT = 86°$.




**(a)** Find the size of angle $TAC$ and give a geometrical reason for your answer.

Angle $TAC =$ ..... , Reason: ..... **[2]**

**(b)** Calculate the size of

**(i)** angle $ABC$,

Angle $ABC =$ ..... ° **[2]**

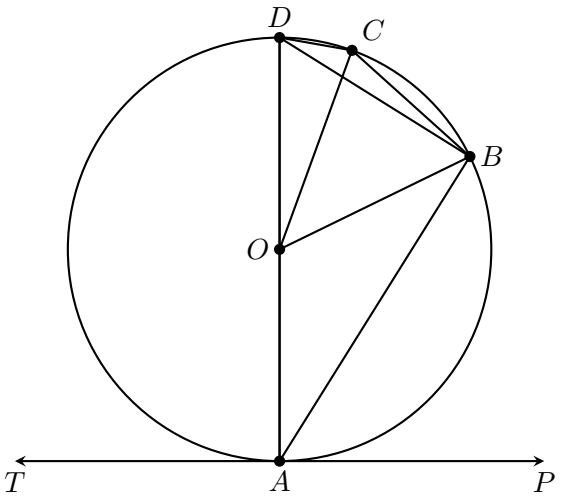
**(ii)** angle $ACB$,

Angle $ACB =$ ..... ° **[2]**

**(iii)** angle $OAT$.

Angle $OAT =$ ..... ° **[3]**

**1094** $A$, $B$, $C$ and $D$ are points on a circle, center $O$.
$AD$ is a diameter of the circle.
$PAT$ is a tangent to the circle at $A$.
Angle $PAB = 58°$ and angle $BDC = 22°$.




**(a)** Calculate angle $ADB$.

Angle $ADB$ = ..... ° **[2]**

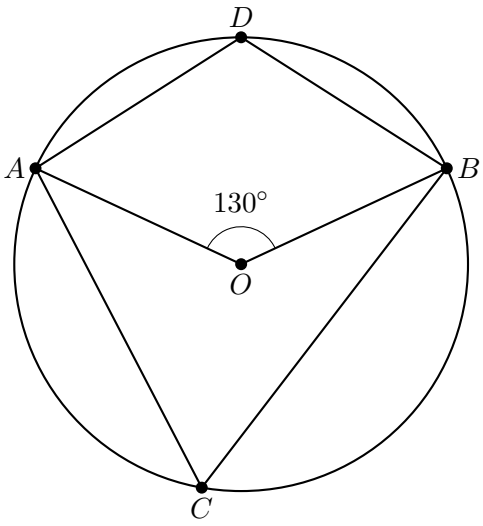
**(b)** Calculate angle $ABC$.

Angle $ABC$ = ..... ° **[2]**

**(c)** Calculate angle $OBC$.

Angle $OBC$ = ..... ° **[3]**

**1095** The points $A$, $B$, $C$, and $D$ lie on a circle with centre $O$.
Angle $AOB = 130°$.




**(a)** Calculate angle $ACB$.

Angle $ACB =$ ..... ° **[1]**

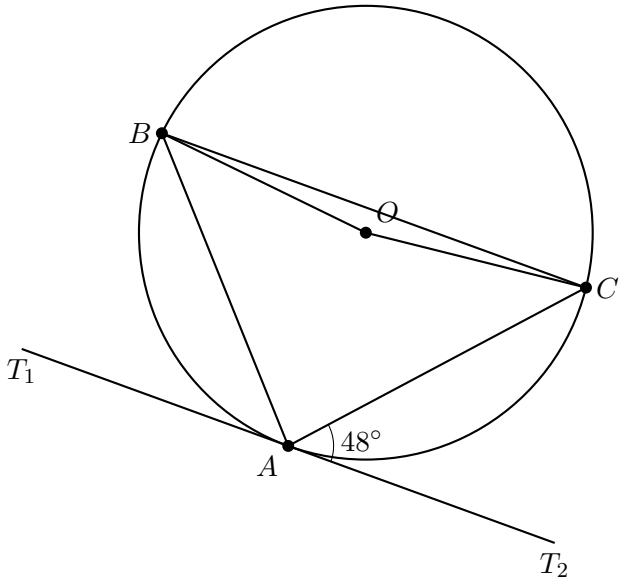
**(b)** Calculate angle $ADB$.

Angle $ADB =$ ..... ° **[1]**

**(c)** Give a geometrical reason for your answer to part (b).

..... **[1]**

**1096** The points $A$, $B$, and $C$ lie on a circle with centre $O$.
The straight line $T_1AT_2$ is a tangent to the circle at $A$.
The chord $BC$ is parallel to the tangent $T_1AT_2$.
Angle $CAT_2 = 48°$.




**(a)** Find angle $ABC$.

Angle $ABC$ = ..... ° **[1]**

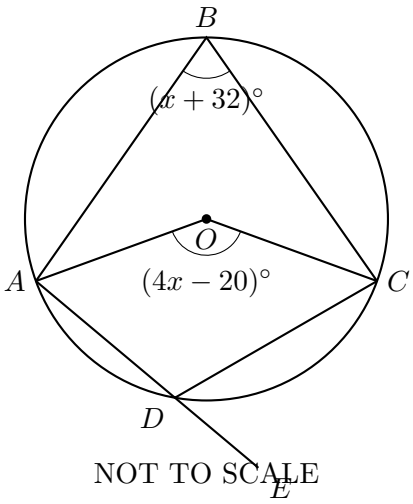
**(b)** Calculate angle $BAC$.

Angle $BAC$ = ..... ° **[2]**

**(c)** Calculate angle $OBC$.

Angle $OBC$ = ..... ° **[2]**

**1097** $A$, $B$, $C$ and $D$ lie on a circle, centre $O$.
Angle $ABC = (x + 32)°$ and the obtuse angle $AOC = (4x - 20)°$.
The line $AD$ is extended to the point $E$.



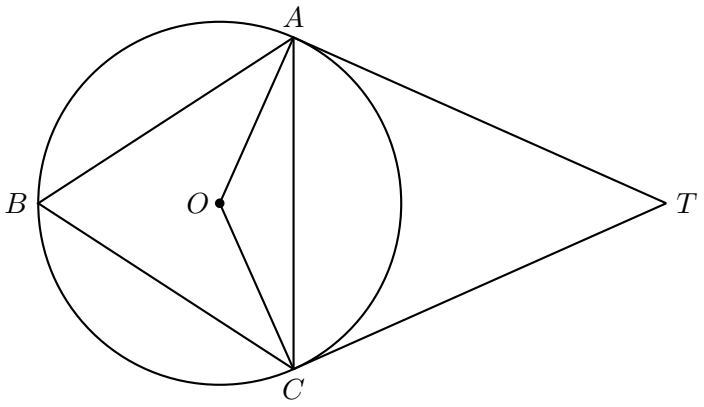

**(a)** Find the value of $x$.

$x =$ ..... **[3]**

**(b)** Work out the size of angle $CDE$.

Angle $CDE =$ ..... ° **[2]**

**1098** $A$, $B$ and $C$ are points on a circle, center $O$.
The lines $TA$ and $TC$ are tangents to the circle at $A$ and $C$ respectively.
Angle $ATC = 48°$.




**(a)** Calculate angle $AOC$.

..... ° **[2]**

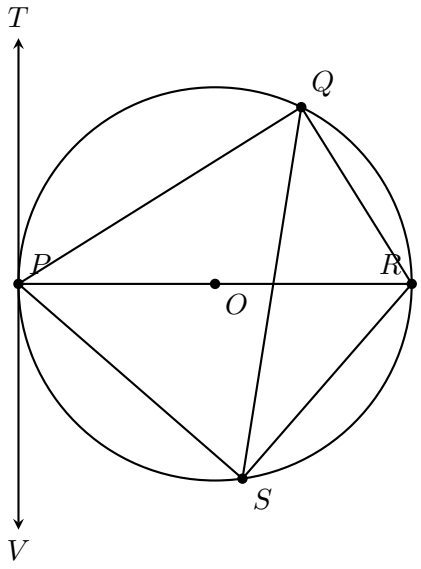
**(b)** Calculate angle $ABC$.

..... ° **[1]**

**(c)** Calculate angle $OAC$.

..... ° **[2]**

**1099** $P$, $Q$, $R$ and $S$ are points on a circle, center $O$.
$PR$ is a diameter of the circle.
$TPV$ is a tangent to the circle at $P$.
Angle $QPR = 32°$ and angle $RPS = 41°$.




**(a)** Calculate angle $PRQ$.

Angle $PRQ =$ ..... ° **[2]**

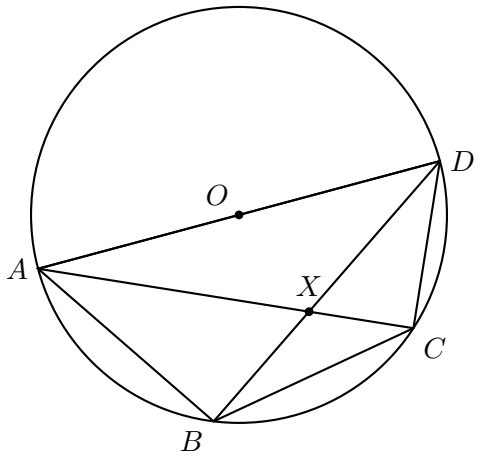
**(b)** Calculate angle $TPQ$.

Angle $TPQ =$ ..... ° **[2]**

**(c)** Calculate angle $PQS$.

Angle $PQS =$ ..... ° **[2]**

**1100** $AD$ is a diameter of the circle, centre $O$.
The points $A, B, C$, and $D$ lie on the circumference.
The chords $AC$ and $BD$ intersect at $X$.
Angle $BAC = 32°$ and angle $CAD = 24°$.




**(a)** Find angle $ADC$.

..... ° **[2]**

**(b)** Find angle $ABC$.

..... ° **[1]**

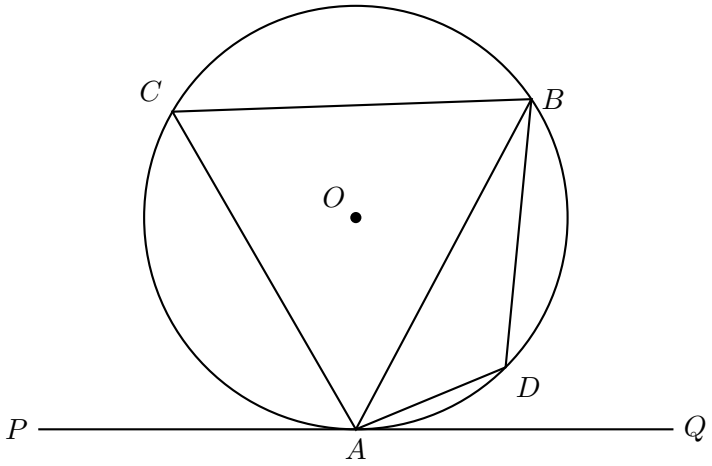
**(c)** Find angle $DBC$.

..... ° **[1]**

**(d)** Find angle $AXB$.

..... ° **[2]**

**1101** The diagram shows a circle with centre $O$.
The line $PAQ$ is a tangent to the circle at the point $A$.
The points $B$, $C$ and $D$ lie on the circle.
Angle $QAB = 62°$.




**(a)** Calculate angle $ACB$ and give a geometrical reason for your answer.

$AngleACB =$ ..... ° , Reason: ..... **[2]**

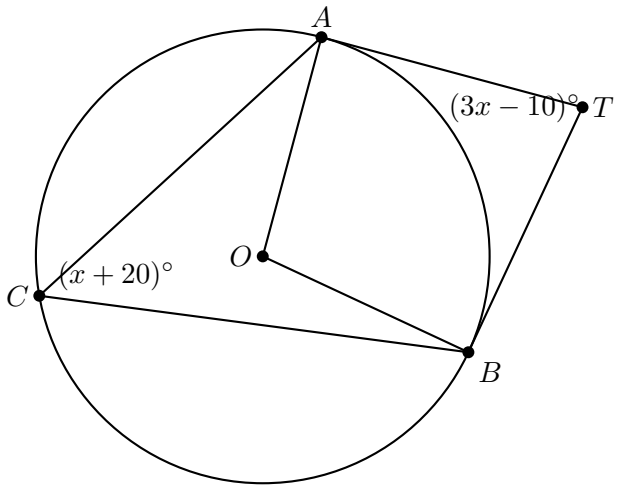
**(b)** Calculate angle $ADB$.

..... ° **[2]**

**(c)** Calculate angle $OBA$.

..... ° **[2]**

**1102** The diagram shows a circle, centre $O$.
$A$, $B$ and $C$ are points on the circumference.
The lines $TA$ and $TB$ are tangents to the circle at $A$ and $B$ respectively.
Angle $ATB = (3x - 10)°$ and angle $ACB = (x + 20)°$.




**(a)** Find the value of $x$.

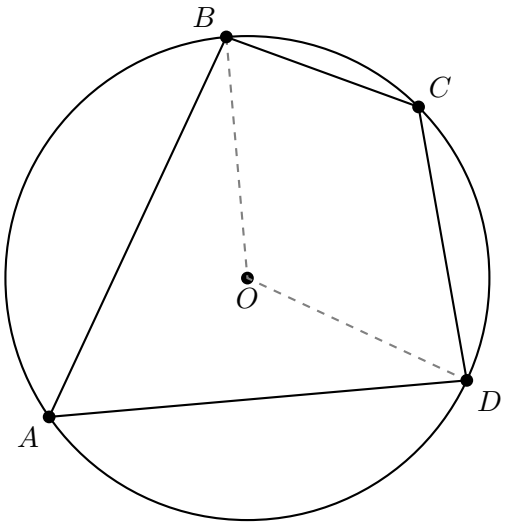
$x =$ ..... **[3]**

**(b)** Calculate the size of angle $OAB$.

Angle $OAB =$ ..... ° **[2]**

**1103** The points $A$, $B$, $C$, and $D$ form a cyclic quadrilateral on a circle with centre $O$.
The interior angles of the quadrilateral are given in terms of $x$ and $y$:

$$\angle DAB = (3x + 2y)^\circ$$
$$\angle ABC = (5x + 3y)^\circ$$
$$\angle BCD = (9x + 2y)^\circ$$
$$\angle CDA = (4x + 3y)^\circ$$



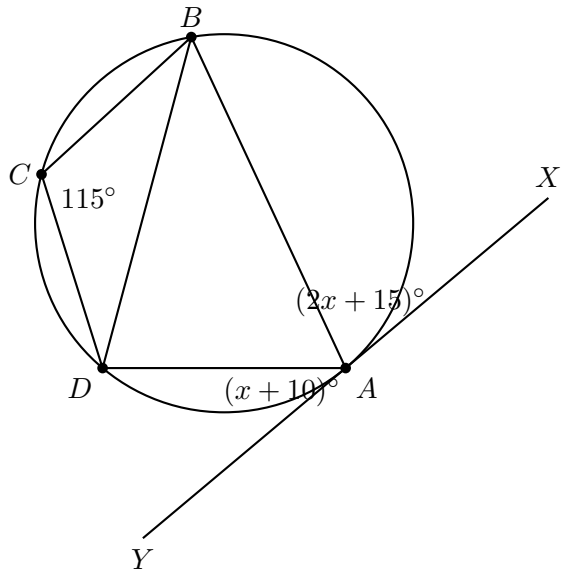

**(a)** By setting up and solving simultaneous equations, find the value of $x$ and the value of $y$.

$x =$ ..... , $y =$ ..... **[4]**

**(b)** Calculate the obtuse angle $BOD$.

Angle $BOD =$ ..... $^\circ$ **[2]**

**1104** The points $A$, $B$, $C$ and $D$ lie on a circle.
$XY$ is a tangent to the circle at $A$.
Angle $BCD = 115°$, angle $XAB = (2x + 15)°$ and angle $YAD = (x + 10)°$.



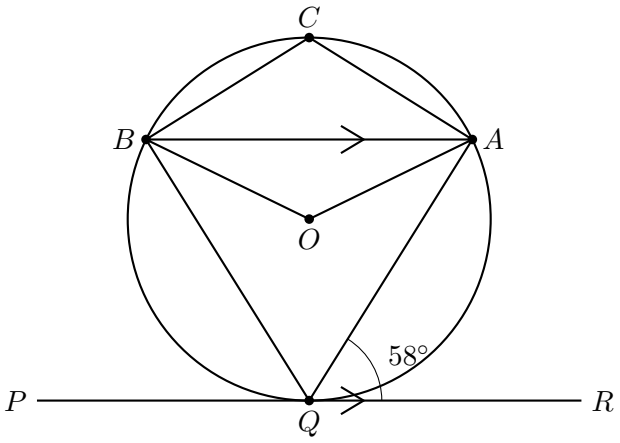

**(a)** Find the value of $x$.

$x =$ ..... **[4]**

**(b)** Calculate angle $ABD$.

Angle $ABD =$ ..... ° **[2]**

**1105** $A$, $B$, $C$ and $Q$ are points on a circle, center $O$.
The line $PQR$ is a tangent to the circle at $Q$.
The chord $AB$ is parallel to $PR$.
Angle $AQR = 58°$.




**(a)** **(i)** Find the size of angle $ABQ$.

..... ° **[1]**

**(ii)** Give a geometric reason for your answer.

..... **[1]**

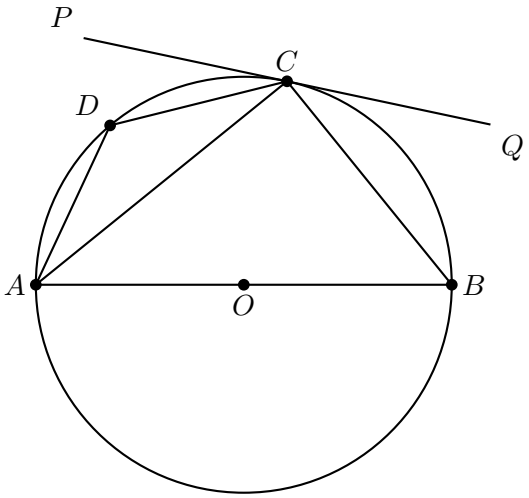
**(b)** Find angle $AOB$.

..... ° **[2]**

**(c)** Find angle $ACB$.

..... ° **[2]**

**1106** $A$, $B$, $C$ and $D$ are points on a circle, centre $O$.
$AB$ is a diameter of the circle.
$PQ$ is a tangent to the circle at $C$.
Angle $CAB = (2x - 5)^\circ$ and Angle $BCQ = (x + 17)^\circ$.




NOT TO SCALE

**(a)** Set up an equation in terms of $x$ and solve it to show that $x = 22$.
Give a geometrical reason to justify your equation.

**[3]**

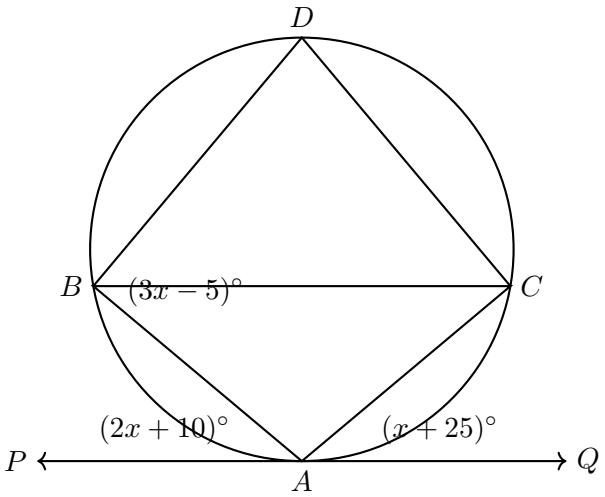
**(b)** Work out the size of angle $CBA$.

Angle $CBA =$ ..... ° **[2]**

**(c)** Find the size of angle $ADC$.

Angle $ADC =$ ..... ° **[2]**

**1107** Points $A$, $B$, $C$ and $D$ lie on a circle.
The straight line $PAQ$ is a tangent to the circle at $A$.
Angle $PAB = (2x + 10)°$, angle $ABC = (3x - 5)°$ and angle $CAQ = (x + 25)°$.
$D$ is a point on the major arc $BC$.




**(a)** Find the value of $x$.
Show all your working.

$x =$ ..... **[3]**

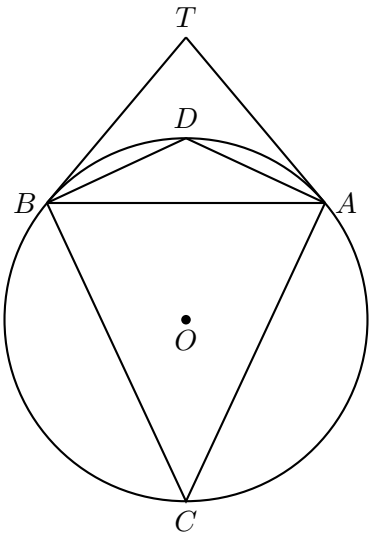
**(b)** Find the size of angle $BAC$.

..... ° **[2]**

**(c)** Find the size of angle $BDC$.

..... ° **[2]**

**1108** $TA$ and $TB$ are tangents to a circle, centre $O$, at the points $A$ and $B$.
$C$ is a point on the major arc $AB$ and $D$ is a point on the minor arc $AB$.
Angle $ATB = (x + 10)^\circ$
Angle $TAB = (2y - 10)^\circ$
Angle $ACB = (3y - 40)^\circ$




**(a)** Find the value of $y$.

$y =$ ..... **[3]**

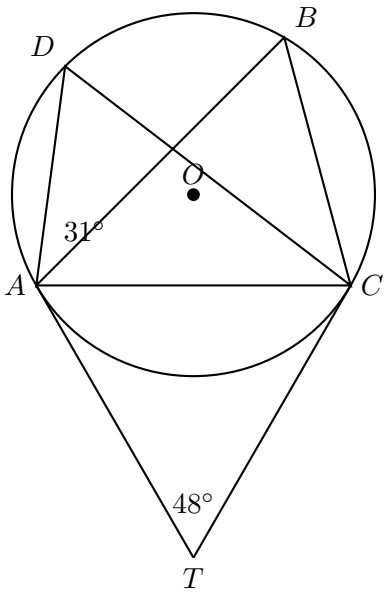
**(b)** Find the value of $x$.

$x =$ ..... **[2]**

**(c)** Find angle $ADB$.

Angle $ADB =$ ..... $^\circ$ **[2]**

**1109** $A$, $B$, $C$ and $D$ are points on a circle, centre $O$.
$TA$ and $TC$ are tangents to the circle at $A$ and $C$.
Angle $ATC = 48°$ and angle $CAD = 31°$.




**(a)** **(i)** Find angle $ABC$.

Angle $ABC$ = ..... ° **[2]**

**(ii)** Find angle $ADC$.

Angle $ADC$ = ..... ° **[2]**

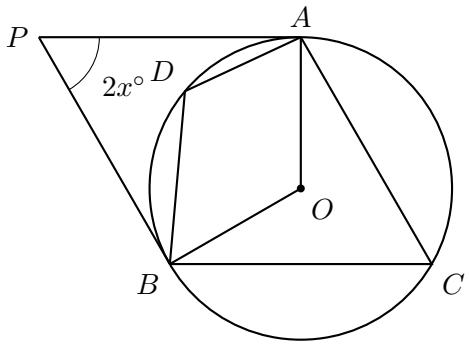
**(iii)** Find angle $ACD$.

Angle $ACD$ = ..... ° **[1]**

**(iv)** Calculate angle $AOC$.

Angle $AOC$ = ..... ° **[2]**

**1110** The points $A, B, C$, and $D$ lie on a circle, centre $O$.
$PA$ and $PB$ are tangents to the circle at $A$ and $B$.
Angle $APB = 2x°$.
$C$ is a point on the major arc $AB$ and $D$ is a point on the minor arc $AB$.




**(a)** Find an expression, in terms of $x$, for each of the following angles.
Give each answer in its simplest form.

Angle $AOB$

..... ° **[1]**

**(b)** Angle $ACB$

..... ° **[1]**

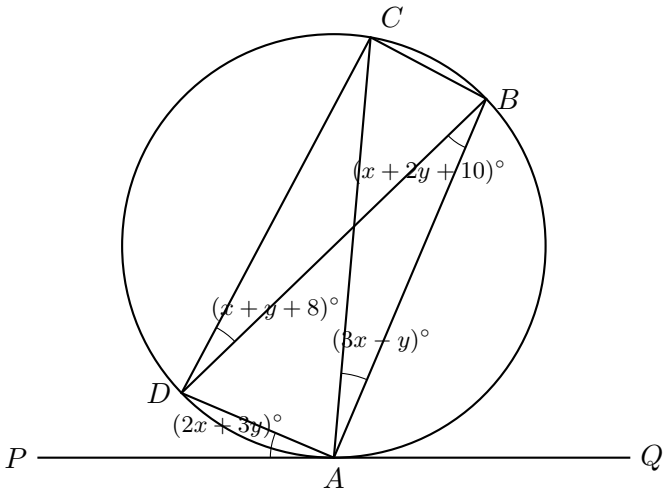
**(c)** Angle $ADB$

..... ° **[2]**

**(d)** Given that angle $ADB = 3 \times$ angle $ACB$, find the value of $x$.

$x =$ ..... **[3]**

**1111** The diagram shows a circle containing the points $A$, $B$, $C$ and $D$.
The straight line $PAQ$ is a tangent to the circle at $A$.
The chords $AC$ and $BD$ intersect.
Angle $PAD = (2x + 3y)^\circ$ and angle $ABD = (x + 2y + 10)^\circ$.
Angle $BAC = (3x - y)^\circ$ and angle $BDC = (x + y + 8)^\circ$.




NOT TO SCALE

**(a)** **(i)** Use the circle theorems to write down two equations in $x$ and $y$.

**[2]**

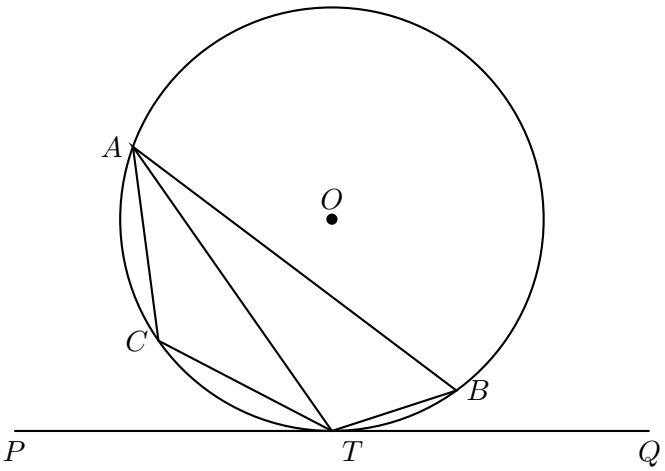
**(ii)** Solve your equations to find the value of $x$ and the value of $y$.
Show all your working.

$x =$ ..... , $y =$ ..... **[3]**

**(b)** Given that angle $ABC = 95^\circ$, work out the size of angle $QAB$.

Angle $QAB =$ ..... $^\circ$ **[3]**

**1112** $A$, $B$ and $C$ are points on a circle, center $O$.
The straight line $PTQ$ is a tangent to the circle at the point $T$.
$\angle ATP = (4x - 5)^\circ$ and $\angle ABT = (2x + 25)^\circ$
$\angle BTQ = (3y)^\circ$ and $\angle BAT = (y + 12)^\circ$




**(a)** Use the alternate segment theorem to find the values of $x$ and $y$.

$x =$ ..... , $y =$ ..... **[4]**

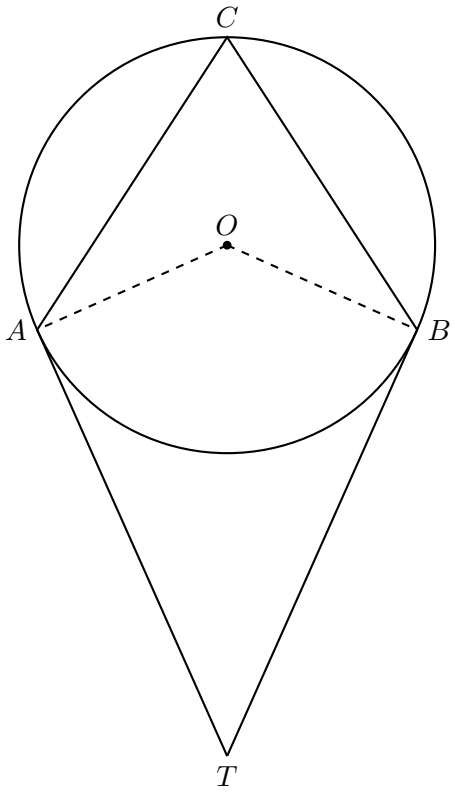
**(b)** Calculate angle $ATB$.

..... ° **[2]**

**(c)** $C$ is a point on the minor arc $AT$.
Calculate angle $ACT$.

..... ° **[2]**

**1113** The diagram shows a circle with centre $O$.
The lines $TA$ and $TB$ are tangents to the circle at the points $A$ and $B$.
$C$ is a point on the major arc $AB$.
Angle $ATB = x°$ and angle $ACB = (2x - 30)°$.




**(a) (i)** Find the value of $x$.

$x =$ ..... **[4]**

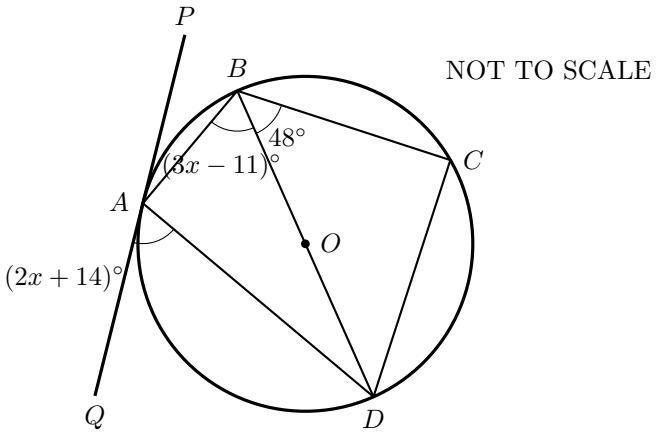
**(ii)** Calculate angle $OAB$.

..... ° **[2]**

**(b)** $D$ is a point on the minor arc $AB$.
Calculate angle $ADB$.

..... ° **[2]**

**1114** The diagram shows a circle, centre $O$, of radius 9.4 cm. Points $A$, $B$, $C$, and $D$ lie on the circumference, and $BD$ is a diameter of the circle.
The straight line $PAQ$ is a tangent to the circle at $A$.
Angle $QAD = (2x + 14)°$, angle $ABD = (3x - 11)°$, and angle $CBD = 48°$.




**(a)** Find the value of $x$ and state the circle theorem you used.

$x =$ ..... , Theorem: ..... **[3]**

**(b)** Calculate the size of angle $ADC$, giving a geometrical reason for your answer.

Angle $ADC =$ ..... , Reason: ..... **[2]**

**(c)** Calculate the length of chord $AC$, giving your answer correct to 3 significant figures.

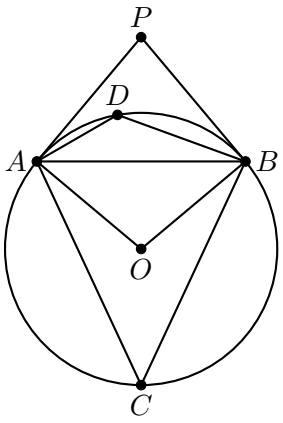
$AC =$ ..... cm **[3]**

**(d)** Calculate the area of the minor segment bounded by chord $AD$ and the minor arc $AD$, giving

your answer correct to 3 significant figures.

..... $cm^2$ **[3]**

**1115** $PA$ and $PB$ are tangents to a circle, centre $O$, from a point $P$.
$C$ is a point on the major arc $AB$.
Angle $APB = 4x°$.
Angle $ACB = (3x - 10)°$.




**(a)** Find an expression for angle $PAB$ in terms of $x$.

..... ° **[3]**

**(b)** Use the alternate segment theorem to form an equation in $x$, and show that $x = 20$.

**[3]**

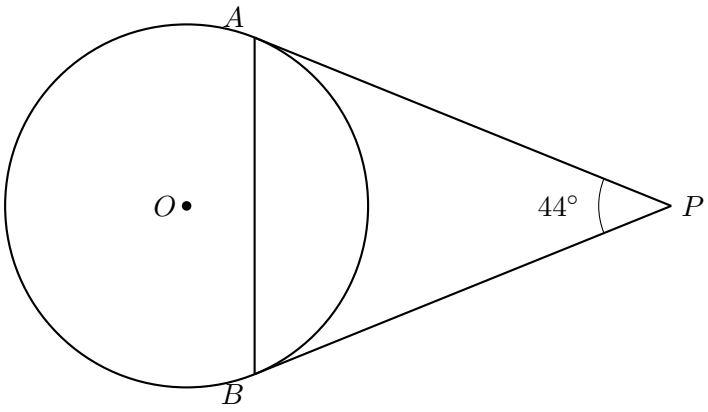
**(c)** Calculate angle $AOB$.

..... ° **[3]**

**(d)** $D$ is a point on the minor arc $AB$.
Calculate angle $ADB$.

..... ° **[3]**

**1116** The diagram shows a circle with centre $O$.
$P$ is an external point.
The lines $PA$ and $PB$ are tangents to the circle at the points $A$ and $B$.
Angle $APB = 44°$.




NOT TO SCALE

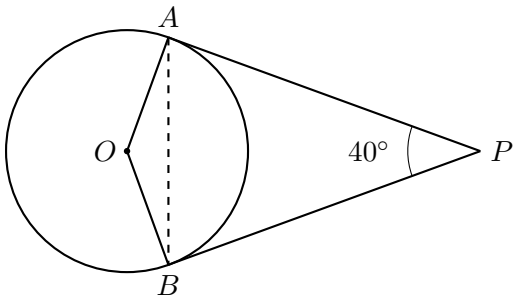
**(a)** Calculate angle $PAB$.

Angle $PAB$ = ..... ° **[2]**

**(b)** Calculate angle $AOB$.

Angle $AOB$ = ..... ° **[2]**

**1117** $P$ is a point outside a circle, centre $O$.
$PA$ and $PB$ are tangents to the circle at $A$ and $B$.
Angle $APB = 40°$.




**(a)** Write down the geometrical reason why triangle $PAB$ is isosceles.

..... **[1]**

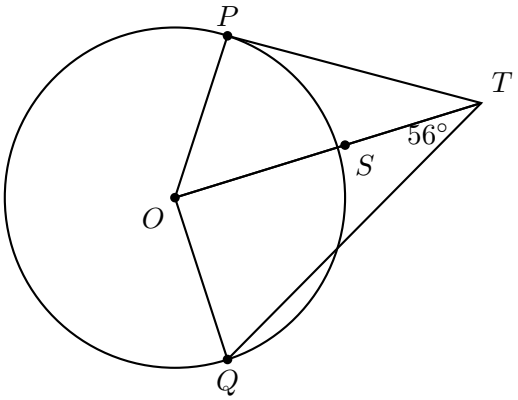
**(b)** Calculate angle $PAB$.

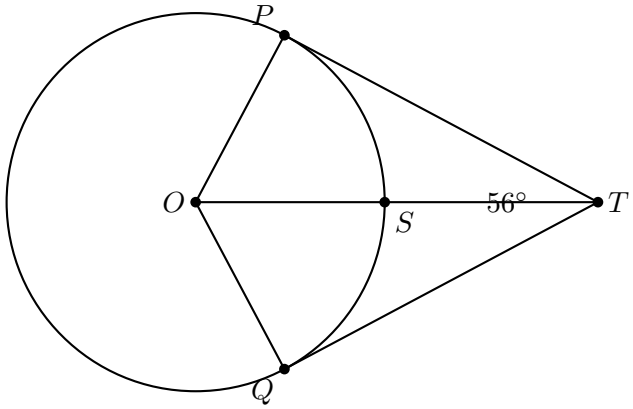
Angle $PAB =$ ..... ° **[1]**

**(c)** Calculate angle $AOB$.

Angle $AOB =$ ..... ° **[2]**

**1118** $P$ and $Q$ are points on a circle, centre $O$.
$TP$ and $TQ$ are tangents to the circle from an external point $T$.
The line $TO$ intersects the circle at $S$.
Angle $PTQ = 56°$.




**(a)** Calculate angle $POT$.

Angle $POT =$ ..... ° **[2]**

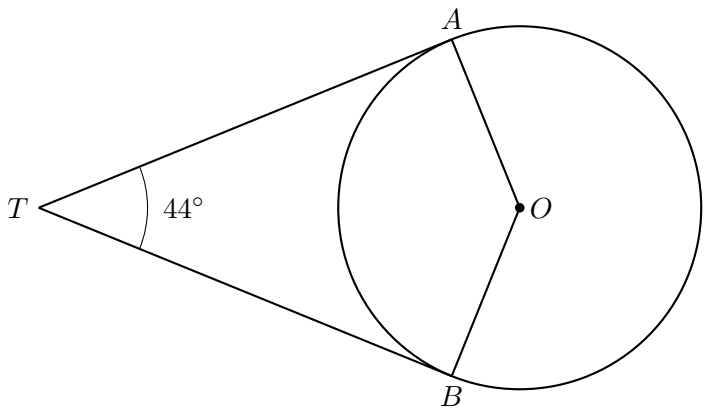
**(b)** Explain why triangle $TPQ$ is isosceles.

..... **[1]**

**(c)** Calculate angle $TPQ$.

Angle $TPQ =$ ..... ° **[1]**

**1119** The diagram shows a circle, centre $O$.
$TA$ and $TB$ are tangents to the circle at $A$ and $B$.
Angle $ATB = 44°$.




**(a)** Work out angle $AOB$.

Angle $AOB =$ ..... ° **[2]**

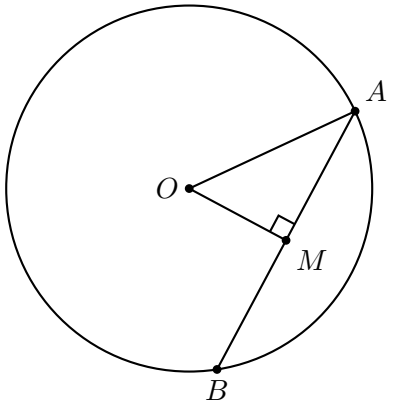
**(b)** Work out angle $OAB$.

Angle $OAB =$ ..... ° **[2]**

**(c)** Write down the geometrical reason why the length of $TA$ is equal to the length of $TB$.

..... **[1]**

**1120** The diagram shows a circle, centre $O$, with radius 10 cm.
$AB$ is a chord of length 16 cm and $M$ is the midpoint of $AB$.



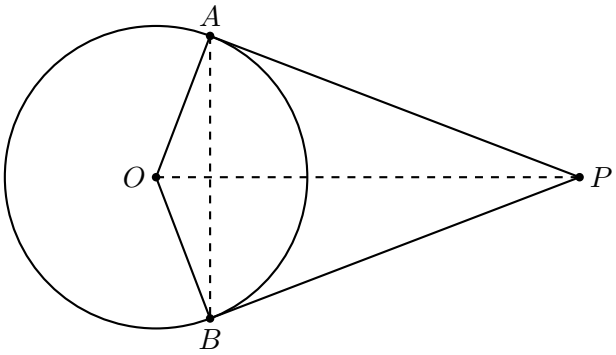

**(a)** State the size of angle $OMA$, giving a geometrical reason for your answer.

$\angle OMA =$ ..... ° **[2]**

**(b)** Calculate the distance $OM$ of the chord $AB$ from the centre $O$.

$OM =$ ..... $cm$ **[3]**

**1121** $P$ is an external point.
$PA$ and $PB$ are tangents to a circle, centre $O$.
The line $OP$ intersects the chord $AB$ at right angles.
$OA = 7$ cm and $PA = 24$ cm.




**(a)** Write down the length of $PB$.

..... cm **[1]**

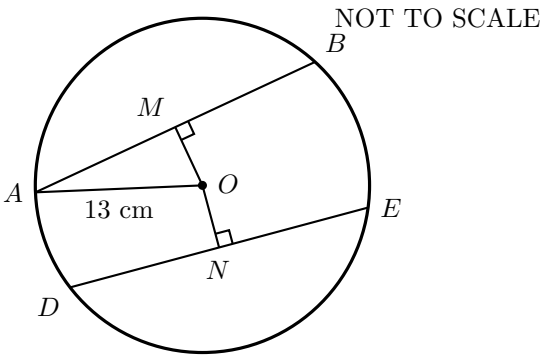
**(b)** Calculate the length of $OP$.

..... cm **[2]**

**(c)** Calculate the shortest distance from $O$ to the chord $AB$.

..... cm **[3]**

**1122** The diagram shows a circle, centre $O$, of radius 13 cm. Chords $AB$ and $DE$ each have length 24 cm. The line $OM$ is perpendicular to $AB$ at $M$, and the line $ON$ is perpendicular to $DE$ at $N$.




**(a)** Write down the length of $AM$ and give a geometrical reason for your answer.

$AM =$ ..... , Reason: ..... **[2]**

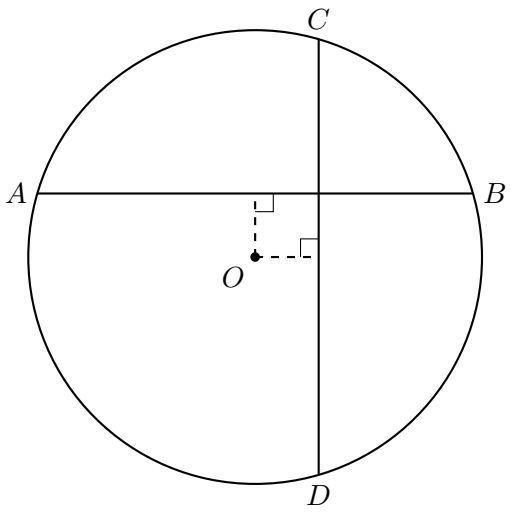
**(b)** Calculate the distance $OM$.

$OM =$ ..... cm **[2]**

**(c)** Write down the length of $ON$ and give a geometrical reason for your answer.

$ON =$ ..... , Reason: ..... **[2]**

**1123** The diagram shows a circle, centre $O$, with radius 25 cm.
$AB$ and $CD$ are two chords of the circle such that $AB = CD$.
The perpendicular distance from $O$ to $AB$ is $(x + 3)$ cm.
The perpendicular distance from $O$ to $CD$ is $(3x - 5)$ cm.



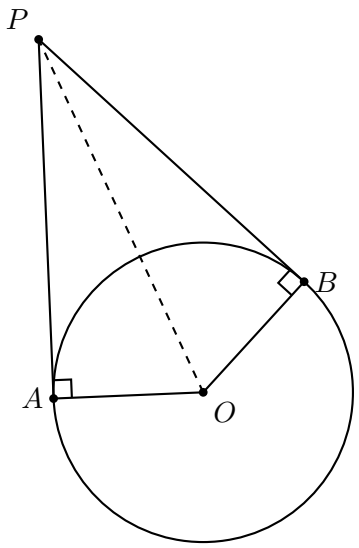

**(a)** Find the value of $x$.

$x =$ ..... **[2]**

**(b)** Calculate the length of the chord $AB$.

$AB =$ ..... cm **[3]**

**1124** The diagram shows a circle, centre $O$, with radius 5 cm.
$PA$ and $PB$ are tangents to the circle at $A$ and $B$ respectively.
The length of $PA$ is 12 cm.




**(a)** Write down the length of $PB$, giving a geometrical reason for your answer.

$PB =$ ..... $cm$ **[2]**

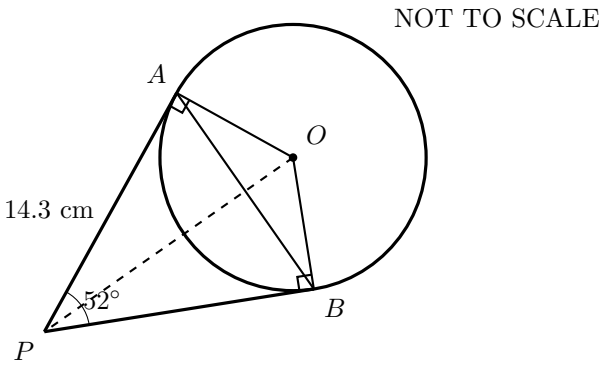
**(b)** Calculate the distance $OP$.

$OP =$ ..... $cm$ **[2]**

**(c)** Calculate the area of the quadrilateral $OAPB$.

Area = ..... $cm^2$ **[2]**

**1125** The diagram shows a circle, centre $O$, and an external point $P$.
The straight lines $PA$ and $PB$ are tangents to the circle at $A$ and $B$ respectively.
Angle $APB = 52°$ and $PA = 14.3$ cm.




**(a)** Explain why triangle $OAP$ is congruent to triangle $OBP$.

.....

.....

..... **[3]**

**(b)** Calculate the size of

**(i)** angle $AOB$,

Angle $AOB =$ ..... ° **[2]**

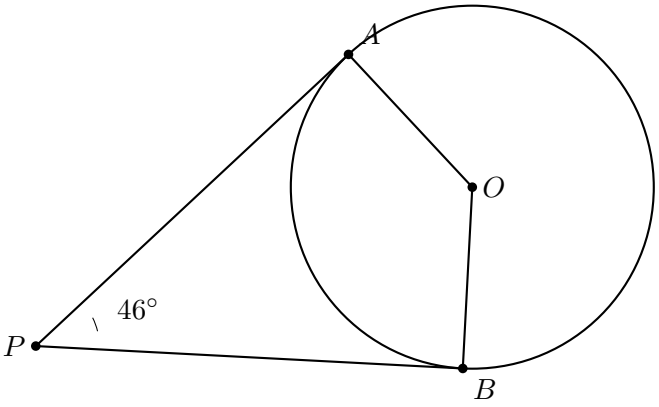
**(ii)** angle $OAB$.

Angle $OAB =$ ..... ° **[2]**

**(c)** Write down the length of $PB$.

$PB =$ ..... cm **[1]**

**1126** $P$ is an external point.
$PA$ and $PB$ are tangents to the circle, centre $O$, at the points $A$ and $B$.
Angle $APB = 46°$.



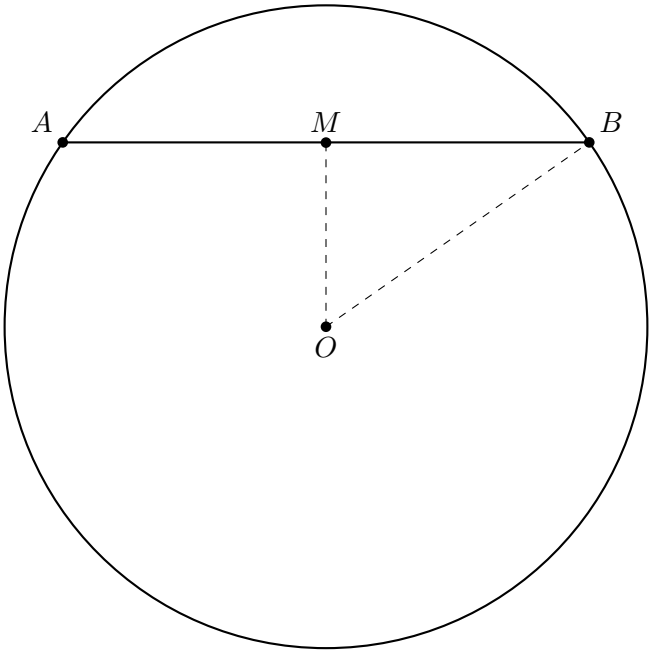

**(a)** Calculate angle $AOB$.

Angle $AOB =$ ..... ° **[2]**

**(b)** Calculate angle $OAB$.

Angle $OAB =$ ..... ° **[2]**

**1127** A circle, centre $O$, has a radius of 8.5 cm.
A chord $AB$ has length 14 cm.
$M$ is the midpoint of $AB$.



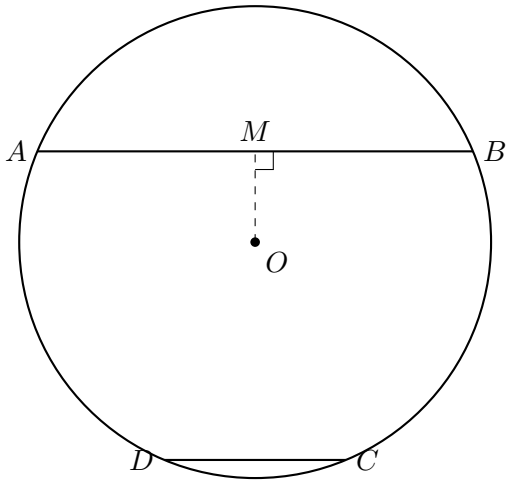

**(a)** Calculate the distance $OM$.

$OM =$ ..... cm **[3]**

**(b)** Another chord $CD$ is drawn such that it is parallel to $AB$ and is at a distance of 3 cm from $O$.
Calculate the length of the chord $CD$.

$CD =$ ..... cm **[2]**

**1128** The diagram shows a circle, centre $O$, with radius 13 cm.
$AB$ is a chord of the circle of length 24 cm.
The point $M$ is the midpoint of the chord $AB$.
Another chord $CD$ is drawn inside the circle.




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**(a)** Calculate the perpendicular distance, $OM$, from the centre of the circle to the chord $AB$.

$OM =$ ..... cm **[3]**

**(b)** The perpendicular distance from the centre $O$ to the chord $CD$ is also 5 cm.
Write down the length of the chord $CD$.

$CD =$ ..... cm **[1]**

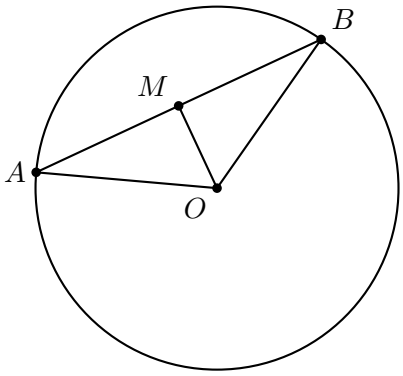
**(c)** Calculate the area of triangle $OAB$.

Area =

.....

$cm^2$ **[2]**

**1129** $A$ and $B$ are points on a circle, centre $O$ and radius 13 cm.
The length of the chord $AB$ is 24 cm.
The point $M$ is the midpoint of $AB$.




**(a)** State the geometrical reason why angle $OMA = 90°$.

..... **[1]**

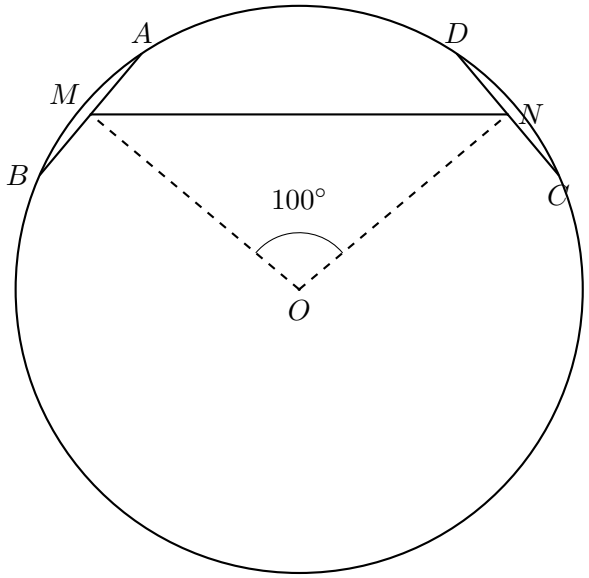
**(b)** Calculate the length of $OM$.

$OM =$ ..... cm **[3]**

**(c)** Calculate angle $AOB$.

Angle $AOB =$ ..... ° **[3]**

**1130** $A$, $B$, $C$ and $D$ are points on a circle, centre $O$ and radius 25 cm.
The chords $AB$ and $CD$ each have a length of 14 cm.
$M$ is the midpoint of $AB$ and $N$ is the midpoint of $CD$.
Angle $MON = 100°$.




**(a)** Show that $OM = 24$ cm.

**[2]**

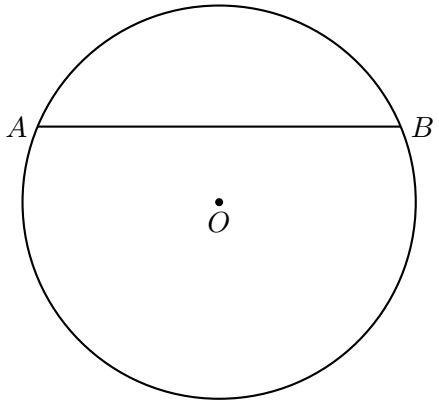
**(b)** Calculate the length of $MN$.

..... cm **[3]**

**(c)** Calculate the area of triangle $MON$.

..... $cm^2$ **[2]**

**1131** The diagram shows a circle, centre $O$, with radius 13 cm.
$AB$ is a chord of the circle of length 24 cm.




**(a)** Calculate the shortest distance from $O$ to the chord $AB$.

..... *cm* **[3]**

**(b)** $CD$ is another chord of the circle.
$CD$ is parallel to $AB$ and has length 24 cm.
Write down the shortest distance from $O$ to $CD$.

..... *cm* **[1]**

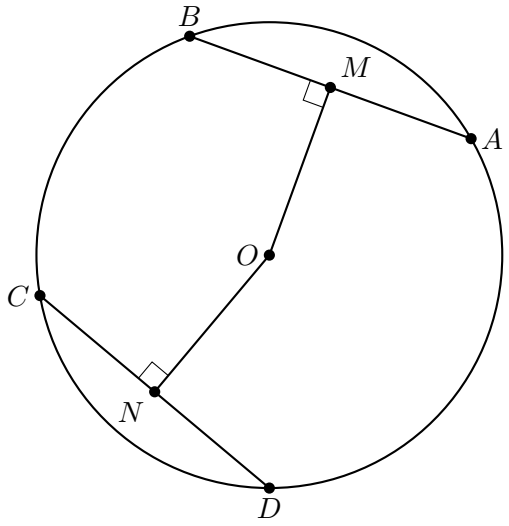
**(c)** $EF$ is a third chord of the circle.
The shortest distance from $O$ to $EF$ is 12 cm.
Calculate the length of the chord $EF$.

..... *cm* **[3]**

**(d)** State the geometrical property of a circle used to find the length of half the chord in part (a).

..... **[1]**

**1132** $AB$ and $CD$ are two chords of a circle, centre $O$.
The chords $AB$ and $CD$ are equal in length.
The line $OM$ is the perpendicular bisector of $AB$, and the line $ON$ is the perpendicular bisector of $CD$.
$OM = (5x - 7)$ cm and $ON = (2x + 2)$ cm.
The radius of the circle is 17 cm.



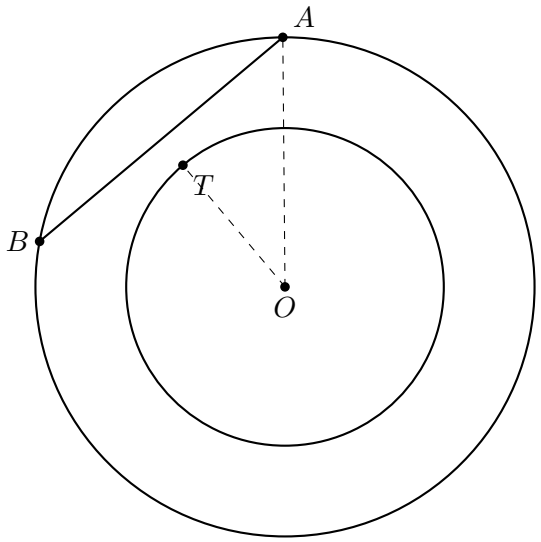

**(a)** Find the value of $x$.

$x =$ ..... **[3]**

**(b)** Calculate the length of the chord $AB$.

$AB =$ ..... cm **[4]**

**1133** Two circles, $C_1$ and $C_2$, have a common centre $O$.
The radius of $C_1$ is $r$ cm and the radius of $C_2$ is $R$ cm, where $R > r$.
A chord $AB$ of the larger circle $C_2$ is tangent to the smaller circle $C_1$ at the point $T$.
The length of the chord $AB$ is 24 cm and the area of the annular region between the two circles is $k\pi$ cm$^2$.




**(a)** Show that $R^2 - r^2 = 144$.

**[3]**

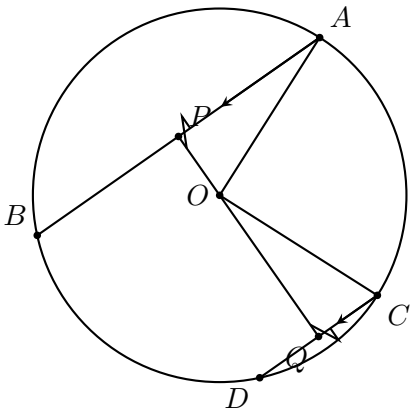
**(b)** Find the value of $k$.

$k =$ ..... **[2]**

**(c)** A second chord $CD$ of the larger circle $C_2$ has length 24 cm.
State, with a reason, the distance from $O$ to the chord $CD$.

..... **[2]**

**1134** The diagram shows a circle, centre $O$, with two parallel chords $AB$ and $CD$ on opposite sides of the centre.
$AB = 24$ cm, $CD = 10$ cm, and the perpendicular distance between the two chords is 17 cm.
$P$ and $Q$ are the midpoints of $AB$ and $CD$ respectively.



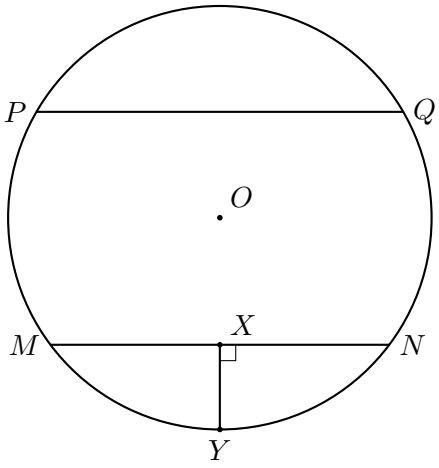

**(a)** Explain why $P$, $O$ and $Q$ lie on a straight line.

..... **[2]**

**(b)** Let $OP = x$ cm and the radius of the circle be $r$ cm.
Calculate the radius, $r$, of the circle.

$r =$ ..... $cm$ **[5]**

**1135** A circle has centre $O$ and radius 10 cm.
$MN$ is a chord of length 16 cm.
The perpendicular bisector of $MN$ intersects $MN$ at $X$ and the minor arc $MN$ at $Y$.
$PQ$ is a second chord which is parallel to $MN$.
The perpendicular distance between $MN$ and $PQ$ is 11 cm.




**(a)** Explain why the line containing $XY$ passes through the centre $O$.

..... **[1]**

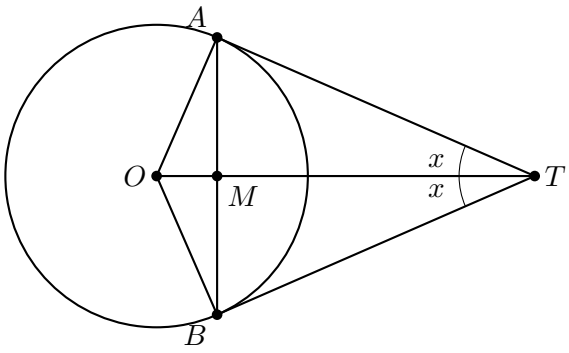
**(b)** Calculate the length of $XY$.

$XY =$ ..... cm **[3]**

**(c)** Calculate the length of the chord $PQ$.

$PQ =$ ..... cm **[4]**

**1136** $TA$ and $TB$ are tangents to a circle, centre $O$ and radius $r$, at the points $A$ and $B$.
The line $TO$ intersects the chord $AB$ at $M$.
Angle $ATO$ = Angle $BTO = x$.




**(a)** Show that the length of the chord $AB$ is $2r\cos x$.

**[2]**

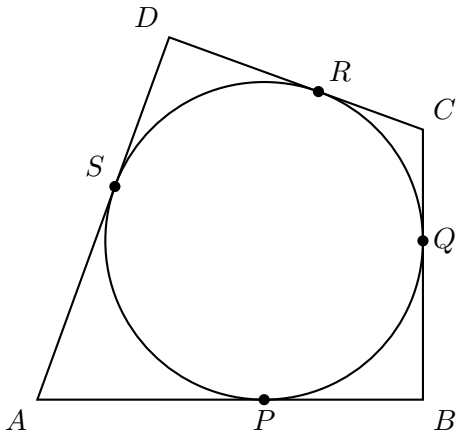
**(b)** A second chord $BC$ is drawn such that $BC = AB$.
Write down the shortest distance from $O$ to the chord $BC$, in terms of $r$ and $x$.

..... **[1]**

**(c)** The area of triangle $OAB$ is equal to the area of triangle $ATB$.
Find the exact value of $x$.

$x =$ ..... $^\circ$ **[5]**

**1137** A circle is inscribed in a quadrilateral $ABCD$.
The circle touches the sides $AB$, $BC$, $CD$, and $DA$ at the points $P$, $Q$, $R$, and $S$ respectively.




**(a)** Write down the geometrical reason why $AP = AS$.

..... **[1]**

**(b)** Show that $AB + CD = AD + BC$.

**[3]**

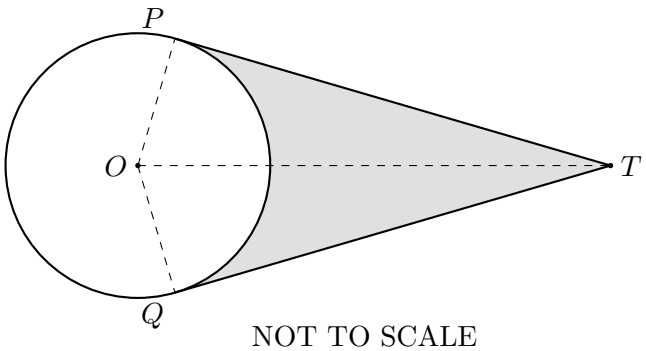
**(c)** Given that $AB = (2x + 5)$ cm, $BC = (3x - 2)$ cm, $CD = (x + 8)$ cm and $AD = (2x + 1)$ cm, work out the value of $x$.

$x =$ ..... **[3]**

**(d)** Calculate the perimeter of the quadrilateral $ABCD$.

Perimeter = ..... $cm$ **[2]**

**1138** The diagram shows a circle with centre $O$ and radius 7 cm.
The lines $TP$ and $TQ$ are tangents to the circle at the points $P$ and $Q$.
The distance from the centre $O$ to the external point $T$ is 25 cm.




NOT TO SCALE

**(a)** Show that the length of the tangent $TP$ is 24 cm.

**[2]**

**(b)** Calculate angle $PTQ$.

Angle $PTQ$ = ..... ° **[3]**

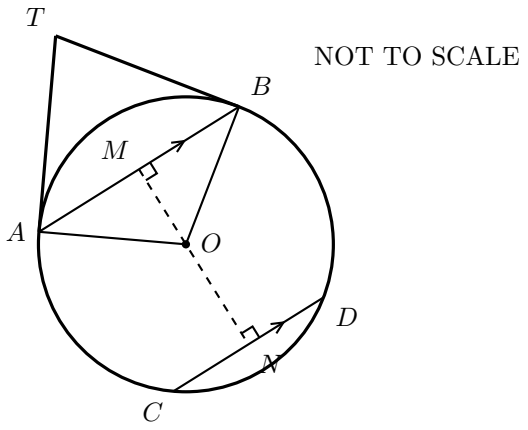
**(c)** Calculate the area of the shaded region bounded by the tangents $TP$, $TQ$ and the minor arc $PQ$.

Area =

.....

$cm^2$ **[4]**

**1139** The diagram shows a circle, centre $O$, of radius $r$ cm. Chords $AB$ and $CD$ are parallel and lie on opposite sides of the centre $O$.
$AB = 16$ cm, $CD = 12$ cm, and the perpendicular distance between $AB$ and $CD$ is 14 cm. The tangents to the circle at $A$ and $B$ meet at the external point $T$.




**(a)** The perpendicular line through $O$ meets $AB$ at $M$ and $CD$ at $N$, with $OM = x$ cm. Write down two equations in terms of $r$ and $x$, and solve them to show that $x = 6$ and $r = 10$.

**[4]**

**(b)** Calculate the exact length of tangent $TA$, giving your answer as an improper fraction.

$TA =$ ..... cm **[3]**

**(c)** Write down the length of $TB$, and calculate the perimeter of quadrilateral $OATB$, giving your answer correct to 1 decimal place.

$TB =$ ..... , Perimeter = ..... **[2]**

**(d)** Calculate the size of angle $ATB$, giving your answer correct to 1 decimal place.

Angle $ATB =$ ..... ° **[2]**

**1140** **(a)** Change 52000 $cm^2$ into $m^2$.

..... $m^2$ **[1]**

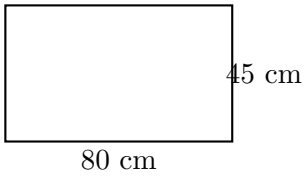
**(b)** Change 4.7 $m^3$ into litres.

..... litres **[1]**

**(c)** Change 850 millilitres into $cm^3$.

..... $cm^3$ **[1]**

**1141** A rectangular piece of card and a water jug are being measured.




**(a)** Calculate the area of the rectangular piece of card in square metres ($m^2$).

..... $m^2$ **[2]**

**(b)** The jug holds 3.2 litres of water.
Write down this volume in $cm^3$.

..... $cm^3$ **[1]**

**1142** **(a)** Change 3.5 $m^2$ into $cm^2$.

..... $cm^2$ **[1]**

**(b)** A rectangular tank measures 60 $cm$ by 45 $cm$ by 30 $cm$.
Calculate the volume of the tank in litres.

..... litres **[3]**

**1143** Complete each of the following metric unit conversions.

**(a)** Convert 4.75 km into metres.

..... $m$ **[1]**

**(b)** Convert 320 g into kilograms.

..... kg **[1]**

**(c)** Convert 2.4 $m^2$ into $cm^2$.

..... $cm^2$ **[1]**

**(d)** Convert 6500 $cm^3$ into litres.

..... litres **[1]**

**1144** A solid wooden block is in the shape of a cuboid.
The dimensions of the block are 40 cm, 250 mm and 0.2 m.

**(a)** Work out the volume of the block in $cm^3$.

..... $cm^3$ **[2]**

**(b)** Convert your answer to part (a) into litres.

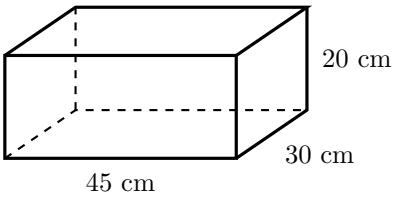
..... litres **[1]**

**(c)** The density of the wood is 0.6 g/$cm^3$.
Calculate the mass of the block in kilograms.

..... kg **[2]**

**1145** The diagram shows a closed rectangular box of length 45 cm, width 30 cm, and height 20 cm.

NOT TO SCALE




**(a)** Calculate

**(i)** the total surface area of the box in $cm^2$,

..... $cm^2$ **[2]**

**(ii)** your answer to part (a)(i) converted into $m^2$.

..... $m^2$ **[1]**

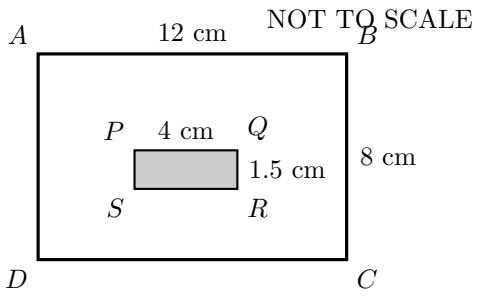
**(b)** Calculate the volume of the box in $cm^3$, and write down its capacity in litres.

Volume = ..... , Capacity = ..... **[2]**

**(c)** The empty box has a mass of 650 g. It is packed with 18 identical books, each of mass 1.15 kg. Calculate the total mass of the packed box in kilograms.

..... kg **[2]**

**1146** The diagram shows the scale plan of a rectangular park $ABCD$ containing a rectangular pond $PQRS$.
The plan is drawn to a scale of $1 : 25\,000$.
On the plan, $AB = 12$ cm, $BC = 8$ cm, $PQ = 4$ cm, and $QR = 1.5$ cm.




**(a)** Calculate the actual perimeter of the park $ABCD$, giving your answer in kilometres.

..... km **[2]**

**(b)** Calculate

**(i)** the area of the park excluding the pond, in $cm^2$, on the plan,

..... $cm^2$ **[2]**

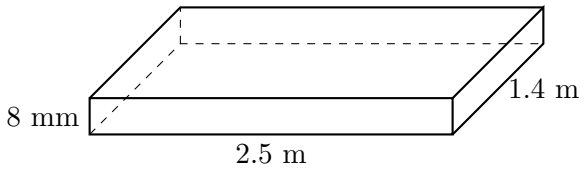
**(ii)** the actual area of the park excluding the pond, giving your answer in $km^2$.

..... $km^2$ **[2]**

**(c)** The actual pond $PQRS$ has a uniform depth of 1.6 m. Calculate the capacity of the pond when it is completely full, giving your answer in litres in standard form.

..... litres **[3]**

**1147** A rectangular block of metal has length 2.5 m, width 1.4 m and thickness 8 mm.




NOT TO SCALE

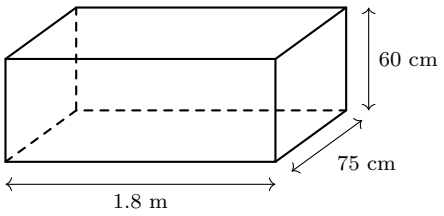
**(a)** Work out the volume of the block of metal in $cm^3$.

..... $cm^3$ **[2]**

**(b)** The mass of 1 $cm^3$ of this metal is 7.8 g.
Work out the mass of the block of metal in kilograms.

..... kg **[2]**

**1148** The diagram shows a rectangular water tank with internal length 1.8 m, internal width 75 cm, and internal height 60 cm.




**(a)** Calculate the total internal capacity of the tank in litres.

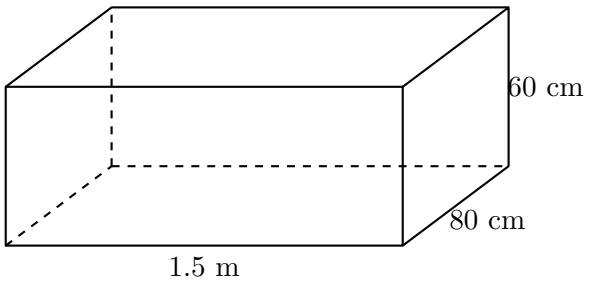
Capacity = ..... litres **[3]**

**(b)** The empty tank is filled with water from a tap flowing at a rate of 450 ml per second.

Calculate the time taken to fill the tank completely.
Give your answer in minutes.

Time = ..... minutes **[3]**

**1149** A metal water tank is made in the shape of a cuboid.




**(a)** Calculate the capacity of the water tank in litres.

..... litres **[3]**

**(b)** A solid metal sphere is melted down to make the tank.
The metal sphere has a volume of 0.024 $m^3$.
The mass of 1 $cm^3$ of this metal is 7.8 g.

Calculate the total mass of the sphere, giving your answer in kilograms.

..... $kg$ **[3]**

**1150** During a storm, 15 $mm$ of rain falls on a flat rectangular roof measuring 15 $m$ by 12 $m$.

**(a)** Calculate the volume of rain that falls on the roof.
Give your answer in litres.

..... litres **[3]**

**(b)** The water from the roof flows into a cylindrical tank with a base radius of 60 $cm$.
Calculate the increase in the depth of the water in the tank.
Give your answer in centimetres, correct to 1 decimal place.

..... $cm$ **[4]**

**1151** A solid metal sphere has a radius of 15 cm.
The volume, $V$, of a sphere with radius $r$ is $V = \frac{4}{3}\pi r^3$.

**(a)** Calculate the volume of the sphere in $m^3$.
Give your answer in standard form.

..... $m^3$ **[3]**

**(b)** The mass of the sphere is 125 kg.
Calculate the density of the metal in g/$cm^3$.

..... $g/cm^3$ **[3]**

**(c)** The sphere is melted down to make completely solid identical cones.
Each cone has a base radius of 24 mm and a height of 5.5 cm.
The volume, $V$, of a cone with radius $r$ and height $h$ is $V = \frac{1}{3}\pi r^2 h$.

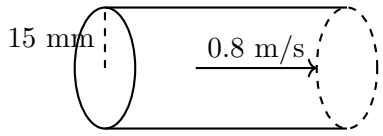
**(i)** Work out the volume of one cone in $cm^3$.

..... $cm^3$ **[3]**

**(ii)** Work out the maximum number of completely solid cones that can be made.

..... **[2]**

**1152** Water flows through a cylindrical pipe into an empty rectangular pool.
The pool measures 4 m by 2.5 m.




**(a)** The cylindrical pipe has a radius of 15 mm.
Water flows through the pipe at a speed of 0.8 m/s.

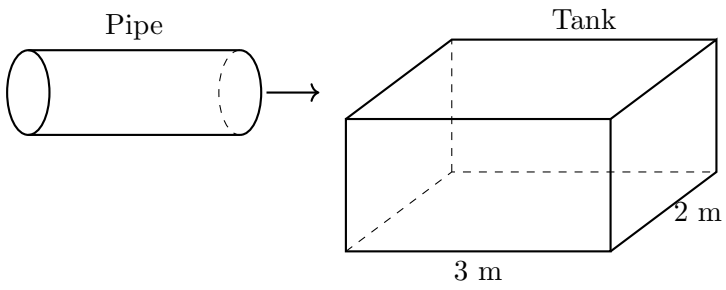
Show that the volume of water flowing out of the pipe in one second is 0.565 litres, correct to 3 significant figures.

**[3]**

**(b)** Calculate the time taken for the depth of water in the pool to reach 1.2 m.
Give your answer in hours and minutes, correct to the nearest minute.

..... hours ..... minutes **[4]**

**1153** Water flows through a cylindrical pipe of internal radius 2.5 cm at a speed of 5.4 km/h.
The water flows into an empty rectangular tank of length 3 m and width 2 m.




NOT TO SCALE

**(a)** Show that the volume of water flowing into the tank in 1 minute is 176.7 litres, correct to 1 decimal place.

**[3]**

**(b)** Calculate the depth of the water in the tank after 45 minutes.
Give your answer in centimetres.

..... cm **[4]**

**1154** A farmer owns a rectangular field measuring 450 m by 320 m.

**(a)** Calculate the area of the field in $km^2$.

Area = ..... $km^2$ **[2]**

**(b)** A liquid fertiliser is sprayed over the entire field at a rate of 25 ml per $m^2$.
The fertiliser is sold in drums each containing 0.18 $m^3$ of liquid.

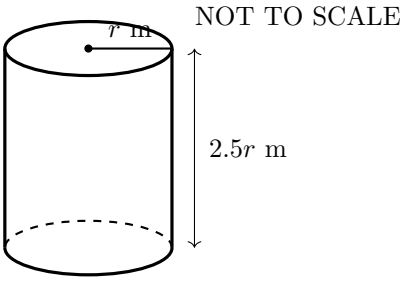
**(i)** Calculate the total volume of fertiliser required for the field, in $m^3$.

Volume = ..... $m^3$ **[3]**

**(ii)** Each drum of fertiliser has a mass of 216 kg when full, and the empty drum has a mass of 9 kg.
Calculate the mass, in grams, of 1 $cm^3$ of the liquid fertiliser.

Mass = ..... $g$ **[3]**

**1155** A closed cylindrical water tank has internal radius $r$ metres and internal height $h$ metres, where $h = 2.5r$.
The internal capacity of the tank is 15 700 litres.




**(a)** Convert 15 700 litres into $m^3$, and hence calculate the value of $r$, giving your answer correct to 3 significant figures.

Volume = ..... , $r$ = ..... **[3]**

**(b)** The entire inside surface of the closed tank is coated with a protective waterproof layer of uniform thickness 1.5 mm. Using $r = 1.26$, estimate the volume of waterproof coating used by multiplying the internal surface area by the thickness.
Give your answer in $cm^3$ correct to 3 significant figures.

..... $cm^3$ **[3]**

**(c)** Water flows out of the full tank through a pipe at a constant rate of 450 ml per second.
Calculate the time taken to empty the 15 700 litres of water from the tank, giving your answer in hours and minutes correct to the nearest minute.

..... hours ..... minutes **[3]**

**(d)** The water in the tank has a density of 1.02 g per $cm^3$.
Calculate the total mass of the 15 700 litres of water, giving your answer in tonnes, where 1 tonne = 1000 kg.

..... tonnes **[2]**

**1156** Water flows through a cylindrical pipe of radius 2.5 $cm$ at a speed of 18 $km/h$.




**(a)** Calculate the volume of water, in litres, that flows through the pipe in 1 minute.

..... litres **[5]**

**(b)** The water from the pipe is used to fill an empty rectangular reservoir measuring 50 $m$ by 30 $m$ by 4 $m$.
Calculate the time it takes to fill the reservoir completely.
Give your answer in days and hours, correct to the nearest hour.

days ..... , hours ..... **[7]**

**1157** Water flows through a cylindrical pipe of radius 2.5 cm at a speed of 1.6 m/s.

**(a)** Calculate the volume of water, in litres, that flows through the pipe in one minute.
Give your answer correct to 3 significant figures.

..... litres **[4]**

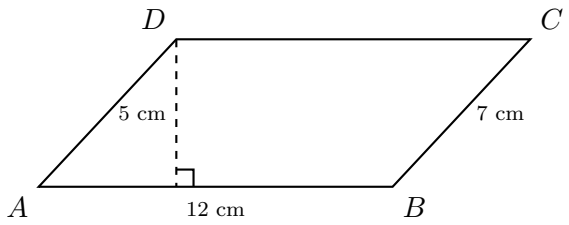
**(b)** This water flows into an empty rectangular tank measuring 4.2 m by 3.5 m by 2.4 m.
Calculate the time taken to fill the tank completely.
Give your answer in hours and minutes, correct to the nearest minute.

hours ..... , minutes ..... **[5]**

**(c)** The outside vertical walls of the rectangular tank are to be painted.
1 litre of paint covers 45000 $cm^2$.
Paint is sold in 2.5-litre tins.
Work out the minimum number of tins of paint needed.

..... tins **[6]**

**1158** The diagram shows a parallelogram $ABCD$ with $AB = 12$ cm and $BC = 7$ cm.
The perpendicular height from $D$ to $AB$ is 5 cm.



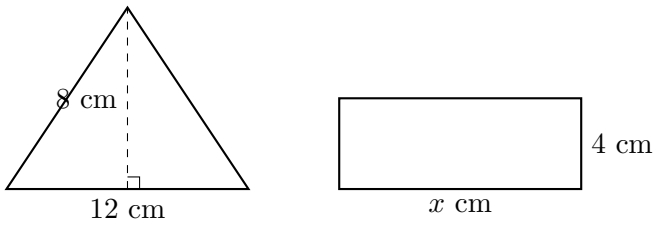

**(a)** Calculate the perimeter of parallelogram $ABCD$.

Perimeter = ..... $cm$ **[2]**

**(b)** Calculate the area of parallelogram $ABCD$.

Area = ..... $cm^2$ **[2]**

**1159** An isosceles triangle and a rectangle are shown below.
The area of the triangle is equal to the area of the rectangle.




NOT TO SCALE

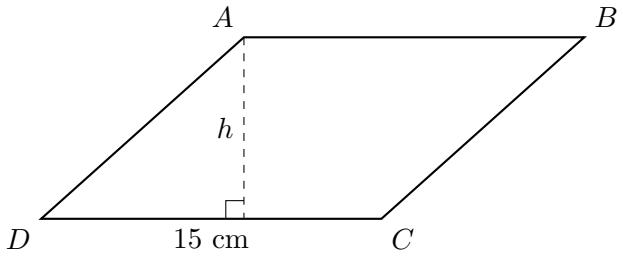
**(a)** Calculate the value of $x$.

$x =$ ..... **[2]**

**(b)** Calculate the perimeter of the isosceles triangle.

..... cm **[3]**

**1160** The diagram shows a parallelogram $ABCD$.
The area of the parallelogram is 120 $cm^2$ and its perimeter is 54 cm.




**(a)** Work out the value of $h$.

$h =$ ..... cm **[2]**

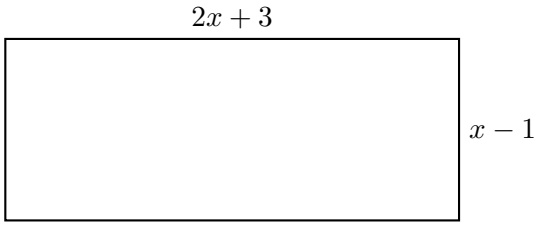
**(b)** Work out the length of $AD$.

$AD =$ ..... cm **[2]**

**(c)** Calculate the area of triangle $ACD$.

Area = ..... $cm^2$ **[1]**

**1161** The diagram shows a rectangle.
All measurements are in centimetres.



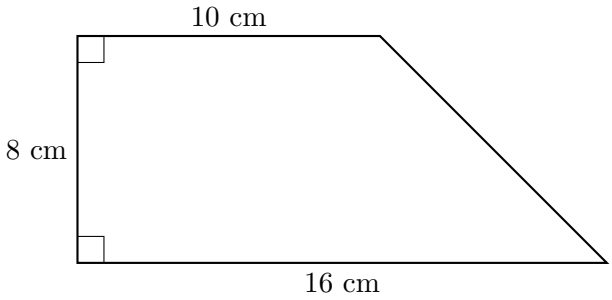

**(a)** The perimeter of the rectangle is 34 cm.
Find the value of $x$.

$x =$ ..... **[3]**

**(b)** Calculate the area of the rectangle.

..... $cm^2$ **[2]**

**1162** The diagram shows a right-angled trapezium.



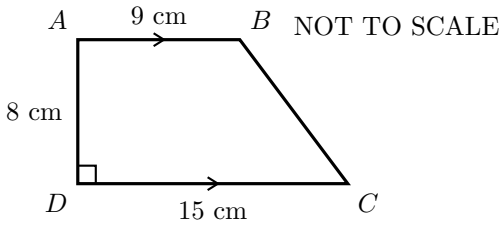

**(a)** Calculate the area of the trapezium.

..... $cm^2$ **[2]**

**(b)** Calculate the perimeter of the trapezium.

..... cm **[3]**

**1163** The diagram shows a trapezium $ABCD$ in which $AB$ is parallel to $DC$ and angle $ADC = 90°$.
$AB = 9$ cm, $DC = 15$ cm, and $AD = 8$ cm.




**(a)** Calculate the area of trapezium $ABCD$.

..... $cm^2$ **[2]**

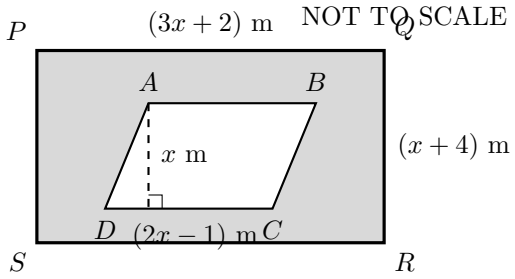
**(b)** Calculate the length of $BC$, and hence write down the perimeter of trapezium $ABCD$.

$BC =$ ..... , Perimeter = ..... **[3]**

**(c)** Point $E$ lies on $DC$ such that $ABCE$ is a parallelogram.
Calculate the area of triangle $ADE$.

..... $cm^2$ **[2]**

**1164** The diagram shows a rectangular garden $PQRS$ of length $(3x + 2)$ m and width $(x + 4)$ m. Inside the garden, a flowerbed in the shape of a parallelogram $ABCD$ has base $(2x - 1)$ m and perpendicular height $x$ m. The shaded region surrounding the flowerbed is covered with grass.




**(a)** Write down and simplify an expression, in terms of $x$, for the perimeter of the rectangular garden $PQRS$.

..... $m$ **[2]**

**(b)** Show that the area of the grass, $A$ $m^2$, is given by $A = x^2 + 15x + 8$.

**[3]**

**(c)** The area of the grass is 62 $m^2$.

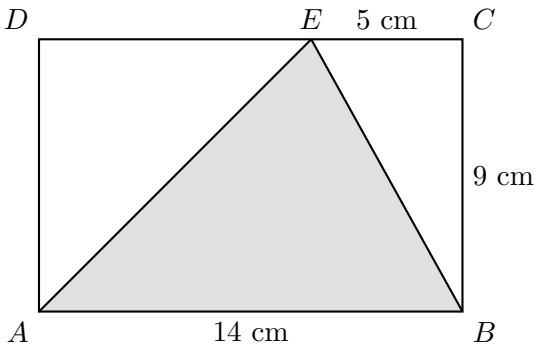
**(i)** Write down a quadratic equation in terms of $x$ and solve it to calculate the value of $x$.

$x =$ ..... **[3]**

**(ii)** Calculate the area of the flowerbed $ABCD$.

..... $m^2$ **[1]**

**1165** $ABCD$ is a rectangle with length 14 cm and width 9 cm.
The point $E$ lies on the edge $CD$ such that $CE = 5$ cm.




NOT TO SCALE

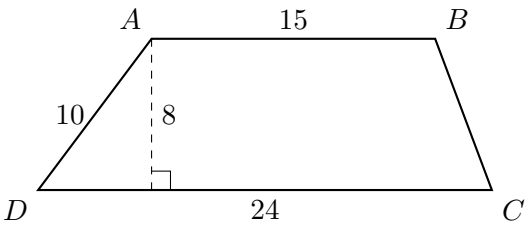
**(a)** Calculate the area of the shaded triangle $ABE$.

..... $cm^2$ **[2]**

**(b)** Calculate the perimeter of the shaded triangle $ABE$.

..... cm **[4]**

**1166** The diagram shows a trapezium $ABCD$ with $AB$ parallel to $DC$.
The perpendicular height of the trapezium is 8 cm.
All measurements are in centimetres.



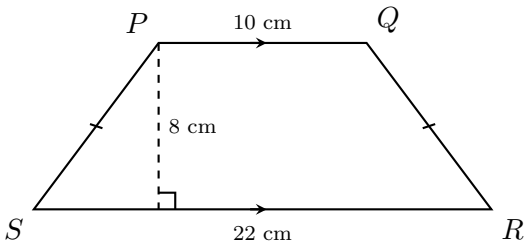

**(a)** Calculate the area of the trapezium.

..... $cm^2$ **[2]**

**(b)** Calculate the perimeter of the trapezium.

..... cm **[4]**

**1167** The diagram shows an isosceles trapezium $PQRS$ with $PQ$ parallel to $SR$.
$PQ = 10$ cm, $SR = 22$ cm, and the perpendicular height of the trapezium is 8 cm.



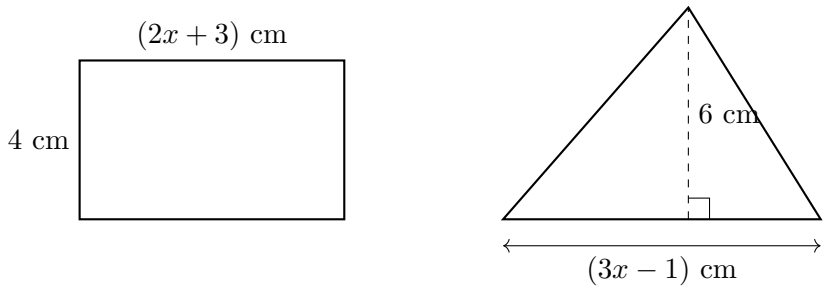

**(a)** Calculate the area of the trapezium $PQRS$.

Area = ..... $cm^2$ **[2]**

**(b)** Calculate the perimeter of the trapezium $PQRS$.

Perimeter = ..... $cm$ **[4]**

**1168** The diagram shows a rectangle and a triangle.
The area of the rectangle is equal to the area of the triangle.



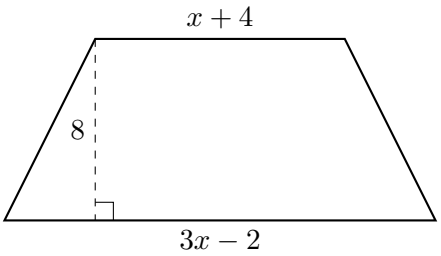

**(a)** Form an equation in $x$ and solve it to find the value of $x$.

$x =$ ..... **[4]**

**(b)** Work out the perimeter of the rectangle.

..... cm **[2]**

**1169** The diagram shows a trapezium.
All measurements are in centimetres.



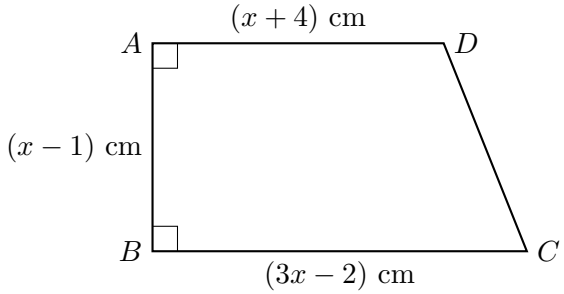

**(a)** The area of the trapezium is 120 $cm^2$.
Work out the value of $x$.

$x =$ ..... **[3]**

**(b)** The trapezium is an isosceles trapezium, with the two slanted sides being equal in length.
Calculate the perimeter of the trapezium.

Perimeter = ..... cm **[4]**

**1170** A trapezium $ABCD$ has right angles at $A$ and $B$.
The side $AD$ is parallel to $BC$.
$AD = (x + 4)$ cm, $BC = (3x - 2)$ cm and $AB = (x - 1)$ cm.




NOT TO SCALE

**(a)** The area of the trapezium is 20 cm$^2$.

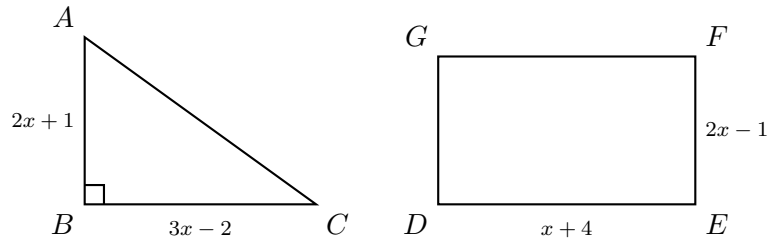
**(i)** Show that $2x^2 - x - 21 = 0$.

**[3]**

**(ii)** Calculate the perimeter of the trapezium.

..... cm **[5]**

**1171** The diagram shows a right-angled triangle $ABC$ and a rectangle $DEFG$.
All lengths are in centimetres.
In triangle $ABC$, $AB = (2x + 1)$ cm and $BC = (3x - 2)$ cm.
In rectangle $DEFG$, $DE = (x + 4)$ cm and $EF = (2x - 1)$ cm.




**(a)** The area of triangle $ABC$ is equal to the area of rectangle $DEFG$.
Show that $2x^2 - 15x + 6 = 0$.

**[3]**

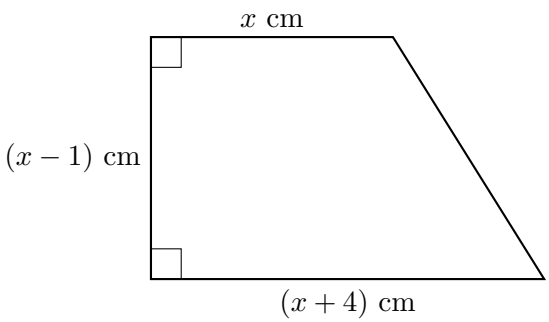
**(b)** Solve the equation $2x^2 - 15x + 6 = 0$.
Give your answers correct to 2 decimal places.

$x =$ ..... , $x =$ ..... **[3]**

**(c)** Calculate the perimeter of rectangle $DEFG$.

Perimeter = ..... $cm$ **[2]**

**1172** The diagram shows a right-angled trapezium with parallel sides of length $x$ cm and $(x + 4)$ cm.
The height of the trapezium is $(x - 1)$ cm.
The area of the trapezium is 40 cm$^2$.




**(a)** Show that $x^2 + x - 42 = 0$.

**[3]**

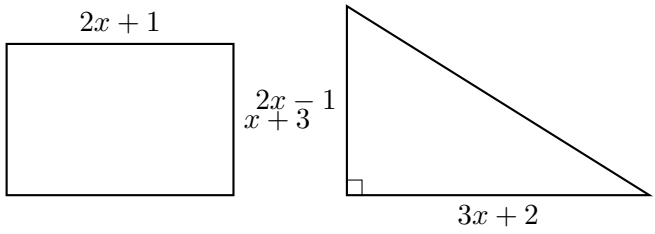
**(b)** Solve the equation $x^2 + x - 42 = 0$ to find the value of $x$.

$x =$ ..... **[2]**

**(c)** Calculate the perimeter of the trapezium.

..... cm **[4]**

**1173** A rectangle and a right-angled triangle are shown.
All measurements are in centimetres.




**(a)** The area of the rectangle is equal to the area of the right-angled triangle.
Show that $2x^2 - 13x - 8 = 0$.

**[4]**

**(b)** Solve the equation $2x^2 - 13x - 8 = 0$.
Show all your working and give your answers correct to 2 decimal places.

$x =$ ..... , $x =$ ..... **[4]**

**(c)** Calculate the perimeter of the rectangle.

Perimeter = ..... cm **[2]**

**1174** A piece of wire of length 60 cm is cut into two pieces.
One piece is bent to form a square.
The other piece is bent to form a rectangle where the length is exactly twice the width.
Let the width of the rectangle be $y$ cm.

**(a)** Write down an expression, in terms of $y$, for the side length of the square.

..... cm **[2]**

**(b)** The total area enclosed by the square and the rectangle is 113 $cm^2$.
Show that $17y^2 - 180y + 448 = 0$.

**[3]**

**(c)** Solve the equation.

$$17y^2 - 180y + 448 = 0$$

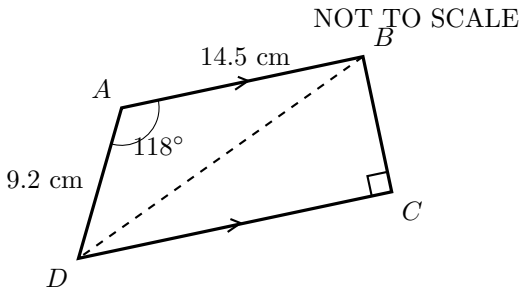
Show all your working.

$y =$ ..... , $y =$ ..... **[3]**

**(d)** Given that the side length of the square must be greater than the width of the rectangle, calculate the area of the square.

..... $cm^2$ **[2]**

**1175** The diagram shows a trapezium $ABCD$ in which $AB$ is parallel to $DC$ and angle $BCD = 90°$. $AB = 14.5$ cm, $AD = 9.2$ cm, and angle $BAD = 118°$.




**(a)** Calculate the length of diagonal $BD$, giving your answer correct to 3 significant figures.

$BD =$ ..... cm **[3]**

**(b)** Show that angle $BDC = 23.3°$, correct to 1 decimal place.

**[2]**

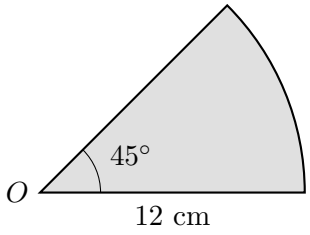
**(c)** Calculate the perimeter of trapezium $ABCD$, giving your answer correct to 3 significant figures.

Perimeter = ..... cm **[3]**

**(d)** Calculate the area of trapezium $ABCD$, giving your answer correct to 3 significant figures.

Area = ..... $cm^2$ **[2]**

**1176** The diagram shows a sector of a circle, center $O$, with radius 12 cm and sector angle 45°.



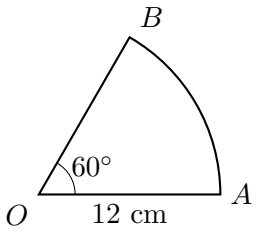

**(a)** Work out the exact length of the arc of the sector.
Give your answer in terms of $\pi$.

..... cm **[2]**

**(b)** Work out the exact area of the sector.
Give your answer in terms of $\pi$.

..... $cm^2$ **[2]**

**1177** The diagram shows a minor sector $AOB$ of a circle with centre $O$ and radius 12 cm.
The sector angle is 60°.



NOT TO SCALE

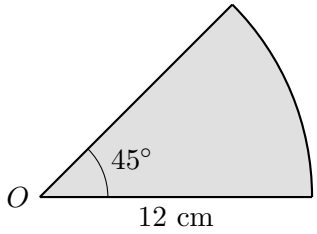
**(a)** Work out the length of the minor arc $AB$.
Give your answer in terms of $\pi$.

..... cm **[2]**

**(b)** Work out the area of the minor sector $AOB$.
Give your answer in terms of $\pi$.

..... $cm^2$ **[2]**

**1178** The diagram shows a sector of a circle with centre $O$ and radius 12 cm.
The sector angle is 45°.



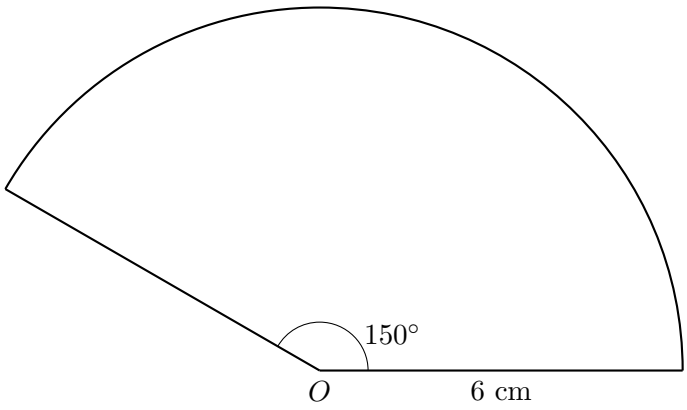

**(a)** Work out the exact area of the sector.
Give your answer in terms of $\pi$.

Area = ..... $cm^2$ **[2]**

**(b)** Work out the exact perimeter of the sector.
Give your answer in terms of $\pi$.

Perimeter = ..... cm **[3]**

**1179** The diagram shows a sector of a circle with centre $O$ and radius 6 cm.
The sector angle is 150°.



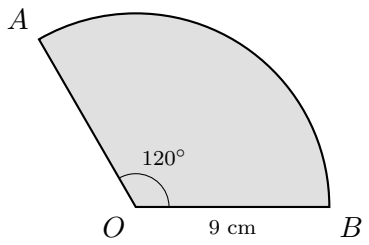

**(a)** Calculate the area of the sector.
Give your answer in terms of $\pi$.

..... $cm^2$ **[2]**

**(b)** Calculate the perimeter of the sector.
Give your answer in terms of $\pi$.

..... cm **[3]**

**1180** The diagram shows a sector $OAB$ of a circle with centre $O$ and radius 9 cm.
The sector angle $\angle AOB = 120°$.



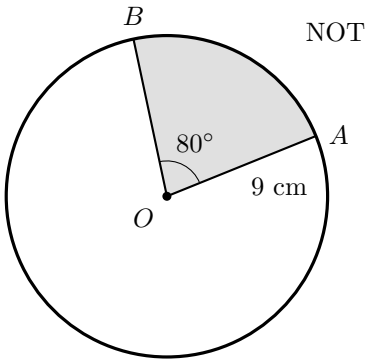

**(a)** Calculate the length of the minor arc $AB$.
Give your answer in terms of $\pi$.

..... $cm$ **[2]**

**(b)** Calculate the area of the shaded minor sector $OAB$.
Give your answer in terms of $\pi$.

Area = ..... $cm^2$ **[3]**

**1181** The diagram shows a circle, centre $O$, of radius 9 cm. Points $A$ and $B$ lie on the circumference such that the minor sector angle $AOB = 80°$.




**(a)** Giving your answers in terms of $\pi$, find

**(i)** the length of the minor arc $AB$,

..... cm **[2]**

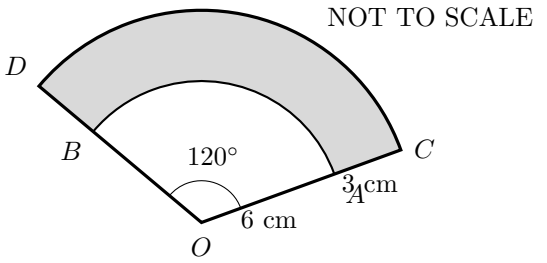
**(ii)** the perimeter of the minor sector $OAB$.

..... cm **[1]**

**(b)** Calculate the area of the major sector $OAB$, giving your answer in terms of $\pi$.

..... $cm^2$ **[3]**

**1182** The diagram shows two sectors $OAB$ and $OCD$ with common centre $O$ and sector angle 120°. $OA = OB = 6$ cm and $AC = BD = 3$ cm. The region $ABDC$ is shaded.




**(a)** Calculate

**(i)** the length of arc $AB$, giving your answer in terms of $\pi$,

..... cm **[2]**

**(ii)** the perimeter of the shaded region $ABDC$, giving your answer in the form $p + q\pi$, where $p$ and $q$ are integers.

..... cm **[3]**

**(b)** Calculate the area of the shaded region $ABDC$, giving your answer in terms of $\pi$.

..... $cm^2$ **[3]**

**(c)** A different sector $OPQ$ has radius $r$ cm and sector angle 150°.
The area of sector $OPQ$ is equal to the area of the shaded region $ABDC$.
Calculate the value of $r$.

$r =$ ..... **[2]**

**1183** **(a)** A sector of a circle, center $O$, has radius $r$ cm and sector angle $\theta°$.
The area of the sector is $10\pi$ cm$^2$ and the length of its arc is $2\pi$ cm.

Find the value of $r$ and the value of $\theta$.

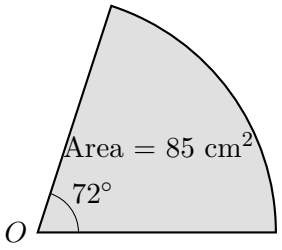
$r =$ ..... , $\theta =$ ..... **[5]**

**(b)** Another sector of a circle has radius $x$ cm and sector angle $(x + 20)°$.
The length of its arc is $\frac{16\pi}{5}$ cm.

Find the value of $x$.
You must show all your working.

$x =$ ..... **[4]**

**1184** The diagram shows a sector of a circle with centre $O$ and angle 72°.
The area of the sector is 85 $cm^2$.




NOT TO SCALE

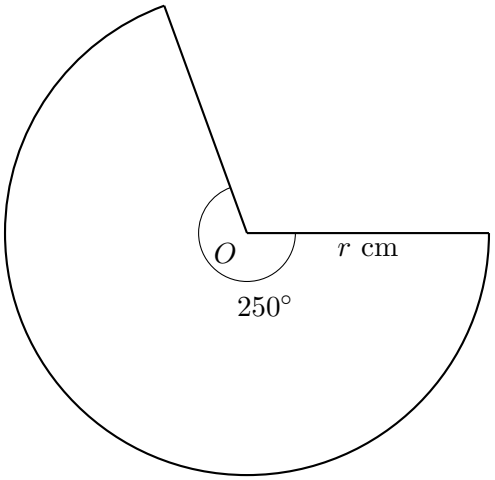
**(a)** Calculate the radius of the sector.

..... cm **[3]**

**(b)** Calculate the perimeter of the sector.

..... cm **[3]**

**1185** The diagram shows a major sector of a circle with centre $O$ and radius $r$ cm.
The reflex angle is 250°.
The length of the major arc is 35 cm.



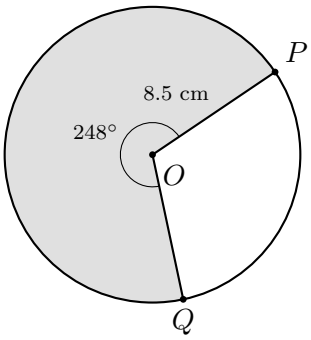

**(a)** Calculate the value of $r$.

$r =$ ..... **[3]**

**(b)** Calculate the area of the major sector.

..... $cm^2$ **[3]**

**1186** The diagram shows a circle, centre $O$, with radius 8.5 cm.
Points $P$ and $Q$ lie on the circumference such that the reflex angle $POQ = 248°$.



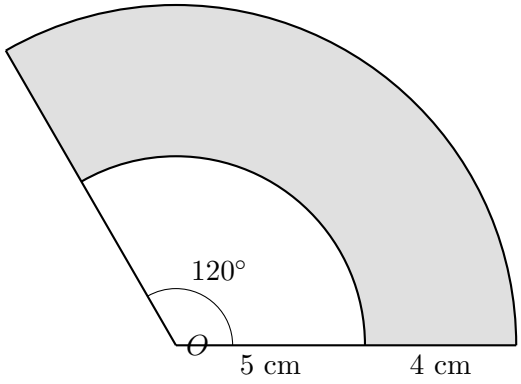

**(a)** Calculate the perimeter of the shaded major sector $OPQ$.

Perimeter = ..... $cm$ **[3]**

**(b)** Calculate the area of the unshaded minor segment bounded by the chord $PQ$ and the minor arc $PQ$.

Area = ..... $cm^2$ **[3]**

**1187** The diagram shows a shaded shape made from two concentric sectors of circles with centre $O$.
The inner radius is 5 cm and the outer radius is 9 cm.
The angle of both sectors is 120°.




**(a)** Calculate the area of the shaded shape.

Area = ..... $cm^2$ **[3]**

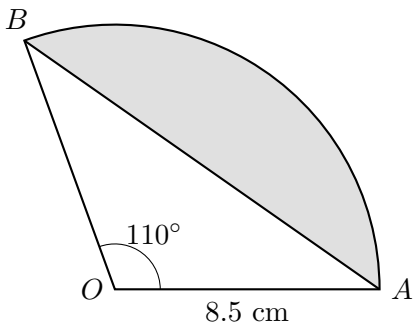
**(b)** Calculate the perimeter of the shaded shape.

Perimeter = ..... cm **[3]**

**(c)** A complete circle has the same area as the shaded shape.
Calculate the radius of this circle.

Radius = ..... cm **[2]**

**1188** The diagram shows a sector $OAB$ of a circle with center $O$ and radius 8.5 cm.
The sector angle is 110°.
A straight line connects $A$ and $B$, forming a triangle $OAB$ and a shaded segment.




**(a)** Calculate the area of the sector $OAB$.

..... $cm^2$ **[2]**

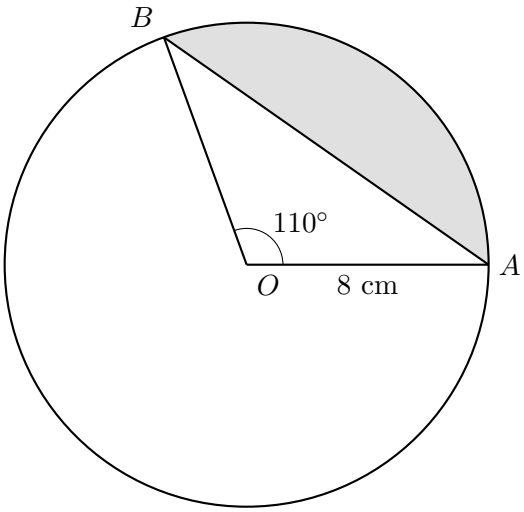
**(b)** Calculate the area of the shaded segment.

..... $cm^2$ **[3]**

**(c)** Calculate the perimeter of the shaded segment.

..... cm **[4]**

**1189** The diagram shows a circle with centre $O$ and radius 8 cm.
$A$ and $B$ are points on the circumference such that angle $AOB = 110°$.



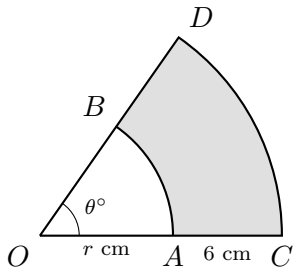

**(a)** Calculate the area of the shaded segment.

..... $cm^2$ **[4]**

**(b)** Calculate the perimeter of the shaded segment.

..... cm **[3]**

**1190** The diagram shows two concentric sectors $OAB$ and $OCD$ with centre $O$ and sector angle $\theta°$.
$OA = r$ cm and $AC = 6$ cm.
The area of the shaded region $ABDC$ is $54\pi$ cm$^2$.




**(a)** Show that $\theta = \dfrac{1620}{r+3}$.

**[4]**

**(b)** The total perimeter of the shaded region $ABDC$ is 54 cm.

**(i)** Calculate the value of $r$.

$r =$ ..... **[3]**

**(ii)** Find the value of $\theta$.

$\theta =$ ..... **[1]**

**1191** A sector of a circle has radius $r$ cm and sector angle $x°$.

**(a)** The arc length of the sector is $6\pi$ cm and the area of the sector is $27\pi$ cm$^2$.

**(i)** Work out the value of $r$.

$r =$ ..... **[3]**

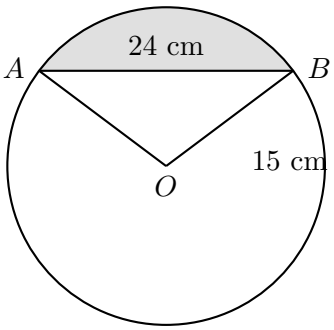
**(ii)** Work out the value of $x$.

$x =$ ..... **[2]**

**(b)** Another sector has radius $R$ cm and sector angle $y°$.
The perimeter of this sector is equal to the circumference of a circle with radius $R$ cm.
Find an expression for $y$ in terms of $\pi$.

$y =$ ..... **[4]**

**1192** The diagram shows a circle, centre $O$, with radius 15 cm.
$AB$ is a chord of the circle with length 24 cm.
The region bounded by the chord $AB$ and the minor arc $AB$ is shaded.




NOT TO SCALE

**(a)** Show that angle $AOB = 106.3°$, correct to 1 decimal place.

**[3]**

**(b)** For the following calculations, use angle $AOB = 106.3°$.

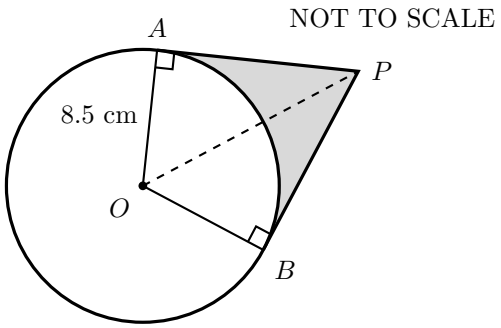
**(i)** Calculate the perimeter of the shaded minor segment.

..... cm **[3]**

**(ii)** Calculate the area of the shaded minor segment.

..... $cm^2$ **[4]**

**1193** The diagram shows a circle, centre $O$, of radius 8.5 cm. From an external point $P$, tangents $PA$ and $PB$ are drawn to touch the circle at $A$ and $B$ respectively, where $OP = 15.2$ cm. The region bounded by $PA$, $PB$, and the minor arc $AB$ is shaded.




**(a)** Show that obtuse angle $AOB = 112.0°$, correct to 1 decimal place.

**[3]**

**(b)** Calculate the length of the major arc $AB$, giving your answer correct to 3 significant figures.

..... cm **[3]**

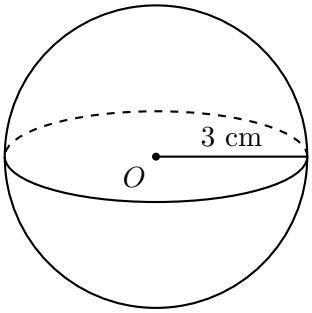
**(c)** Calculate the area of the shaded region, giving your answer correct to 3 significant figures.

..... $cm^2$ **[4]**

**(d)** A sector of a different circle has radius $R$ cm, arc length 18.4 cm, and area 115 $cm^2$. Calculate the value of $R$ and the sector angle $\theta$, giving $\theta$ correct to 1 decimal place.

$R =$ ..... , $\theta =$ ..... **[3]**

**1194** The diagram shows a solid sphere with center $O$ and radius 3 cm.



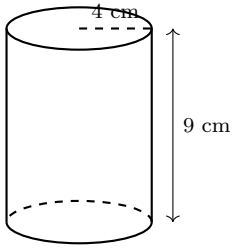

**(a)** Calculate the exact volume of the sphere.
Give your answer in terms of $\pi$.

..... $cm^3$ **[2]**

**(b)** Calculate the exact surface area of the sphere.
Give your answer in terms of $\pi$.

..... $cm^2$ **[2]**

**1195** The diagram shows a solid cylinder of base radius 4 cm and height 9 cm.



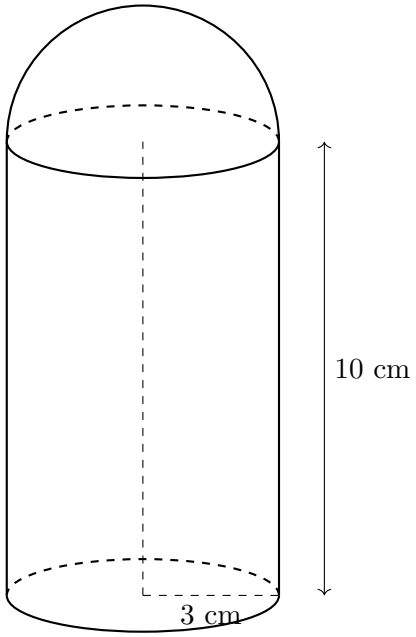

**(a)** Calculate the volume of the cylinder.
Give your answer in terms of $\pi$.

Volume = ..... $cm^3$ **[2]**

**(b)** Calculate the total surface area of the solid cylinder.
Give your answer in terms of $\pi$.

Surface area = ..... $cm^2$ **[3]**

**1196** A solid is made by joining a solid hemisphere of radius 3 cm to a solid cylinder of radius 3 cm and height 10 cm.




NOT TO SCALE

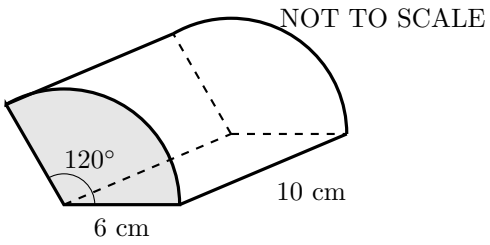
**(a)** Work out the volume of the solid.
Give your answer in terms of $\pi$.

..... $cm^3$ **[3]**

**(b)** Work out the total surface area of the solid.
Give your answer in terms of $\pi$.

..... $cm^2$ **[3]**

**1197** The diagram shows a solid prism of length 10 cm. The uniform cross-section of the prism is a sector of a circle of radius 6 cm and sector angle 120°.




**(a)** Giving your answers in terms of $\pi$, calculate

**(i)** the area of the sector cross-section,

..... $cm^2$ **[2]**

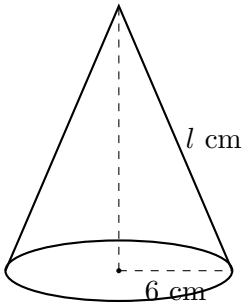
**(ii)** the volume of the prism.

..... $cm^3$ **[1]**

**(b)** Calculate the total surface area of the prism, giving your answer in the form $a + b\pi$, where $a$ and $b$ are integers.

..... $cm^2$ **[4]**

**1198** The diagram shows a solid cone.
The base radius of the cone is 6 cm and the slant height is $l$ cm.
The total surface area of the cone is $96\pi$ cm$^2$.



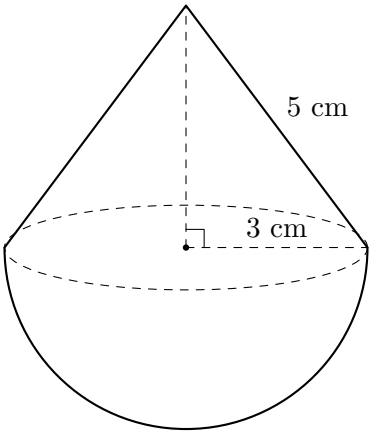

**(a)** Show that the slant height, $l$, is 10 cm.

**[3]**

**(b)** Work out the volume of the cone.
Give your answer in terms of $\pi$.

Volume = ..... $cm^3$ **[4]**

**1199** The diagram shows a solid formed by joining a cone to a hemisphere with the same radius.
The radius of the hemisphere is 3 cm and the slant height of the cone is 5 cm.



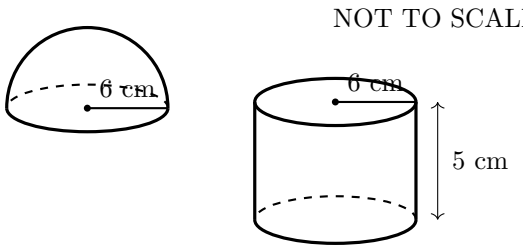

**(a)** Calculate the exact volume of the solid.
Give your answer in terms of $\pi$.

..... $cm^3$ **[4]**

**(b)** Calculate the total surface area of the solid.
Give your answer in terms of $\pi$.

..... $cm^2$ **[3]**

**1200** The diagram shows a solid metal hemisphere of radius 6 cm and a solid metal cylinder of base radius 6 cm and height 5 cm.




**(a)** Giving your answers in terms of $\pi$, calculate

**(i)** the volume of the solid hemisphere,

..... $cm^3$ **[2]**

**(ii)** the total volume of metal in the hemisphere and the cylinder.

..... $cm^3$ **[2]**

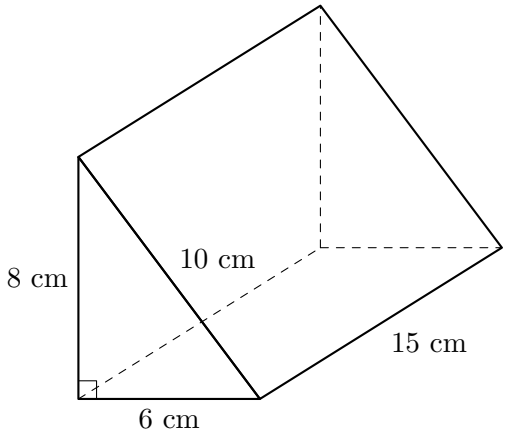
**(b)** The hemisphere and the cylinder are melted down and recast, with no loss of metal, into a single solid cone of base radius 9 cm. Show that the vertical height of the cone is 12 cm.

**[2]**

**(c)** Calculate the total surface area of the solid cone, giving your answer in terms of $\pi$.

..... $cm^2$ **[3]**

**1201** The diagram shows a solid triangular prism.
The cross-section of the prism is a right-angled triangle with sides 6 cm, 8 cm and 10 cm.
The length of the prism is 15 cm.



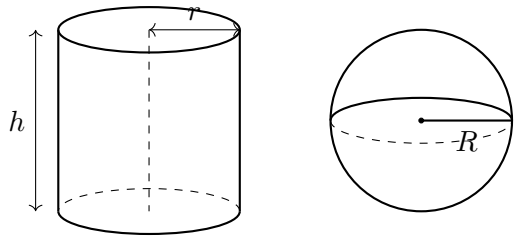

**(a)** Calculate the volume of the prism.

..... $cm^3$ **[2]**

**(b)** Calculate the total surface area of the prism.

..... $cm^2$ **[3]**

**1202** The diagram shows a solid cylinder and a solid sphere.



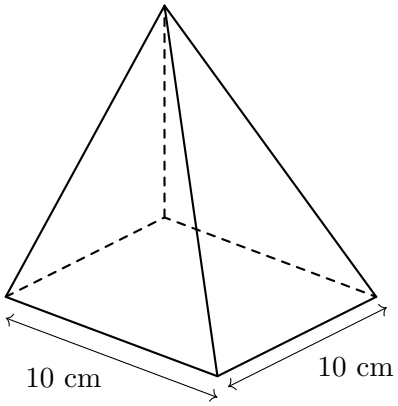

**(a)** The cylinder has radius $r = 5$ cm and height $h = 12$ cm.
Work out the total surface area of the cylinder.

Surface area = ..... $cm^2$ **[3]**

**(b)** The sphere has the same volume as the cylinder.
Calculate the radius, $R$, of the sphere.

$R =$ ..... cm **[3]**

**1203** The diagram shows a solid pyramid with a square base.
The base has sides of length 10 cm.
The volume of the pyramid is 400 $cm^3$.




NOT TO SCALE

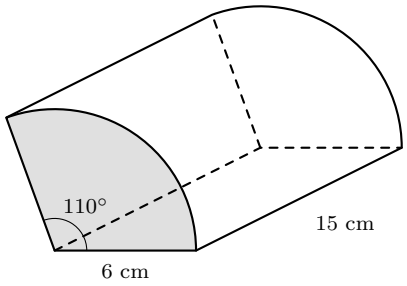
**(a)** Calculate the perpendicular height of the pyramid.

..... cm **[2]**

**(b)** Calculate the total surface area of the pyramid.

..... $cm^2$ **[4]**

**1204** The diagram shows a prism of length 15 cm.
The cross-section of the prism is a sector of a circle with radius 6 cm and sector angle 110°.



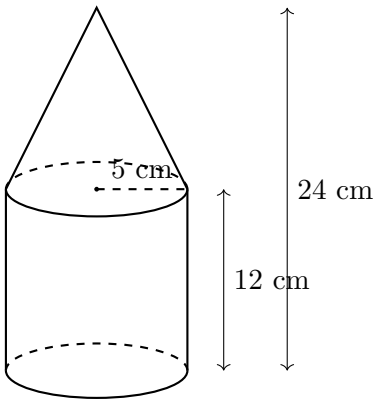

**(a)** Calculate the volume of the prism.

Volume = ..... $cm^3$ **[3]**

**(b)** Calculate the total surface area of the prism.

Surface area = ..... $cm^2$ **[3]**

**1205** A solid is formed by attaching a right circular cone to the top of a solid cylinder.
The cylinder and the base of the cone both have a radius of 5 cm.
The height of the cylinder is 12 cm and the total height of the solid is 24 cm.




**(a)** Calculate the volume of the solid.

..... $cm^3$ **[4]**

**(b)** Calculate the total surface area of the solid.

..... $cm^2$ **[5]**

**1206** A solid metal cylinder has radius $x$ cm and height $3x$ cm.
The cylinder is melted down and completely recast into solid spheres.
Each sphere has a radius of $\frac{x}{2}$ cm.

**(a)** Find an expression, in terms of $x$ and $\pi$, for the volume of the cylinder.
Give your answer in its simplest form.

..... $\text{cm}^3$ **[2]**

**(b)** Find an expression, in terms of $x$ and $\pi$, for the volume of one sphere.
Give your answer in its simplest form.

..... $\text{cm}^3$ **[2]**

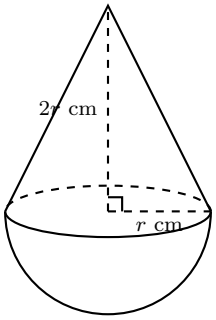
**(c)** Work out the number of complete spheres that can be made.

..... **[1]**

**(d)** The total surface area of the original cylinder is 140 $\text{cm}^2$.
Work out the value of $x$.

$x =$ ..... **[3]**

**1207** A solid toy is made by joining a solid right circular cone to the flat circular face of a solid hemisphere.
Both the cone and the hemisphere have radius $r$ cm.
The perpendicular height of the cone is $2r$ cm.



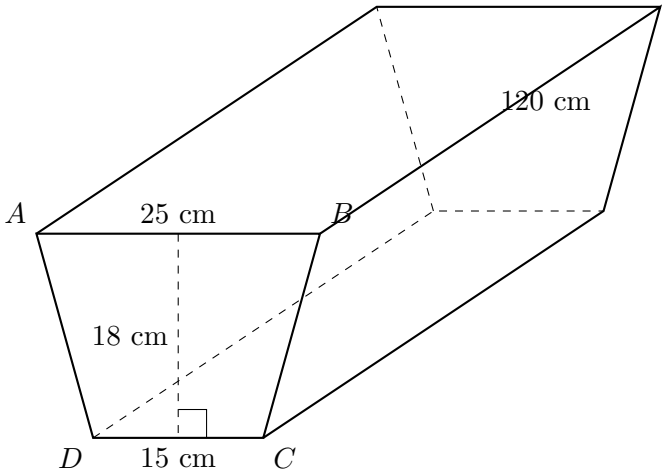

**(a)** The total volume of the solid toy is $450\pi$ cm$^3$.
Show that $r = 3\sqrt[3]{12.5}$.

**[4]**

**(b)** Calculate the total surface area of the solid toy.
Give your answer correct to 3 significant figures.

Surface area = ..... $cm^2$ **[4]**

**1208** The diagram shows a water trough in the shape of a prism of length 120 cm.
The cross-section of the prism is a trapezium $ABCD$.
$AB$ is parallel to $DC$.
$AB = 25$ cm, $DC = 15$ cm and the perpendicular distance between $AB$ and $DC$ is 18 cm.



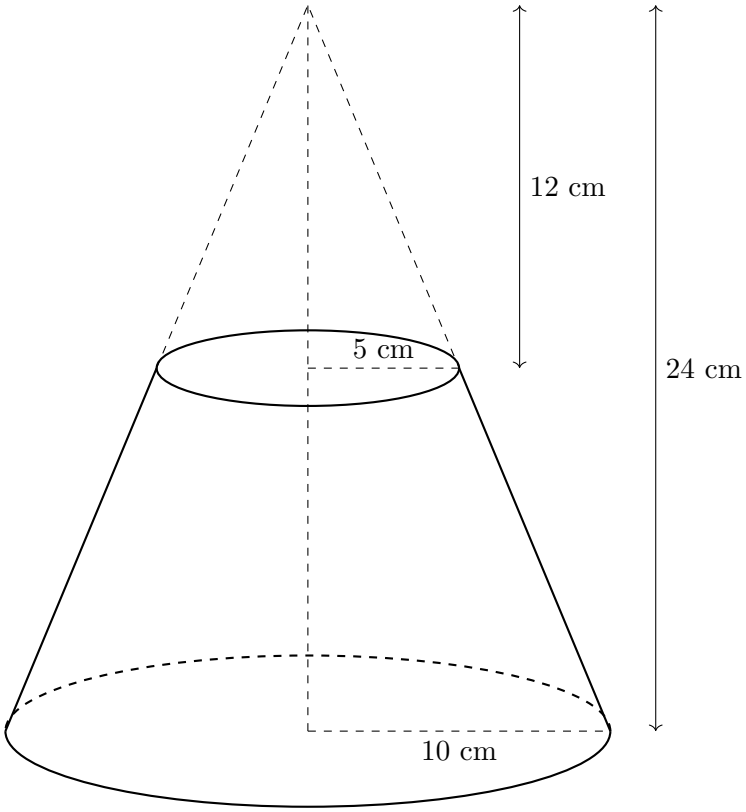

**(a)** Calculate the volume of the trough in litres.

Volume = ..... litres **[4]**

**(b)** The trough is initially completely empty.
Water flows into the trough at a constant rate of 0.8 litres per second.
Calculate the depth of the water in the trough after 30 seconds.

Depth = ..... cm **[5]**

**1209** A solid is made by removing a small cone from a large cone to form a frustum.
The large cone has a base radius of 10 cm and a perpendicular height of 24 cm.
The small cone has a base radius of 5 cm and a perpendicular height of 12 cm.




NOT TO SCALE

**(a)** Calculate the volume of the frustum.

..... $cm^3$ **[3]**

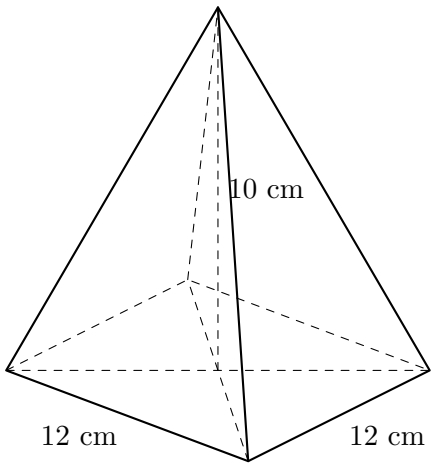
**(b)** Calculate the total surface area of the solid frustum.

..... $cm^2$ **[4]**

**(c)** The solid frustum is melted down to form solid spheres of radius 3 cm.
Calculate the maximum number of complete spheres that can be made.

..... **[3]**

**1210** A solid metal pyramid has a square base of side 12 cm and a perpendicular height of 10 cm.




**(a)** Calculate the volume of the pyramid.

..... $cm^3$ **[2]**

**(b)** Calculate the total surface area of the pyramid.

..... $cm^2$ **[4]**

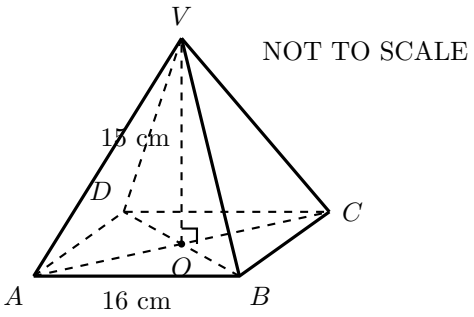
**(c)** The metal pyramid is melted down and recast to make solid spheres of radius 2.5 cm.
Calculate the number of complete spheres that can be made.

..... **[3]**

**(d)** A mathematically similar pyramid has a volume of 1620 $cm^3$.
Calculate the perpendicular height of this similar pyramid.

..... cm **[3]**

**1211** The diagram shows a solid right pyramid $VABCD$ with a horizontal square base $ABCD$ of side 16 cm.
$O$ is the centre of the base and the vertical height $VO = 15$ cm.




**(a)** Calculate the volume of pyramid $VABCD$.

..... $cm^3$ **[2]**

**(b)** Calculate the total surface area of pyramid $VABCD$.

..... $cm^2$ **[3]**

**(c)** The pyramid is cut by a horizontal plane parallel to the base $ABCD$ at a vertical height of 9 cm above the base.
The top small pyramid is removed, leaving a solid frustum.

**(i)** Calculate the volume of the frustum.

..... $cm^3$ **[3]**

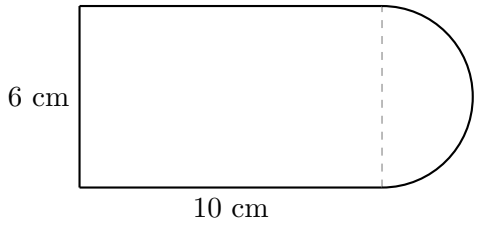
**(ii)** Calculate the total surface area of the frustum.

..... $cm^2$ **[3]**

**(d)** Calculate the angle that the slant edge $VA$ makes with the horizontal base $ABCD$, giving your answer correct to 1 decimal place.

..... $^\circ$ **[3]**

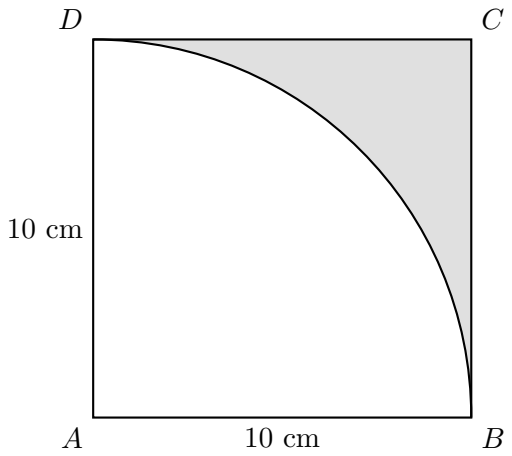
**1212** A shape is made from a rectangle and a semicircle.
The rectangle has a length of 10 cm and a width of 6 cm.
The semicircle is attached to one of the 6 cm sides.




Find the exact perimeter of the shape.
Give your answer in the form $a + b\pi$.

..... cm **[3]**

**1213** The diagram shows a square $ABCD$ with side length 10 cm.
A quarter circle with center $A$ and radius 10 cm is drawn inside the square.
The region inside the square but outside the quarter circle is shaded.



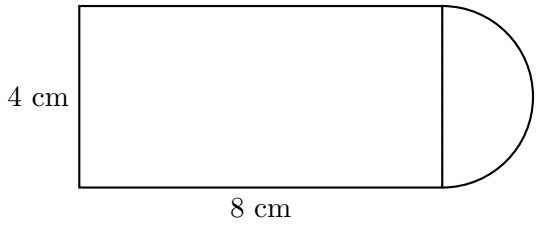

**(a)** Find the exact perimeter of the shaded region.
Give your answer in terms of $\pi$.

..... cm **[2]**

**(b)** Find the exact area of the shaded region.
Give your answer in terms of $\pi$.

..... $cm^2$ **[2]**

**1214** The diagram shows a compound shape formed by joining a semi-circle to a rectangle.
The rectangle has a length of 8 cm and a width of 4 cm.
The diameter of the semi-circle is 4 cm.



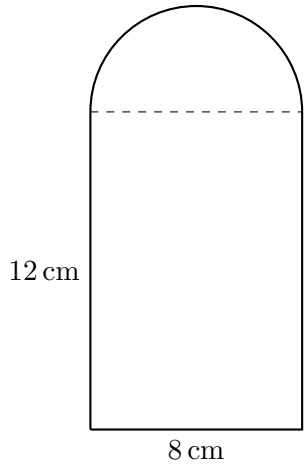

**(a)** Calculate the perimeter of the shape.
Give your answer in its simplest form, in terms of $\pi$.

..... *cm* **[2]**

**(b)** Calculate the area of the shape.
Give your answer in its simplest form, in terms of $\pi$.

..... $cm^2$ **[2]**

**1215** A compound shape is made from a rectangle and a semicircle.
The rectangle has width 8 cm and length 12 cm. The semicircle is attached to one of the sides of width 8 cm.



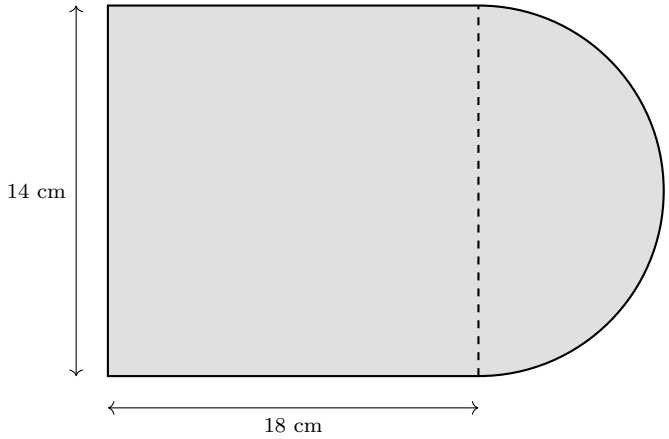

**(a)** Calculate the exact perimeter of the compound shape.
Give your answer in its simplest form, in terms of $\pi$.

..... cm **[2]**

**(b)** Calculate the exact area of the compound shape.
Give your answer in its simplest form, in terms of $\pi$.

..... $cm^2$ **[2]**

**1216** The diagram shows a compound shape made by attaching a semicircle of diameter 14 cm to one end of a rectangle of length 18 cm and width 14 cm.



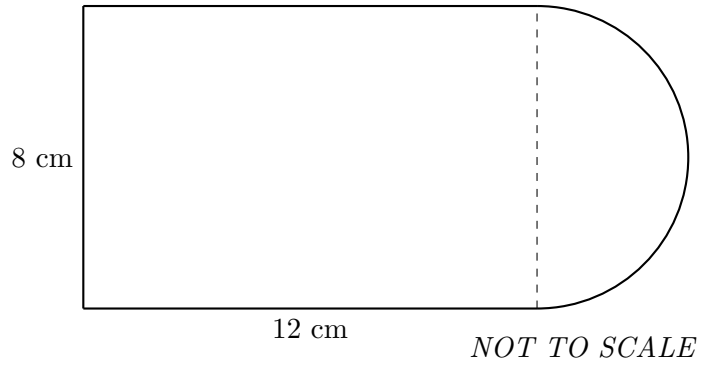

**(a)** Calculate the perimeter of the compound shape.
Give your answer in terms of $\pi$.

Perimeter = ..... $cm$ **[2]**

**(b)** Calculate the total area of the compound shape.
Give your answer in terms of $\pi$.

Area = ..... $cm^2$ **[3]**

**1217** The diagram shows a shape made from a rectangle and a semicircle.
The rectangle has length 12 cm and width 8 cm.
The semicircle has a diameter of 8 cm.



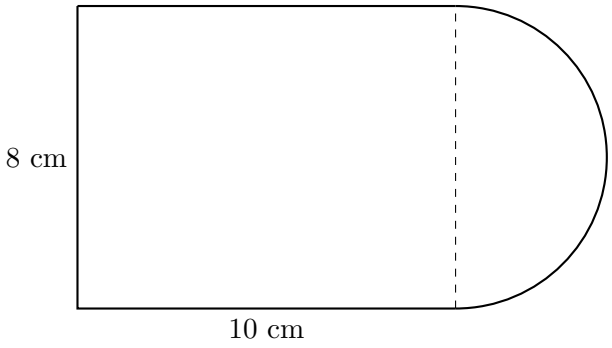

**(a)** Find the perimeter of the shape.
Give your answer in its simplest form in terms of $\pi$.

Perimeter = ..... cm **[2]**

**(b)** Find the area of the shape.
Give your answer in its simplest form in terms of $\pi$.

Area = ..... $cm^2$ **[3]**

**1218** The diagram shows a compound shape made from a rectangle and a semicircle.
The rectangle has a length of 10 cm and a width of 8 cm.
The semicircle is attached to one of the 8 cm sides.



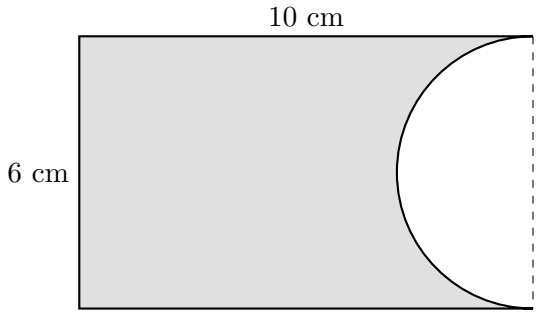

**(a)** Work out the exact area of the shape.
Give your answer in the form $a + b\pi$, where $a$ and $b$ are integers.

Area = ..... $cm^2$ **[3]**

**(b)** Work out the exact perimeter of the shape.
Give your answer in the form $c + d\pi$, where $c$ and $d$ are integers.

Perimeter = ..... cm **[3]**

**1219** The diagram shows a compound shape made from a rectangle and a semicircle.
The rectangle has a length of 10 cm and a width of 6 cm.
A semicircle of diameter 6 cm has been removed from one end of the rectangle.




NOT TO SCALE

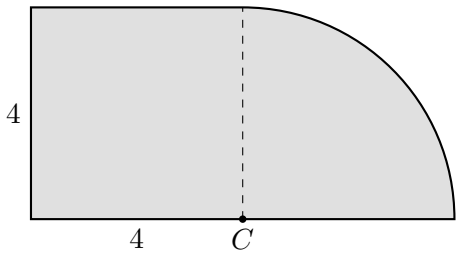
**(a)** Work out the perimeter of the shape.
Give your answer in the form $a + b\pi$, where $a$ and $b$ are integers.

..... cm **[3]**

**(b)** Work out the area of the shape.
Give your answer in the form $c - d\pi$, where $c$ is an integer and $d$ is a decimal.

..... $cm^2$ **[3]**

**1220** The diagram shows a shape formed by a square of side 4 cm and a quarter of a circle with center $C$.



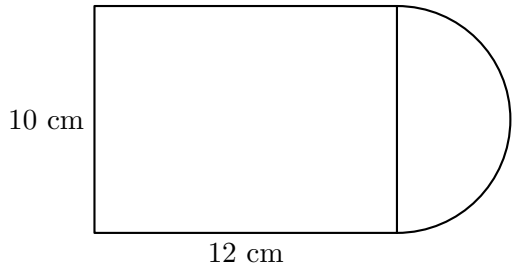

**(a)** Find the area of the shape.
Give your answer in terms of $\pi$.

Area = ..... $cm^2$ **[3]**

**(b)** Find the perimeter of the shape.
Give your answer in terms of $\pi$.

Perimeter = ..... $cm$ **[3]**

**1221** The diagram shows a compound shape made from a rectangle and a semicircle.
The rectangle has length 12 cm and width 10 cm.
The semicircle has a diameter of 10 cm.



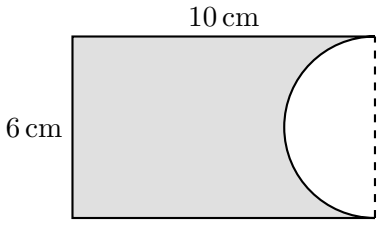

**(a)** Calculate the perimeter of the shape.
Give your answer in the form $a + b\pi$.

..... cm **[3]**

**(b)** Calculate the area of the shape.
Give your answer in the form $c + d\pi$.

..... $cm^2$ **[3]**

**1222** The diagram shows a shape made from a rectangle with dimensions 10 cm by 6 cm. A semicircle of diameter 6 cm has been removed from one end.




NOT TO SCALE

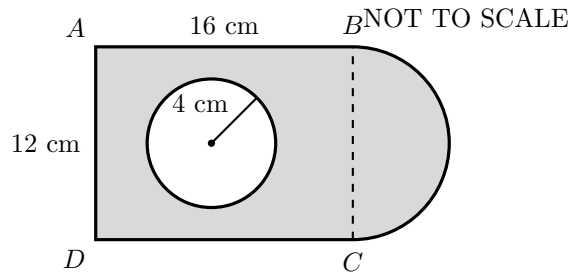
**(a)** Calculate the area of the shape.
Give your answer in terms of $\pi$.

..... $cm^2$ **[3]**

**(b)** Calculate the perimeter of the shape.
Give your answer in terms of $\pi$.

..... cm **[3]**

**1223** The diagram shows a shaded metal plate formed from a rectangle $ABCD$ with a semicircle of diameter $BC$ attached to side $BC$.
Inside the rectangle, a circular hole of radius 4 cm is cut out.
$AB = 16$ cm and $BC = 12$ cm.




**(a)** Giving your answers in terms of $\pi$, calculate

**(i)** the outer perimeter of the metal plate,

..... cm **[2]**

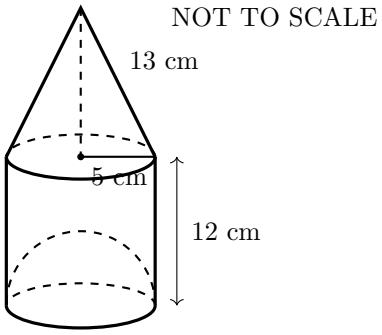
**(ii)** the area of the shaded region.

..... $cm^2$ **[3]**

**(b)** The shaded metal plate is the uniform cross-section of a solid prism of thickness 5 cm. Calculate the volume of the prism, giving your answer in the form $p + q\pi$, where $p$ and $q$ are integers.

..... $cm^3$ **[2]**

**1224** The diagram shows a compound solid formed from a right circular cylinder of base radius 5 cm and height 12 cm. A right circular cone of base radius 5 cm and slant height 13 cm is joined to the top face of the cylinder, and a hemisphere of radius 5 cm is hollowed out of the bottom face of the cylinder.




**(a)** Show that the vertical height of the conical part is 12 cm.

**[2]**

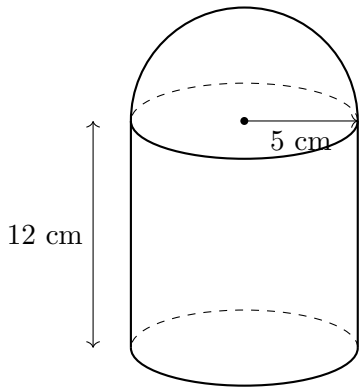
**(b)** Calculate the total surface area of the compound solid, giving your answer in terms of $\pi$.

..... $cm^2$ **[4]**

**(c)** Calculate the volume of the compound solid, giving your answer as a single fraction in terms of $\pi$.

..... $cm^3$ **[4]**

**1225** A solid is made by attaching a hemisphere of radius 5 cm to one end of a solid cylinder.
The cylinder has a radius of 5 cm and a height of 12 cm.



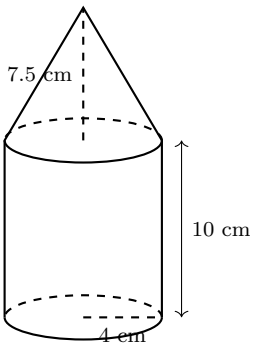

**(a)** Calculate the total volume of the solid.
Give your answer correct to 3 significant figures.

..... $cm^3$ **[3]**

**(b)** Calculate the total surface area of the solid.
Give your answer correct to 3 significant figures.

..... $cm^2$ **[3]**

**1226** A solid metal trophy consists of a right circular cone placed on top of a cylinder, both of base radius 4 cm.
The cylinder has height 10 cm and the cone has perpendicular height 7.5 cm.



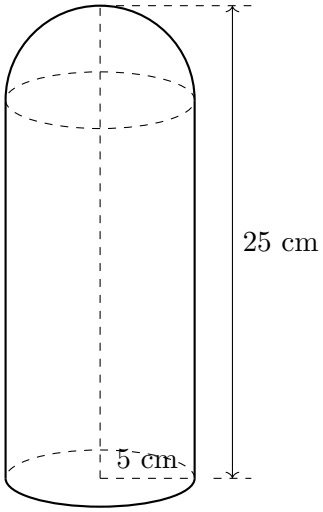

**(a)** Calculate the total volume of the solid trophy.

Volume = ..... $cm^3$ **[3]**

**(b)** Calculate the total surface area of the solid trophy.

Surface area = ..... $cm^2$ **[3]**

**1227** The diagram shows a solid in the shape of a cylinder with a hemisphere attached to the top.
The radius of both the cylinder and the hemisphere is 5 cm.
The total height of the solid is 25 cm.




NOT TO SCALE

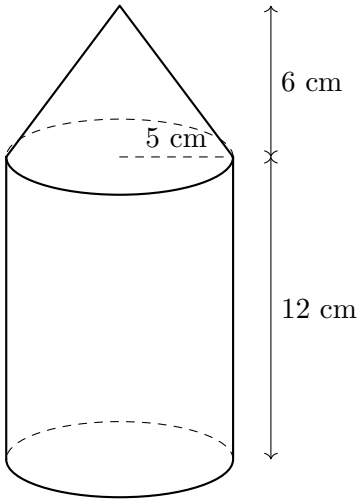
**(a)** Calculate the total volume of the solid.

..... $cm^3$ **[3]**

**(b)** Calculate the total surface area of the solid.

..... $cm^2$ **[4]**

**1228** The diagram shows a solid compound shape made from a cylinder and a cone.
The cylinder and the cone both have a base radius of 5 cm.
The height of the cylinder is 12 cm and the height of the cone is 6 cm.



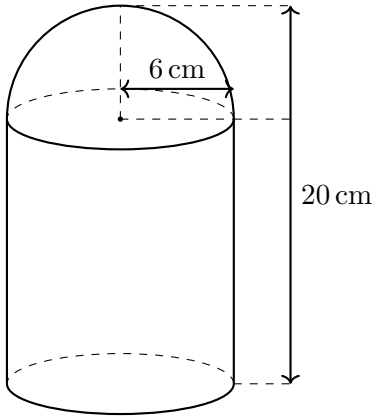

**(a)** Calculate the volume of the solid.

..... $cm^3$ **[3]**

**(b)** Calculate the total surface area of the solid.

..... $cm^2$ **[4]**

**1229** A solid is made by placing a solid hemisphere on top of a solid cylinder.
The cylinder and the hemisphere both have radius 6 cm.
The total height of the solid is 20 cm.



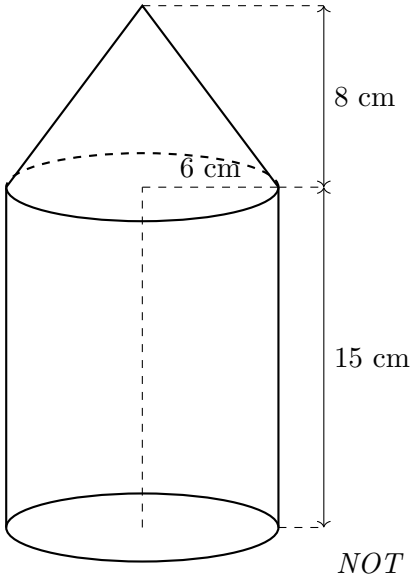

**(a)** Calculate the volume of the solid.

..... $cm^3$ **[3]**

**(b)** Calculate the total surface area of the solid.

..... $cm^2$ **[4]**

**1230** A solid is made from a cylinder and a cone.
The cylinder has radius 6 cm and height 15 cm.
The cone has radius 6 cm and height 8 cm.




*NOT TO SCALE*

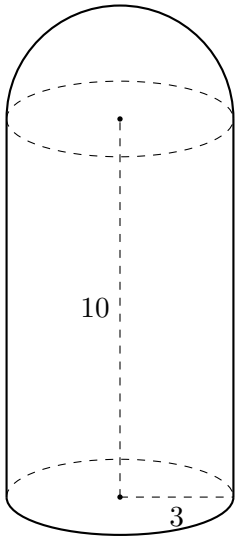
**(a)** Calculate the total volume of the solid.

Volume = ..... $cm^3$ **[3]**

**(b)** Calculate the total surface area of the solid.

Surface area = ..... $cm^2$ **[4]**

**1231** A solid is made by attaching a hemisphere of radius 3 cm to one end of a solid cylinder of radius 3 cm and height 10 cm.



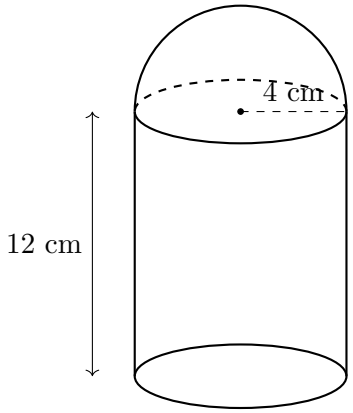

**(a)** Calculate the volume of the solid.

Volume = ..... $cm^3$ **[3]**

**(b)** Calculate the total surface area of the solid.

Total surface area = ..... $cm^2$ **[4]**

**1232** A solid object is formed by joining a hemisphere to one end of a solid cylinder.
The cylinder has a radius of 4 cm and a length of 12 cm.
The hemisphere has the same radius as the cylinder.



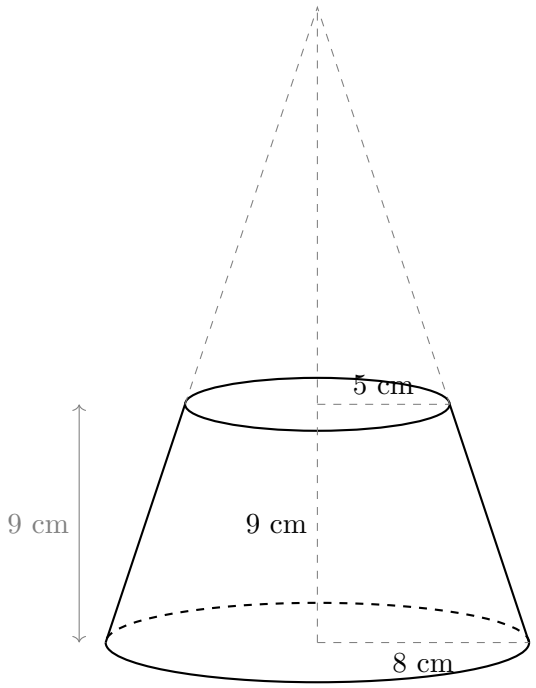

**(a)** Calculate the volume of the solid.

Volume = ..... $cm^3$ **[3]**

**(b)** Calculate the total surface area of the solid.

Surface area = ..... $cm^2$ **[4]**

**1233** The diagram shows a solid frustum.
The frustum is made by removing a smaller cone from a larger cone.
The larger cone has a base radius of 8 cm and a vertical height of 24 cm.
The smaller cone has a base radius of 5 cm and a vertical height of 15 cm.



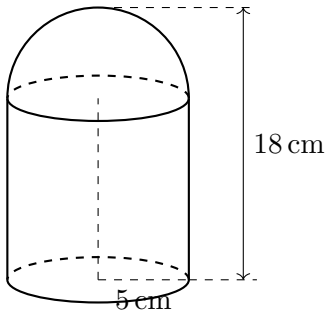

**(a)** Calculate the volume of the solid frustum.

..... $cm^3$ **[3]**

**(b)** Calculate the total surface area of the solid frustum.

..... $cm^2$ **[5]**

**1234** A solid is formed by placing a hemisphere of radius 5 cm on top of a cylinder of radius 5 cm.
The total height of the solid is 18 cm.




NOT TO SCALE

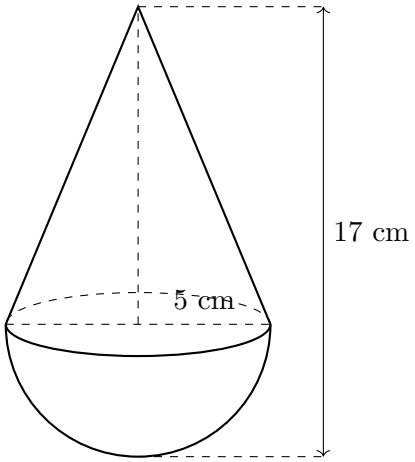
**(a)** Calculate the volume of the solid.
Give your answer correct to 3 significant figures.

..... $cm^3$ **[4]**

**(b)** Calculate the total surface area of the solid.
Give your answer correct to 3 significant figures.

..... $cm^2$ **[4]**

**1235** The diagram shows a solid made by joining a solid hemisphere to a solid cone.
The common radius of the hemisphere and the base of the cone is 5 cm.
The total height of the solid is 17 cm.




**(a)** Calculate the volume of the solid.

Volume = ..... $cm^3$ **[3]**

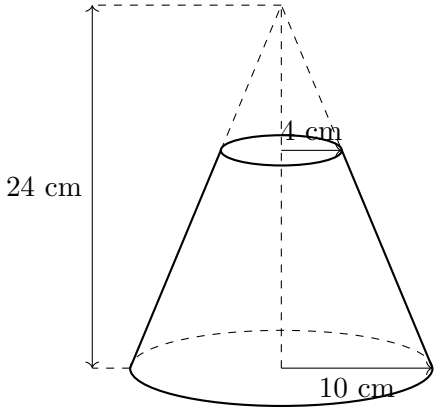
**(b)** Calculate the total surface area of the solid.

Surface area = ..... $cm^2$ **[3]**

**(c)** The solid is melted down and completely recast into a single solid cylinder of radius 4 cm.
Calculate the height of the cylinder.

Height = ..... cm **[3]**

**1236** A solid cone of height 24 cm and base radius 10 cm is cut parallel to its base to form a frustum and a smaller cone.
The radius of the top circular face of the frustum is 4 cm.




**(a)** Show that the height of the removed small cone is 9.6 cm.

**[2]**

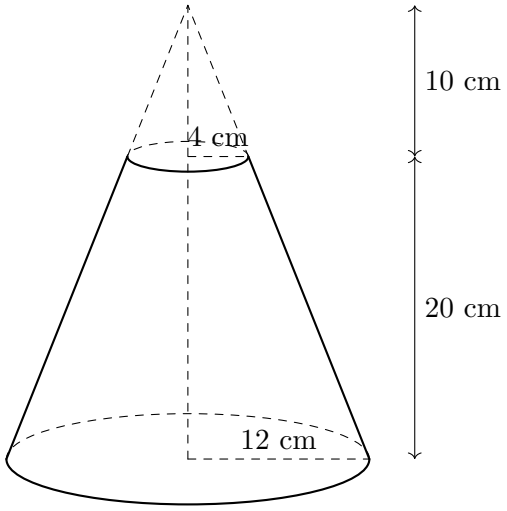
**(b)** Calculate the volume of the frustum.

..... $cm^3$ **[3]**

**(c)** The solid frustum is melted down and recast entirely into a solid sphere.
Calculate the radius of this sphere.

..... $cm$ **[3]**

**1237** The diagram shows a solid metal frustum.
The frustum is formed by removing a small cone of height 10 cm from a larger cone.
The radius of the circular base of the frustum is 12 cm and the radius of the circular top is 4 cm.
The height of the frustum is 20 cm.



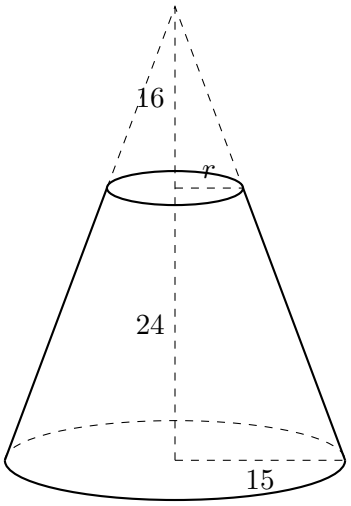

**(a)** Calculate the volume of the solid frustum.

..... $cm^3$ **[4]**

**(b)** Calculate the total surface area of the solid frustum.

..... $cm^2$ **[4]**

**1238** A container is in the shape of a frustum of a cone.
The original cone has a base radius of 15 cm and a vertical height of 40 cm.
A smaller cone of vertical height 16 cm is removed from the top to form the frustum.




**(a)** Show that the radius $r$ of the top circular face of the frustum is 6 cm.

**[2]**

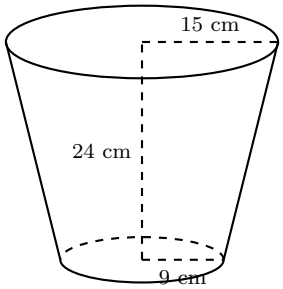
**(b)** Calculate the volume of the frustum.

Volume = ..... $cm^3$ **[3]**

**(c)** The solid frustum is melted down and recast entirely into a solid sphere.
Calculate the radius of the sphere.

Radius = ..... $cm$ **[3]**

**1239** The diagram shows an open bucket in the shape of a frustum of a right circular cone.
The circular base has radius 9 cm and the open circular top has radius 15 cm.
The perpendicular height of the bucket is 24 cm.




**(a)** Show that the perpendicular height of the original cone before the bottom was removed is 60 cm.

**[2]**

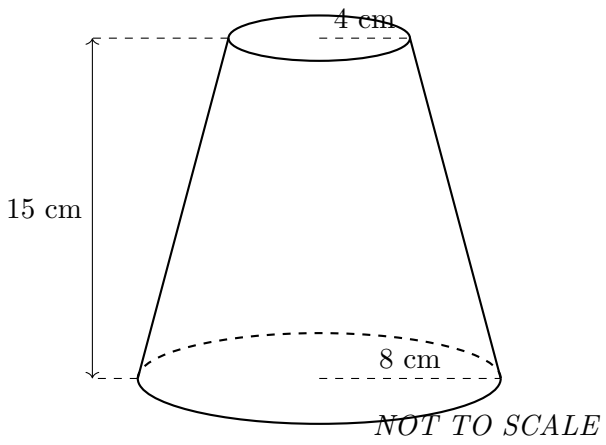
**(b)** Calculate the capacity of the bucket in litres.

Capacity = ..... litres **[3]**

**(c)** Calculate the total external surface area of the open bucket.

Surface area = ..... $cm^2$ **[3]**

**1240** A solid frustum of a cone has top radius 4 cm, bottom radius 8 cm and vertical height 15 cm.
It is formed by removing a smaller cone from the top of a larger cone.




**(a)** Show that the original height of the cone before the top was removed was 30 cm.

**[2]**

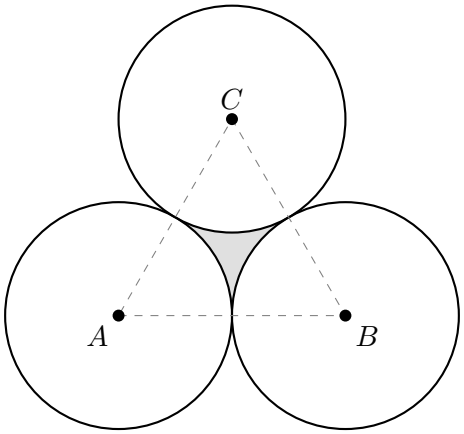
**(b)** Calculate the volume of the frustum.

Volume = ..... $cm^3$ **[3]**

**(c)** Calculate the total surface area of the solid frustum.

Surface area = ..... $cm^2$ **[4]**

**1241** The diagram shows three identical circles, each of radius $r$ cm.
The circles touch each other externally.
The region enclosed between the three circles is shaded.




**(a)** Show that the area of the shaded region, $A$, is given by the formula

$$A = r^2\left(\sqrt{3} - \frac{\pi}{2}\right)$$

**[4]**

**(b)** The area of the shaded region is 15 cm$^2$.

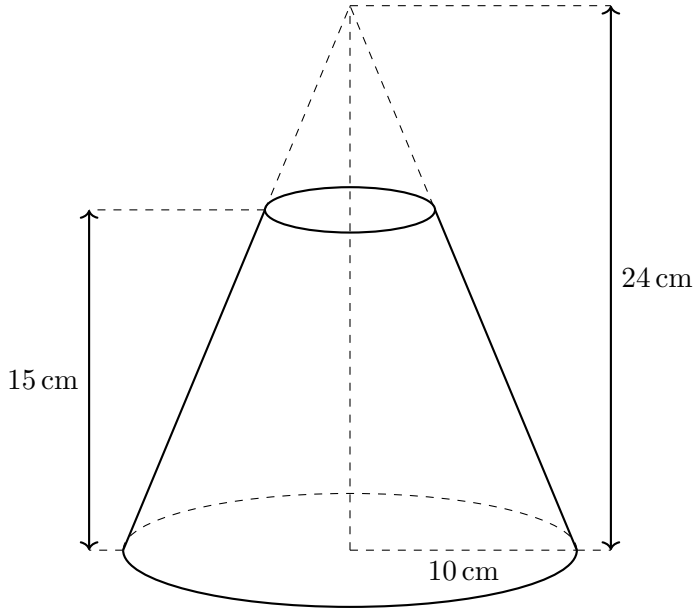
Work out the value of $r$.

$r =$ ..... **[3]**

**(c)** Calculate the total perimeter of the shaded region.

..... cm **[3]**

**1242** A frustum is made by removing a smaller cone from the top of a larger cone.
The larger cone has base radius 10 cm and vertical height 24 cm.
The frustum has vertical height 15 cm.




**(a)** Find the radius of the top of the frustum.

..... cm **[2]**

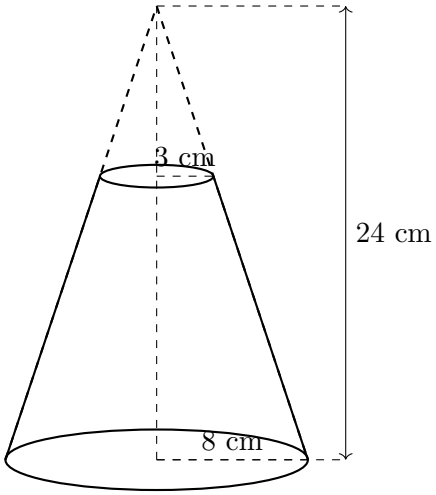
**(b)** Calculate the volume of the frustum.

..... $cm^3$ **[3]**

**(c)** Calculate the total surface area of the frustum.

..... $cm^2$ **[5]**

**1243** A frustum is made by removing a small cone from the top of a large cone.
The large cone has a base radius of 8 cm and a total height of 24 cm.
The removed small cone has a base radius of 3 cm.




**(a)** Show that the height of the small cone removed is 9 cm.

**[2]**

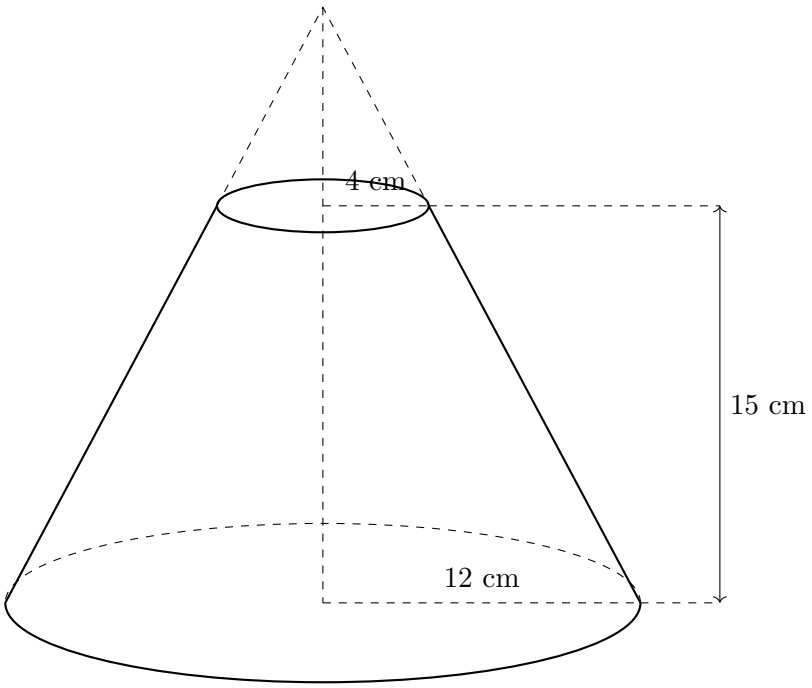
**(b)** Calculate the volume of the frustum.

Volume = ..... $cm^3$ **[3]**

**(c)** Calculate the total surface area of the frustum.

Surface area = ..... $cm^2$ **[5]**

**1244** The diagram shows a solid frustum of a cone.
The frustum is made by removing a small cone from the top of a larger solid cone.
The base radius of the frustum is 12 cm and the top radius is 4 cm.
The height of the frustum is 15 cm.




**(a)** Calculate the height of the small cone that was removed.

Height = ..... cm **[3]**

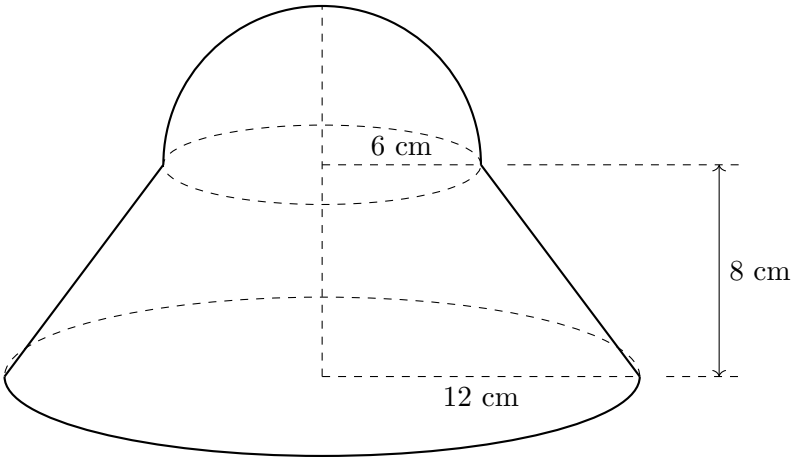
**(b)** Calculate the volume of the solid frustum.

Volume = ..... $cm^3$ **[3]**

**(c)** Calculate the total surface area of the solid frustum.

Surface area = ..... $cm^2$ **[5]**

**1245** The diagram shows a compound solid made from a frustum of a cone and a solid hemisphere.
The frustum is created by removing a small cone from a large cone.
The base radius of the frustum is 12 cm and its top radius is 6 cm.
The height of the frustum is 8 cm.
A solid hemisphere of radius 6 cm is attached to the top face of the frustum.




NOT TO SCALE

**(a)** Show that the perpendicular height of the large cone from which the frustum is made is 16 cm.

**[3]**

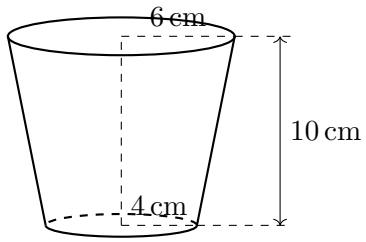
**(b)** Calculate the total volume of the compound solid.

..... $cm^3$ **[4]**

**(c)** Calculate the total surface area of the compound solid.

..... $cm^2$ **[4]**

**1246** The diagram shows a closed container in the shape of a frustum of a cone.
The radius of the top circular face is 6 cm and the radius of the base is 4 cm.
The vertical height of the container is 10 cm.




NOT TO SCALE

**(a)** The frustum is made by removing a smaller cone from a larger cone.
Calculate the vertical height of the larger cone.

..... cm **[3]**

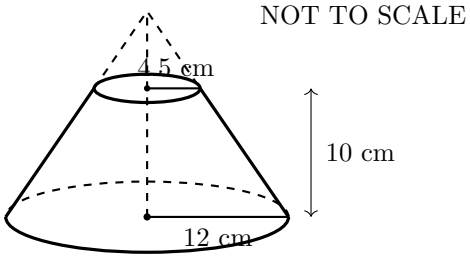
**(b)** Calculate the volume of the container.
Give your answer correct to 3 significant figures.

..... $cm^3$ **[3]**

**(c)** Calculate the total surface area of the closed container.
Give your answer correct to 3 significant figures.

..... $cm^2$ **[5]**

**1247** The diagram shows a solid frustum formed by removing a small right circular cone of base radius 4.5 cm from the top of a larger right circular cone of base radius 12 cm. The vertical height of the frustum is 10 cm.




**(a)** Use similar triangles to show that the vertical height of the removed cone is 6 cm.

**[2]**

**(b)** Calculate the total surface area of the solid frustum, giving your answer correct to 3 significant figures.

..... $cm^2$ **[4]**

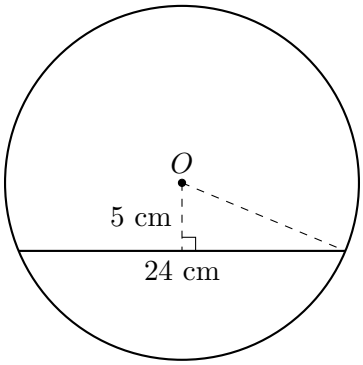
**(c)** Calculate the volume of the solid frustum, giving your answer correct to 3 significant figures.

..... $cm^3$ **[3]**

**(d)** The solid frustum is melted down and recast into $n$ identical solid spheres, each of radius 2.4 cm. Calculate the greatest possible integer value of $n$.

$n =$ ..... **[3]**

**1248** A chord of a circle, centre $O$, has length 24 cm. The perpendicular distance from the centre of the circle to the chord is 5 cm.



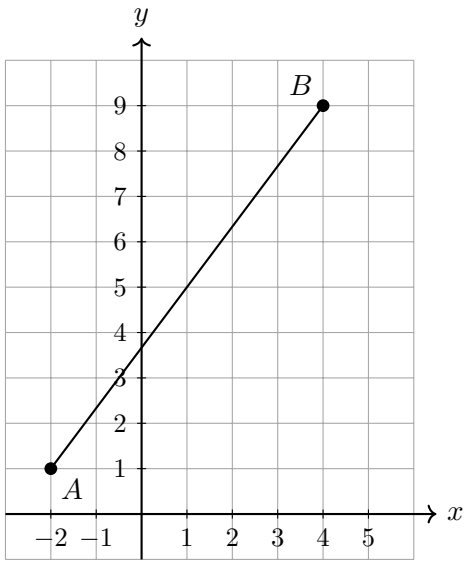

Calculate the radius of the circle.

..... cm **[3]**

**1249** The points $A(-4, 6)$ and $B(8, 1)$ lie on a coordinate grid.
Work out the exact length of the line segment $AB$.

..... **[3]**

**1250** Points $A(-2, 1)$ and $B(4, 9)$ are plotted on a coordinate grid.




Calculate the exact length of the line segment $AB$.

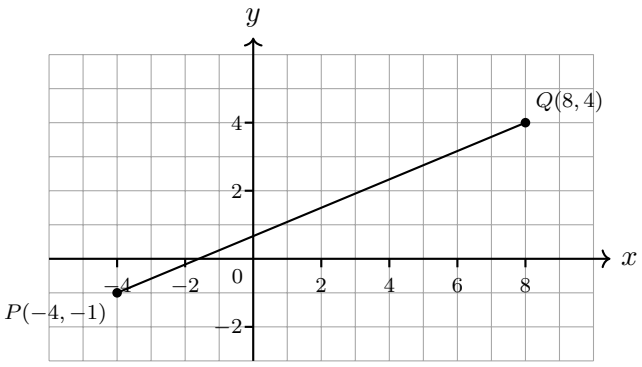
$AB =$ ..... units **[3]**

**1251** The point $A$ has coordinates $(1, 4)$ and the point $B$ has coordinates $(k, 19)$.
The exact distance between point $A$ and point $B$ is 17.

Find the two possible values of $k$.

$k =$ ..... , $k =$ ..... **[4]**

**1252** The diagram shows a coordinate grid with points $P(-4, -1)$ and $Q(8, 4)$ plotted.




**(a)** Calculate the distance $PQ$.

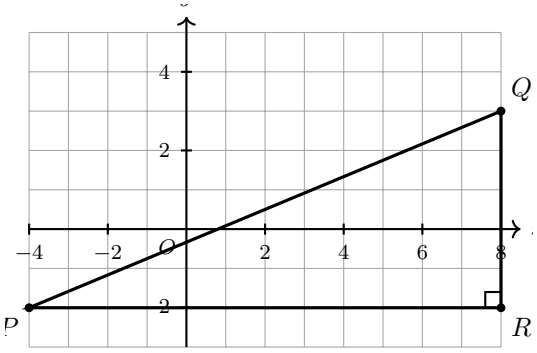
$PQ =$ ..... units **[3]**

**(b)** Point $R$ has coordinates $(8, -6)$.

Determine whether triangle $PQR$ is right-angled at $P$.
Show working to support your answer.

..... **[2]**

**1253** The diagram shows a coordinate grid with points $P(-4, -2)$, $Q(8, 3)$, and $R(8, -2)$ plotted.




**(a)** Calculate the length of $PQ$.

$PQ =$ ..... units **[3]**

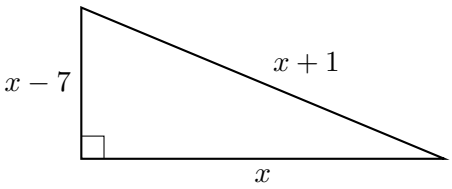
**(b)** Point $S$ has coordinates $(2, 4)$.
Calculate the exact length of $PS$, giving your answer in the form $k\sqrt{2}$, where $k$ is an integer.

$PS =$ ..... units **[3]**

**(c)** Point $T$ has coordinates $(-3, 3)$.
Show that triangle $PST$ is isosceles.

**[3]**

**1254** The diagram shows a right-angled triangle.
All lengths are in centimetres.



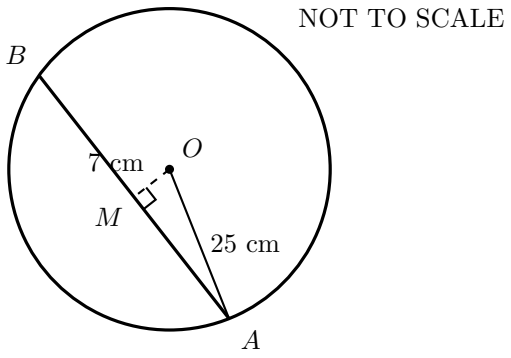

**(a)** Show that $x^2 - 16x + 48 = 0$.

**[3]**

**(b)** Solve the equation to find the value of $x$, and hence calculate the perimeter of the triangle.

Perimeter = ..... cm **[4]**

**1255** The diagram shows a circle, centre $O$, of radius 25 cm.
$AB$ is a chord at a perpendicular distance of 7 cm from $O$, where $M$ is the midpoint of $AB$.




**(a)** Calculate the length of chord $AB$.

$AB =$ ..... cm **[3]**

**(b)** A second chord $CD$ is parallel to $AB$ and lies on the same side of the centre $O$ as $AB$.
The length of chord $CD$ is 30 cm, and $N$ is the midpoint of $CD$.

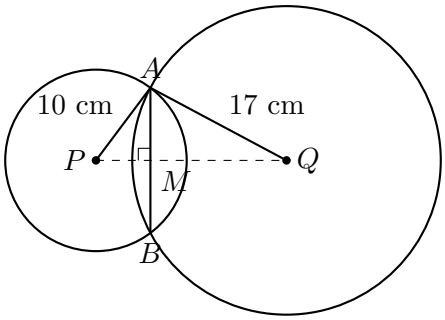
**(i)** Calculate the perpendicular distance $ON$ from the centre $O$ to the chord $CD$.

$ON =$ ..... cm **[3]**

**(ii)** Calculate the area of trapezium $ABDC$.

Area = ..... $cm^2$ **[3]**

**1256** The diagram shows two intersecting circles.
The circle with centre $P$ has radius 10 cm.
The circle with centre $Q$ has radius 17 cm.
The points $A$ and $B$ are the intersections of the two circles.
The length of the common chord $AB$ is 16 cm.
$M$ is the point of intersection of the chord $AB$ and the line $PQ$.




NOT TO SCALE

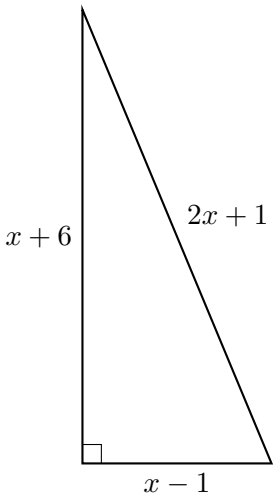
**(a)** Show that the distance $PM$ is 6 cm.

**[2]**

**(b)** Calculate the distance $PQ$.

..... cm **[3]**

**1257** The diagram shows a right-angled triangle.
The lengths of the sides are $x - 1$, $x + 6$, and $2x + 1$, where $x$ is an integer.
All lengths are in centimetres.




**(a)** Show that $x^2 - 3x - 18 = 0$.

**[4]**

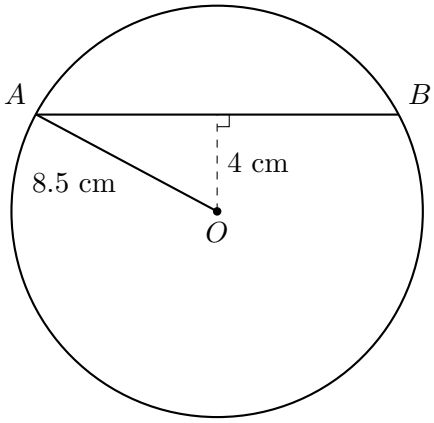
**(b)** Solve the equation by factorising to find the value of $x$.
You must show your working.

$x =$ ..... **[3]**

**(c)** Work out the area of the triangle.

..... $cm^2$ **[3]**

**1258** The diagram shows a circle, centre $O$, with radius 8.5 cm.
$AB$ is a chord of the circle.
The perpendicular distance from $O$ to the chord $AB$ is 4 cm.

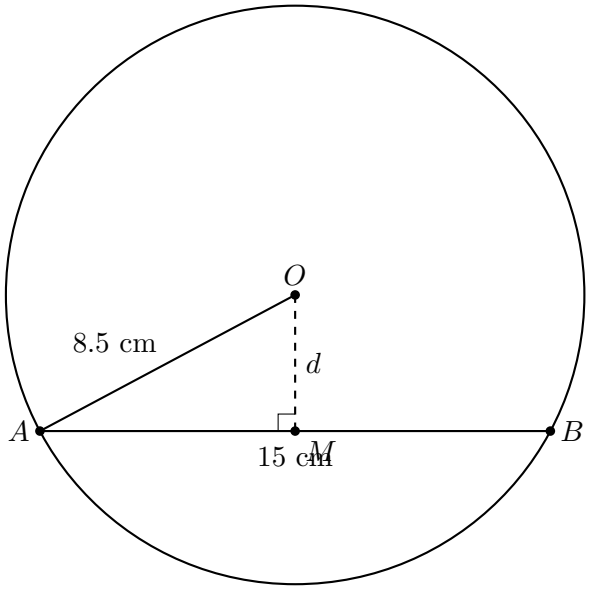



NOT TO SCALE

Calculate the length of the chord $AB$.

..... cm **[3]**

**1259** The diagram shows a circle, centre $O$, with radius 8.5 cm.
$AB$ is a chord of length 15 cm.
$M$ is the midpoint of $AB$.




**(a)** Calculate the distance $OM$, labeled $d$ in the diagram.

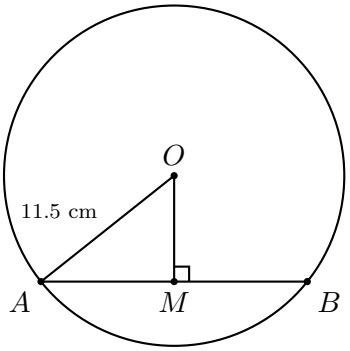
$d =$ ..... cm **[3]**

**(b)** A second chord, $CD$, is drawn in the same circle such that its perpendicular distance from the centre $O$ is 7.5 cm.

Calculate the length of the chord $CD$.

Length = ..... cm **[3]**

**1260** The diagram shows a circle, centre $O$, with radius 11.5 cm.
$AB$ is a chord of length 18 cm and $M$ is the midpoint of $AB$.




**(a)** Calculate the perpendicular distance $OM$ from the centre $O$ to the chord $AB$.

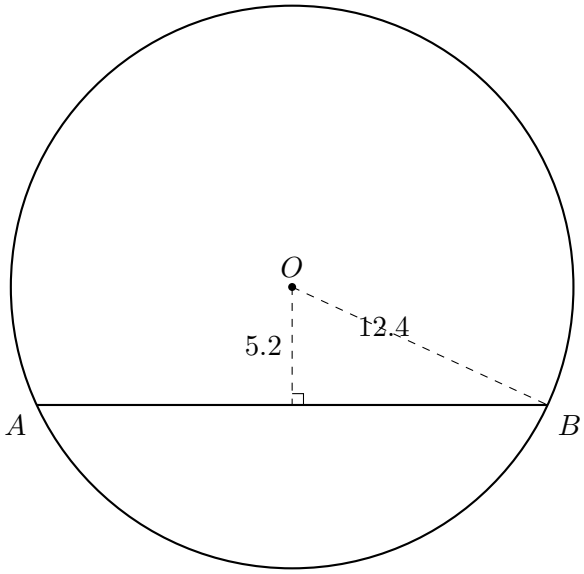
$OM =$ ..... $cm$ **[3]**

**(b)** A second chord $CD$ in the same circle is at a perpendicular distance of 6.4 cm from the centre $O$.

Calculate the length of the chord $CD$.

$CD =$ ..... $cm$ **[3]**

**1261** The diagram shows a circle, centre $O$, with radius 12.4 cm.
$AB$ is a chord of the circle.
The perpendicular distance from $O$ to the chord $AB$ is 5.2 cm.




**(a)** Calculate the length of the chord $AB$.

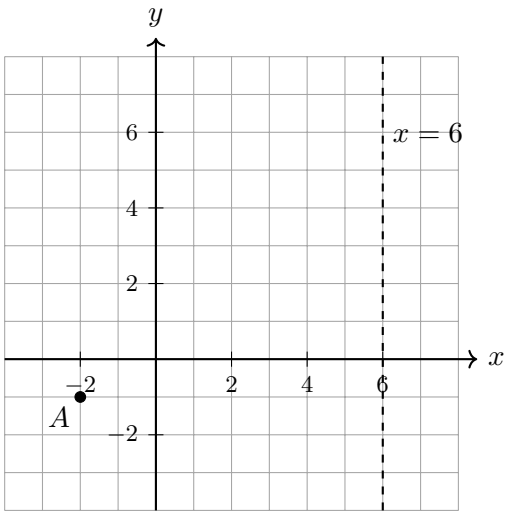
..... cm **[3]**

**(b)** Another chord $CD$ is drawn parallel to $AB$ and on the opposite side of the centre $O$.
The length of the chord $CD$ is 18 cm.

Calculate the perpendicular distance between the two chords $AB$ and $CD$.

..... cm **[4]**

**1262** The diagram shows a coordinate grid.
Point $A$ is $(-2, -1)$ and point $B$ lies on the line $x = 6$.
The $y$-coordinate of $B$ is positive.
The distance $AB$ is 10 units.



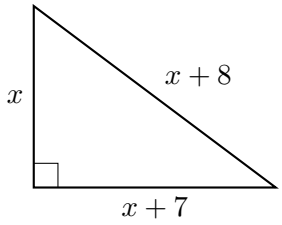

**(a)** Find the $y$-coordinate of point $B$.

$y =$ ..... **[3]**

**(b)** Point $C$ lies on the $x$-axis.
The triangle $ABC$ is isosceles with $AC = BC$.
Calculate the $x$-coordinate of $C$.

$x =$ ..... **[4]**

**1263** A right-angled triangle has sides of length $x$ cm, $(x + 7)$ cm and a hypotenuse of length $(x + 8)$ cm.




**(a)** Use Pythagoras' theorem to write down an equation in $x$ and show that it simplifies to

$x^2 - 2x - 15 = 0$

**[3]**

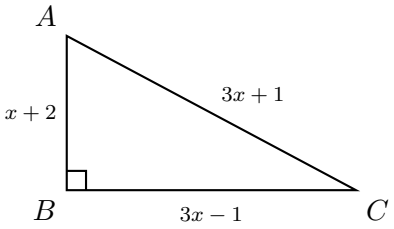
**(b)** Solve the equation to find the value of $x$.

$x =$ ..... **[2]**

**(c)** Calculate the area of the triangle.

Area = ..... $cm^2$ **[2]**

**1264** The diagram shows a right-angled triangle $ABC$ with $\angle ABC = 90°$.
$AB = (x + 2)$ cm, $BC = (3x - 1)$ cm, and $AC = (3x + 1)$ cm.




**(a)** Show that $x^2 - 8x + 4 = 0$.

**[3]**

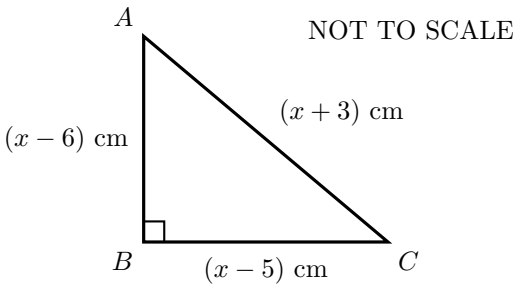
**(b)** Solve the equation $x^2 - 8x + 4 = 0$.
Give your answers in the form $p \pm q\sqrt{r}$, where $p$, $q$ and $r$ are integers.

$x =$ ..... **[3]**

**(c)** Explain why $x = 4 - 2\sqrt{3}$ is not a valid solution for this triangle.

..... **[2]**

**1265** The diagram shows a right-angled triangle $ABC$ with angle $ABC = 90°$.
$AB = (x - 6)$ cm, $BC = (x - 5)$ cm, and $AC = (x + 3)$ cm.




**(a)** Show that $x^2 - 28x + 52 = 0$.

**[3]**

**(b)** Solve the equation $x^2 - 28x + 52 = 0$ and hence write down the lengths of $AB$, $BC$, and $AC$.

$AB =$ ..... , $BC =$ ..... , $AC =$ ..... **[3]**

**(c)** A rectangle $ABCD$ is formed from triangle $ABC$ and forms the horizontal base of a cuboid $ABCDEFGH$ of vertical height 12 cm, where $AE = BF = CG = DH = 12$ cm.

Calculate

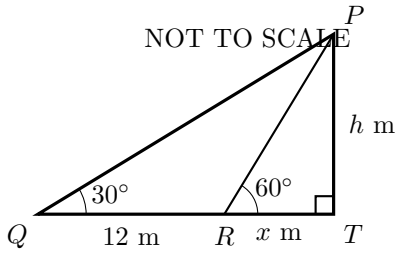
**(i)** the length of the space diagonal $AG$ of the cuboid, giving your answer correct to 3 significant figures,

$AG =$ ..... cm **[2]**

**(ii)** the shortest distance from vertex $C$ to the space diagonal $AG$, giving your answer correct to 3 significant figures.

Distance = ..... cm **[3]**

**1266** The diagram shows a vertical tower $PT$ of height $h$ metres standing on horizontal ground $QRT$.
From point $Q$, the angle of elevation of the top of the tower $P$ is 30°.
Point $R$ lies on the straight line $QT$ such that $QR = 12$ m and $RT = x$ m. From point $R$, the angle of elevation of $P$ is 60°.




**(a)** Express $x$ in terms of $h$, and hence show that the height of the tower is $h = 6\sqrt{3}$ m.

**[4]**

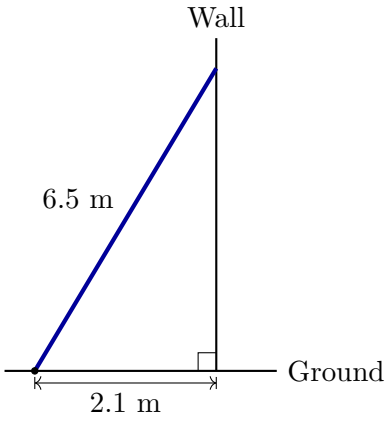
**(b)** Calculate the exact perimeter of triangle $PQR$, giving your answer in the form $a + b\sqrt{3}$, where $a$ and $b$ are integers.

Perimeter = ..... $m$ **[3]**

**(c)** Calculate the exact shortest distance from the base of the tower $T$ to the line $PR$, giving your answer in the form $k\sqrt{3}$, where $k$ is an integer.

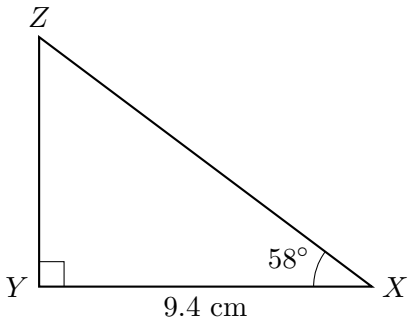
Distance = ..... $m$ **[3]**

**1267** A straight ladder of length 6.5 m leans against a vertical wall.
The base of the ladder is 2.1 m from the bottom of the wall on horizontal ground.




Calculate the angle the ladder makes with the horizontal ground.

..... ° **[3]**



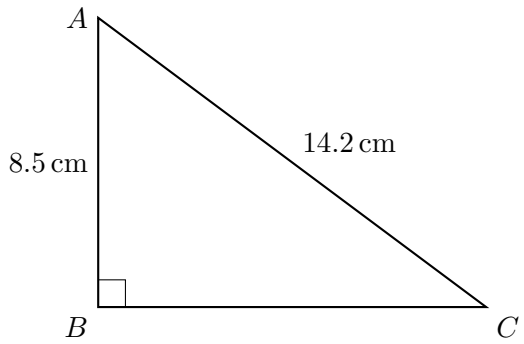

NOT TO SCALE

**1268** Triangle $XYZ$ is right-angled at $Y$.
Angle $ZXY = 58°$ and $XY = 9.4$ cm.

Calculate the length of $XZ$.

..... cm **[3]**

**1269** The diagram shows a right-angled triangle $ABC$ with the right angle at $B$.
$AB = 8.5$ cm and $AC = 14.2$ cm.



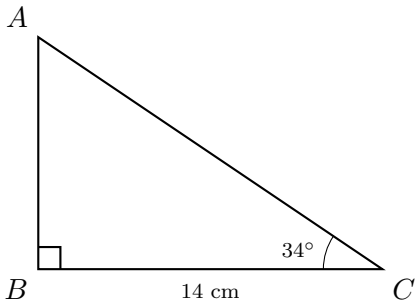

**(a)** Calculate the length of $BC$.

$BC =$ ..... cm **[3]**

**(b)** Calculate angle $BAC$.

Angle $BAC =$ ..... ° **[2]**

**1270** The diagram shows a right-angled triangle $ABC$ with $\angle ABC = 90°$, $BC = 14$ cm and $\angle ACB = 34°$.



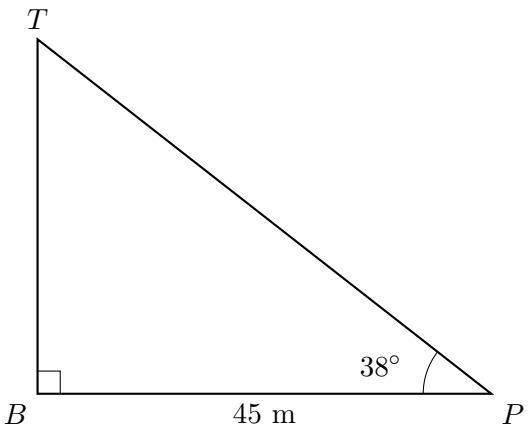

**(a)** Calculate the length of $AB$.

$AB =$ ..... $cm$ **[2]**

**(b)** Point $D$ lies on $AC$ such that $BD$ is the shortest distance from $B$ to $AC$.
Calculate the length of $BD$.

$BD =$ ..... $cm$ **[3]**

**1271** A surveyor stands at a point $P$, which is 45 m away from the base, $B$, of a vertical tower $TB$. The angle of elevation of the top of the tower, $T$, from $P$ is 38°.




**(a)** Calculate the height of the tower, $TB$.

..... $m$ **[2]**

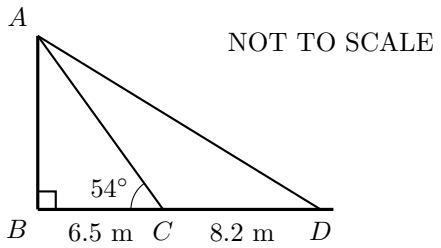
**(b)** Calculate the straight-line distance $PT$.

..... $m$ **[2]**

**(c)** A bird sits at a point $X$ on the tower, where $BX = 20$ m.
Calculate the angle of elevation of the bird from $P$.

..... ° **[3]**

**1272** The diagram shows a vertical flagpole $AB$ standing on horizontal ground $BCD$.
A straight wire $AC$ is attached to the top of the flagpole $A$ and anchored at $C$, where $BC = 6.5$ m and angle $ACB = 54°$.
A second straight wire $AD$ is anchored at $D$, where $CD = 8.2$ m.




**(a)** Calculate the height $AB$ of the flagpole, giving your answer correct to 3 significant figures.

$AB =$ ..... $m$ **[2]**

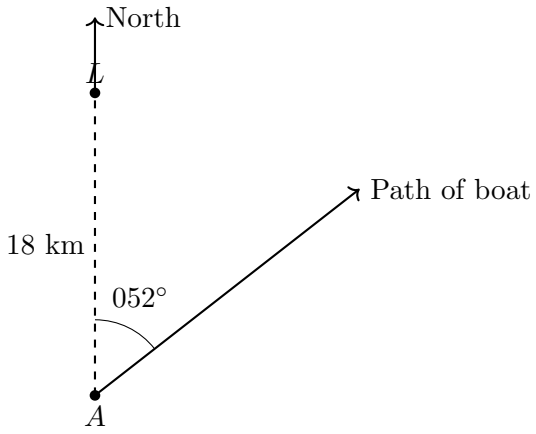
**(b)** Calculate the angle of depression of point $D$ from the top of the flagpole $A$, giving your answer correct to 1 decimal place.

..... $°$ **[3]**

**(c)** Calculate the shortest distance from point $B$ to the wire $AD$, giving your answer correct to 3 significant figures.

Distance = ..... $m$ **[3]**

**1273** A boat sails from a point $A$ in a straight line on a bearing of 052°.
A lighthouse, $L$, is located 18 km due North of $A$.




NOT TO SCALE

**(a)** Calculate the shortest distance from the lighthouse $L$ to the path of the boat.

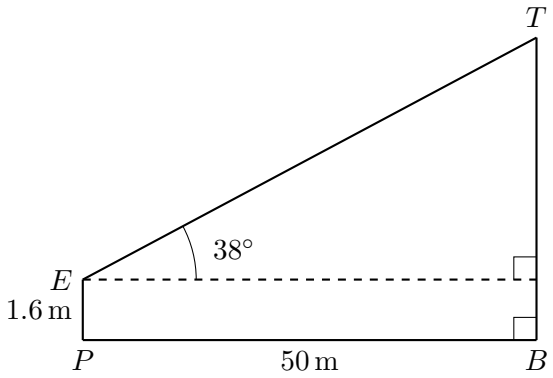
..... km **[2]**

**(b)** The boat travels at a constant speed of 14 km/h.

Calculate the time it takes for the boat to reach the point on its path closest to the lighthouse.
Give your answer in minutes, correct to the nearest minute.

..... minutes **[3]**

**1274** A vertical tower $TB$ stands on horizontal ground.
A person stands at point $P$, 50 m from the base of the tower $B$.
The person's eyes are at point $E$, 1.6 m vertically above $P$.
The angle of elevation of the top of the tower, $T$, from $E$ is 38°.



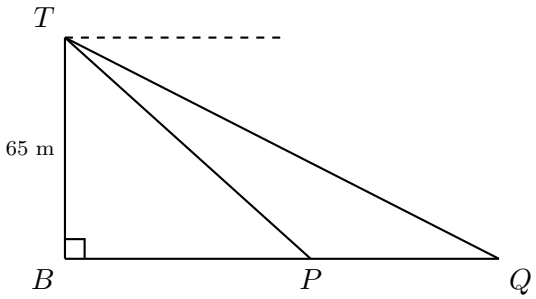

**(a)** Calculate the total height of the tower, $TB$.

$TB =$ ..... $m$ **[3]**

**(b)** A bird on top of the tower at $T$ looks down at a coin on the ground at point $X$.
The point $B$ lies on the straight line between $P$ and $X$.
The angle of depression of $X$ from $T$ is 25°.
Calculate the distance $BX$.

$BX =$ ..... $m$ **[3]**

**1275** From the top $T$ of a vertical cliff $TB$ of height 65 m, a lighthouse keeper observes two boats, $P$ and $Q$, at sea.
Points $B$, $P$ and $Q$ lie in a straight horizontal line with $P$ closer to the base of the cliff $B$ than $Q$.
The angle of depression of boat $P$ from $T$ is 42° and the angle of depression of boat $Q$ from $T$ is 27°.



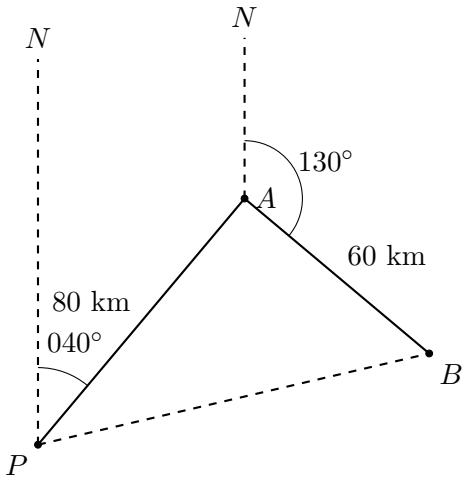

**(a)** Calculate the distance $BP$ of boat $P$ from the base of the cliff.

$BP =$ ..... $m$ **[2]**

**(b)** Calculate the distance $PQ$ between the two boats.

$PQ =$ ..... $m$ **[4]**

**1276** A ship sails from port $P$ on a bearing of 040° for 80 km to point $A$.
It then changes course and sails on a bearing of 130° for 60 km to point $B$.




**(a)** Calculate the direct distance from $P$ to $B$.

$PB =$ ..... km **[3]**

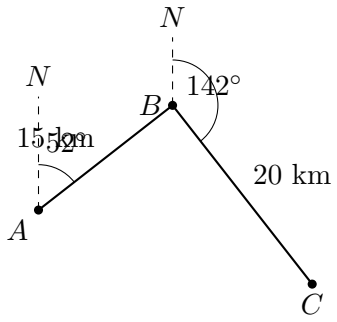
**(b)** Calculate the bearing of $B$ from $P$.

Bearing = ..... ° **[3]**

**(c)** Calculate the shortest distance from point $A$ to the path $PB$.

..... km **[3]**

**1277** A ship sails from port $A$ to point $B$ on a bearing of 052° for 15 km.
It then sails from point $B$ to point $C$ on a bearing of 142° for 20 km.




**(a)** Show that angle $ABC = 90°$.

**[2]**

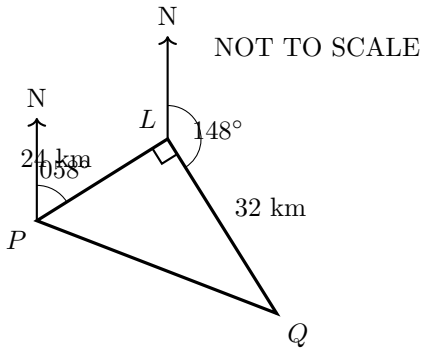
**(b)** Calculate the distance $AC$.

..... km **[3]**

**(c)** Calculate the bearing of $C$ from $A$.

..... ° **[4]**

**1278** A ship sails from port $P$ on a bearing of 058° for 24 km to a lighthouse $L$.
It then changes course and sails on a bearing of 148° for 32 km to an island $Q$.




**(a)** Show that angle $PLQ = 90°$.

**[2]**

**(b)** Calculate

**(i)** the direct distance $PQ$ from the port to the island,

$PQ =$ ..... km **[2]**

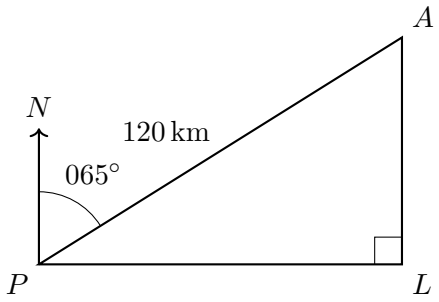
**(ii)** the bearing of the island $Q$ from the port $P$, giving your answer correct to 1 decimal place.

..... ° **[3]**

**(c)** A rescue boat is stationed at lighthouse $L$.
Calculate the shortest distance from the rescue boat to the ship if the ship sails directly along the straight line $PQ$.

Distance = ..... km **[2]**

**1279** A ship sails from a port $P$ on a bearing of 065° for 120 km to point $A$.
A lighthouse is situated at point $L$, which is due East of $P$.
Point $A$ is due North of the lighthouse $L$.



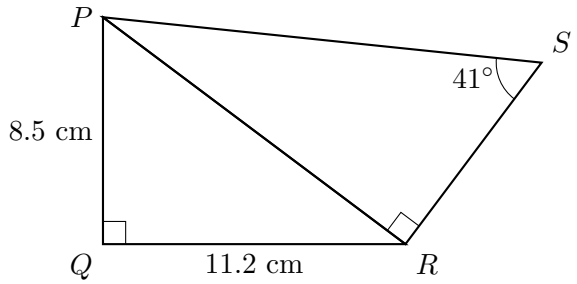

**(a)** Calculate the distance $PL$.

$PL =$ ..... km **[3]**

**(b)** Find the shortest distance from the lighthouse $L$ to the path of the ship $PA$.

..... km **[3]**

**1280** The diagram shows a quadrilateral $PQRS$ formed by two right-angled triangles, $PQR$ and $PRS$.
Angle $PQR = 90°$ and angle $PRS = 90°$.
$PQ = 8.5$ cm, $QR = 11.2$ cm and angle $PSR = 41°$.




NOT TO SCALE

**(a)** Calculate the length of $PR$.

..... cm **[2]**

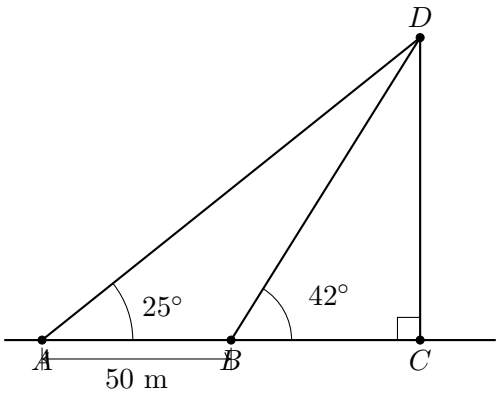
**(b)** Calculate the length of $PS$.

..... cm **[2]**

**(c)** Calculate the perpendicular distance from $R$ to the line $PS$.

..... cm **[2]**

**1281** Point $A$ and point $B$ are on horizontal ground, in a straight line with the base $C$ of a vertical building $CD$.
The distance between $A$ and $B$ is 50 m.
The angle of elevation of the top of the building, $D$, from $A$ is 25° and from $B$ is 42°.



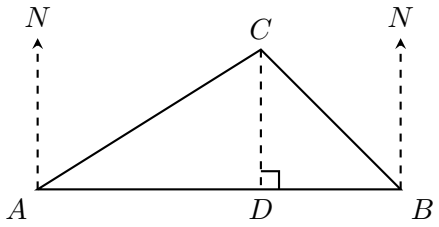

**(a)** Calculate the height of the building $CD$.

$CD =$ ..... $m$ **[5]**

**(b)** A bird flies directly along the straight path from $A$ to $D$.
Calculate the shortest distance from point $B$ to the flight path $AD$.

..... $m$ **[3]**

**1282** Port $B$ is 48 km due east of port $A$.
A ship sails from port $A$ on a bearing of 058° until it reaches a point $C$ which is directly north-west of port $B$ (on a bearing of 315° from $B$).
Point $D$ lies on the line $AB$ such that $CD$ is perpendicular to $AB$.




**(a)** Write down the size of angle $CAB$ and the size of angle $CBA$.

$\angle CAB =$ ..... , $\angle CBA =$ ..... **[2]**

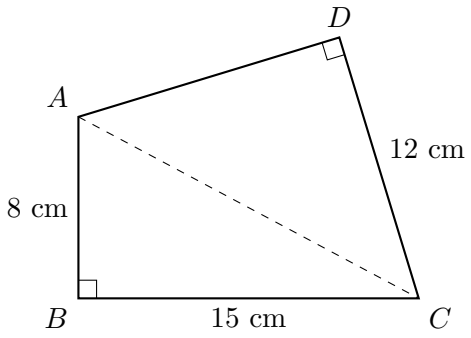
**(b)** Calculate the shortest distance, $CD$, of the ship from the straight line path $AB$.

$CD =$ ..... km **[4]**

**(c)** Calculate the distance $AC$ sailed by the ship from port $A$ to point $C$.

$AC =$ ..... km **[2]**

**1283** The diagram shows a quadrilateral $ABCD$.
Angle $ABC = 90°$ and angle $ADC = 90°$.
$AB = 8$ cm, $BC = 15$ cm, and $CD = 12$ cm.




**(a)** Calculate the length of $AC$.

..... cm **[2]**

**(b)** Calculate the length of $AD$.

..... cm **[3]**

**(c)** Calculate the shortest distance from $D$ to the line $AC$.

..... cm **[3]**

**(d)** Calculate the size of angle $DAB$.

..... ° **[4]**

**1284** Without using a calculator, answer the following questions.

**(a)** Write down the exact value of $\cos 30°$.

..... **[1]**

**(b)** Show that

$$\frac{1 - \sin 30°}{\tan^2 60°} = \frac{1}{6}$$

**[2]**

**1285** Without using a calculator, write down the exact value of each of the following expressions.
You must show all your working.

**(a)** $\sin 90^\circ + \cos 0^\circ$

..... **[1]**

**(b)** $\tan 45^\circ \times \sin 30^\circ$

..... **[1]**

**(c)** $\dfrac{\tan 60^\circ}{\cos 30^\circ}$

..... **[2]**

**1286** **(a)** Work out the exact value of

$$2\sin 60^\circ \times \tan 30^\circ$$

..... **[2]**

**(b)** Find the value of $x$, where $0^\circ \leq x \leq 90^\circ$, such that

$$\sqrt{2}\cos x = \tan 45^\circ$$

$x =$ ..... $^\circ$ **[2]**

**1287** Without using a calculator, work out the exact value of each of the following.

**(a)**

$\sin 30^\circ + \cos 60^\circ$

..... **[1]**

**(b)**

$\tan 60^\circ \times \sin 60^\circ - \cos 0^\circ$

..... **[3]**

**1288** Do not use a calculator in this question.

**(a)** Work out the exact value of

**(i)** $\sin 30^\circ + \cos 60^\circ$,

..... **[2]**

**(ii)** $\dfrac{\tan 60^\circ}{\cos 30^\circ} - (\sin 45^\circ)^2$.

..... **[3]**

**(b)** For $0^\circ \leq x \leq 90^\circ$, solve the equation

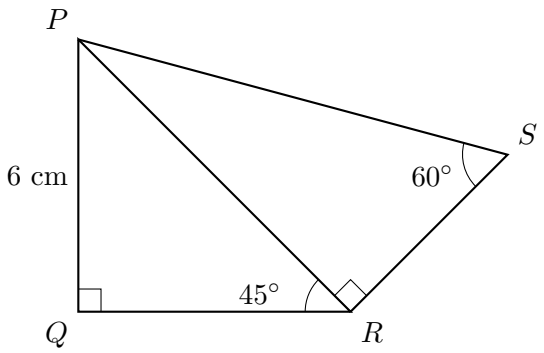
$$2\cos x - \sqrt{3} = 0$$

$x =$ ..... $^\circ$ **[2]**

**(c)** Given that $\tan y = \cos 0^\circ - \sin 90^\circ$ and $0^\circ \leq y \leq 90^\circ$, find the value of $y$.

$y =$ ..... $^\circ$ **[2]**

**1289** The diagram shows two right-angled triangles, $PQR$ and $PRS$.
$PQ = 6$ cm, angle $PQR = 90°$ and angle $PRQ = 45°$.
Angle $PRS = 90°$ and angle $PSR = 60°$.



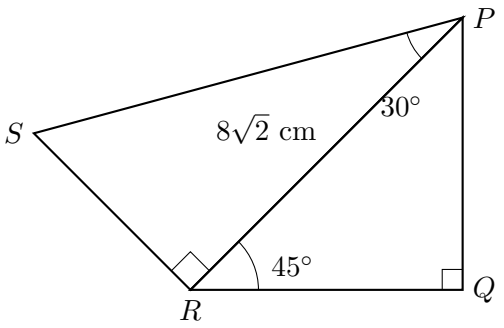

**(a)** Calculate the exact length of $PR$.

..... cm **[2]**

**(b)** Calculate the exact length of $PS$.
Give your answer in the form $a\sqrt{b}$ where $a$ and $b$ are integers.

..... cm **[3]**

**1290** The diagram shows two right-angled triangles, $PQR$ and $PRS$.
Angle $PQR = 90°$ and angle $PRS = 90°$.
Angle $PRQ = 45°$ and angle $SPR = 30°$.
$PR = 8\sqrt{2}$ cm.




NOT TO SCALE

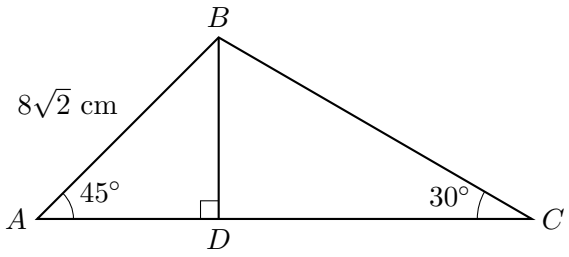
**(a)** Without using a calculator, show that the exact length of $PQ$ is 8 cm.

**[2]**

**(b)** Without using a calculator, work out the exact length of $PS$.
Give your answer in the form $\dfrac{a\sqrt{b}}{c}$, where $a, b$ and $c$ are integers.

$PS =$ ..... cm **[3]**

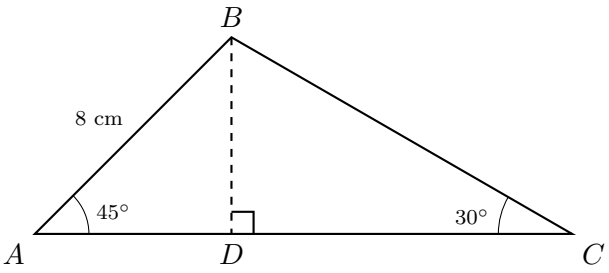
**1291** The diagram shows a triangle $ABC$ with a point $D$ on the side $AC$.
$BD$ is perpendicular to $AC$.
$AB = 8\sqrt{2}$ cm, $\angle BAC = 45°$ and $\angle BCA = 30°$.




Calculate the exact perimeter of triangle $ABC$.
Give your answer in the form $a + b\sqrt{2} + c\sqrt{3}$ where $a, b$, and $c$ are integers.

Perimeter = ..... cm **[6]**

**1292** The diagram shows a triangle $ABC$ with $AB = 8$ cm, $\angle BAC = 45°$ and $\angle BCA = 30°$.
$BD$ is perpendicular to $AC$.




**(a)** Find the exact length of $BD$.
Give your answer in its simplest surd form.

$BD =$ ..... *cm* **[2]**

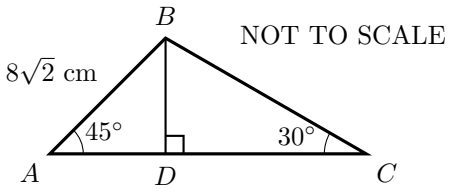
**(b)** Find the exact length of $BC$.
Give your answer in its simplest surd form.

$BC =$ ..... *cm* **[2]**

**(c)** Show that the exact length of $AC$ is $(4\sqrt{2} + 4\sqrt{6})$ cm.

**[2]**

**1293** The diagram shows triangle $ABC$ with $BD$ perpendicular to $AC$, where $D$ lies on $AC$.
$AB = 8\sqrt{2}$ cm, angle $BAD = 45°$, and angle $BCD = 30°$.




**(a)** Show that $BD = 8$ cm and $AD = 8$ cm.

**[2]**

**(b)** Calculate the exact length of $CD$, giving your answer in the form $k\sqrt{3}$, where $k$ is an integer.

$CD =$ ..... cm **[2]**

**(c)** Calculate the exact area of triangle $ABC$, giving your answer in the form $a + b\sqrt{3}$, where $a$ and $b$ are integers.

Area = ..... $cm^2$ **[3]**

**(d)** Calculate the exact perimeter of triangle $ABC$, giving your answer in the form $p + q\sqrt{2} + r\sqrt{3}$, where $p$, $q$, and $r$ are integers.

Perimeter = ..... cm **[3]**

**1294** Do not use a calculator for this question.
All working must be clearly shown.

**(a)** Show that

$$\frac{1 - \tan^2 30^\circ}{1 + \tan^2 30^\circ} = \cos 60^\circ$$

**[3]**

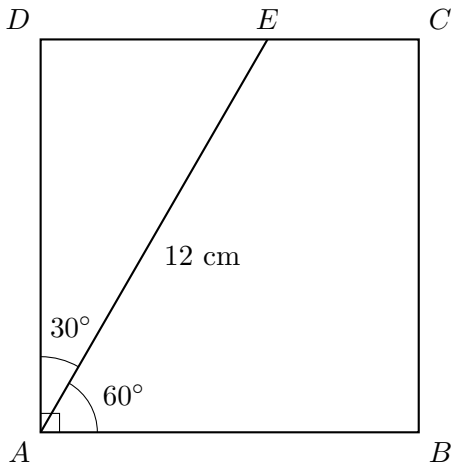
**(b)** Solve the equation.

$$x \sin 60^\circ = \tan 60^\circ - x \cos 60^\circ$$

Give your answer in the form $a - \sqrt{b}$, where $a$ and $b$ are integers.

$x =$ ..... **[4]**

**1295** The diagram shows a rectangle $ABCD$.
$E$ is a point on the side $CD$.
$AE = 12$ cm, angle $DAE = 30°$ and angle $EAB = 60°$.




**(a)** Show that the exact length of $AD$ is $6\sqrt{3}$ cm.

**[2]**

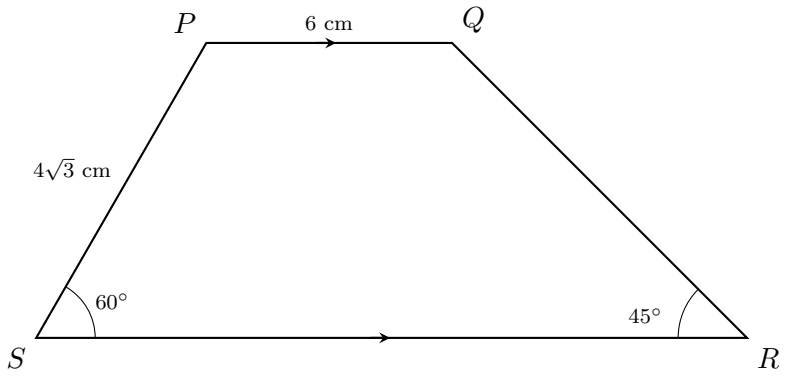
**(b)** The exact area of the rectangle $ABCD$ is $60\sqrt{3}$ cm$^2$.
Work out the exact length of $CE$.

..... cm **[3]**

**(c)** Find the exact value of $\tan(\angle EBC)$.
Give your answer in the form $\dfrac{a\sqrt{b}}{c}$ where $a$, $b$, and $c$ are integers.

..... **[3]**

**1296** The diagram shows a trapezium $PQRS$ with $PQ$ parallel to $SR$.
$PQ = 6$ cm, $PS = 4\sqrt{3}$ cm, $\angle PSR = 60°$ and $\angle QRS = 45°$.




**(a)** Find the exact perpendicular height of the trapezium $PQRS$.

Height = ..... $cm$ **[2]**

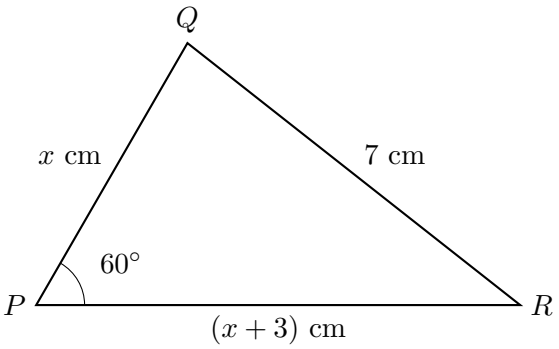
**(b)** Calculate the exact area of the trapezium $PQRS$.
Give your answer in the form $a + b\sqrt{3}$, where $a$ and $b$ are integers.

Area = ..... $cm^2$ **[4]**

**(c)** Show that the exact perimeter of the trapezium $PQRS$ is $(18 + 6\sqrt{3} + 6\sqrt{2})$ cm.

**[2]**

**1297** The diagram shows a triangle $PQR$.
$PQ = x$ cm, $PR = (x + 3)$ cm and $QR = 7$ cm.
Angle $QPR = 60°$.




**(a)** Show that $x^2 + 3x - 40 = 0$.

**[4]**

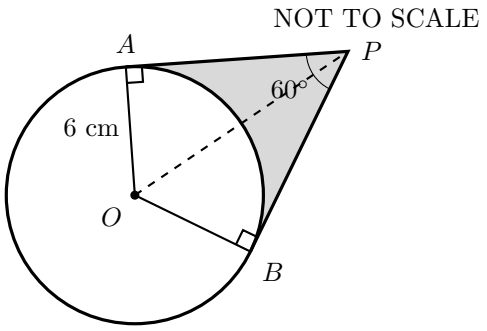
**(b)** Solve the equation to find the value of $x$, and hence find the perimeter of triangle $PQR$.

Perimeter = ..... cm **[3]**

**(c)** Find the exact area of triangle $PQR$.

Area = ..... $cm^2$ **[3]**

**1298** The diagram shows a circle, centre $O$, of radius 6 cm. From an external point $P$, tangents $PA$ and $PB$ are drawn to touch the circle at $A$ and $B$ respectively, such that angle $APB = 60°$.
The region bounded by $PA$, $PB$, and the minor arc $AB$ is shaded.




**(a)** Find the exact length of the tangent $PA$, giving your answer in the form $k\sqrt{3}$, where $k$ is an integer.

$PA =$ ..... cm **[3]**

**(b)** Calculate the exact area of the shaded region, giving your answer in the form $a\sqrt{3} - b\pi$, where $a$ and $b$ are integers.

Area = ..... $cm^2$ **[4]**

**(c)** Calculate

**(i)** the exact length of the chord $AB$, giving your answer in the form $m\sqrt{3}$, where $m$ is an integer,

$AB =$ ..... cm **[2]**

**(ii)** the exact shortest distance from point $P$ to the chord $AB$.

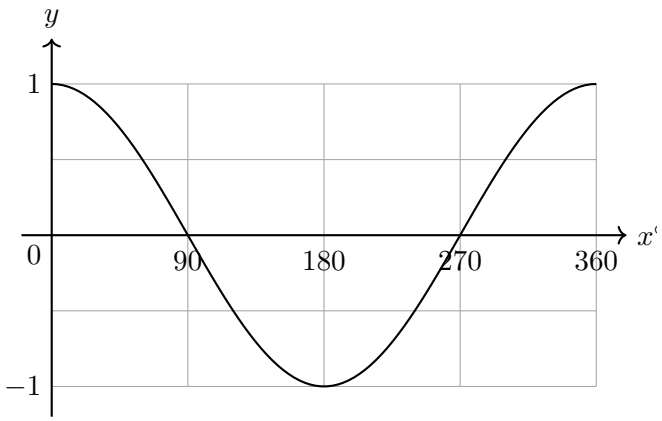
Distance = ..... cm **[2]**

**1299** Solve the equation for $0° \leq x \leq 360°$.

$\sqrt{2}\sin x - 1 = 0$

$x =$ ..... **[3]**

**1300** The diagram shows the graph of $y = \cos x$ for $0° \leq x \leq 360°$.



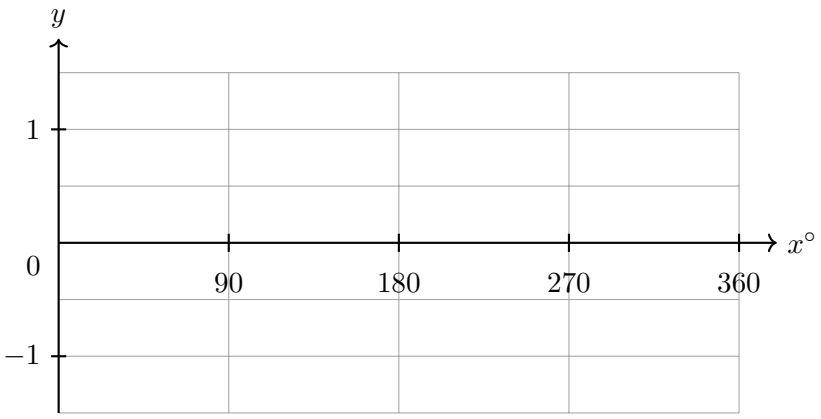

**(a)** Write down the coordinates of the lowest point on the graph.

( ..... , ..... ) **[1]**

**(b)** Use the graph, or otherwise, to solve the equation $\cos x = 0.5$ for $0° \leq x \leq 360°$.

$x =$ ..... **[2]**

**1301** **(a)** On the grid, sketch the graph of $y = \sin x$ for $0° \leq x \leq 360°$.




**[2]**

**(b)** Write down the exact solutions to the equation $\sin x = 0.5$ for $0° \leq x \leq 360°$.

$x =$ ..... , $x =$ ..... **[2]**

**1302** Answer the following questions for $0° \leq x \leq 360°$.

**(a)** Solve the equation

$$2\cos x - 1 = 0$$

$x =$ ..... **[3]**

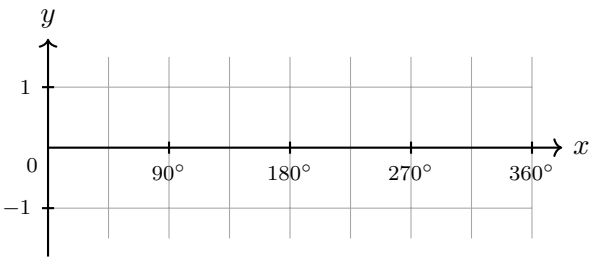
**(b)** Write down the number of solutions to the equation

$$3\sin x = 5$$

..... **[1]**

**1303** Consider the trigonometric function $y = \cos x$ for $0° \leq x \leq 360°$.

**(a)** On the grid below, sketch the graph of $y = \cos x$ for $0° \leq x \leq 360°$.




**[2]**

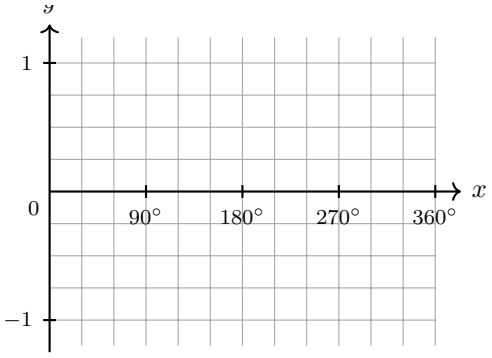
**(b)** Solve the equation for $0° \leq x \leq 360°$.

$$2\cos x + 1 = 0$$

$x =$ ..... , $x =$ ..... **[3]**

**1304** Do not use a calculator in this question.

**(a)** On the axes below, sketch the graph of $y = \sin x$ for $0° \leq x \leq 360°$.




**[2]**

**(b)** Write down the coordinates of the maximum point and the minimum point of $y = \sin x$ for $0° \leq x \leq 360°$.

Maximum = ..... , Minimum = ..... **[2]**

**(c)** Solve the equation for $0° \leq x \leq 360°$:

**(i)** $\sin x = \dfrac{\sqrt{3}}{2}$

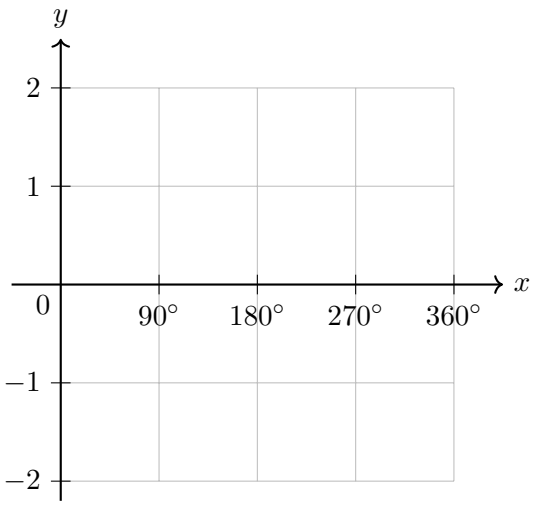
$x =$ ..... , $x =$ ..... **[2]**

**(ii)** $2\cos x + 1 = 0$

$x =$ ..... , $x =$ ..... **[3]**

**1305** The properties of trigonometric graphs can be used to solve equations.

**(a)** On the grid, sketch the graph of $y = \cos x$ for $0° \leq x \leq 360°$.




**[2]**

**(b)** Write down the coordinates of the minimum point on this graph.

( ..... , ..... ) **[1]**

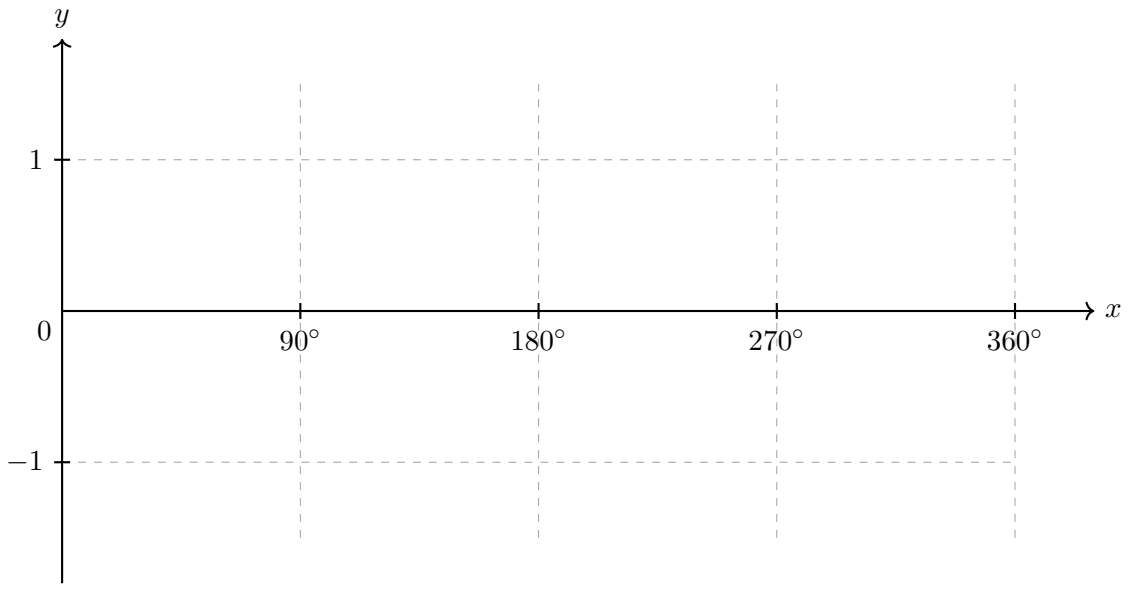
**(c)** Solve the equation for $0° \leq x \leq 360°$.

$2\cos x + 1 = 0$

$x =$ ..... **[3]**

**1306** The functions $g(x) = \sin x$ and $h(x) = \cos x$ are defined for $0° \leq x \leq 360°$.

**(a)** On the grid provided, sketch the graphs of $y = g(x)$ and $y = h(x)$.




**[3]**

**(b)** Solve the equation $g(x) = h(x)$ for $0° \leq x \leq 360°$.

$x =$ ..... **[2]**

**(c)** Use your graphs to write down the range of values of $x$ for which $g(x) > h(x)$ for $0° \leq x \leq 360°$.

..... **[2]**

**1307** Solve the following equations for $0° \leq x \leq 360°$.

**(a)** $2\sin x + \sqrt{3} = 0$

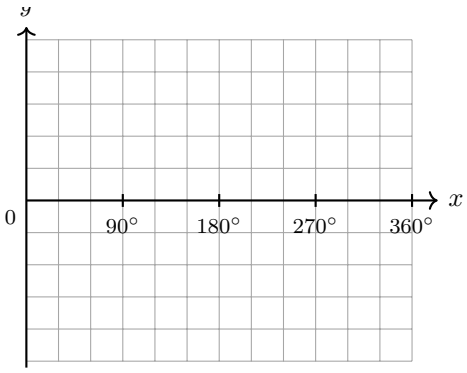
$x =$ ..... **[3]**

**(b)** $4\cos^2 x - 3 = 0$

$x =$ ..... **[4]**

**1308** Do not use a calculator in this question.

**(a)** On the axes below, sketch the graph of $y = \tan x$ for $0° \leq x \leq 360°$.
Draw dashed lines to show any asymptotes.




**[3]**

**(b)** Write down the equations of the two asymptotes of the graph of $y = \tan x$ for $0° \leq x \leq 360°$.

$x =$ ..... , $x =$ ..... **[2]**

**(c)** Solve the equation for $0° \leq x \leq 360°$:

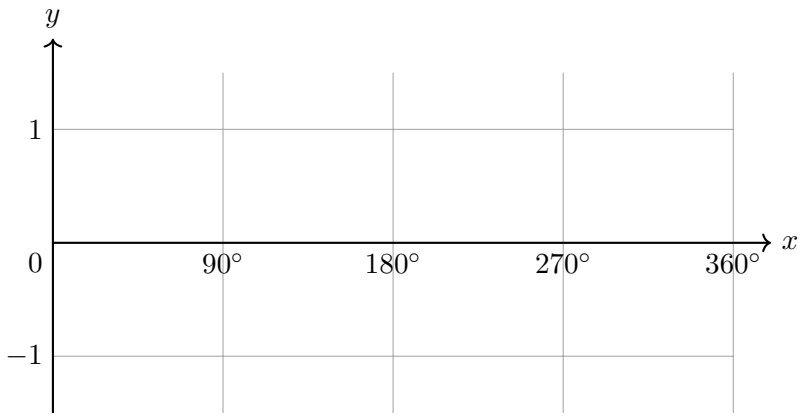
**(i)** $\sqrt{3}\tan x + 1 = 0$

$x =$ ..... , $x =$ ..... **[3]**

**(ii)** $2\sin^2 x - \sin x - 1 = 0$

$x =$ ..... , $x =$ ..... , $x =$ ..... **[4]**

**1309** **(a)** On the grid, sketch the graph of $y = \sin x$ for $0° \leq x \leq 360°$.




**[2]**

**(b)** Solve the equation

$\sin x = 0.7$ for $0° \leq x \leq 360°$.

$x =$ ..... **[3]**

**1310** Consider the trigonometric function $f(x) = 4\sin x + 3$ for $0° \leq x \leq 360°$.

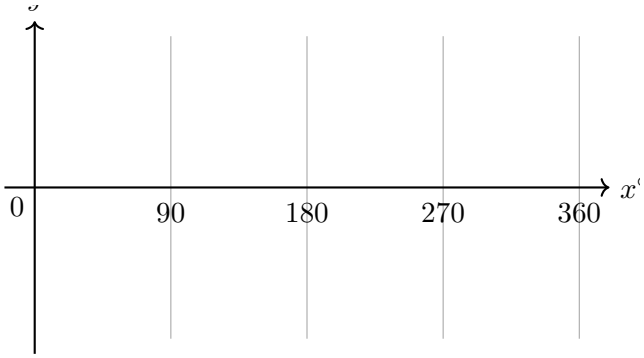
**(a)** Solve the equation $f(x) = 0$.
Give your answers correct to 1 decimal place.

$x =$ ..... **[3]**

**(b)** Write down the exact coordinates of the minimum point on the graph of $y = f(x)$.

( ..... , ..... ) **[2]**

**1311** **(a)** On the axes provided, sketch the graph of $y = \tan x$ for $0° \leq x \leq 360°$.




**[2]**

**(b)** Solve the equation $3\tan x + 5 = 0$ for $0° \leq x \leq 360°$.
Give your answers correct to 1 decimal place.

$x =$ ..... **[3]**

**1312** **(a)** Solve the equation for $0° \leq x \leq 360°$.

$4\cos x + 1 = 0$

Give your answers correct to 1 decimal place.

$x =$ ..... , $x =$ ..... **[3]**

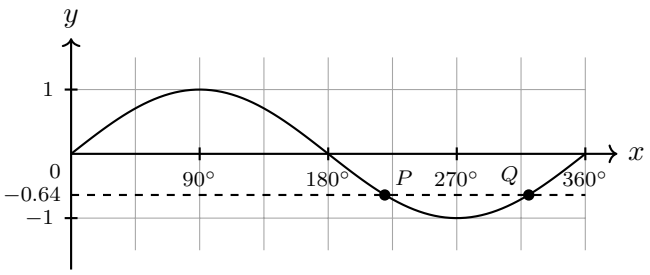
**(b)** Solve the equation for $0° \leq x \leq 360°$.

$2\tan x - 5 = 0$

Give your answers correct to 1 decimal place.

$x =$ ..... , $x =$ ..... **[3]**

**1313** The diagram shows the graph of $y = \sin x$ for $0° \leq x \leq 360°$.
The line $y = -0.64$ intersects the curve at points $P$ and $Q$.




**(a)** Calculate the $x$-coordinates of $P$ and $Q$.
Give your answers correct to 1 decimal place.

$x$-coordinate of $P =$ ..... , $x$-coordinate of $Q =$ ..... **[3]**

**(b)** Solve the equation for $0° \leq x \leq 360°$.

$3 \tan x + 5 = 0$ Give your answers correct to 1 decimal place.

$x =$ ..... , $x =$ ..... **[3]**

**1314** **(a)** Solve the equation

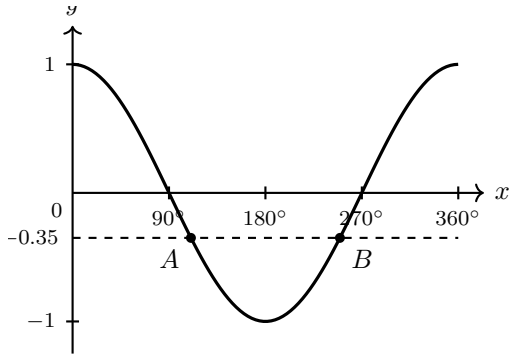
$3\cos x + 1 = 0$ for $0° \leq x \leq 360°$.

$x =$ ..... **[4]**

**(b)** The graph of $y = 3\cos x + 1$ crosses the $x$-axis at points $A$ and $B$.
Calculate the distance between $A$ and $B$.

Distance = ..... **[3]**

**1315** The diagram shows the graph of $y = \cos x$ and the line $y = -0.35$ for $0° \leq x \leq 360°$.
The line intersects the curve at points $A$ and $B$.




**(a)** Calculate the $x$-coordinate of $A$ and the $x$-coordinate of $B$, giving your answers correct to 1 decimal place.

$x$-coordinate of $A =$ ..... , $x$-coordinate of $B =$ ..... **[3]**

**(b)** Solve the equation for $0° \leq x \leq 360°$, giving your answers correct to 1 decimal place:

$5 \sin x + 3 = 0$

$x =$ ..... , $x =$ ..... **[3]**

**(c)** Solve the equation for $0° \leq x \leq 360°$, giving your answers correct to 1 decimal place:

$3 \tan x - 7 = 0$

$x =$ ..... , $x =$ ..... **[3]**

**1316** **(a)** The equation $a \sin x - b = 0$ has a solution $x = 30°$.
Given that $a$ and $b$ are positive integers with no common factors, find the value of $a$ and the value of $b$.

$a =$ ..... , $b =$ ..... **[2]**

**(b)** Write down the second solution to the equation $a \sin x - b = 0$ for $0° \leq x \leq 360°$.

$x =$ ..... ° **[1]**

**(c)** A horizontal line $y = k$ intersects the graph of $y = \cos x$ at exactly two points in the interval $0° \leq x \leq 360°$.
The $x$-coordinate of one of these points is $242°$.
Find the value of $k$ correct to 3 significant figures and find the $x$-coordinate of the second point of intersection.

$k =$ ..... , $x =$ ..... **[3]**

**1317** Algebraic techniques can be applied to solve trigonometric equations.

**(a)** Solve the equation for $0° \leq x \leq 360°$.

$5 \tan x + 14 = 0$

Give your answers correct to 1 decimal place.

$x =$ ..... **[3]**

**(b)** Solve the equation for $0° \leq x \leq 360°$.

$2 \sin^2 x - \sin x = 0$

$x =$ ..... **[4]**

**1318** Solve the equation

$$5\sin x = 2\tan x \text{ for } 0° \leq x \leq 360°.$$

Show all your working.

$x =$ ..... **[7]**

**1319** Solve the following trigonometric equations for $0° \leq x \leq 360°$.

**(a)**

$5\sin x - 2\cos x = 0$ Give your answers correct to 1 decimal place.

$x =$ ..... , $x =$ ..... **[3]**

**(b)**

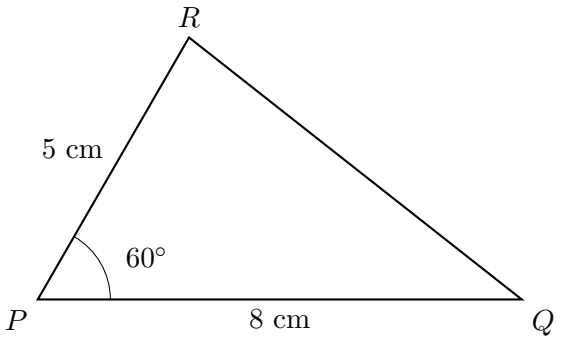
$2\sin^2 x + \sin x - 1 = 0$

$x =$ .....

$x =$ .....

$x =$ ..... **[5]**

**1320** The diagram shows triangle $PQR$.
$PQ = 8$ cm, $PR = 5$ cm and angle $QPR = 60°$.




NOT TO SCALE

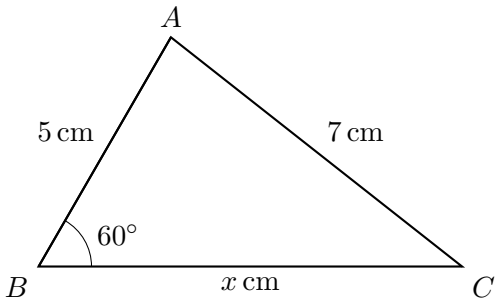
**(a)** Calculate the exact area of triangle $PQR$.
Give your answer in the form $a\sqrt{3}$.

..... cm$^2$ **[2]**

**(b)** Calculate the exact length of $QR$.

$QR =$ ..... cm **[3]**

**1321** In triangle $ABC$, $AB = 5\,\text{cm}$, $BC = x\,\text{cm}$ and $AC = 7\,\text{cm}$.
Angle $ABC = 60°$.




NOT TO SCALE

**(a)** Use the cosine rule to show that $x^2 - 5x - 24 = 0$.

**[3]**

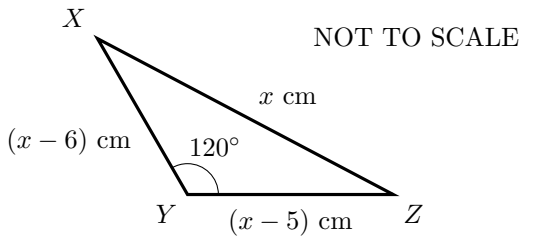
**(b)** Solve the equation $x^2 - 5x - 24 = 0$ to find the length of $BC$.

$BC =$ ..... cm **[2]**

**(c)** Calculate the exact area of triangle $ABC$.
Give your answer in the form $a\sqrt{b}$, where $a$ and $b$ are integers.

Area = ..... $cm^2$ **[2]**

**1322** The diagram shows triangle $XYZ$ with $XY = (x - 6)$ cm, $YZ = (x - 5)$ cm, $XZ = x$ cm, and angle $XYZ = 120°$.




**(a)** Use the cosine rule to show that $2x^2 - 33x + 91 = 0$.

**[3]**

**(b)** Solve the equation $2x^2 - 33x + 91 = 0$ and hence write down the lengths of $XY$, $YZ$, and $XZ$.

$XY =$ ..... , $YZ =$ ..... , $XZ =$ ..... **[3]**

**(c)** Calculate

**(i)** the exact area of triangle $XYZ$, giving your answer in the form $k\sqrt{3}$, where $k$ is an integer,

Area = ..... $cm^2$ **[2]**

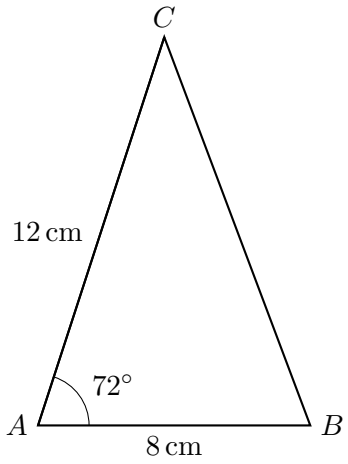
**(ii)** the exact shortest distance from vertex $Y$ to the side $XZ$, giving your answer in the form $\frac{p\sqrt{3}}{q}$, where $p$ and $q$ are integers.

Distance = ..... cm **[2]**

**(d)** Find the exact value of $\sin(\angle YXZ)$, giving your answer in the form $\dfrac{m\sqrt{3}}{n}$, where $m$ and $n$ are integers.

$\sin(\angle YXZ) =$ ..... **[2]**

**1323** The diagram shows triangle $ABC$.



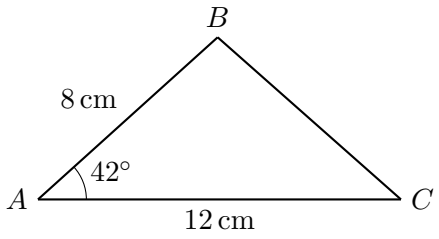

**(a)** Calculate the length of $BC$.

$BC =$ ..... cm **[3]**

**(b)** Calculate the area of triangle $ABC$.

Area = ..... $cm^2$ **[2]**

**1324** The diagram shows triangle $ABC$.
$AB = 8\,\text{cm}$, $AC = 12\,\text{cm}$ and angle $BAC = 42°$.



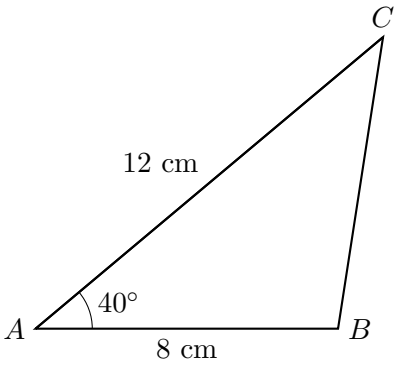

**(a)** Calculate the area of triangle $ABC$.

..... $cm^2$ **[2]**

**(b)** Calculate the length of $BC$.

$BC =$ ..... cm **[3]**

**1325** The diagram shows a triangle $ABC$.



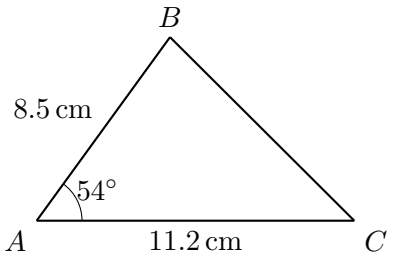

**(a)** Calculate the area of triangle $ABC$.

..... $cm^2$ **[2]**

**(b)** Calculate the length of $BC$.

..... cm **[3]**

**1326** The diagram shows triangle $ABC$.



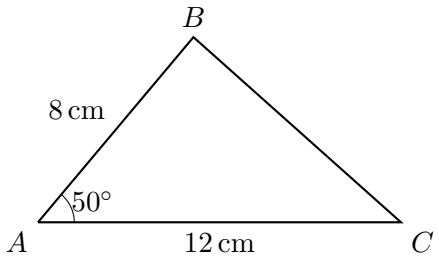

**(a)** Calculate the area of triangle $ABC$.

..... $cm^2$ **[2]**

**(b)** Calculate the length of $BC$.

..... cm **[3]**

**1327** A triangle $ABC$ is shown below.




NOT TO SCALE

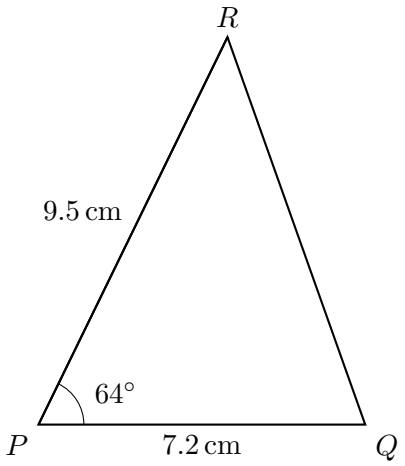
**(a)** Calculate the area of triangle $ABC$.

..... $cm^2$ **[2]**

**(b)** Calculate the length of $BC$.

..... $cm$ **[3]**

**1328** In triangle $PQR$, $PQ = 7.2\,\text{cm}$, $PR = 9.5\,\text{cm}$ and angle $QPR = 64°$.




NOT TO SCALE

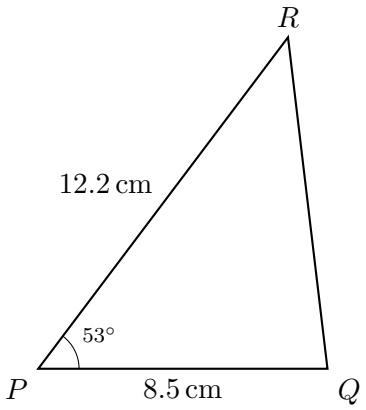
**(a)** Calculate the area of triangle $PQR$.

Area = ..... $cm^2$ **[2]**

**(b)** Calculate the length of $QR$.

$QR$ = ..... cm **[3]**

**1329** The diagram shows a triangle $PQR$.
$PQ = 8.5$ cm, $PR = 12.2$ cm and angle $QPR = 53°$.



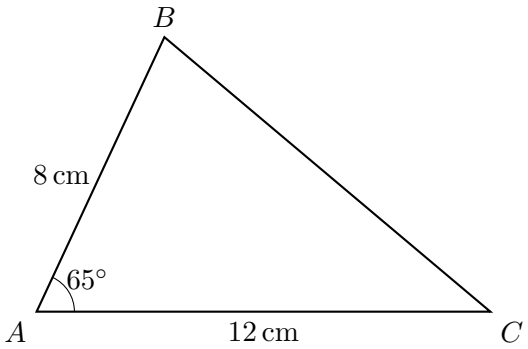

**(a)** Calculate the area of triangle $PQR$.

Area = ..... $cm^2$ **[2]**

**(b)** Calculate the length of $QR$.

$QR$ = ..... $cm$ **[3]**

**1330** The diagram shows a triangle $ABC$ with $AB = 8\,\text{cm}$, $AC = 12\,\text{cm}$ and angle $BAC = 65°$.




NOT TO SCALE

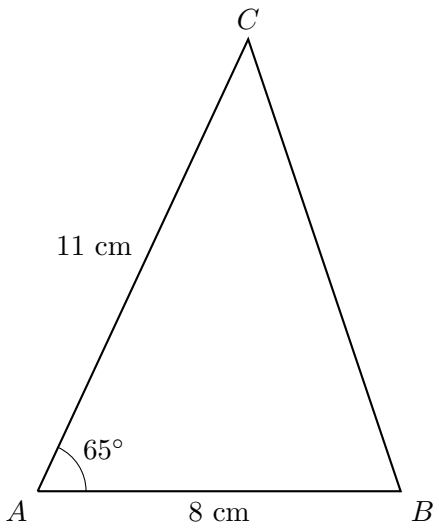
**(a)** Calculate the length of $BC$.

$BC =$ ..... cm **[3]**

**(b)** Calculate the area of triangle $ABC$.

Area = ..... $cm^2$ **[2]**

**1331** The diagram shows a triangle $ABC$.
$AB = 8$ cm, $AC = 11$ cm and angle $BAC = 65°$.



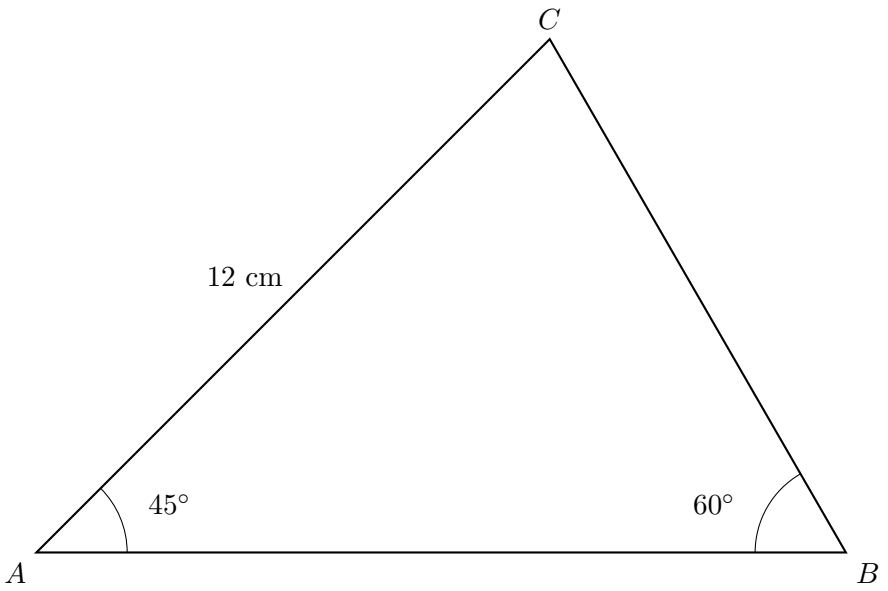

**(a)** Calculate the area of triangle $ABC$.

Area = ..... $cm^2$ **[2]**

**(b)** Calculate the length of $BC$.

$BC$ = ..... cm **[3]**

**1332** In triangle $ABC$, $AC = 12$ cm, angle $BAC = 45°$ and angle $ABC = 60°$.



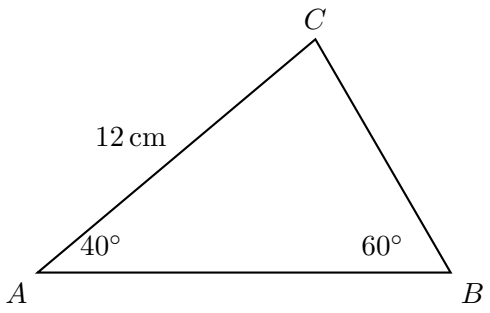

**(a)** Calculate the length of $BC$.

..... cm **[3]**

**(b)** Calculate the area of triangle $ABC$.

..... $cm^2$ **[3]**

**1333** The diagram shows triangle $ABC$.
$AC = 12$ cm, angle $BAC = 40°$ and angle $ABC = 60°$.



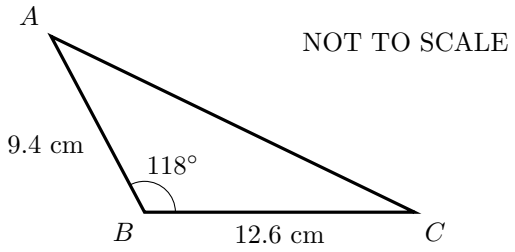

**(a)** Calculate the length of $BC$.

$BC =$ ..... cm **[3]**

**(b)** Calculate the area of triangle $ABC$.

Area = ..... $cm^2$ **[3]**

**1334** The diagram shows triangle $ABC$ with $AB = 9.4$ cm, $BC = 12.6$ cm, and obtuse angle $ABC = 118°$.




**(a)** Calculate the length of $AC$, giving your answer correct to 3 significant figures.

$AC =$ ..... cm **[3]**

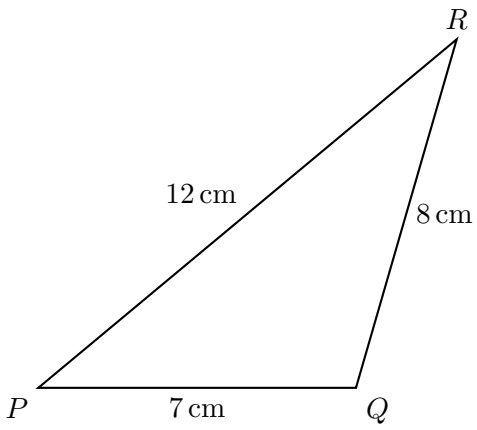
**(b)** Calculate angle $BAC$, giving your answer correct to 1 decimal place.

..... ° **[3]**

**(c)** Calculate the area of triangle $ABC$, giving your answer correct to 3 significant figures.

Area = ..... $cm^2$ **[2]**

**1335** The diagram shows triangle $PQR$.
$PQ = 7$ cm, $QR = 8$ cm and $PR = 12$ cm.



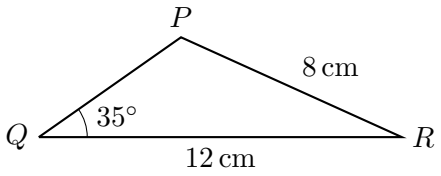

**(a)** Calculate the largest angle in triangle $PQR$.

..... ° **[3]**

**(b)** Calculate the area of triangle $PQR$.

Area = ..... $cm^2$ **[2]**

**1336** In triangle $PQR$, $PR = 8$ cm, $QR = 12$ cm and angle $PQR = 35°$.
Angle $QPR$ is an obtuse angle.




NOT TO SCALE

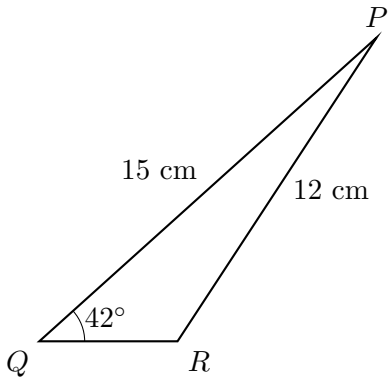
**(a)** Calculate the size of angle $QPR$.

..... ° **[4]**

**(b)** Calculate the area of triangle $PQR$.

..... $cm^2$ **[2]**

**1337** The diagram shows a triangle $PQR$.
The length of $PQ$ is 15 cm and the length of $PR$ is 12 cm.
Angle $PQR = 42°$.
Angle $PRQ$ is obtuse.



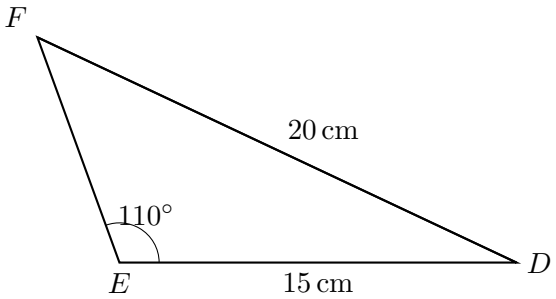

**(a)** Calculate angle $PRQ$.

..... ° **[3]**

**(b)** Calculate the length of $QR$.

..... cm **[3]**

**1338** The diagram shows triangle $DEF$.
$DE = 15$ cm, $DF = 20$ cm, and angle $DEF = 110°$.




NOT TO SCALE

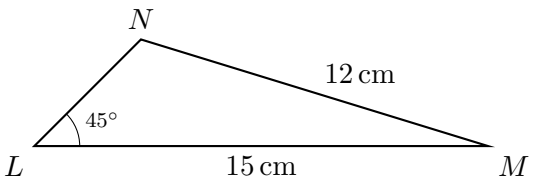
**(a)** Calculate angle $DFE$.

Angle $DFE =$ ..... ° **[3]**

**(b)** Calculate the length of $EF$.

$EF =$ ..... cm **[3]**

**1339** The diagram shows a triangle $LMN$.
$LM = 15$ cm, $MN = 12$ cm and angle $MLN = 45°$.
Angle $LNM$ is obtuse.



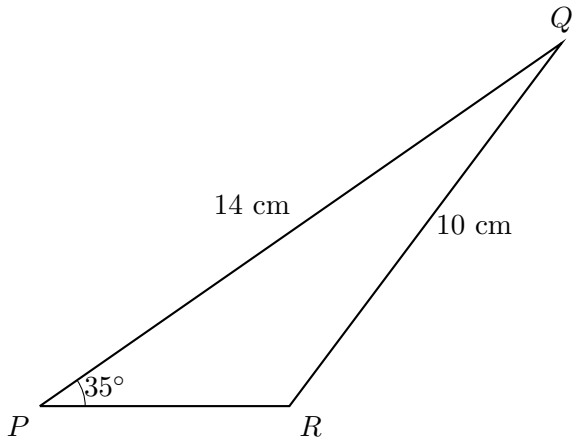

**(a)** Calculate the size of angle $LNM$.

Angle $LNM =$ ..... ° **[3]**

**(b)** Calculate the area of triangle $LMN$.

Area = ..... $cm^2$ **[3]**

**1340** The diagram shows triangle $PQR$.
$PQ = 14$ cm, $QR = 10$ cm and angle $QPR = 35°$.
Angle $PRQ$ is obtuse.



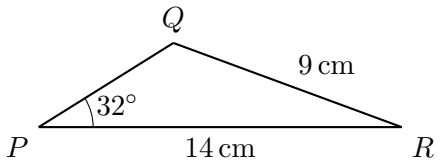

**(a)** Calculate angle $PRQ$.

Angle $PRQ =$ ..... ° **[3]**

**(b)** Calculate the area of triangle $PQR$.

Area = ..... $cm^2$ **[3]**

**1341** The diagram shows triangle $PQR$.
The angle $PQR$ is obtuse.



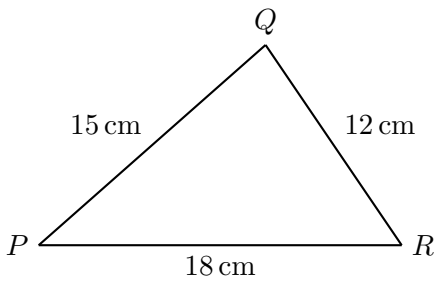

**(a)** Calculate the obtuse angle $PQR$.

..... ° **[4]**

**(b)** Calculate the area of triangle $PQR$.

..... $cm^2$ **[3]**

**1342** The diagram shows triangle $PQR$.
$PQ = 15\,\text{cm}$, $QR = 12\,\text{cm}$ and $PR = 18\,\text{cm}$.




**(a)** Use the cosine rule to show that angle $PQR = 82.8°$, correct to 1 decimal place.

**[3]**

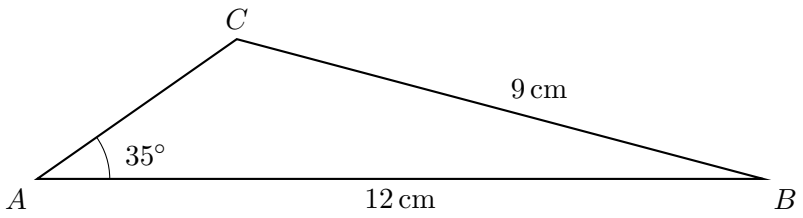
**(b)** Calculate the area of triangle $PQR$.

..... $cm^2$ **[2]**

**(c)** Calculate the shortest distance from $R$ to the line $PQ$.

..... cm **[2]**

**1343** In triangle $ABC$, $AB = 12\,\text{cm}$, $BC = 9\,\text{cm}$ and angle $BAC = 35°$.




NOT TO SCALE

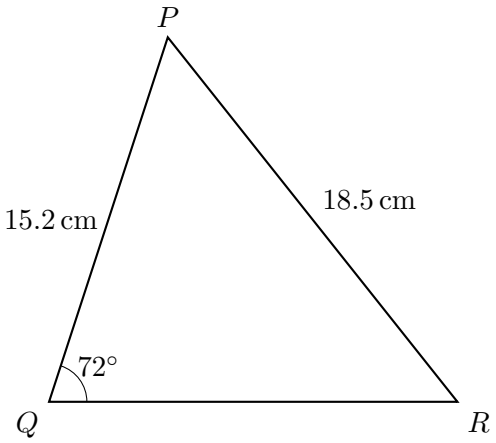
**(a)** Calculate the two possible values of angle $ACB$.
Give your answers correct to 1 decimal place.

..... ° **[4]**

**(b)** Given that angle $ACB$ is obtuse, calculate the area of triangle $ABC$.

Area = ..... $cm^2$ **[3]**

**1344** The diagram shows a triangle $PQR$.
$PQ = 15.2\,\text{cm}$, $PR = 18.5\,\text{cm}$ and angle $PQR = 72°$.




NOT TO SCALE

**(a)** Calculate the size of angle $PRQ$.

Angle $PRQ =$ ..... ° **[3]**

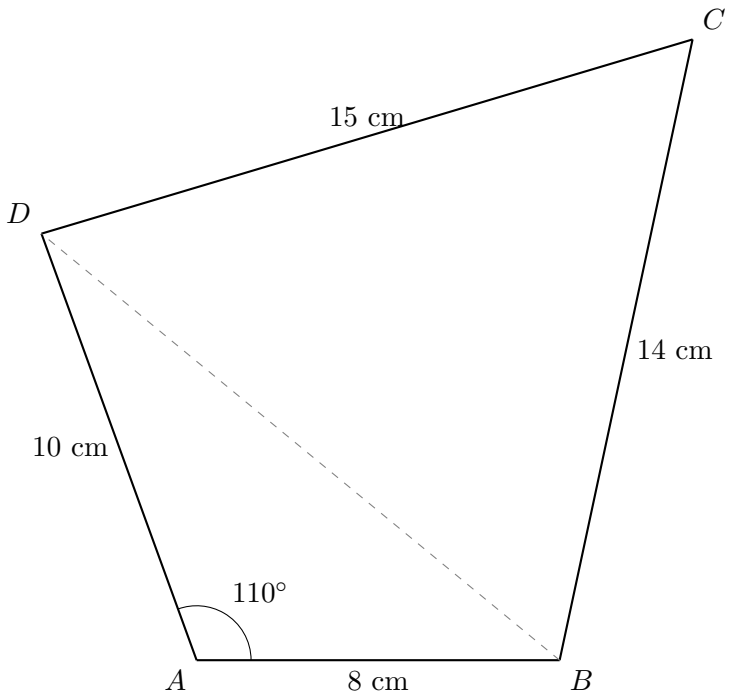
**(b)** Calculate the length of $QR$.

$QR =$ ..... cm **[3]**

**(c)** Calculate the shortest distance from $P$ to the line $QR$.

Distance = ..... cm **[2]**

**1345** The diagram shows a quadrilateral $ABCD$.
$AB = 8$ cm, $AD = 10$ cm, $BC = 14$ cm and $CD = 15$ cm.
Angle $BAD = 110°$.




**(a)** Calculate the length of the diagonal $BD$.

..... cm **[3]**

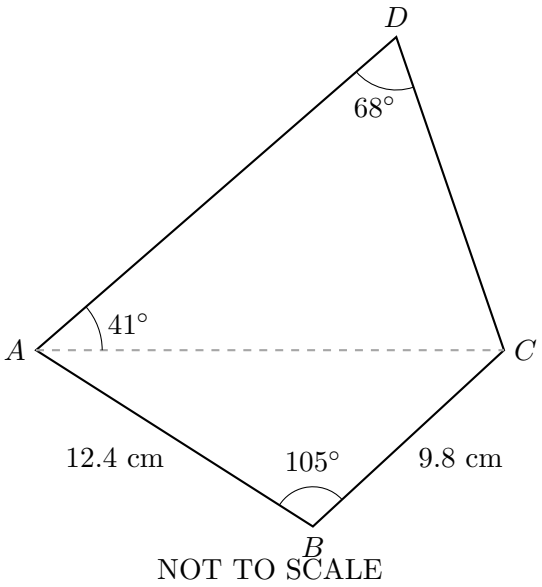
**(b)** Calculate the angle $BCD$.

..... ° **[3]**

**(c)** Calculate the area of the quadrilateral $ABCD$.

..... $cm^2$ **[3]**

**1346** The diagram shows a quadrilateral $ABCD$.
$AB = 12.4$ cm, $BC = 9.8$ cm and angle $ABC = 105°$.
Angle $CAD = 41°$ and angle $ADC = 68°$.



NOT TO SCALE

**(a)** Calculate the length of $AC$.
Give your answer correct to 3 significant figures.

$AC =$ ..... cm **[3]**

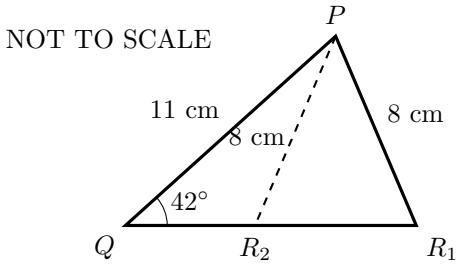
**(b)** Calculate the length of $CD$.
Give your answer correct to 3 significant figures.

$CD =$ ..... cm **[3]**

**(c)** Calculate the total area of the quadrilateral $ABCD$.

..... $cm^2$ **[3]**

**1347** In triangle $PQR$, $PQ = 11$ cm, $PR = 8$ cm, and angle $PQR = 42°$.
As shown in the diagram, there are two possible positions, $R_1$ and $R_2$, for vertex $R$.




**(a)** Calculate the two possible sizes of angle $PRQ$, giving your answers correct to 1 decimal place.

Acute angle = ..... , Obtuse angle = ..... **[4]**

**(b)** Write down the size of angle $QPR_1$ and the size of angle $QPR_2$, giving your answers correct to 1 decimal place.

$AngleQPR_1$ = ..... , $AngleQPR_2$ = ..... **[2]**

**(c)** Calculate the difference between the area of triangle $PQR_1$ and the area of triangle $PQR_2$, giving your answer correct to 3 significant figures.

Difference = ..... $cm^2$ **[4]**

**1348** In triangle $ABC$, $AB = 15$ cm, $BC = 12$ cm and angle $BAC = 48°$.

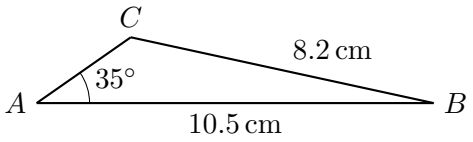
**(a)** Calculate the two possible values of angle $BCA$.

and ..... **[3]**

**(b)** Hence, calculate the difference between the two possible areas of triangle $ABC$.

..... $cm^2$ **[4]**

**1349** In triangle $ABC$, $AB = 10.5\,\text{cm}$, $BC = 8.2\,\text{cm}$ and angle $BAC = 35°$.
Angle $BCA$ is obtuse.




**(a)** Calculate the size of angle $BCA$.

..... ° **[3]**

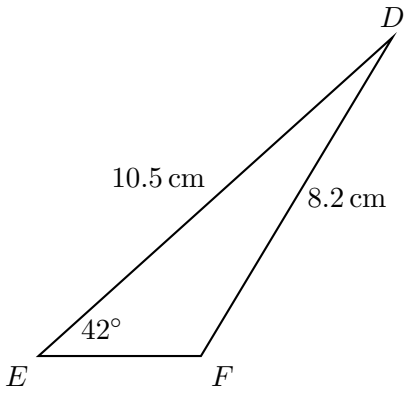
**(b)** Calculate the length of $AC$.

$AC =$ ..... cm **[3]**

**(c)** A point $D$ lies outside the triangle $ABC$ such that $CD = 6.4\,\text{cm}$ and angle $ACD$ is acute.
The area of triangle $ACD$ is $8.5\,\text{cm}^2$.
Calculate the size of angle $ACD$.

..... ° **[2]**

**1350** The diagram shows triangle $DEF$.
$DE = 10.5$ cm, $DF = 8.2$ cm and angle $DEF = 42°$.
Angle $DFE$ is an obtuse angle.




**(a)** Calculate angle $DFE$.

Angle $DFE =$ ..... ° **[3]**

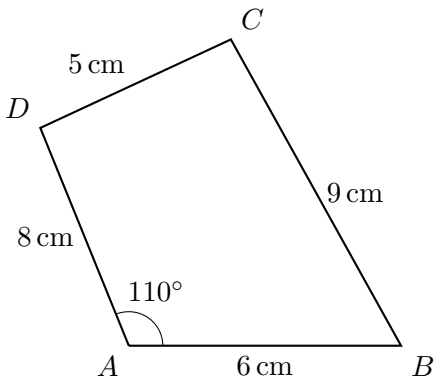
**(b)** Calculate the length of $EF$.

$EF =$ ..... cm **[3]**

**(c)** Calculate the shortest distance from $D$ to the line $EF$.

..... cm **[2]**

**1351** The diagram shows a quadrilateral $ABCD$.
$AB = 6$ cm, $AD = 8$ cm, $DC = 5$ cm, and $BC = 9$ cm.
Angle $DAB = 110°$.




NOT TO SCALE

**(a)** Calculate the length of the diagonal $BD$.

..... *cm* **[3]**

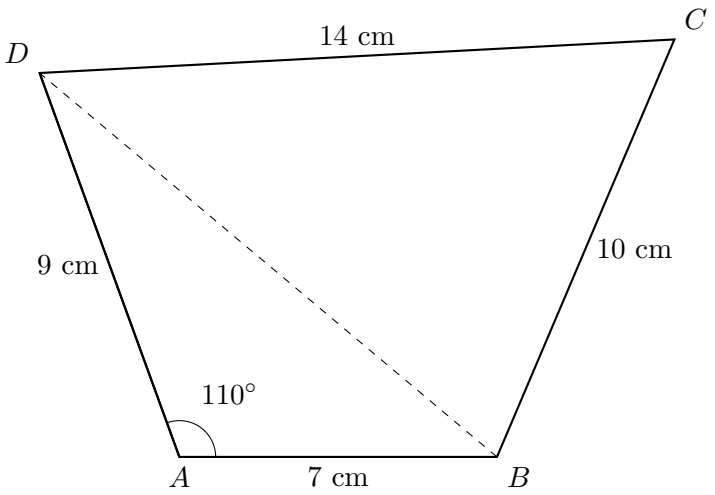
**(b)** Calculate the size of angle $BCD$.

..... ° **[3]**

**(c)** Calculate the total area of the quadrilateral $ABCD$.

..... $cm^2$ **[3]**

**1352** The diagram shows a quadrilateral $ABCD$.
$AB = 7$ cm, $AD = 9$ cm, $BC = 10$ cm and $CD = 14$ cm.
Angle $BAD = 110°$.




**(a)** Calculate the length of $BD$.

..... cm **[3]**

**(b)** Calculate angle $BCD$.

..... ° **[3]**

**(c)** Calculate the total area of the quadrilateral $ABCD$.

..... $cm^2$ **[3]**

**1353** The diagram shows a quadrilateral $ABCD$.
$AB = 7.4$ cm, $BC = 9.1$ cm, and $CD = 10.5$ cm.
Angle $ABC = 112°$ and angle $CAD = 43°$.

**(a)** Calculate the length of $AC$.

..... cm **[3]**

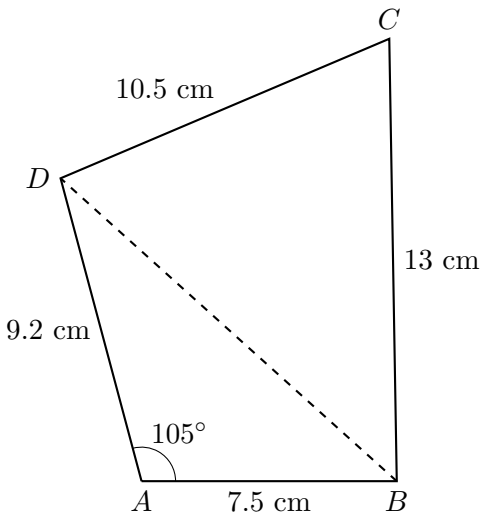
**(b)** Calculate the acute angle $ADC$.

..... ° **[3]**

**(c)** Calculate the shortest distance from $D$ to the line $AC$.

..... cm **[3]**

**1354** The diagram shows a quadrilateral $ABCD$.
$AB = 7.5$ cm, $AD = 9.2$ cm and angle $BAD = 105°$.
$BC = 13$ cm and $CD = 10.5$ cm.




**(a)** Calculate the length of $BD$.

$BD =$ ..... cm **[3]**

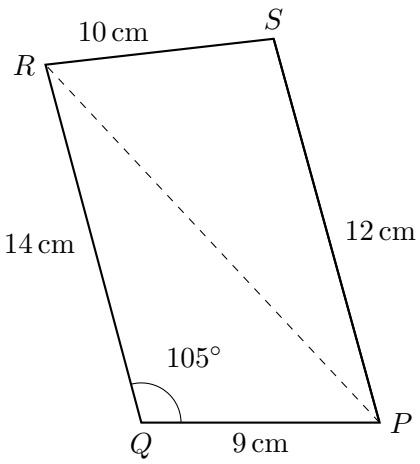
**(b)** Calculate angle $BCD$.

Angle $BCD =$ ..... ° **[3]**

**(c)** Calculate the area of the quadrilateral $ABCD$.

Area = ..... $cm^2$ **[3]**

**1355** The diagram shows a quadrilateral $PQRS$.
$PQ = 9$ cm, $QR = 14$ cm, and angle $PQR = 105°$.
$PS = 12$ cm and $RS = 10$ cm.



NOT TO SCALE

**(a)** Calculate the length of the diagonal $PR$.

$PR =$ ..... cm **[3]**

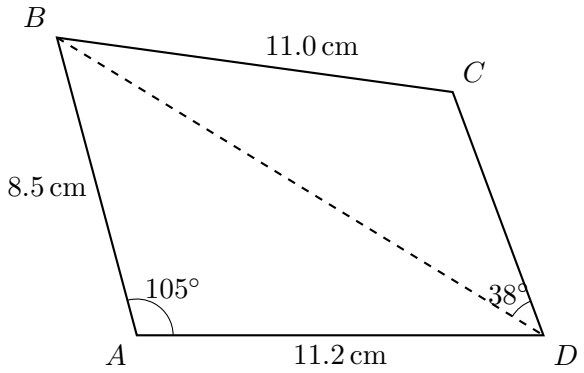
**(b)** Calculate angle $PSR$.

Angle $PSR =$ ..... ° **[4]**

**(c)** Calculate the total area of the quadrilateral $PQRS$.

Area = ..... $cm^2$ **[3]**

**1356** The diagram shows a quadrilateral $ABCD$.
$AB = 8.5\,\text{cm}$, $AD = 11.2\,\text{cm}$ and angle $BAD = 105°$.
$BC = 11.0\,\text{cm}$ and angle $BDC = 38°$.
Angle $BCD$ is obtuse.




NOT TO SCALE

**(a)** Calculate the length of $BD$.

$BD =$ ..... cm **[3]**

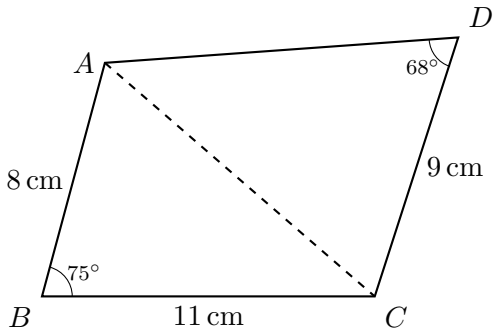
**(b)** Calculate the size of angle $BCD$.

Angle $BCD =$ ..... ° **[3]**

**(c)** Calculate the area of the quadrilateral $ABCD$.

Area = ..... $cm^2$ **[4]**

**1357** The diagram shows a quadrilateral $ABCD$.
$AB = 8$ cm, $BC = 11$ cm and $CD = 9$ cm.
Angle $ABC = 75°$ and angle $ADC = 68°$.




**(a)** Calculate the length of $AC$.

$AC =$ ..... $cm$ **[3]**

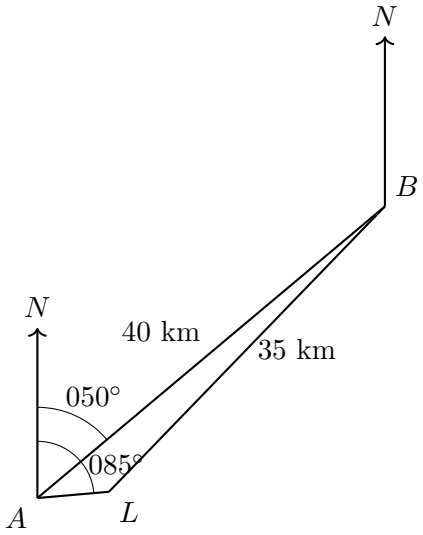
**(b)** Calculate the size of angle $CAD$.

Angle $CAD =$ ..... $°$ **[3]**

**(c)** Calculate the area of the quadrilateral $ABCD$.

Area = ..... $cm^2$ **[4]**

**1358** Ship $A$ and Ship $B$ are at sea.
Ship $B$ is 40 km from Ship $A$ on a bearing of 050°.
A lighthouse $L$ is 35 km from Ship $B$.
The bearing of $L$ from $A$ is 085°.
Given that the angle $ALB$ is obtuse:




**(a)** Calculate the obtuse angle $ALB$.

..... ° **[4]**

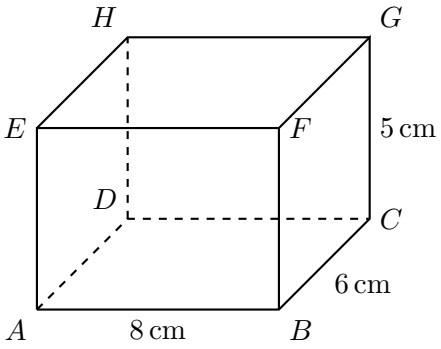
**(b)** Calculate the distance $AL$.

..... km **[3]**

**(c)** Calculate the bearing of $L$ from $B$.

..... ° **[3]**

**1359** The diagram shows a cuboid $ABCDEFGH$.
$AB = 8$ cm, $BC = 6$ cm and $CG = 5$ cm.



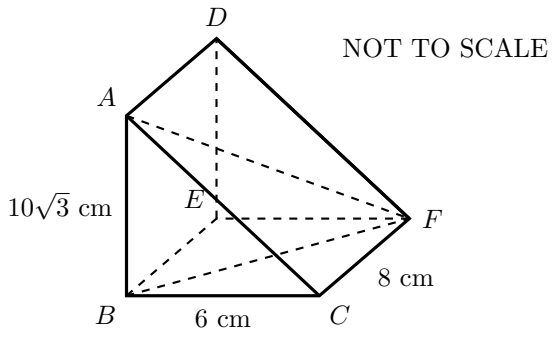

**(a)** Calculate the exact length of the diagonal $AC$.

$AC =$ ..... $cm$ **[2]**

**(b)** Calculate the exact length of the space diagonal $AG$.
Give your answer in the form $a\sqrt{b}$, where $a$ and $b$ are integers.

$AG =$ ..... $cm$ **[3]**

**1360** The diagram shows a triangular prism $ABCDEF$ resting on a horizontal rectangular base $BCFE$, where $BC = 6$ cm and $BE = 8$ cm. The rectangular face $ABED$ is vertical with $AB = 10\sqrt{3}$ cm.




**(a)** Calculate the length of $BF$.

$BF =$ ..... cm **[2]**

**(b)** Calculate the length of $AF$.

$AF =$ ..... cm **[2]**

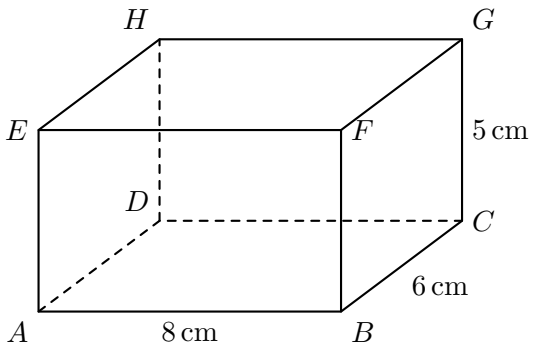
**(c)** Calculate the size of the angle between the line $AF$ and the horizontal base $BCFE$.

..... ° **[2]**

**(d)** Calculate the exact area of the rectangular face $ACFD$, giving your answer in the form $k\sqrt{21}$, where $k$ is an integer.

Area = ..... $cm^2$ **[3]**

**1361** The diagram shows a cuboid $ABCDEFGH$.
$AB = 8$ cm, $BC = 6$ cm and $CG = 5$ cm.




**(a)** Calculate the length of the diagonal $AC$.

..... cm **[2]**

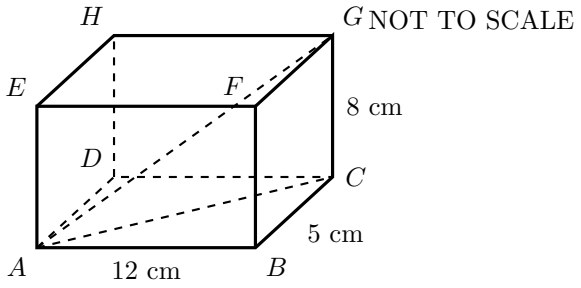
**(b)** Calculate the length of the space diagonal $AG$.

..... cm **[2]**

**(c)** Calculate the angle between the line $AG$ and the base $ABCD$.

..... ° **[2]**

**1362** The diagram shows a cuboid $ABCDEFGH$ with a horizontal rectangular base $ABCD$.
$AB = 12$ cm, $BC = 5$ cm, and $CG = 8$ cm.




**(a)** Calculate the length of $AC$.

$AC =$ ..... cm **[2]**

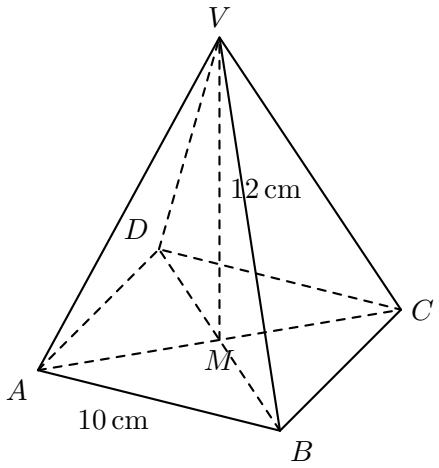
**(b)** Calculate the length of the diagonal $AG$, giving your answer correct to 3 significant figures.

$AG =$ ..... cm **[2]**

**(c)** Calculate the angle that the diagonal $AG$ makes with the horizontal base $ABCD$, giving your answer correct to 1 decimal place.

..... ° **[3]**

**1363** The diagram shows a pyramid $VABCD$ with a square base $ABCD$ of side 10 cm.
The point $M$ is the centre of the base and $V$ is vertically above $M$.
The vertical height $VM$ of the pyramid is 12 cm.



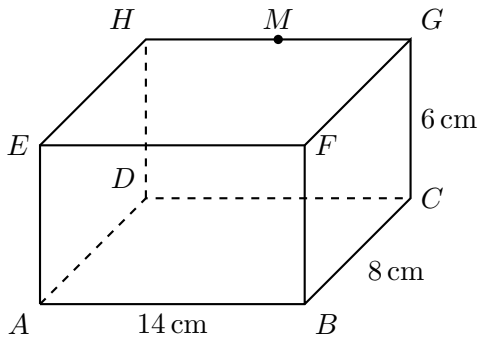

**(a)** Calculate the length of the edge $VA$.

..... cm **[3]**

**(b)** Calculate the angle between the edge $VA$ and the base $ABCD$.

..... ° **[2]**

**1364** The diagram shows a cuboid $ABCDEFGH$.
$AB = 14$ cm, $BC = 8$ cm and $CG = 6$ cm.
The point $M$ is the midpoint of the edge $GH$.




**(a)** Calculate the length of the diagonal $BD$.

$BD =$ ..... $cm$ **[2]**

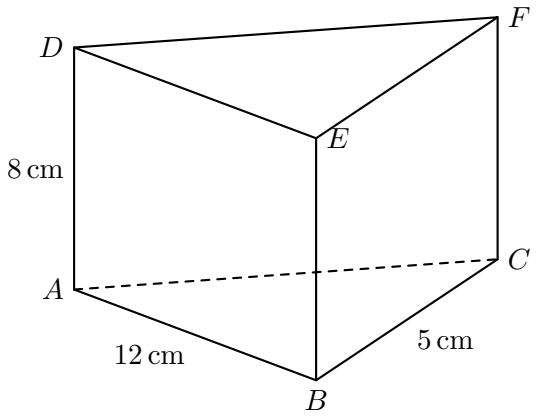
**(b)** Calculate the angle between the line $DF$ and the base $ABCD$.

Angle = ..... ° **[3]**

**(c)** Calculate the length of $AM$.

$AM =$ ..... $cm$ **[3]**

**1365** The diagram shows a right triangular prism $ABCDEF$.
The base $ABC$ is a horizontal right-angled triangle with angle $ABC = 90°$.
$AB = 12$ cm, $BC = 5$ cm and the vertical height $AD = 8$ cm.




**(a)** Calculate the length of $AF$.

..... cm **[3]**

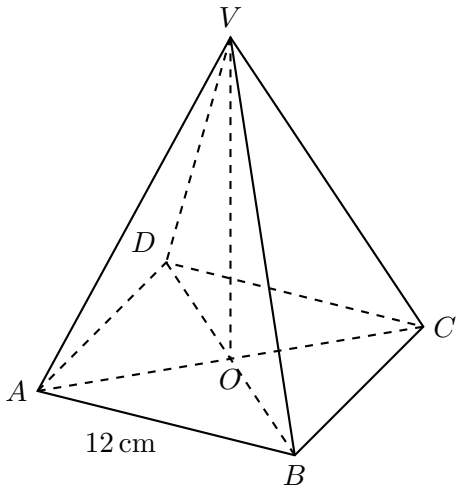
**(b)** Calculate the angle between the line $AF$ and the base $ABC$.

..... ° **[2]**

**(c)** Calculate the angle between the line $AF$ and the vertical plane $ABED$.

..... ° **[3]**

**1366** The diagram shows a solid right pyramid $VABCD$ with a square base $ABCD$.
The vertex $V$ is directly above the centre $O$ of the base.
The sides of the square base are 12 cm and the volume of the pyramid is 384 $cm^3$.




**(a)** Show that the vertical height, $VO$, is 8 cm.

**[2]**

**(b)** Calculate the length of the slant edge $VA$.

$VA =$ ..... $cm$ **[3]**

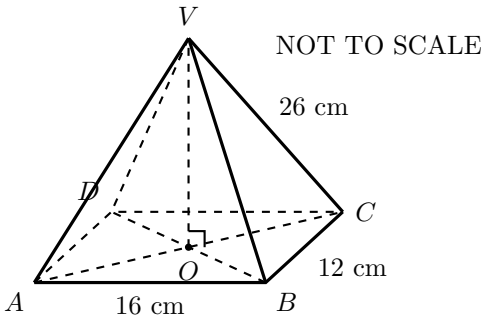
**(c)** Calculate the angle between the edge $VA$ and the base $ABCD$.

Angle = ..... ° **[2]**

**(d)** Calculate the angle between the triangular face $VAB$ and the base $ABCD$.

Angle = ..... ° **[3]**

**1367** The diagram shows a solid right pyramid $VABCD$ with a horizontal rectangular base $ABCD$ of length $AB = 16$ cm and width $BC = 12$ cm. The apex $V$ is vertically above the centre $O$ of the base, and each slant edge $VA = VB = VC = VD = 26$ cm.




**(a)** Show that the vertical height $VO$ of the pyramid is 24 cm.

**[3]**

**(b)** Calculate the angle that the slant edge $VA$ makes with the horizontal base $ABCD$, giving your answer correct to 1 decimal place.

..... ° **[2]**

**(c)** $M$ is the midpoint of $BC$.

Calculate

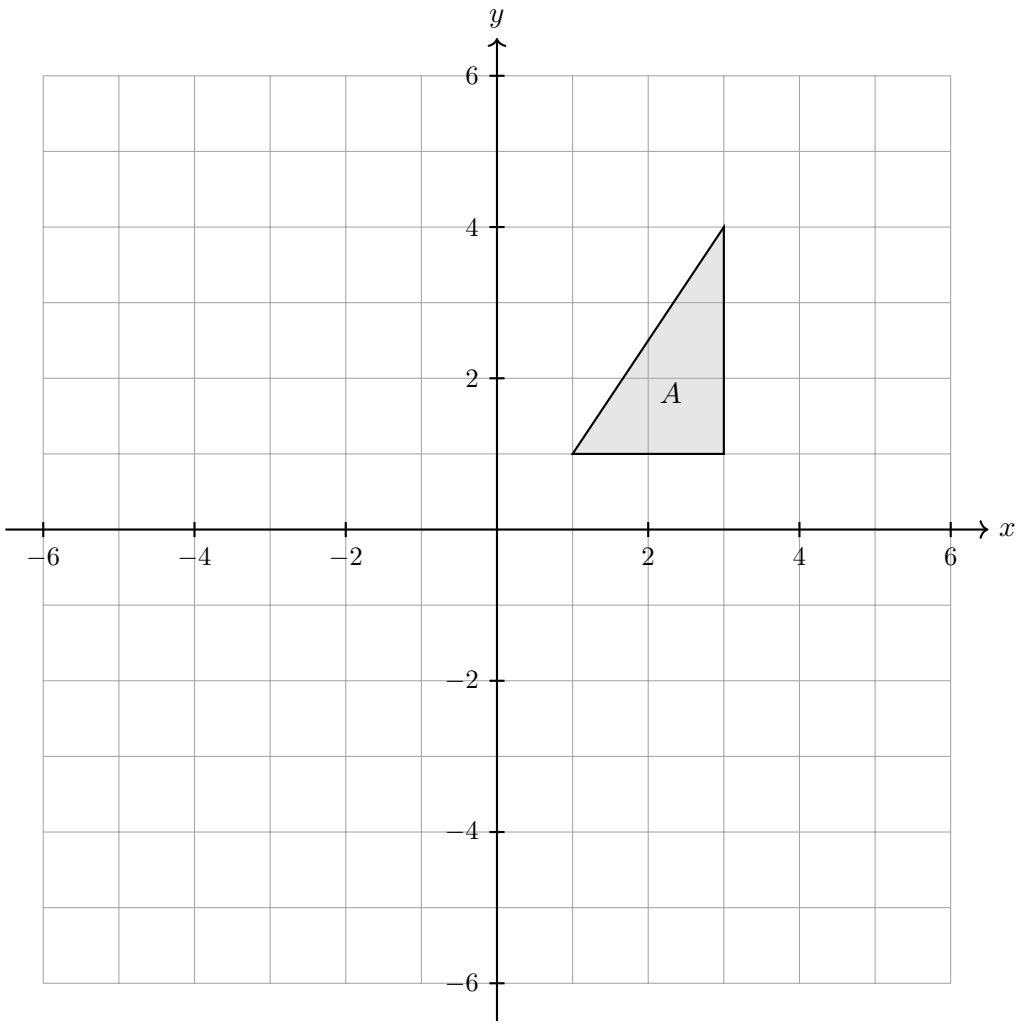
**(i)** the angle that the triangular face $VBC$ makes with the horizontal base $ABCD$, giving your answer correct to 1 decimal place,

..... ° **[3]**

**(ii)** the shortest distance from the centre $O$ to the triangular face $VBC$, giving your answer correct to 3 significant figures.

Distance = ..... cm **[3]**

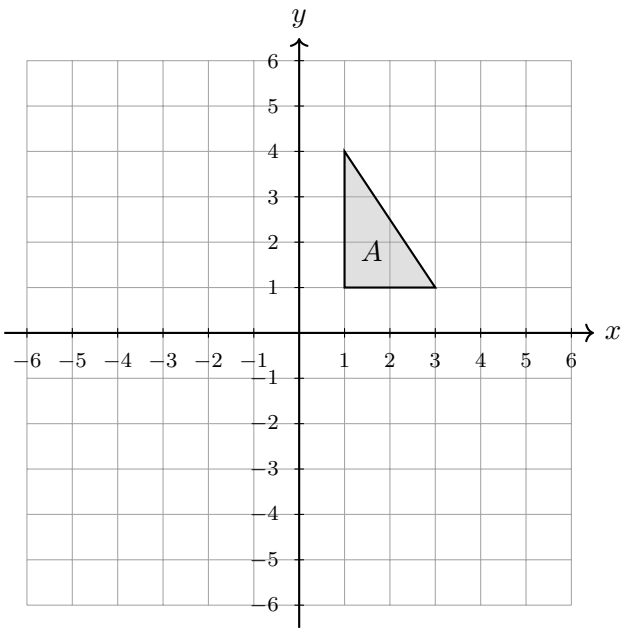
**1368** The grid shows triangle $A$.




**(a)** On the grid, draw the image of triangle $A$ after a translation by the vector $\begin{pmatrix} -4 \\ 2 \end{pmatrix}$.
Label the image $B$. **[2]**

**(b)** On the grid, draw the image of triangle $A$ after a reflection in the line $x = -1$.
Label the image $C$. **[2]**

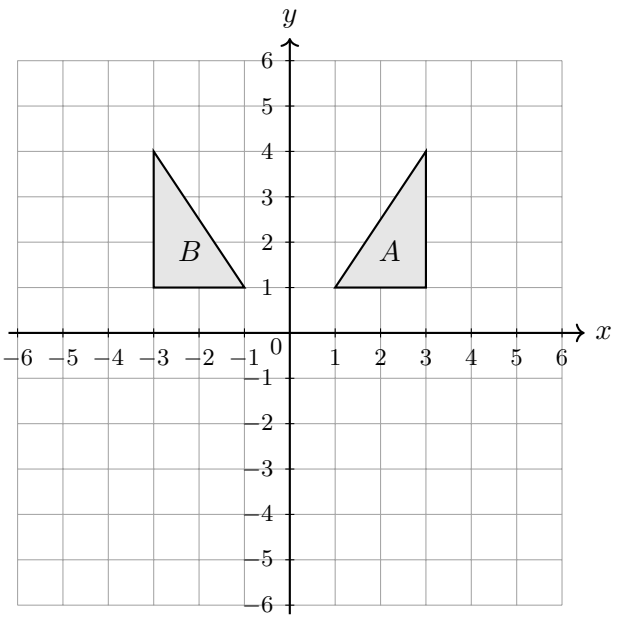
**1369** The diagram shows a triangle, Shape $A$, drawn on a coordinate grid.




**(a)** On the grid, draw the image of Shape $A$ after a translation by the vector $\begin{pmatrix} -4 \\ 2 \end{pmatrix}$.
Label this image $B$. **[2]**

**(b)** On the grid, draw the image of Shape $A$ after a reflection in the line $y = -1$.
Label this image $C$. **[2]**

**1370** The diagram shows two triangles, $A$ and $B$, plotted on a coordinate grid.




**(a)** On the grid, draw the image of triangle $A$ after a translation by the vector $\begin{pmatrix} -4 \\ 2 \end{pmatrix}$. **[2]**

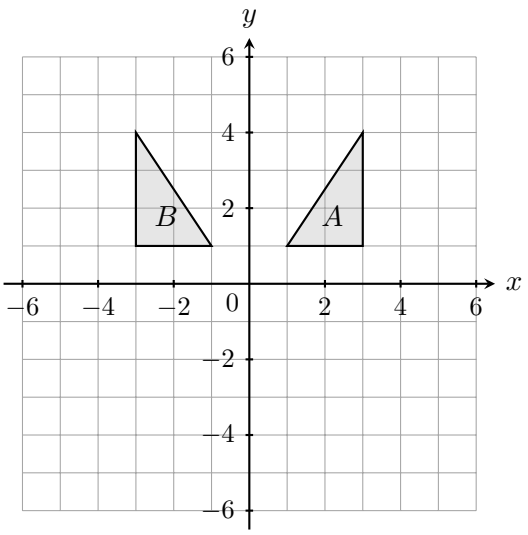
**(b)** Describe fully the single transformation that maps triangle $A$ onto triangle $B$.

.....

.....

..... **[2]**

**1371** The grid shows triangle $A$ and triangle $B$.




**(a)** On the grid, draw the image of triangle $A$ after a translation by the vector $\begin{pmatrix} -2 \\ -5 \end{pmatrix}$. **[2]**

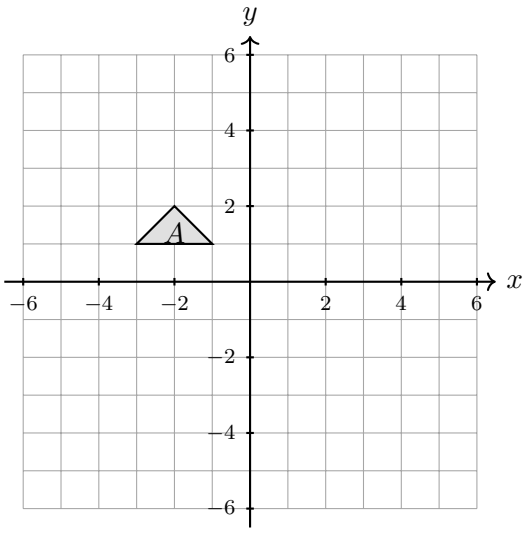
**(b)** Describe fully the single transformation that maps triangle $A$ onto triangle $B$.

.....
.....
.....

**[2]**

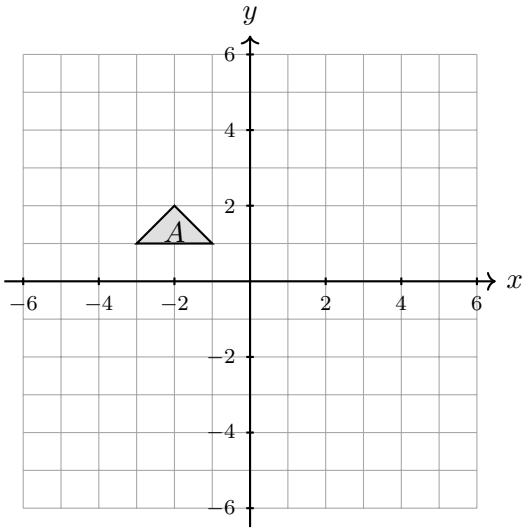
**1372** A triangle $A$ is drawn on a grid.

**(a)** On the grid above, draw the image of triangle $A$ after a translation by the vector $\begin{pmatrix} 4 \\ -3 \end{pmatrix}$.



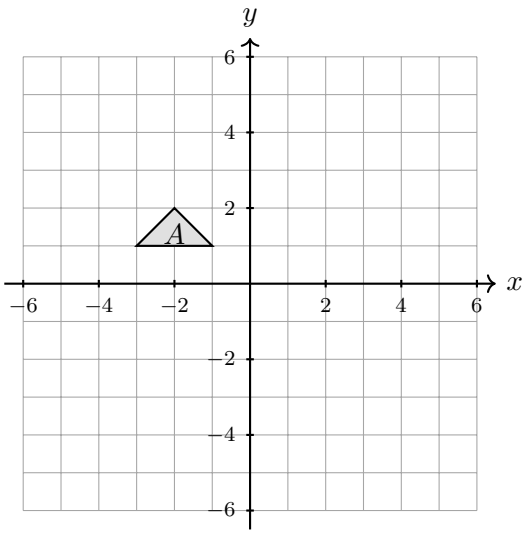

**[2]**

**(b)** On the grid above, draw the image of triangle $A$ after a reflection in the line $y = x$.



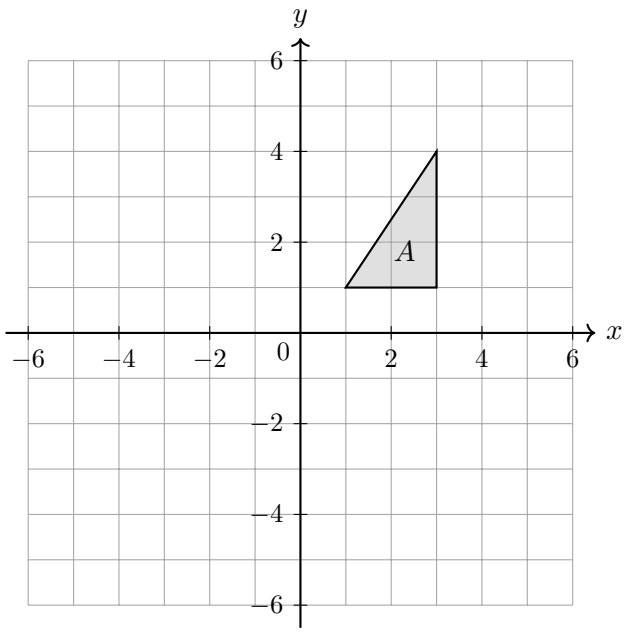

**[2]**

**(c)** On the grid above, draw the image of triangle $A$ after an enlargement with scale factor 2 and centre $(-1, 0)$.




**[2]**

**1373** The diagram shows triangle $A$ on a coordinate grid.

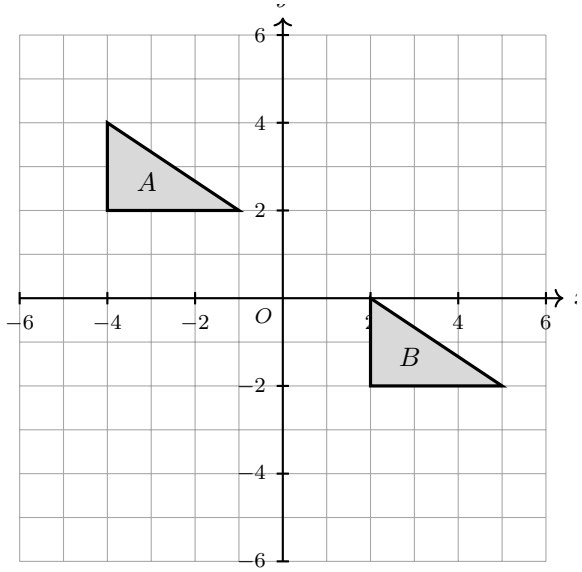



**(a)** On the grid, translate shape $A$ by the vector $\begin{pmatrix} -5 \\ 2 \end{pmatrix}$.
Label the image $B$. **[2]**

**(b)** On the grid, reflect shape $A$ in the $x$-axis.
Label the image $C$. **[2]**

**(c)** On the grid, rotate shape $A$ by 180° about the origin $(0, 0)$.
Label the image $D$. **[2]**

**1374** The diagram shows triangles $A$ and $B$ on a coordinate grid.




**(a)** Describe fully the single transformation that maps triangle $A$ onto triangle $B$.

.....

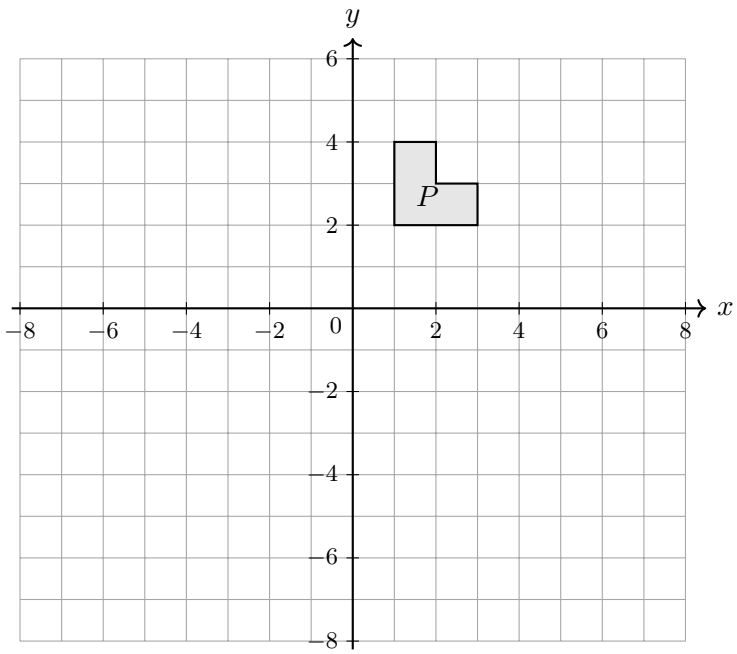
.....

..... **[2]**

**(b)** On the grid, draw the image of triangle $A$ after a reflection in the line $y = 1$.
Label the image $C$. **[2]**

**(c)** On the grid, draw the image of triangle $A$ after a rotation of 90° clockwise about the origin $(0, 0)$.
Label the image $D$. **[3]**

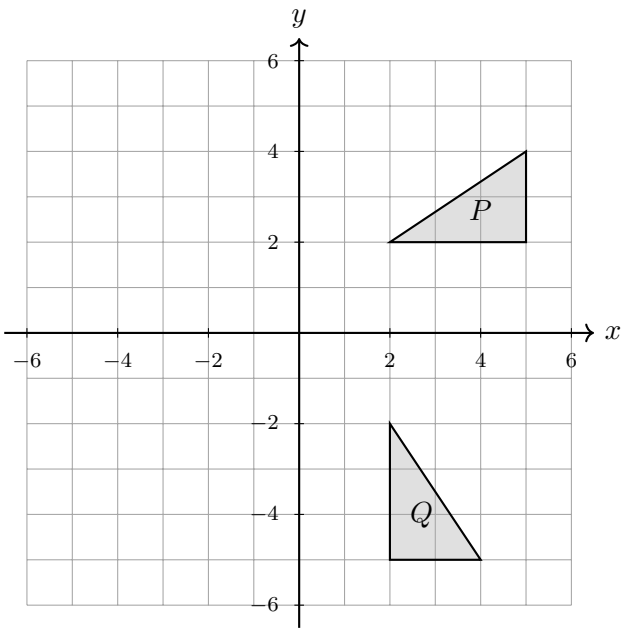
**1375** The diagram shows a shape $P$ plotted on a coordinate grid.




**(a)** On the grid, draw the image of shape $P$ after an enlargement, scale factor $-2$, centre $(0, 0)$. **[2]**

**(b)** On the grid, draw the image of shape $P$ after a rotation of $90°$ clockwise about the point $(1, -1)$. **[2]**

**1376** The diagram shows two triangles, Shape $P$ and Shape $Q$, drawn on a grid.




**(a)** Describe fully the single transformation that maps Shape $P$ onto Shape $Q$.

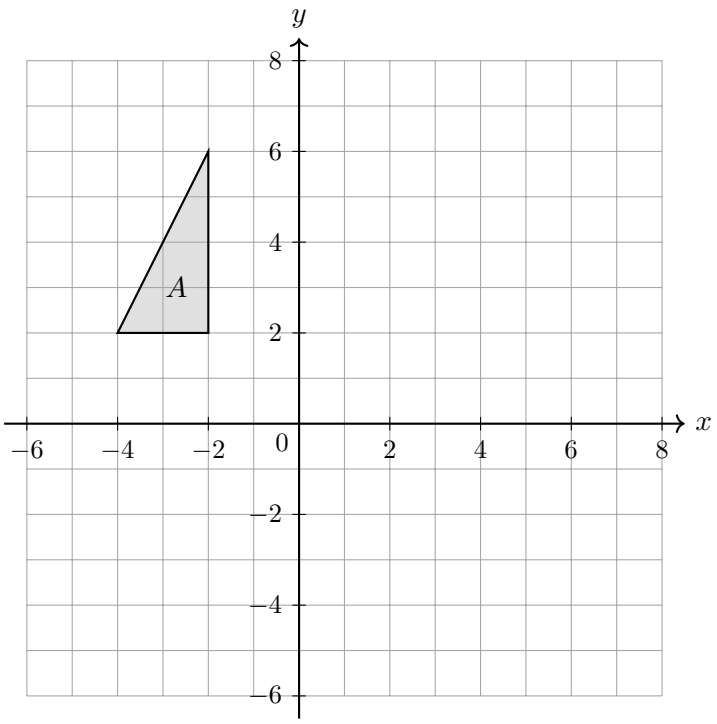
.....

.....

..... **[3]**

**(b)** Shape $P$ is enlarged by a scale factor of $\frac{1}{2}$ with centre of enlargement $(-2, 2)$.
On the grid, draw the image of Shape $P$ and label it $R$. **[2]**

**1377** The diagram shows shape $A$ on a coordinate grid.




**(a)** Shape $A$ is reflected in the line $y = x$ to give shape $B$.
On the grid, draw and label shape $B$. **[2]**

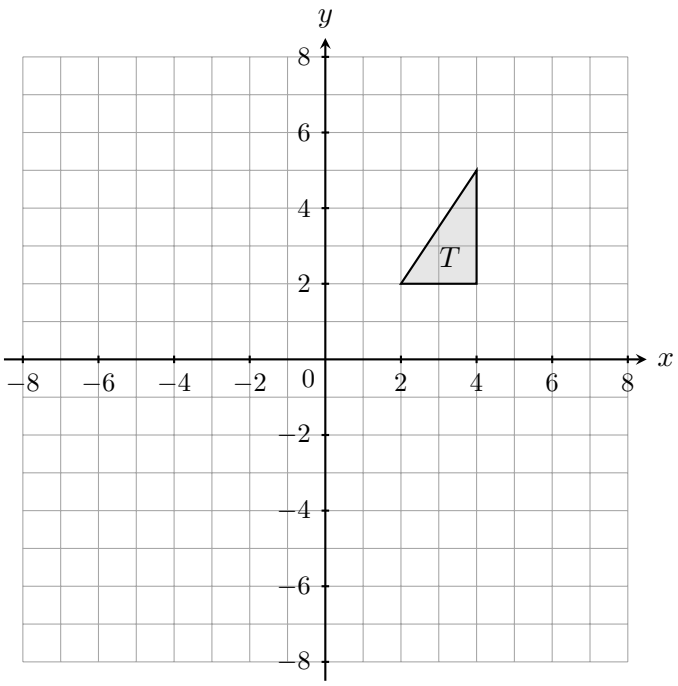
**(b)** Shape $B$ is rotated by 90° clockwise about the point $(2, 2)$ to give shape $C$.
On the grid, draw and label shape $C$. **[2]**

**(c)** Describe fully the single transformation that maps shape $A$ onto shape $C$.

.....

..... **[2]**

**1378** The grid shows triangle $T$.




**(a)** On the grid, draw the image of triangle $T$ after an enlargement with scale factor $-\frac{1}{2}$ and centre $(0, 2)$. **[2]**

**(b)** Triangle $T$ is reflected in the line $x = 1$ to give triangle $U$.
Triangle $U$ is then reflected in the line $y = 2$ to give triangle $V$.
Describe fully the single transformation that maps triangle $T$ onto triangle $V$.

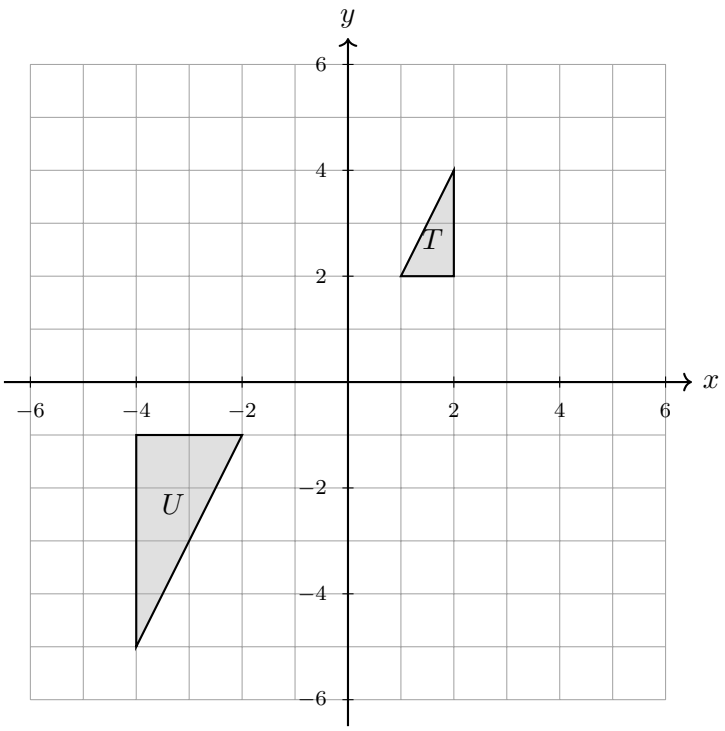
.....
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**[3]**

**(c)** Triangle $T$ is enlarged by a scale factor of $k$ to give triangle $W$.
The area of triangle $W$ is 48 square units.
Find the value of $k$, given that $k > 0$.

$k =$ ..... **[2]**

**1379** The diagram shows two triangles, Shape $T$ and Shape $U$.




**(a)** Describe fully the single transformation that maps Shape $T$ onto Shape $U$.

.....

.....

..... **[3]**

**(b)** Describe fully the single transformation that maps Shape $U$ onto Shape $T$.

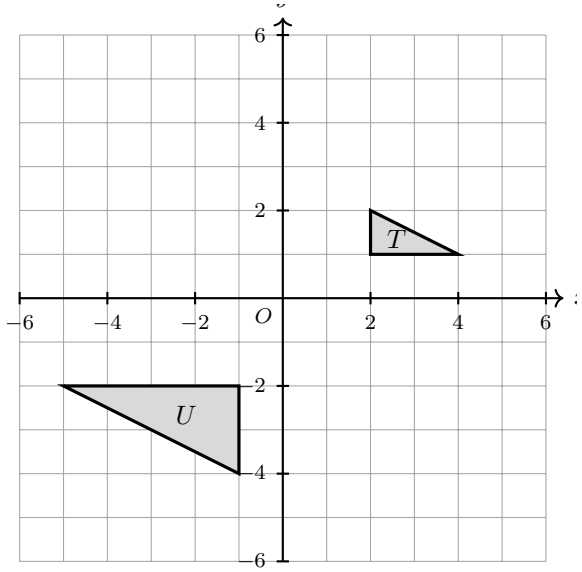
.....

..... **[2]**

**(c)** Shape $T$ is translated by the vector $\begin{pmatrix} -3 \\ 1 \end{pmatrix}$ to give Shape $V$.
Shape $V$ is then reflected in the line $y = x$ to give Shape $W$.
Find the coordinates of the three vertices of Shape $W$.

..... **[3]**

**1380** The diagram shows triangles $T$ and $U$ on a coordinate grid.




**(a)** Describe fully the single transformation that maps triangle $T$ onto triangle $U$.

.....

.....

..... **[3]**

**(b)** Triangle $T$ is reflected in the line $x = 1$ to give triangle $V$.
Triangle $V$ is then reflected in the line $y = 2$ to give triangle $W$.

**(i)** On the grid, draw and label triangle $W$. **[3]**

**(ii)** Describe fully the single transformation that maps triangle $T$ directly onto triangle $W$.

.....

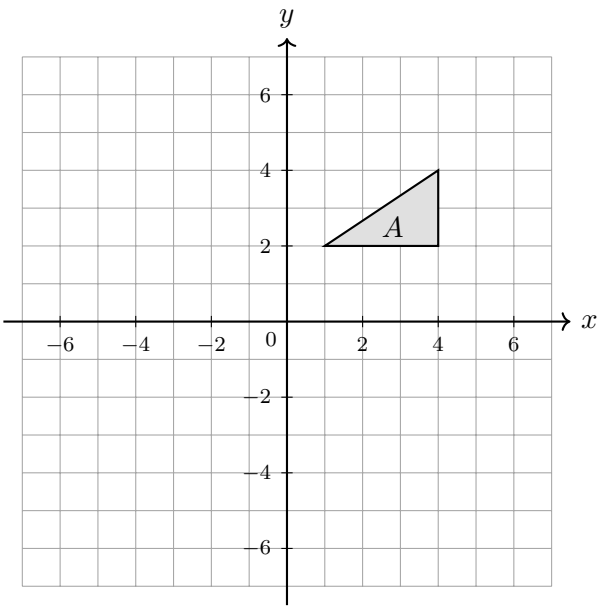
.....

.....
**[3]**

**(c)** Point $P(a, b)$ is mapped onto the point $P'(5, -3)$ by a reflection in the line $y = x$ followed by a translation by the vector $\begin{pmatrix} 2 \\ -4 \end{pmatrix}$.
Find the coordinates of $P$.

$P = (\ldots\ldots\ ,\ \ldots\ldots)$ **[3]**

**1381** The diagram shows triangle $A$ on a coordinate grid.

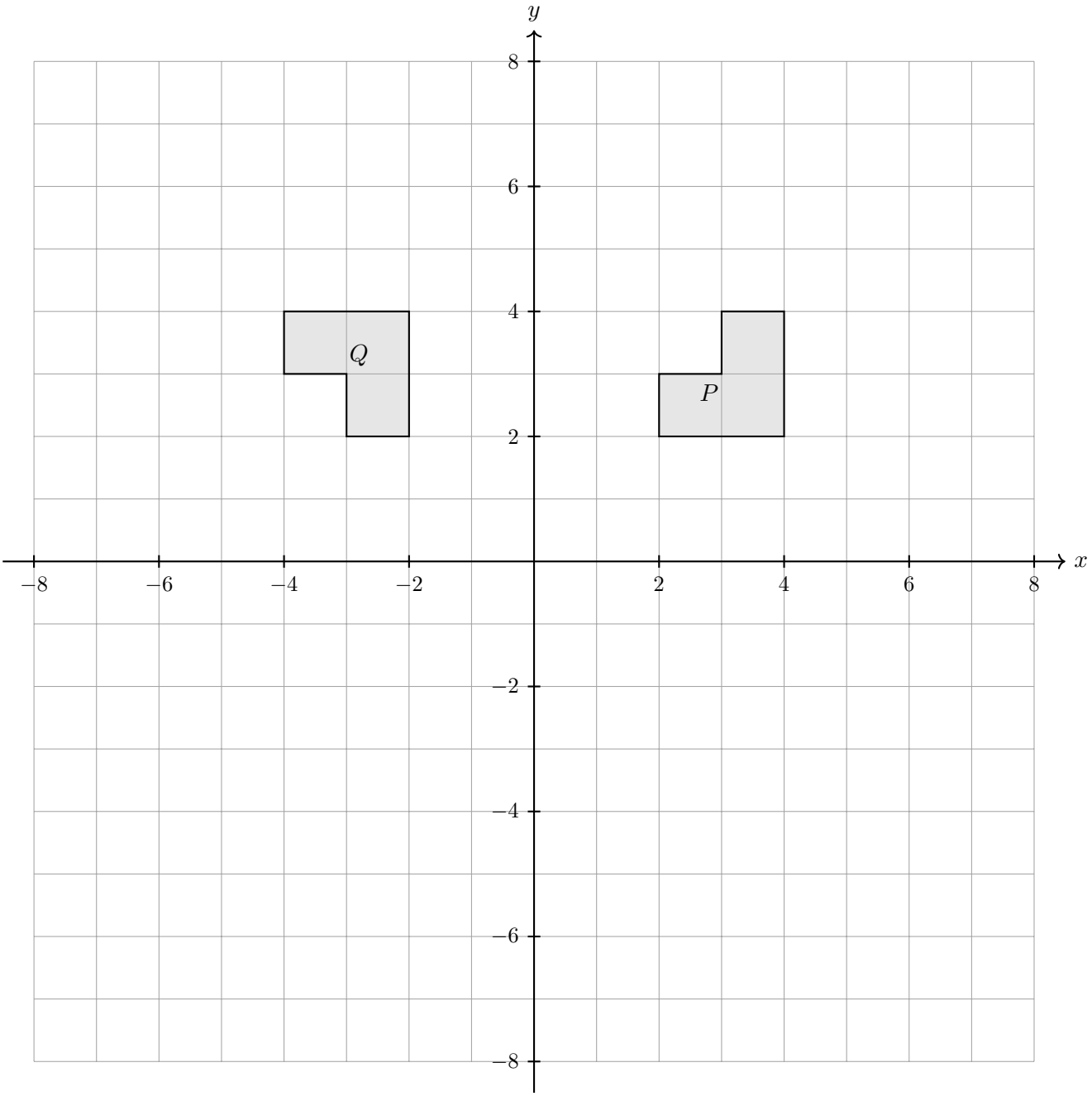



**(a)** On the grid, draw the image of triangle $A$ after a translation by the vector $\begin{pmatrix} -6 \\ -3 \end{pmatrix}$. **[2]**

**(b)** On the grid, draw the image of triangle $A$ after a reflection in the line $y = 1$. **[2]**

**(c)** On the grid, draw the image of triangle $A$ after a rotation of $180°$ about the origin $(0, 0)$. **[2]**

**1382** The grid shows shapes $P$ and $Q$.




**(a)** Describe fully the single transformation that maps shape $P$ onto shape $Q$.

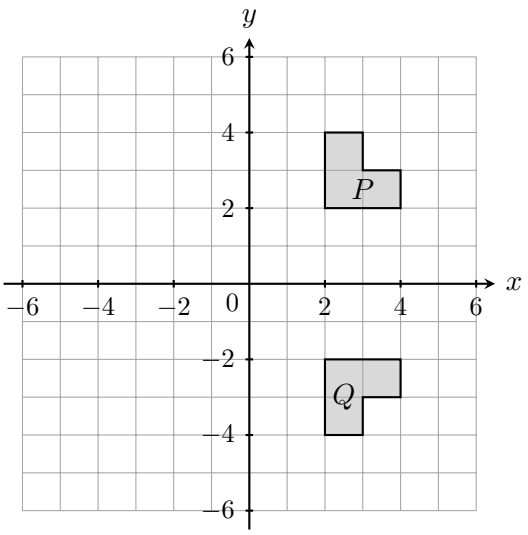
.....

.....

..... **[3]**

**(b)** On the grid, enlarge shape $P$ by scale factor $\frac{1}{2}$, centre $(6, 6)$.
Label the image $R$. **[2]**

**1383** The grid shows shapes $P$ and $Q$.



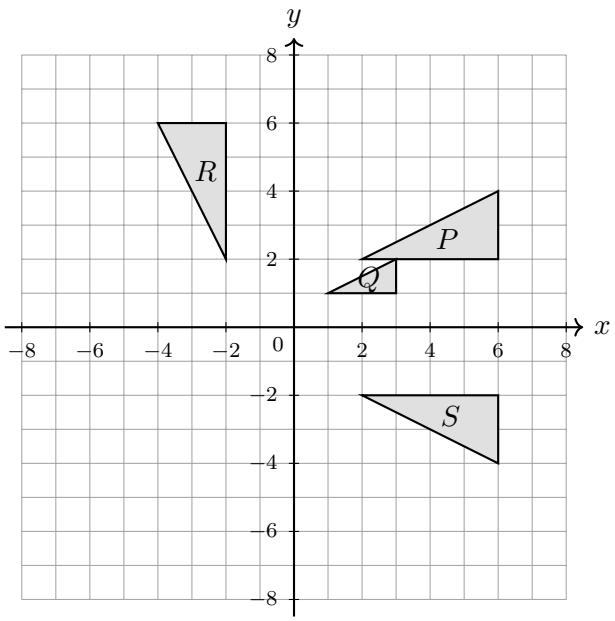

**(a)** On the grid, draw the image of shape $P$ after an enlargement with scale factor 0.5 and centre $(0, 0)$. **[2]**

**(b)** Describe fully the single transformation that maps shape $P$ onto shape $Q$.

.....
.....
.....

**[3]**

**1384** The diagram shows triangles $P$, $Q$, $R$, and $S$ on a coordinate grid.




**(a)** Describe fully the single transformation that maps triangle $P$ onto triangle $Q$.

.....

.....

..... **[3]**

**(b)** Describe fully the single transformation that maps triangle $P$ onto triangle $R$.

.....

.....

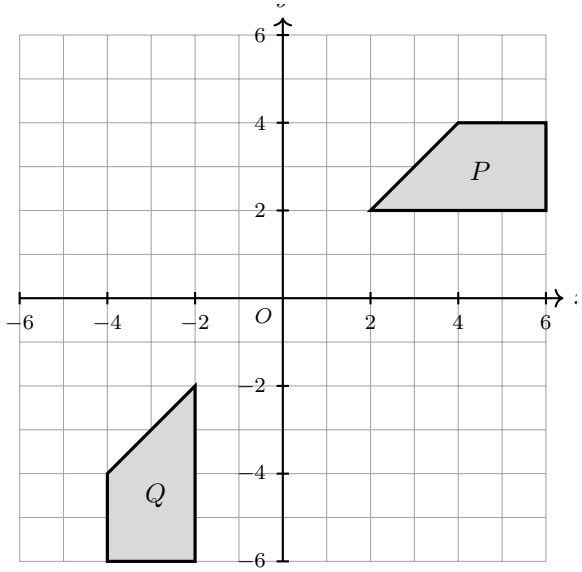
..... **[3]**

**(c)** Describe fully the single transformation that maps triangle $P$ onto triangle $S$.

.....

..... **[2]**

**1385** The diagram shows shapes $P$ and $Q$ on a coordinate grid.




**(a)** Describe fully the single transformation that maps shape $P$ onto shape $Q$.

.....

.....

..... **[2]**

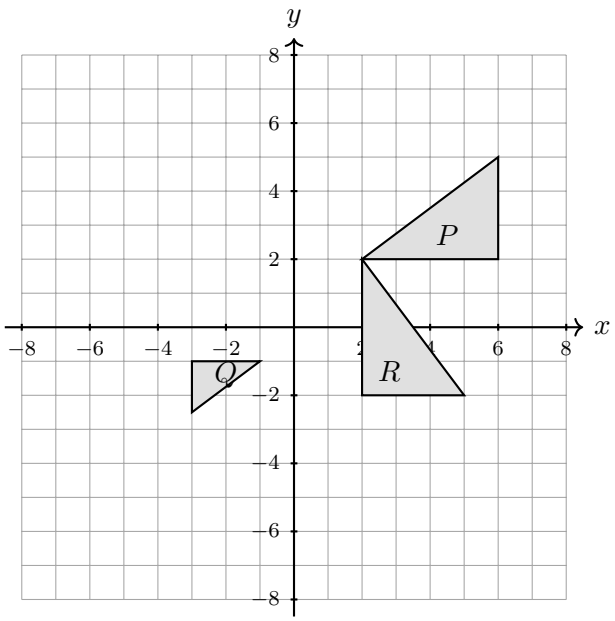
**(b)** On the grid, draw the image of shape $P$ after an enlargement with scale factor $\frac{1}{2}$ and centre $(-2, -2)$.
Label the image $R$. **[3]**

**(c)** On the grid, draw the image of shape $P$ after a rotation of 90° anticlockwise about the point $(1, 1)$.
Label the image $S$. **[3]**

**(d)** Find the ratio of the area of shape $R$ to the area of shape $P$, giving your answer in its simplest form.

..... **[1]**

**1386** The diagram shows triangles $P$, $Q$ and $R$.




**(a)** Describe fully the single transformation that maps triangle $P$ onto triangle $Q$.

.....

.....

..... **[3]**

**(b)** Describe fully the single transformation that maps triangle $P$ onto triangle $R$.

.....

.....

..... **[3]**

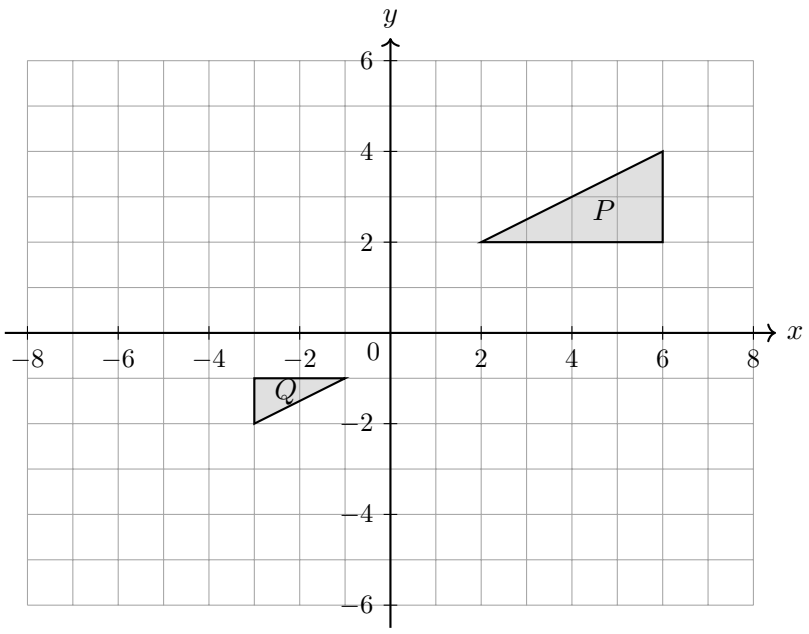
**(c)** Triangle $Q$ is reflected in the origin $(0, 0)$ to give triangle $S$.
Describe fully the single transformation that maps triangle $P$ onto triangle $S$.

.....

.....

..... **[3]**

**1387** The diagram shows triangles $P$ and $Q$ on a coordinate grid.




**(a)** Describe fully the single transformation that maps shape $P$ onto shape $Q$.

.....

.....

..... **[3]**

**(b)** On the grid, enlarge shape $P$ by scale factor $\frac{1}{2}$ with centre $(0, 4)$.
Label the image $R$. **[3]**

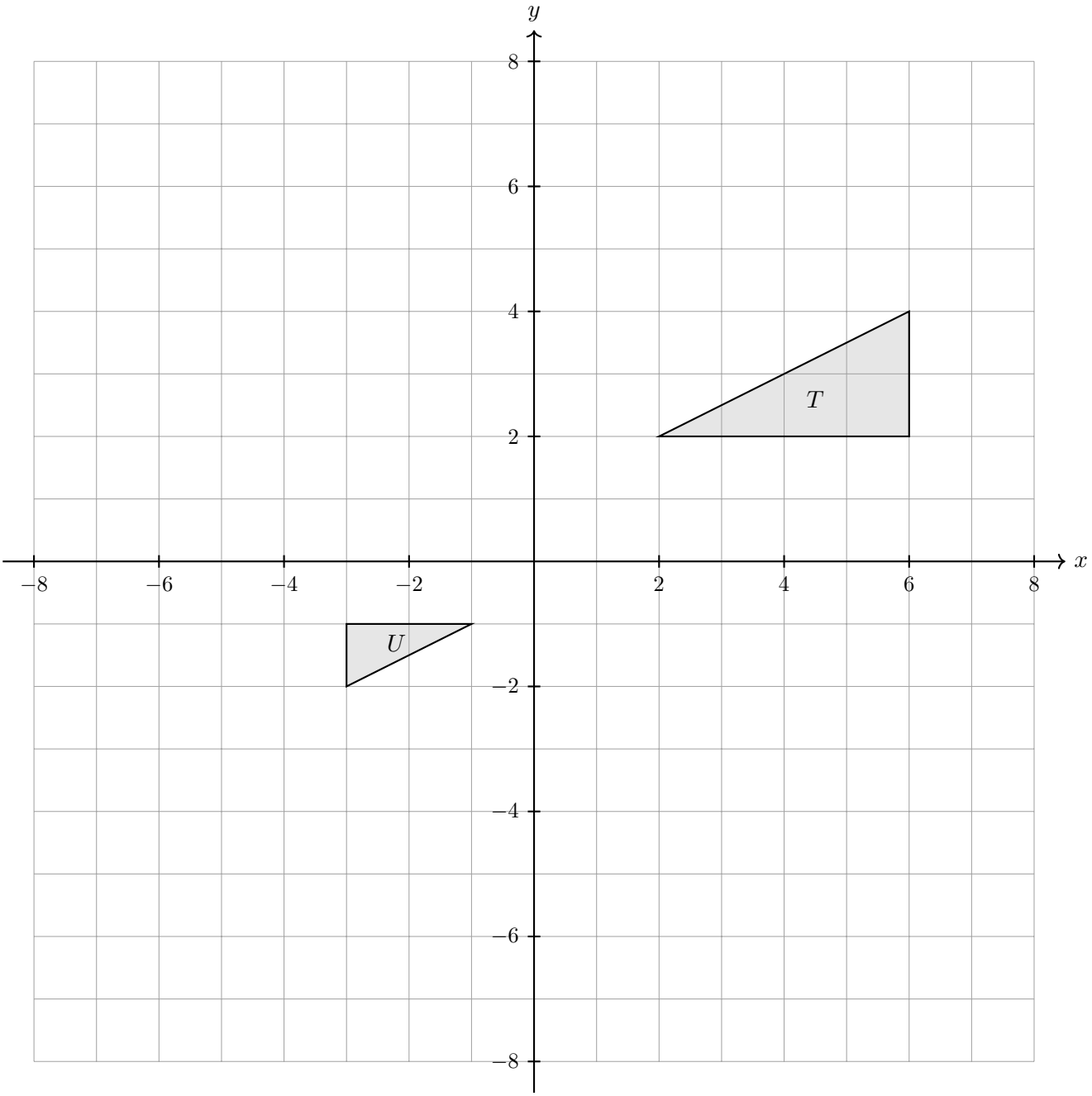
**(c)** Describe fully the single transformation that maps shape $R$ onto shape $Q$.

.....

.....

..... **[3]**

**1388** The grid shows triangles $T$ and $U$.




**(a)** Describe fully the single transformation that maps triangle $T$ onto triangle $U$.

.....

.....

..... **[3]**

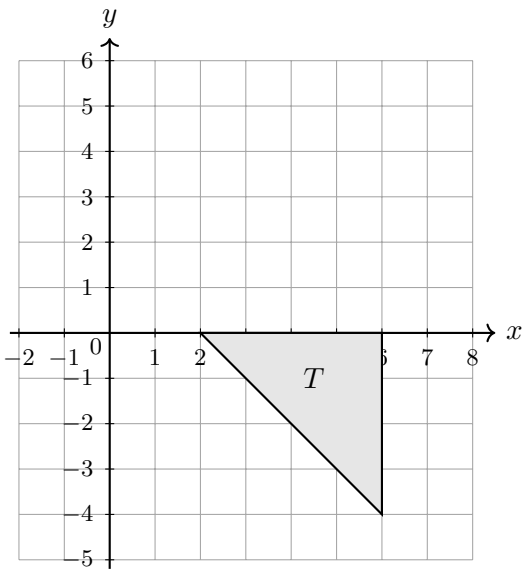
**(b)** Triangle $T$ is reflected in the line $y = x$ to give triangle $V$.
Triangle $V$ is then rotated 90° clockwise about the origin to give triangle $W$.
Describe fully the single transformation that maps triangle $T$ onto triangle $W$.

.....

.....

..... **[2]**

**1389** The diagram shows triangle $T$ plotted on a coordinate grid.




**(a)** Triangle $T$ is mapped onto triangle $U$ by an enlargement, scale factor $\frac{1}{2}$, centre $(0, 4)$.
On the grid, draw triangle $U$. **[2]**

**(b)** Triangle $U$ is mapped onto triangle $V$ by a reflection in the line $y = x$.
On the grid, draw triangle $V$. **[2]**

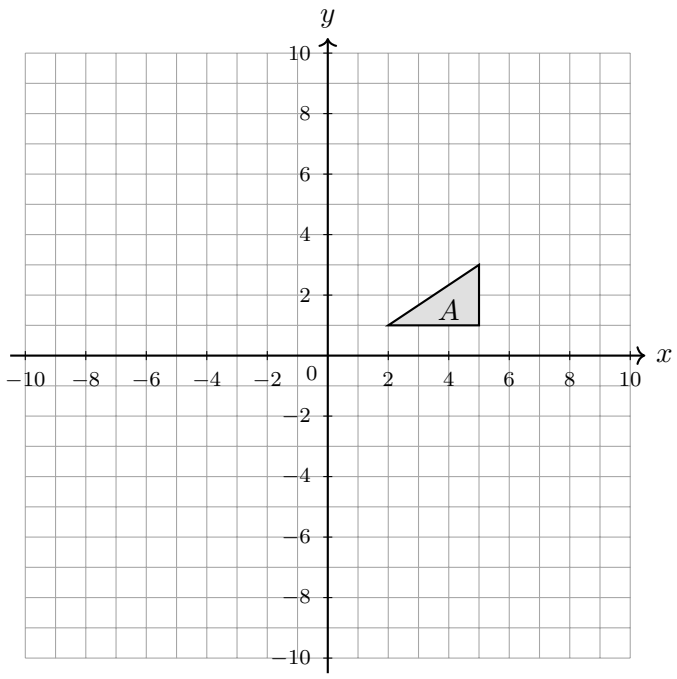
**(c)** Describe fully the single transformation that maps triangle $U$ back onto triangle $T$.

.....

.....

..... **[3]**

**1390** The diagram shows triangle $A$ on a coordinate grid.




**(a)** On the grid, draw the image of triangle $A$ after an enlargement by scale factor $-2$, centre $(1, 0)$. Label this image $B$. **[3]**

**(b)** Triangle $A$ is mapped onto triangle $C$ by a reflection in the line $y = -x$.
Write down the coordinates of the vertices of triangle $C$.

..... **[2]**

**(c)** Triangle $A$ is first translated by the vector $\begin{pmatrix} 0 \\ 4 \end{pmatrix}$.

The resulting image is then rotated 90° clockwise about the point $(5, 5)$ to form triangle $D$.
Describe fully the single transformation that maps triangle $A$ onto triangle $D$.

.....

.....

..... **[3]**

**1391** A triangle $T$ has vertices at $(a, b)$, $(c, d)$ and $(e, f)$.
The translation by the vector $\begin{pmatrix} 3 \\ -2 \end{pmatrix}$ maps triangle $T$ onto triangle $U$ with vertices at $(5, 2)$, $(8, 2)$ and $(7, 6)$.

**(a)** Find the coordinates of the three vertices of triangle $T$.

..... **[2]**

**(b)** Triangle $U$ is mapped onto triangle $V$ by a reflection in the line $y = x$.
Write down the coordinates of the three vertices of triangle $V$.

..... **[2]**

**(c)** Triangle $V$ is mapped onto triangle $W$ by a rotation of $90°$ clockwise about the point $(2, 5)$.
Find the coordinates of the three vertices of triangle $W$.

..... **[3]**

**(d)** Describe fully the single transformation that maps triangle $T$ onto triangle $X$, where the vertices of $X$ are $(-4, -8)$, $(-10, -8)$ and $(-8, -16)$.

.....

.....

..... **[3]**

**1392** Vectors $\mathbf{p}$ and $\mathbf{q}$ are given as

$$\mathbf{p} = \begin{pmatrix} 4 \\ -3 \end{pmatrix} \qquad \mathbf{q} = \begin{pmatrix} -2 \\ 5 \end{pmatrix}$$

**(a)** Work out

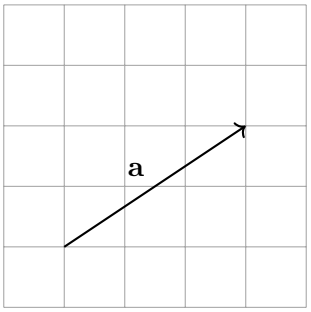
$$2\mathbf{p} + \mathbf{q}$$

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

**(b)** Point $A$ has coordinates $(1, 4)$.
The translation from $A$ to $B$ is given by the vector $\mathbf{p}$.
Work out the coordinates of $B$.

( ..... , ..... ) **[1]**

**1393** The diagram shows a vector **a**.




**(a)** Write **a** as a column vector.

$\mathbf{a} = \begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[1]**

**(b)** $\mathbf{b} = \begin{pmatrix} -1 \\ 4 \end{pmatrix}$.
Work out $2\mathbf{a} - \mathbf{b}$.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

**1394** The vectors **a** and **b** are given by

$\mathrm{a} = \begin{pmatrix} 4 \\ -3 \end{pmatrix}$ and $\mathrm{b} = \begin{pmatrix} -2 \\ 5 \end{pmatrix}$

**(a)** Work out $\mathbf{a} + 2\mathbf{b}$ as a single column vector.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

**(b)** Point $P$ has coordinates $(3, -1)$.
Point $Q$ is translated from $P$ by the vector **a**.
Find the coordinates of point $Q$.

( ..... , ..... ) **[2]**

**1395** Given the vectors $\mathbf{p} = \begin{pmatrix} 4 \\ -3 \end{pmatrix}$ and $\mathbf{q} = \begin{pmatrix} -2 \\ 5 \end{pmatrix}$.

**(a)** Work out $3\mathbf{p}$.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

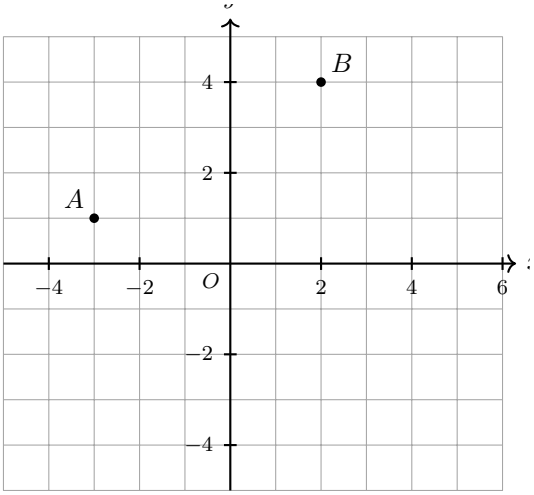
**(b)** Work out $\mathbf{p} - 2\mathbf{q}$.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

**(c)** Point $M$ is translated to point $N$ by the vector $\mathbf{q}$.
Given that the coordinates of $N$ are $(6, -1)$, find the coordinates of $M$.

( ..... , ..... ) **[2]**

**1396** The diagram shows points $A(-3, 1)$ and $B(2, 4)$ on a coordinate grid.




**(a)** Write down $\vec{AB}$ as a column vector.

$\vec{AB} = \begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

**(b)** $\mathbf{p} = \begin{pmatrix} 3 \\ -2 \end{pmatrix}$ and $\mathbf{q} = \begin{pmatrix} -1 \\ 4 \end{pmatrix}$.

Calculate

**(i)** $\mathbf{p} + 2\mathbf{q}$,

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

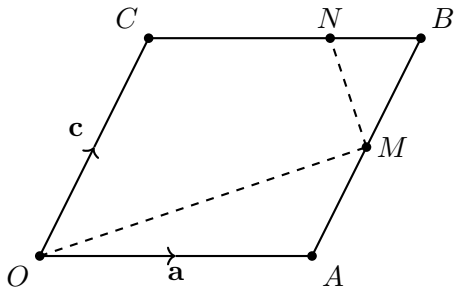
**(ii)** $3\mathbf{p} - \mathbf{q}$.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

**(c)** Point $E$ is such that $\vec{BE} = \begin{pmatrix} -4 \\ -6 \end{pmatrix}$.
Find the coordinates of $E$.

$E = (\ldots\ldots\ldots\ ,\ \ldots\ldots\ldots)$ **[2]**

**1397** The diagram shows a parallelogram $OABC$, where $O$ is the origin.
$\vec{OA} = \mathbf{a}$ and $\vec{OC} = \mathbf{c}$.
$M$ is the midpoint of $AB$ and $N$ is the point on $BC$ such that $BN : NC = 2 : 1$.




**(a)** Find each of the following vectors in terms of **a** and **c**.
Give each answer in its simplest form.

**(i)** $\vec{OM}$

$\vec{OM} =$ ..... **[2]**

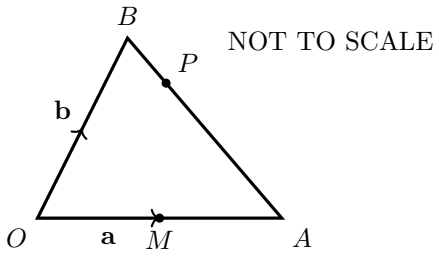
**(ii)** $\vec{MN}$

$\vec{MN} =$ ..... **[2]**

**(b)** $\vec{ON} = k\mathbf{a} + \mathbf{c}$.
Write down the value of $k$.

$k =$ ..... **[2]**

**1398** The diagram shows triangle $OAB$ with $\vec{OA} = \mathbf{a}$ and $\vec{OB} = \mathbf{b}$.
$M$ is the midpoint of $OA$, and $P$ is the point on $AB$ such that $AP : PB = 1 : 3$.




**(a)** Express each of the following vectors in terms of **a** and **b**, giving your answers in their simplest form:

**(i)** $\vec{AB}$,

$\vec{AB} =$ ..... **[1]**

**(ii)** $\vec{OP}$,

$\vec{OP} =$ ..... **[2]**

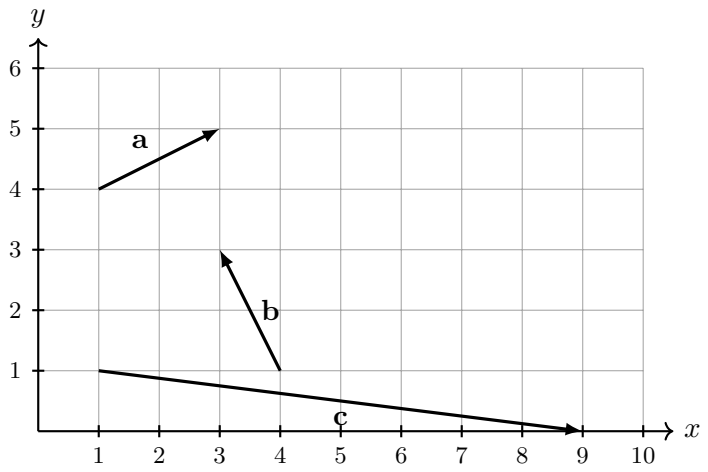
**(iii)** $\vec{MP}$.

$\vec{MP} =$ ..... **[2]**

**(b)** The line $OP$ is extended to meet the line through $B$ parallel to $OA$ at the point $C$, where $\vec{BC} = k\mathbf{a}$ and $\vec{OC} = m\vec{OP}$.
Find the value of $k$ and the value of $m$.

$k =$ ..... , $m =$ ..... **[4]**

**1399** The diagram shows vectors **a**, **b** and **c** plotted on a Cartesian coordinate grid.




**(a)** Write **c** as a column vector.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[1]**

**(b)** Vector **c** can be written in the form $p\mathbf{a} + q\mathbf{b}$, where $p$ and $q$ are integers.
Find the value of $p$ and the value of $q$.

$p =$ ..... , $q =$ ..... **[4]**

**1400** Vectors $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$ are given by

$$\mathbf{u} = \begin{pmatrix} 3 \\ m \end{pmatrix} \qquad \mathbf{v} = \begin{pmatrix} n \\ -2 \end{pmatrix} \qquad \mathbf{w} = \begin{pmatrix} 10 \\ 7 \end{pmatrix}$$

**(a)** Given that $2\mathbf{u} - \mathbf{v} = \mathbf{w}$, find the value of $m$ and the value of $n$.

$m =$ ..... , $n =$ ..... **[3]**

**(b)** Vector $\mathbf{z} = \begin{pmatrix} k \\ 12 \end{pmatrix}$.
Given that $\mathbf{z}$ is parallel to $\mathbf{u}$, find an expression for $k$ in terms of $m$.

$k =$ ..... **[3]**

**1401** $\mathbf{p} = \begin{pmatrix} 4 \\ -3 \end{pmatrix}$ $\mathbf{q} = \begin{pmatrix} -2 \\ 5 \end{pmatrix}$

**(a)** Work out $3\mathbf{p} - 2\mathbf{q}$.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

**(b)** Point $A$ has coordinates $(3, -1)$.
$\vec{AB} = \mathbf{p} + \mathbf{q}$.
Find the coordinates of $B$.

( ..... , ..... ) **[2]**

**1402** Vectors $\mathbf{p}$ and $\mathbf{q}$ are given as:

$$\mathbf{p} = \begin{pmatrix} 4 \\ -3 \end{pmatrix} \qquad \mathbf{q} = \begin{pmatrix} -2 \\ 5 \end{pmatrix}$$

**(a)** Work out $3\mathbf{p}$.

$$3\mathbf{p} = \begin{pmatrix} \quad \\ \quad \end{pmatrix}$$ **[1]**

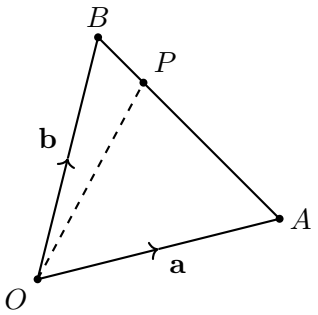
**(b)** Work out $\mathbf{p} - 2\mathbf{q}$.

$$\mathbf{p} - 2\mathbf{q} = \begin{pmatrix} \quad \\ \quad \end{pmatrix}$$ **[2]**

**(c)** Point $A$ has coordinates $(3, -1)$.
Point $B$ is the image of $A$ after a translation by the vector $\mathbf{p}$.
Work out the coordinates of $B$.

( ..... , ..... ) **[2]**

**1403** In triangle $OAB$, $\vec{OA} = \mathbf{a}$ and $\vec{OB} = \mathbf{b}$.
$P$ is the point on $AB$ such that $AP : PB = 3 : 1$.




**(a)** Find $\vec{OP}$ in terms of **a** and **b**.
Give your answer in its simplest form.

$\vec{OP} =$ ..... **[3]**

**(b)** $M$ is the midpoint of $OA$.
Find $\vec{MP}$ in terms of **a** and **b**.
Give your answer in its simplest form.

$\vec{MP} =$ ..... **[2]**

**1404** Vectors **u**, **v** and **w** are given by:

$$\mathbf{u} = \begin{pmatrix} 3 \\ -2 \end{pmatrix} \qquad \mathbf{v} = \begin{pmatrix} x \\ 5 \end{pmatrix} \qquad \mathbf{w} = \begin{pmatrix} -1 \\ y \end{pmatrix}$$

**(a)** Given that $2\mathbf{u} + \mathbf{v} = \mathbf{w}$, find the value of $x$ and the value of $y$.

$x =$ ..... , $y =$ ..... **[3]**

**(b)** Point $M$ has coordinates $(4, 7)$ and $\vec{MN} = \begin{pmatrix} -2 \\ 6 \end{pmatrix}$.

**(i)** Find the coordinates of $N$.

( ..... , ..... ) **[2]**

**(ii)** Write down the vector $\vec{NM}$.

$\vec{NM} = \begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[1]**

**1405** **(a)** $\mathbf{u} = \begin{pmatrix} 4 \\ -3 \end{pmatrix}$, $\mathbf{v} = \begin{pmatrix} -1 \\ y \end{pmatrix}$ and $\mathbf{w} = \begin{pmatrix} 10 \\ 6 \end{pmatrix}$.

Given that $3\mathbf{u} + k\mathbf{v} = \mathbf{w}$, work out the value of $k$ and the value of $y$.

$k =$ ..... , $y =$ ..... **[3]**

**(b)** Points $P$, $Q$ and $R$ have coordinates $(2, -1)$, $(5, 5)$ and $(11, 17)$ respectively.

**(i)** Write $\vec{PQ}$ as a column vector.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[1]**

**(ii)** Show that $P$, $Q$ and $R$ lie on a straight line.

**[2]**

**1406** The coordinates of point $A$ are $(2, 3)$ and the coordinates of point $B$ are $(-1, 5)$.

**(a)** Write down the column vector $\vec{AB}$.

$\vec{AB} = \begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[1]**

**(b)** Point $C$ is such that $\vec{BC} = 3\vec{AB}$.
Find the coordinates of $C$.

( ..... , ..... ) **[3]**

**(c)** $\mathbf{p} = \begin{pmatrix} 4 \\ -1 \end{pmatrix}$.
Find the column vector $\mathbf{q}$ such that $\vec{AB} + \mathbf{q} = \mathbf{p}$.

$\mathbf{q} = \begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

**1407** Points $A$, $B$ and $C$ have coordinates $(1, -2)$, $(7, 10)$ and $(4, 4)$ respectively.

**(a)** Write $\vec{AB}$ as a column vector.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

**(b)** Show that $\vec{AC} = \frac{1}{2}\vec{AB}$.

**[2]**

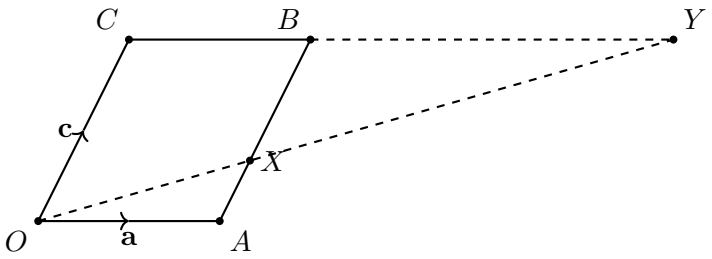
**(c)** Point $C$ is translated by the vector $\begin{pmatrix} -5 \\ 2 \end{pmatrix}$ to point $D$.
Find the coordinates of $D$.

( ..... , ..... ) **[2]**

**(d)** Work out the vector $\vec{BD}$ as a column vector.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

**1408** $OABC$ is a parallelogram.
$\vec{OA} = \mathbf{a}$ and $\vec{OC} = \mathbf{c}$.
$X$ is a point on $AB$ such that $AX : XB = 1 : 2$.
$Y$ is a point on the line $CB$ extended such that $CB : BY = 1 : 2$.




**(a)** Find $\vec{OX}$ in terms of **a** and **c**.

$\vec{OX} =$ ..... **[2]**

**(b)** Find $\vec{OY}$ in terms of **a** and **c**.

$\vec{OY} =$ ..... **[2]**

**(c)** Show that the points $O$, $X$ and $Y$ lie on a straight line.

**[2]**

**(d)** Find the ratio $OX : XY$.

$OX : XY =$ ..... **[1]**

**1409** The position vectors of points $A$, $B$, and $C$ relative to an origin $O$ are given by

$= \begin{pmatrix} 2 \\ -1 \end{pmatrix}$, $= \begin{pmatrix} 6 \\ 7 \end{pmatrix}$, and $= \begin{pmatrix} k+1 \\ 3k-2 \end{pmatrix}$

where $k$ is a constant.

**(a)** Find $\vec{AB}$ as a column vector.

$\vec{AB} = \begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

**(b)** Given that points $A$, $B$, and $C$ are collinear, find the value of $k$.

$k =$ ..... **[3]**

**(c)** Using your value of $k$ from part (b), write down $\vec{AC}$ in terms of $\vec{AB}$.

$\vec{AC} =$ ..... **[2]**

**1410** Given the vectors $\mathbf{u} = \begin{pmatrix} 3 \\ -2 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} -1 \\ 4 \end{pmatrix}$.

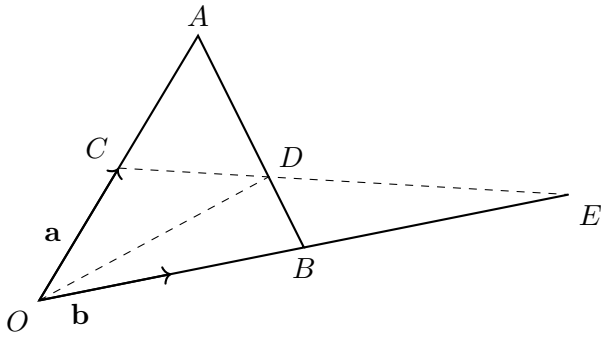
**(a)** Point $P$ has coordinates $(5, -1)$.
The vector $\vec{PQ} = 2\mathbf{u} - \mathbf{v}$.
Find the coordinates of $Q$.

( ..... , ..... ) **[3]**

**(b)** Point $R$ has coordinates $(14, 3)$.
The vector $\vec{PR}$ can be written in the form $h\mathbf{u} + k\mathbf{v}$.
Work out the values of the constants $h$ and $k$.

$h =$ ..... , $k =$ ..... **[5]**

**1411** The diagram shows a triangle $OAB$.
$O$ is the origin, $\vec{OA} = \mathbf{a}$ and $\vec{OB} = \mathbf{b}$.
$C$ is the midpoint of $OA$.
$D$ is a point on $AB$ such that $AD : DB = 2 : 1$.
$E$ is a point on $OB$ extended such that $\vec{OE} = 2\mathbf{b}$.




NOT TO SCALE

**(a)** Find $\vec{AB}$ in terms of **a** and **b**.

$\vec{AB} =$ ..... **[1]**

**(b)** Find $\vec{OD}$ in terms of **a** and **b**.
Give your answer in its simplest form.

$\vec{OD} =$ ..... **[3]**

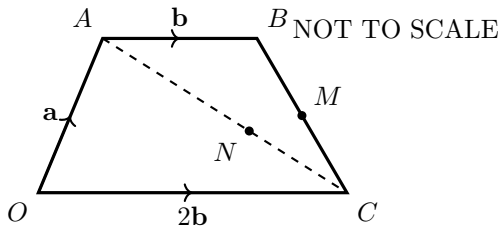
**(c)** Find $\vec{CE}$ in terms of **a** and **b**.

$\vec{CE} =$ ..... **[2]**

**(d)** Show that the points $C$, $D$ and $E$ lie on a straight line.

**[2]**

**1412** The diagram shows trapezium $OABC$ with $\vec{OA} = \mathbf{a}$, $\vec{AB} = \mathbf{b}$, and $\vec{OC} = 2\mathbf{b}$.
$M$ is the midpoint of $BC$, and $N$ is the point on $AC$ such that $AN : NC = 3 : 2$.




**(a)** Express each of the following vectors in terms of **a** and **b**, giving your answers in their simplest form:

**(i)** $\vec{OB}$,

$\vec{OB} =$ ..... **[1]**

**(ii)** $\vec{BC}$,

$\vec{BC} =$ ..... **[2]**

**(iii)** $\vec{OM}$.

$\vec{OM} =$ ..... **[2]**

**(b)** Express $\vec{ON}$ in terms of **a** and **b**, giving your answer in its simplest form.

$\vec{ON} =$ ..... **[3]**

**(c)** Use your answers to part (a)(iii) and part (b) to

**(i)** explain why $O$, $N$, and $M$ lie on a straight line,

**[2]**

**(ii)** write down the ratio $ON : NM$ in its simplest form.

$ON : NM =$ ..... **[2]**

**1413** Vector $\mathbf{a} = \begin{pmatrix} -12 \\ 5 \end{pmatrix}$.

Calculate $|\mathbf{a}|$.

$|\mathbf{a}| =$ ..... **[2]**

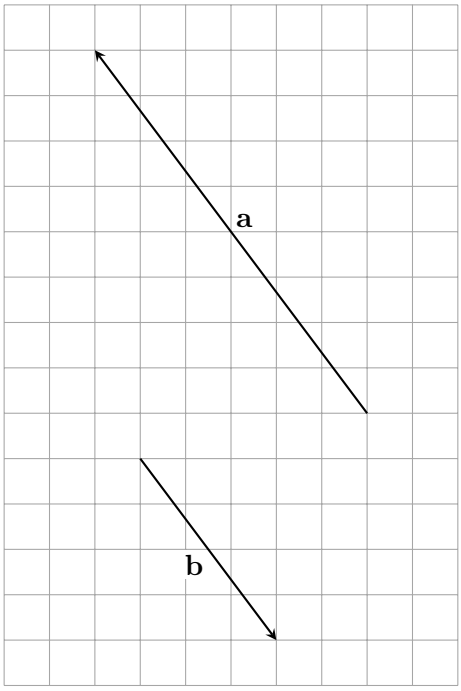
**1414** Vector $\mathbf{v}$ is given as

$$\mathbf{v} = \begin{pmatrix} 5 \\ -12 \end{pmatrix}$$

Work out $|\mathbf{v}|$, the magnitude of vector $\mathbf{v}$.

..... **[2]**

**1415** The diagram shows vectors **a** and **b** drawn on a square grid.




**(a)** Work out $|\mathbf{a}|$.

$|\mathbf{a}| =$ ..... **[2]**

**(b)** Work out $|2\mathbf{a} + 3\mathbf{b}|$.

$|2\mathbf{a} + 3\mathbf{b}| =$ ..... **[3]**

**1416** Given the vectors $\mathbf{p} = \begin{pmatrix} -8 \\ 15 \end{pmatrix}$ and $\mathbf{q} = \begin{pmatrix} 2 \\ -4 \end{pmatrix}$.

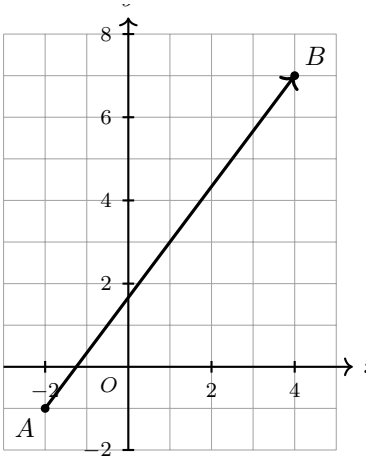
**(a)** Calculate $|\mathbf{p}|$.

..... **[2]**

**(b)** Calculate $|\mathbf{p} + 2\mathbf{q}|$.

..... **[3]**

**1417** The diagram shows points $A(-2, -1)$ and $B(4, 7)$ on a coordinate grid.




**(a)** Write down $\vec{AB}$ as a column vector.

$\vec{AB} = \begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[1]**

**(b)** Calculate $|\vec{AB}|$.

$|\vec{AB}| =$ ..... units **[2]**

**(c)** The vector $\mathbf{v}$ is given by $\mathbf{v} = \begin{pmatrix} -12 \\ -5 \end{pmatrix}$.

Calculate

**(i)** $|\mathbf{v}|$,

$|\mathbf{v}| =$ ..... units **[2]**

**(ii)** $|\vec{AB} + \mathbf{v}|$, giving your answer in the form $k\sqrt{5}$, where $k$ is an integer.

$|\vec{AB} + \mathbf{v}| =$ ..... units **[2]**

**1418** **(a)** The vector $\mathbf{a} = \begin{pmatrix} p \\ p \end{pmatrix}$, where $p > 0$.

Given that $|\mathbf{a}| = \sqrt{18}$, find the value of $p$.

$p =$ ..... **[2]**

**(b)** The vectors $\mathbf{u}$ and $\mathbf{v}$ are given by

$$\mathbf{u} = \begin{pmatrix} 4 \\ 1 \end{pmatrix}, \qquad \mathbf{v} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$$

A vector $\mathbf{w}$ is defined as $\mathbf{w} = \mathbf{u} + t\mathbf{v}$.

Given that $|\mathbf{w}| = \sqrt{29}$, find the two possible values of $t$.

$t =$ ..... , or $t =$ ..... **[5]**

**1419** Vector $\mathbf{u} = \begin{pmatrix} x \\ x+2 \end{pmatrix}$.

**(a)** The magnitude of vector **u** is 10.
Show that $x^2 + 2x - 48 = 0$.

**[3]**

**(b)** Solve the equation

$$x^2 + 2x - 48 = 0$$

$x =$ ..... , $x =$ ..... **[2]**

**(c)** Hence write down the two possible column vectors for **u**.

**[2]**

**1420** Relative to an origin $O$, the position vectors of points $A$ and $B$ are $\vec{OA} = \begin{pmatrix} 6 \\ 2 \end{pmatrix}$ and $\vec{OB} = \begin{pmatrix} -2 \\ 6 \end{pmatrix}$.

**(a)** Calculate the exact value of

**(i)** $|\vec{OA}|$, giving your answer in the form $a\sqrt{10}$, where $a$ is an integer,

$|\vec{OA}| =$ ..... **[1]**

**(ii)** $|\vec{AB}|$, giving your answer in the form $b\sqrt{5}$, where $b$ is an integer.

$|\vec{AB}| =$ ..... **[2]**

**(b)** Show that triangle $OAB$ is both isosceles and right-angled.

**[3]**

**(c)** Point $P$ lies on the line segment $AB$ such that $AP : PB = 3 : 1$.

**(i)** Find $\vec{OP}$ as a column vector.

$\vec{OP} = \begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

**(ii)** Point $Q$ has position vector $\vec{OQ} = \begin{pmatrix} 8 \\ m \end{pmatrix}$, where $m < 0$.

Given that $|\vec{PQ}| = 17$, find the value of $m$.

$m =$ ..... **[3]**

**1421** Points $P$ and $Q$ have coordinates $(4, -2)$ and $(-1, 10)$ respectively.

**(a)** Find $\vec{PQ}$.

$\vec{PQ} = \begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[1]**

**(b)** Calculate $|\vec{PQ}|$.

$|\vec{PQ}| =$ ..... **[2]**

**(c)** Vector $\mathbf{m} = \begin{pmatrix} 3 \\ c \end{pmatrix}$.

Given that $|\mathbf{m}| = |\vec{PQ}|$, find the possible values of $c$.

$c =$ ..... , $c =$ ..... **[3]**

**1422** Points $A$, $B$, and $C$ have coordinates $(3, -2)$, $(-5, y)$, and $(7, 1)$ respectively.

**(a)** Calculate $|\vec{AC}|$.

..... **[2]**

**(b)** The magnitude of vector $\vec{AB}$ is 17.
Work out the two possible values of $y$.

$y =$ ..... , $y =$ ..... **[5]**

**1423** The points $P$, $Q$ and $R$ have coordinates $(2, -3)$, $(k, 5)$ and $(4, 1)$ respectively.

**(a)** Given that $|\vec{PQ}| = 10$, find the two possible values of $k$.

$k =$ ..... , or $k =$ ..... **[4]**

**(b)** Using the positive value of $k$ from part (a), calculate the magnitude of $\vec{QR}$.
Give your answer correct to 3 significant figures.

$|\vec{QR}| =$ ..... **[3]**

**1424** **(a)** The points $A$ and $B$ have coordinates $(2, -3)$ and $(9, 4)$ respectively.
Work out the column vector $\vec{AB}$.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

**(b)** Calculate $|\vec{AB}|$.
Give your answer correct to 3 significant figures.

..... **[2]**

**(c)** Vector $\mathbf{p} = \begin{pmatrix} k \\ 4 \end{pmatrix}$.
Given that $|\mathbf{p}| = 5$, find the two possible values of $k$.

$k =$ ..... , $k =$ ..... **[3]**

**1425** $\mathbf{p} = \begin{pmatrix} 4 \\ -3 \end{pmatrix}$ and $\mathbf{q} = \begin{pmatrix} -1 \\ 5 \end{pmatrix}$.

**(a)** Calculate $|2\mathbf{p} - \mathbf{q}|$, giving your answer correct to 3 significant figures.

$|2\mathbf{p} - \mathbf{q}| =$ ..... **[3]**

**(b)** The vector $\mathbf{r} = k\mathbf{p}$, where $k < 0$ and $|\mathbf{r}| = 35$.
Find the column vector $\mathbf{r}$.

$\mathbf{r} = \begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[3]**

**(c)** The vector $\mathbf{w} = \begin{pmatrix} t+3 \\ 2t-1 \end{pmatrix}$ has magnitude $\sqrt{34}$.
Find the two possible values of $t$.

$t =$ ..... , $t =$ ..... **[4]**

**1426** Vector $\mathbf{v} = \begin{pmatrix} k+3 \\ 2k-1 \end{pmatrix}$, where $k$ is a constant.

The magnitude of $\mathbf{v}$ is $\sqrt{98}$.

**(a)** Show that $5k^2 + 2k - 88 = 0$.

**[3]**

**(b)** Solve the equation $5k^2 + 2k - 88 = 0$ to find the possible values of $k$.

$k =$ ..... , $k =$ ..... **[3]**

**(c)** Given that $k > 0$, write down the column vector $\mathbf{v}$.

$\mathbf{v} = \begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[1]**

**1427** Given the vectors $\mathbf{u} = \begin{pmatrix} 2k \\ k-3 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} k+1 \\ 2 \end{pmatrix}$.

**(a)** Given that $|\mathbf{u} + \mathbf{v}| = \sqrt{50}$, show that $5k^2 + 2k - 24 = 0$.

**[4]**

**(b)** Solve the equation

$$5k^2 + 2k - 24 = 0$$

$k =$ ..... , $k =$ ..... **[4]**

**(c)** Hence, find the two possible column vectors for $\mathbf{u}$.

$\mathbf{u} = \begin{pmatrix} \quad \\ \quad \end{pmatrix}$ ..... , $\mathbf{u} = \begin{pmatrix} \quad \\ \quad \end{pmatrix}$ ..... **[2]**

**1428** Event $A$ is that a train arrives on time.

$P(A) = 0.83$

**(a)** Work out $P(A')$.

..... **[1]**

**(b)** State in words what the event $A'$ represents.

..... **[1]**

**1429** A bag contains only red, blue, and green marbles.
The probability of picking a red marble is $\frac{2}{7}$ and the probability of picking a blue marble is $\frac{1}{3}$.

**(a)** Find the probability of picking a green marble.

..... **[2]**

**(b)** Let $R$ be the event of picking a red marble.
Write down the value of $P(R')$.

$P(R') =$ ..... **[1]**

**1430** A biased spinner can land on 1, 2, 3, 4, or 5.
The table shows the probabilities of the spinner landing on each of the numbers.

| Outcome | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Probability | 0.15 | $2x$ | 0.31 | $x$ | 0.24 |

**(a)** Find the value of $x$.

$x =$ ..... **[2]**

**(b)** The spinner is spun once.
Find the probability that it does not land on 2.

..... **[2]**

**1431** A biased spinner can land on Red, Blue, Green, or Yellow.
The table shows the probabilities of the spinner landing on Red, Blue, and Green.

| Colour | Red | Blue | Green | Yellow |
|---|---|---|---|---|
| Probability | 0.15 | 0.3 | 0.45 | |

**(a)** Work out the probability that the spinner lands on Yellow.

..... **[2]**

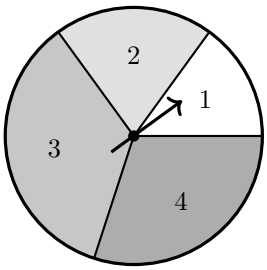
**(b)** Work out the probability that the spinner lands on Red or Blue.

..... **[1]**

**(c)** Let $G$ be the event that the spinner lands on Green.
Find $P(G')$.

$P(G') =$ ..... **[2]**

**1432** The diagram shows a biased 4-sided spinner with sectors numbered 1, 2, 3, and 4.
When the spinner is spun once, the probabilities of landing on each number are $P(1) = 0.15$, $P(2) = 2x$, $P(3) = 0.35$, and $P(4) = 3x$.




**(a)** Find the value of $x$.

$x =$ ..... **[2]**

**(b)** Event $A$ is that the spinner lands on an even number.
Find

**(i)** $P(A)$,

$P(A) =$ ..... **[2]**

**(ii)** $P(A')$.

$P(A') =$ ..... **[1]**

**(c)** The spinner is spun 200 times.
Calculate an estimate for the number of times the spinner lands on a prime number.

..... **[2]**

**1433** A biased spinner can land on 1, 2, 3 or 4.
The table shows the probabilities of the spinner landing on each number.

| Number | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| Probability | $\frac{1}{5}$ | 0.35 | $x$ | $2x$ |

**(a)** Work out the value of $x$.

$x =$ ..... **[2]**

**(b)** Find the probability that the spinner does not land on 4.

..... **[2]**

**1434** A biased four-sided die is rolled.
The possible scores are 1, 2, 3 and 4.
The table shows the probabilities of rolling each score.

| Score | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| Probability | $p^2$ | $p$ | 0.15 | 0.1 |

**(a)** Show that $4p^2 + 4p - 3 = 0$.

**[2]**

**(b)** Find the value of $p$.

$p =$ ..... **[2]**

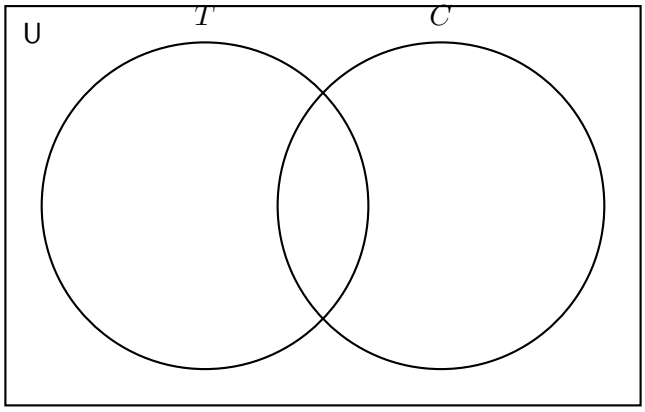
**(c)** Let $E$ be the event of rolling an even number.
Work out $P(E')$.

$P(E') =$ ..... **[2]**

**1435** In a group of 60 people, some like tea ($T$) and some like coffee ($C$).

- The total number of people who like tea is 38.
- The total number of people who like coffee is 24.
- The number of people who like neither is $x$.
- The number of people who like both is $2x$.

**(a)** Use this information to complete the Venn diagram in terms of $x$.




**[3]**

**(b)** Find the value of $x$.

$x =$ ..... **[2]**

**(c)** A person is chosen at random from the group.
Find $P(T \cap C')$.

$P(T \cap C') =$ ..... **[2]**

**1436** A bag contains only red counters, blue counters, and green counters.
The number of red counters is $x^2$.
The number of blue counters is $3x$.
The number of green counters is 2.

**(a)** Write down an expression, in its simplest form, for the total number of counters in the bag.

..... **[1]**

**(b)** A counter is chosen at random.
The probability that it is red is $\frac{8}{15}$.
Show that $7x^2 - 24x - 16 = 0$.

**[2]**

**(c)** Solve the equation $7x^2 - 24x - 16 = 0$ to find the value of $x$.

$x =$ ..... **[3]**

**(d)** Find the probability that a randomly chosen counter is not blue.
Give your answer as a fraction in its simplest form.

..... **[2]**

**1437** A box contains $x$ red counters, $(2x - 1)$ blue counters, and $(x^2 - 4)$ green counters, where $x > 2$.
A counter is taken at random from the box.
Event $R$ is taking a red counter, and event $G$ is taking a green counter.
It is given that $P(G) = \frac{3}{5}$.

**(a)** Show that $2x^2 - 9x - 5 = 0$.

**[3]**

**(b)** Solve the equation $2x^2 - 9x - 5 = 0$ to find the number of red counters in the box.

$x =$ ..... **[2]**

**(c)** Find, as a fraction in its simplest form,

**(i)** $P(R)$,

$P(R) =$ ..... **[2]**

**(ii)** $P(G')$.

$P(G') =$ ..... **[2]**

**(d)** A further $k$ red counters are added to the box.
When a counter is now taken at random from the box, $P(R') = \frac{3}{4}$.
Find the value of $k$.

$k =$ ..... **[3]**

**1438** A bag contains only red marbles, blue marbles, and green marbles.
The probability of choosing a red marble at random is $\frac{2}{5}$.
The probability of choosing a blue marble at random is $\frac{1}{3}$.
There are 12 green marbles in the bag.

**(a)** Calculate the probability of choosing a green marble at random.

..... **[2]**

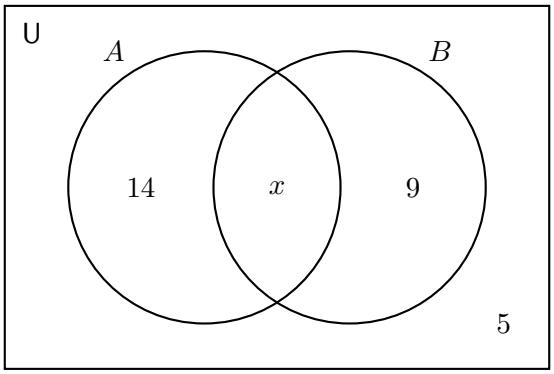
**(b)** Calculate the total number of marbles in the bag.

..... **[3]**

**(c)** Write down the probability that a marble chosen at random is not blue.

..... **[1]**

**1439** The Venn diagram shows the number of elements in the universal set U and sets $A$ and $B$.
There are 36 elements in total in the universal set U.




**(a)** Find the value of $x$.

$x =$ ..... **[2]**

**(b)** An element is chosen at random from the universal set.
Find $P(A)$.

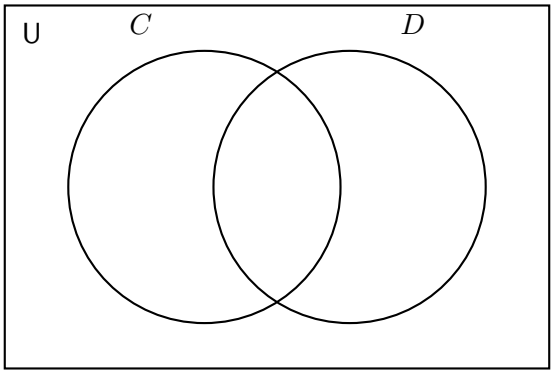
$P(A) =$ ..... **[2]**

**(c)** Find $P(B')$.

$P(B') =$ ..... **[2]**

**1440** A group of 120 people were asked about the pets they own.
$C = \{\text{people who own a cat}\}$
$D = \{\text{people who own a dog}\}$
$n(\mathsf{U}) = 120$
$n(C) = 55$
$n(D) = 68$
18 people own neither a cat nor a dog.

**(a)** Complete the Venn diagram to show the number of people in each region.




**[3]**

**(b)** A person is chosen at random.
Find $P(C \cap D)$.

..... **[1]**

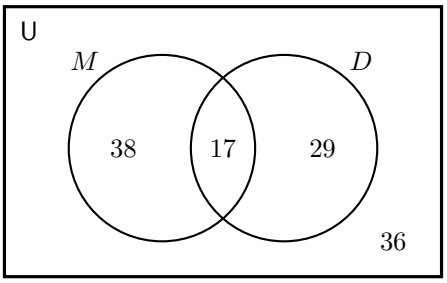
**(c)** Find $P(D')$.

..... **[2]**

**(d)** Find the probability that a person chosen at random owns a cat or a dog, but not both.

..... **[2]**

**1441** In a survey of 120 students, $M$ is the set of students who study Music and $D$ is the set of students who study Drama.
The Venn diagram shows the number of students in each subset, where U is the set of all 120 students.




**(a)** Write down $n((M \cup D)')$.

..... **[1]**

**(b)** A student is chosen at random from the 120 students.
Find, as a fraction in its simplest form,

**(i)** $P(M)$,

$P(M) =$ ..... **[2]**

**(ii)** $P(M' \cap D)$,

$P(M' \cap D) =$ ..... **[2]**

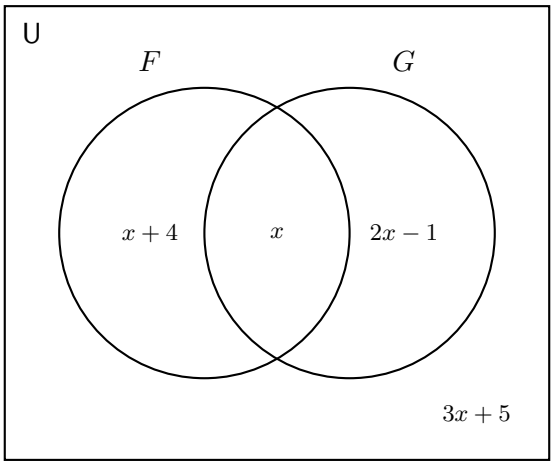
**(iii)** $P((M \cap D)')$.

$P((M \cap D)') =$ ..... **[2]**

**(c)** One of the students who studies Music is chosen at random.
Write down the probability that this student also studies Drama.

..... **[2]**

**1442** The Venn diagram shows information about the number of students in a class of 50 who study French ($F$) and German ($G$).




**(a)** Work out the value of $x$.

$x =$ ..... **[2]**

**(b)** A student is chosen at random.
Find $P(F')$.

..... **[2]**

**(c)** A student is chosen at random.
Find the probability that this student studies exactly one of these languages.

..... **[2]**

**1443** A biased five-sided spinner is numbered 1, 2, 3, 4, and 5.
The table shows the probability of the spinner landing on each number.

| **Number** | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| **Probability** | 0.15 | 0.25 | $x$ | 0.2 | $3x$ |

**(a)** Work out the value of $x$.

$x =$ ..... **[2]**

**(b)** The spinner is spun 250 times.
Calculate the expected number of times the spinner lands on a number greater than 3.

..... **[2]**

**1444** A spinner has five sections, labelled $A$, $B$, $C$, $D$ and $E$.
The table shows the relative frequencies of the spinner landing on each section after 200 spins.

| Section | $A$ | $B$ | $C$ | $D$ | $E$ |
|---|---|---|---|---|---|
| Relative frequency | 0.15 | 0.3 | $x$ | 0.25 | 0.1 |

**(a)** Work out the value of $x$.

$x =$ ..... **[2]**

**(b)** Estimate the number of times the spinner lands on section $B$ in 200 spins.

..... **[2]**

**1445** A biased 4-sided spinner, numbered 1, 2, 3, and 4, is spun 200 times.
The table shows the frequencies of each outcome.

| Number | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| Frequency | 45 | 35 | 70 | 50 |

**(a)** Write down the relative frequency of the spinner landing on 3.

..... **[1]**

**(b)** The spinner is spun 1000 times.
Estimate the number of times it will land on 1.

..... **[2]**

**(c)** Is the spinner fair?
Give a reason for your answer.

..... **[1]**

**(d)** Explain why spinning the spinner 1000 times gives a more reliable estimate of the probability than spinning it 200 times.

..... **[1]**

**1446** A biased 5-sided dice has faces numbered 1, 2, 3, 4, and 5.
Maya rolls the dice 200 times and records the frequency of each score in the table below.

| Score on dice | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Frequency | 32 | 28 | 54 | 46 | 40 |

**(a)** Write down the relative frequency of rolling a score of 3, giving your answer as a decimal.

..... **[1]**

**(b)** Explain why the results in the table suggest that the dice is biased.

..... **[1]**

**(c)** Maya rolls the dice a further 500 times.
Calculate an estimate for the number of times she will roll an odd score.

..... **[3]**

**(d)** Maya rolls the dice $n$ times and expects to roll a score of 5 exactly 36 times.
Find the value of $n$.

$n =$ ..... **[2]**

**1447** A biased four-sided die is rolled 60 times.
The scores are 1, 2, 3, and 4.
The frequencies of each score are shown in the table.

| Score | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| Frequency | $x$ | $2x + 5$ | $3x - 2$ | $x + 8$ |

**(a)** Find the value of $x$.

$x =$ ..... **[2]**

**(b)** Find the relative frequency of rolling a 3.

..... **[1]**

**(c)** The die is rolled 1200 times.
Calculate the expected number of times it rolls a 2.

..... **[2]**

**1448** A biased spinner has three sectors labelled $A$, $B$, and $C$.
In an experiment, the spinner is spun $(x^2 + 10x)$ times, where $x > 0$.
Sector $A$ occurs $(2x + 4)$ times and sector $B$ occurs $(5x + 2)$ times.
The relative frequency of the spinner landing on sector $A$ is $\frac{1}{6}$.

**(a)** Show that $x^2 - 2x - 24 = 0$.

**[2]**

**(b)** Solve the equation $x^2 - 2x - 24 = 0$ and hence find the total number of times the spinner was spun.

Total spins = ..... **[3]**

**(c)** Using the results of this experiment, find, as a fraction in its simplest form, the relative frequency of the spinner landing on

**(i)** sector $B$,

..... **[2]**

**(ii)** sector $C$.

..... **[2]**

**(d)** In a game, a player spins the spinner 600 times.
The player wins \$6 each time the spinner lands on sector $A$, wins \$2 each time it lands on sector $B$, and loses \$3 each time it lands on sector $C$. Calculate the player's expected net profit

after 600 spins.

\$ ..... **[3]**

**1449** A spinner has four colours: Red, Blue, Green, and Yellow.
A student spins it 50 times and records the results.
The table shows the relative frequencies for three of the colours.

| Colour | Red | Blue | Green | Yellow |
|---|---|---|---|---|
| Relative frequency | 0.15 | 0.45 | 0.10 | $x$ |

**(a)** Find the value of $x$.

$x =$ ..... **[1]**

**(b)** The spinner is spun 300 times in total.
Estimate the number of times it lands on Blue.

..... **[2]**

**(c)** The spinner is meant to be fair.
Comment on whether the results show it is fair or biased.

..... **[1]**

**1450** A biased coin is tossed $n$ times.
The probability of getting a head on each toss is $p$.
The expected number of heads is 30 and the expected number of tails is 45.

**(a)** Find the value of $n$.

$n =$ ..... **[1]**

**(b)** Calculate the value of $p$.

$p =$ ..... **[2]**

**(c)** The same coin is now tossed 300 times.
Work out the expected number of tails.

..... **[3]**

**1451** A bag contains a large number of coloured beads.
A bead is chosen at random, its colour recorded, and then returned to the bag.
This experiment is repeated 300 times.
The table shows the frequencies of the colours drawn.

| Colour | Red | Blue | Green | Yellow |
|---|---|---|---|---|
| Frequency | 84 | 105 | $x$ | $y$ |

**(a)** The relative frequency of choosing a Green bead is 0.22.
Work out the value of $x$ and the value of $y$.

$x =$ ..... , $y =$ ..... **[3]**

**(b)** There are exactly 15 Yellow beads in the bag.
Estimate the total number of beads in the bag.

..... **[2]**

**(c)** A new experiment is done where a bead is chosen and replaced 800 times.
Calculate the expected number of times a Blue or Green bead is chosen.

..... **[2]**

**1452** A researcher records the colours of cars passing a checkpoint.
Over one week, 800 cars pass the checkpoint.
The frequencies of the colours are shown in the table.

| **Colour** | Silver | Black | White | Other |
|---|---|---|---|---|
| **Frequency** | 260 | 180 | $w$ | $w + 40$ |

**(a)** Find the relative frequency of a car being Silver.
Give your answer as a fraction in its simplest form.

..... **[2]**

**(b)** Find the value of $w$.

$w =$ ..... **[2]**

**(c)** The researcher uses these results to estimate the probability that the next car passing the checkpoint is White.
Work out this estimate.

..... **[1]**

**(d)** In one month, 12000 cars are expected to pass the checkpoint.
Calculate the expected number of Black cars.

..... **[2]**

**1453** A quality control inspector tests a random sample of 250 light bulbs produced by a factory.
The table shows the lifetimes, $t$ hours, of these 250 light bulbs.
The relative frequency of a light bulb having a lifetime of more than 1500 hours is 0.42.

| Lifetime ($t$ hours) | $0 < t \leq 500$ | $500 < t \leq 1000$ | $1000 < t \leq 1500$ | $1500 < t \leq 2000$ | $2000 < t \leq 2500$ |
|---|---|---|---|---|---|
| Frequency | 15 | $x$ | 95 | $y$ | 35 |

**(a)** Find the value of $x$ and the value of $y$.

$x =$ ..... , $y =$ ..... **[3]**

**(b)** A batch of 4800 light bulbs is produced by the factory.
Calculate an estimate for the number of light bulbs in this batch that have a lifetime of 1000 hours or less.

..... **[3]**

**(c)** Explain why it is important that the sample of 250 light bulbs is chosen at random.

..... **[1]**

**(d)** The inspector tests a second random sample of 750 light bulbs from the same factory, and finds that 115 of them have a lifetime of more than 2000 hours.
Combine the results of both samples to find the best estimate for the probability that a light bulb has a lifetime of more than 2000 hours.

..... **[2]**

**1454** A factory produces light bulbs.
The probability that a light bulb is defective is 0.025.
A sample of $N$ light bulbs is taken.
The expected number of defective light bulbs in this sample is 40.

**(a)** Calculate the value of $N$.

$N =$ ..... **[2]**

**(b)** A new quality control process is introduced.
The factory takes a new sample of 2400 light bulbs.
The expected number of non-defective light bulbs in this new sample is 2364.
Calculate the percentage decrease in the probability of a light bulb being defective.

..... % **[5]**

**1455** A biased 5-sided spinner has sides numbered 1, 2, 3, 4, and 5.
The table shows the probability of the spinner landing on each number in terms of $x$.

| Number | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Probability | 0.15 | $2x$ | $x + 0.05$ | 0.2 | $x$ |

**(a)** Find the value of $x$.

$x =$ ..... **[2]**

**(b)** The spinner is spun 600 times.
Work out the expected number of times it lands on a prime number.

..... **[2]**

**(c)** A different spinner has 3 sections: Red, Blue, and Green.
The probability of landing on Red is $p$.
The probability of landing on Blue is $2p$.
The probability of landing on Green is $1 - 3p$.
This spinner is spun $N$ times.

The expected number of times it lands on Red is 54 and the expected number of times it lands on Green is 36.
Work out the value of $N$ and the value of $p$.

$N =$ ..... , $p =$ ..... **[4]**

**1456** A biased four-sided spinner is numbered 1, 2, 3 and 4.
The table shows the probability of scoring each number.
The spinner is spun 400 times.
Let $E_1$ be the expected number of times the spinner lands on 1.
Let $E_3$ be the expected number of times the spinner lands on 3.
You are given that $E_1 \times E_3 = 5600$.
Also, the probability of scoring a 3 is greater than the probability of scoring a 1.

| **Score** | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| **Probability** | $x$ | $2x$ | $y$ | $y$ |

**(a)** Show that $300x^2 - 100x + 7 = 0$.

**[3]**

**(b)** Solve the equation to find the value of $x$ and the value of $y$.

$x =$ ..... , $y =$ ..... **[4]**

**(c)** Calculate the expected number of times the spinner lands on an even number.

..... **[2]**

**1457** Two biased coins, Coin A and Coin B, are tossed.
The probability of Coin A landing on Heads is $p$.
The probability of Coin B landing on Heads is $p + 0.15$.
Coin A is tossed 300 times and Coin B is tossed 200 times.
The total expected number of Heads from these 500 tosses is 255.

**(a)** Form an equation in $p$ and show that $p = 0.45$.

**[3]**

**(b)** Both coins are tossed together 1000 times.
Work out the expected number of times they land on different outcomes.

..... **[3]**

**(c)** In a new experiment, Coin A is tossed $N$ times.
The relative frequency of landing on Heads is 0.48 and it lands on Heads 72 times.

**(i)** Work out the value of $N$.

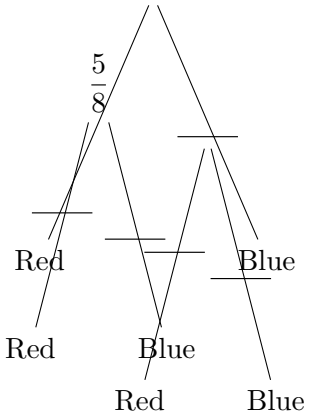
$N =$ ..... **[2]**

**(ii)** Calculate the expected number of Tails in these $N$ tosses, using the theoretical probability $p = 0.45$.

..... **[2]**

**1458** A bag contains 5 red counters and 3 blue counters.
A counter is taken at random from the bag, its colour is noted and it is replaced.
A second counter is then taken at random.

**(a)** Complete the tree diagram.



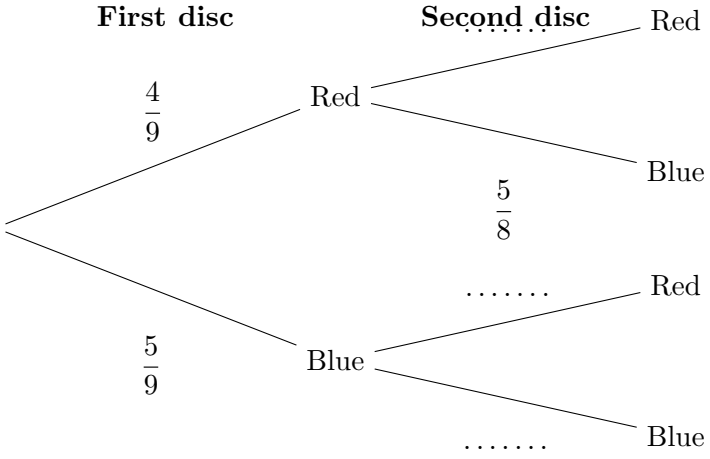

**[2]**

**(b)** Calculate the probability that both counters are red.

..... **[2]**

**1459** A bag contains 4 red discs and 5 blue discs.
Jamil takes one disc at random, notes its colour, and does not replace it.
He then takes a second disc at random.

**(a)** Complete the tree diagram.



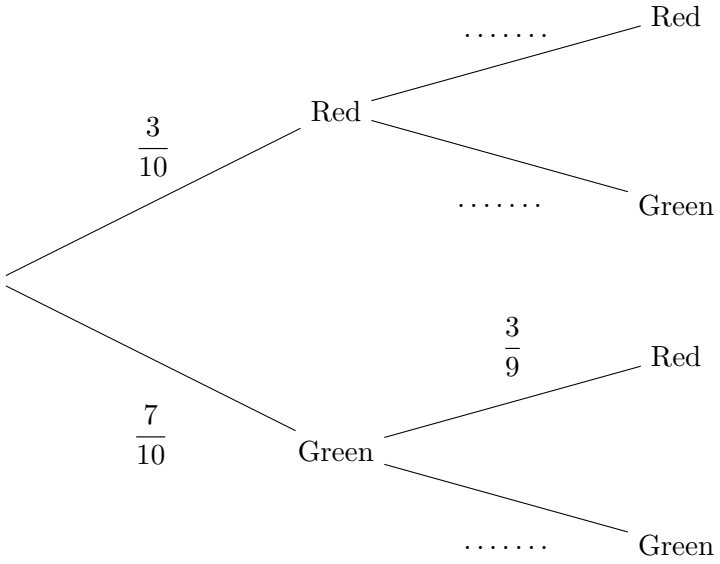

**[2]**

**(b)** Calculate the probability that Jamil takes one red disc and one blue disc.

..... **[3]**

**1460** A basket contains 7 green apples and 3 red apples.
Two apples are chosen at random from the basket, without replacement.

**(a)** Complete the tree diagram.



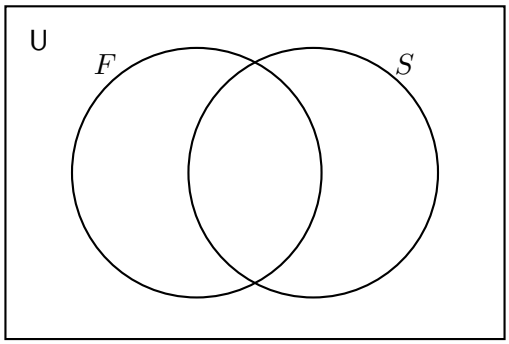

**[2]**

**(b)** Calculate the probability that both apples chosen are the same colour.

..... **[3]**

**1461** A class of 30 students were surveyed about the subjects they study.
17 study French ($F$), 14 study Spanish ($S$), and 5 study neither subject.

**(a)** Complete the Venn diagram to represent this information.




**[2]**

**(b)** A student is chosen at random from the class.
Work out the probability that this student:

**(i)** studies both French and Spanish,

..... **[1]**

**(ii)** studies French but not Spanish,

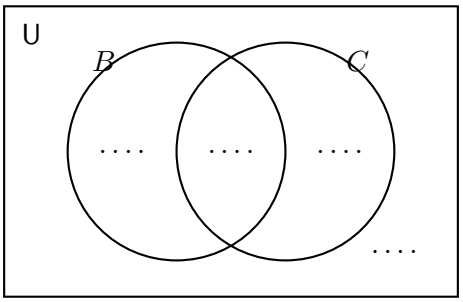
..... **[1]**

**(iii)** is a member of the set $F \cup S$.

..... **[1]**

**1462** In a group of 30 students, 18 study Biology ($B$), 15 study Chemistry ($C$), and 4 study neither Biology nor Chemistry.

**(a)** Complete the Venn diagram to show the number of students in each region.




**[2]**

**(b)** A student is chosen at random from the group.

**(i)** Find the probability that this student studies Chemistry but not Biology.

..... **[1]**

**(ii)** Find $\mathrm{P}(B \cup C)$.

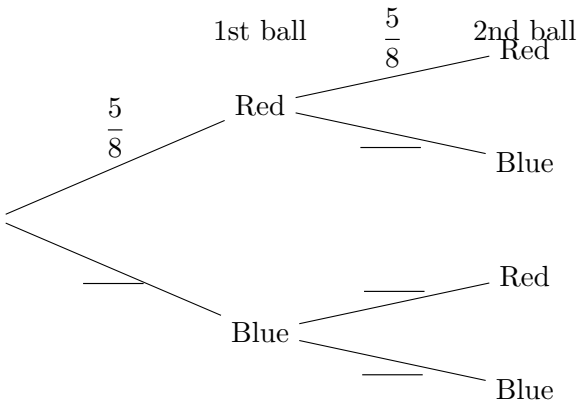
..... **[1]**

**(iii)** Given that the student chosen studies Biology, find the probability that this student also studies Chemistry.

..... **[1]**

**1463** A bag contains 8 balls.
5 of the balls are Red and 3 of the balls are Blue.
A ball is taken at random from the bag, its colour is noted, and it is replaced.
A second ball is then taken at random from the bag.

**(a)** Complete the tree diagram.




**[2]**

**(b)** Calculate the probability that both balls are Red.

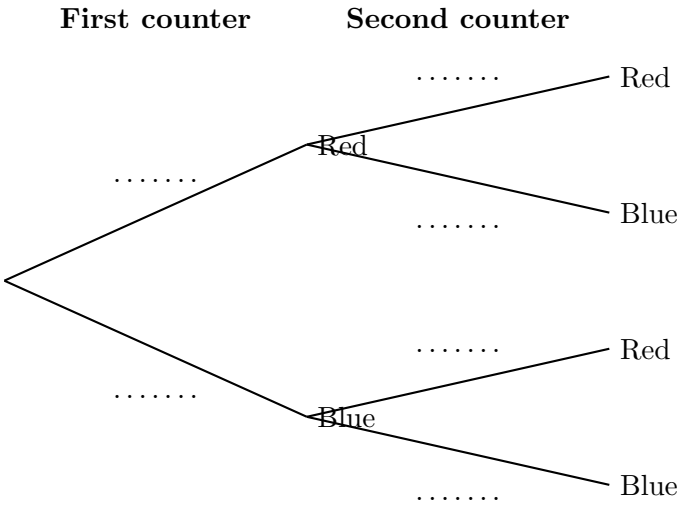
..... **[1]**

**(c)** Calculate the probability of picking one Red ball and one Blue ball in any order.

..... **[2]**

**1464** A bag contains 5 red counters and 3 blue counters.
A counter is picked at random and its colour is noted.
It is then replaced in the bag.
A second counter is then picked at random.

**(a)** Complete the tree diagram.




**[2]**

**(b)** Work out the probability that both counters are red.

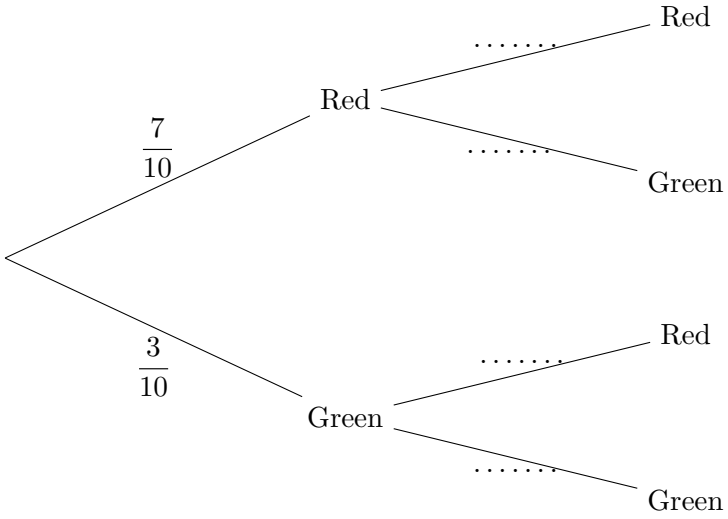
..... **[1]**

**(c)** Work out the probability that one counter is red and one counter is blue.

..... **[2]**

**1465** A bag contains 7 red apples and 3 green apples.
Liam takes two apples at random, without replacement.

**(a)** Complete the tree diagram.



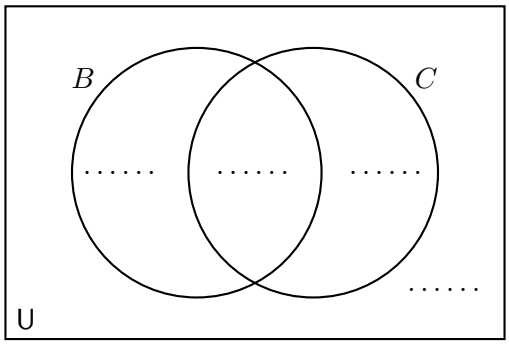

**[2]**

**(b)** Work out the probability that Liam takes two apples of the same colour.

..... **[3]**

**1466** In a group of 30 students, 18 study Biology ($B$) and 15 study Chemistry ($C$).
5 students study neither Biology nor Chemistry.

**(a)** Complete the Venn diagram to show the number of students in each region.




**[2]**

**(b)** A student is chosen at random from the group.

**(i)** Find the probability that this student studies Biology but not Chemistry.

..... **[1]**

**(ii)** Find $\mathrm{P}(B \cup C)$.

..... **[1]**

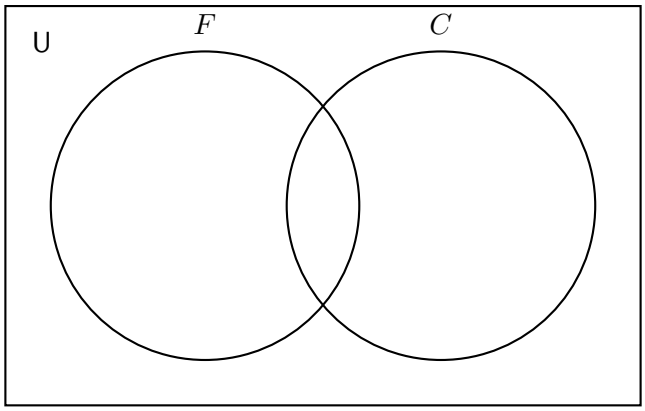
**(c)** Find the probability that a student chosen at random from those who study Chemistry also studies Biology.

..... **[2]**

**1467** There are 40 students in a group.

- 24 students play Football ($F$).
- 15 students play Cricket ($C$).
- 8 students play both Football and Cricket.

**(a)** Complete the Venn diagram to show this information.




**[2]**

**(b)** A student is chosen at random.
Calculate the probability that the student plays neither Football nor Cricket.

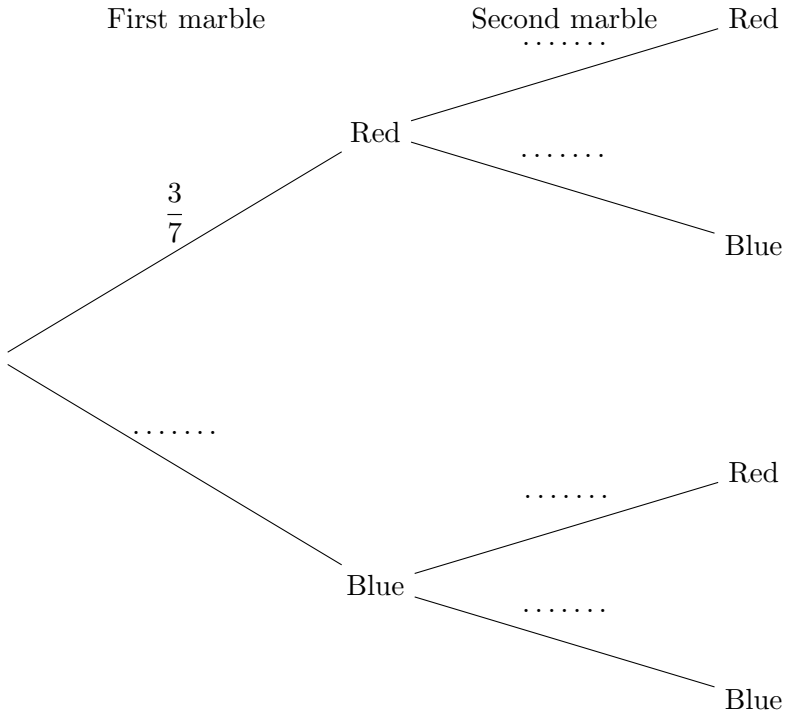
..... **[2]**

**(c)** Find $n(F \cup C)$.

..... **[2]**

**1468** A bag contains 3 red marbles and 4 blue marbles.
A marble is chosen at random and replaced.
A second marble is chosen.

**(a)** Complete the tree diagram.




**[2]**

**(b)** Find the probability that both marbles are red.

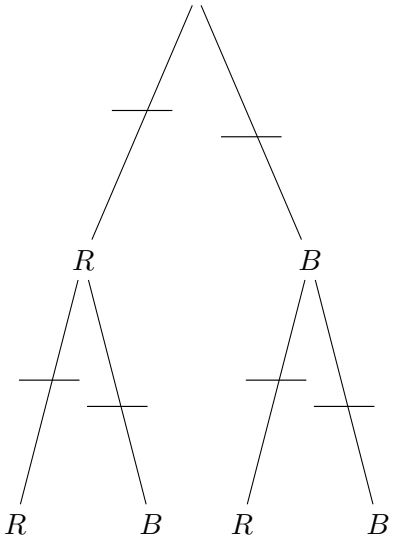
..... **[2]**

**(c)** Find the probability that the two marbles are different colours.

..... **[2]**

**1469** A bag contains 7 red counters and 5 blue counters.
A counter is taken at random, its colour is noted, and it is NOT replaced in the bag.
A second counter is then taken at random and its colour is noted.

**(a)** Complete the tree diagram by writing the missing probabilities on the branches.




**[3]**

**(b)** Calculate the probability that both counters are red.

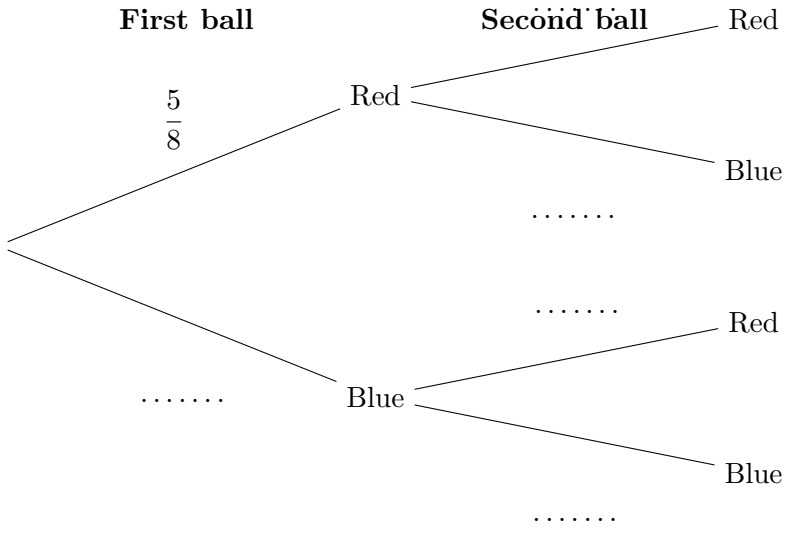
..... **[2]**

**(c)** Calculate the probability that one counter is red and one counter is blue.

..... **[2]**

**1470** A bag contains 8 balls.
There are 5 Red balls and 3 Blue balls.
A ball is taken at random from the bag, its colour is recorded, and it is replaced.
A second ball is then taken at random from the bag.

**(a)** Complete the tree diagram.




**[2]**

**(b)** Calculate the probability that both balls are Red.

..... **[2]**

**(c)** Calculate the probability that one ball is Red and one ball is Blue.

..... **[3]**

**1471** A fair 4-sided spinner has edges numbered 1, 2, 3, and 4.
A fair 6-sided dice has faces numbered 1, 2, 3, 4, 5, and 6.
Both the spinner and the dice are spun and rolled together, and the two scores are added to give the total score.

**(a)** Complete the sample space diagram below to show all possible total scores.

| Spin \ Dice | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | | | |
| 2 | 3 | 4 | | | | |
| 3 | 4 | | | | | |
| 4 | | | | | | |

**[2]**

**(b)** Find the probability, as a fraction in its simplest form, that the total score is

**(i)** equal to 7,

..... **[1]**

**(ii)** a prime number,

..... **[2]**

**(iii)** greater than 6 and even.

..... **[2]**

**(c)** The spinner and the dice are spun and rolled together twice.
Calculate the probability that the total score is equal to 7 on both occasions.

..... **[2]**

**1472** A box contains $x$ blue counters and 4 green counters.
Two counters are chosen at random from the box, one after another, without replacement.

**(a)** Show that the probability that both counters chosen are blue is $\dfrac{x(x-1)}{x^2+7x+12}$.

**[2]**

**(b)** The probability that both counters chosen are blue is $\dfrac{1}{3}$.

**(i)** Show that $x^2 - 5x - 6 = 0$.

**[3]**

**(ii)** Solve the equation $x^2 - 5x - 6 = 0$ to find the number of blue counters in the box.

$x =$ ..... **[3]**

**1473** A bag contains $x$ red counters and 5 blue counters.
Two counters are taken at random from the bag without replacement.
The probability that both counters are blue is $\frac{2}{15}$.

**(a)** Show that $x^2 + 9x - 130 = 0$.

**[3]**

**(b)** Find the value of $x$.

$x =$ ..... **[2]**

**(c)** Calculate the probability that the two counters chosen are different colours.

..... **[3]**

**1474** A box contains $x$ red balls and 4 blue balls.
Two balls are taken from the box at random, without replacement.
The probability that both balls are red is $\frac{1}{3}$.

**(a)** Show that $x^2 - 5x - 6 = 0$.

**[3]**

**(b)** Work out the number of red balls in the box.

..... **[2]**

**(c)** Calculate the probability that the two balls taken have different colours.

..... **[3]**

**1475** A bag contains $n$ counters.
3 of the counters are red and the rest are blue.
Two counters are chosen at random, without replacement.
The probability that both counters are red is $\frac{1}{15}$.

**(a)** Show that $n^2 - n - 90 = 0$.

**[3]**

**(b)** Find the number of blue counters in the bag.

..... **[2]**

**(c)** Find the probability that the two counters chosen have different colours.

..... **[3]**

**1476** A bag contains $n$ balls.
3 of the balls are red and the rest are blue.
Two balls are chosen at random, without replacement.
The probability that both balls are red is $\frac{1}{15}$.

**(a)** Show that $n^2 - n - 90 = 0$.

**[3]**

**(b)** Find the value of $n$.

$n =$ ..... **[2]**

**(c)** Find the probability that the two balls have different colours.

..... **[3]**

**1477** A box contains $n$ counters.
3 of the counters are Red and all the remaining counters are Blue.
Two counters are taken at random from the box, without replacement.
The probability that both counters are Red is $\frac{1}{15}$.

**(a)** Show that $n^2 - n - 90 = 0$.

**[3]**

**(b)** Find the value of $n$.

$n =$ ..... **[2]**

**(c)** Calculate the probability that exactly one of the two counters chosen is Red.

..... **[3]**

**1478** A bag contains $x$ red discs and 3 blue discs.
Two discs are chosen at random without replacement.
The probability that both discs are red is $\frac{5}{14}$.

**(a)** Show that $3x^2 - 13x - 10 = 0$.

**[4]**

**(b)** Find the value of $x$.

$x =$ ..... **[2]**

**(c)** Two discs are chosen at random without replacement from the original bag.
Work out the probability that they are different colours.

..... **[2]**

**1479** A box contains $n$ black beads and 6 white beads, where $n > 0$.
Two beads are taken at random from the box without replacement.
The probability that both beads are black is $\frac{1}{2}$.

**(a)** Show that $n^2 - 13n - 30 = 0$.

**[3]**

**(b)** Solve the equation $n^2 - 13n - 30 = 0$ to find the number of black beads in the box.

$n =$ ..... **[2]**

**(c)** Calculate the probability, as a fraction in its simplest form, that

**(i)** the two beads are of different colours,

..... **[2]**

**(ii)** at least one of the two beads is black.

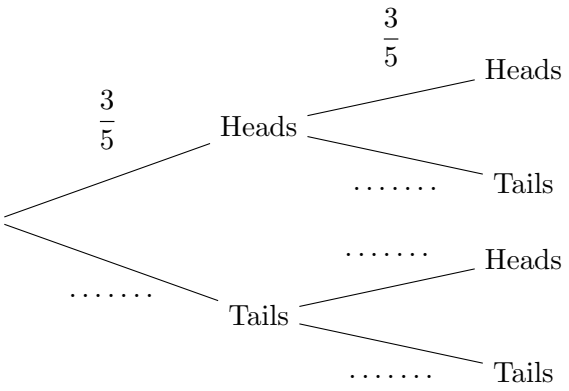
..... **[2]**

**(d)** Instead of two beads, three beads are taken at random from the original box without replacement.
Calculate the probability, as a fraction in its simplest form, that two of the beads are black and one is white.

..... **[3]**

**1480** A biased coin has a probability of showing Heads of $\frac{3}{5}$.
The coin is flipped twice.

**(a)** Complete the tree diagram.




**[2]**

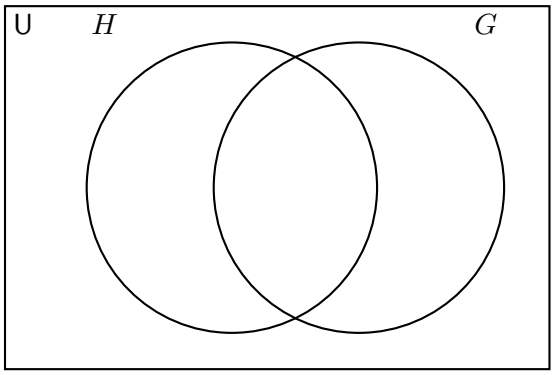
**(b)** Calculate the probability that the coin shows exactly one Head.

..... **[2]**

**1481** In a survey, 40 students are asked if they study History ($H$) or Geography ($G$).

- 23 students study History.
- 19 students study Geography.
- 7 students study neither subject.

**(a)** Complete the Venn diagram to show this information.




**[2]**

**(b)** A student is chosen at random from the 40 students.
Find the probability that the student studies History but not Geography.

..... **[1]**

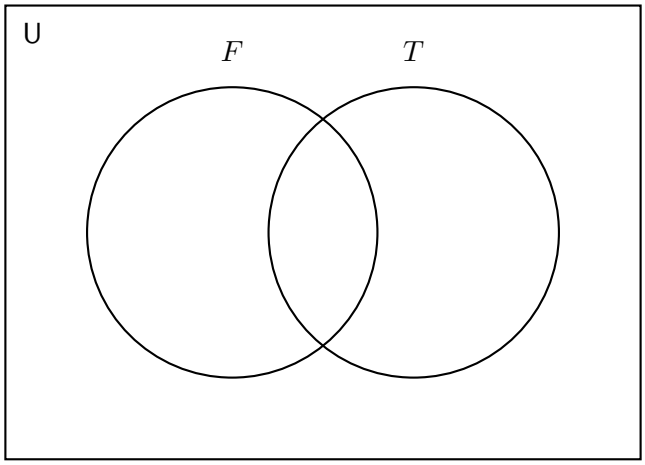
**(c)** Two students are chosen at random from those who study History.
Calculate the probability that they both also study Geography.

..... **[2]**

**1482** There are 30 students in a class.
$\mathsf{U}$ = {students in the class}
$F$ = {students who play Football}
$T$ = {students who play Tennis

18 students play Football. 14 students play Tennis. 4 students do not play Football and do not play Tennis.

**(a)** Complete the Venn diagram to show the number of students in each region.




**[2]**

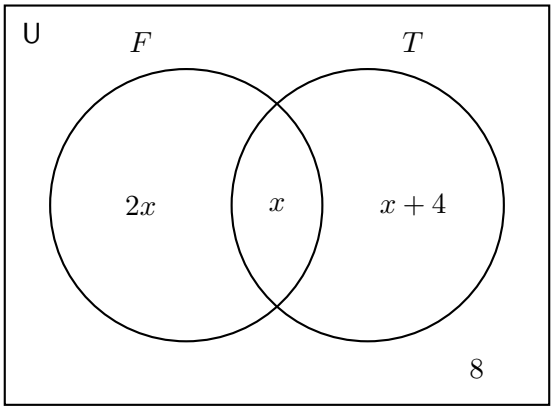
**(b)** A student is chosen at random from the class.
Find $P(F \cap T)$.

$P(F \cap T) =$ ..... **[1]**

**(c)** Two students are chosen at random from the class.
Calculate the probability that both students play Football but not Tennis.

..... **[3]**

**1483** A group of 40 students are asked if they play Football ($F$) or Tennis ($T$).
The Venn diagram shows the number of students in each region in terms of $x$.




**(a)** Find the value of $x$.

$x =$ ..... **[2]**

**(b)** Find $P(F \cup T)$.

..... **[1]**

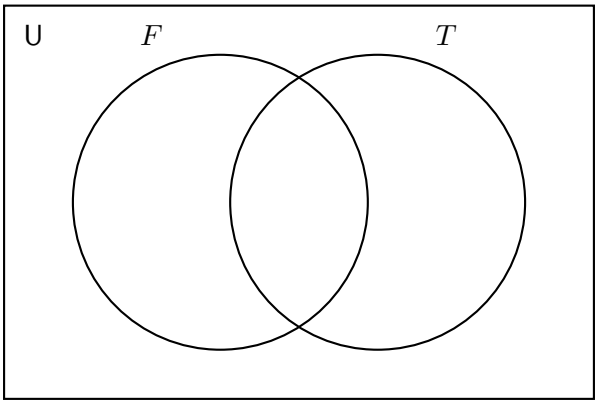
**(c)** Two students are chosen at random from the group of 40.
Calculate the probability that both students play Football only.

..... **[3]**

**1484** In a group of 40 students:

- 24 play Football ($F$)
- 19 play Tennis ($T$)
- 7 do not play either sport.

**(a)** Complete the Venn diagram to show this information.




**[2]**

**(b)** A student is chosen at random from the group.
Find the probability that the student plays both Football and Tennis.

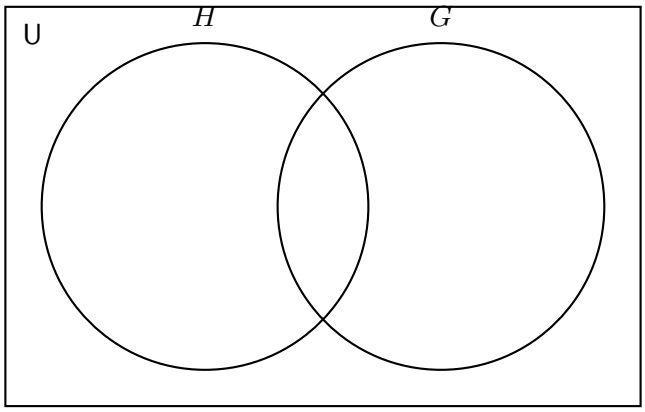
..... **[1]**

**(c)** Two students who play Football are chosen at random.
Find the probability that exactly one of them also plays Tennis.

..... **[3]**

**1485** In a group of 50 students, 32 study History ($H$), 21 study Geography ($G$), and 6 study neither subject.

**(a)** Complete the Venn diagram to show the number of students in each region.




**[3]**

**(b)** A student is chosen at random from the group of 50.

Find:

**(i)** $P(H \cap G)$

$P(H \cap G) =$ ..... **[1]**

**(ii)** $P(H \cup G')$

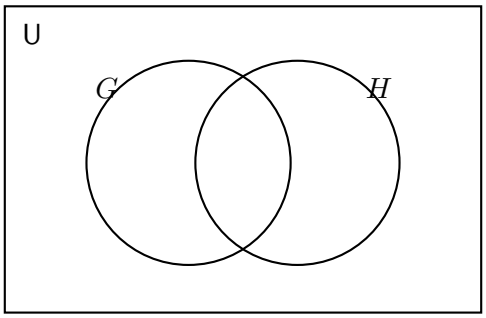
$P(H \cup G') =$ ..... **[1]**

**(c)** Two students who study History are chosen at random.
Calculate the probability that both students also study Geography.

..... **[2]**

**1486** In a group of 40 students, 24 study Geography ($G$), 18 study History ($H$), and 6 study neither subject.

**(a)** Complete the Venn diagram to show the number of students in each region.




**[2]**

**(b)** A student is chosen at random from the group of 40 students.

**(i)** Find the probability that the student studies History but not Geography.

..... **[1]**

**(ii)** Find $P(G \cup H')$.

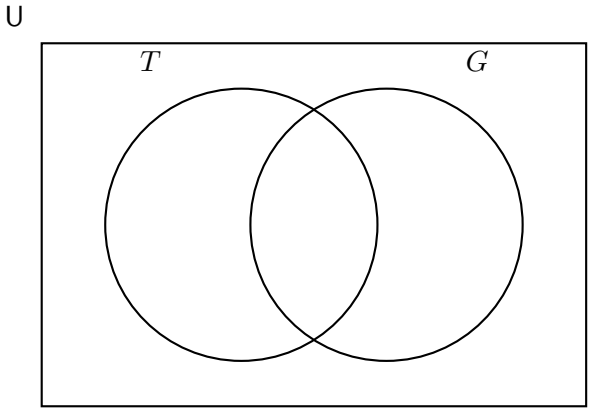
..... **[2]**

**(c)** Two students are chosen at random without replacement from the group of 40 students. Work out the probability that both students study Geography.

..... **[2]**

**1487** In a group of 40 students, 24 play tennis ($T$) and 18 play golf ($G$).
5 students play neither tennis nor golf.

**(a)** Complete the Venn diagram to show this information.




**[2]**

**(b)** Two students are chosen at random from the group of 40.
Calculate the probability that both students play tennis but not golf.

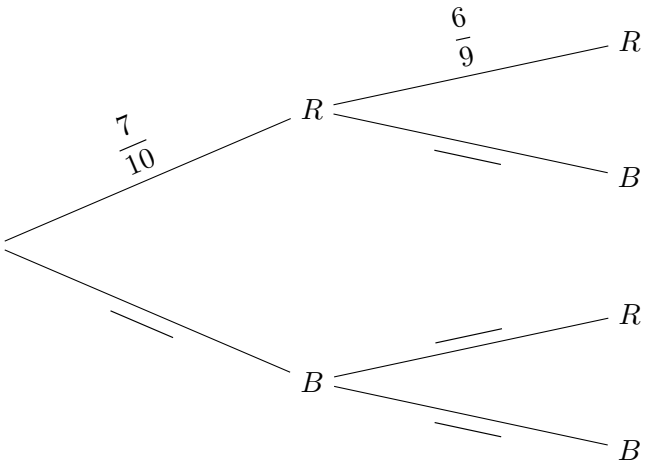
..... **[3]**

**(c)** A student who plays golf is chosen at random.
Find the probability that they also play tennis.

..... **[2]**

**1488** A box contains 7 red counters and 3 blue counters.
A counter is chosen at random from the box and its colour is recorded.
It is not replaced.
A second counter is then chosen at random from the box.

**(a)** Complete the tree diagram.




**[3]**

**(b)** Calculate the probability that both counters chosen are red.

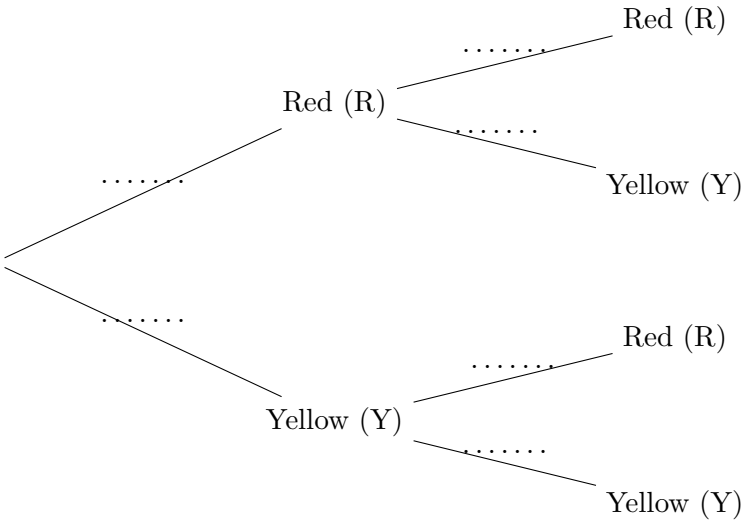
..... **[2]**

**(c)** Calculate the probability that the two counters chosen are of different colours.

..... **[2]**

**1489** A bag contains 5 red sweets and 4 yellow sweets.
Maya takes a sweet at random from the bag and eats it.
She then takes a second sweet at random from the bag and eats it.

**(a)** Complete the tree diagram.




**[2]**

**(b)** Calculate the probability that Maya eats two sweets of the same colour.

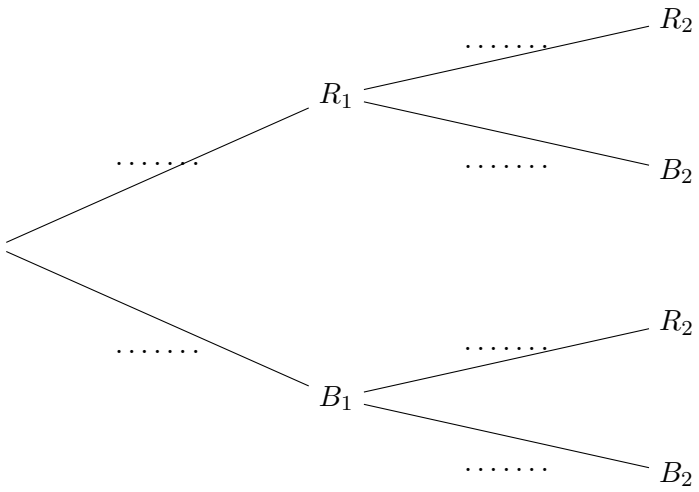
..... **[2]**

**(c)** Calculate the probability that Maya eats at least one yellow sweet.

..... **[3]**

**1490** A bag contains 5 red counters and 3 blue counters.
Two counters are taken from the bag at random, without replacement.

**(a)** Complete the tree diagram by writing the missing probabilities on the branches.




**[3]**

**(b)** Calculate the probability that both counters taken are the same colour.

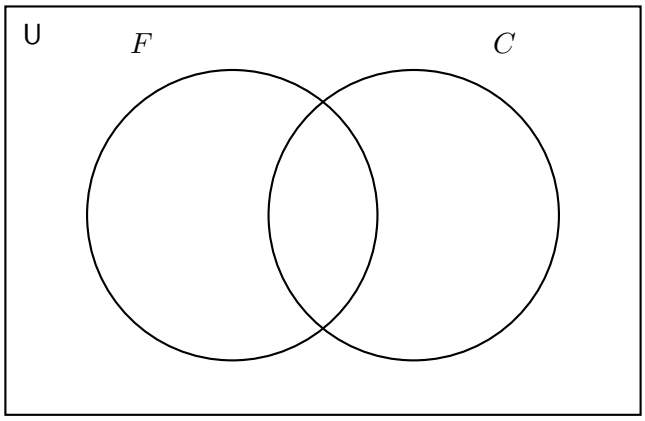
..... **[2]**

**(c)** Calculate the probability that at least one of the counters taken is red.

..... **[2]**

**1491** In a group of 40 students, 24 like football ($F$) and 18 like cricket ($C$).
$x$ students like both sports.
3 students do not like either sport.

**(a)** Complete the Venn diagram to show this information in terms of $x$.




**[2]**

**(b)** Work out the value of $x$.

$x =$ ..... **[2]**

**(c)** Two students are chosen at random from the group of 40.
Work out the probability that one student likes football only and the other student likes cricket only.

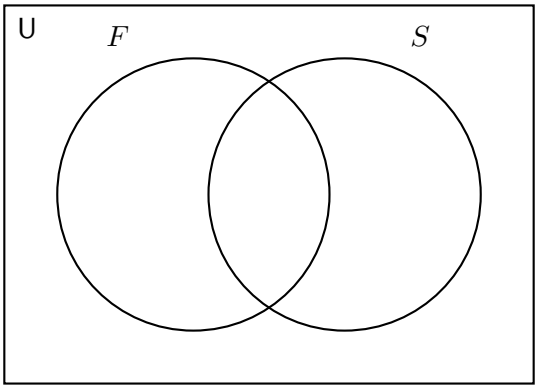
..... **[3]**

**1492** In a class of 30 students, 18 study French ($F$) and 14 study Spanish ($S$).
3 students study neither French nor Spanish.
Let $x$ be the number of students who study both French and Spanish.

**(a)** Find the value of $x$.

$x =$ ..... **[2]**

**(b)** Complete the Venn diagram to show the number of students in each region.



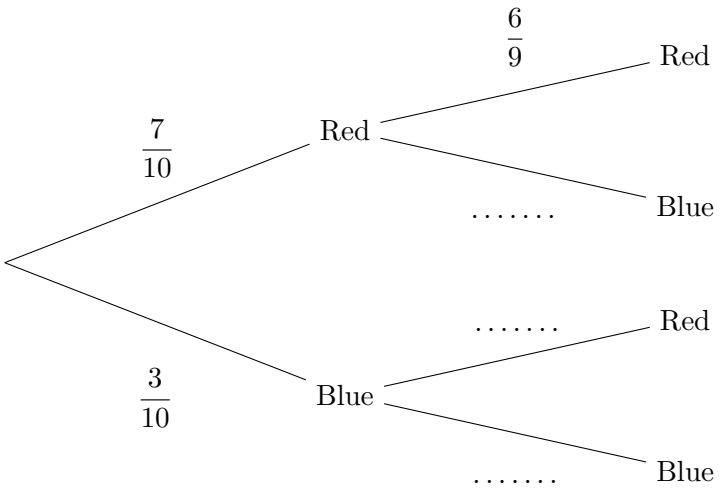

**[2]**

**(c)** Two students are chosen at random from the class.
Calculate the probability that both students study French but do not study Spanish.

..... **[3]**

**1493** A box contains 7 red counters and 3 blue counters.
Two counters are taken at random from the box, without replacement.

**(a)** Complete the tree diagram.




**[3]**

**(b)** Calculate the probability that both counters are red.

..... **[2]**

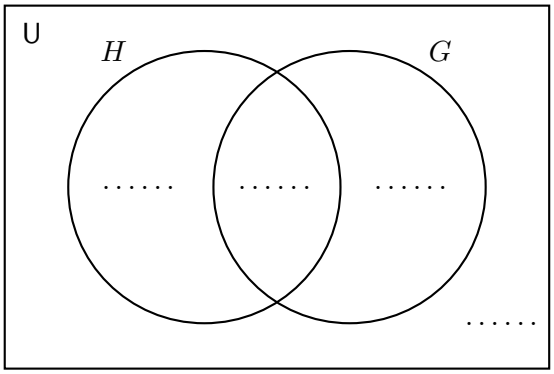
**(c)** Calculate the probability that the two counters are different colours.

..... **[2]**

**1494** In a group of 30 students,

- 18 study History ($H$)
- 15 study Geography ($G$)
- 4 study neither subject.

**(a)** Complete the Venn diagram.




**[2]**

**(b)** A student is chosen at random.
Find $P(H \cap G')$.

$P(H \cap G') =$ ..... **[1]**

**(c)** Two students are chosen at random from the group of 30 students.
Find the probability that both students study History.

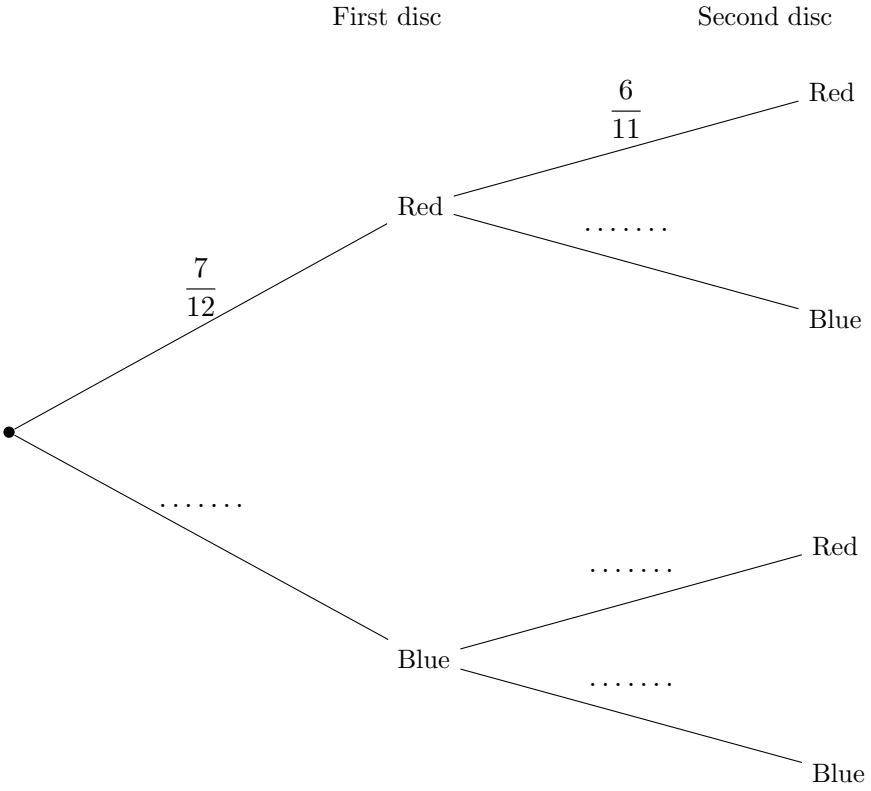
..... **[2]**

**(d)** Two students are chosen at random from those who study History.
Find the probability that exactly one of them studies Geography.

..... **[3]**

**1495** A bag contains 7 red discs and 5 blue discs.
Priya takes a disc at random from the bag, notes its colour, and does not replace it.
She then takes a second disc at random from the bag.

**(a)** Complete the tree diagram below by writing the missing probabilities on the branches.




**[3]**

**(b)** Calculate the probability, as a fraction in its simplest form, that Priya takes

**(i)** two red discs,

..... **[2]**

**(ii)** one red disc and one blue disc.

..... **[3]**

**(c)** If Priya had replaced the first disc before taking the second disc, calculate the probability, as a

fraction in its simplest form, that she would take two discs of the same colour.

..... **[3]**

**1496** A bag contains $n$ sweets.
5 of the sweets are orange and the rest are lemon.
Two sweets are taken from the bag at random, without replacement.
The probability that both sweets are orange is $\frac{2}{21}$.

**(a)** Show that $n^2 - n - 210 = 0$.

**[3]**

**(b)** Solve the equation $n^2 - n - 210 = 0$ to find the number of sweets in the bag.

$n =$ ..... **[2]**

**(c)** Find the probability that the two sweets taken have different flavours.

..... **[3]**

**1497** Box A contains $x$ blue counters and 3 red counters.
Box B contains 4 blue counters and 2 red counters.
A counter is taken at random from Box A and placed into Box B.
A counter is then taken at random from Box B.

**(a)** Show that the probability of taking a blue counter from Box B is

$$\frac{5x+12}{7(x+3)}$$

**[4]**

**(b)** The probability of taking a blue counter from Box B is $\frac{3}{4}$.

**(i)** Work out the value of $x$.

$x =$ ..... **[3]**

**(ii)** Using your value of $x$, find the probability that the counter transferred from Box A to Box B was red, given that the counter taken from Box B is blue.

..... **[1]**

**1498** A bag contains 4 red discs and some blue discs.
Let the total number of discs in the bag be $x$.
Two discs are taken at random from the bag, without replacement.
The probability that both discs are red is $\frac{1}{11}$.

**(a)** Show that $x^2 - x - 132 = 0$.

**[3]**

**(b)** Find the number of blue discs in the bag.

..... **[3]**

**(c)** Calculate the probability that one disc is red and one disc is blue.

..... **[3]**

**1499** A box contains $n$ chocolates.
There are 4 milk chocolates and the rest are dark chocolates.
Two chocolates are taken at random from the box, without replacement.
The probability that both chocolates are milk chocolate is $\frac{1}{14}$.

**(a)** Show that $n^2 - n - 168 = 0$.

**[3]**

**(b)** Solve the equation to find the number of dark chocolates in the box.

..... **[3]**

**(c)** Calculate the probability that the two chocolates taken are of different types.

..... **[3]**

**1500** A box contains $n$ sweets. 6 of the sweets are lemon flavour and the rest are orange flavour.

**(a)** Hannah takes two sweets at random from the box without replacement.
The probability that both sweets are lemon flavour is $\frac{1}{3}$.

**(i)** Show that $n^2 - n - 90 = 0$.

**[3]**

**(ii)** Solve the equation $n^2 - n - 90 = 0$ to find the total number of sweets in the box.

$n =$ ..... **[3]**

**(b)** Three sweets are now taken at random without replacement from the original box of 10 sweets.
Calculate the probability that at least one of the sweets taken is orange flavour.

..... **[3]**

**1501** A box contains $n$ chocolates, of which 4 are dark chocolates and the rest are milk chocolates.

**(a)** Two chocolates are taken from the box at random, without replacement.
The probability that both chocolates are dark is $\frac{2}{15}$.
Show that $n^2 - n - 90 = 0$.

**[3]**

**(b)** Solve the equation $n^2 - n - 90 = 0$ to find the total number of chocolates in the box.

$n =$ ..... **[2]**

**(c)** Three chocolates are now taken at random from the original box of chocolates, without replacement.
Work out the probability that exactly two of the three chocolates taken are dark chocolates.

..... **[4]**

**1502** A bag contains $x$ green apples and 4 red apples.
Two apples are taken at random, without replacement.

**(a)** The probability that both apples are green is $\frac{5}{18}$.

Show that $13x^2 - 53x - 60 = 0$.

**[4]**

**(b)** Solve the equation $13x^2 - 53x - 60 = 0$ to find the value of $x$.

$x =$ ..... **[3]**

**(c)** Work out the probability that exactly one of the two apples taken is green.

..... **[3]**

**1503** 70 students are asked to choose their favourite sport from Football, Tennis, or Basketball.
The incomplete two-way table shows some information about their choices.

**(a)** Complete the two-way table.

| | **Football** | **Tennis** | **Basketball** | **Total** |
|---|---|---|---|---|
| **Boys** | 15 | | 12 | |
| **Girls** | | 14 | | 30 |
| **Total** | 23 | | 20 | 70 |

**[3]**

**(b)** A student is chosen at random.
Find the probability that the student is a boy whose favourite sport is Tennis.

..... **[1]**

**1504** 50 students each choose one sport to play: Tennis, Football, or Swimming.
The students are classified as Boys or Girls.

**(a)** Complete the two-way table to show this information.

| | **Tennis** | **Football** | **Swimming** | **Total** |
|---|---|---|---|---|
| **Boys** | 7 | | | 28 |
| **Girls** | 8 | | 6 | |
| **Total** | | 20 | | 50 |

**[3]**

**(b)** A student is chosen at random.
Find the probability that the student is a boy who chose Football.

..... **[1]**

**1505** A group of 50 students were asked which sport they prefer.
Some information is shown in the two-way table.

**(a)** Complete the two-way table.

| | Football | Tennis | Swimming | Total |
|---|---|---|---|---|
| Boys | 15 | 8 | | 28 |
| Girls | | 8 | | |
| Total | 20 | | | 50 |

**[3]**

**(b)** A student is chosen at random.
Find the probability that the student is a girl who prefers Swimming.

..... **[1]**

**1506** 50 students were asked to choose their favourite sport from Football, Basketball, and Tennis.
The two-way table shows some information about their choices.

**(a)** Complete the two-way table.

| | **Football** | **Basketball** | **Tennis** | **Total** |
|---|---|---|---|---|
| **Boys** | 15 | | 5 | 28 |
| **Girls** | | 10 | 6 | |
| **Total** | 21 | | | 50 |

**[3]**

**(b)** A student who chose Tennis is chosen at random.
Work out the probability that the student is a boy.

..... **[2]**

**1507** 80 students were asked to choose their favourite sport from Football, Basketball, and Tennis.

- There are 35 girls in total.
- 38 students chose Football, of which 24 were boys.
- 12 girls chose Basketball.
- 20 students chose Tennis.

**(a)** Complete the two-way table.

| | **Football** | **Basketball** | **Tennis** | **Total** |
|---|---|---|---|---|
| **Boys** | | | | |
| **Girls** | | | | |
| **Total** | | | | |

**[3]**

**(b)** A student is chosen at random.
Find the probability that the student is a boy whose favourite sport is Basketball.

..... **[1]**

**(c)** A girl is chosen at random.
Find the probability that her favourite sport is Tennis.

..... **[2]**

**1508** A fitness instructor collects statistical data from 24 members of a sports club.

**(a)** Classify each of the following types of data as qualitative, discrete, or continuous:

**(i)** favourite sport,

..... **[1]**

**(ii)** height, in metres,

..... **[1]**

**(iii)** number of hours spent exercising each week, rounded to the nearest integer.

..... **[1]**

**(b)** The resting heart rates, $r$ beats per minute, of the 24 members are listed below.

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 64 | 72 | 58 | 81 | 67 | 75 | 60 | 78 | 54 | 69 | 84 | 71 |
| 63 | 77 | 59 | 68 | 70 | 82 | 65 | 74 | 56 | 66 | 73 | 80 |

Complete the grouped tally and frequency table below.

| Resting heart rate ($r$) | Tally | Frequency |
|---|---|---|
| $50 \leq r < 60$ | \|\|\|\| | 4 |
| $60 \leq r < 70$ | | |
| $70 \leq r < 80$ | | |
| $80 \leq r < 90$ | | |

**[3]**

**(c)** Calculate the percentage of these 24 members who have a resting heart rate of less than 70 beats per minute.

..... % **[2]**

**1509** A survey of adults recorded their preferred hot drink, either Tea or Coffee.
Let $x$ be the number of males who prefer Coffee.
The number of males who prefer Tea is $2x + 10$.
The number of females who prefer Coffee is $x + 5$.
The number of females who prefer Tea is $3x - 11$.

**(a)** Write down an expression, in terms of $x$, for the total number of adults.
Give your answer in its simplest form.

..... **[2]**

**(b)** The survey included 130 adults.
Find the value of $x$.

$x =$ ..... **[2]**

**(c)** Complete the two-way table.

| | **Coffee** | **Tea** | **Total** |
|---|---|---|---|
| **Male** | | | |
| **Female** | | | |
| **Total** | | | |

**[3]**

**(d)** One of the adults who prefers Coffee is chosen at random.
Find the probability that this adult is female.

..... **[2]**

**1510** A researcher classifies 150 participants in a clinical trial according to their blood group ($A$, $B$, or $O$) and whether they responded positively ($Yes$) or did not respond positively ($No$) to a new treatment. The two-way table below shows the number of participants in each category, where $x$ and $y$ are positive integers.
The total number of participants with blood group $B$ is 28.

| | Blood group $A$ | Blood group $B$ | Blood group $O$ |
|---|---|---|---|
| $Yes$ | $x^2 + 2x$ | $2x + y$ | $4x - y$ |
| $No$ | $2x + 3y$ | $x + 6$ | $3x + 3y$ |

**(a)** **(i)** Use the total number of participants with blood group $B$ to express $y$ in terms of $x$.

$y =$ ..... **[2]**

**(ii)** Show that $x^2 - 4x - 12 = 0$.

**[3]**

**(b)** Solve the equation $x^2 - 4x - 12 = 0$ to find the value of $x$ and the value of $y$.

$x =$ ..... , $y =$ ..... **[3]**

**(c)** **(i)** Complete the numerical two-way table below using your values of $x$ and $y$.

| | Group $A$ | Group $B$ | Group $O$ | Total |
|---|---|---|---|---|
| $Yes$ | | | | |
| $No$ | | | | |
| Total | | 28 | | 150 |

**[2]**

**(ii)** Calculate the percentage of all 150 participants who responded positively ($Yes$) to the treatment.

..... % **[2]**

**1511** A group of 120 adults were asked whether they prefer Tea or Coffee.

- 55 of the adults are women.
- 70 of the adults prefer Tea.
- 25 of the men prefer Coffee.

**(a)** Complete the two-way table to show this information.

| | **Tea** | **Coffee** | **Total** |
|---|---|---|---|
| **Men** | | | |
| **Women** | | | |
| **Total** | | | |

**[3]**

**(b)** Work out the number of women who prefer Tea as a percentage of the total number of adults.

..... % **[2]**

**1512** A company has 120 employees. There are 75 men and 45 women.
They travel to work by bus, train, or car.

- 20% of the men travel by bus.
- $\frac{1}{3}$ of the women travel by bus.
- A total of 38 employees travel by train, of which 14 are women.
- The remaining employees travel by car.

**(a)** Construct and complete the two-way table to show this information.

| | **Bus** | **Train** | **Car** | **Total** |
|---|---|---|---|---|
| **Men** | | | | |
| **Women** | | | | |
| **Total** | | | | |

**[4]**

**(b)** An employee who travels by car is chosen at random.
Work out the probability that this employee is a man.

..... **[2]**

**1513** In a school of 400 students, the ratio of juniors to seniors is 3 : 2.
Each student travels to school by Bus, by Car, or Walks.

- 45% of the 400 students take the Bus.
- Of the seniors, 30 take the Car and 50 Walk.
- There are 80 juniors who take the Car.

**(a)** Complete the two-way table to classify this data.

| | **Bus** | **Car** | **Walk** | **Total** |
|---|---|---|---|---|
| **Juniors** | | | | |
| **Seniors** | | | | |
| **Total** | | | | 400 |

**[3]**

**(b)** A student is chosen at random.
Find the probability that they are a junior who walks.

..... **[1]**

**(c)** Two seniors are chosen at random.
Find the probability that they both take the bus.

..... **[2]**

**1514** A group of 85 people were asked about their favourite drink.
The two-way table shows the number of people in terms of $x$.

| | Tea | Coffee | Water |
|---|---|---|---|
| Men | $x$ | $2x - 5$ | $x + 6$ |
| Women | $x - 2$ | $x + 5$ | $2x + 1$ |

**(a)** Find the value of $x$.

$x =$ ..... **[3]**

**(b)** Find the probability that a person chosen at random is a man who prefers Water.

..... **[2]**

**(c)** A person who prefers Coffee is chosen at random.
Find the probability that this person is a woman.

..... **[2]**

**1515** A school surveys 300 students in Year 9, Year 10, and Year 11 about their main method of travel to school: Walk, Bus, or Car.

- 32% of the 300 students are in Year 9.
- There are 18 more students in Year 10 than in Year 9.
- $\frac{3}{8}$ of the Year 9 students travel by Bus, and 25 Year 9 students travel by Car.
- Among the Year 11 students, the ratio of those who Walk to those who travel by Bus is 5 : 7.

**(a)** Complete the two-way table below to show the results of the survey.

| | Walk | Bus | Car | Total |
|---|---|---|---|---|
| Year 9 | | | | |
| Year 10 | | | | |
| Year 11 | | | 18 | |
| Total | 115 | | 85 | 300 |

**[5]**

**(b)** Calculate the percentage of Year 10 students who travel to school by Car, giving your answer correct to 1 decimal place.

..... % **[2]**

**(c)** One of the students who travels by Bus is chosen at random.
Find the probability, as a fraction in its simplest form, that this student is in Year 11.

..... **[2]**

**1516** 80 students travel to school either by Bus or by Walking.
The two-way table shows the number of boys and girls using each method of transport, in terms of $x$.

| | **Bus** | **Walk** | **Total** |
|---|---|---|---|
| **Boys** | $x$ | $2x$ | |
| **Girls** | $x + 5$ | $3x - 2$ | |
| **Total** | | | 80 |

**(a)** Work out the value of $x$.

$x =$ ..... **[3]**

**(b)** Complete the two-way table with the number of students in each category.

| | **Bus** | **Walk** | **Total** |
|---|---|---|---|
| **Boys** | | | |
| **Girls** | | | |
| **Total** | | | 80 |

**[2]**

**(c)** A student is chosen at random from those who walk to school.
Find the probability that this student is a girl.

..... **[2]**

**1517** 300 students each choose exactly one activity: Art, Music, or Drama.

- The ratio of boys to girls is 2 : 3.
- 20% of the boys choose Art.
- The number of boys who choose Drama is twice the number of boys who choose Art.
- $\frac{1}{3}$ of the girls choose Drama.
- The ratio of girls who choose Art to girls who choose Music is 3 : 5.

**(a)** Complete the two-way table to show this information.

| | Art | Music | Drama | Total |
|---|---|---|---|---|
| Boys | | | | |
| Girls | | | | |
| Total | | | | |

**[5]**

**(b)** A student is chosen at random.
Find the probability that the student is a girl who does not choose Drama.

..... **[2]**

**1518** A company has 300 employees. Each employee is either a Manager, an Office Worker, or a Technician.
Each employee works either Full-Time or Part-Time.

- The ratio of Managers : Office Workers : Technicians is 2 : 5 : 3.
- 20% of the Managers work Part-Time.
- 40% of the Office Workers work Part-Time.
- The number of Part-Time Technicians is equal to the number of Part-Time Managers.

**(a)** Complete the two-way table for this information.

| | **Full-Time** | **Part-Time** | **Total** |
|---|---|---|---|
| **Managers** | | | |
| **Office Workers** | | | |
| **Technicians** | | | |
| **Total** | | | |

**[4]**

**(b)** A Part-Time employee is chosen at random.
Find the probability that they are an Office Worker.

..... **[2]**

**(c)** Two Managers are chosen at random.
Find the probability that exactly one of them works Part-Time.

..... **[3]**

**1519** A sports club has 200 members. Each member plays either Golf or Tennis.
There are $x$ male members in the club.

- The number of male members who play Golf is $\frac{1}{4}x$.
- The number of female members who play Tennis is 30 more than the number of male members who play Golf.
- A total of 110 members play Tennis.

**(a)** Write down an expression, in terms of $x$, for the number of male members who play Tennis.
Give your answer in its simplest form.

..... **[1]**

**(b)** Use the information about the members who play Tennis to form an equation in $x$, and show that $x = 80$.

**[2]**

**(c)** Complete the two-way table to show the number of members in each category.

| | **Golf** | **Tennis** | **Total** |
|---|---|---|---|
| **Male** | | | |
| **Female** | | | |
| **Total** | | | |

**[3]**

**(d)** Two members are chosen at random from those who play Golf.
Calculate the probability that they are both female.

..... **[3]**

**1520** A group of 120 students are classified by gender (Boys, Girls) and handedness (Left-handed, Right-handed).
The two-way table shows the number of students in each category in terms of $x$ and $y$.
Two students are chosen at random from the 120 students.
The probability that both students are left-handed is $\frac{3}{34}$.

| | **Left-handed** | **Right-handed** |
|---|---|---|
| **Boys** | $x$ | $y$ |
| **Girls** | $2x$ | $y - 6$ |

**(a)** Show that $3x^2 - x - 420 = 0$.

**[4]**

**(b)** Find the value of $x$ and the value of $y$.

$x =$ ..... , $y =$ ..... **[4]**

**(c)** A right-handed student is chosen at random.
Find the probability that the student is a girl.

..... **[2]**

**1521** The table shows information about the marks scored by two different classes, Class A and Class B, in a mathematics examination.

| **Statistic** | **Class A** | **Class B** |
|---|---|---|
| Mean | 62 | 70 |
| Median | 65 | 72 |
| Range | 40 | 25 |
| Interquartile Range | 15 | 10 |

**(a)** Which class had the higher average mark?
Give a reason for your answer.

..... **[1]**

**(b)** Which class had marks that were more consistent?
Give a reason for your answer.

..... **[1]**

**(c)** A student says, "Every student in Class B scored a higher mark than every student in Class A."
Explain why this statement cannot be confirmed using only the information in the table.

.....

.....

..... **[2]**

**1522** The number of goals scored by Team A in each of their last 6 matches is recorded below.

3 1 0 4 2 2

Team B also played 6 matches.
For Team B, the median number of goals scored was 1.5 and the range was 5.

**(a)** Work out the median and the range of the number of goals scored by Team A.

Median = ..... , Range = ..... **[2]**

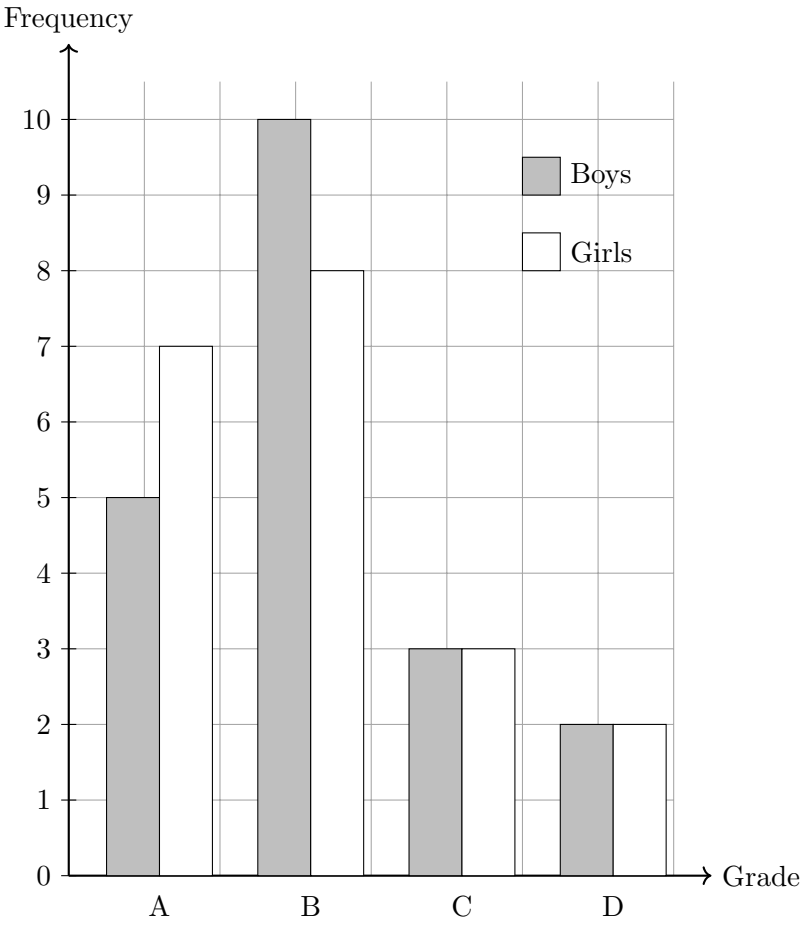
**(b)** Make two comments comparing the performance of Team A with the performance of Team B.

.....

.....

..... **[2]**

**1523** The dual bar chart shows the grades achieved by a group of boys and a group of girls in an examination.




**(a)** Write down the modal grade for the boys.

..... **[1]**

**(b)** Find the number of girls who scored Grade C.

..... **[1]**

**(c)** Calculate the percentage of all students who scored Grade A.

..... % **[2]**

**1524** The stem-and-leaf diagram shows the heights, in centimetres, of 15 plants.

| | |
|---|---|
| 1 | 2 5 8 |
| 2 | 0 1 1 4 7 |
| 3 | 2 3 5 9 |
| 4 | 0 4 |
| 5 | 1 |

Key: 1 | 2 represents 12 cm

**(a)** Find the median height.

..... cm **[1]**

**(b)** Find the interquartile range of the heights.

..... cm **[2]**

**(c)** Find the probability that a plant chosen at random has a height greater than 30 cm.

..... **[1]**

**1525** The stem-and-leaf diagram shows the ages, in years, of 15 members of a chess club.

| | |
|---|---|
| 1 | 2 5 8 |
| 2 | 0 1 1 4 7 9 |
| 3 | 2 5 6 9 |
| 4 | 1 3 |

Key: 2 | 1 represents 21 years

**(a)** Find the median age.

..... years **[1]**

**(b)** Find the range of the ages.

..... years **[1]**

**(c)** Find the interquartile range.

..... years **[2]**

**(d)** A member is chosen at random.
Find the probability that they are older than 30 years.

..... **[1]**

**1526** The back-to-back stem-and-leaf diagram shows the scores, out of 50, achieved by 15 students in Class A and 15 students in Class B in a mathematics test.

| Class A | \$tem | Class B |
|---|---|---|
| 9 8 | 1 | 4 6 8 |
| 8 7 5 3 | 2 | 0 2 4 5 9 |
| 9 6 4 2 1 | 3 | 0 3 5 8 |
| 7 5 2 0 | 4 | 1 6 9 |

Key: 3 | 2 | 0 represents 23 for Class A and 20 for Class B

**(a)** Write down the median score and the interquartile range for Class A.

Median = ..... , Interquartile range = ..... **[2]**

**(b)** For Class B, the median score is 29 and the interquartile range is 18.

Compare the test scores of Class A and Class B. Make two comparative statements, one about average performance and one about the spread of the scores.

.....

..... **[2]**

**(c)** A student claims: "More students in Class B scored above 40 than in Class A." Explain whether the diagram supports this claim.

..... **[2]**

**(d)** Explain one restriction on using these results to conclude that Class A is better at mathematics than Class B.

..... **[1]**

**1527** A set of 5 integers has the following statistical properties:

- Mode = 3
- Median = 7
- Mean = 8
- Range = 11

**(a)** Find the 5 integers.

..... **[4]**

**(b)** A sixth integer, $x$, is added to the set.
The new mean is 9.
Work out the value of $x$.

$x =$ ..... **[2]**

**(c)** A student claims: "If another completely different set of 5 integers has the same range of 11, it must also have the same standard deviation."

State whether the student is correct.
Give a statistical reason for your answer.

.....

..... **[1]**

**1528** Two branches of a call centre, Branch North and Branch South, record the duration, $t$ minutes, of customer support calls during one morning.
The table shows the distribution of call durations for both branches.
Branch North recorded 80 calls in total.
In Branch South, the relative frequency of a call lasting more than 8 minutes is $\frac{13}{20}$.

| Call duration ($t$ minutes) | Branch North | Branch South |
|---|---|---|
| $0 < t \leq 4$ | 12 | 15 |
| $4 < t \leq 8$ | 28 | $x$ |
| $8 < t \leq 12$ | $y$ | 35 |
| $12 < t \leq 16$ | 14 | 20 |
| $16 < t \leq 20$ | 6 | 10 |

**(a)** Find the value of $y$ and the value of $x$.

$y =$ ..... , $x =$ ..... **[4]**

**(b)** Calculate an estimate of the mean call duration for Branch North and for Branch South.

Branch North = ..... , Branch South = ..... **[4]**

**(c)** A supervisor inspects the table and makes two statements:

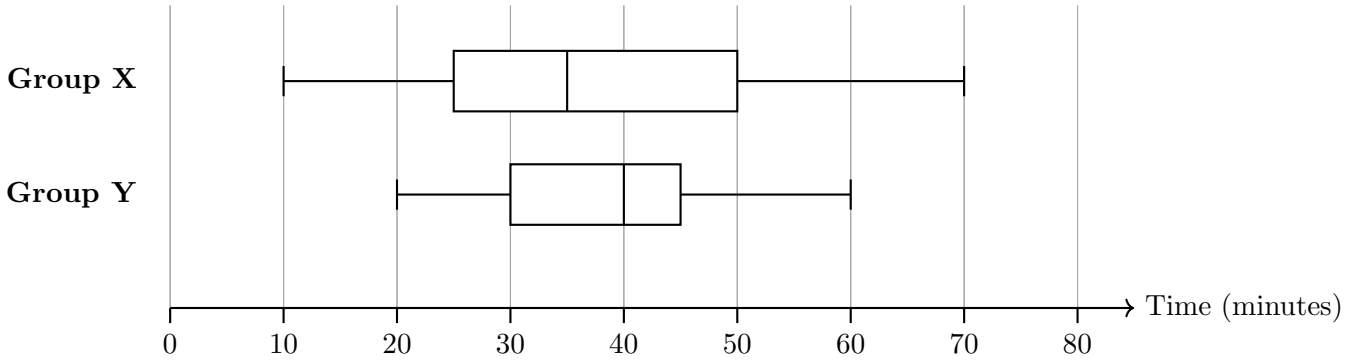
- Statement 1: "The maximum possible range of call durations in Branch North is 20 minutes."
- Statement 2: "At least 18 calls in Branch South lasted between 9 minutes and 11 minutes."

Explain whether each statement is valid, giving a reason based on the grouped data.

.....

..... **[4]**

**1529** The box-and-whisker plots show information about the times, in minutes, taken by two groups of people (Group X and Group Y) to complete a puzzle.




**(a)** Find the median time for Group Y.

..... minutes **[1]**

**(b)** Find the interquartile range for Group X.

..... minutes **[1]**

**(c)** Make two different comments comparing the distributions of the times taken by Group X and Group Y.

.....

.....

.....

..... **[2]**

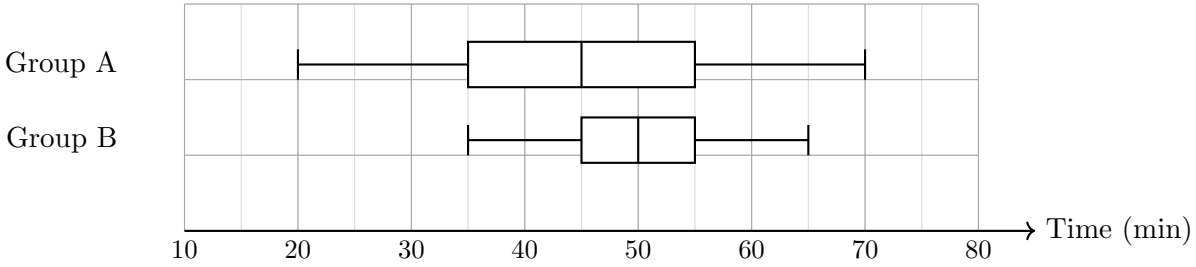
**(d)** Which group had a larger proportion of people completing the puzzle in less than 30 minutes? Explain your reasoning.

.....

.....

..... **[2]**

**1530** The box-and-whisker plots show information about the times, in minutes, taken by two groups of students, Group A and Group B, to complete a mathematics test.




**(a)** Write down the median time for Group A.

..... minutes **[1]**

**(b)** Work out the interquartile range for Group B.

..... minutes **[1]**

**(c)** Make two comments comparing the distributions of the times taken by Group A and Group B.

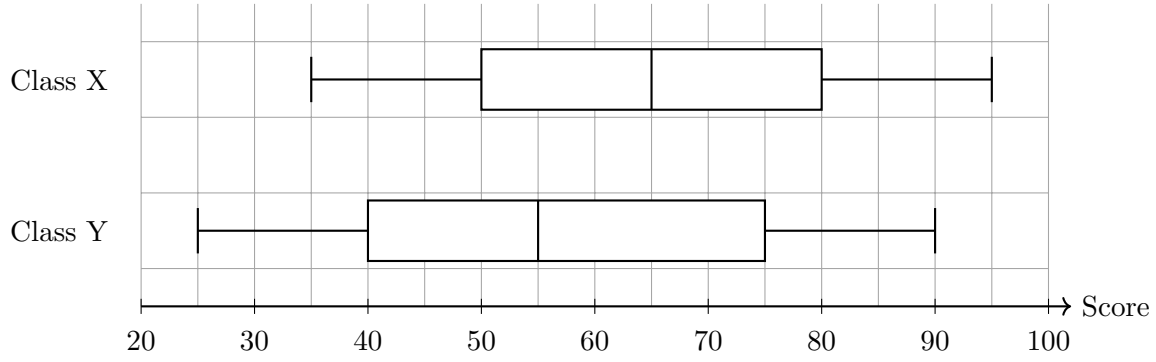
Comment 1:

Comment 2: **[2]**

**(d)** Group A has 80 students.
Estimate the number of students in Group A who took longer than 55 minutes.

..... students **[2]**

**1531** The box-and-whisker plots show the distribution of test scores for two classes, Class X and Class Y.




**(a)** Write down the median score for Class X.

..... **[1]**

**(b)** Work out the interquartile range for Class Y.

..... **[2]**

**(c)** Make two different comments comparing the distribution of test scores in Class X with Class Y.

**[2]**

**(d)** A student states: "Because the median score for Class X is higher, every student in Class X scored more than every student in Class Y."

Is this statement correct?
Give a statistical reason for your answer.

.....

..... **[1]**

**1532** The table shows information about the mass, $m$ grams, of 60 mangoes collected by a farmer.

| **Mass ($m$ grams)** | $100 < m \leq 120$ | $120 < m \leq 140$ | $140 < m \leq 160$ | $160 < m \leq 180$ | $180 < m \leq 200$ |
|---|---|---|---|---|---|
| **Frequency** | 10 | 20 | 15 | 10 | 5 |

**(a)** State the modal class.

..... **[1]**

**(b)** Calculate an estimate of the mean mass of the mangoes.

..... $g$ **[4]**

**(c)** The farmer states that exactly 15 mangoes have a mass of 150 grams because 150 is the midpoint of the class $140 < m \leq 160$.
Explain why the farmer's reasoning is incorrect.

.....

.....

..... **[2]**

**1533** The table shows the number of goals scored by a football team in each of 40 matches.

| Number of goals | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| Frequency | 8 | 12 | $x$ | 6 | $y$ | 2 |

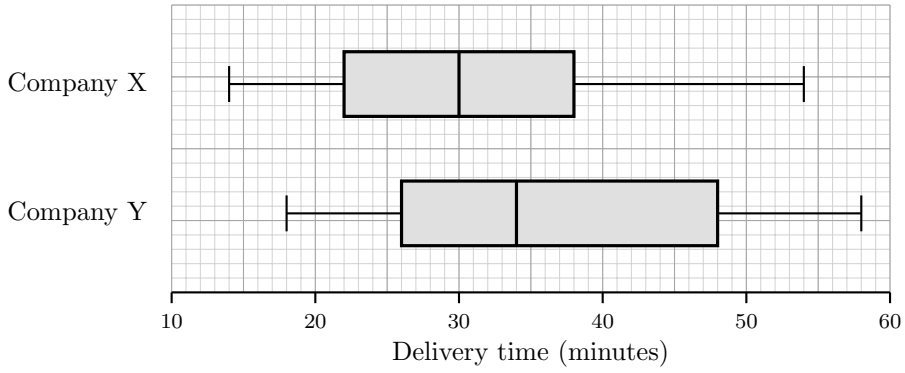
**(a)** The mean number of goals scored per match is 1.7.
Show that $2x + 4y = 30$.

**[3]**

**(b)** Find the value of $x$ and the value of $y$.

$x =$ ..... , $y =$ ..... **[4]**

**1534** The box-and-whisker plots show information about the delivery times, in minutes, of parcels delivered by two courier companies, Company X and Company Y.




**(a)** For Company Y, find

**(i)** the median delivery time,

..... minutes **[1]**

**(ii)** the interquartile range of the delivery times.

..... minutes **[2]**

**(b)** Company X recorded 160 deliveries and Company Y recorded 240 deliveries.

**(i)** Calculate the number of deliveries by Company Y that took 26 minutes or less.

..... **[2]**

**(ii)** A manager claims: "More deliveries took between 22 minutes and 38 minutes for Company X than for Company Y." Explain why it is not possible to be certain from the box-and-whisker plots whether this claim is true.

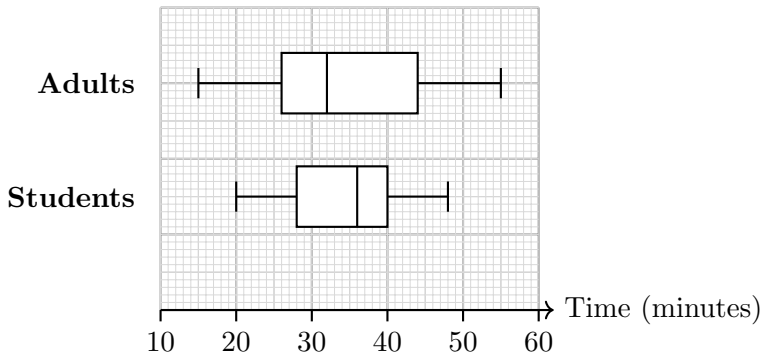
..... **[3]**

**(c)** A customer wants a courier company that is both faster on average and more consistent in its delivery times.
State which company the customer should choose, giving two statistical reasons to justify your decision.

.....

..... **[3]**

**1535** The box-and-whisker plots show information about the times, in minutes, taken by a group of Adults and a group of Students to complete a logic puzzle.




**(a)** Use the diagram to find the interquartile range of the times taken by the Adults.

..... minutes **[2]**

**(b)** Make two different statements comparing the distribution of the times taken by the Adults with the times taken by the Students.

.....

.....

..... **[2]**

**(c)** A student claims: "Because the range of the Students' times is smaller, this proves that every student completed the puzzle faster than the slowest Adult."

Comment on the validity of this reasoning.

.....

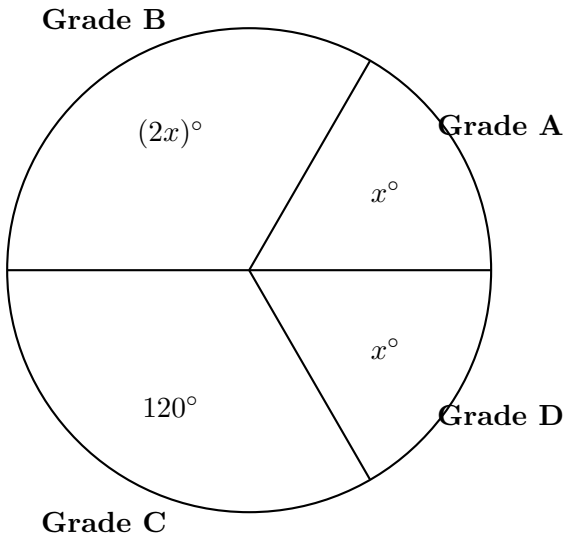
.....

..... **[2]**

**1536** The pie chart shows the distribution of grades (A, B, C, and D) achieved by 120 students in Class P.

The frequency table shows the distribution of the same grades achieved by the 80 students in Class Q.

The ratio of the number of students in Class Q achieving Grade A : Grade B : Grade C : Grade D is 1 : 3 : 2 : 2.




**Class P**

**(a)** Find the number of students who achieved Grade A in Class P.

..... **[3]**

**(b)** A student claims: "A higher percentage of students in Class Q achieved Grade B than in Class P."

Show that this claim is correct.

**[3]**

**(c)** The school principal states that Class P performed better overall because more students achieved Grade C in Class P than in Class Q.

Give one statistical reason why comparing the absolute frequencies (number of students) for

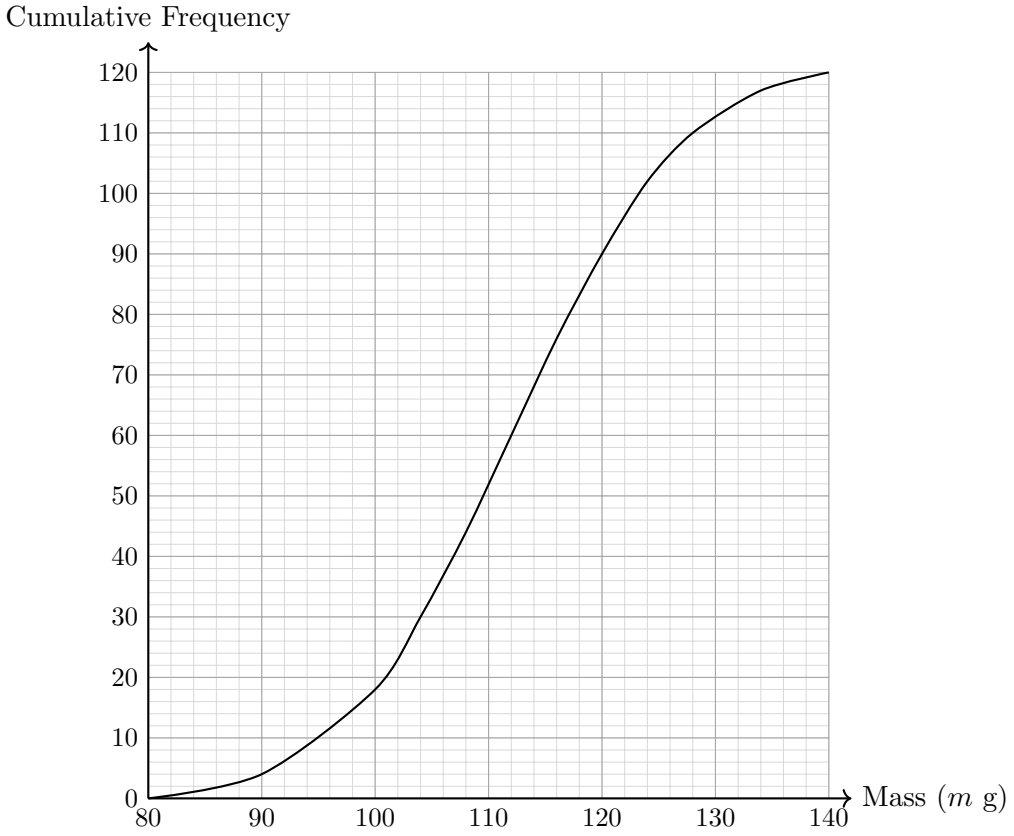
each grade is not an appropriate way to determine which class performed better.

.....

.....

..... **[1]**

**1537** The cumulative frequency curve shows information about the masses, $m$ grams, of 120 apples from Farm X.




**(a)** Use the graph to find an estimate for the median mass.

..... $g$ **[1]**

**(b)** Use the graph to find an estimate for the interquartile range.

..... $g$ **[2]**

**(c)** Apples with a mass greater than 128 g are classed as Large.
Find the percentage of these apples that are Large.

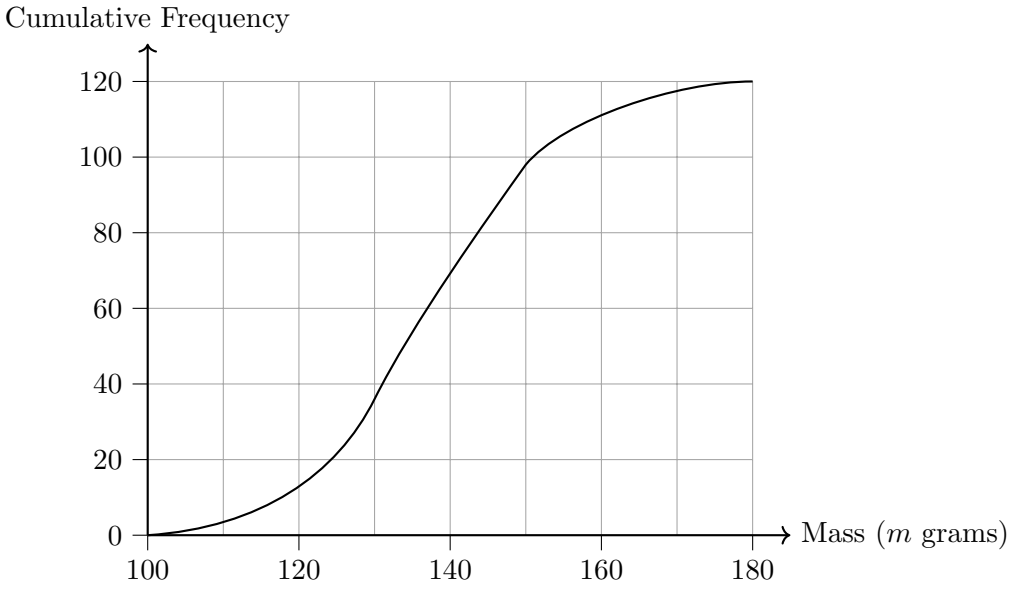
..... % **[2]**

**(d)** 120 apples from Farm Y have a median mass of 105 g and an interquartile range of 22 g.
Compare the distribution of the masses of the apples from Farm X and Farm Y.

.....

..... **[2]**

**1538** The cumulative frequency graph shows information about the masses, $m$ grams, of 120 apples.




**(a)** Use the graph to find an estimate for

**(i)** the median,

..... $g$ **[1]**

**(ii)** the interquartile range.

..... $g$ **[2]**

**(b)** Apples with a mass greater than 150 grams are classified as 'large'.
Find an estimate for the number of large apples.

..... **[2]**

**(c)** Two of the 120 apples are chosen at random.
Find the probability that both apples have a mass of 150 grams or less.

..... **[3]**

**1539** A weather station records the following data each day.
$A$: The number of times lightning strikes.
$B$: The maximum wind speed.
$C$: The volume of water collected in a rain gauge.
$D$: The number of days it snows in a year.

**(a)** Write down the letters of the variables that represent discrete data.

..... **[1]**

**(b)** Write down the letters of the variables that represent continuous data.

..... **[1]**

**1540** A scientist records different types of weather data.
State whether each of the following types of data is discrete or continuous.

**(a)** The daily maximum temperature.

..... **[1]**

**(b)** The number of rainy days in a month.

..... **[1]**

**(c)** The amount of rainfall in one hour.

..... **[1]**

**1541** Statistical data can be classified as either discrete or continuous.

**(a)** State whether each of the following variables is discrete or continuous data.

**(i)** The number of cars in a car park.

..... **[1]**

**(ii)** The temperature of a cup of coffee.

..... **[1]**

**(iii)** The time it takes to complete a jigsaw puzzle.

..... **[1]**

**(iv)** The number of keys on a piano.

..... **[1]**

**1542** A sports analyst collects data during a football match.

**(a)** State whether the number of goals scored in the match is discrete or continuous.

..... **[1]**

**(b)** State whether the mass of the football is discrete or continuous.

..... **[1]**

**(c)** State whether the time taken to complete the match is discrete or continuous.

..... **[1]**

**(d)** State whether the temperature on the pitch is discrete or continuous.

..... **[1]**

**1543** A student collects data for a statistics project.
For each of the variables below, state whether the data collected is discrete or continuous.

**(a)** The number of goals scored in a football match.

..... **[1]**

**(b)** The mass of a football.

..... **[1]**

**(c)** The volume of water in a swimming pool.

..... **[1]**

**(d)** The number of people swimming in a pool.

..... **[1]**

**1544** An environmental science club records four different types of statistical data during a field trip to a local river.

**(a)** State whether each of the following sets of data is discrete or continuous:

**(i)** the number of fish counted in each net sample,

..... **[1]**

**(ii)** the water temperature, in degrees Celsius,

..... **[1]**

**(iii)** the width of the river, in metres,

..... **[1]**

**(iv)** the shoe size of each student attending the field trip.

..... **[1]**

**(b)** Explain why the width of the river is classified as continuous data.

..... **[1]**

**(c)** A student measures the widths, $w$ metres, of the river at 10 points and groups the continuous data into intervals of width 2 metres starting at 4 metres.
Explain why writing the first two intervals as $4 \leq w \leq 6$ and $6 \leq w \leq 8$ is not suitable, and write down the correct inequality notation for the first interval.

.....

..... **[2]**

**1545** A hardware store sells nails in large boxes.

**(a)** The mass of one box of nails is measured as 40 kg, correct to the nearest 5 kg.
Mass is a continuous variable.
Find the lower bound and the upper bound for the mass of the box.

Lower bound = ..... , Upper bound = ..... **[2]**

**(b)** The number of nails in the box is 1250, correct to the nearest 5.
The number of nails is a discrete variable.
Write down the smallest possible number of nails and the largest possible number of nails in the box.

Smallest possible number = ..... , Largest possible number = .....
**[2]**

**1546** Two sets of data are collected during a fitness test.
Data Set A is the exact time, $t$ seconds, taken by athletes to run a race.
Data Set B is the number of push-ups, $n$, completed by the athletes.

**(a)** Data Set A is grouped into classes such as $10 \leq t < 12$.
Data Set B is grouped into classes such as $10 - 19$.
Explain why the classes for Data Set A use inequalities while the classes for Data Set B use a hyphen.

.....

..... **[2]**

**(b)** A student decides to draw a histogram to represent Data Set B. One of the classes is $20 - 29$.

**(i)** State the lower and upper class boundaries that must be used for this class on the horizontal axis of the histogram.

Lower = ..... , Upper = ..... **[1]**

**(ii)** Calculate the class width used for the frequency density calculation.

..... **[1]**

**(c)** Continuous data can theoretically take any value within a range.
Give an example of a continuous variable that is practically recorded as discrete data in everyday life.

.....

..... **[1]**

**1547** A factory manufactures metal rods and packs them into boxes. For each box, two variables are recorded:

- $k$: the number of rods in the box.
- $m$: the total mass, in kilograms, of the rods in the box.

**(a)** State whether $k$ is discrete or continuous, and state whether $m$ is discrete or continuous.

$k$ is ..... , $m$ is ..... **[2]**

**(b)** Each rod has a mass of 2.0 kg, correct to 1 decimal place.
A box of $k$ rods has a total mass of 20.0 kg, correct to 1 decimal place.
By considering the lower and upper bounds of the continuous mass measurements, show that $k = 10$.

**[4]**

**(c)** The factory inspects 50 boxes and records the number of defective rods, $d$, in each box and the length, $L$ cm, of a sample of 50 rods.
The results are shown in the two tables below.

| Defective ($d$) | Frequency |
|---|---|
| 0 | 24 |
| 1 | 15 |
| 2 | 7 |
| 3 | 3 |
| 4 | 1 |

| Length ($L$ cm) | Frequency |
|---|---|
| $49.6 \leq L < 49.8$ | 8 |
| $49.8 \leq L < 50.0$ | 18 |
| $50.0 \leq L < 50.2$ | 17 |
| $50.2 \leq L < 50.4$ | 7 |

**(i)** Find the exact mean number of defective rods per box, and explain why this value is exact whereas the mean length calculated from the second table is only an estimate.

..... **[3]**

**(ii)** A rod is accepted if its length satisfies $49.8 \leq L < 50.2$.
Calculate the percentage of the 50 sampled rods that are accepted, and explain why it is not possible from the table to determine the exact number of rods with length greater than 50.1 cm.

.....

..... **[2]**

**1548** A botanist studies a new species of plant in a greenhouse.
She records the height, $h$ cm, of each plant and the total number of leaves on each plant.

**(a)** State whether the height of a plant is discrete or continuous data.

..... **[1]**

**(b)** The botanist wants to group the heights of the plants into a frequency table.
She suggests using the following class intervals:

$$10 - 19 \qquad 20 - 29 \qquad 30 - 39$$

Explain why these class intervals are not appropriate for the height of the plants.

.....

..... **[2]**

**(c)** State whether the number of leaves on a plant is discrete or continuous data.

..... **[1]**

**1549** A student collects data about a local park for a science project.

**(a)** Explain the mathematical difference between discrete and continuous data.

.....

.....

..... **[2]**

**(b)** The student measures the lengths of 50 leaves to the nearest millimetre.

**(i)** State whether the true length of a leaf is discrete or continuous.

..... **[1]**

**(ii)** Explain why measuring the lengths to the nearest millimetre does not make the variable discrete.

.....

..... **[1]**

**1550** A farmer weighs 60 apples.
The masses, $m$ grams, are recorded in the table.

| **Mass ($m$ grams)** | **Frequency** |
|---|---|
| $80 < m \leq 100$ | 12 |
| $100 < m \leq 120$ | 26 |
| $120 < m \leq 140$ | 15 |
| $140 < m \leq 160$ | 7 |

**(a)** State whether the mass of an apple is discrete or continuous data.

..... **[1]**

**(b)** Calculate an estimate of the mean mass.

..... $g$ **[3]**

**(c)** Explain why your answer to part (b) is an estimate.

..... **[1]**

**1551** A scientist measures the lengths of 50 worms and counts the number of spots on each worm.
The frequency table shows the data collected for the number of spots.

| **Number of spots** | $0-4$ | $5-9$ | $10-14$ | $15-19$ |
|---|---|---|---|---|
| **Frequency** | 12 | 18 | 14 | 6 |

**(a)** State which of the two variables, length or number of spots, is continuous.

..... **[1]**

**(b)** The length of one worm is recorded as 14.5 cm, correct to the nearest millimetre.

**(i)** Write down the lower bound of this length.

..... *cm* **[1]**

**(ii)** Write down the upper bound of this length.

..... *cm* **[1]**

**(c)** Using the frequency table, write down the class width for the $5-9$ class.

..... **[2]**

**1552** A factory tests a sample of 40 light bulbs.
It records the lifetime, $t$ hours, of each bulb.

**(a)** State whether the lifetime of a light bulb is discrete or continuous.

..... **[1]**

**(b)** The results are shown in the table.

| Lifetime ($t$ hours) | $100 < t \leq 200$ | $200 < t \leq 300$ | $300 < t \leq 400$ | $400 < t \leq 500$ |
|---|---|---|---|---|
| Frequency | 4 | 14 | 16 | 6 |

Calculate an estimate of the mean lifetime.

..... hours **[4]**

**1553** A sports analyst collects data on 40 athletes competing in an athletics meet. Two variables are recorded for each athlete:

- $N$: the number of training sessions attended in the previous month.
- $T$: the time taken, in seconds, to complete a 400 m sprint.

The two tables below summarise the results.

| Sessions ($N$) | Frequency |
|---|---|
| $10 \leq N \leq 12$ | 6 |
| $13 \leq N \leq 15$ | 14 |
| $16 \leq N \leq 18$ | 12 |
| $19 \leq N \leq 21$ | 8 |

| Time ($T$ seconds) | Frequency |
|---|---|
| $48 \leq T < 50$ | 5 |
| $50 \leq T < 52$ | 15 |
| $52 \leq T < 54$ | 13 |
| $54 \leq T < 56$ | 7 |

**(a)** Explain how the inequality notation used for the class intervals in the two tables reflects the difference between discrete data and continuous data.

.....

..... **[2]**

**(b)** Calculate an estimate of the mean number of training sessions attended, $N$, and an estimate of the mean sprint time, $T$.

Mean of $N$ = ..... , Mean of $T$ = ..... **[4]**

**(c)** **(i)** A student says: "The estimated mean of $N$ is 15.65, which is impossible because $N$ is discrete data and an athlete cannot attend 15.65 training sessions." Explain whether the student is correct about the validity of the mean.

..... **[2]**

**(ii)** State a suitable type of statistical diagram to represent the distribution of discrete data $N$ and a suitable type of statistical diagram to represent the distribution of continuous data $T$.

Diagram for $N =$ ..... , Diagram for $T =$ ..... **[2]**

**1554** Data is collected about the members of an athletics club.

**(a)** State whether each of the following variables is discrete or continuous.

**(i)** The number of members in the club.

..... **[1]**

**(ii)** The exact distance a member jumps in the long jump.

..... **[1]**

**(b)** The shoe sizes of the members are recorded.
Shoe sizes are available in sizes such as $7, 7.5, 8, 8.5, 9$.
A student claims that because these shoe sizes include decimal values, the data is continuous.
Explain why the student is incorrect.

.....

.....

..... **[2]**

**1555** Measurements and quantities in statistics are defined by their ability to take certain values.

**(a)** Explain the mathematical difference between continuous data and discrete data.

.....

..... **[2]**

**(b)** A person's exact foot length is measured in centimetres.
This measurement is then used to determine their shoe size, which can take values such as 7, 7.5, 8, or 8.5.

**(i)** State whether exact foot length is discrete or continuous data.

..... **[1]**

**(ii)** A student argues that shoe size must be continuous because it includes decimal values like 7.5.
Explain why the student is incorrect.

.....

.....
**[2]**

**1556** A factory packs identical books into boxes.
Each box contains exactly 24 books.

**(a)** State whether each of the following variables is discrete or continuous.

**(i)** The number of books in a box.

..... **[1]**

**(ii)** The mass of a book.

..... **[1]**

**(b)** The mass of one book is 850 g, correct to the nearest 10 g.
Find the upper bound for the total mass of the 24 books.
Give your answer in kilograms.

..... kg **[2]**

**(c)** An empty box has a mass of 1.2 kg, correct to 1 decimal place.
Calculate the lower bound for the total mass of a box containing 24 books.

..... kg **[3]**

**1557** A student records the number of hours she spends reading each week for seven weeks.
The results are:

7 4 8 4 12 9 5

**(a)** **(a)** Write down the mode.

..... **[1]**

**(b)** Find the median.

..... **[1]**

**(c)** Work out the range.

..... **[1]**

**1558** The number of goals scored by a football team in each of their last 10 matches is shown below.

2 0 1 4 1 0 3 1 2 1

**(a)** Find the mode.

..... **[1]**

**(b)** Find the median.

..... **[2]**

**(c)** Calculate the mean.

..... **[2]**

**1559** A student records the number of points scored by a basketball team in 11 matches.

14 20 8 24 16 5 26 13 22 10 18

**(a)** Find the median.

..... **[1]**

**(b)** Find the range.

..... **[1]**

**(c)** Find the interquartile range.

..... **[2]**

**(d)** Calculate the mean.

..... **[2]**

**1560** The list shows the number of points scored by a basketball player in 7 games.

$$7 \quad 12 \quad 8 \quad 7 \quad 12 \quad x \quad 10$$

The mode of the number of points is 7.

**(a)** Write down the value of $x$.

$x =$ ..... **[1]**

**(b)** Find the range of the number of points.

..... **[1]**

**(c)** Find the median.

..... **[1]**

**(d)** Calculate the mean.

..... **[2]**

**(e)** In the 8th game, the player scores $y$ points.
The new mean number of points for all 8 games is 9.5.
Find the value of $y$.

$y =$ ..... **[2]**

**1561** The waiting times, in minutes, for 11 patients at a dental clinic are listed below.

15 11 23 8 50 14 18 10 15 21 13

**(a)** Write down the mode.

..... minutes [**1**]

**(b)** Find

**(i)** the median,

..... minutes [**2**]

**(ii)** the interquartile range.

..... minutes [**2**]

**(c)** Calculate the mean waiting time.

..... minutes [**2**]

**(d)** Explain why the median is a better measure of average than the mean for these data.

..... [**1**]

**1562** A botanist measures the heights, $h$ cm, of 50 plants in a greenhouse.
The grouped frequency table shows the results, where $x$ and $y$ are positive integers.

| Height ($h$ cm) | Frequency |
|---|---|
| $0 < h \leq 10$ | 4 |
| $10 < h \leq 20$ | $x$ |
| $20 < h \leq 30$ | 18 |
| $30 < h \leq 40$ | $y$ |
| $40 < h \leq 50$ | 6 |

**(a)** Use the total number of plants to show that $x + y = 22$.

**[1]**

**(b)** The estimated mean height of the 50 plants is 27 cm.

**(i)** Show that $3x + 7y = 122$.

**[3]**

**(ii)** Solve the simultaneous equations $x + y = 22$ and $3x + 7y = 122$ to find the value of $x$ and the value of $y$.

$x =$ ..... , $y =$ ..... **[3]**

**(c)** Write down the modal class of the distribution.

..... **[1]**

**(d)** Ten more plants are measured, and all ten have heights in the interval $30 < h \leq 40$.

**(i)** State the new modal class.

..... **[1]**

**(ii)** Calculate the new estimated mean height of all 60 plants, giving your answer as a mixed number.

..... cm **[3]**

**1563** The table shows the heights, $h$ cm, of 80 plants.

| **Height ($h$ cm)** | **Frequency** |
|---|---|
| $10 < h \leq 20$ | 12 |
| $20 < h \leq 30$ | 25 |
| $30 < h \leq 40$ | 28 |
| $40 < h \leq 50$ | 11 |
| $50 < h \leq 60$ | 4 |

**(a)** Write down the modal class.

..... **[1]**

**(b)** Calculate an estimate of the mean height.

..... cm **[4]**

**1564** The table shows information about the heights, $h$ cm, of 50 plants.

| Height ($h$ cm) | $0 < h \leq 10$ | $10 < h \leq 20$ | $20 < h \leq 30$ | $30 < h \leq 40$ | $40 < h \leq 50$ |
|---|---|---|---|---|---|
| Frequency | 4 | 12 | 20 | 10 | 4 |

**(a)** **(a)** Write down the modal class.

..... **[1]**

**(b)** Calculate an estimate of the mean height.

..... cm **[4]**

**1565** A survey was conducted on the masses, $m$ grams, of 80 apples. The results are shown in the frequency table.

| Mass ($m$ grams) | $80 < m \leq 90$ | $90 < m \leq 100$ | $100 < m \leq 110$ | $110 < m \leq 120$ | $120 < m \leq 130$ |
|---|---|---|---|---|---|
| Frequency | 8 | 22 | 31 | 14 | 5 |

**(a)** Write down the modal class.

..... **[1]**

**(b)** Calculate an estimate of the mean mass of the apples.

..... $g$ **[4]**

**(c)** Explain why your answer to part (b) is an estimate.

.....

..... **[2]**

**1566** The table shows information about the time, $t$ seconds, taken by 100 students to complete a puzzle.

| Time ($t$ seconds) | Frequency |
|---|---|
| $10 < t \leq 20$ | 14 |
| $20 < t \leq 30$ | 28 |
| $30 < t \leq 40$ | 32 |
| $40 < t \leq 50$ | 18 |
| $50 < t \leq 60$ | 8 |

**(a)** Write down the modal time interval.

..... **[1]**

**(b)** Calculate an estimate of the mean time.

..... seconds **[4]**

**(c)** Calculate the percentage of students who took strictly more than 40 seconds to complete the puzzle.

..... % **[2]**

**(d)** Write down the interval that contains the upper quartile.

..... **[1]**

**1567** The table shows information about the masses, $m$ grams, of 80 apples harvested from an orchard.

| Mass ($m$ grams) | Frequency |
|---|---|
| $100 \leq m < 120$ | 10 |
| $120 \leq m < 140$ | 22 |
| $140 \leq m < 160$ | 28 |
| $160 \leq m < 180$ | 14 |
| $180 \leq m < 200$ | 6 |

**(a)** Write down the modal class.

..... **[1]**

**(b)** Calculate an estimate of the mean mass of the 80 apples.

..... $g$ **[4]**

**(c)** A second crate contains 40 apples with a mean mass of 155 g. Calculate an estimate of the mean mass of all 120 apples combined.

..... $g$ **[3]**

**(d)** State the class interval that contains the median mass of the 80 apples.

..... **[1]**

**1568** A factory produces 60 items.
The masses, $m$ kg, of these items are recorded in the table.
The estimate of the mean mass is 48 kg.

| **Mass ($m$ kg)** | **Frequency** |
|---|---|
| $0 < m \leq 20$ | $p$ |
| $20 < m \leq 40$ | 14 |
| $40 < m \leq 60$ | 16 |
| $60 < m \leq 80$ | $q$ |
| $80 < m \leq 100$ | 5 |

**(a)** Find the value of $p$ and the value of $q$.

$p =$ ..... , $q =$ ..... **[5]**

**(b)** State the class interval which contains the median mass.
You must show your working.

..... **[2]**

**1569** The mean of five numbers is 12.
The median of the five numbers is 11.
The mode of the five numbers is 8.
The range of the five numbers is 15.

**(a)** Find the five numbers.

..... **[5]**

**(b)** A sixth number, $k$, is added to the list.
The new mean is 13.
Work out the value of $k$.

$k =$ ..... **[2]**

**1570** Six numbers are given in terms of $x$.

$$x-3 \qquad x \qquad x+2 \qquad 2x \qquad 2x+5 \qquad 3x$$

The mean of these six numbers is 9.

**(a)** **(a)** Calculate the value of $x$.

$x =$ ..... **[3]**

**(b)** Find the interquartile range of the six numbers.

..... **[2]**

**(c)** A seventh number, $y$, is added to the original six numbers.
The new median of the seven numbers is 8.
Find the value of $y$.

$y =$ ..... **[2]**

**1571** The table shows information about the masses, $m$ kg, of 50 parcels.

| Mass ($m$ kg) | Frequency |
|---|---|
| $0 < m \leq 2$ | 15 |
| $2 < m \leq 4$ | $x$ |
| $4 < m \leq 6$ | $y$ |
| $6 < m \leq 8$ | 8 |
| $8 < m \leq 10$ | 2 |

The estimate of the mean mass of the parcels is 3.8 kg.

**(a)** Use the total number of parcels to write down an equation in $x$ and $y$, and show that it simplifies to $x + y = 25$.

**[1]**

**(b)** Use the estimate of the mean to show that $3x + 5y = 101$.

**[4]**

**(c)** Solve the simultaneous equations to find the value of $x$ and the value of $y$.
You must show all your working.

$x =$ ..... , $y =$ ..... **[3]**

**(d)** Write down the interval that contains the median mass.

..... **[1]**

**1572** The number of text messages received by 15 students in one hour is shown below.

| | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 12 | 23 | 15 | 18 | 21 | 14 | 17 | 15 | 22 | 19 | 15 | 24 | 11 | 16 | 20 |

**(a)** Find the mean.

..... **[1]**

**(b)** Find the median.

..... **[1]**

**(c)** Find the interquartile range.

..... **[2]**

**1573** A survey was conducted to find the number of books read by 21 students during a month.
The results are shown in the table.

| Number of books | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Frequency | 2 | 5 | 7 | 4 | 3 |

**(a)** Use a graphic display calculator to find the following statistics for this data.

**(i)** Calculate the mean number of books read.
Give your answer correct to 3 significant figures.

Mean = ..... **[2]**

**(ii)** Find the median.

Median = ..... **[1]**

**(iii)** Find the interquartile range.

Interquartile range = ..... **[2]**

**1574** The marks scored by 15 students in a mathematics test are shown below.

| | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 15 | 19 | 14 | 28 | 11 | 16 | 9 | 20 | 15 | 13 | 17 | 21 | 10 | 18 | 14 |

**(a)** Find the mean.

..... **[2]**

**(b)** Find the median.

..... **[1]**

**(c)** Find the interquartile range.

..... **[2]**

**1575** A student records the number of hours he spends playing sports each week for 14 weeks.
The results are:

5 3 8 4 5 6 7 3 4 7 5 2 6 5

**(a)** Find the mode.

..... **[1]**

**(b)** Find the mean.

..... **[1]**

**(c)** Find the median.

..... **[1]**

**(d)** Find the lower quartile.

..... **[1]**

**(e)** Find the upper quartile.

..... **[1]**

**1576** The times, in minutes, taken by 15 students to complete a logic puzzle are listed below.

28 17 35 22 46 14 31 25
18 33 26 20 38 23 29

**(a)** Use your graphic display calculator to find

**(i)** the mean,

..... minutes **[1]**

**(ii)** the median,

..... minutes **[1]**

**(iii)** the lower quartile and the upper quartile.

Lower quartile = ..... , Upper quartile = ..... **[2]**

**(b)** Write down the interquartile range.

..... minutes **[1]**

**(c)** Another student also completes the puzzle in $x$ minutes.
When this time is included, the mean time for all 16 students becomes 27.5 minutes.
Calculate the value of $x$.

$x =$ ..... **[2]**

**1577** The table shows the heights, $h$ cm, of 120 plants in a greenhouse.

| Height ($h$ cm) | $130 < h \leq 140$ | $140 < h \leq 150$ | $150 < h \leq 160$ | $160 < h \leq 170$ | $170 < h \leq 180$ |
|---|---|---|---|---|---|
| Frequency | 15 | 38 | 42 | 18 | 7 |

**(a)** Write down the modal class.

..... **[1]**

**(b)** Use your graphic display calculator to find an estimate of the mean height.

..... cm **[2]**

**1578** The table shows information about the time, $t$ minutes, taken by 100 people to travel to work.

| Time ($t$ minutes) | Frequency |
|---|---|
| $0 < t \leq 10$ | 14 |
| $10 < t \leq 20$ | 28 |
| $20 < t \leq 30$ | 35 |
| $30 < t \leq 40$ | 18 |
| $40 < t \leq 60$ | 5 |

**(a)** Write down the modal class.

..... **[1]**

**(b)** Use your graphic display calculator to calculate an estimate of the mean time.

..... minutes **[3]**

**(c)** Explain why your answer to part (b) is an estimate.

..... **[1]**

**1579** The table shows the time, $t$ minutes, taken by 80 people to complete a crossword puzzle.

| Time ($t$ minutes) | $0 < t \leq 10$ | $10 < t \leq 20$ | $20 < t \leq 30$ | $30 < t \leq 40$ | $40 < t \leq 60$ |
|---|---|---|---|---|---|
| Frequency | 12 | 25 | 28 | 11 | 4 |

**(a)** Calculate an estimate of the mean time.

..... minutes **[3]**

**(b)** Explain why your answer to part (a) is an estimate.

..... **[1]**

**(c)** An 81st person completes the puzzle in 12 minutes.
Without calculating the new mean, explain how this affects the mean time.

..... **[1]**

**1580** A survey was conducted on the time, $t$ minutes, taken by 80 students to travel to school. The results are shown in the frequency table.

| Time ($t$ minutes) | $0 < t \leq 10$ | $10 < t \leq 20$ | $20 < t \leq 30$ | $30 < t \leq 40$ | $40 < t \leq 50$ |
|---|---|---|---|---|---|
| Frequency | 12 | 25 | 28 | 11 | 4 |

**(a)** Calculate an estimate of the mean time.

..... minutes **[2]**

**(b)** Explain why your answer to part (a) is an estimate.

.....

..... **[1]**

**(c)** The mean time for a different group of 120 students to travel to school is 24.5 minutes. Calculate the overall mean time for the 200 students.

..... minutes **[3]**

**1581** A delivery company records the number of parcels, $n$, delivered by each of 60 vans on Monday. The results are shown in the table below.

| Parcels ($n$) | 18 | 19 | 20 | 21 | 22 | 23 | 24 |
|---|---|---|---|---|---|---|---|
| Frequency | 4 | 9 | 14 | 15 | 10 | 5 | 3 |

**(a)** Use your graphic display calculator to find

**(i)** the mean number of parcels delivered per van,

..... **[1]**

**(ii)** the median,

..... **[1]**

**(iii)** the interquartile range.

..... **[2]**

**(b)** The company also records the masses, $m$ kilograms, of a sample of 50 parcels. The grouped frequency table shows the results.

| Mass ($m$ kg) | Frequency |
|---|---|
| $0 < m \leq 2$ | 8 |
| $2 < m \leq 5$ | 17 |
| $5 < m \leq 10$ | 15 |
| $10 < m \leq 20$ | 10 |

**(i)** Use your graphic display calculator to calculate an estimate of the mean mass of these 50 parcels.

..... kg **[3]**

**(ii)** Write down the modal class.

..... **[1]**

**1582** The table shows the number of goals scored by a football team in 35 matches.

| Number of goals | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| Frequency | 3 | $x$ | $y$ | 6 | 4 | 1 |

**(a)** The mean number of goals scored is 2.2.
Find the value of $x$ and the value of $y$.

$x =$ ..... , $y =$ ..... **[4]**

**(b)** Find the median number of goals.

..... **[1]**

**1583** The table shows information about the number of goals scored by a football team in a number of matches.

| Number of goals | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| Frequency | 6 | $x$ | 8 | 4 | 2 |

The mean number of goals scored is 1.5.

**(a)** Show that $x = 12$.

**[3]**

**(b)** Find the median.

..... **[1]**

**(c)** Find the interquartile range.

..... **[2]**

**1584** A game is played where players can score $1, 2, 3, 4$, or $5$ points.
The table shows the frequencies of the scores obtained by a group of players.
The mean score is 2.8.

| Score | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Frequency | 10 | $k$ | $2k + 3$ | 7 | 5 |

**(a)** Show algebraically that $k = 5$.

**[4]**

**(b)** Use your graphic display calculator to find the following statistics for the scores.

**(i)** Find the median.

Median = ..... **[1]**

**(ii)** Find the lower quartile.

Lower quartile = ..... **[1]**

**(iii)** Find the upper quartile.

Upper quartile = ..... **[1]**

**1585** The number of days of rain per week in a city was recorded for 40 weeks.
The table shows the results.

| Number of days | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| Frequency | 8 | $x$ | $y$ | 6 | 2 |

The mean number of days of rain per week is 1.55.

**(a)** Show that $x + y = 24$ and $x + 2y = 36$.

**[3]**

**(b)** Find the value of $x$ and the value of $y$.

$x =$ ..... , $y =$ ..... **[2]**

**(c)** Use your graphic display calculator to find:

**(i)** the median

..... **[1]**

**(ii)** the interquartile range

..... **[1]**

**1586** A weather station records the daily rainfall, $r$ millimetres, over 100 days.
The grouped frequency table shows the results, where $p$ and $q$ are positive integers.

| Rainfall ($r$ mm) | Frequency |
|---|---|
| $0 \leq r < 5$ | 34 |
| $5 \leq r < 10$ | $p$ |
| $10 \leq r < 20$ | 26 |
| $20 \leq r < 30$ | $q$ |
| $30 \leq r < 50$ | 8 |

**(a)** Use the total number of days to show that $p + q = 32$.

**[1]**

**(b)** Using the mid-interval values, the estimated mean daily rainfall for the 100 days is 12.45 mm.

**(i)** Show that $3p + 10q = 180$.

**[3]**

**(ii)** Solve the simultaneous equations $p + q = 32$ and $3p + 10q = 180$ to find the value of $p$ and the value of $q$.

$p =$ ..... , $q =$ ..... **[2]**

**(c)** For the 12 days in the interval $20 \leq r < 30$, the exact daily rainfall values, in millimetres, are listed below.

24 27 21 28 22 25 20 27 23 28 22 24

Use your graphic display calculator to find, for these 12 days,

**(i)** the mean,

..... mm **[1]**

**(ii)** the median,

..... mm **[1]**

**(iii)** the interquartile range.

..... mm **[2]**

**(d)** When the exact values from part (c) are used instead of the mid-interval value for the interval $20 \leq r < 30$, calculate the revised estimate of the mean daily rainfall for all 100 days.

..... mm **[2]**

**1587** The stem-and-leaf diagram shows the ages of 15 people.

| | |
|---|---|
| 1 | 2 5 8 |
| 2 | 0 1 4 4 7 |
| 3 | 3 5 6 9 |
| 4 | 1 2 8 |

Key: 2 | 4 represents 24 years

**(a)** Find the mode.

..... **[1]**

**(b)** Find the median.

..... **[1]**

**(c)** Find the range.

..... **[1]**

**1588** The stem-and-leaf diagram shows the masses, in kg, of 15 parcels.

| | | | | | |
|---|---|---|---|---|---|
| 1 | 2 | 4 | 7 | | |
| 2 | 0 | 1 | 1 | 1 | 5 |
| 3 | 0 | 2 | 6 | 8 | |
| 4 | 1 | 4 | 5 | | |

**Key:** 1 | 2 represents 12 kg

**(a)** Find the mode of the masses.

..... kg **[1]**

**(b)** Find the median mass.

..... kg **[1]**

**(c)** Work out the range of the masses.

..... kg **[1]**

**(d)** A parcel is chosen at random.
Find the probability that its mass is strictly greater than 30 kg.

..... **[2]**

**1589** The masses, in kilograms, of 15 parcels are recorded below.

27 15 31 42 23 12 53 38 49 27 45 40 34 51 42

**(a)** Complete the ordered stem-and-leaf diagram to show this information.

| | |
|---|---|
| 1 | |
| 2 | |
| 3 | |
| 4 | |
| 5 | |

Key: 1 | 2 represents 12 kg

**[3]**

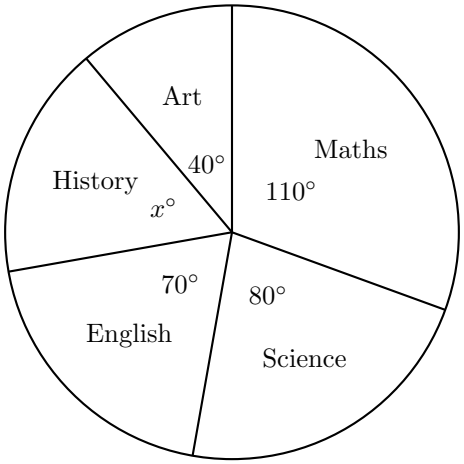
**(b)** Find the median mass.

..... kg **[1]**

**(c)** Work out the range.

..... kg **[1]**

**1590** The pie chart shows information about the favourite subjects of 180 students.




**(a)** Find the value of $x$.

$x =$ ..... **[1]**

**(b)** Find the number of students whose favourite subject is Science.

..... **[2]**

**(c)** Write down the fraction of students whose favourite subject is Maths.
Give your fraction in its simplest form.

..... **[2]**

**1591** The times, in seconds, taken by 16 athletes to complete an obstacle course are listed below.

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 48 | 35 | 52 | 41 | 63 | 39 | 44 | 57 |
| 41 | 50 | 36 | 47 | 61 | 45 | 54 | 49 |

**(a)** Complete the ordered stem-and-leaf diagram below, including the key.

| Stem | Leaf |
|---|---|
| 3 | |
| 4 | |
| 5 | |
| 6 | |

Key: ..... | ..... represents ..... seconds

**[3]**

**(b)** Find

**(i)** the median time,

..... seconds **[2]**

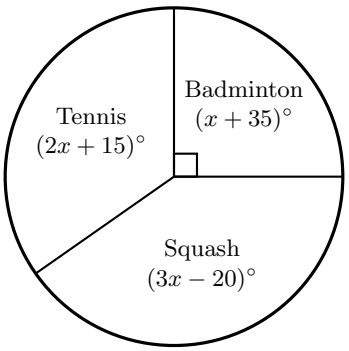
**(ii)** the interquartile range.

..... seconds **[2]**

**(c)** Athletes who complete the course in less than 45 seconds qualify for the final.
Calculate the percentage of these 16 athletes who qualify for the final.

..... % **[2]**

**1592** A sports club surveys $N$ members about their preferred racket sport: Tennis, Badminton, or Squash. The pie chart shows the distribution of the $N$ members, where the sector angles are $(2x + 15)°$, $(x + 35)°$, and $(3x - 20)°$, and the angle for Badminton is a right angle.




**(a)** Find the value of $x$.

$x =$ ..... **[2]**

**(b)** In the survey, 75 members prefer Tennis.

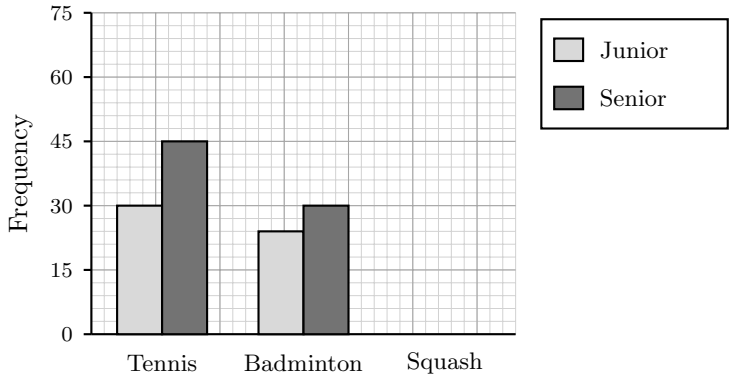
**(i)** Show that the total number of members surveyed is 216.

**[2]**

**(ii)** Calculate the number of members who prefer Badminton and the number of members who prefer Squash.

Badminton = ..... , Squash = ..... **[2]**

**(c)** The dual bar chart below shows the number of Junior and Senior members who prefer Tennis and Badminton.
Among all 216 members surveyed, the ratio of Junior members to Senior members is 3 : 5.
Complete the dual bar chart by drawing the bars for Squash.




**[4]**

**(d)** In a pictogram representing the 216 members, one full racket symbol represents 12 members. Explain why a partial symbol is needed to represent the members who prefer Badminton, and state what fraction of a symbol must be drawn.

.....

..... **[2]**

**1593** The times, in minutes, taken by 17 students to complete a mathematics puzzle are listed below.

23 14 35 18 41 27 19 31 22
38 12 25 29 18 33 27 15

**(a)** Complete the ordered stem-and-leaf diagram to show this information.

| **Stem** | **Leaf** |
|---|---|
| 1 | |
| 2 | |
| 3 | |
| 4 | |

Key: 2 | 3 represents 23 minutes

**[3]**

**(b)** Write down the median time.

Median = ..... minutes **[1]**

**(c)** Calculate the interquartile range.

Interquartile range = ..... minutes **[2]**

**1594** The masses, $m$ kilograms, of 15 parcels are recorded.

| | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 3.2 | 1.5 | 4.6 | 2.1 | 1.8 | 3.5 | 2.9 | 4.1 | 3.2 | 1.9 | 2.4 | 3.7 | 4.0 | 2.6 | 1.7 |

**(a)** Complete the ordered stem-and-leaf diagram to show this information.

| | |
|---|---|
| 1 | |
| 2 | |
| 3 | |
| 4 | |

Key: 1 | 5 represents 1.5 kg

**[3]**

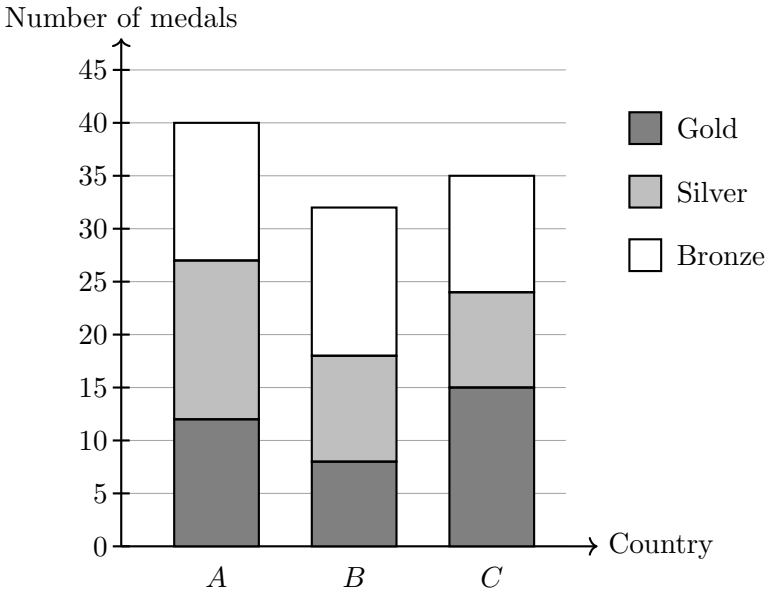
**(b)** Find the median mass.

..... kg **[1]**

**(c)** A parcel is chosen at random.
Find the probability that its mass is greater than 3.5 kg.

..... **[1]**

**1595** The composite bar chart shows the number of gold, silver and bronze medals won by three countries, $A$, $B$ and $C$, at an international sports competition.




**(a)** Find the number of silver medals won by Country $A$.

..... **[1]**

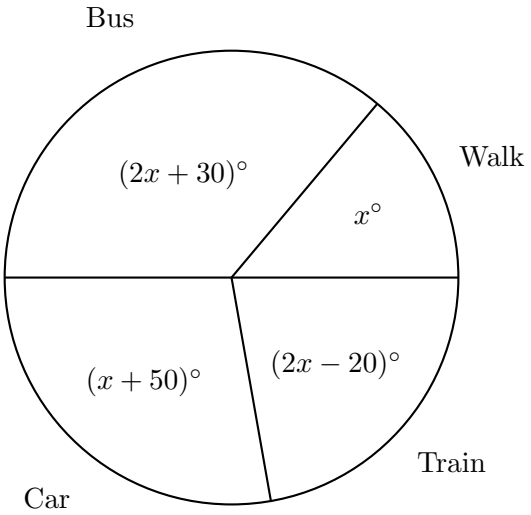
**(b)** Calculate the percentage of Country $C$'s total medals that are gold.

..... % **[2]**

**(c)** A pie chart is drawn to represent all the medals won by the three countries combined.
Calculate the sector angle for the total number of bronze medals won by all three countries.

..... ° **[3]**

**1596** The pie chart shows information about how 72 students travel to school.
The sectors represent Walk, Bus, Car, and Train.




**(a)** Show that $x = 50$.

**[2]**

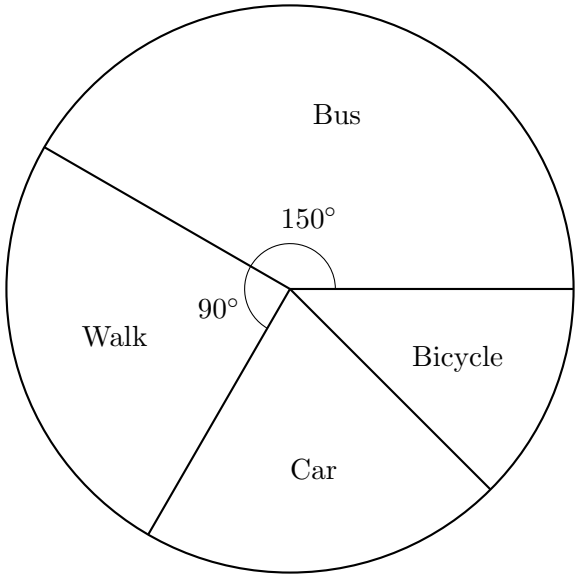
**(b)** Work out the number of students who travel by Car.

..... **[2]**

**(c)** A student is chosen at random.
Find the probability that the student travels by Bus or Train.

..... **[2]**

**1597** The pie chart shows information about how 120 students travel to school.




**(a)** Calculate the number of students who travel by Bus.

..... **[2]**

**(b)** There are 25 students who travel by Car.
Calculate the sector angle for Car.

..... ° **[2]**

**(c)** Calculate the percentage of students who travel by Bicycle.

..... % **[3]**

**1598** A survey asked 120 students about their favourite sport.
The results are shown in the table.

| **Sport** | **Frequency** |
|---|---|
| Football | 45 |
| Basketball | 35 |
| Tennis | 25 |
| Swimming | 15 |

**(a)** A pie chart is drawn to show this information.
Calculate the sector angle for Basketball.

..... ° **[2]**

**(b)** The pie chart has a radius of 6 cm.

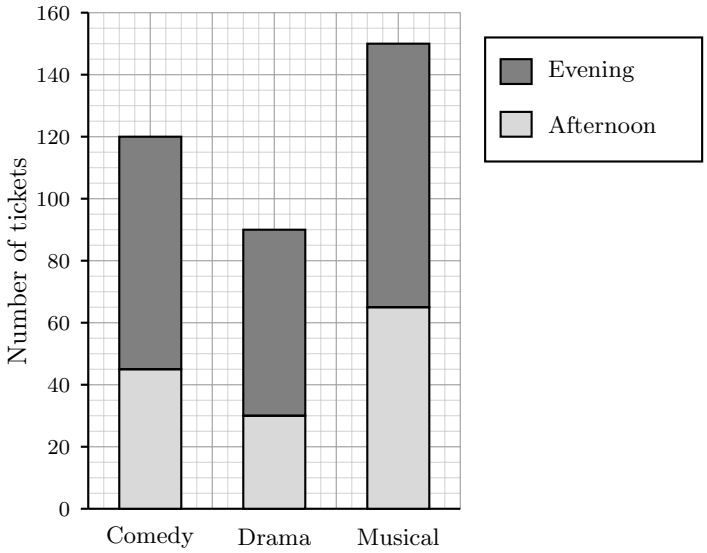
Calculate the area of the sector representing Tennis.

..... $cm^2$ **[3]**

**(c)** In a different school, the pie chart for favourite sports has a sector angle of 108° for Football.
This sector represents 54 students.
Calculate the total number of students in this school.

..... **[2]**

**1599** The composite bar chart shows the number of tickets sold by a theatre for three shows (Comedy, Drama, and Musical) during the afternoon and evening performances on a Saturday.




**(a)** Write down the number of evening tickets sold for the Musical.

..... **[1]**

**(b)** Calculate the percentage of all tickets sold that were for afternoon performances, giving your answer correct to 1 decimal place.

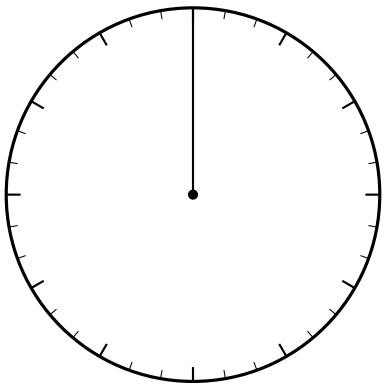
..... % **[3]**

**(c)** The total number of tickets sold for each show is to be represented on a pie chart.

**(i)** Calculate the sector angle for Comedy and the sector angle for Musical.

Comedy = ..... , Musical = ..... **[2]**

**(ii)** Complete the pie chart below to represent the total number of tickets sold for the three shows.
Label each sector clearly.



**[2]**

**(d)** An adult ticket costs \$24 and a child ticket costs \$15.
For the Drama show, the ratio of adult tickets sold to child tickets sold was 7 : 3.
Calculate the total ticket revenue for the Drama show.

\$ ..... **[3]**

**1600** A pie chart is drawn to show the favourite sport of a group of students.
The angles of the sectors for each sport are given in terms of $x$.

Football: $(2x + 30)^\circ$ Basketball: $x^\circ$ Tennis: $(x + 50)^\circ$ Swimming: $(2x - 20)^\circ$

**(a)** Find the value of $x$.

$x =$ ..... **[2]**

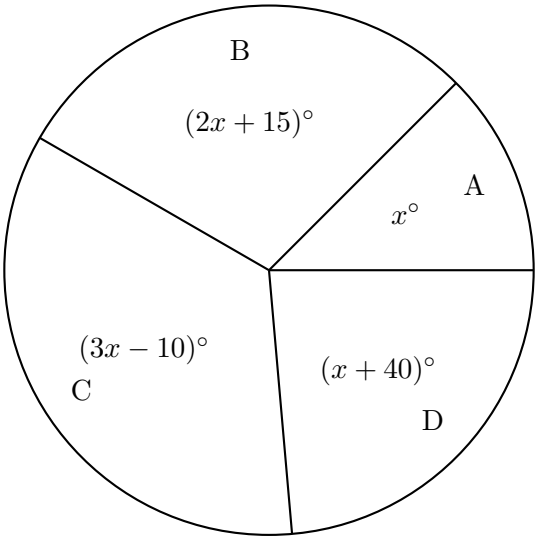
**(b)** 10 students chose Tennis as their favourite sport.
Calculate the total number of students in the group.

..... **[2]**

**(c)** Two students are chosen at random from the group.
Calculate the probability that one student chose Football and the other student chose Basketball.

..... **[3]**

**1601** A pie chart shows the grades (A, B, C, and D) achieved by students in a mathematics test.
The angles of the sectors are given in terms of $x$.




**(a)** Show that $x = 45$.

**[3]**

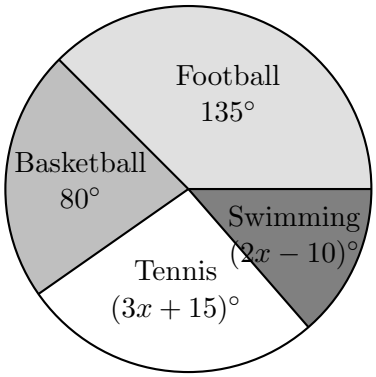
**(b)** There are 24 students who achieved Grade A.
Calculate the total number of students who took the test.

..... **[2]**

**(c)** Calculate the percentage of students who achieved Grade C.

..... % **[3]**

**1602** The pie chart shows the distribution of favourite sports chosen by a group of students.
The sector angle for Tennis is $(3x + 15)^\circ$ and the sector angle for Swimming is $(2x - 10)^\circ$.
The number of students who chose Football is 108.




**(a)** Find the value of $x$.

$x =$ ..... **[3]**

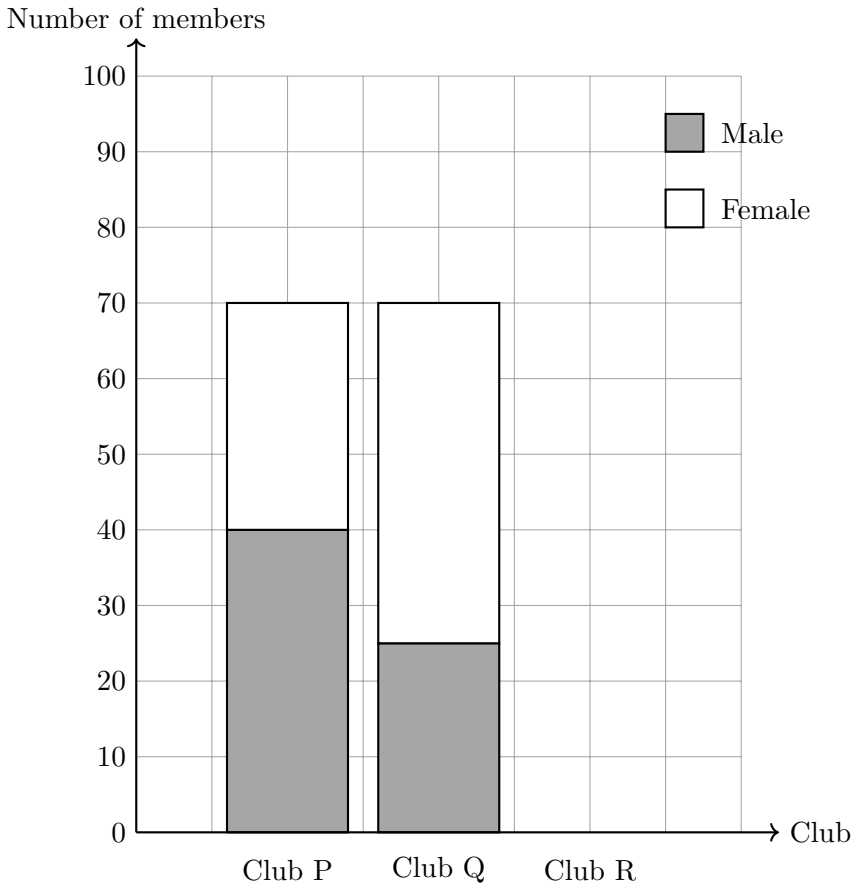
**(b)** Calculate the total number of students in the group.

..... **[2]**

**(c)** A dual bar chart is drawn to compare the number of boys and girls who chose each sport.
For Tennis, the ratio of boys to girls is 5 : 3.
Calculate the difference between the number of boys and the number of girls who chose Tennis.

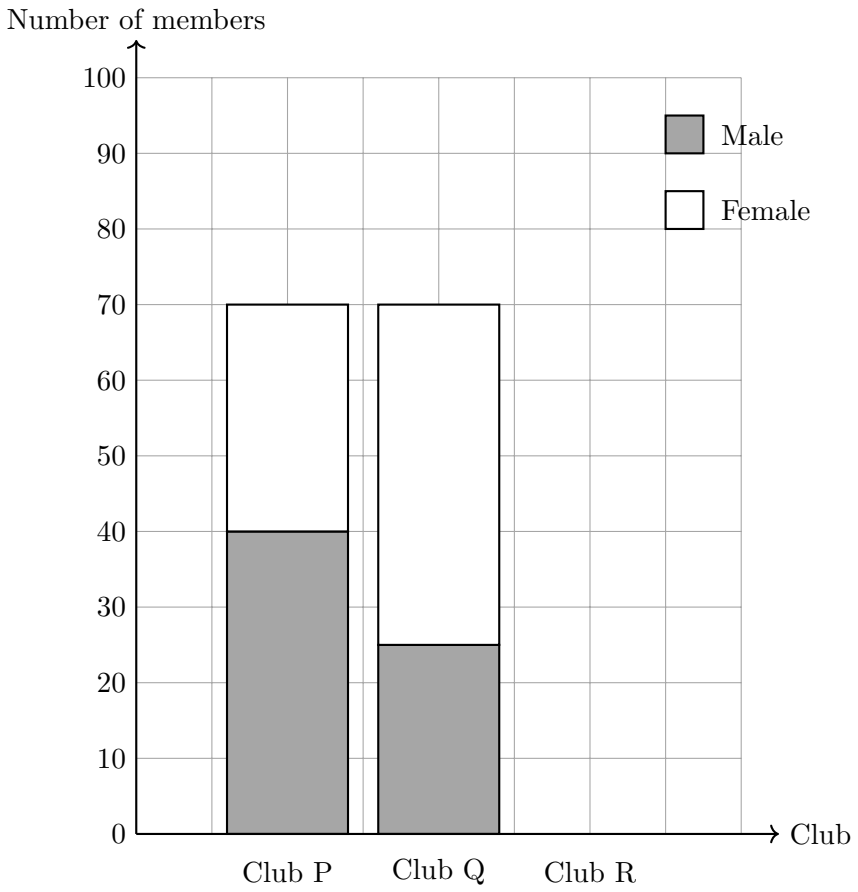
..... **[3]**

**1603** The composite bar chart shows information about the number of Male and Female members in three sports clubs, P, Q, and R.
The data for Club R is incomplete on the diagram.




**(a)** Club R has a total of 60 members.
The ratio of Male to Female members in Club R is 1 : 2.

Complete the composite bar chart for Club R.




**[3]**

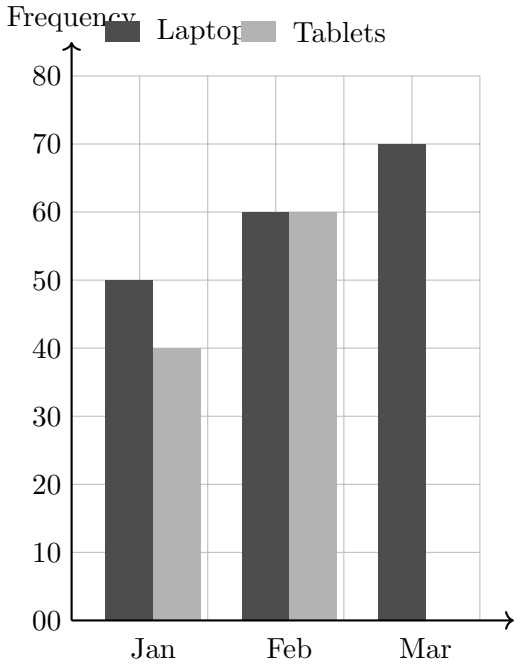
**(b)** Calculate the percentage of total members across all three clubs who are Female.

..... % **[3]**

**(c)** Two members from Club P are chosen at random.
Calculate the probability that they are both Male.

..... **[3]**

**1604** A company records the number of Laptops and Tablets sold over three months.
The dual bar chart shows the sales for January, February, and March.
The bar for Tablets sold in March is incomplete and not shown.




**(a)** The total number of Laptops sold over the three months is 20% more than the total number of Tablets sold.
Calculate the number of Tablets sold in March.

..... **[4]**

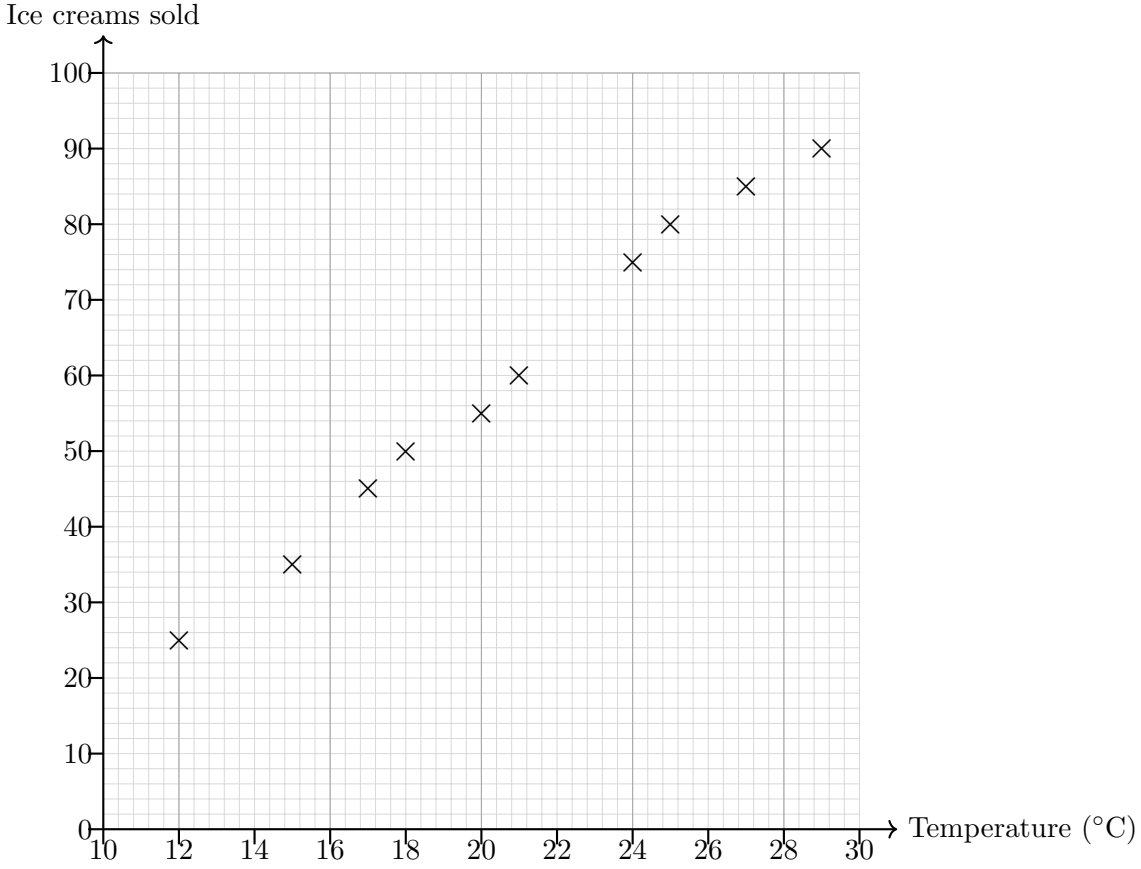
**(b)** In April, the total number of items sold is $\frac{1}{3}$ of the total number of items sold in January, February, and March combined.
The ratio of Laptops to Tablets sold in April is 6 : 5.
Work out the number of Laptops sold in April.

..... **[4]**

**(c)** The price of a Laptop is \$450 and the price of a Tablet is \$200.
Calculate the percentage of the total revenue in January that came from Tablet sales.

..... % **[3]**

**1605** The scatter diagram shows the relationship between the maximum daily temperature, $x$°C, and the number of ice creams sold, $y$, on 10 different days.




**(a)** State the type of correlation shown in the scatter diagram.

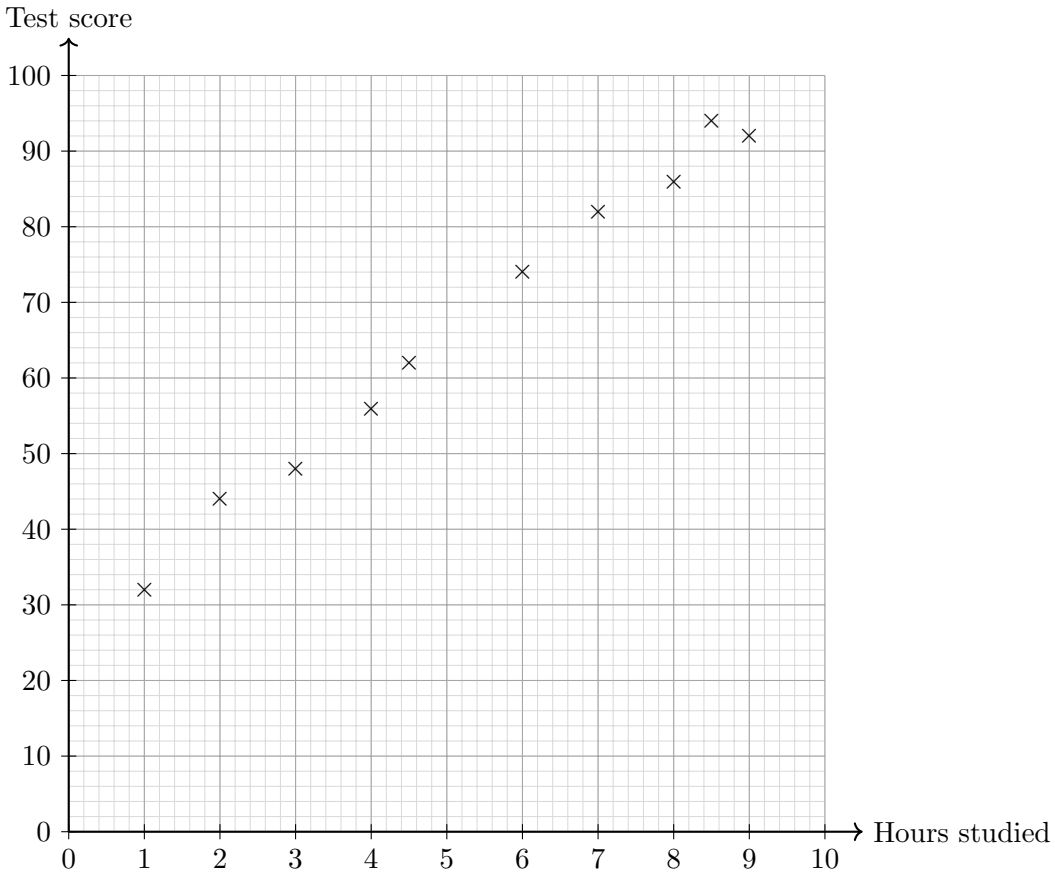
..... **[1]**

**(b)** On the scatter diagram, draw a line of best fit. **[1]**

**(c)** Use your line of best fit to estimate the number of ice creams sold on a day when the maximum temperature is 22°C.

..... **[1]**

**1606** The scatter diagram shows the number of hours studied and the test score achieved by 10 students.




**(a)** What type of correlation is shown by the scatter diagram?

..... **[1]**

**(b)** On the scatter diagram, draw a line of best fit. **[1]**

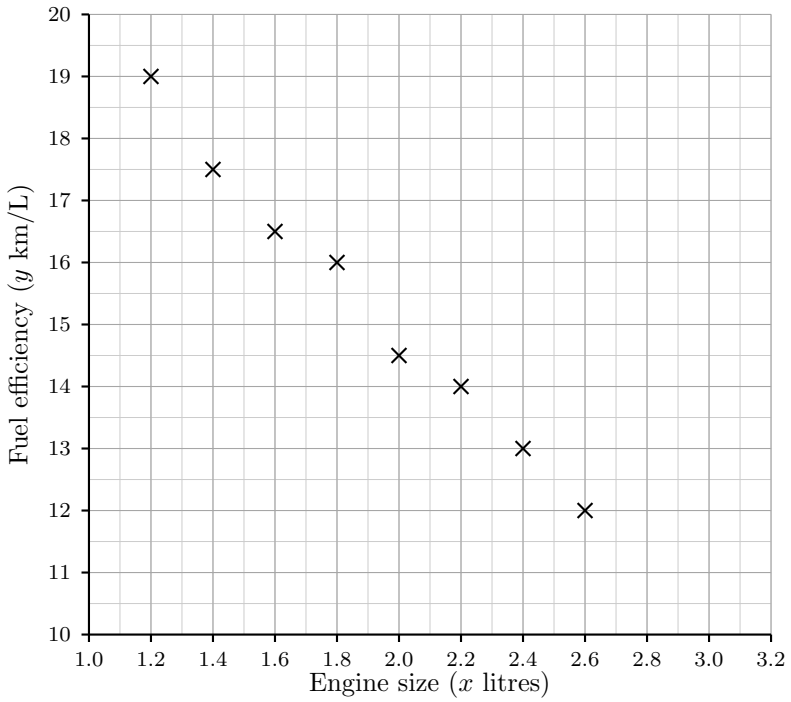
**(c)** A student studied for 5.5 hours.
Use your line of best fit to estimate their test score.

..... **[1]**

**1607** A motoring magazine records the engine size, $x$ litres, and the fuel efficiency, $y$ km/L, for 10 cars. The table shows the results.

| Engine size ($x$ L) | 1.2 | 1.4 | 1.6 | 1.8 | 2.0 | 2.2 | 2.4 | 2.6 | 2.8 | 3.0 |
|---|---|---|---|---|---|---|---|---|---|---|
| Efficiency ($y$ km/L) | 19.0 | 17.5 | 16.5 | 16.0 | 14.5 | 14.0 | 13.0 | 12.0 | 11.5 | 11.0 |

**(a)** The first 8 points are plotted on the scatter diagram below.
Plot the remaining 2 points, $(2.8, 11.5)$ and $(3.0, 11.0)$, on the grid.




**[2]**

**(b)** State the type of correlation shown by the scatter diagram.

..... **[1]**

**(c)** **(i)** Calculate the mean engine size, $\bar{x}$, and the mean fuel efficiency, $\bar{y}$.

$\bar{x} =$ ..... , $\bar{y} =$ ..... **[2]**

**(ii)** On the scatter diagram in part (a), plot the mean point $(\bar{x}, \bar{y})$ and draw a straight line of best fit by eye. **[2]**

**(d)** Use your line of best fit to estimate the fuel efficiency of a car with an engine size of 2.5 litres.

..... km/L **[1]**

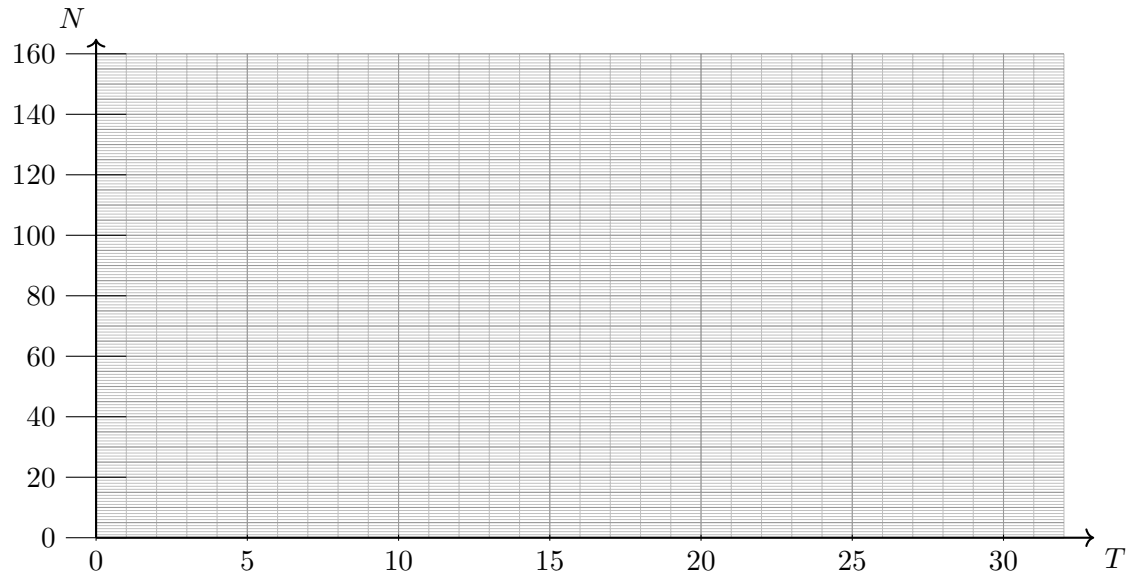
**(e)** Explain why your line of best fit should not be used to estimate the fuel efficiency of a car with an engine size of 5.0 litres.

..... **[1]**

**1608** The table shows the outside temperature, $T$°C, and the number of hot drinks, $N$, sold by a cafe on 10 different days.

| Temperature ($T$°C) | 5 | 8 | 12 | 15 | 18 | 20 | 22 | 25 | 28 | 30 |
|---|---|---|---|---|---|---|---|---|---|---|
| Number of hot drinks ($N$) | 140 | 125 | 100 | 95 | 70 | 65 | 50 | 40 | 25 | 15 |

**(a)** Complete the scatter diagram.
The first four points have been plotted for you.




**[2]**

**(b)** What type of correlation is shown by the scatter diagram?

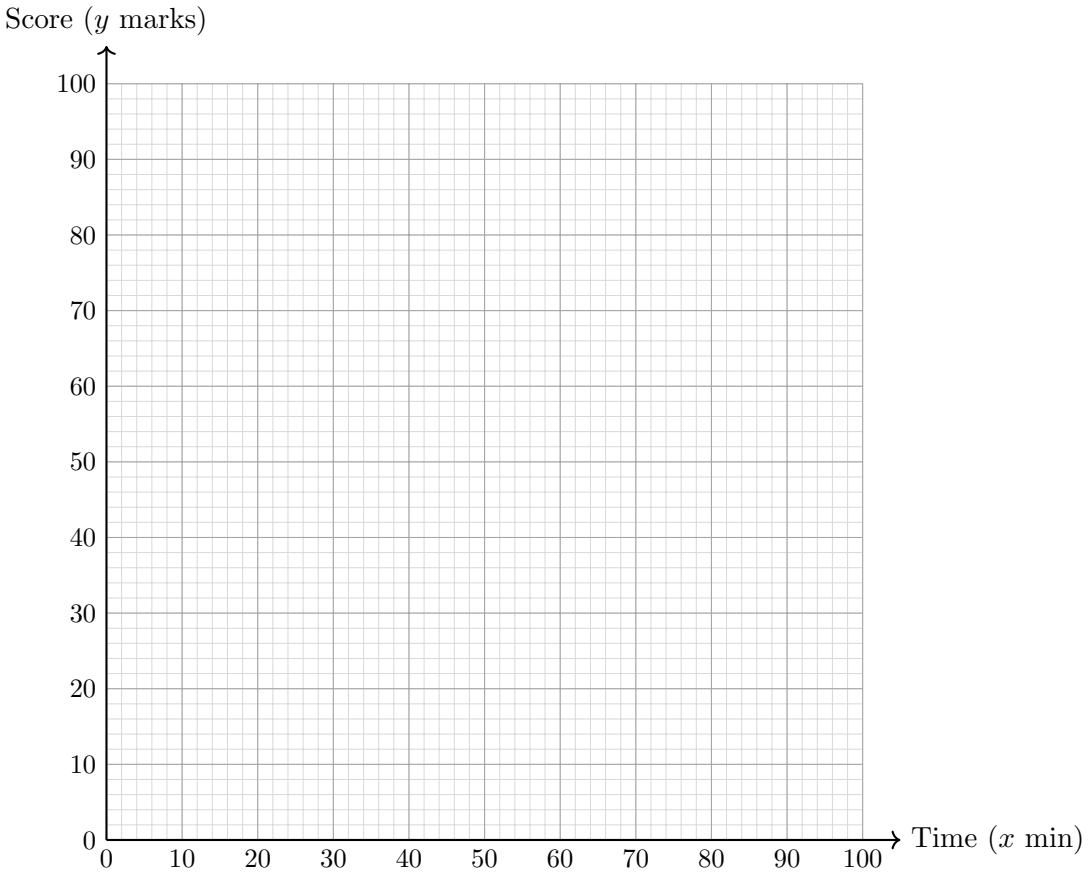
..... **[1]**

**(c)** Draw a line of best fit on the scatter diagram. **[1]**

**1609** A teacher records the time, $x$ minutes, spent studying and the score, $y$ marks, for 8 students.
The table shows the results.

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Time ($x$ min) | 10 | 20 | 30 | 40 | 50 | 60 | 70 | 80 |
| Score ($y$ marks) | 15 | 25 | 20 | 40 | 50 | 70 | 60 | 80 |

**(a)** On the grid, complete the scatter diagram to show this information.




**[2]**

**(b)** What type of correlation is shown by the scatter diagram?

..... **[1]**

**(c)** On the scatter diagram, draw a line of best fit. **[1]**

**(d)** Another student spent 45 minutes studying.
Use your line of best fit to estimate their score.

..... **[1]**

**1610** The table shows the time spent studying, $x$ hours, and the test score, $y$ marks, for a group of 8 students.

| Study time ($x$ hours) | 2 | 3.5 | 4 | 5 | 6.5 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|
| Test score ($y$ marks) | 45 | 52 | 60 | 68 | 75 | 82 | 85 | 95 |

**(a)** Use your graphic display calculator to find the equation of the linear regression line, giving $y$ in terms of $x$.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[2]**

**(b)** Use your equation to estimate the test score for a student who studies for 7 hours.

..... **[2]**

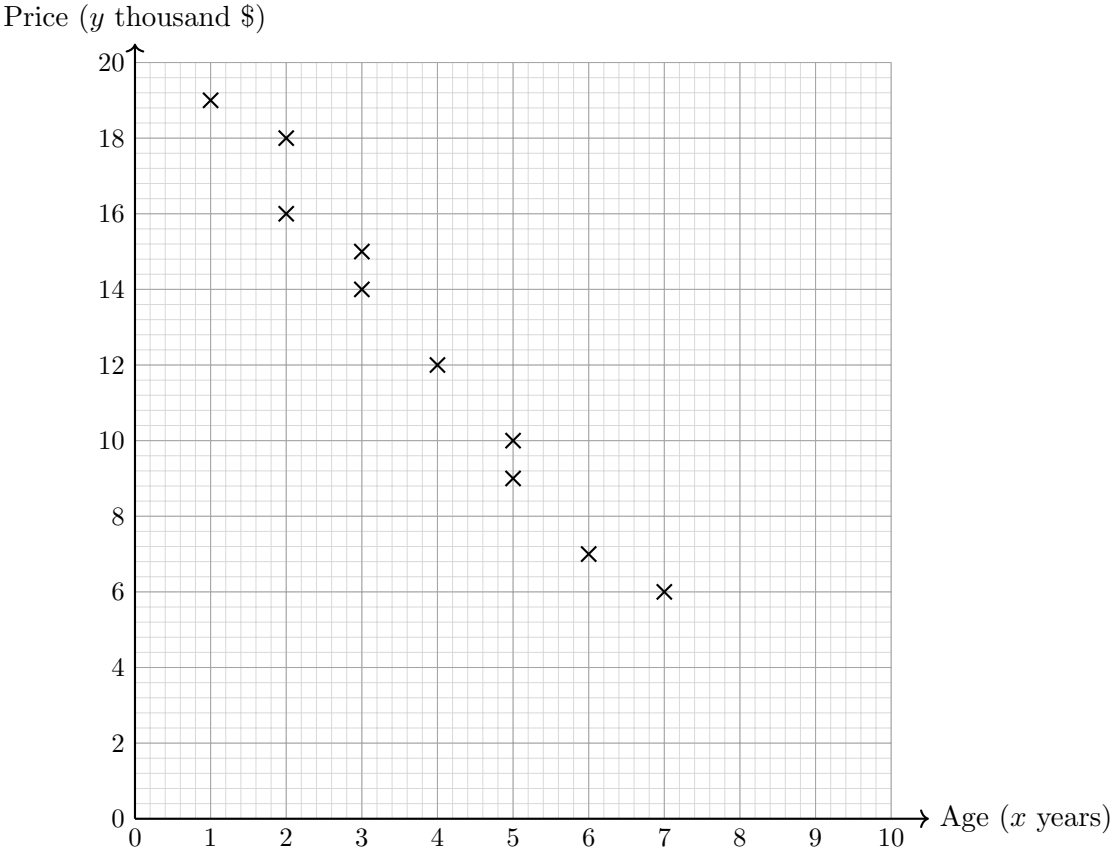
**(c)** Explain why it may not be reliable to use this equation to estimate the test score for a student who studies for 15 hours.

..... **[1]**

**1611** A car dealer records the age, $x$ years, and the price, $y$ thousand dollars, of 12 used cars of the same model.
The table shows the data.

| Age ($x$ years) | 1 | 2 | 2 | 3 | 3 | 4 | 5 | 5 | 6 | 7 | 7 | 9 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Price ($y$ thousand \$) | 19 | 18 | 16 | 15 | 14 | 12 | 10 | 9 | 7 | 6 | 5 | 1 |

**(a)** The scatter diagram shows the first 10 points from the table.
Plot the remaining 2 points on the scatter diagram.




**[1]**

**(b)** Calculate the mean age and the mean price.

Mean age = ..... , Mean price = ..... **[2]**

**(c)** On the scatter diagram, plot the mean point and draw a line of best fit. **[2]**

**(d)** State the type of correlation shown.

..... **[1]**

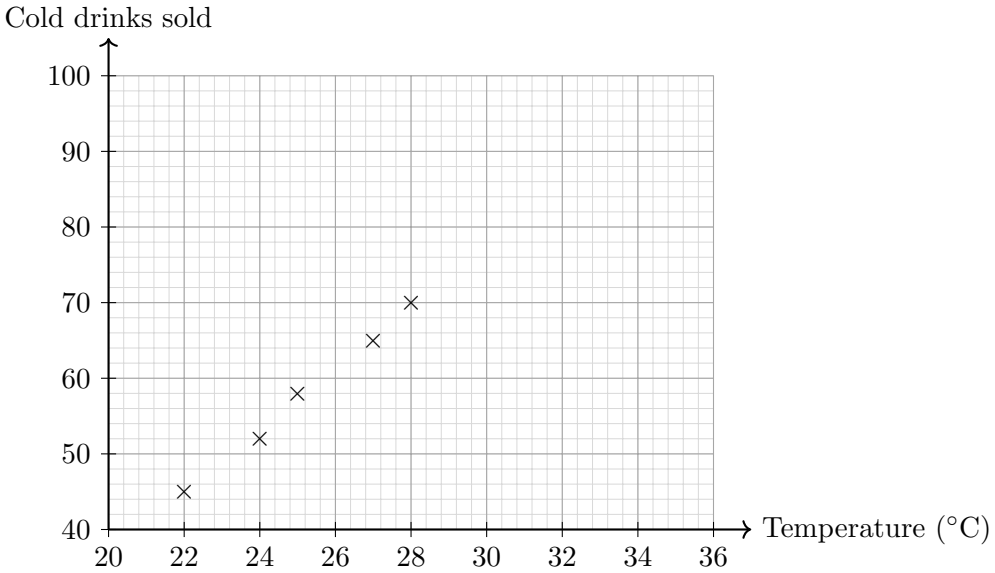
**(e)** Use your line of best fit to estimate the price of a car that is 8 years old.

..... thousand dollars **[1]**

**1612** The table shows the maximum temperature, $x$°C, and the number of cold drinks sold, $y$, at a shop on 8 different days.

| Temperature, $x$ (°C) | 22 | 24 | 25 | 27 | 28 | 31 | 32 | 35 |
|---|---|---|---|---|---|---|---|---|
| Cold drinks sold, $y$ | 45 | 52 | 58 | 65 | 70 | 81 | 85 | 96 |

The first 5 points have been plotted on the scatter diagram.




**(a)** Plot the remaining 3 points on the scatter diagram. **[2]**

**(b)** Calculate the mean temperature and the mean number of cold drinks sold.

Mean temperature = ..... , Mean drinks sold = ..... **[2]**

**(c)** On the scatter diagram, plot the mean point and draw a line of best fit. **[2]**

**(d)** Predict the number of cold drinks sold on a day when the temperature is 29°C.

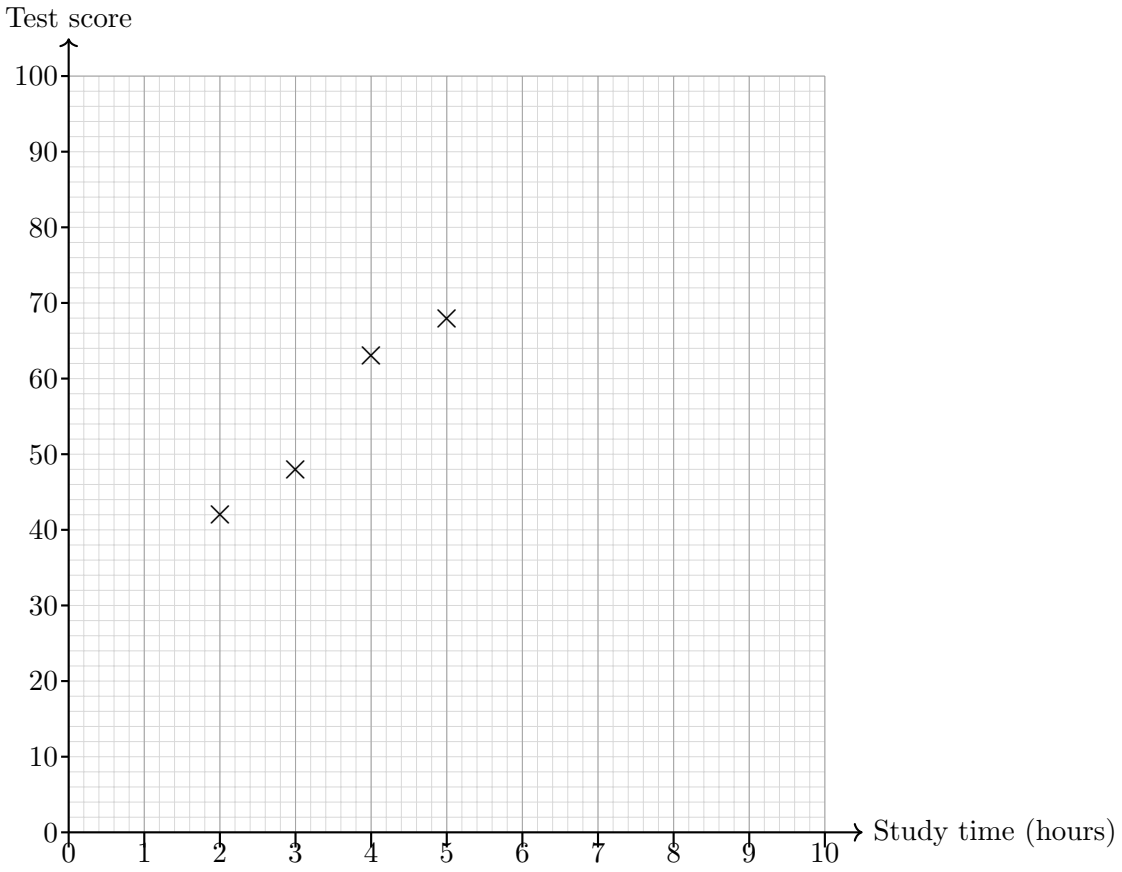
..... **[1]**

**1613** A teacher records the number of hours spent studying, $x$, and the test score, $y$, for 7 students.

| Study time ($x$ hours) | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|
| Test score ($y$) | 42 | 48 | 63 | 68 | 81 | 88 | 100 |

The scatter diagram for this data is partially completed.

**(a)** Plot the remaining three points on the scatter diagram.




**[2]**

**(b)** Calculate the mean study time, $\bar{x}$, and the mean test score, $\bar{y}$.

$\bar{x} =$ ..... , $\bar{y} =$ ..... **[2]**

**(c)** On the scatter diagram, plot the mean point and draw a line of best fit. **[2]**

**(d)** Another student studied for 5.5 hours.
Use your line of best fit to estimate their test score.

..... **[1]**

**1614** A teacher records the daily revision time, $x$ minutes, and the test score, $y$ marks, for 8 students. The results are shown in the table below.

| Revision time ($x$ min) | 10 | 15 | 20 | 25 | 30 | 35 | 40 | 45 |
|---|---|---|---|---|---|---|---|---|
| Test score ($y$ marks) | 34 | 40 | 44 | 51 | 55 | 62 | 65 | 73 |

**(a) (i)** Use your graphic display calculator to find the equation of the linear regression line of $y$ on $x$, giving the coefficients correct to 3 significant figures.

$y =$ ..... **[2]**

**(ii)** Write down the coordinates of the mean point $(\bar{x}, \bar{y})$.

( ..... , ..... ) **[2]**

**(b) (i)** Use your regression equation to estimate the test score of a student who revises for 32 minutes per day.

..... marks **[2]**

**(ii)** Explain why using the regression equation to estimate the test score of a student who revises for 90 minutes per day would not be valid.

..... **[2]**

**(c)** A 9th student revises for 25 minutes per day and scores 51 marks.
State, with a reason, how adding this student's data affects the coordinates of the mean point $(\bar{x}, \bar{y})$.

.....

..... **[2]**

**1615** The table shows the age, $x$ years, and the value, $y$ dollars, of 6 cars of the same model.

| Age, $x$ (years) | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| Value, $y$ (\$) | 12600 | 10800 | 9600 | 8100 | 6300 | 5100 |

**(a)** Use a graphic display calculator to find the equation of the linear regression line. Give your answer in the form $y = mx + c$.

$y =$ ..... **[2]**

**(b)** Give an interpretation of the value of $m$ in this context.

..... **[1]**

**(c)** Use your equation to estimate the value of a car that is 4.5 years old.

% ..... **[2]**

**(d)** Explain why it may not be reliable to use this equation to estimate the value of a car that is 15 years old.

..... **[1]**

**1616** A researcher measures the height, $x$ cm, and the mass, $y$ kg, of 12 adult males.
The results are summarised as follows:

$$\sum x = 2100 \qquad \sum y = 924$$

The equation of the linear regression line is $y = 0.85x - c$, where $c$ is a constant.

**(a)** Calculate the mean height, $\bar{x}$, and the mean mass, $\bar{y}$.

$\bar{x} =$ ..... , $\bar{y} =$ ..... **[2]**

**(b)** The point $(\bar{x}, \bar{y})$ lies on the linear regression line.
Find the value of $c$.

$c =$ ..... **[2]**

**(c)** A new adult male is added to the study.
His height is 180 cm.
His mass is exactly equal to the value predicted by the regression line.
Calculate the new mean mass for all 13 men.

..... kg **[2]**

**1617** An automotive engineer tests 8 different cars to investigate the relationship between engine size, $x$ litres, and fuel consumption, $y$ km/l.
The results are shown in the table.

| Engine size ($x$ litres) | 1.0 | 1.2 | 1.5 | 1.8 | 2.0 | 2.4 | 2.5 | 3.0 |
|---|---|---|---|---|---|---|---|---|
| Fuel consumption ($y$ km/l) | 21.5 | 20.0 | 18.2 | 16.5 | 15.0 | 13.0 | 12.5 | 9.8 |

**(a) (i)** Use a graphic display calculator to find the equation of the linear regression line. Give your answer in the form $y = mx + c$.

$y =$ ..... **[2]**

**(ii)** Explain what the value of the gradient represents in this context.

..... **[1]**

**(b)** Use your equation from part (a)(i) to estimate the fuel consumption of a car with an engine size of 1.6 litres.

..... km/l **[2]**

**(c)** Explain why it may not be reliable to use this equation to estimate the fuel consumption of a car with an engine size of 5.0 litres.

..... **[1]**

**1618** The table shows the temperature, $x$°C, and the number of ice creams sold, $y$, at a shop over 8 days.

| Temperature ($x$°C) | 12 | 15 | 18 | 20 | 22 | 25 | 28 | 30 |
|---|---|---|---|---|---|---|---|---|
| Ice creams sold ($y$) | 40 | 52 | 68 | 75 | 84 | 98 | 110 | 115 |

**(a)** Calculate the mean temperature and the mean number of ice creams sold.

Mean temperature = ..... , Mean ice creams sold = ..... **[2]**

**(b)** Use a graphic display calculator to find the equation of the linear regression line for this data. Give your answer in the form $y = mx + c$, with values of $m$ and $c$ correct to 3 significant figures.

$y =$ ..... **[2]**

**(c)** The shop owner predicts the number of ice creams sold on a day when the temperature is 24°C. Use your equation to estimate this number.

..... **[2]**

**(d)** Comment on the reliability of using your equation to predict the number of ice creams sold on a day when the temperature is 42°C.

..... **[1]**

**1619** A company records its monthly advertising spend, $x$ thousand dollars, and its monthly sales, $y$ thousand dollars, over 8 months.
Two entries in the table, $p$ and $q$, are missing.
Over the 8 months, the mean advertising spend is $\bar{x} = 11$ thousand dollars and the mean monthly sales is $\bar{y} = 45$ thousand dollars.

| Advertising ($x$ thousand \$) | 4 | 7 | 8 | $p$ | 12 | 14 | 15 | 18 |
|---|---|---|---|---|---|---|---|---|
| Sales ($y$ thousand \$) | 26 | 34 | 38 | 43 | 47 | 53 | $q$ | 62 |

**(a)** Find the value of $p$ and the value of $q$.

$p =$ ..... , $q =$ ..... **[2]**

**(b)** **(i)** Use your graphic display calculator to find the equation of the linear regression line of $y$ on $x$ in the form $y = ax + b$.

$y =$ ..... **[2]**

**(ii)** Interpret the value of $a$ in the context of this question.

..... **[1]**

**(c)** Calculate the difference between the actual monthly sales when $x = 14$ and the monthly sales predicted by the regression line, giving your answer in dollars.

\$ ..... **[3]**

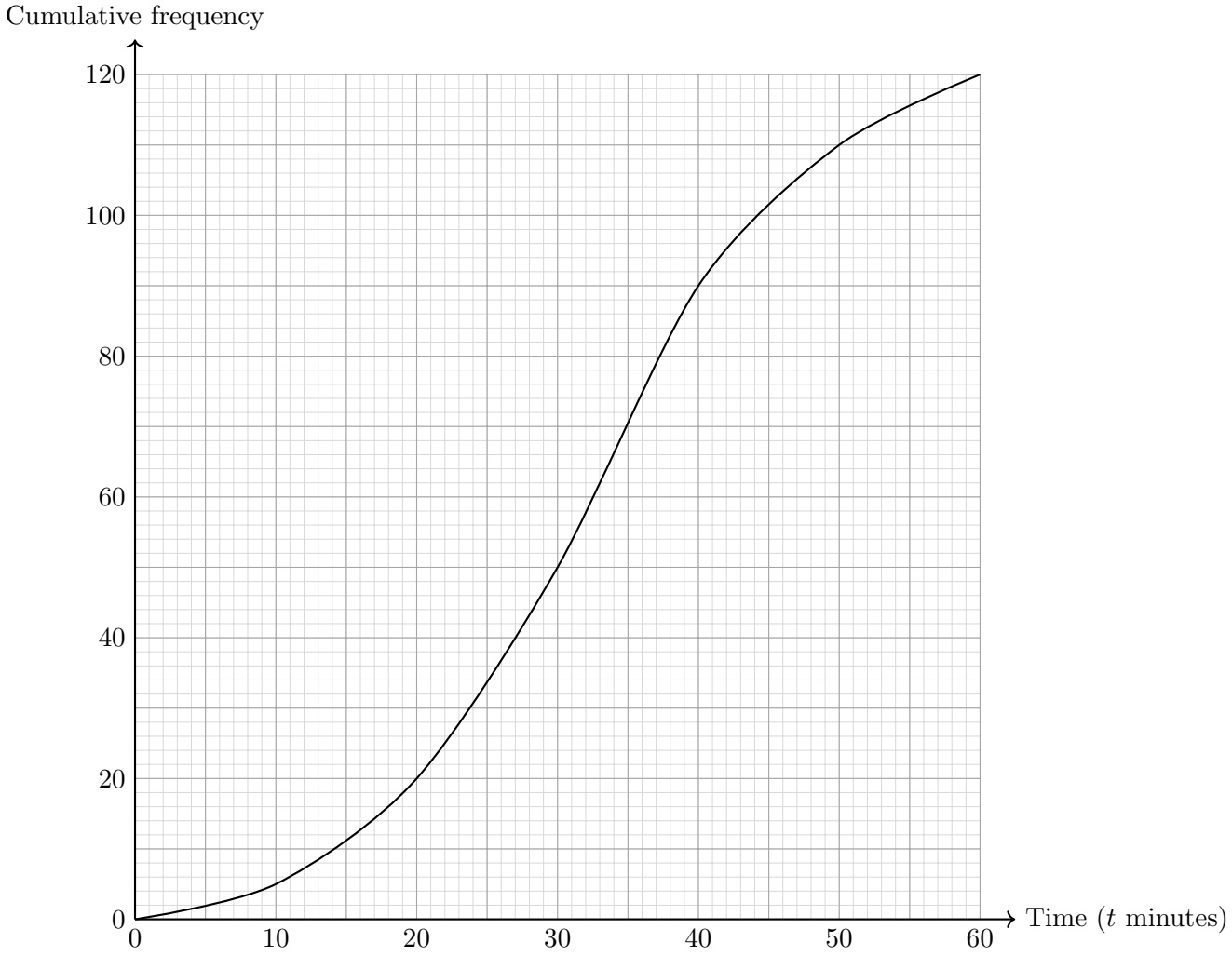
**(d)** **(i)** Use the regression equation to estimate the monthly advertising spend required to achieve monthly sales of \$68 000, giving your answer in thousand dollars correct to 3 significant figures.

..... thousand dollars **[2]**

**(ii)** Explain why this estimate may be unreliable.

..... **[1]**

**1620** The cumulative frequency diagram shows the times taken, $t$ minutes, by 120 students to complete a test.




**(a)** Use the diagram to estimate the median time.

..... minutes **[1]**

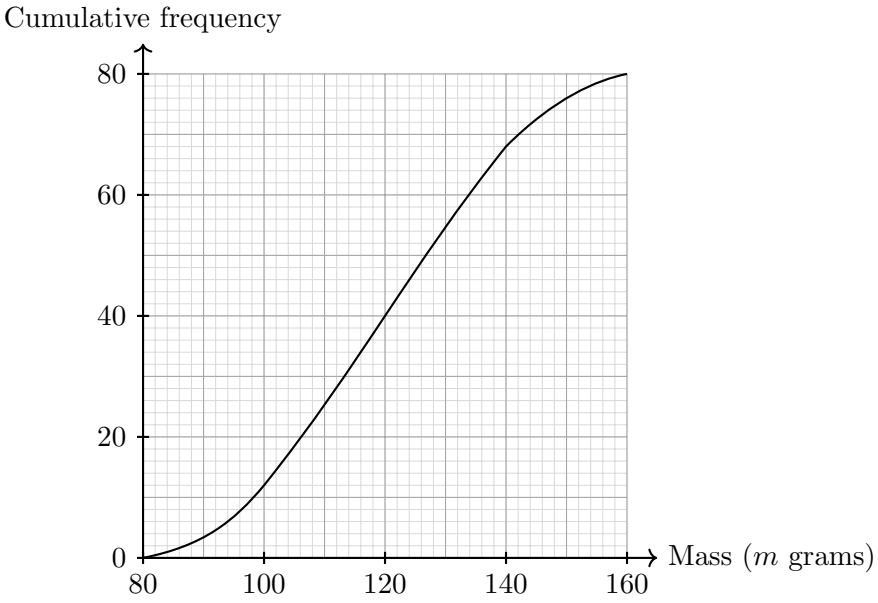
**(b)** Use the diagram to estimate the interquartile range.

..... minutes **[2]**

**(c)** Students who take longer than 45 minutes are given a penalty.
Estimate the number of students who are given a penalty.

. . . . . **[2]**

**1621** The cumulative frequency diagram shows information about the masses, $m$ grams, of 80 apples.




**(a)** Use the cumulative frequency diagram to find an estimate for

**(i)** the median,

..... $g$ **[1]**

**(ii)** the lower quartile,

..... $g$ **[1]**

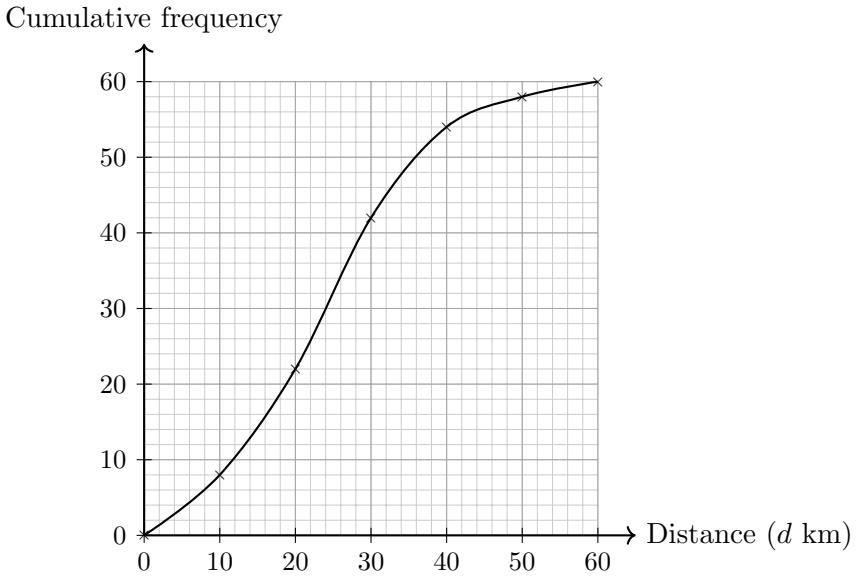
**(iii)** the interquartile range.

..... $g$ **[2]**

**(b)** Find an estimate for the number of apples with a mass greater than 140 g.

..... **[1]**

**1622** The cumulative frequency diagram shows information about the distance, $d$ km, travelled by 60 people to work.




**(a)** Use the diagram to find an estimate for:

**(i)** the median,

..... km **[1]**

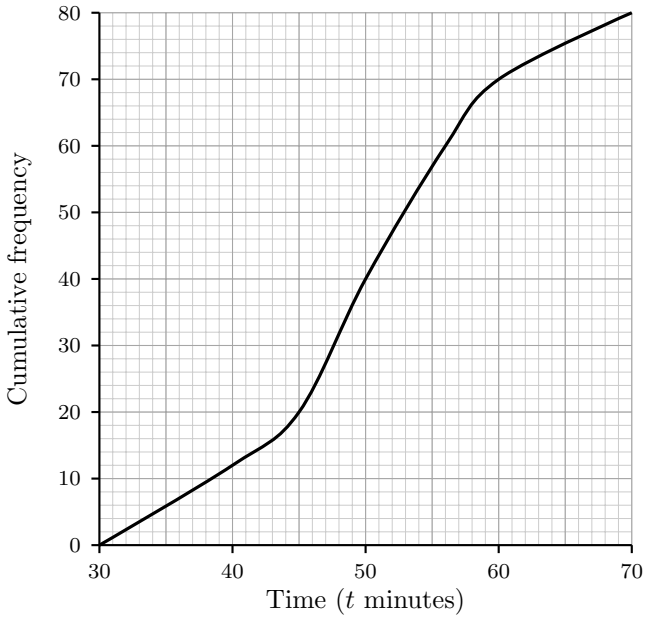
**(ii)** the interquartile range.

..... km **[2]**

**(b)** Find the number of people who travelled more than 45 km to work.

..... **[3]**

**1623** The cumulative frequency diagram shows information about the times, $t$ minutes, taken by 80 runners to complete a 10 km race.




**(a)** Use the cumulative frequency diagram to find

**(i)** the median time,

..... minutes **[1]**

**(ii)** the lower quartile and the upper quartile,

Lower quartile = ..... , Upper quartile = ..... **[2]**

**(iii)** the interquartile range.

..... minutes **[1]**

**(b)** Find the number of runners who took more than 60 minutes to complete the race.

..... **[2]**

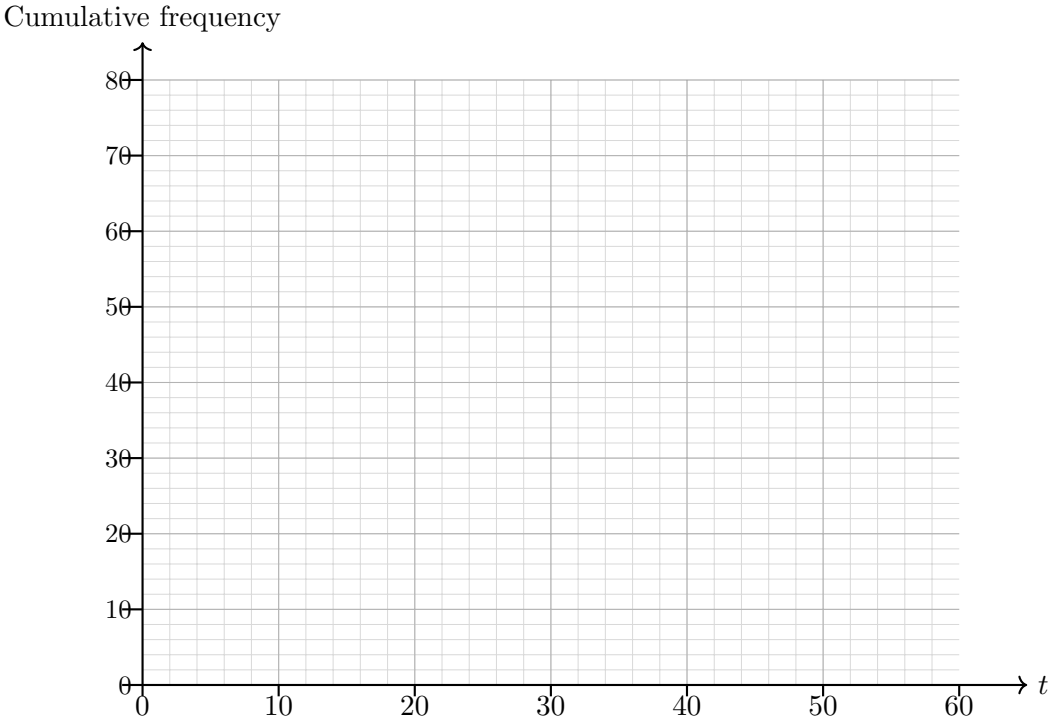
**(c)** Runners who finished in the fastest 15% of times qualified for a regional event.
Estimate the maximum qualifying time.

..... minutes **[2]**

**1624** The table shows the cumulative frequencies for the times, $t$ minutes, taken by 80 students to complete a puzzle.

| Time ($t$ minutes) | $t \leq 10$ | $t \leq 20$ | $t \leq 30$ | $t \leq 40$ | $t \leq 50$ | $t \leq 60$ |
|---|---|---|---|---|---|---|
| Cumulative frequency | 6 | 22 | 48 | 68 | 78 | 80 |

**(a)** On the grid, draw a cumulative frequency diagram to show this information.

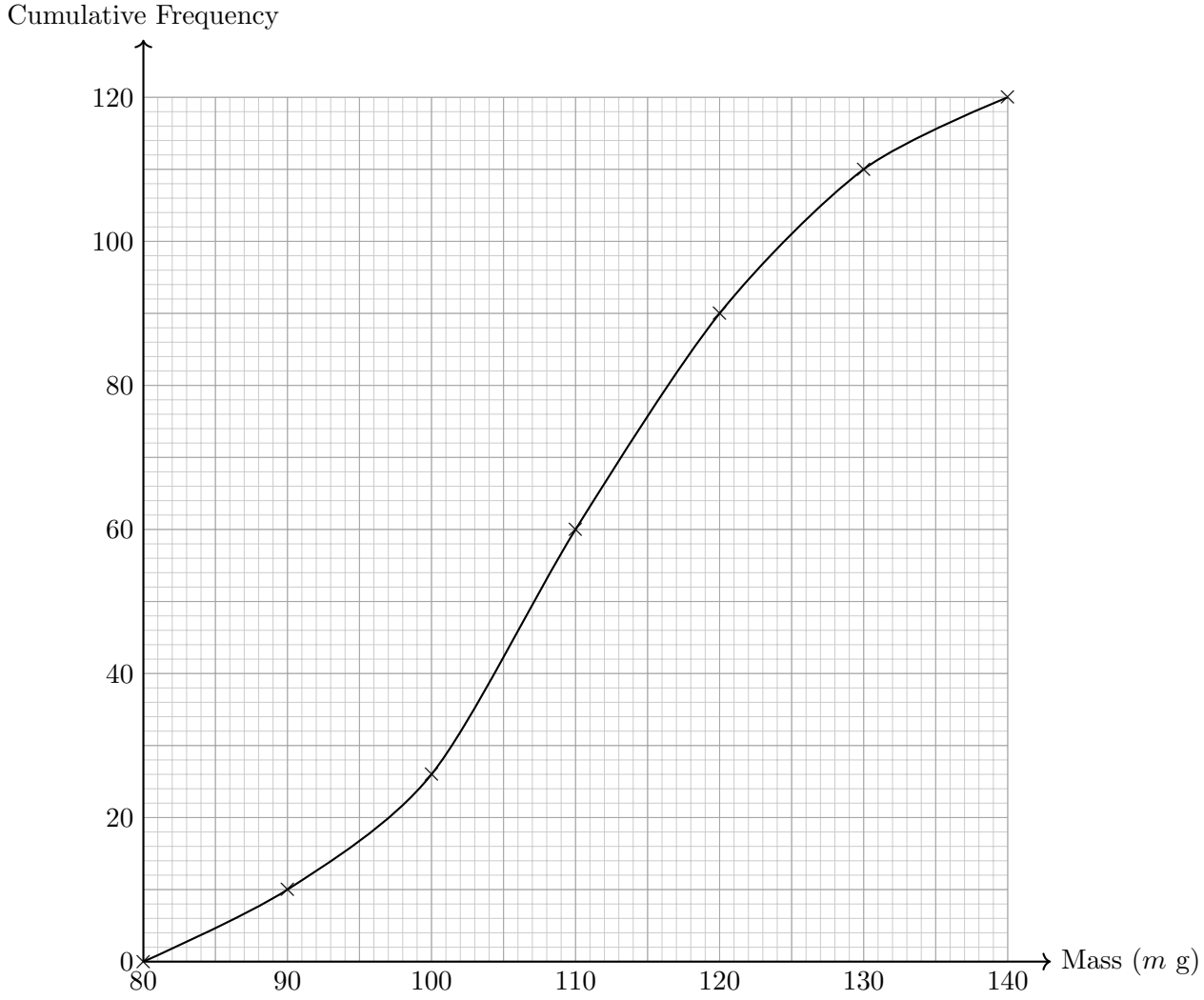



**[3]**

**(b)** Use your diagram to estimate the median time.

..... minutes **[2]**

**1625** The cumulative frequency diagram shows information about the masses, $m$ grams, of 120 apples.




**(a)** Find the median mass.

..... $g$ **[1]**

**(b)** Find the interquartile range.

..... $g$ **[2]**

**(c)** Find the number of apples with a mass greater than 125 g.

..... **[2]**

**1626** The frequency table shows information about the times, $t$ minutes, taken by 100 students to complete a puzzle.

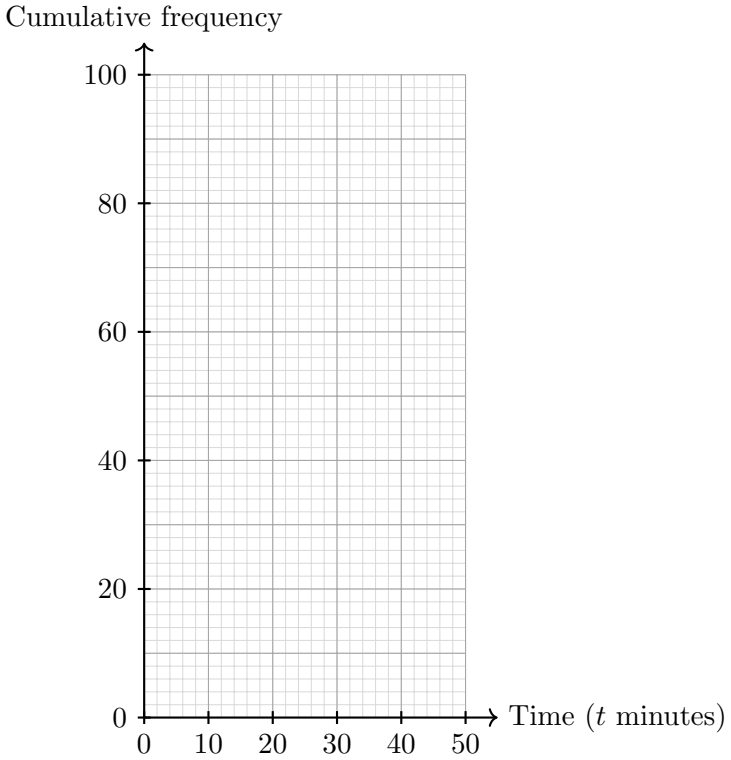
| Time ($t$ minutes) | $0 < t \leq 10$ | $10 < t \leq 20$ | $20 < t \leq 30$ | $30 < t \leq 40$ | $40 < t \leq 50$ |
|---|---|---|---|---|---|
| Frequency | 12 | 24 | 36 | 18 | 10 |

**(a)** Complete the cumulative frequency table.

| Time ($t$ minutes) | $t \leq 10$ | $t \leq 20$ | $t \leq 30$ | $t \leq 40$ | $t \leq 50$ |
|---|---|---|---|---|---|
| Cumulative frequency | 12 | | | | 100 |

**[1]**

**(b)** On the grid, draw a cumulative frequency diagram to show this information.




**[3]**

**(c)** Use your diagram to find an estimate for the 80th percentile.

..... minutes **[1]**

**(d)** Students who take more than 35 minutes must take a second test.
Use your diagram to estimate the percentage of students who must take a second test.

..... % **[2]**

**1627** The grouped frequency table shows the time, $t$ minutes, taken by 80 students to complete a science test.

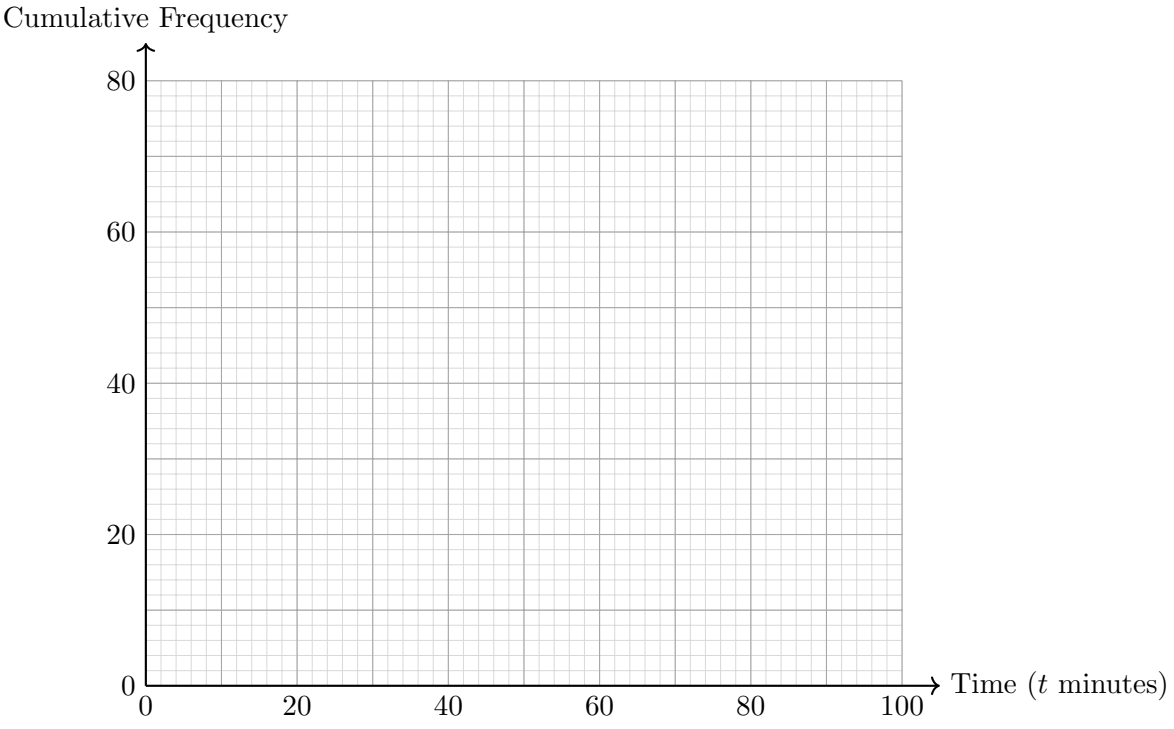
| Time ($t$ minutes) | $0 < t \leq 20$ | $20 < t \leq 40$ | $40 < t \leq 60$ | $60 < t \leq 80$ | $80 < t \leq 100$ |
|---|---|---|---|---|---|
| Frequency | 8 | 22 | 34 | 12 | 4 |

**(a)** Complete the cumulative frequency table.

| Time ($t$ minutes) | $t \leq 20$ | $t \leq 40$ | $t \leq 60$ | $t \leq 80$ | $t \leq 100$ |
|---|---|---|---|---|---|
| Cumulative freq. | 8 | | | | |

**[2]**

**(b)** On the grid, draw a cumulative frequency diagram to show this information.




**[3]**

**(c)** Use your graph to estimate the number of students who took more than 55 minutes to complete the test.

..... **[2]**

**1628** A company records the salaries, $s$ thousand dollars, of its 200 employees.
The frequency table shows the results.

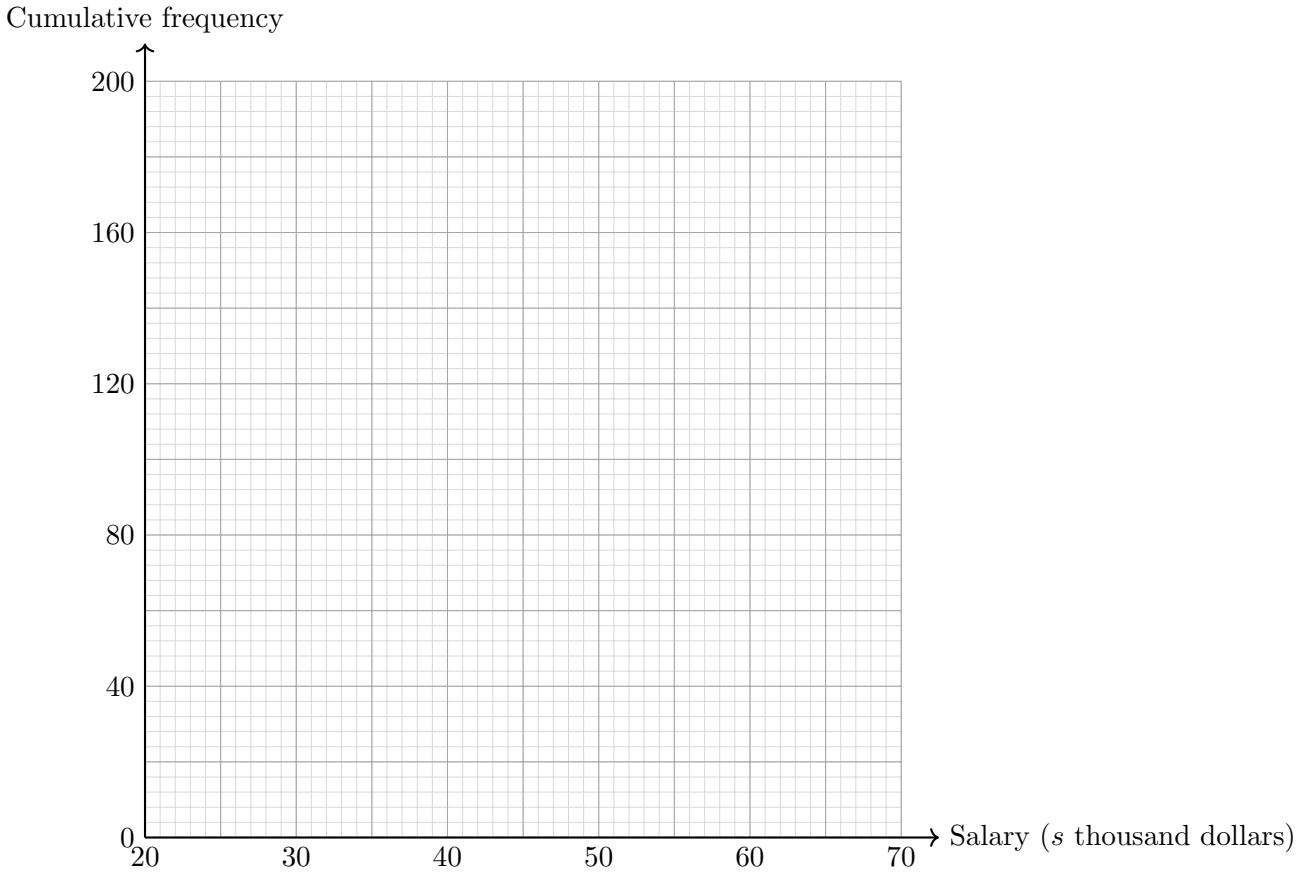
| Salary ($s$ thousand dollars) | $20 < s \leq 30$ | $30 < s \leq 40$ | $40 < s \leq 50$ | $50 < s \leq 60$ | $60 < s \leq 70$ |
|---|---|---|---|---|---|
| Frequency | 24 | 66 | 70 | 28 | 12 |

**(a)** Complete the cumulative frequency table.

| Salary ($s$ thousand dollars) | $s \leq 30$ | $s \leq 40$ | $s \leq 50$ | $s \leq 60$ | $s \leq 70$ |
|---|---|---|---|---|---|
| Cumulative frequency | 24 | | | | |

**[2]**

**(b)** On the grid, draw a cumulative frequency diagram to show this information.

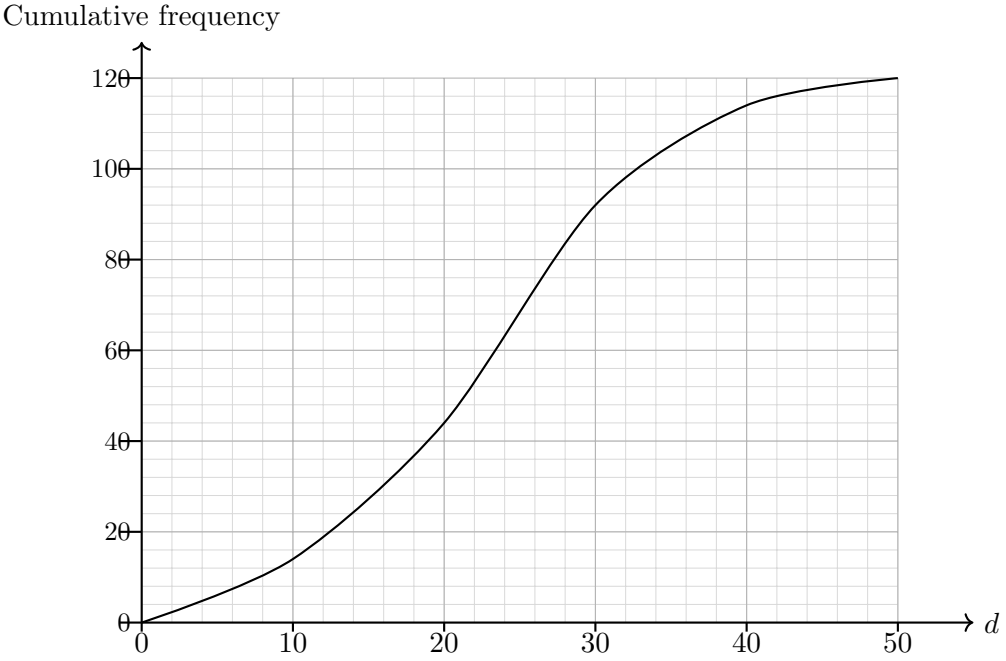



**[3]**

**(c)** Use your diagram to estimate the 70th percentile of the salaries.

..... thousand **[2]**

**1629** The cumulative frequency diagram shows information about the distances, $d$ km, travelled by 120 people to work.




**(a)** Use the diagram to estimate the median distance.

..... km **[2]**

**(b)** Use the diagram to estimate the interquartile range.

..... km **[3]**

**(c)** Estimate the number of people who travelled more than 35 km to work.

..... **[3]**

**1630** A table shows information about the masses of 120 apples.

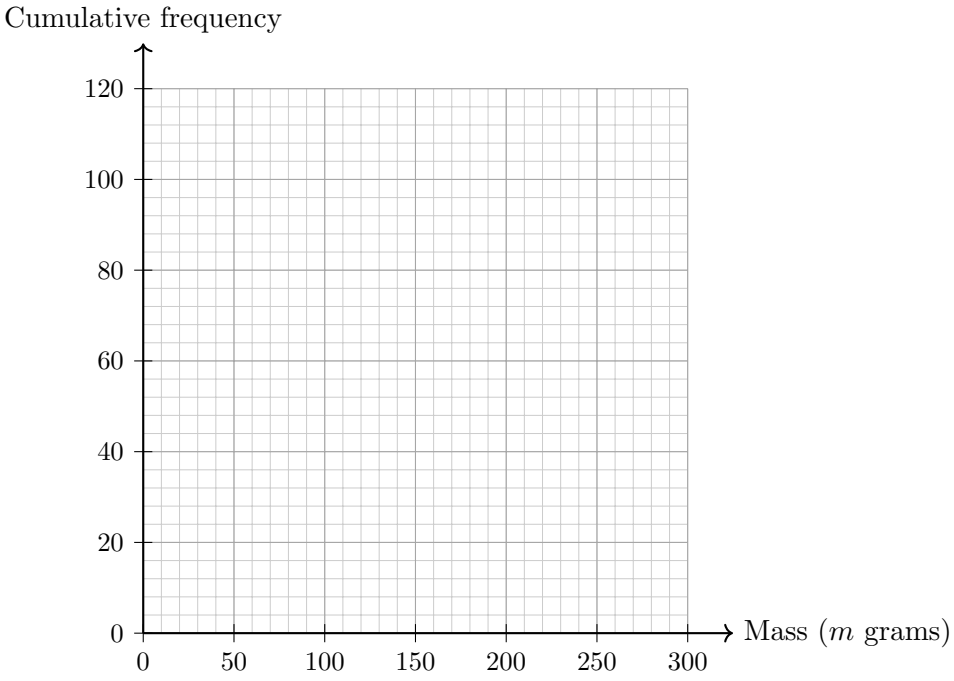
| Mass ($m$ grams) | $50 < m \leq 100$ | $100 < m \leq 150$ | $150 < m \leq 200$ | $200 < m \leq 250$ | $250 < m \leq 300$ |
|---|---|---|---|---|---|
| Frequency | 12 | 34 | 46 | 20 | 8 |

**(a)** Complete the cumulative frequency table.

| Mass ($m$ grams) | $m \leq 100$ | $m \leq 150$ | $m \leq 200$ | $m \leq 250$ | $m \leq 300$ |
|---|---|---|---|---|---|
| Cumulative frequency | 12 | | | | 120 |

**[2]**

**(b)** On the grid, draw a cumulative frequency diagram to show this information.




**[3]**

**(c)** Use your diagram to estimate the 60th percentile.

..... $g$ **[2]**

**(d)** Apples with a mass greater than 220 grams are classified as large.
Estimate the number of large apples.

..... **[3]**

**1631** A supermarket records the checkout waiting time, $w$ seconds, for 120 customers.
The grouped frequency table shows the results.

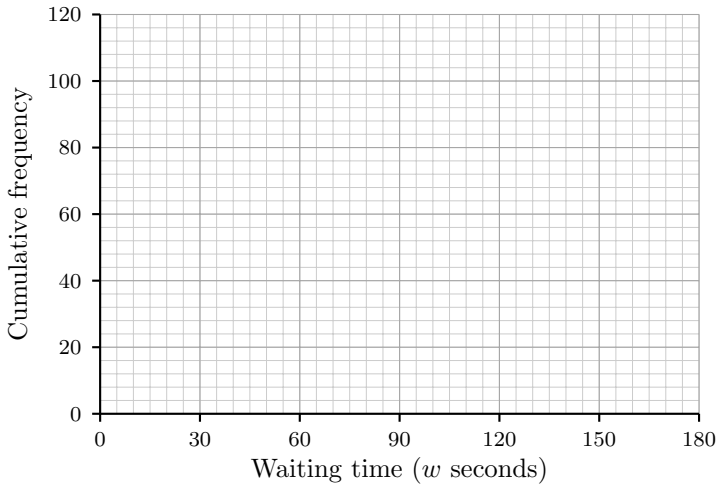
| Waiting time ($w$ s) | Frequency |
|---|---|
| $0 < w \leq 30$ | 10 |
| $30 < w \leq 60$ | 22 |
| $60 < w \leq 90$ | 44 |
| $90 < w \leq 120$ | 28 |
| $120 < w \leq 150$ | 12 |
| $150 < w \leq 180$ | 4 |

**(a)** Complete the cumulative frequency table below.

| Waiting time ($w$ s) | Cumulative frequency |
|---|---|
| $w \leq 30$ | 10 |
| $w \leq 60$ | |
| $w \leq 90$ | |
| $w \leq 120$ | |
| $w \leq 150$ | |
| $w \leq 180$ | |

**[2]**

**(b)** On the grid below, draw a cumulative frequency diagram to represent this information.
Mark your points clearly with small crosses ($\times$) and join them with a smooth curve.




**[3]**

**(c)** Use your cumulative frequency diagram to estimate

**(i)** the median waiting time,

..... seconds **[1]**

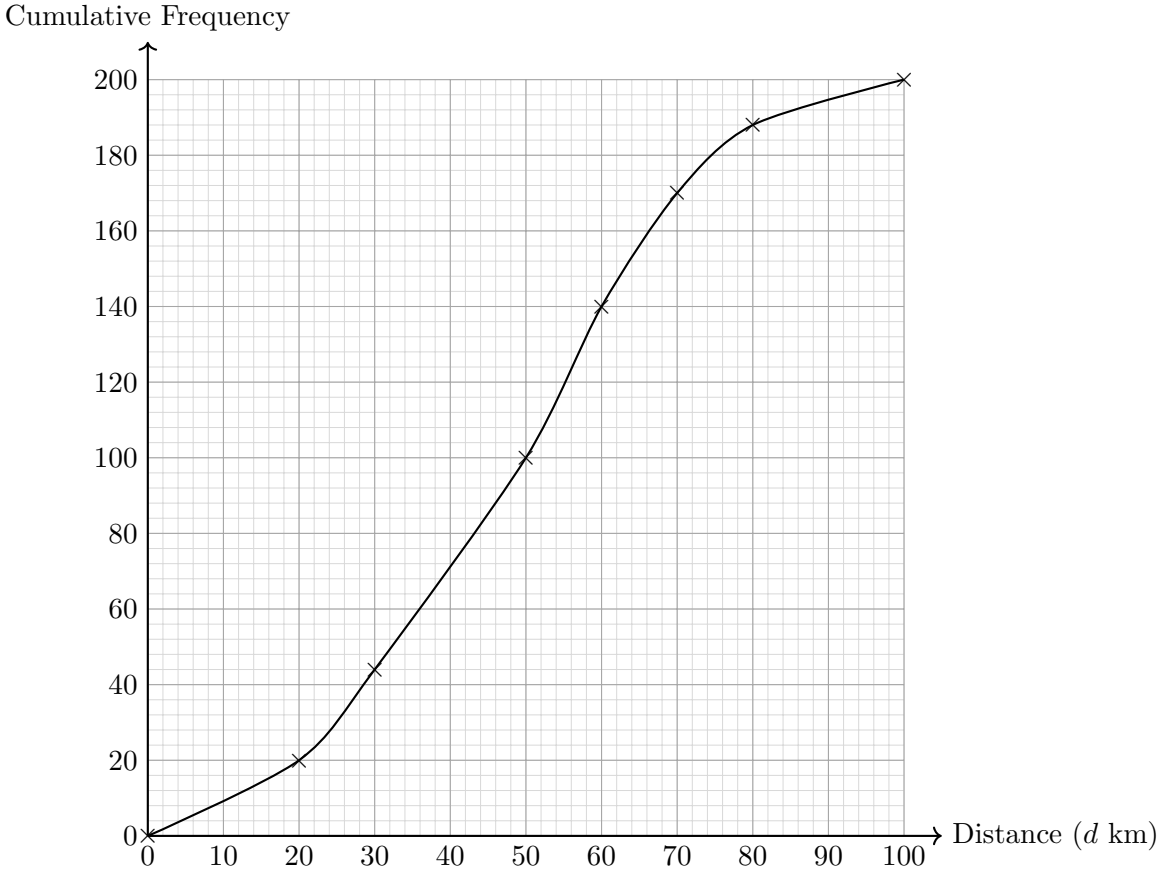
**(ii)** the 80th percentile.

..... seconds **[2]**

**(d)** The supermarket manager sets a target that at least 85% of customers must wait no more than 115 seconds.
Use your diagram to determine whether this target is met, showing your working clearly.

..... **[3]**

**1632** The cumulative frequency diagram shows information about the distance, $d$ kilometres, travelled by 200 delivery vans in a single day.




**(a)** Find an estimate for the 60th percentile.

..... km **[2]**

**(b)** 15% of the vans travelled further than $k$ km.
Find the value of $k$.

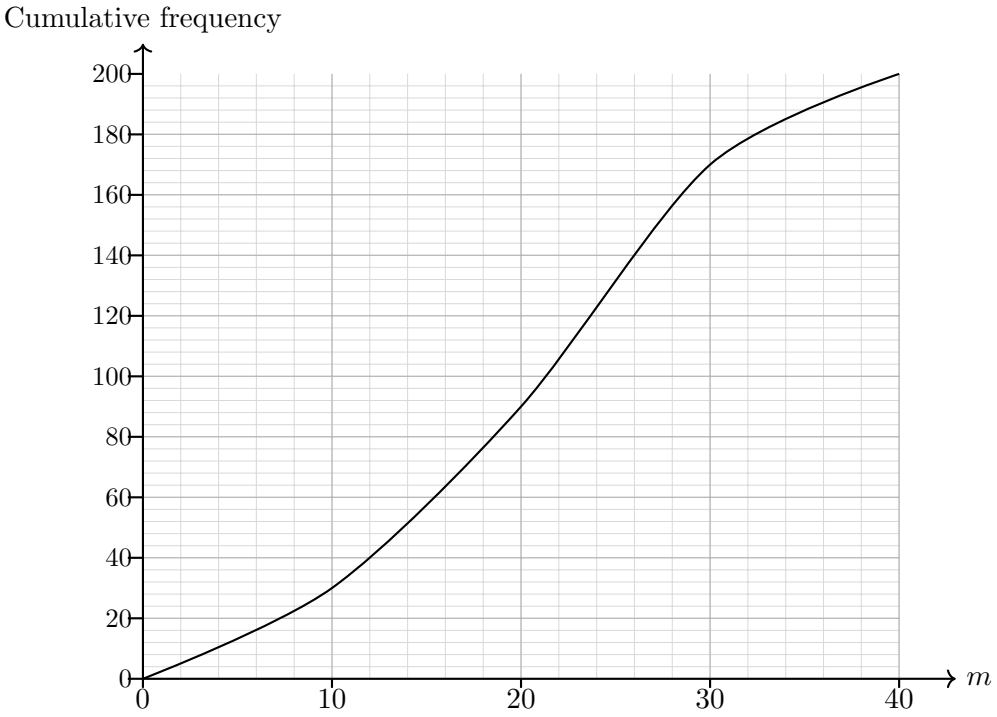
$k =$ ..... **[2]**

**(c)** Use the cumulative frequency diagram to complete the frequency table.

| Distance ($d$ km) | $0 < d \leq 30$ | $30 < d \leq 50$ | $50 < d \leq 80$ | $80 < d \leq 100$ |
|---|---|---|---|---|
| Frequency | | | | |

**[3]**

**1633** The cumulative frequency diagram shows information about the masses, $m$ grams, of 200 parcels.




**(a)** Find an estimate for the 80th percentile.

..... grams **[2]**

**(b)** Use the cumulative frequency diagram to complete the frequency table.

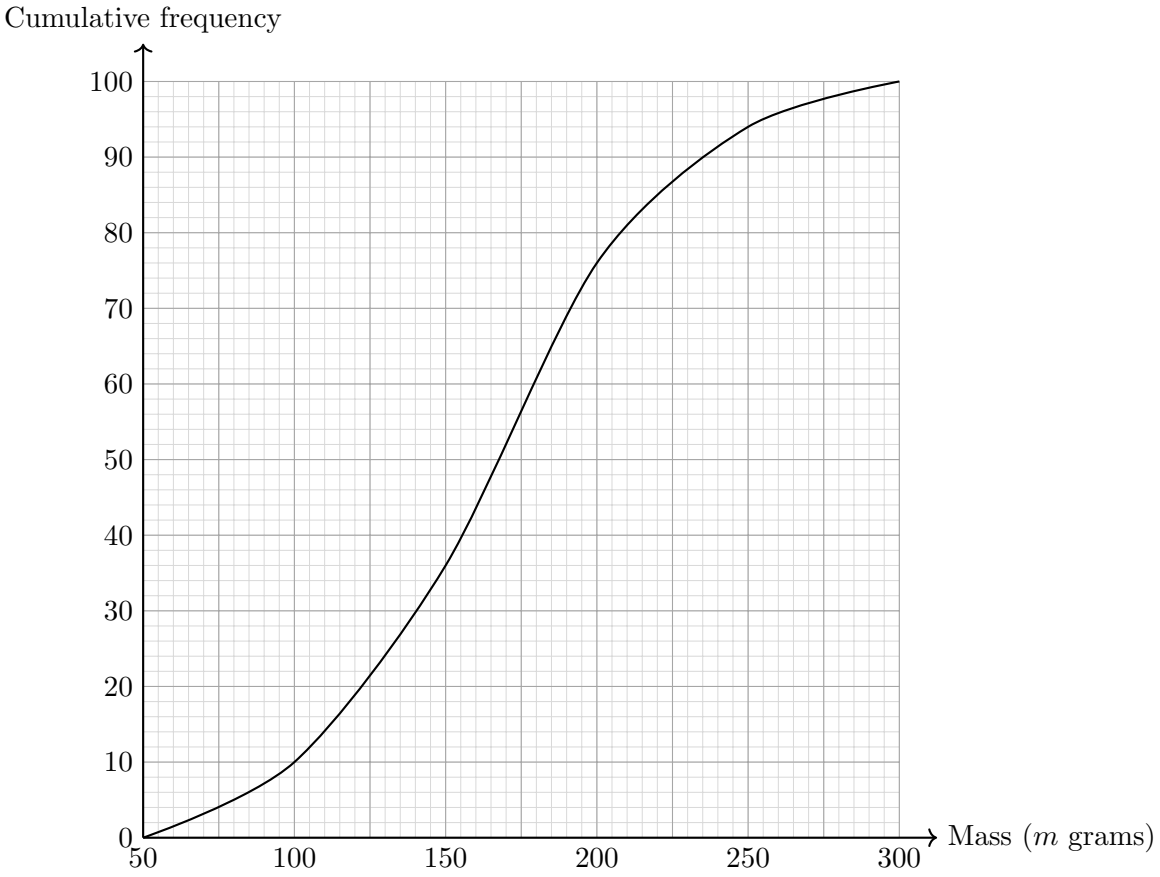
| Mass ($m$ grams) | $0 < m \leq 10$ | $10 < m \leq 20$ | $20 < m \leq 30$ | $30 < m \leq 40$ |
|---|---|---|---|---|
| Frequency | | | | |

**[2]**

**(c)** Calculate an estimate of the mean mass.

..... grams **[4]**

**1634** The cumulative frequency curve shows information about the masses, $m$ grams, of 100 potatoes.




**(a)** Use the curve to estimate the number of potatoes with a mass greater than 220 grams.

..... **[3]**

**(b)** The curve was drawn using grouped data.
Find the values of $p$, $q$, and $r$.

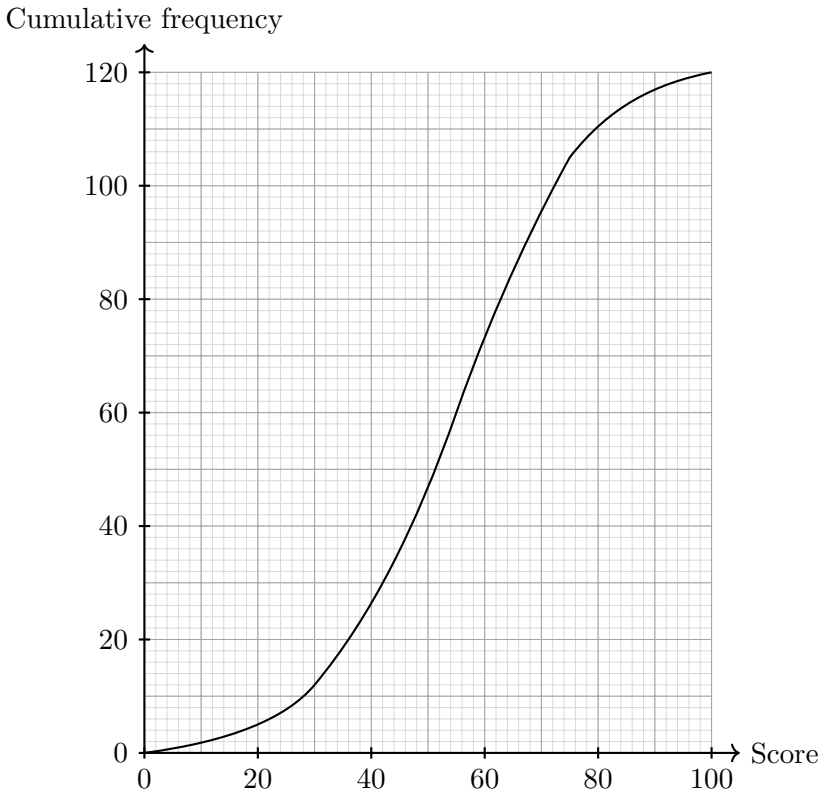
| Mass ($m$ grams) | $50 < m \leq 100$ | $100 < m \leq 150$ | $150 < m \leq 200$ | $200 < m \leq 250$ | $250 < m \leq 300$ |
|---|---|---|---|---|---|
| Frequency | $p$ | $q$ | 40 | $r$ | 6 |

$p =$ ..... , $q =$ ..... , $r =$ ..... **[3]**

**(c)** Calculate an estimate of the mean mass of the potatoes.

..... $g$ **[3]**

**1635** The cumulative frequency diagram shows information about the test scores of 120 students in Mathematics.
The maximum possible score is 100.




**(a)** Find an estimate for the interquartile range of the Mathematics scores.

..... **[2]**

**(b)** The top 15% of students are awarded a distinction.
Use the diagram to estimate the minimum score needed for a distinction.

..... **[2]**

**(c)** The same 120 students also took a Physics test.
For the Physics scores, the median is 62 and the interquartile range is 16.
Make two comparisons between the Mathematics scores and the Physics scores.

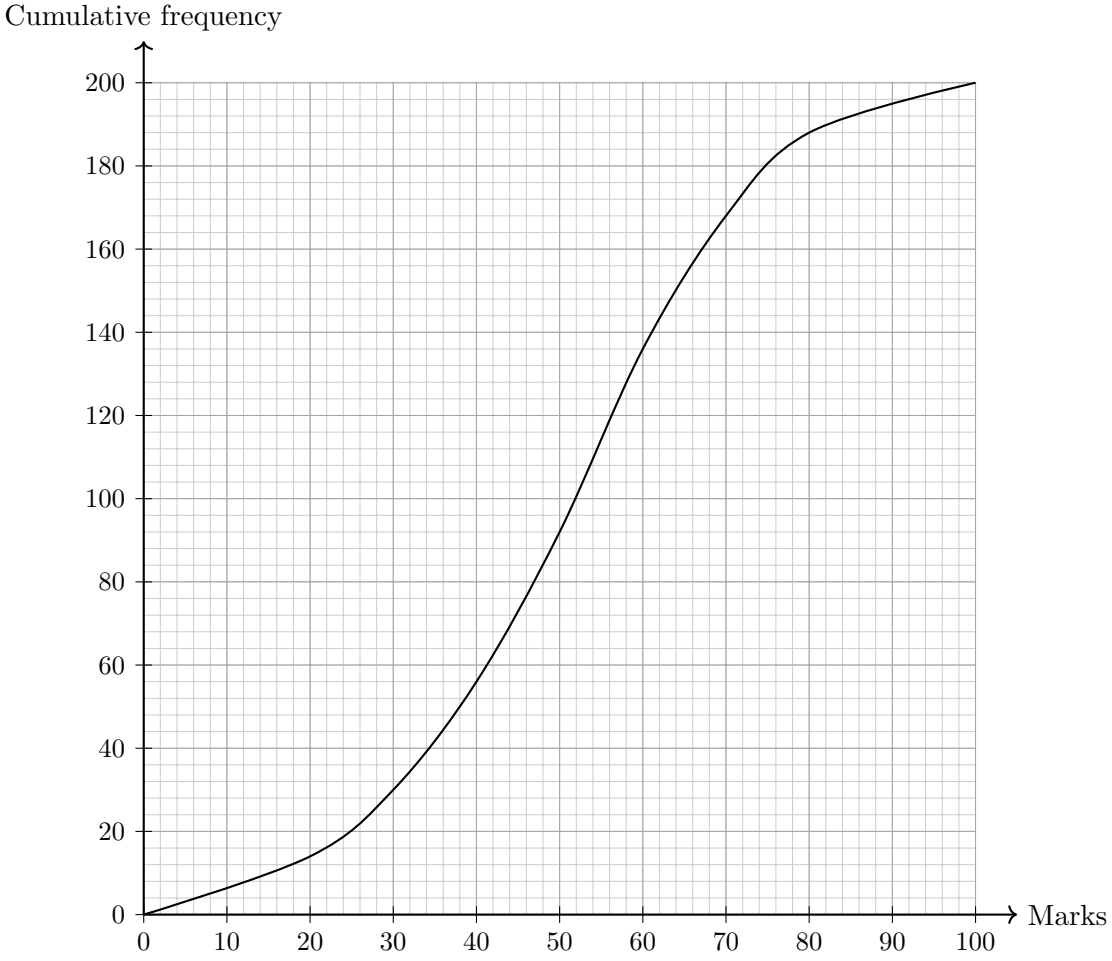
.....

..... **[2]**

**(d)** Two of the 120 students are chosen at random.
Using the cumulative frequency diagram, calculate an estimate for the probability that both students scored more than 60 in Mathematics.

..... **[3]**

**1636** The cumulative frequency diagram shows the distribution of marks scored by 200 candidates in an examination.




**(a)** Use your graph to estimate the interquartile range.

..... **[2]**

**(b)** 15% of the candidates score less than $m$ marks.
Find the value of $m$.

$m =$ ..... **[2]**

**(c)** Candidates scoring more than $P$ marks are awarded a Distinction.
Given that 30 candidates are awarded a Distinction, find the value of $P$.

$P =$ ..... **[2]**

**(d)** Candidates who score less than 45 marks have to take a supplementary test.
Calculate the percentage of candidates who do NOT have to take this test.

..... % **[4]**